



The DFSA Rulebook

Prudential – Investment, Insurance
Intermediation and
Banking Module

(PIB)

Contents

The contents of this module are divided into the following chapters, sections and appendices:

1	APPLICATION, INTERPRETATION AND CATEGORISATION	1
1.1	Application	1
1.2	Glossary for PIB	7
1.3	Categories of Authorised Firms	21
1.4	Systemically Important Banks (SIBs)	24
2	GENERAL REQUIREMENTS	27
	Introduction	27
2.1	Application	27
2.2	Trading Book.....	27
2.3	Reporting to the DFSA	28
2.4	Prudent valuation practices	29
2.5	Risk appetite statement.....	30
3	CAPITAL	31
	Introduction	31
	PART 1 – Application.....	31
3.1	Application	31
	PART 2 – Basic Requirements	32
3.2	Requirements.....	32
	PART 3 – Calculating the Capital Requirement	36
3.3	Capital Requirements for Categories 1 and 5.....	36
3.4	Capital Requirements for Categories 2 and 3A	36
3.5	Capital and Liquidity Requirements for Categories 3B, 3C, 3D and 4.....	37
3.6	Base Capital Requirement	39
3.7	Expenditure Based Capital Minimum	41
3.8	Risk Capital Requirement.....	45
3.8A	Stored Value Capital Requirement	46
3.8B	Transaction Based Capital Requirement	47
3.9	Capital Conservation Buffer	47
3.9A	Countercyclical Capital Buffer (CCyB).....	48
3.9B	HLA Capital Buffer	51
3.9C	Failure to meet a Capital Buffer Requirement.....	53
	PART 4 – Calculating Capital Resources	57
3.10	Application	57
3.11	Calculation of Capital Resources	57
3.12	Tier 1 capital (T1 Capital)	58
3.13	Common Equity Tier 1 capital (CET1 Capital).....	58
3.14	Additional Tier 1 capital (AT1 Capital)	67

3.15	Tier 2 capital (T2 Capital)	72
3.16	Minority interests and instruments issued by Subsidiaries	76
3.17	Qualifying Holdings outside the financial sector	79
PART 5 – Calculating the Leverage Ratio		81
3.18	Leverage Ratios	81
Minimum Leverage Ratios		83
4	CREDIT RISK	85
Introduction		85
PART 1 – Application		85
4.1	Application	85
PART 2 – Credit Risk systems and controls		87
4.2	Application of this part	87
4.3	Credit Risk management systems	87
4.4	Credit Risk strategy, policy, and procedures manual	90
4.5	Processes for credit assessment	94
PART 3 - CRCOM		98
4.6	Application	98
4.7	Simplified Approach	98
4.8	Calculation of the CRCOM	99
4.9	Methodology for measurement of Exposures	102
4.10	Categorisation of Credit Risk Exposures (CR Exposures)	106
4.11	Credit Quality Grade and External Credit Assessments	107
4.12	Risk weights	108
4.13	Credit Risk mitigation	128
4.14	Securitisation	143
4.14A	Counterparty Credit Risk	160
4.15	Concentration Risk	161
5	MARKET RISK	168
Introduction		168
5.1	Application	168
5.2	Market Risk systems and controls	169
5.3	Calculation of the Market Risk Capital Requirement	169
5.4	Interest Rate Risk Capital Requirement	170
5.5	Equity Risk Capital Requirement	170
5.6	Foreign Exchange Risk Capital Requirement	170
5.7	Commodities Risk Capital Requirement	171
5.8	Option Risk Capital Requirement	171
5.9	Collective Investment Fund Risk Capital Requirement	171
5.10	Securities Underwriting Capital Requirement	171
5.10A	Credit Valuation Adjustment Risk Capital Requirement	173
5.11	Use of internal Market Risk models	173

6 OPERATIONAL RISK.....176

Introduction176

6.1	Application	176
6.2	Risk management framework and governance	177
6.3	Risk identification and assessment	179
6.4	Risk monitoring and reporting	181
6.5	Control and mitigation	182
6.6	Information Technology (IT) systems	182
6.7	[Deleted]	183
6.8	Outsourcing.....	183
6.9	Business continuity and disaster recovery.....	184
6.10	Management of Operational Risks in trading activities	184
6.11	Operational Risk Capital Requirement	185
6.12	Professional indemnity insurance.....	186
6.13	Management of Operational Risk in Money Services	187

7 INTEREST RATE RISK IN THE NON-TRADING BOOK192

Introduction192

7.1	Application	192
7.2	Measuring Non-Trading Book interest rate risk	193
7.3	Standardised approach for interest rate risk in the Non-Trading Book (SA-IRRBB)	196
7.4	Non-Trading Book interest rate risk under chapter 10	208
7.5	Systems and controls for Non-Trading Book interest rate risk	209

8 GROUP RISK213

Introduction213

8.1	Application	213
8.2	Systems and controls requirements	214
8.3	Financial Group Capital Requirements and Financial Group Capital Resources	215
8.4	Financial Group Concentration Risk limits.....	217
8.5	Restrictions on ownership or control	217

9 LIQUIDITY RISK219

Introduction219

9.1	Application	219
9.2	Liquidity Risk policy, systems and controls.....	219
9.2A	Funding strategy, stress testing and contingency funding plan.....	228
9.3	Liquidity requirements	234

10 SUPERVISORY REVIEW AND EVALUATION PROCESSES.....240

Introduction240

10.1	Application	240
------	-------------------	-----

10.2	Overview	240
10.3	IRAP	241
10.4	ICAAP	242
10.5	SREP	243
10.6	Imposition of an Individual Capital Requirement.....	243
11	DISCLOSURE REQUIREMENTS	245
	Introduction	245
11.1	Application and general obligation of disclosure	245
11.2	Disclosure policy	246
11.3	Disclosure frequency, locations and process.....	246
APP1	CATEGORIES OF AUTHORISED FIRMS.....	248
A1.1	Categorisation of Authorised Firms	248
APP2	GENERAL REQUIREMENTS	251
A2.1	Detail in the Trading Book	251
A2.2	Trading Book policy.....	257
A2.3	Risk management systems and controls for Trading Book	259
A2.4	Reporting to the DFSA	259
A2.5	Prudent Valuation Practices	269
APP3	CAPITAL	272
A3.1	Stress and scenario testing	272
APP4	CREDIT RISK.....	274
A4.1	Credit Risk systems and controls	274
A4.2	Credit conversion factors (CCFs) for calculating Exposures.....	284
A4.3	Collateral calculations and haircuts	286
A4.4	Qualifying securities financing transactions (SFTs)	292
A4.5	[Deleted]	293
A4.6	Credit RWA - Unsettled Transactions.....	293
A4.7	[Deleted]	295
A4.8	Standardised approach to Counterparty Credit Risk (SA-CCR).....	295
A4.9	Exposures to Central Counterparties (CCPs)	316
A4.10	Securitisation	317
A4.11	Concentration Risk.....	322
A4.12	The Simplified Approach for Category 2 and 3A firms	330
APP5	MARKET RISK.....	334
A5.1	Market Risk systems and controls	334
A5.2	Interest Rate Risk Capital Requirement	336
A5.3	Equity Risk Capital Requirement.....	349
A5.4	Foreign Exchange Risk Capital Requirement	354
A5.5	Commodities Risk Capital Requirement	356
A5.6	Option Risk Capital Requirement	360
A5.7	Collective Investment Fund Risk Capital Requirement	364

A5.8	Securities Underwriting Risk Capital Requirement	366
A5.9	Use of internal models for Market Risk.....	372
A5.10	Credit Valuation Adjustment Risk Capital Requirement.....	382
APP6	CALCULATING THE OPERATIONAL RISK CAPITAL REQUIREMENT	389
A6.1	Basic Indicator Approach	390
A6.2	Standardised Approach.....	392
A6.3	Alternative Standardised Approach	393
APP7	NOT USED	396
APP8	NOT USED	397
APP9	LIQUIDITY	398
A9.1	Application for a global liquidity concession	398
A9.2	The Liquidity Coverage Ratio	399
A9.3	The Maturity Mismatch approach	415
A9.4	The Net Stable Funding Ratio (NSFR)	416
APP10	SUPERVISORY REVIEW AND Evaluation PROCESSES.....	424
A10.1	IRAP	424
A10.2	ICAAP	425
A10.3	Supervisory Review and Evaluation Process (SREP)	427
APP11	PUBLIC DISCLOSURE REQUIREMENTS.....	430

1 APPLICATION, INTERPRETATION AND CATEGORISATION

1.1 Application

- 1.1.1** (1) This module (PIB) applies to every Authorised Firm other than:
- (a) an Insurer;
 - (b) a Representative Office; and
 - (c) a Credit Rating Agency.
- (2) The Rules in PIB apply to an Authorised Firm in accordance with its Category determined under section 1.3.
- (3) Where a chapter, part or section of PIB applies to a limited scope of Categories of Authorised Firm, the term “Authorised Firm” used in those provisions is to be read accordingly.
- (4) The Rules in PIB apply to the whole business of an Authorised Firm except in relation to Client Assets and Insurance Monies that are held or controlled by an Authorised Firm which are not included in any prudential calculation.

Branches

- 1.1.2** Unless otherwise directed by the DFSA, an Authorised Firm that is a Branch is required to comply with the Rules in chapters 2 to 11 as specifically provided in “Application Table A” which forms part of this Rule.

Guidance

1. The effect of Rule 1.1.1(1) is that these Rules apply to all Authorised Firms, except those carrying on Insurance Business (that is, Insurers), that operate a Representative Office and those that are CRAs. Those Authorised Firms that are authorised to effect or carry out Contracts of Insurance should refer to the PIN module. Authorised Firms that are authorised to Operate a Representative Office should refer to the REP module.
2. These Rules apply both to Domestic Firms and, to the extent specified in Rule 1.1.2, to Authorised Firms conducting Financial Services through a Branch in the DIFC.
3. The Rules in PIB reinforce the fitness and propriety requirements found in GEN chapter 5 – Management, Systems and Controls and Principle 4 for Authorised Firms. The PIB module is set out in:
 - a. an initial chapter establishing a categorisation of firms for the application of PIB;
 - b. two general chapters setting overall requirements: General Requirements and Capital;
 - c. six chapters setting specific requirements relating to the following particular risks or issues: Credit Risk, Market Risk, Operational Risk, Interest Rate Risk in the Non-Trading Book, Group Risk and Liquidity Risk;
 - d. a chapter imposing processes for supervisory assessment, by which an Authorised Firm which is a Domestic Firm in Category 1, 2 (other than a Matched Principal) or 5 has

obligations to establish and conduct an IRAP and an ICAAP, and to provide documented assessments to the DFSA; and

- e. a final chapter imposing public disclosure requirements.

Domestic Firms

- 4. To assist Authorised Firms that are Domestic Firms there is a table “Application Table B” which sets out in general the application of the provisions in this module to different Categories of Authorised Firms. This table is for Guidance purposes only. The Rules in this module apply to Authorised Firms in accordance with this Chapter and as specified in Rules elsewhere in this module.
- 5. With regards to Authorised Firms carrying on Islamic Financial Business, there are additional matters that should be included in their report to the DFSA which are in the Islamic Finance Rules (IFR) module (see section 5.4 of IFR module).

APPLICATION TABLE A FOR AUTHORISED FIRMS THAT OPERATE AS A BRANCH IN THE DIFC

PIB Chapters	Category 1	Category 2	Category 3A	Category 3B	Category 3C	Category 3D	Category 4	Category 5
Chapter 1: Application, Interpretation and Categorisation	Whole Chapter	Whole Chapter	Whole Chapter	Whole Chapter	Whole Chapter	Whole Chapter	Whole Chapter	Whole Chapter
Chapter 2: General Requirements	Whole Chapter	Whole Chapter	Whole Chapter	Whole Chapter, except section 2.5	Whole Chapter, except section 2.5	Whole Chapter, except section 2.5	Whole Chapter, except section 2.5	Whole Chapter
Chapter 3: Capital	Only Rule 3.2.4	Only Rule 3.2.4	Only Rule 3.2.4	Only Rule 3.2.4	Only Rule 3.2.4	Only Rule 3.2.4	Only Rule 3.2.4	Only Rule 3.2.4
Chapter 4: Credit Risk	Only sections 4.1 to 4.4 and Rules 4.5.1 to 4.5.7 and 4.5.9	Only sections 4.1 to 4.4 and Rules 4.5.1 to 4.5.7 and 4.5.9	Only sections 4.1 to 4.4 and Rules 4.5.1 to 4.5.7 and 4.5.9					Only sections 4.1 to 4.4 and Rules 4.5.1 to 4.5.7 and 4.5.9
Chapter 5: Market Risk	Only sections 5.1 and 5.2	Only sections 5.1 and 5.2	Only sections 5.1 and 5.2					Only sections 5.1 and 5.2
Chapter 6: Operational Risk	Whole Chapter, except sections 6.11 and 6.12	Whole Chapter, except sections 6.11 and 6.12	Whole Chapter, except sections 6.11 and 6.12	Whole Chapter, except sections 6.10, 6.11 and 6.12	Whole Chapter, except sections 6.10, 6.11 and 6.12	Whole Chapter, except sections 6.10, 6.11 and 6.12	Whole Chapter, except sections 6.10, 6.11 and 6.12	Whole Chapter, except sections 6.11 and 6.12
Chapter 7: Interest Rate Risk In the Non-Trading Book	Only Rules 7.2.2A and 7.2.4, and section 7.5	Only Rules 7.2.2A and 7.2.4, and section 7.5						
Chapter 8: Group Risk	Only sections 8.1, 8.2 and 8.5	Only sections 8.1, 8.2 and 8.5	Only sections 8.1 and 8.5	Only sections 8.1 and 8.5	Only sections 8.1 and 8.5	Only sections 8.1 and 8.5	Only sections 8.1 and 8.5	Only sections 8.1, 8.2 and 8.5
Chapter 9: Liquidity Risk	Whole Chapter, except Rules 9.2.2(3), 9.3.12 and 9.3.13	Only Rule 9.2.2 (3)						Whole Chapter, except Rules 9.2.2(3), 9.3.12 and 9.3.13

PRUDENTIAL – INVESTMENT, INSURANCE INTERMEDIATION AND BANKING (PIB)

PIB Chapters	Category 1	Category 2	Category 3A	Category 3B	Category 3C	Category 3D	Category 4	Category 5
Chapter 10:Supervisory Review and Evaluation Processes								
Chapter 11:Disclosure Requirements								

APPLICATION TABLE B FOR AUTHORISED FIRMS THAT OPERATE AS A DOMESTIC FIRM

PIB Chapters	Category 1	Category 2	Category 3A	Category 3B	Category 3C	Category 3D	Category 4	Category 5
Chapter 1: Application, Interpretation and Categorisation	Whole Chapter	Whole Chapter	Whole Chapter	Whole Chapter	Whole Chapter	Whole Chapter	Whole Chapter	Whole Chapter
Chapter 2: General Requirements	Whole Chapter	Whole Chapter	Whole Chapter	Whole Chapter, except section 2.5	Whole Chapter, except section 2.5	Whole Chapter, except section 2.5	Whole Chapter, except section 2.5	Whole Chapter
Chapter 3:Capital	Whole Chapter, except Rules 3.2.4 and 3.2.6, and sections 3.4, 3.5 and 3.7	Whole Chapter, except Rules 3.2.4 and 3.2.6, and sections 3.3 and 3.5	Whole Chapter, except Rules 3.2.4 and 3.2.6, except sections 3.3, 3.5, 3.9, 3.9A, 3.9B, 3.9C and 3.18	Whole Chapter, except Rule 3.2.4 and sections 3.3, 3.4, 3.8, 3.9, 3.9A, 3.9B, 3.9C and 3.18	Whole Chapter, except Rule 3.2.4 and sections 3.3, 3.4, 3.8, 3.9, 3.9A, 3.9B, 3.9C and 3.18	Whole Chapter, except Rule 3.2.4 and sections 3.3, 3.4, 3.8, 3.9, 3.9A, 3.9B, 3.9C and 3.18	Whole Chapter, except Rule 3.2.4 and sections 3.3, 3.4, 3.8, 3.9, 3.9A, 3.9B, 3.9C and 3.18	Whole Chapter, except Rules 3.2.4 and 3.2.6, and sections 3.4, 3.5 and 3.7
Chapter 4:Credit Risk	Whole Chapter	Whole Chapter	Whole Chapter					Whole Chapter
Chapter 5:Market Risk	Whole Chapter	Whole Chapter	Only sections 5.1, 5.2 and 5.6					Whole Chapter, except section 5.4
Chapter 6:Operational Risk	Whole Chapter, except section 6.12	Whole Chapter, except section 6.12	Whole Chapter, except section 6.12	Whole Chapter, except sections 6.10 and 6.11	Whole Chapter, except sections 6.10 and 6.11	Whole Chapter, except sections 6.10 and 6.11	Whole Chapter, except sections 6.10 and 6.11	Whole Chapter, except section 6.12
Chapter 7:Interest Rate Risk In the Non-Trading Book	Whole Chapter, except Rule 7.2.2A	Only Rules 7.2.2A and 7.2.4, and section 7.5						

PIB Chapters	Category 1	Category 2	Category 3A	Category 3B	Category 3C	Category 3D	Category 4	Category 5
Chapter 8:Group Risk	Whole Chapter	Whole Chapter	Only sections 8.1 and 8.5	Only sections 8.1 and 8.5	Only sections 8.1 and 8.5	Only sections 8.1 and 8.5	Only sections 8.1 and 8.5	Whole Chapter
Chapter 9:Liquidity Risk	Whole Chapter, except Rule 9.2.2(3)	Only Rule 9.2.2 (3).						Whole Chapter, except Rule 9.2.2(3)
Chapter 10:Supervisory Review and Evaluation Processes	Whole Chapter	Whole Chapter						Whole Chapter
Chapter 11:Disclosure Requirements	Whole Chapter	Whole Chapter						Whole Chapter

1.2 Glossary for PIB

Guidance

Set out under Rule 1.2.1 are a number of mainly technical definitions used solely in PIB. Such definitions do not also appear in GLO unless they are used elsewhere in the Rulebook. GLO also contains definitions of abbreviations, terms and phrases used in PIB and those are also included in 1.2.1 for convenience purposes where such definitions are embedded in PIB specific definitions. Commonly used definitions such as “Authorised Firms”, “Domestic Firms”, and “Financial Services” appear only in GLO.

1.2.1 The following terms and abbreviations bear the following meanings for the purpose of this module:

Acquisition of land, development and construction (ADC) Exposure	An Exposure to a company or SPV that is financing any land acquisition for development and construction purposes or development and construction of any residential or commercial property.
Alternative Standardised Approach	The manner in which the Operation Risk Capital Requirement is calculated in accordance with sections 6.11 and A6.3.
Annual Audited Expenditure	The expenditure calculated in accordance with Rule 3.7.3 or, if applicable, Rule 3.7.3A.
Asset-Backed Commercial Paper (ABCP) Programme	A programme that predominately issues commercial paper with an Original Maturity of one year or less that is backed by assets or other Exposures held in a bankruptcy-remote SPE.
AT1	Additional tier 1
AT1 Capital	Has the meaning given in section 3.14.
Available Stable Funding (ASF)	The amount, calculated in accordance with Rule A9.4.1, representing the relative stability of an Authorised Firm’s available funding sources.
ASF Category	The applicable category, listed in the table to Rule A9.4.1, to which the carrying value of a liability or capital instrument of an Authorised Firm is assigned.
ASF Factor	The applicable factor, listed in the table to Rule A9.4.1, used to multiply the carrying value of a liability or capital instrument of an Authorised Firm.
Base Capital Requirement	Has the meaning given in section 3.6.
Basel Committee	The Basel Committee on Banking Supervision.

Basic Indicator Approach	The manner in which the Operational Risk Capital Requirement is calculated in accordance with sections 6.11 and A6.1.
Capital Buffer	In relation to an Authorised Firm, means the sum of the following (to the extent applicable to the firm): (a) Capital Conservation Buffer; (b) Countercyclical Capital Buffer; and (c) HLA Capital Buffer.
Capital Buffer Requirement	Means a requirement to maintain any one or more of the following: (a) a Capital Conservation Buffer; (b) a Countercyclical Capital Buffer; or (c) an HLA Capital Buffer.
Capital Conservation Buffer	The Capital Buffer that an Authorised Firm must maintain under section 3.9.
Capital Conservation Buffer Requirement	The requirement to maintain a Capital Conservation Buffer under section 3.9.
Capital Requirement	The amount of capital an Authorised Firm must hold, calculated in accordance with sections 3.3, 3.4, 3.5, 3.9, 3.9A or 3.9B, as applicable.
Capital Resources	The total Capital Resources of an Authorised Firm calculated in accordance with section 3.11.
Category	A prudential grouping of Authorised Firms which determines the application of the Rules in this Module.
CCF	Credit conversion factor
CCP	Central Counterparty
CCR Exposure	The Exposure value or amount for a Counterparty Credit Risk Exposure.
CET1	Common equity tier 1
CET1 Capital	Has the meaning given in section 3.13.

Clean-Up Call	An option that permits the SE Exposures (e.g. asset-backed Securities) to be called before all of the underlying Exposures or SE Exposures have been repaid. In the case of Traditional Securitisations, this is generally accomplished by repurchasing the remaining SE Exposures once the pool balance or outstanding Securities have fallen below some specified level. In the case of a synthetic Exposure, the Clean-Up Call may take the form of a clause that extinguishes the credit protection.
Close Links	A Person (Person A) has Close Links with a Person (Person B) if: (a) Person B: (i) is a Holding Company of Person A; (ii) is a Subsidiary of Person A; (iii) is a Holding Company of the Subsidiary of Person A; (iv) is a Subsidiary of a Holding Company of Person A; or (v) owns and controls 20% or more of the voting rights or shares of Person A; or (b) Person A owns and controls 20% or more of the voting rights or shares of Person B.
Closely Related	Has the meaning given in Rule A4.11.5.
Collateral	Any form of asset, guarantee, or indemnity which is held or controlled by an Authorised Firm and is subject to a security interest or arrangement in favour of that firm.
Collective Investment Fund Risk Capital Requirement	A component of Market Risk Capital Requirement calculated in accordance with section 5.9.
Commercial Real Estate Exposure	A Real Estate Exposure that is not a Residential Real Estate Exposure or an ADC Exposure.
Commodities Risk Capital Requirement	A component of the Market Risk Capital Requirement to cover the risk of holding or taking positions in commodities, including precious metals, but excluding gold, calculated in accordance with section 5.7.
Concentration Risk	The risk faced by an Authorised Firm arising out of its Large Exposures.

Connected	In relation to a Person (A), a Person which has or has at any relevant time had the following relationship to A: (a) a member of A's Group; (b) a Controller of A; (c) a member of a partnership of which A is a member; (d) an employee or former employee of A; (e) if A is a company: (i) an officer or manager of A or of a parent of A; (ii) an agent of A or of a parent of A; (f) if A is a partnership is or has been a member, manager or agent of A; or (g) if A is an unincorporated association of persons which is not a partnership, is or has been an officer, manager or agent of A.
Connected Counterparties	Has the meaning given in Rule A4.11.7.
Contingency Funding Plan (CFP)	The plan referred to in Rule 9.2A.6.
Controlled Early Amortisation	Early Amortisation that meets the conditions in Rule 4.14.58.
Countercyclical Capital Buffer (CCyB)	The countercyclical capital buffer that an Authorised Firm must maintain under section 3.9A.
CCyB Authority	(1) For the State, means the Central Bank; and (2) For any other jurisdiction, means the authority in that jurisdiction responsible for setting CCyB Rates.
CCyB Rate	A rate expressed as a percentage of Risk Weighted Assets that applies under Rules 3.9A.7 to 3.9A.9.
CCyB Requirement	The requirement to maintain a Countercyclical Capital Buffer under section 3.9A.
Counterparty	Means any person with or for whom an Authorised Firm carries on, or intends to carry on, any regulated business or associated business. In this context, Counterparty includes an individual, unincorporated body, company, government, local authority or other public body.
Counterparty Credit Risk	The risk that the Counterparty to a transaction defaults before the final settlement of the transaction's cash flows.
CR Exposure	The Exposure value or amount for a Credit Risk Exposure.

Credit Derivative	Any contract which transfers the Credit Risk of a reference obligation or set of reference obligations from the protection buyer to the protection seller, such that the protection seller has an Exposure to the reference obligation(s).
Credit Enhancement	A contractual arrangement in which the Authorised Firm retains or assumes an SE Exposure and, in substance, provides some degree of added protection to other parties to the transaction.
Credit Quality Grade	A credit quality step in a credit quality assessment scale. A credit quality assessment scale is a scale onto the credit assessments of an ECAI or an expert credit agency are mapped.
Credit Risk	In relation to an Authorised Firm, the risk of loss if another party fails to perform on its financial obligation to the Authorised Firm.
Credit Risk Capital Requirement (CRCOM)	The requirement calculated in accordance with section 4.6.
Credit Valuation Adjustment Risk Capital Requirement	A component of the Market Risk Capital Requirement calculated in accordance with section 5.10A to cover the risk of losses arising from changing credit valuation adjustment values in response to changes in Counterparty credit spreads and market risk factors that drive prices of derivative transactions and SFTs.
Credit-Enhancing Interest-Only Strip	An on-balance sheet asset that: (i) represents a valuation of cash flows related to future margin income; and (ii) is subordinated.
CRW	Credit risk weight for an Exposure.
CV	Contracted value for delivery.
Delta	The measure of an Option's sensitivity to a change in value of the underlying investment, asset or property.
Displaced Commercial Risk Capital Requirement	The requirement calculated in accordance with chapter 5 of the IFR Module.
Domestic Systemically Important Bank (D-SIB)	An Authorised Firm designated by the DFSA under section 1.4 as a domestic systemically important bank.
Duration Method	A measure of General Market Risk calculated in accordance with Rule A5.2.19.

E	An Exposure value or amount
E*	An Exposure value or amount adjusted in the manner provided in the relevant Rule
EAE	The Exposure value or amount for an Early Amortisation Exposure
Early Amortisation	A mechanism that, once triggered, allows investors to be paid out prior to the originally stated maturity of the securities issued. For risk-based capital purposes, an Early Amortisation provision will be considered either controlled or non-controlled.
ECAI	A CRA or an external credit rating agency approved by the DFSA for the purpose of this Module.
Equity Risk Capital Requirement	A component of Market Risk Capital Requirement and calculated in accordance with section 5.5.
Evergreening	Evergreening refers to the practice by some banks to roll over or renew their non-performing loans or potentially nonperforming loans, so that they can avoid recognising them as non-performing loans in their accounts and consequently avoid provisioning for them.
Excess Spread	Gross finance charge collections and other income received by the trust or SPE minus certificate interest, servicing fees, charge-offs, and other senior trust or SPE expenses.
Expenditure Based Capital Minimum	A Capital Requirement calculated in accordance with section 3.7.
Exposure	The maximum loss that an Authorised Firm (and where applicable its PSIA holders) might suffer if: (a) a Counterparty or a group of Connected Counterparties fail to meet their obligations; or (b) it realises assets or off-balance sheet positions. An Exposure also includes any asset or off-balance sheet item, which could result in a potential loss to the Authorised Firm due to Market Risk or Operational Risk or any other risk factor.
FCCA	Financial Collateral Comprehensive Approach as described in Rule 4.9.5.
FCSA	Financial Collateral Simplified Approach as described in Rule 4.9.5.

Financial Group	A group of entities which includes an Authorised Firm and: (a) any Parent incorporated in the DIFC; (b) any Financial Institution subsidiaries (whether direct or indirect) of the Parent or Parents in (a) or of the Authorised Firm; (c) any Financial Institution in which the Parent or Parents in (a), the Financial Institution subsidiaries in (b) or the Authorised Firm (whether direct or indirect) hold 20% or more of the voting rights or capital; and (d) any entity which the DFSA directs the Authorised Firm to include in accordance with Rule 8.1.2 or PIN Rule 8.1.2.
Financial Group Capital Requirement	The Capital Requirement of a Financial Group calculated in accordance with Rule 8.3.3 or PIN Rule 8.3.3.
Financial Group Capital Resources	The Capital Resources of a Financial Group calculated in accordance with Rule 8.3.4 or PIN 8.3.4.
First Loss Position	Represents the first level of support provided to the special purpose entity or vehicle that should bear all, or a significant part of, the risk associated with the items held by the special purpose entity or vehicle.
Foreign Exchange Risk Capital Requirement	A component of the Market Risk Capital Requirement and as calculated in accordance with section 5.6.
Gamma	The measure of the rate of change of Delta.
General Market Risk	(1) For the purposes of the Interest Rate Risk Capital Requirement, means the risk that losses may arise from price changes in Securities caused by parallel or non-parallel shifts in the yield curve or from price movements in the equity market for a given country; (2) For the purposes of the Equity Risk Capital Requirement, means the risk that losses may arise from a price movement in the equity market for a given country; or (3) For the purposes of internal models, means both of the above risks.
Global Systemically Important Bank (G-SIB)	An Authorised Firm designated by the DFSA under section 1.4 as a global systemically important bank.

Group	Means a group of entities which includes an entity (the 'first entity') and: (a) any Parent of the first entity; and (b) any subsidiaries (direct or indirect) of the Parent or Parents in (a) or the first entity.
Group Risk	The risk of loss to the Authorised Firm as a result of its membership of, or linkages within a Group.
High Quality Liquid Assets (HQLA)	Liquid assets that meet the conditions in Rules A9.2.2 to A9.2.9 of App 9.
HLA Capital Buffer	A higher loss absorbency capital buffer that a SIB must maintain under section 3.9B.
HLA Capital Buffer Requirement	The requirement to maintain a higher loss absorbency capital buffer under section 3.9B.
HLA Ratio	In relation to a SIB, means the ratio determined by the DFSA under Rule 3.9B.6 for that SIB.
ICR (Individual Capital Requirement)	Individual capital requirement.
Implicit Support	Arises when an Authorised Firm provides support to a securitisation in excess of its predetermined contractual obligation.
Individual Liquidity Requirement	An individual liquidity requirement imposed by the DFSA under Rule 9.3.6.
Interest Rate Risk Capital Requirement	A component of Market Risk Capital Requirement and calculated in accordance with section 5.4.
Internal Capital Adequacy Assessment Process (ICAAP)	The internal capital adequacy assessment process prescribed in chapter 10.
Internal Risk Assessment Process (IRAP)	The internal risk assessment process prescribed in section 10.3.
Investment Grade	A credit rating which is a Credit Quality Grade of 1, 2 or 3.
Large Exposure	Has the meaning given in Rule 4.15.4.
LCR Requirement	The LCR required to be maintained by an Authorised Firm under Rule 9.3.4 or, if applicable, under Rule 9.3.6.
Leverage Ratio	The amount expressed as a percentage that is calculated in accordance with the Rules in section 3.18.

Leverage Ratio Exposure Measure	The value of an Authorised Firm's exposures calculated in accordance with Rule 3.18.3 to determine its Leverage Ratio.
Liquidity Coverage Ratio (LCR)	The amount expressed as a percentage that is calculated in accordance with Rule 9.3.5 and section A9.2 of App 9.
Liquidity Risk	The risk of loss to an Authorised Firm as a result of inability to meet its obligations as they fall due.
Market Risk	The risk of loss that arises from fluctuations in the values of, or income from, assets or in interest or exchange rates.
Market Risk Capital Requirement	The requirement calculated in accordance with Rule 5.3.1.
Maturity Ladder	A table that ordinaly ranks the maturity time bands and assets and liabilities within them.
Maturity Method	This is an advanced approach that an Authorised Firm may use to measure the risk of holding or taking positions in debt Securities and other interest rate-related instruments, calculated in accordance with Rule A5.2.17.
MDB	Multilateral Development Bank
Modified Duration	The time period calculation for the purposes of the Duration Method in accordance with Rule A5.2.21.
Multilateral Development Bank	An institution created by a group of countries that: (a) provides financing and professional advice for economic and social development projects; (b) has a large sovereign membership; and (c) has its own independent legal and operational status.
MV	Market value
Net Stable Funding Ratio (NSFR)	The amount, expressed as a percentage, calculated in accordance with Rule 9.3.12, or, if applicable, Rule 9.3.6.
Netting	A process by which the claims and obligations between two Counterparties are offset against each other to leave a single net sum.
Netting Set	Has the meaning given in PIB Rule A4.8.1(g).

Non-Financial Private Sector Credit Exposure	A Credit Risk Exposure to a party other than: (a) a sovereign; (b) a regional, provincial or municipal government; (c) a public sector entity; (d) a multilateral development bank; or (e) a bank.
Non-Trading Book	Describes positions, Exposures and on-and off-balance sheet items, which are not in the Trading Book.
NP	Nominal principal amount.
NSFR Requirement	The requirement for an Authorised Firm to maintain a minimum NSFR under Rule 9.3.12(1) or, if applicable, Rule 9.3.6.
OBS	Off-balance sheet.
OBS Exposure	An Exposure of an Authorised Firm that is off-balance sheet.
OBS-RSF Category	A category, listed in Table 2 to Rule A9.4.2, to which carrying value of an OBS Exposure (or potential liquidity Exposure) is assigned.
OBS-RSF Factor	The factor, listed in Table 2 to Rule A9.4.2, used to multiply the carrying value of an OBS Exposure (or a potential liquidity Exposure).
Operational Risk	The risk of loss resulting from inadequate or failed internal processes, people and systems or from external events, including legal risk.
Operational Risk Capital Requirement	The requirement calculated in accordance with section 6.11.
Option Risk Capital Requirement	A component of the Market Risk Capital Requirement and calculated in accordance with section 5.8.
Original Maturity	(1) The time period between the date an offer is made and the date it expires or lapses. (2) In relation to Debentures, the interval between its issue date and the date on which it becomes due and payable.

Originator	(a) an entity which, either itself or through related entities, directly or indirectly, creates the Exposure being securitised; or (b) any entity which purchases or advises or causes an SPE to purchase the Exposures of a third party, which are then used in a securitisation (for avoidance of doubt, selling credit protection such that the entity or the SPE has a long position in the Credit Risk of the obligor is equivalent to purchasing Exposures); Where an entity lends to an SPE with a view to enabling that SPE to make loans which are then used in a securitisation, the entity will generally be deemed to be acting as an Originator.
OTC	Over the counter
Other Real Estate Exposure	A Real Estate Exposure that is not a Regulatory Real Estate Exposure or an ADC Exposure.
PSE	Public Sector Entity
PSIA	Profit sharing investment account.
Qualifying Bilateral Netting Agreement	A Netting agreement that meets the requirements in Rule A4.3.4.
Qualifying CCP	Means a CCP: (a) that is operated by an Authorised Person in accordance with its Licence; or (b) in respect of which all of the following conditions are met: (i) the CCP is authorised and supervised by a Financial Services Regulator in a jurisdiction other than the DIFC; (ii) the Financial Services Regulator has publicly stated that it applies on an ongoing basis rules and regulations that are consistent with the CPSS-IOSCO Principles for Financial Market Infrastructures; and (iii) the CCP satisfies the Basel Committee requirements for calculating its capital requirements for its default fund exposures.
Qualifying Holding	Any holding in the capital of a non-financial Undertaking of which the Authorised Firm is a controller.
Related Person	Has the meaning given in Rule 4.4.6.
Real Estate Exposure	An Exposure secured by a lien on immovable property.

Regulatory Commercial Real Estate Exposure	A Commercial Real Estate Exposure that is a Regulatory Real Estate Exposure.
Regulatory Real Estate Exposure	A Real Estate Exposure that meets the conditions in Rule 4.12.20.
Regulatory Residential Real Estate Exposure	A Residential Real Estate Exposure that is a Regulatory Real Estate Exposure.
Relevant Entity	Means any of the following: (a) another Authorised Firm; (b) a Regulated Financial Institution; (c) any unregulated Financial Institution; or (d) any financial holding company.
Required Stable Funding (RSF)	The amount, calculated in accordance with Rule A9.4.2, representing the stable funding required by an Authorised Firm.
Residential Real Estate	Property that is, or will be, a dwelling and that satisfies all applicable laws and regulations enabling the property to be occupied for housing purposes.
Residential Real Estate Exposure	A Real Estate Exposure secured by Residential Real Estate.
Retail Exposure	An Exposure to: (a) one or more individuals; or (b) a SME that meets the requirements in Rule 4.12.16(2).
RSF Category	A category, listed in Table 1 to Rule A9.4.2, to which the carrying value of an Authorised Firm's asset is assigned.
RSF Factor	The applicable factor, listed in Table 1 to Rule A9.4.2, used to multiply the carrying value of an asset of an Authorised Firm.
Re-securitisation	Has the meaning given in section 4.14.
Restricted PSIA (PSIAr)	A PSIA in respect of the investment account holder imposes certain restrictions as to where, how and for what purpose his funds are to be invested.

Revolving Securitisation	A Traditional or Synthetic Securitisation in which the specified items consist of revolving assets such as loan facilities or credit card balances which permit borrowers to vary the drawn amount within an agreed limit, or the scheme itself is revolving.
Rho	The measure of an Option's sensitivity to a change in interest rates.
Risk Capital Requirement	Has the meaning given in Rule 3.8.1A.
Risk Weighted Assets (RWA)	The risk weighted assets of an Authorised Firm calculated in accordance with Rule 3.8.2.
SE Exposure	The Exposure value or amount for a securitisation Exposure.
Securities Underwriting Capital Requirement	A component of the Market Risk Capital Requirement defined in section 5.10.
Servicer	A Person that administers the securitised items.
SFT	Securities financing transactions, such as repurchase agreements, reverse repurchase agreements, security lending and borrowing and margin lending transactions, where the value of the transactions are often subject to margin agreements.
Simplified Approach	An alternative application of the provisions of chapter 4 for an Authorised Firm in Category 2 and 3A, as described in sections 4.7 and A4.12.
Special Purpose Entity (SPE)	A corporation, trust, or other entity organised for a specific purpose, the activities of which are limited to those appropriate to accomplish the purpose of the SPE, and the structure of which is intended to isolate the SPE from the Credit Risk of an Originator or seller of Exposures. SPEs are commonly used as financing vehicles in which Exposures are sold to a trust or similar entity in exchange for cash or other assets funded by debt issued by the trust.
Specific Risk	The risk that losses on an Authorised Firm's net long or short position in an individual equity or Security may arise from a negative or positive price movement of that equity or Security relative to the relevant market generally.
Sponsor	An Authorised Firm that repackages third party assets directly into a securitisation scheme. Where an Authorised Firm repackages non-Investment Grade third party assets, it may fall within the definition of an Originator unless it originates or repackages no more than 10% of the scheme's total assets.

Standardised Approach	The manner in which the Operational Risk Capital Requirement is calculated in accordance with sections 6.11 and A6.2.
Stored Value Capital Requirement	Has the meaning given in section 3.8A.
Supervisory Review and Evaluation Process (SREP)	The supervisory review and evaluation process prescribed in chapter 10.
Synthetic Securitisation	Has the meaning given in Rule 4.14.2(b).
Systemically Important Bank (SIB)	An Authorised Firm designated by the DFSA under section 1.4 to be either or both of the following: (a) a D-SIB; or (b) a G-SIB.
T	Trade date, which is the date on which a transaction is entered into.
T1	Tier 1
T1 Capital	Has the meaning given in Rule 3.12.1.
T2	Tier 2
T2 Capital	Has the meaning given in Rule 3.15.1.
Theta	The ratio of the change in an Option price to the decrease in time to expiration. Theta can also be referred to as time decay.
Total Net Cash Outflow	Has the meaning given in Rule A9.2.13
Total Return Swap	A contract under which two parties exchange their positive or negative returns on a notional amount of a reference asset for a specified period of time.
Trading Book	The positions and Exposures including, on and off-balance sheet items eligible for inclusion in the Trading Book, as described in section 2.2.
Traditional Securitisation	Has the meaning given in Rule 4.14.2(a).
Transaction Based Capital Requirement	Has the meaning given in section 3.8B.

Transactor	An obligor in relation to: (a) facilities such as credit cards and charge cards where the balance has been repaid in full at each scheduled repayment date for the previous 12 months; or (b) overdraft facilities if there has been no drawdown over the previous 12 months.
Unrestricted PSIA (PSIAu)	A PSIA in respect of which the investment account holder authorises the Authorised Firm to invest the account holder's funds in a manner which the Authorised Firm deems appropriate without laying down any restrictions as to where, how and for what purpose the funds should be invested.
Unsettled Transaction	A transaction where: (a) delivery of an instrument is due to take place against the receipt of cash but remains outstanding; (b) physical delivery of the underlying asset or security is not required and the final cash settlement of the transaction remains outstanding past its due date; or (c) deliverables were delivered without receipt of the corresponding cash payment.
Vega	The measure of the sensitivity of the value of the Option to a change in the volatility of the underlying asset.
Walkaway Clause	A provision which permits a non-defaulting party to make payments, or no payments at all, to the estate of the defaulter, even if the defaulter is a net creditor.

1.3 Categories of Authorised Firms

Guidance

1. Authorised Firms are divided into Categories to provide a clear framework for determining which specific Rules in PIB apply to each Authorised Firm. The Rules in this section enable an Authorised Firm to determine into which Category it falls.
2. The table in A1.1 of App1 sets out the categorisation process diagrammatically. In that table, an emboldened box indicates the Financial Service that is determinative of the Category into which an Authorised Firm falls. An Authorised Firm may, if authorised under its Licence to do so, conduct any number of Financial Services specified under any lower Category than the one that applies to the Authorised Firm in accordance with this section (For this purpose Category 5 is considered to be equivalent to Category 1). For example, a Category 1 firm could conduct any of the Financial Services specified under Categories 2, 3A, 3B, 3C, 3D or 4 (if authorised to do

so). However, a Category 4 firm may only conduct any of the Financial Services listed under Category 4 for which it is authorised.

Category 1

1.3.1 An Authorised Firm is in Category 1 if:

- (a) its Licence authorises it to carry on one or more of the Financial Services of Accepting Deposits or Managing a PSIAu; and
- (b) it does not meet the criteria of Category 5.

Guidance

A Category 1 Authorised Firm may be authorised to conduct other Financial Services, but it is the authorisation for Accepting Deposits or Managing a PSIAu that is determinative of its belonging to Category 1.

Category 2

1.3.2 An Authorised Firm is in Category 2 if:

- (a) its Licence authorises it to carry on one or more of the Financial Services of Providing Credit or Dealing in Investments as Principal; and
- (b) it does not meet the criteria of Categories 1 or 5.

Guidance

A Category 2 Authorised Firm may be authorised to conduct other Financial Services, but it is the authorisation for Dealing in Investments as Principal or authorisation for Providing Credit, and the absence of authorisation for the activities specified in Rule 1.3.1, that are determinative of its belonging to Category 2.

Category 3A

1.3.3 An Authorised Firm is in Category 3A if:

- (a) its Licence authorises it to carry the Financial Services of Dealing in Investments as Agent; and
- (b) it does not meet the criteria of Categories 1, 2 or 5.

Guidance

A Category 3A Authorised Firm may be authorised to conduct other Financial Services, but it is the authorisation for Dealing in Investments as Agent and the absence of authorisation for the activities specified in Rules 1.3.1 and 1.3.2 that are determinative of its belonging to Category 3A.

Category 3B

1.3.4 An Authorised Firm is in Category 3B if:

- (a) its Licence authorises it to carry on one or more of the Financial Services of:
 - (i) Providing Custody (where it does so for a Fund or Provides Custody of Crypto Assets); or

- (ii) Acting as the Trustee of a Fund;
- (iii) Operating an Employee Money Purchase Scheme; or
- (iv) Acting as the Administrator of an Employee Money Purchase Scheme; and
- (b) it does not meet the criteria of Categories 1, 2, 3A or 5.

Guidance

A Category 3B Authorised Firm may be authorised to conduct other Financial Services, but it is the authorisation for Providing Custody for a Fund, Providing Custody of Crypto Assets, Acting as Trustee of a Fund or Operating or Acting as the Administrator of an Employee Money Purchase Scheme, and the absence of authorisation for the activities specified in Rules 1.3.1, 1.3.2 and 1.3.3 that are determinative of its belonging to Category 3B.

Category 3C

1.3.5 An Authorised Firm is in Category 3C if:

- (a) its Licence authorises it to carry on one or more of the Financial Services of:
 - (i) Managing Assets;
 - (ii) Managing a Collective Investment Fund;
 - (iii) Providing Custody (where it does so other than for a Fund and other than in relation to Crypto Assets);
 - (iv) Managing a PSIA (which is a PSIAr);
 - (v) Providing Trust Services (where it is acting as trustee in respect of at least one express trust); or
 - (vi) Providing Money Services (where it issues Stored Value); and
- (b) it does not meet the criteria of Categories 1, 2, 3A, 3B or 5.

Guidance

A Category 3C Authorised Firm may be authorised to conduct other Financial Services, but it is the authorisation for Managing Assets, Managing a Collective Investment Fund, Providing Custody other than for a Fund or in relation to Crypto Assets, Managing a PSIA which is a PSIAr, Providing Trust Services (where it is acting as a trustee in respect of at least one express trust), or Providing Money Services (where it issues Stored Value) and the absence of authorisation for the activities specified in Rules 1.3.1, 1.3.2, 1.3.3 and 1.3.4 that are determinative of its belonging to Category 3C.

Category 3D

1.3.5A An Authorised Firm is in Category 3D if:

- (a) its Licence authorises it to Provide Money Services and it:
 - (i) provides or operates a Payment Account;

- (ii) executes a Payment Transaction on a Payment Account provided or operated by another Person; or
 - (iii) issues a Payment Instrument; and
- (b) it does not meet the criteria of Categories 1, 2, 3A, 3B, 3C or 5.

Category 4

1.3.6 An Authorised Firm is in Category 4 if:

- (a) its Licence authorises it to carry on one or more of the Financial Services of Arranging Deals in Investments, Advising on Financial Products, Arranging Custody, Insurance Intermediation, Insurance Management, Operating an Alternative Trading System, Providing Fund Administration, Providing Trust Services (where it is not acting as trustee in respect of an express trust), Providing Money Services (where it only provides Money Transmission), Arranging Credit and Advising on Credit, Operating a Crowdfunding Platform or Arranging or Advising on Money Services; and
- (b) it does not meet the criteria of Categories 1, 2, 3A, 3B, 3C, 3D or 5.

Guidance

An Authorised Firm in Category 4 may not be authorised to conduct any other Financial Service beyond those listed in Rule 1.3.6(a); if it were so authorised it would belong to another Category.

Category 5

1.3.7 An Authorised Firm is in Category 5 if it:

- (a) is an Islamic Financial Institution; and
- (b) Manages a PSIAu.

Guidance

Authorised Firms in Categories 1 to 4 may also carry out Islamic Financial Business, but only those Authorised Firms in Categories 1 or 5 may Manage a PSIAu. They will not fall within Category 5 unless the whole of the business is conducted in accordance with Shari'a and they Manage a PSIAu.

1.4 Systemically Important Banks (SIBs)

Guidance

1. This section provides for the DFSA to be able to designate an Authorised Firm as a Systemically Important Bank (SIB). A SIB is a bank whose impact, if it were to fail, could cause significant disruption to the financial system and the broader economy. To address the additional risks that SIBs pose, SIBs are subject to specific regulatory and supervisory measures, including higher capital charges in the form of an additional buffer (an HLA Capital Buffer) and more intensive supervision.

2. The DFSA will base its approach to identifying SIBs, and the measures that it applies to SIBs, on the framework issued by the Basel Committee.
3. In accordance with the Basel Committee framework, this section provides for the DFSA to designate two types of SIBs: global systemically important banks (G-SIBs) and domestic systemically important banks (D-SIBs). The main difference between the two types of SIBs, is that G-SIBs are capable of having a significant impact on the effective working and stability of the global financial system, while the potential impact of D-SIBs is on the local or regional financial system. The differences in the requirements that apply to each are explained later in this section.
4. In appropriate cases, the DFSA may designate an Authorised Firm as both a G-SIB and a D-SIB if its failure could have a significant impact on both the global and regional financial systems. If the DFSA designates a firm as both a G-SIB and a D-SIB, the HLA Capital Buffer that applies will be whichever is the higher of the amount calculated for it as a G-SIB and the amount calculated for it as a D-SIB (see Rule 3.9B.3). If a firm is a G-SIB, it does not automatically mean that it will be a D-SIB. For example, a firm may be systemically important globally, but it may not conduct significant business locally or regionally.

Global Systemically Important Banks (G-SIBs)

1.4.1 The DFSA will designate an Authorised Firm as a G-SIB if:

- (a) it is a Domestic Firm in Category 1 or 5;
- (b) it is a member of a Group that is included on the list of global systemically important banks published by the Financial Stability Board; and
- (c) the DFSA is the overall consolidated supervisor of that Group.

1.4.2 The DFSA may designate an Authorised Firm as a G-SIB if:

- (a) the conditions in paragraphs (a) and (b) of Rule 1.4.1 are met in relation to the Authorised Firm; and
- (b) the DFSA is not the overall consolidated supervisor of the Group but it considers that it is appropriate to treat the Authorised Firm as a G-SIB.

Guidance

1. The Financial Stability Board (FSB), in consultation with the Basel Committee, identifies and periodically publishes a list of global systemically important banks.
2. If the DFSA is the consolidated supervisor of a Group included on the FSB list of global systemically important banks, it will designate an Authorised Firm in the Group that is a bank as a G-SIB (see Rule 1.4.1).
3. If an Authorised Firm that is a bank is part of a Group included on the FSB list, but the DFSA is not the consolidated supervisor of that Group, the DFSA may designate that Authorised Firm as a G-SIB if it considers it is appropriate to do so in all the circumstances (see Rule 1.4.2). If the DFSA proposes to designate an Authorised Firm as a G-SIB where it is not the consolidated supervisor of the relevant Group, it will normally consult first with the consolidated supervisor to ensure that the supervisors are taking a co-ordinated approach.
4. A G-SIB, in addition to being subject to more intensive supervision, must maintain an extra capital buffer (the HLA Capital Buffer) under section 3.9B. At the consolidated Group level, a G-SIB

also has to publish quantitative indicators relating to its systemic importance (see Rule 11.1.3 and App 11 Table 15) and carry out recovery and resolution planning.

Domestic Systemically Important Banks (D-SIBs)

1.4.3 The DFSA may designate an Authorised Firm as a D-SIB if:

- (a) it is a Domestic Firm or Branch in Category 1, 2 (other than a Matched Principal) or 5; and
- (b) the DFSA considers that the risks associated with the Authorised Firm are such that, if it were to fail, it could have a significant impact on the effective working and stability of the banking or financial system locally or regionally.

Guidance

1. The DFSA will, in assessing if it should designate an Authorised Firm as a D-SIB, take into account the factors suggested by the Basel Committee in its framework for domestic systemically important banks. Those factors include the size, interconnectedness, substitutability and complexity of the firm. The DFSA will publish the general assessment methodology it applies in assessing if it should designate a firm as a D-SIB.
2. The DFSA may designate an Authorised Firm as a D-SIB whether it is a Domestic Firm or a Branch. However, the measures that apply to a D-SIB that is a Branch will be less extensive than those that apply to a D-SIB that is a Domestic Firm. For example, while both are subject to more intensive supervision, a D-SIB that is a Branch will not be subject to the HLA Capital Buffer Requirement. This is because most capital requirements in chapter 3 do not apply to a Branch as it is subject to capital requirements applied by its home state regulator.

Procedures for designating SIBs

- 1.4.4**
- (1) The DFSA may amend or cancel a designation made under this section.
 - (2) The DFSA must publish a copy of any designation made under this section or any amendment or cancellation of that designation.
 - (3) The procedures in Schedule 3 to the Regulatory Law apply to a DFSA decision to designate an Authorised Firm as a SIB or to amend the designation.
 - (4) If the DFSA decides to designate an Authorised Firm as a SIB or to amend the designation, the Authorised Firm may refer the matter to the FMT for review.
 - (5) Paragraphs (3) and (4) do not apply to a DFSA decision to designate a G-SIB under Rule 1.4.1 or to amend the designation.

Guidance

The Schedule 3 procedures and the right of review by the FMT do not apply to the designation of a G-SIB under Rule 1.4.1 (see Rule 1.4.4(5)). This is because the DFSA, under Rule 1.4.1, will designate a bank as a G-SIB if it is included on the list of global systemically important banks published by the Financial Stability Board.

2 GENERAL REQUIREMENTS

Introduction

Guidance

This chapter details the threshold conditions for the mandatory maintenance of a Trading Book, periodic prudential reporting requirements to the DFSA, guidance on prudent valuation practices, and requirements to produce and maintain a risk appetite statement. Appendix 2 includes detailed Rules on the positions to be included in the Trading Book, the valuation of such positions, prudent valuation practices and associated issues related to the identification and treatment of Trading Book positions. Appendix 2 also specifies the DFSA's expectations with regard to the need for a documented Trading Book policy and risk management systems and controls for the Trading Book. Appendix 2 also presents in a tabular fashion, detailed specifications on periodic prudential reporting requirements for different categories of Authorised Firms.

2.1 Application

2.1.1 This chapter applies to an Authorised Firm in any Category.

2.2 Trading Book

2.2.1 An Authorised Firm must have a Trading Book if:

- (a) it has positions that must be included in a Trading Book in accordance with section A2.1 of App2;
- (b) those positions are held with trading intent in accordance with Rule A2.1.5; and
- (c) the total value of the positions eligible for inclusion in the Trading Book pursuant to (a) and (b):
 - (i) normally exceeds \$15 million or 5% of its combined on and off-balance sheet positions; or
 - (ii) has exceeded \$20 million or 6% of its combined on and off-balance sheet positions at any time in the preceding twelve month period.

2.2.2 An Authorised Firm that must have a Trading Book in accordance with Rule 2.2.1 must:

- (a) comply with the requirements of section A2.1 of App2; and
- (b) differentiate its business between Trading Book activity and Non-Trading Book activity on a consistent basis.

2.2.3 An Authorised Firm which has a Trading Book must have adequate systems and controls to:

- (a) monitor the size of its Trading Book; and
- (b) ensure that positions are included consistently in its Trading Book and Non-Trading Book so that:
 - (i) the inclusion of hedging positions in the Trading Book or the Non-Trading Book at all times reflects the intent of the Authorised Firm in holding the position; and
 - (ii) adequate records are made if positions are transferred between Trading and Non-Trading Books so that the transfers may be identified.

2.3 Reporting to the DFSA

2.3.1 (1) An Authorised Firm must comply with the accounting and prudential reporting requirements set out in this chapter and PRU which apply to it.

(2) The DFSA may impose additional reporting requirements on an Authorised Firm.

2.3.2 An Authorised Firm must, subject to Rule 2.3.3:

- (a) prepare its returns in accordance with the Rules in this chapter, the instructional guidelines in PRU, and the requirements of the DFSA's electronic prudential reporting system; and
- (b) submit the returns to the DFSA using the electronic prudential reporting system.

Guidance

The returns and instructional guidelines are provided in PRU and the DFSA's electronic prudential reporting system.

2.3.3 The DFSA may by way of a written notice direct an Authorised Firm to submit its returns in a form, manner or frequency other than as prescribed in Rule 2.3.2. An Authorised Firm must continue to submit its returns in accordance with this direction until the DFSA by way of written notice directs otherwise.

2.3.4 (1) The submission of any return must be accompanied by a Form B100 (Declaration by Authorised Firms) signed by the Authorised Firm in the manner set out in (2) or (3) as applicable.

(2) In relation to an annual return the form must be signed by two officers of the Authorised Firm each of whom is a Director, Partner or individual previously approved by the DFSA for that purpose.

- (3) In relation to a quarterly return the form must be signed by one officer of the Authorised Firm who is a Director, Partner or individual previously approved by the DFSA for that purpose.

2.3.5 An original signed hard copy of Form B100 (Declaration by Authorised Firms), together with a copy of the return submitted to the DFSA must be kept for at least 6 years for inspection by the DFSA.

2.3.6 If the DFSA notifies an Authorised Firm, or the Authorised Firm itself forms the view, that a return that has been submitted to the DFSA appears to be inaccurate or incomplete, the Authorised Firm must consider the matter and within a reasonable time it must correct any inaccuracies and make good any omissions, and re-submit the relevant parts of the return.

2.3.7 (1) An Authorised Firm must prepare and submit returns in accordance with Table 1 in section A2.4 of App2, which forms part of these Rules.

(2) All returns must be completed in thousands of dollars (\$).

(3) The requirement in (1) does not apply to an Authorised Firm if the only Financial Service it carries on is Managing a Venture Capital Fund.

2.3.8 (1) An Authorised Firm must submit to the DFSA any annual return required by Table 1 in section A2.4 of App2, within four months of the end of the Authorised Firm's financial year.

(2) An Authorised Firm must submit to the DFSA any other return required by Table 1 in section A2.4 of App2, within one month after the end of the reporting period to which the return relates.

2.4 Prudent valuation practices

Guidance

1. This section and related section A2.5 in App2 provide Authorised Firms with Guidance on prudent valuation for positions that are accounted for at fair value, whether they are in the Trading Book or in the Non-Trading Book (also known as the banking book).
2. A framework for prudent valuation practices should at a minimum include adequate systems and controls and valuation methodologies. The DFSA's expectations in this regard are set out in section A2.5 App2.
3. The Guidance is especially important for positions without actual market prices or observable inputs to valuation, as well as less liquid positions which raise supervisory concerns about prudent valuation. The Guidance is not intended to require Authorised Firms to change valuation procedures for financial reporting purposes.
4. The DFSA will assess an Authorised Firm's valuation procedures for consistency with the Guidance. The DFSA may impose a valuation adjustment if there is a material degree of inconsistency between the Authorised Firm's valuation procedures and the Guidance.

2.5 Risk appetite statement

- 2.5.1** (1) This Rule applies to an Authorised Firm in Category 1, 2, 3A or 5.
- (2) An Authorised Firm must produce and maintain a clearly articulated risk appetite statement which:
- (a) prescribes the level of aggregate risk that it is willing to assume and manage in the pursuit of its strategic objectives and business plan; and
 - (b) sets appropriate limits that are consistent with the Authorised Firm's level of aggregate risk under (1)(a), as well as its risk profile and capital and liquidity position.
- (3) The risk appetite statement in (1) must be:
- (a) documented;
 - (b) approved by the Governing Body;
 - (c) communicated to all relevant Employees; and
 - (d) regularly reviewed and appropriately adjusted to reflect changes in its risk profile or market and macroeconomic conditions.

Guidance

1. The Authorised Firm should ensure that the level of aggregate risk that it is willing to assume allows it to manage its risks prudently in normal times and in a way that allows it to withstand periods of stress. The level of aggregate risk should include both quantitative and qualitative elements, as appropriate, and encompass a range of measures. These should be expressed in terms that are clear enough for all levels of management to be able to understand the trade-off between risks and profits.
2. The DFSA expects that the risk appetite statement also outlines actions to be taken when risk limits are breached, including disciplinary actions for excessive risk-taking, escalation procedures and Governing Body notification.
3. For an Authorised Firm that is a Branch the capital and liquidity position referred to in Rule 2.5.1(2)(b) should be determined in accordance with Rule 3.2.4(a).

3 CAPITAL

Introduction

Guidance

1. This chapter deals with all aspects of prudential requirements relating to capital adequacy. The chapter aims to ensure that an Authorised Firm maintains adequate capital to support the risks associated with its activities and that it can absorb potential unexpected losses to its capital. It also includes provisions forming part of the framework for assessing the capital adequacy of an Authorised Firm.
2. Part 1 of this chapter deals with the application provisions. Part 2 outlines the fundamental capital adequacy obligations and systems and controls requirements to ensure compliance with this critical regulatory obligation. Part 3 includes all the Rules and associated guidance for the calculation of minimum Capital Requirement for different Categories of Authorised Firms. This part also specifies the requirements in respect of Capital Buffers and associated obligations. It specifies three types of Capital Buffers: the Capital Conservation Buffer, the Countercyclical Capital Buffer and the HLA Capital Buffer. Part 4 of this chapter specifies detailed Rules on the calculation of Capital Resources of an Authorised Firm, including detailed Rules on the eligibility criteria for different components of Capital Resources which correspond to varying levels of quality. Part 5 prescribes requirements for a non-risk sensitive Leverage Ratio which acts as a supplementary measure to risk-based capital requirements.
3. Appendix 3 provides detailed guidance on various aspects of stress and scenario testing which are required to be considered by an Authorised Firm to effectively comply with the Rules in this chapter.

PART 1 – Application

3.1 Application

- 3.1.1** The parts, sections and Rules in this chapter apply to an Authorised Firm as stated in those provisions.

Guidance

1. Part 2 of this chapter imposes a number of basic requirements, including the core requirement that the amount of a firm's Capital Resources must at all times exceed the amount of its Capital Requirement.
2. In particular, note that:
 - a. Part 3 (Calculating Capital Requirements) applies to all firms, but with differentiated calculations for Capital Requirements for the various Categories of Authorised Firms as prescribed in sections 3.3, 3.4 and 3.5;
 - b. Within Part 3, an exemption from the calculation of Tier 2 (T2) Capital in relation to firms authorised to Manage a PSIAu is prescribed in Rule 3.15.9; and
 - c. Part 4 (Calculating Capital Resources) applies to all firms, but in a differentiated manner for different Categories of firms as demonstrated in the table in section 3.11.

PART 2 – Basic Requirements

3.2 Requirements

Application

3.2.1 In this section:

- (a) Rules 3.2.2 to 3.2.5 and 3.2.7 apply to an Authorised Firm in any Category; and
- (b) Rule 3.2.6 applies only to an Authorised Firm in Category 3B, 3C, 3D or 4.

Maintaining Capital Resources

3.2.2 An Authorised Firm that is a Domestic Firm must:

- (a) have and maintain, at all times, Capital Resources of the kinds and amounts specified in, and calculated in accordance with, the Rules in PIB; and
- (b) ensure that it maintains capital and liquid assets in addition to the requirement in (a) which are adequate in relation to the nature, size and complexity of its business to ensure that there is no significant risk that liabilities cannot be met as they fall due.

Guidance

1. For the purposes of Rule 3.2.2, an Authorised Firm's Governing Body should assess whether the Capital Resources which are required by the DFSA as set out in PIB are adequate in relation to the Authorised Firm's specific business. Additional resources should be maintained by the Authorised Firm where its Governing Body has considered that the required Capital Resources do not adequately reflect the nature and risks of the Authorised Firm's business.
2. The liabilities referred to in Rule 3.2.2(b) include an Authorised Firm's contingent and prospective liabilities, such as liabilities arising from a change in business strategy or claims made against the Authorised Firm, but not liabilities that might arise from prospective transactions which the Authorised Firm could avoid, for example by ceasing its operations. Liabilities from prospective transactions refers to the potential liabilities which can be avoided by either adequate risk management, risk transfer or avoiding the transaction completely. This refers to any prospective transaction, for example, lending money to a borrower or entering into a contract for the provision of services by a service provider.
3. An Authorised Firm subject to the requirements in chapter 10 may be required to meet Individual Capital Requirements under those Rules.

3.2.3 An Authorised Firm must have, at all times, Capital Resources which exceed the amount of its Capital Requirement.

Guidance

The specific Capital Requirements for the various Categories of Authorised Firms are dealt with in sections 3.3, 3.4, 3.5, 3.9, 3.9A and 3.9B.

Systems and controls

Guidance

For the purposes of section 3.2, an Authorised Firm is required to have systems and controls in place to enable it to be certain that it has adequate Capital Resources to comply with Rule 3.2.3 at all times. An Authorised Firm's systems and controls should be such as to allow it to demonstrate its capital adequacy at any particular time if required to do so by the DFSA. Where through the operation of those systems and controls an Authorised Firm forms the view that it may not be able to satisfy the requirements of Rule 3.2.3 in the future, that Authorised Firm is required to immediately inform the DFSA in accordance with Rule 3.2.5.

3.2.4 An Authorised Firm that is a Branch must:

- (a) ensure that it has and maintains, at all times, liquid assets and access to financial resources which are adequate in relation to the nature, size and complexity of its business to ensure that there is no significant risk that liabilities cannot be met as they fall due;
- (b) ensure that it complies with its home state Financial Services Regulator's prudential requirements;
- (c) submit to the DFSA a copy of every capital adequacy summary report and leverage ratio report submitted to its home state Financial Services Regulator within ten business days of the due date for submission to that regulator; and
- (d) in the event of any anticipated or actual breach of any prudential requirements set by its home state Financial Services Regulator, notify the DFSA forthwith with any relevant documents.

3.2.5 (1) An Authorised Firm must have systems and controls to enable it to determine and monitor:

- (a) its Capital Requirement; and
 - (b) whether the amount of its Capital Resources is, and is likely to remain, greater than the amount of its Capital Requirement.
- (2) Such systems and controls must include an analysis of:
- (a) realistic scenarios which are relevant to the circumstances of the Authorised Firm; and
 - (b) the effects on the Capital Requirement of the Authorised Firm and on its Capital Resources if those scenarios occurred.
- (3) An Authorised Firm must notify the DFSA immediately and confirm in writing any breach, or expected breach, of any of the provisions of this chapter by the Authorised Firm.

Guidance

1. App3 provides Guidance on the nature and type of stress and scenario testing that Authorised Firms should be undertaking to support their view that they have adequate financial resources to meet their obligations.
2. The requirements in this chapter apply to Authorised Firms on a solo basis. An Authorised Firm may also be subject to Capital Resources requirements at a Group level. Group requirements are addressed in chapter 8.

Notifications to the DFSA

- 3.2.6**
- (1) This Rule applies to an Authorised Firm in Category 3B, 3C, 3D or 4.
 - (2) An Authorised Firm must notify the DFSA immediately and confirm in writing if its Capital Resources fall below 120% of its Capital Requirement.

Requirements as to composition of Capital Resources

- 3.2.7**
- (1) This Rule applies to an Authorised Firm in any Category.
 - (2) Subject to Rule 3.6.3 and (3), an Authorised Firm must at all times maintain the following components of capital:
 - (a) where the Risk Capital Requirement forms part of the Capital Requirement of the firm under section 3.3 or 3.4:
 - (i) CET1 Capital equating to at least 6.0% of the firm's Risk Weighted Assets; and
 - (ii) T1 Capital equating to at least 8.0% of the firm's Risk Weighted Assets;
 - (b) where the Expenditure Based Capital Minimum forms the Capital Requirement of the firm under section 3.4:
 - (i) CET1 Capital equating to at least 60% of the firm's Expenditure Based Capital Minimum; and
 - (ii) T1 Capital equating to at least 80% of the firm's Expenditure Based Capital Minimum; or
 - (c) where the Expenditure Based Capital Minimum, the Stored Value Capital Requirement or the Transaction Based Capital Requirement forms the Capital Requirement of the firm under Rule 3.5.2:
 - (i) CET1 Capital equating to at least 60% of the firm's Capital Requirement under Rule 3.5.2; and
 - (ii) T1 Capital equating to at least 80% of the firm's Capital Requirement under Rule 3.5.2.
 - (3) The CET1 Capital used to meet the requirement in (2)(a) must not also be used as a component of a Capital Buffer.

Guidance

1. It follows from Rule 3.2.7(2)(a) and Rule 3.2.3 that an Authorised Firm cannot use T2 Capital of more than 2% of its Risk Weighted Assets to meet its Risk Capital Requirement.
2. It follows from Rule 3.2.7(2)(b) and Rule 3.2.3 that an Authorised Firm cannot use T2 Capital to meet more than 20% of its Expenditure Based Capital Minimum.
3. It follows from Rule 3.2.7(2)(c) and Rule 3.2.3 that an Authorised Firm cannot use T2 Capital to meet more than 20% of its Expenditure Based Capital Minimum, Stored Value Capital Requirement or Transaction Based Capital Requirement, as applicable.
4. In accordance with Rules 3.9.5, 3.9A.3 and 3.9B.4, the CET1 Capital used for a Capital Buffer cannot constitute CET1 Capital for meeting the Risk Capital Requirement.

PART 3 – Calculating the Capital Requirement

3.3 Capital Requirements for Categories 1 and 5

3.3.1 This section applies to an Authorised Firm in Category 1 or 5.

- 3.3.2** (1) The Capital Requirement for an Authorised Firm is calculated, subject to (2), as the higher of:
- (a) the applicable Base Capital Requirement; or
 - (b) its Risk Capital Requirement plus applicable Capital Buffer Requirements.
- (2) Where the Authorised Firm has an ICR imposed on it then the Capital Requirement is its ICR plus Risk Capital Requirement plus applicable Capital Buffer Requirements.

Guidance

1. An Authorised Firm should refer to chapters 4, 5 and 6 to determine whether it is required to calculate a Credit Risk Capital Requirement (also referred to in this module as CRCOM), a Market Risk Capital Requirement or an Operational Risk Capital Requirement, respectively.
2. The Displaced Commercial Risk Capital Requirement will only apply to an Authorised Firm Managing a PSIAu.
3. An Authorised Firm will also need to consider the relevant provisions in IFR chapter 5 when calculating its Credit Risk and Market Risk for Islamic Contracts.
4. If an Individual Capital Requirement is imposed on an Authorised Firm under Chapter 10, such a requirement is additional to the Risk Capital Requirement and is, therefore, a component of the Authorised Firms Capital Requirement.

3.4 Capital Requirements for Categories 2 and 3A

3.4.1 This section applies to an Authorised Firm in Category 2 or 3A.

- 3.4.2** (1) The Capital Requirement for an Authorised Firm is calculated, subject to (2), as the highest of:
- (a) the applicable Base Capital Requirement;
 - (b) the Expenditure Based Capital Minimum; or
 - (c) its Risk Capital Requirement plus applicable Capital Buffer Requirements.
- (2) Where the Authorised Firm has an ICR imposed on it then the Capital Requirement is its ICR plus Risk Capital Requirement plus applicable Capital Buffer Requirements.

Guidance

1. An Authorised Firm should refer to chapters 4, 5 and 6 to determine whether it is required to calculate a Credit Risk Capital Requirement (also referred to in this module as CRCOM), a Market Risk Capital Requirement or an Operational Risk Capital Requirement, respectively.
2. An Authorised Firm will also need to consider the relevant provisions in IFR chapter 5 when calculating its Credit Risk and Market Risk for Islamic Contracts.
3. If the DFSA imposes an Individual Capital Requirement on an Authorised Firm under chapter 10, such a requirement is additional to the Risk Capital Requirement and is, therefore, a component of the Authorised Firms Capital Requirement.

3.5 Capital and Liquidity Requirements for Categories 3B, 3C, 3D and 4

Application

- 3.5.1** (1) This section applies to an Authorised Firm in Category 3B, 3C, 3D or 4.
- (2) Rule 3.5.2 does not apply to an Authorised Firm if the only Financial Service it carries on is Managing a Venture Capital Fund.

Capital Requirement

- 3.5.2** The Capital Requirement for an Authorised Firm is calculated as the highest of:
- (a) the applicable Base Capital Requirement as set out in section 3.6;
 - (b) where applicable, the Expenditure Based Capital Minimum as set out in section 3.7; or
 - (c) in the case of a Money Services Provider:
 - (i) the Stored Value Capital Requirement, if it issues Stored Value;
 - (ii) the Transaction Based Capital Requirement, if it provides Payment Services; or
 - (iii) the aggregate of the sums referred to in (i) and (ii), if it carries on both activities.

Liquidity requirements

- 3.5.3** (1) An Authorised Firm to which this section applies must, at all times, maintain an amount in the form of liquid assets:
- (a) if section 3.7 applies, exceeds the higher of its:
 - (i) Base Capital Requirement; or
 - (ii) Expenditure Based Capital Minimum;

- (b) if section 3.7 does not apply, exceeds its Base Capital Requirement; and
 - (c) if the only Financial Service it carries on is Managing a Venture Capital Fund, are adequate in relation to the nature, size and complexity of its business to ensure that there is no significant risk that its liabilities cannot be met as they fall due.
- (2) For the purpose of this Rule, and subject to (3) and (4), liquid assets comprise any of the following:
 - (a) cash in hand;
 - (b) money deposited with a regulated bank or deposit-taker which has a short-term credit rating of A2 or P2 or above, or a long-term credit rating of BBB or Baa2 or above, from an ECAI;
 - (c) demand deposits with a tenor of 1 year or less with a bank or deposit-taker in (b);
 - (d) time deposits with a tenor of 1 year or less which have an option to redeem the deposit at any time. In such cases, the deposit amount eligible to be included as liquid assets must be calculated as net of any costs associated with such early redemption;
 - (e) cash receivable from a regulated clearing house and cash deposits with such clearing houses, net of any fees or contributions to guarantee or reserve funds of such clearing houses;
 - (f) cash receivable from regulated merchant acquirers for card based payment transactions, net of any fees or charges owed to such merchant acquirers; or
 - (g) debt Securities representing claims on a sovereign, Central Bank, PSE or MDB provided these Securities satisfy all of the following conditions:
 - (i) they are assigned Credit Quality Grade 1 according to chapter 4;
 - (ii) they are denominated in United States Dollars, U.A.E. Dirhams, Euros or Pounds Sterling;
 - (iii) they have a remaining maturity of 12 months or less; and
 - (iv) they are administered and safeguarded by a custodian who meets one of the criteria listed in CIR 8.2.4 (a) to (h).
- (3) For the purpose of this Rule, liquid assets do not include:
 - (a) any investment, asset or deposit which has been pledged as security or Collateral for any obligations or liabilities assumed by it or by any other third party; or

- (b) cash held in Client Money or Insurance Monies accounts.
- (4) Liquid assets referred to in (2)(g) can only comprise two-thirds of the total stock of liquid assets held to comply with (1).
- (5) An Authorised Firm must value the liquid assets referred to in (2)(g) which it holds to comply with (1):
 - (a) conservatively, taking into account the likely deterioration in the value of assets under market-wide stress conditions; and
 - (b) on an ongoing basis.

Guidance

The DFSA may require an Authorised Firm to comply with any specified additional capital or liquidity requirements under Article 75A of the Regulatory Law, where it considers that this is warranted by the firm's business model or risk profile.

3.6 Base Capital Requirement

- 3.6.1** This section applies to an Authorised Firm in any Category (other than an Authorised Firm if the only Financial Service it carries on is Managing a Venture Capital Fund).

Guidance

The Base Capital Requirement is a component of the calculation of the Capital Requirement under sections 3.3, 3.4 and Rule 3.5.2.

- 3.6.2** The table below sets out the Base Capital Requirement for each Category of an Authorised Firm.

Category	Base Capital Requirement
Category 1	US \$10 million
Category 2	US \$2 million Except if the Authorised Firm is Dealing in Investments as Principal (where it does so only as a Matched Principal), in which case its Base Capital Requirement is US \$500,000.
Category 3A	US \$200,000
Category 3B	1. If the only Financial Service referred to in Rule 1.3.4(a) which the Authorised Firm is authorised to carry on is Providing Custody of Crypto Assets its Base Capital Requirement is US \$1 million. 2. If the only Financial Service referred to in Rule 1.3.4(a) which the Authorised Firm is authorised to carry on is Operating an Employee Money Purchase Scheme or Acting as the Administrator of an Employee Money Purchase Scheme its Base Capital Requirement is: (a) US \$500,000 for an Operator; or (b) US \$1 million for an Administrator. 3. If the only Financial Service referred to in Rule 1.3.4(a) which the Authorised Firm is authorised to carry on is Providing Custody (where it does so for a Fund and other than in relation to Crypto Assets) its Base Capital Requirement is US \$500,000. 4. If the only Financial Service referred to in Rule 1.3.4(a) which the Authorised Firm is authorised to carry on is Acting as the Trustee of a Fund its Base Capital Requirement is US \$2 million. 5. If the Authorised Firm is authorised to carry on more than one of the Financial Services referred to in Rule 1.3.4(a) its Base Capital Requirement is the highest Base Capital Requirement that applies under paragraphs 1 to 4.

Category 3C	US \$500,000 Except: (1) if the only Financial Service referred to in Rule 1.3.5(a) which the Authorised Firm is authorised to carry on is Managing a Collective Investment Fund, in which case its Base Capital Requirement is: (a) US \$140,000, if it manages any Public Fund or Credit Fund; or (b) US \$40,000 otherwise; or (2) if the only Financial Service referred to in Rule 1.3.5(a) which the Authorised Firm is authorised to carry on is Managing Assets or Managing a Profit Sharing Investment Account (which is a PSIAr), in which case its Base Capital Requirement is US \$140,000; or (3) if an Authorised Firm is authorised to carry on more than one of the Financial Services referred to in (1) or (2) and it is not authorised to carry on any other Financial Service referred to in Rule 1.3.5(a), in which case its Base Capital Requirement is US \$140,000.
Category 3D	US \$200,000
Category 4	1. If an Authorised Firm is authorised to Operate a Crowdfunding Platform, Provide Money Transmission its Base Capital Requirement is \$140,000. 2. In any other case, the Authorised Firm's Base Capital Requirement is \$30,000.
Category 5	US \$10 million

3.6.3 An Authorised Firm must have Common Equity Tier 1 Capital (CET1 Capital), as defined in section 3.13, of not less than its relevant Base Capital Requirement at the time that it obtains authorisation and at all times thereafter.

3.7 Expenditure Based Capital Minimum

3.7.1 This section applies to an Authorised Firm if it is:

- (a) in Category 2;
- (b) in Category 3A, 3B, 3C (other than an Authorised Firm that only Manages a Venture Capital Fund) or 3D and meets any of the following conditions:

- (i) it holds Client Assets in any of the circumstances specified in COB Rule 6.11.4(a) to (c); or
- (ii) its Licence authorises it to:
 - (A) Provide Custody;
 - (B) Act as the Trustee of a Fund;
 - (C) Manage a Collective Investment Fund
 - (D) Provide Money Services;
 - (E) Operate an Employee Money Purchase Scheme; or
 - (F) Act as the Administrator of an Employee Money Purchase Scheme; or
- (c) in Category 4 and meets any of the following conditions:
 - (i) its Licence authorises it to carry on the Financial Service of Operating a Crowdfunding Platform and it holds Clients Assets in any of the circumstances specified in COB Rule 6.11.4(a) to (c);
 - (ii) its Licence authorises it to carry on the Financial Service of Providing Money Services (where it only provides Money Transmissions); or
 - (iii) it holds Insurance Monies and its Licence authorises it to carry on any of the following Financial Services:
 - (A) Insurance Intermediation; or
 - (B) Insurance Management.

Guidance

The Expenditure Based Capital Minimum is a component of the calculation of the Capital Requirement under sections 3.4 and 3.5 and is a key factor in the calculation of the capital components under Rule 3.2.7.

- 3.7.2** An Authorised Firm must calculate its Expenditure Based Capital Minimum by multiplying its Annual Audited Expenditure, calculated in accordance with Rule 3.7.3, with the applicable ratio, as set out in the following table:

Category	Applicable ratio
Category 2	1. If an Authorised Firm holds Client Assets in any of the circumstances specified in COB Rule 6.11.4(a) to (c), 18/52. 2. In any other case, 13/52.

Category 3	1. Subject to points (2) to (6), if an Authorised Firm holds Client Assets in any of the circumstances specified in COB Rule 6.11.4(a) to (c), or its Licence authorises it to Provide Custody, Act as the Trustee of a Fund or Manage a Collective Investment Fund, 13/52. 2. If an Authorised Firm's Licence authorises it to Provide Money Services and it holds Client Assets in any of the circumstances specified in COB Rule 6.11.4(a) to (c), 18/52. 3. If an Authorised Firm's Licence authorises it to Provide Money Services (where it does not issue Stored Value) and it does not hold Client Assets in any of the circumstances specified in COB Rule 6.11.4(a) to (c), 9/52. 4. If an Authorised Firm's Licence authorises it to Operate an Employee Money Purchase Scheme and it holds Client Assets in any of the circumstances specified in COB Rule 6.11.4(a) to (c), 18/52. 5. If an Authorised Firm's Licence authorises it to Operate an Employee Money Purchase Scheme or Provide Money Services (where it issues Stored Value) and it does not hold Client Assets in any of the circumstances specified in COB Rule 6.11.4(a) to (c), 13/52. 6. If an Authorised Firm's Licence authorises it to Act as the Administrator of an Employee Money Purchase Scheme, 18/52.
Category 4	1. If an Authorised Firm's Licence authorises it to Operate a Crowdfunding Platform and it holds Client Money in any of the circumstances specified in COB Rule 6.11.4(a) to (c), 13/52. 2. If an Authorised Firm's Licence authorises it to Provide Money Services (where it only provides Money Transmission) and it holds Client Money in any of the circumstances specified in COB Rule 6.11.4(a) to (c), 18/52. 3. If an Authorised Firm's Licence authorises it to Provide Money Services (where it only provides Money Transmission) and it does not hold Client Money in any of the circumstances specified in COB Rule 6.11.4(a) to (c), 6/52. 4. If an Authorised Firm's Licence authorises it to Provide Insurance Intermediation or Insurance Management and it holds Insurance Monies, 9/52.

Annual Audited Expenditure

- 3.7.3** (1) Subject to Rules 3.7.3A and 3.7.4, Annual Audited Expenditure constitutes all expenses and losses that arise in the Authorised Firm's normal course of business in a twelve month accounting period (excluding exceptional items) which are recorded in the Authorised Firm's audited profit and loss account, less the following items (if they are included in the Authorised Firm's audited profit and loss account):
- (a) staff bonuses, except to the extent that they are non-discretionary;
 - (b) employees' and directors' shares in profits, including share options, except to the extent that they are non-discretionary;
 - (c) other appropriations of profits, except to the extent that they are automatic;
 - (d) shared commissions and fees payable that are directly related to commissions and fees receivable, which are included with total revenue;
 - (e) fees, brokerage and other charges paid to clearing houses, exchanges and intermediate brokers for the purposes of executing, registering or clearing transactions;
 - (f) any expenses for which pre-payments or advances have already been made to the respective claimant (e.g. pre-paid rent, pre-paid communication charges etc.);
 - (g) charges for the amortisation of intangible assets;
 - (h) charges for the depreciation of fixed assets;
 - (i) foreign exchange losses; and
 - (j) contributions to charities.
- (2) For the purposes of (1)(c), a management charge must not be treated as an appropriation of profits.
- 3.7.3A** If a Fund Manager uses a Fund Platform, the Annual Audited Expenditure of the Fund Manager is to be calculated as the aggregate of the expenses and losses of the Authorised Firm and the expenses and losses of the Incorporated Cell Company (the Fund Platform) calculated in accordance with Rule 3.7.4.
- 3.7.4** (1) For the purposes of Rule 3.7.3, an Authorised Firm must calculate its relevant Annual Audited Expenditure with reference to the Authorised Firm's most recent audited financial statements.
- (2) If the Authorised Firm's most recent audited financial statements do not represent a twelve month accounting period, it must calculate its Annual Audited Expenditure on a pro rata basis so as to produce an equivalent annual amount.

- (3) If an Authorised Firm has not completed its first twelve months of business operations, it must calculate its Annual Audited Expenditure based on forecast expenditure as reflected in the budget for the first twelve months of business operations, as submitted with its application for authorisation.
- (4) (a) If an Authorised Firm:
 - (i) has a material change in its expenditure (either up or down); or
 - (ii) has varied its authorised activities;
 it must recalculate its Annual Audited Expenditure and Expenditure Based Capital Minimum accordingly.
- (b) Where an Authorised Firm has recalculated its Annual Audited Expenditure and Expenditure Based Capital Minimum in accordance with (a), it must submit this recalculation to the DFSA within 7 days of its completion and seek agreement/approval from the DFSA. The DFSA may within 30 days of receiving the recalculation object to the recalculation and require the Authorised Firm to revise its Expenditure Based Capital Minimum.

3.8 Risk Capital Requirement

3.8.1 This section applies to an Authorised Firm in Category 1, 2, 3A or 5.

Guidance

The Risk Capital Requirement is a component of the calculation of the Capital Requirement under sections 3.3 and 3.4.

Calculation of Risk Capital Requirement

3.8.1A An Authorised Firm must calculate its Risk Capital Requirement as 10% of its Risk Weighted Assets.

Risk Weighted Assets

3.8.2 An Authorised Firm must calculate its Risk Weighted Assets as 12.5 multiplied by the sum of the following:

- (a) the CRCOM;
- (b) the Market Risk Capital Requirement;
- (c) the Operational Risk Capital Requirement; and
- (d) the Displaced Commercial Risk Capital Requirement, where applicable.

CRCOM

3.8.3 An Authorised Firm must calculate its Credit Risk Capital Requirement in accordance with the applicable Rules in chapter 4.

Guidance

1. Detailed Rules and Guidance in respect of the CRCOM are specified in chapter 4. The CRCOM includes the risk weighted assets (RWA) for all Credit Risk Exposures and securitisation Exposures.
2. Rules and Guidance in respect of calculating the CRCOM for Islamic Contracts are contained in IFR chapter 5.

Market Risk Capital Requirement

- 3.8.4** An Authorised Firm must calculate its Market Risk Capital Requirement in accordance with the applicable Rules in chapter 5.

Guidance

1. Detailed Rules and Guidance in respect of the Market Risk Capital Requirement and each of its components are contained in chapter 5.
2. Rules and Guidance in respect of calculating Market Risk for Islamic Contracts are contained in IFR chapter 5.

Operational Risk Capital Requirement

- 3.8.5** An Authorised Firm must calculate its Operational Risk Capital Requirement in accordance with the applicable Rules in chapter 6.

Displaced Commercial Risk Capital Requirement

- 3.8.6** An Authorised Firm Managing a PSIAu must calculate its Displaced Commercial Risk Capital Requirement in accordance with IFR chapter 5.

3.8A Stored Value Capital Requirement

- 3.8A.1** This section applies to an Authorised Firm in Category 3C that issues Stored Value.

Calculation of Stored Value Capital Requirement

- 3.8A.2**
- (1) An Authorised Firm must calculate its Stored Value Capital Requirement as an amount equal to 3% of the average daily outstanding Stored Value of the firm.
 - (2) In (1), the average daily outstanding Stored Value of a firm means the average total of financial liabilities related to Stored Value in issue at the end of each calendar day where that average is calculated over the previous six calendar months.
 - (3) The Authorised Firm must calculate its Stored Value Capital Requirement on the first calendar day of each calendar month and apply the resulting sum as its requirement for that month.

- (4) If the Authorised Firm has not completed its first six months of business, it must make the calculation under (2) and (3) based on the outstanding Stored Value projected in the business plan included with its application for authorisation, subject to any adjustment to that plan required by the DFSA.

3.8B Transaction Based Capital Requirement

3.8B.1 This section applies to an Authorised Firm that is a Payment Service Provider.

Calculation of Transaction Based Capital Requirement

- 3.8B.2** (1) An Authorised Firm must calculate its Transaction Based Capital Requirement as the sum of the following elements multiplied by the scaling factor:
- (a) 4% of the first \$5 million of payment volume;
 - (b) 2.5% of the next \$5 million of payment volume;
 - (c) 1% of the next \$90 million of payment volume;
 - (d) 0.5% of the next \$150 million of payment volume; and
 - (e) 0.25% of any remaining payment volume.
- (2) In (1), payment volume means the total value of Payment Transactions executed by the Authorised Firm in the previous financial year divided by 12.
- (3) If an Authorised Firm has not completed a full financial year's business, references in (2) to the previous financial year are to be read as the equivalent figure projected in the business plan provided in its application for authorisation, subject to any adjustment to that plan required by the DFSA.
- (4) A Payment Service Provider that also issues Stored Value may exclude from the payment volume in (1), payments directly related to issuing Stored Value.
- (5) The scaling factor in (1) is:
- (a) 0.5 for an Authorised Firm that is authorised to provide only Money Transmission; or
 - (b) 1 for an Authorised Firm that is authorised to provide other types of Payment Services.

3.9 Capital Conservation Buffer

3.9.1 This section applies to an Authorised Firm in Category 1, 2 (other than a Matched Principal) or 5.

3.9.2 Where, under section 3.3 or 3.4, the Risk Capital Requirement in section 3.8 applies to an Authorised Firm, then the firm is subject to a Capital Conservation Buffer Requirement.

3.9.3 The Capital Conservation Buffer Requirement is equivalent to 2.5% of an Authorised Firm's Risk Weighted Assets and must constitute only CET1 Capital.

3.9.4 (1) An Authorised Firm must maintain the required buffer amount, calculated in accordance with Rule 3.9.3, at all times.

(2) The Capital Conservation Buffer Requirement applies on both a solo and a consolidated basis for Authorised Firms forming part of Financial Groups.

3.9.5 An Authorised Firm must not apply CET1 Capital that it maintains to meet the Capital Conservation Buffer Requirement towards meeting:

- (a) any Individual Capital Requirement the DFSA may impose on it pursuant to chapter 10;
- (b) its Risk Capital Requirement;
- (c) its Countercyclical Capital Buffer Requirement; or
- (d) its HLA Capital Buffer Requirement.

3.9A Countercyclical Capital Buffer (CCyB)

Guidance

- 1. This section sets out when an Authorised Firm must maintain a Countercyclical Capital Buffer (CCyB) and how the buffer is calculated.
- 2. A Countercyclical Capital Buffer is intended to take into account the macro-financial environment in which firms operate. If national authorities consider that excess credit growth has led to a build-up of system-wide risk, they can impose this measure to ensure the financial system has a buffer of capital to protect it against future potential losses.
- 3. An Authorised Firm will need to maintain a Countercyclical Capital Buffer only if it has a credit exposure in a jurisdiction where a CCyB Authority has imposed a CCyB Rate.
- 4. The Countercyclical Capital Buffer is in addition to the capital required under the Risk Capital Requirement and the Capital Conservation Buffer Requirement.

3.9A.1 This section applies to an Authorised Firm if it:

- (a) is in Category 1, 2 (other than a Matched Principal) or 5; and
- (b) has a Non-Financial Private Sector Credit Exposure in a jurisdiction for which a CCyB Rate applies.

Countercyclical Capital Buffer Requirement

3.9A.2 An Authorised Firm must maintain a Countercyclical Capital Buffer of CET1 Capital that is calculated using the formula:

$$\text{CCyB} = \text{CCyB Rate} \times \text{RWA}$$

where:

- (a) “CCyB” is the Countercyclical Capital Buffer that the Authorised Firm must maintain;
- (b) “CCyB Rate” is the weighted average of Countercyclical Capital Buffer Rates, calculated in accordance with Rule 3.9A.5, that apply in jurisdictions in which the Authorised Firm has Non-Financial Private Sector Credit Exposures; and
- (c) “RWA” is the value of the Authorised Firm’s Risk Weighted Assets.

Guidance

1. The CCyB Requirement applies to credit exposures of an Authorised Firm that are ‘Non-Financial Private Sector Risk Exposures’. Rule 1.2.1 defines that expression to exclude credit exposures to other banks or to sovereigns, government bodies or agencies, or multilateral development banks.
2. An Authorised Firm will need to follow the following steps to calculate its CCyB Requirement:
 - a. identify the jurisdictions in which it has Non- Financial Private Sector Credit Exposures (Rule 3.9A.6 sets out how to determine the location of an exposure);
 - b. identify if a CCyB Rate applies in that jurisdiction and, if so, the date on which it takes effect (see Rules 3.9A.7 to 3.9A.9);
 - c. determine the weighted average of CCyB Rates applying to it (see Rule 3.9A.5); and
 - d. multiply the weighted average by the value of its Risk Weighted Assets.

3.9A.3 An Authorised Firm must not apply CET1 Capital that it maintains to meet the Countercyclical Capital Buffer Requirement towards meeting:

- (a) its Risk Capital Requirement;
- (b) its Capital Conservation Buffer Requirement;
- (c) an HLA Capital Buffer Requirement; or
- (d) an Individual Capital Requirement that the DFSA may impose on it under chapter 10.

3.9A.4 The Countercyclical Capital Buffer Requirement applies on both a solo and a consolidated basis for Authorised Firms forming part of a Group.

Weighted Average of CCyB Rates

- 3.9A.5** (1) The rate to be used to calculate an Authorised Firm’s Countercyclical Capital Buffer is the weighted average of the CCyB Rates that apply in jurisdictions in which it has Non-Financial Private Sector Credit Exposures.
- (2) The weighting applied to the CCyB Rate in each jurisdiction is the risk-weighted amount of an Authorised Firm’s Non-Financial Private Sector Credit Exposures in that jurisdiction, divided by the risk-weighted amount of its Non-Financial Private Sector Credit Exposures in all jurisdictions.

Determining the location of credit exposures

- 3.9A.6** (1) This Rule specifies how an Authorised Firm must determine the jurisdiction in which it has a Non-Financial Private Sector Credit Exposure.
- (2) The jurisdiction in which an Authorised Firm has an exposure is to be determined by allocating the exposure to the jurisdiction where, to the best of the Authorised Firm's knowledge and information, the risk ultimately lies.
- (3) If it is not reasonably possible to determine the jurisdiction of an exposure under (2), then the jurisdiction in which the Authorised Firm has the exposure is the jurisdiction where the exposure is booked.

Guidance

1. The location of an Authorised Firm's credit exposure is determined according to the concept of 'ultimate risk', i.e. the location where the risk ultimately lies. This is usually the location of the counterparties, irrespective of the Authorised Firm's own physical location or place of incorporation.
2. The following examples illustrate how the concept of ultimate risk applies:
 - a. if a firm has an exposure to a borrower in country A, and the risk mitigant (e.g. a guarantor) is in country B, then the ultimate risk is in country B;
 - b. if, in the example in a, the exposure is only partly mitigated, then the ultimate risk would be split between the uncovered portion in country A and a covered portion in country B;
 - c. if a firm has an exposure to a borrower that is a Branch in country A, and the head office of the Branch is in country B, then the ultimate risk is in country B; and
 - d. if a firm has an exposure to a borrower in country A, and the exposure is to finance a project in country B, then the ultimate risk is in country B.

The CCyB Rate that applies in a jurisdiction

- 3.9A.7** (1) The Countercyclical Capital Buffer Rate for an exposure:
- (a) in the DIFC or elsewhere in the State, is the rate set by the Central Bank; and
 - (b) outside the State, is the rate set by the CCyB Authority for that jurisdiction, unless the DFSA has specified a rate under Rule 3.9A.8, in which case that specified rate applies.
- (2) If the rate specified by a CCyB Authority is more than 2.5% then it is taken to be equal to 2.5%, unless the DFSA specifies otherwise.
- 3.9A.8** (1) If the DFSA considers that the CCyB Rate in a jurisdiction outside the State is not sufficient to protect Authorised Firms from the risks of excessive credit growth in that jurisdiction, it may, for credit exposures in that jurisdiction:
- (a) specify a CCyB Rate even though no rate is imposed by the CCyB Authority for that jurisdiction; or

- (b) specify a CCyB Rate that is higher than the rate imposed by the CCyB Authority for that jurisdiction.
- (2) If the DFSA specifies a rate under this Rule, then that rate applies for Non-Financial Private Sector Credit Exposures in the jurisdiction.
- (3) The DFSA may vary or cancel a specified rate under this Rule.
- (4) The DFSA must notify affected Authorised Firms if it specifies a rate, or if it varies or cancels a rate, under this Rule.

Effective date of CCyB Rates

- 3.9A.9** (1) This Rule specifies when a CCyB Rate takes effect for the purposes of calculating a CCyB Buffer under this section.
- (2) A CCyB Rate for a jurisdiction takes effect from whichever is the later of:
- (a) 12 months after the CCyB Authority announces the rate or the DFSA notifies the rate under Rule 3.9A.8 (as the case may be); or
 - (b) 1 July 2018.
- (3) In exceptional circumstances, the DFSA may specify that a CCyB Rate is to take effect from a date earlier or later than that specified in (2).

Guidance

1. CCyB Rates are usually specified to apply after an advance announcement period i.e. a period between when it is announced and when it takes effect, which gives Authorised Firms sufficient time to adopt the new capital buffer. The effect of Rule 3.9A.9(2)(a) is that Authorised Firms will usually have 12 months from the announcement to adopt a buffer.
2. As a transitional measure, Rule 3.9A.9(2)(b) has the effect that Authorised Firms will have at least 6 months from the day on which this section commences (1 January 2018) to adopt a buffer, even if the relevant rate was announced 12 months before the day the section commences.

For example: If a CCyB Authority announced on 1 February 2017 a CCyB Rate of 1% that would apply to credit exposures in its jurisdiction, this would usually take effect on 1 February 2018. However, under Rule 3.9A.9(2)(b), instead an Authorised Firm has until 1 July 2018 (6 months after the commencement of this Rule) to adopt the buffer.

3.9B HLA Capital Buffer

Guidance

Under section 1.4, the DFSA may designate an Authorised Firm as a systemically important bank (SIB). This section requires a SIB to maintain a further capital buffer, a higher loss absorbency capital buffer (HLA Capital Buffer), and sets out how the HLA Capital Buffer is calculated.

- 3.9B.1** This section applies to an Authorised Firm in Category 1, 2 (other than a Matched Principal) or 5 that the DFSA has designated as a SIB.

HLA Capital Buffer Requirement

3.9B.2 A SIB must maintain an HLA Capital Buffer of CET1 Capital that is calculated using the following formula:

$$\text{HLA Capital Buffer} = \text{HLA Ratio} \times \text{Relevant RWA}$$

where:

“HLA Capital Buffer” is the HLA Capital Buffer that the Authorised Firm must maintain;

“HLA Ratio” is the ratio determined by the DFSA for that Authorised Firm under Rule 3.9B.6; and

“Relevant RWA”:

- (a) for a G-SIB, is the value of the its Risk Weighted Assets; or
- (b) for a D-SIB, is the value of its Risk Weighted Assets in jurisdictions for which it is considered to be systemically important.

3.9B.3 If an Authorised Firm is both a G-SIB and a D-SIB, the HLA Capital Buffer that applies under this section is the higher of the amount calculated under Rule 3.9B.2 for the firm as a G-SIB and the amount calculated under that Rule for the firm as a D-SIB.

3.9B.4 An Authorised Firm must not apply CET1 Capital that it maintains to meet an HLA Capital Buffer Requirement towards meeting:

- (a) its Risk Capital Requirement;
- (b) its Capital Conservation Buffer Requirement;
- (c) its Countercyclical Capital Buffer Requirement; or
- (d) an Individual Capital Requirement that the DFSA has imposed on it under chapter 10.

3.9B.5 The HLA Capital Buffer Requirement applies on both a solo and a consolidated basis for Authorised Firms forming part of a Group.

HLA ratio

- 3.9B.6**
- (1) The DFSA must determine an HLA Ratio for each Authorised Firm that it designates as a G-SIB or D-SIB.
 - (2) The HLA Ratio determined under (1) for a D-SIB must be not less than 1% and not more than 3.5%.
 - (3) The DFSA may vary the HLA Ratio determined under this Rule, provided that for a D-SIB the ratio as varied is within the range specified in (2).
 - (4) The procedures in Schedule 3 to the Regulatory Law apply to a DFSA decision to set or vary an HLA Ratio for an Authorised Firm.
 - (5) If the DFSA decides to set or vary an HLA Ratio, the Authorised Firm may refer the matter to the FMT for review.

- (6) Paragraphs (4) and (5) do not apply to a decision relating to the HLA Ratio for a G-SIB designated under Rule 1.4.1.

Guidance

1. The DFSA is likely to base the HLA Ratio it determines for a G-SIB on the rate specified for that G-SIB by the Financial Stability Board, in consultation with the Basel Committee. For a D-SIB, the DFSA will determine an HLA Ratio that is between 1% and 3.5% (see Rule 3.9B.6(2)).
2. The Schedule 3 procedures and the right of review by the FMT do not apply to a rate applied to a G-SIB designated under Rule 1.4.1. This is because the rate specified by the DFSA for such a G-SIB will be the rate recommended by the FSB and Basel Committee.

3.9C Failure to meet a Capital Buffer Requirement

Guidance

This section sets out measures that an Authorised Firm must take if it is not meeting a Capital Buffer Requirement, i.e. its Capital Conservation Buffer Requirement, CCyB Requirement or HLA Capital Buffer Requirement. The measures, such as not distributing capital and preparing a plan to restore capital, do not limit other action that the DFSA may take against the firm for failing to meet the requirement.

- 3.9C.1** This section applies to an Authorised Firm in Category 1, 2 (other than a Matched Principal) or 5.

Restrictions on distributions

- 3.9C.2** Where an Authorised Firm fails to meet a Capital Buffer Requirement, it must:

- (a) calculate the maximum distributable amount in accordance with Rule 3.9C.5;
- (b) ensure that it does not undertake any of the following actions until it has calculated the maximum distributable amount and notified the DFSA under Rule 3.9C.6:
 - (i) make a distribution in connection with CET1 Capital;
 - (ii) create an obligation to pay variable remuneration or discretionary pension benefits or pay variable remuneration if the obligation to pay was created at a time when the institution failed to meet a Capital Buffer Requirement; or
 - (iii) make payments on AT1 and T2 Capital instruments.

- 3.9C.3** An Authorised Firm must:

- (1) in subsequently taking any of the actions described in Rule 3.9C.2(b)(i) to (iii), ensure that it distributes no more than its calculated maximum distributable amount; and

- (2) prepare and submit a capital conservation plan pursuant to Rule 3.9C.8.

3.9C.4 For the purposes of Rule 3.9C.2(b)(i), a distribution in connection with CET1 Capital includes any of the following:

- (a) payment of cash dividends;
- (b) distribution of fully or partly paid bonus shares or other capital instruments;
- (c) a redemption or purchase by an institution of its own shares or other capital instruments;
- (d) a repayment of amounts paid up in connection with capital; or
- (e) a distribution of other items referred to in section 3.13 as eligible for inclusion as CET1 Capital.

3.9C.5 (1) In this section, a reference to a “maximum distributable amount” means the maximum amount that an Authorised Firm may distribute in connection with CET1 Capital as specified in Rules 3.9C.2 and 3.9C.3.

- (2) Subject to (3), an Authorised Firm must determine the maximum distributable amount by multiplying the sum specified in (a) by the factor determined under (b):

- (a) the total of interim or year-end profits that were not included in CET1 Capital pursuant to Rule 3.13.2 and which have accrued after the most recent distribution of profits and after any of the actions referred to in Rule 3.9C.2(b);
- (b) where the CET1 Capital of the Authorised Firm (which is not used to meet the Capital Requirement), expressed as a percentage of the firm’s RWA, is:
 - (i) within the first quartile (0%-25%) of its Capital Buffer, the factor is 0;
 - (ii) within the second quartile (25%-50%) of its Capital Buffer, the factor is 0.2;
 - (iii) within the third quartile (50%-75%) of its Capital Buffer, the factor is 0.4; and
 - (iv) within the fourth quartile (75%-100%) of its Capital Buffer, the factor is 0.6.

- (3) If an Authorised Firm undertakes any action under Rule 3.9C.2(b), it must take that into account and reduce the maximum distributable amount accordingly.

3.9C.6 For the purpose of Rule 3.9C.2(b), where an Authorised Firm intends to distribute any of its distributable profits or intends to undertake an action referred to in Rule 3.9C.2(b)(i) to (iii), the Authorised Firm must notify the DFSA and provide the following information:

- (a) the amount of capital maintained by the Authorised Firm, subdivided as follows:
 - (i) CET1 Capital,
 - (ii) AT1 Capital, and
 - (iii) T2 Capital;
- (b) the amount of its interim and year-end profits;
- (c) the maximum distributable amount calculated in accordance with this section; and
- (d) the amount of distributable profits it intends to allocate between the following:
 - (i) dividend payments,
 - (ii) share buybacks,
 - (iii) payments on AT1 Capital instruments, and
 - (iv) the payment of variable remuneration or discretionary pension benefits, whether by creation of a new obligation to pay, or by payment pursuant to an obligation to pay created at a time when the institution failed to meet a Capital Buffer Requirement.

Guidance

Upon receiving a notification under this Rule, the DFSA will make an assessment of the firm's ability to meet and maintain its Capital Requirement on a sustainable basis going forward.

3.9C.7 An Authorised Firm must maintain systems and processes to ensure that the amount of distributable profits and the maximum distributable amount are calculated accurately, and must be able to demonstrate that accuracy to the DFSA on request.

Capital conservation plan

3.9C.8 Where an Authorised Firm fails to meet a Capital Buffer Requirement, it must prepare a capital conservation plan and submit it to the DFSA no later than 5 business days after it identified its failure to meet the Capital Buffer Requirement. The capital conservation plan must include the following:

- (a) estimates of income and expenditure and a forecast balance sheet;
- (b) measures to increase the Capital Resources of the Authorised Firm;
- (c) a plan and timeframe for the increase of own funds with the objective of restoring the Capital Buffer; and
- (d) any other information the DFSA might need in order effectively to carry out its considerations referred to in Rule 3.9C.9.

3.9C.9 (1) Following assessment, the DFSA will approve the capital conservation plan only if it considers that the plan, if implemented, would be reasonably likely

to conserve or raise sufficient capital to enable the Authorised Firm to meet its Capital Requirement and Capital Buffer Requirement, within a period that the DFSA considers appropriate.

- (2) If the DFSA does not approve the capital conservation plan, the DFSA may require the Authorised Firm to increase its CET1 Capital to meet the Capital Requirement and the Capital Buffer Requirement, within a specified period of time.

PART 4 – Calculating Capital Resources

3.10 Application

3.10.1 This part applies to an Authorised Firm in any Category.

Guidance

The earlier section 3.2 imposes a number of basic requirements on an Authorised Firm, including requirements to:

- a. have and maintain Capital Resources in accordance with these Rules (see Rule 3.2.2); and
- b. maintain an amount of Capital Resources that exceeds the amount of the firm's Capital Requirement (see Rule 3.2.3).

3.11 Calculation of Capital Resources

3.11.1 The total of Capital Resources is derived according to the following formula:

$$\text{T1 Capital} + \text{T2 Capital} = \text{Capital Resources}$$

where:

- (a) "T1 Capital" represents Tier 1 capital as the sum of CET1 Capital and AT1 Capital;
- (b) "CET1 Capital" represents Common Equity Tier 1 capital assessed in accordance with section 3.13;
- (c) "AT1 Capital" represents Additional Tier 1 capital assessed in accordance with section 3.14; and
- (d) "T2 Capital" represents Tier 2 capital assessed in accordance with section 3.15.

3.11.2 An Authorised Firm must calculate its Capital Resources in accordance with the table below and the provisions in sections 3.12 to 3.15.

(A1) Elements of Common Equity Tier 1 (CET1) Capital
(A2) Adjustments to/deductions from CET1 Capital
(A3) CET1 Capital = A1 – A2
(A4) Elements of Additional Tier 1 (AT1) Capital
(A5) Deductions from AT1 Capital
(A6) AT1 Capital = A4 – A5
(A7) Tier 1 (T1) Capital = A3 + A6
(A8) Elements of Tier 2 (T2) Capital
(A9) Deductions from T2 Capital
(A10) Tier 2 (T2) Capital = A8 – A9
(A11) Capital Resources = A7 + A10

3.12 Tier 1 capital (T1 Capital)

3.12.1 The Tier 1 capital (referred to in these Rules as T1 Capital) of an Authorised Firm must be calculated as the total of its Common Equity Tier 1 capital (referred to in these Rules as CET1 Capital) and its Additional Tier 1 capital (referred to in these Rules as AT1 Capital).

3.13 Common Equity Tier 1 capital (CET1 Capital)

3.13.1 The CET1 Capital constitutes the sum of CET1 capital elements in Rule 3.13.2, subject to the adjustments, deductions and exemptions stipulated later in this section.

3.13.2 CET1 Capital consists of the sum of the following capital elements:

- (a) capital instruments, provided the conditions laid down in Rule 3.13.3 are fully met;
- (b) share premium accounts related to the instruments referred to in (a);
- (c) retained earnings;
- (d) accumulated other comprehensive income, as defined in the International Financial Reporting Standards; and

- (e) other reserves which are required to be disclosed under International Financial Reporting Standards, excluding any amounts already included in accumulated other comprehensive income or retained earnings.
- 3.13.3** (1) For the purposes of Rule 3.13.2(a), a capital instrument is eligible for inclusion in CET1 Capital where all the following conditions are met:
- (a) the instruments are issued directly by the Authorised Firm with the prior approval of the shareholders of the Authorised Firm;
 - (b) the instruments are fully paid up and their purchase is not funded directly or indirectly by the Authorised Firm;
 - (c) the instruments meet all the following conditions as regards their classification:
 - (i) they qualify as equity capital within the meaning of the DIFC Companies Law;
 - (ii) they are classified as equity within the meaning of the International Financial Reporting Standards; and
 - (iii) they are classified as equity capital for the purposes of determining balance sheet insolvency, under the DIFC Insolvency Law;
 - (d) the instruments are clearly and separately disclosed on the balance sheet in the financial statements of the Authorised Firm;
 - (e) the instruments are perpetual;
 - (f) the principal amount of the instruments may not be reduced or repaid, except in either of the following cases:
 - (i) the liquidation of the Authorised Firm; or
 - (ii) discretionary repurchases of the instruments or other discretionary means of reducing capital, where the Authorised Firm has notified the DFSA of its intention to do so, in writing, at least 30 days prior to taking such steps;
 - (g) the provisions governing the instruments do not indicate expressly or implicitly that the principal amount of the instruments would or might be reduced or repaid other than in the liquidation of the Authorised Firm, and the Authorised Firm does not otherwise provide such an indication prior to or at issuance of the instruments;
 - (h) the instruments meet the following conditions as regards distributions:
 - (i) there are no preferential distributions, including in relation to other CET1 Capital instruments, and the terms governing the instruments do not provide preferential rights to payment of distributions;

- (ii) distributions to holders of the instruments may be paid only out of distributable items;
 - (iii) the conditions governing the instruments do not include a cap or other restriction on the maximum level of distributions;
 - (iv) the level of distributions is not determined on the basis of the amount for which the instruments were purchased at issuance;
 - (v) the conditions governing the instruments do not include any obligation for the Authorised Firm to make distributions to their holders and the Authorised Firm is not otherwise subject to such an obligation; and
 - (vi) non-payment of distributions does not constitute an event of default of the Authorised Firm;
- (i) compared to all the capital instruments issued by the Authorised Firm, the instruments absorb the first and proportionately greatest share of losses as they occur, and each instrument absorbs losses to the same degree as all other CET1 Capital instruments;
- (j) the instruments rank below all other claims in the event of insolvency or liquidation of the Authorised Firm;
- (k) the instruments entitle their owners to a claim on the residual assets of the Authorised Firm, which, in the event of its liquidation and after the payment of all senior claims, is proportionate to the amount of such instruments issued and is not fixed or subject to a cap;
- (l) the instruments are not secured, or guaranteed by any of the following:
 - (i) the Authorised Firm or its Subsidiaries;
 - (ii) any Parent of the Authorised Firm or its Subsidiaries; or
 - (iii) any member of its Financial Group; and
- (m) the instruments are not subject to any arrangement, contractual or otherwise, that enhances the seniority of claims under the instruments in insolvency or liquidation.
- (2) The conditions in (1)(i) must be complied with notwithstanding a write down on a permanent basis of the principal amount of AT1 Capital instruments.
- (3) Where any of the conditions in (1) cease to be met:
 - (a) the instrument must cease to qualify as a CET1 Capital instrument; and
 - (b) the share premium accounts that relate to that instrument must cease to qualify as a CET1 element.

3.13.4 For the purposes of Rule 3.13.2(c), an Authorised Firm may include interim or year-end net profits in CET1 Capital before the Authorised Firm has approved its annual audited accounts confirming its final profit or loss for the year, but only where:

- (a) those profits have been reviewed by the External Auditor of the Authorised Firm, who is responsible for auditing its accounts; and
- (b) the Authorised Firm is fully satisfied that any foreseeable charge or dividend has been deducted from the amount of those net profits.

Guidance

The review of the interim or year-end profits of the Authorised Firm referred to in Rule 3.13.4 should provide an adequate level of assurance that those profits have been evaluated in accordance with the principles set out in the International Financial Reporting Standards. The DFSA may request an Authorised Firm to provide it with a copy of its external auditor's opinion on whether the interim profits are reasonably stated.

CET1 Adjustments

3.13.5 An Authorised Firm must, in the calculation of CET1 Capital, exclude the following:

- (a) any increase in its equity under the International Financial Reporting Standards; including:
 - (i) where such an increase is associated with future margin income that results in a gain on sale for the Authorised Firm; and
 - (ii) where the Authorised Firm is the Originator of a securitisation, net gains that arise from the capitalisation of future income from the securitised assets that provide Credit Enhancement to positions in the securitisation;
- (b) the amount of cash flow hedge reserve related to gains or losses on cash flow hedges of financial instruments that are not valued at fair value, including projected cash flows;
- (c) all unrealised gains or losses on liabilities of the Authorised Firm that are valued at fair value, and which result from changes in the Authorised Firm's own credit quality, except when such gains or losses are offset by a change in the fair value of another financial instrument which is measured at fair value and resulting from changes in the Authorised Firm's own credit quality; and
- (d) all accounting valuation adjustments to an Authorised Firm's liabilities which result from the Authorised Firm's own credit quality.

3.13.6 Except for the items referred to in Rule 3.13.5, an Authorised Firm must not make any adjustments to remove from its Capital Resources unrealised gains or losses on their assets or liabilities measured at fair value.

Guidance

1. An Authorised Firm is expected to follow the guidance provided in respect of prudent valuation in section 2.4 and in App2, in valuing all its assets measured at fair value while calculating its Capital Resources.

2. An Authorised Firm should not under Rule 3.13.5(d) offset valuation adjustments arising from its own credit quality with those arising from its Counterparty Credit Risk.

CET1 deductions

3.13.7 Subject to the following Rules in this section, an Authorised Firm must deduct the following from the calculation of its CET1 Capital:

- (a) losses for the current financial year;
- (b) goodwill and other intangible assets as defined in the International Financial Reporting Standards;
- (c) deferred tax assets that rely on future profitability;
- (d) defined benefit pension fund assets of the Authorised Firm;
- (e) the applicable amount, by reference to Rule 3.13.12, of direct and indirect holdings by an Authorised Firm of its own CET1 Capital instruments including instruments under which an Authorised Firm is under an actual or contingent obligation to effect a purchase by virtue of an existing contractual obligation;
- (f) holdings of the CET1 Capital instruments of Relevant Entities where those entities have a reciprocal cross holding with the Authorised Firm which have the effect of artificially inflating the Capital Resources of the Authorised Firm;
- (g) the applicable amount, by reference to Rule 3.13.13, of direct and indirect holdings by the Authorised Firm of CET1 Capital instruments of Relevant Entities where the Authorised Firm does not have a significant investment in those entities;
- (h) the applicable amount, by reference to Rules 3.13.13 and 3.13.18, of direct and indirect holdings by the Authorised Firm of the CET1 Capital instruments of Relevant Entities where the Authorised Firm has a significant investment in those entities;
- (i) the amount of items required to be deducted from the calculation of AT1 Capital in accordance with the relevant Rules under section 3.14, that exceeds the AT1 Capital of the Authorised Firm;
- (j) the Exposure amount of the following items which qualify for a risk weight of 1000%, where the Authorised Firm deducts that Exposure amount from CET1 Capital as an alternative to applying a risk weight of 1000%;
 - (i) Qualifying Holdings;
 - (ii) securitisation positions, in accordance with relevant Rules in chapter 4; and
 - (iii) free deliveries, in accordance with the Rules in section A4.6; and
- (k) for an Authorised Firm which is a Partnership or Limited Liability Partnership, the amount by which the aggregate of the amounts withdrawn by its partners or members exceeds the profits of that firm.

CET1 deductions relating to intangible assets

3.13.8 For the purposes of Rule 3.13.7(b), an Authorised Firm must determine the intangible assets to be deducted in accordance with the following:

- (a) the amount to be deducted must be reduced by the amount of associated deferred tax liabilities that would be extinguished if the intangible assets became impaired or were derecognised under the International Financial Reporting Standards; and
- (b) the amount to be deducted must include goodwill included in the valuation of significant investments of the Authorised Firm.

CET1 deductions relating to deferred tax assets

- 3.13.9** (1) For the purposes of Rule 3.13.7(c), and subject to (2), the amount of deferred tax assets that rely on future profitability must be calculated without reducing it by the amount of the associated deferred tax liabilities of the Authorised Firm.
- (2) The amount of deferred tax assets that rely on future profitability may be reduced by the amount of the associated deferred tax liabilities of the Authorised Firm, provided the following conditions are met:
- (a) those deferred tax assets and associated deferred tax liabilities both arise from the tax law of the same tax jurisdiction; and
 - (b) the taxation authority of that tax jurisdiction permits the offsetting of deferred tax assets and the associated deferred tax liabilities.

Guidance

1. Associated deferred tax liabilities of the Authorised Firm used for the purposes of Rule 3.13.9 may not include deferred tax liabilities that reduce the amount of intangible assets or defined benefit pension fund assets required to be deducted. The amount of associated deferred tax liabilities referred to in this guidance should be allocated between the following:
 - a. deferred tax assets that rely on future profitability and arise from temporary differences that are not deducted as part of a threshold exemption for deductions from CET 1 Capital; and
 - b. all other deferred tax assets that rely on future profitability.
2. An Authorised Firm should allocate the associated deferred tax liabilities according to the proportion of deferred tax assets that rely on future profitability that the items referred to in Guidance note 1a and b represent.

- 3.13.10** (1) An Authorised Firm must apply a risk weight in accordance with chapter 4 as applicable, to deferred tax assets that do not rely on future profitability.
- (2) For the purpose of (1), deferred tax assets that do not rely on future profitability comprise the following:
- (a) overpayments of tax by the Authorised Firm for the current year;

- (b) current year tax losses of the Authorised Firm carried back to previous years that give rise to a claim on, or a receivable from, a central government, regional government or local tax authority; and
- (c) deferred tax assets arising from temporary differences which, in the event the Authorised Firm incurs a loss, becomes insolvent or enters liquidation, are replaced, on a mandatory and automatic basis in accordance with the applicable national law, with a claim on the central government of the jurisdiction in which the Authorised Firm is incorporated which must absorb losses to the same degree as CET1 Capital instruments on a going concern basis and in the event of insolvency or liquidation of the Authorised Firm.

Deductions relating to defined benefit pension fund assets

3.13.11 For the purposes of Rule 3.13.7(d), the amount of defined benefit pension fund assets to be deducted from CET1 Capital must be reduced by the following:

- (a) the amount of any associated deferred tax liability which could be extinguished if the assets became impaired or were derecognised under the International Financial Reporting Standards; and
- (b) the amount of assets in the defined benefit pension fund which the Authorised Firm has an unrestricted ability to use where the Authorised Firm has provided adequate advance notification of its intention to use those assets to the DFSA. Those assets used to reduce the amount to be deducted must receive a risk weight in accordance with chapter 4 of PIB.

Deductions relating to holdings of own CET1 Capital instruments

3.13.12 For the purposes of Rule 3.13.7(e), an Authorised Firm must calculate holdings of its own CET1 Capital instruments on the basis of gross long positions subject to the following exceptions:

- (a) an Authorised Firm must calculate the amount of holdings of own CET1 Capital instruments in the Trading Book on the basis of the net long position, provided the long and short positions are in the same underlying Exposure and the short positions involve no Counterparty Credit Risk;
- (b) an Authorised Firm must determine the amount to be deducted for indirect holdings in the Trading Book that take the form of holdings of index Securities by calculating the underlying Exposure to own CET1 Capital instruments included in the indices; and
- (c) an Authorised Firm must net gross long positions in own CET1 Capital instruments in its Trading Book resulting from holdings of index Securities against short positions in own CET1 Capital instruments resulting from short positions in the underlying indices, including where those short positions involve Counterparty Credit Risk.

CET1 deductions relating to significant investment in a Relevant Entity

3.13.13 For the purposes of Rules 3.13.7(g) and (h), an investment by an Authorised Firm in a Relevant Entity must be considered as a significant investment if it meets any of the following conditions:

- (a) the Authorised Firm owns more than 10% of the CET1 Capital instruments issued by that entity;
- (b) the Authorised Firm has Close Links with that entity and owns CET1 Capital instruments issued by that entity; and
- (c) the Authorised Firm owns CET1 Capital instruments issued by that entity and the entity is not included in consolidation pursuant to chapter 8 of PIB but is included in the same accounting consolidation as the Authorised Firm for the purposes of financial reporting under the International Financial Reporting Standards.

Deductions relating to CET1 Capital instruments in Relevant Entities

3.13.14 For the purposes of Rule 3.13.7(f), (g) and (h), the amount of holdings of CET1 Capital instruments and other capital instruments of Relevant Entities to be deducted, must be calculated, subject to Rule 3.13.15, on the basis of the gross long positions.

3.13.15 For the purposes of Rule 3.13.7(g) and (h), an Authorised Firm must make the deductions in accordance with the following:

- (a) the holdings in the Trading Book of the capital instruments of Relevant Entities must be calculated on the basis of the net long position in the same underlying Exposure provided the maturity of the short position matches the maturity of the long position or has a residual maturity of at least one year; and
- (b) the amount to be deducted for indirect holdings in the Trading Book of the capital instruments of Relevant Entities that take the form of holdings of index Securities must be determined by calculating the underlying Exposure to the capital instruments of the Relevant Entities in the indices.

3.13.16 (1) For the purposes of Rule 3.13.7(g), the amount to be deducted is calculated by multiplying the amount referred to in (a) by the factor derived from the calculation referred to in (b):

- (a) the aggregate amount by which the direct, indirect and synthetic holdings by the Authorised Firm of the CET1, AT1 and T2 Capital instruments of Relevant Entities, in which the Authorised Firm does not have a significant investment, exceeds 10% of the CET1 items of the Authorised Firm calculated after applying the following to CET1 items:
 - (i) all of the adjustments referred to in Rules 3.13.5 and 3.13.6;
 - (ii) the deductions referred to in Rules 3.13.7(a) to (f) and (h) to (j), excluding the amount to be deducted for deferred tax assets that

rely on future profitability and arise from temporary differences;
and

- (iii) the deductions referred to in Rules 3.13.14 and 3.13.15;
 - (b) the amount of direct and indirect holdings by the Authorised Firm of the CET1 Capital instruments of Relevant Entities divided by the aggregate amount of direct and indirect holdings by the Authorised Firm of the CET1, AT1 and T2 Capital instruments issued by those Relevant Entities.
- (2) An Authorised Firm must exclude Underwriting positions held for 5 working days or fewer from the amount referred to in (1)(a) and from the calculation of the factor referred to in (1)(b).
- (3) The amount to be deducted pursuant to (1) must be apportioned across each CET1 Capital instrument held. An Authorised Firm must determine the portion of holdings of CET1 Capital instruments that is to be deducted pursuant to (1) by dividing the amount specified in (a) by the amount specified in (b):
- (a) the amount of holdings required to be deducted pursuant to (1)(a);
 - (b) the aggregate amount of direct and indirect holdings by the Authorised Firm of all the capital instruments of Relevant Entities in which the Authorised Firm does not have a significant investment.
- 3.13.17** (1) The amount of holdings referred to in Rule 3.13.7(g) that is equal to or less than 10% of the CET1 items of the Authorised Firm after applying the provisions laid down in (1)(a)(i) to (iii) must not be deducted and must be subject to the applicable risk weights in accordance with chapter 4.
- (2) An Authorised Firm must determine the portion of holdings of all the capital instruments that is risk weighted by dividing the amount specified in (a) by the amount specified in (b):
- (a) the amount of holdings required to be risk weighted pursuant to Rule 3.13.17(1);
 - (b) the aggregate amount of direct and indirect holdings by the Authorised Firm of all the capital instruments of Relevant Entities in which the Authorised Firm does not have a significant investment.
- 3.13.18** For the purposes of Rule 3.13.7(h), the amount to be deducted from CET1 elements must exclude Underwriting positions held for 5 working days or fewer and must be determined in accordance with Rules 3.13.14 and 3.13.15.

CET1 exemptions from deductions

- 3.13.19** (1) In making the deductions required pursuant to Rules 3.13.7(c) and (h), an Authorised Firm must not deduct the items listed in (a) and (b), where in aggregate they are equal to or less than 15% of CET1 Capital:

- (a) deferred tax assets that are dependent on future profitability and arise from temporary differences, and in aggregate are equal to or less than 10% of the CET1 items of the Authorised Firm calculated after applying the following:
 - (i) adjustments referred in Rules 3.13.5 and 3.13.6; and
 - (ii) deductions referred to in (a) to (g) and (i) to (j) of Rule 3.13.7, excluding deferred tax assets that rely on future profitability and arise from temporary differences.
- (b) where an Authorised Firm has a significant investment in a Relevant Entity, the direct and indirect holdings of that Authorised Firm of the CET1 Capital instruments of those entities that in aggregate are equal to or less than 10% of the CET1 items of the Authorised Firm calculated after applying the following:
 - (i) adjustments referred in Rules 3.13.5 and 3.13.6; and
 - (ii) deductions referred to in (a) to (h) and (i) to (j) of Rule 3.13.7 excluding deferred tax assets that rely on future profitability and arise from temporary differences.
- (2) Items that are not deducted pursuant to (1) must be risk weighted at 200% and subject to the requirements of chapter 4, as applicable.

3.14 Additional Tier 1 capital (AT1 Capital)

3.14.1 The AT1 Capital constitutes the sum of AT1 Capital elements in Rule 3.14.2, subject to the deductions stipulated later in this section.

3.14.2 AT1 Capital consists of the sum of the following capital elements:

- (a) capital instruments which meet the eligibility criteria laid down in Rule 3.14.3; and
- (b) the share premium accounts related to the instruments referred to in (a).

3.14.3 (1) For the purposes of Rule 3.14.2(a), a capital instrument is eligible for inclusion in AT1 Capital where all the following conditions are met:

- (a) the instruments are issued and paid up;
- (b) the instruments are not purchased by any of the following:
 - (i) the Authorised Firm or its Subsidiaries; or
 - (ii) an Undertaking in which the Authorised Firm has participation in the form of ownership, direct or by way of control, of 20% or more of the voting rights or capital of that Undertaking;
- (c) the purchase of the instruments is not funded directly or indirectly by the Authorised Firm;

- (d) the instruments rank below T2 Capital instruments in the event of the insolvency of the Authorised Firm;
- (e) the instruments are not secured, or guaranteed by any of the following:
 - (i) the Authorised Firm or its Subsidiaries;
 - (ii) any Parent of the Authorised Firm or their Subsidiaries;
 - (iii) any member of its Financial Group in accordance with chapter 8; or
 - (iv) any Undertaking that has Close Links with entities referred to in (i) to (iii);
- (f) the instruments are not subject to any arrangement, contractual or otherwise that enhances the seniority of the claim under the instruments in insolvency or liquidation;
- (g) the instruments are perpetual and the provisions governing them include no incentive for the Authorised Firm to redeem them;
- (h) where the provisions governing the instruments include one or more call options, the option to call may be exercised at the sole discretion of the issuer;
- (i) the instruments may be called, redeemed or repurchased only where the Authorised Firm has notified the DFSA of its intention to call, redeem or repurchase the instruments in writing and well in advance, and not before 5 years after the date of issuance of the respective instruments;
- (j) the provisions governing the instruments do not indicate explicitly or implicitly that the instruments would or might be called, redeemed or repurchased and the Authorised Firm does not otherwise provide such an indication;
- (k) the Authorised Firm does not indicate explicitly or implicitly that the DFSA would not object to a plan to call, redeem or repurchase the instruments;
- (l) distributions under the instruments meet the following conditions:
 - (i) they are paid out of distributable items;
 - (ii) the level of distributions made on the instruments will not be modified based on the credit standing of the Authorised Firm or any of its Parents or any entities in its Financial Group;
 - (iii) the provisions governing the instruments give the Authorised Firm full discretion at all times to cancel the distributions on the instruments for an unlimited period and on a non-cumulative basis, and the Authorised Firm may use such cancelled payments without restriction to meet its obligations as they fall due;

- (iv) cancellation of distributions does not constitute an event of default of the Authorised Firm; and
 - (v) the cancellation of distributions imposes no restrictions on the Authorised Firm;
- (m) the instruments do not contribute to a determination that the liabilities of an Authorised Firm exceed its assets, where such a determination constitutes a test of insolvency under the DIFC Insolvency Law;
- (n) the provisions governing the instruments require the principal amount of the instruments to be written down, or the instruments to be converted to CET1 Capital instruments, upon the occurrence of a trigger event;
- (o) the provisions governing the instruments include no feature that could hinder the recapitalisation of the Authorised Firm; and
- (p) where the instruments are not issued directly by the Authorised Firm or by an operating entity within the Financial Group to which the Authorised Firm belongs, or by the Parent of the Authorised Firm, the proceeds are immediately available without limitation in a form that satisfies the conditions laid down in this Rule to any of the following:
 - (i) the Authorised Firm;
 - (ii) an operating entity within the Financial Group to which the Authorised Firm belongs; or
 - (iii) any Parent of the Authorised Firm.
- (2) For the purposes of (1)(l)(v) and (1)(o), the provisions governing AT1 Capital instruments must not include the following:
 - (a) a requirement for distributions on the instruments to be made in the event of a distribution being made on an instrument issued by the Authorised Firm that ranks to the same degree as, or more junior than, an AT1 Capital instrument;
 - (b) a requirement for the payment of distributions on CET1, AT1 or T2 Capital instruments to be cancelled in the event that distributions are not made on those AT1 Capital instruments; or
 - (c) an obligation to substitute the payment of interest or dividend by a payment in any other form.
- (3) For the purposes of (1)(n), the following provisions apply to AT1 Capital instruments:
 - (a) a trigger event occurs when the CET1 Capital of the Authorised Firm falls below either of the following:
 - (i) 66.25% of its Capital Requirement; or

- (ii) a level higher than 66.25%, where determined by the Authorised Firm and specified in the provisions governing the instrument;
- (b) where the provisions governing the instruments require them to be converted into CET1 Capital instruments upon the occurrence of a trigger event, those provisions must specify either of the following:
 - (i) the rate of such conversion and a limit on the permitted amount of conversion; or
 - (ii) a range within which the instruments will convert into CET1 Capital instruments;
- (c) where the provisions governing the instruments require their principal amount to be written down upon the occurrence of a trigger event, the write down must reduce all the following:
 - (i) the claim of the holder of the instrument in the liquidation of the Authorised Firm;
 - (ii) the amount required to be paid in the event of the call of the instrument; and
 - (iii) the distributions made on the instrument.
- (4) The following must apply where, in the case of an AT1 Capital instrument, the conditions laid down in this Rule cease to be met:
 - (a) that instrument must cease to qualify as an AT1 Capital instrument; and
 - (b) the part of the share premium accounts that relates to that instrument must cease to qualify as an AT1 Capital element.

AT1 regulatory deductions

3.14.4 Subject to the following Rules in this section, an Authorised Firm must deduct the following from the calculation of its AT1 Capital:

- (a) direct and indirect holdings by an Authorised Firm of own AT1 Capital instruments including instruments under which an Authorised Firm is under an actual or contingent obligation to effect a purchase by virtue of an existing contractual obligation;
- (b) holdings of the AT1 Capital instruments of Relevant Entities where those entities have a reciprocal cross holding with the Authorised Firm which have the effect of artificially inflating the Capital Resources of the Authorised Firm;
- (c) the amount determined in accordance with Rule 3.14.8 of direct and indirect holdings by the Authorised Firm of the AT1 Capital instruments of Relevant Entities where the Authorised Firm does not have a significant investment in those entities ;
- (d) direct and indirect holdings by the Authorised Firm of the AT1 Capital instruments of Relevant Entities where the Authorised Firm has a significant

investment in those entities, excluding Underwriting positions held for 5 working days or fewer; and

- (e) the amounts required to be deducted from T2 Capital pursuant to Rule 3.15.4 that exceed the T2 Capital of the Authorised Firm.

Deductions relating to holdings of own AT1 Capital instruments

3.14.5 For the purposes of Rule 3.14.4(a), an Authorised Firm must calculate holdings of its own AT1 Capital instruments on the basis of gross long positions subject to the following exceptions:

- (a) an Authorised Firm must calculate the amount of holdings of own AT1 Capital instruments in the Trading Book on the basis of the net long position provided the long and short positions are in the same underlying Exposure and the short positions involve no Counterparty Credit Risk;
- (b) an Authorised Firm must determine the amount to be deducted for indirect holdings in the Trading Book of own AT1 Capital instruments that take the form of holdings of index Securities by calculating the underlying Exposure to own AT1 Capital instruments in the indices; and
- (c) an Authorised Firm must net gross long positions in own AT1 Capital instruments in the Trading Book resulting from holdings of index Securities may be netted by the Authorised Firm against short positions in own AT1 instruments resulting from short positions in the underlying indices, including where those short positions involve Counterparty Credit Risk.

Deductions relating to AT1 Capital instruments in Relevant Entities

3.14.6 For the purposes of Rule 3.14.4(b), (c) and (d), the amount of holdings of AT1 Capital instruments of Relevant Entities to be deducted, must be calculated, subject to 3.14.7, on the basis of the gross long positions.

3.14.7 For the purposes of Rule 3.14.4(c) and (d), an Authorised Firm must make the deductions in accordance with the following:

- (a) the holdings in the Trading Book of the capital instruments of Relevant Entities must be calculated on the basis of the net long position in the same underlying Exposure provided the maturity of the short position matches the maturity of the long position or has a residual maturity of at least one year; and
- (b) the amount to be deducted for indirect holdings in the Trading Book of the capital instruments of Relevant Entities that take the form of holdings of index Securities must be determined by calculating the underlying Exposure to the capital instruments of the Relevant Entities in the indices.

AT1 deductions relating to significant investment in a Relevant Entity

3.14.8 (1) For the purposes of Rule 3.14.4(c), an Authorised Firm must calculate the applicable amount to be deducted by multiplying the amount referred to in (a) by the factor derived from the calculation referred to in (b):

- (a) the amount referred to in Rule 3.13.16(1)(a);
 - (b) the amount of direct and indirect holdings by the Authorised Firm of the AT1 Capital instruments of Relevant Entities divided by the aggregate amount of all direct and indirect holdings by the Authorised Firm of the CET1, AT1 and T2 Capital instruments of those Relevant Entities.
- (2) An Authorised Firm must exclude Underwriting positions held for 5 working days or fewer from the amount referred to in Rule 3.13.16(1)(a) and from the calculation of the factor referred to in (1)(b).
- (3) An Authorised Firm must determine the portion of holdings of AT1 Capital instruments that is to be deducted pursuant to (1) by dividing the amount specified in (a) by the amount specified in (b):
 - (a) the amount of holdings required to be deducted pursuant to (1)(a);
 - (b) the aggregate amount of direct and indirect holdings by the Authorised Firm of all the capital instruments of Relevant Entities in which the Authorised Firm does not have a significant investment.

3.15 Tier 2 capital (T2 Capital)

3.15.1 The T2 Capital constitutes the sum of T2 Capital elements in Rule 3.15.2, subject to the deductions stipulated later in this section.

3.15.2 T2 Capital consists of the sum of the following elements:

- (a) capital instruments which meet the eligibility criteria laid down in Rule 3.15.3 ; and
- (b) the share premium accounts related to the instruments referred to in (a).

3.15.3 (1) For the purpose of Rule 3.15.2(a), a capital instrument is eligible for inclusion in T2 Capital where all the following conditions are met:

- (a) the instruments are issued and fully paid-up;
- (b) the instruments are not purchased by any of the following:
 - (i) the Authorised Firm or its Subsidiaries;
 - (ii) an Undertaking in which the Authorised Firm has participation in the form of ownership, direct or by way of control, of 20% or more of the voting rights or capital of that Undertaking;
- (c) the purchase of the instruments is not funded directly or indirectly by the Authorised Firm;
- (d) the claim on the principal amount of the instruments under the provisions governing the instruments is wholly subordinated to claims of all non-subordinated creditors;

- (e) the instruments are not secured, or guaranteed by any of the following:
 - (i) the Authorised Firm or its Subsidiaries;
 - (ii) any Parent of the Authorised Firm or their Subsidiaries;
 - (iii) any member of the Financial Group to which the Authorised Firm belongs; or
 - (iv) any Undertaking that has Close Links with entities referred to in (i) to (iii);
- (f) the instruments are not subject to any arrangement that otherwise enhances the seniority of the claim under the instruments;
- (g) the instruments have an Original Maturity of at least 5 years;
- (h) the provisions governing the instruments do not include any incentive for them to be redeemed by the Authorised Firm;
- (i) where the instruments include one or more call options, the options are exercisable at the sole discretion of the Issuer;
- (j) the instruments may be called, redeemed or repurchased only where the Authorised Firm has notified the DFSA of its intention to call, redeem or repurchase the instruments in writing and well in advance, and not before 5 years after the date of issuance of the respective instruments;
- (k) the provisions governing the instruments do not indicate or suggest that the instruments would or might be redeemed or repurchased other than at maturity and the Authorised Firm does not otherwise provide such an indication or suggestion;
- (l) the provisions governing the instruments do not give the holder the right to accelerate the future scheduled payment of interest or principal, other than in the insolvency or liquidation of the Authorised Firm;
- (m) the level of interest or dividend payments due on the instruments will not be modified based on the credit standing of the Authorised Firm, its Parent or any member of its Financial Group; and
- (n) where the instruments are not issued directly by the Authorised Firm or by an operating entity within its Financial Group, or by its Parent, the proceeds are immediately available without limitation in a form that satisfies the conditions laid down in this Rule to any of the following:
 - (i) the Authorised Firm;
 - (ii) an operating entity within its Financial Group; or
 - (iii) any Parent of the Authorised Firm.

- (2) The extent to which T2 Capital instruments can be considered as eligible for inclusion in T2 Capital during the final 5 years of maturity of those instruments is calculated by multiplying the result derived from the calculation in (a) by the amount referred to in (b):
 - (a) the nominal amount of the instruments on the first day of the final 5 year period of their contractual maturity divided by the number of calendar days in that period;
 - (b) the number of remaining calendar days of contractual maturity of the instruments.
- (3) The following must apply where, in the case of a T2 Capital instrument, the conditions laid down in this Rule cease to be met:
 - (a) that instrument must cease to qualify as a T2 Capital instrument; and
 - (b) the part of the share premium accounts that relates to that instrument must cease to qualify as a T2 Capital element.

T2 regulatory deductions and exclusions

3.15.4 Subject to the following Rules in this section, an Authorised Firm must deduct the following from the calculation of its T2 Capital:

- (a) direct and indirect holdings by an Authorised Firm of own T2 Capital instruments, including own T2 instruments that an Authorised Firm could be obliged to purchase as a result of existing contractual obligations;
- (b) holdings of the T2 Capital instruments of Relevant Entities where those entities have a reciprocal cross holding with the Authorised Firm which have the effect of artificially inflating the Capital Resources of the Authorised Firm;
- (c) the amount of direct and indirect holdings by the Authorised Firm of the T2 Capital instruments of Relevant Entities where the Authorised Firm does not have a significant investment in those entities; and
- (d) direct and indirect holdings by the Authorised Firm of the T2 Capital instruments of Relevant Entities where the Authorised Firm has a significant investment in those entities, excluding Underwriting positions held for fewer than 5 working days.

Deductions relating to holdings of own T2 Capital instruments

3.15.5 For the purposes of Rule 3.15.4(a), an Authorised Firm must calculate holdings of its own T2 Capital instruments on the basis of the gross long positions subject to the following exceptions:

- (a) an Authorised Firm may calculate the amount of holdings in the Trading Book on the basis of the net long position provided the long and short positions are in the same underlying Exposure and the short positions involve no Counterparty Risk;
- (b) an Authorised Firm must determine the amount to be deducted for indirect holdings in the Trading Book of own T2 Capital instruments that take the

form of holdings of index Securities by calculating the underlying Exposure to own T2 Capital instruments in the indices; and

- (c) an Authorised Firm may net gross long positions in own T2 Capital instruments in the Trading Book resulting from holdings of index Securities against short positions in own T2 instruments resulting from short positions in the underlying indices, including where those short positions involve Counterparty Risk.

Deductions relating to T2 Capital instruments in Relevant Entities

3.15.6 For the purposes of Rule 3.15.4(b), (c) and (d), the amount of holdings of T2 Capital instruments and other capital instruments of Relevant Entities to be deducted, must be calculated, subject to 3.15.7, on the basis of the gross long positions.

3.15.7 For the purposes of Rule 3.15.4(c) and (d), an Authorised Firm must make the deductions in accordance with the following:

- (a) the holdings in the Trading Book of the capital instruments of Relevant Entities must be calculated on the basis of the net long position in the same underlying Exposure provided the maturity of the short position matches the maturity of the long position or has a residual maturity of at least one year; and
- (b) the amount to be deducted for indirect holdings in the Trading Book of the capital instruments of Relevant Entities that take the form of holdings of index Securities must be determined by calculating the underlying Exposure to the capital instruments of the Relevant Entities in the indices.

T2 deductions relating to insignificant investment in a Relevant Entity

3.15.8 (1) For the purposes of Rule 3.15.4(c), an Authorised Firm must calculate the applicable amount to be deducted by multiplying the amount referred to in (a) by the factor derived from the calculation referred to in (b):

- (a) the amount referred to in Rule 3.13.16(1)(a);
 - (b) the amount of direct and indirect holdings by the Authorised Firm of the T2 Capital instruments of Relevant Entities divided by the aggregate amount of all direct and indirect holdings by the Authorised Firm of the CET1, AT1 and T2 Capital instruments of those Relevant Entities.
- (2) An Authorised Firm must exclude Underwriting positions held for 5 working days or fewer from the amount referred to in Rule 3.13.16(1)(a) and from the calculation of the factor referred to in (1)(b).
 - (3) An Authorised Firm must determine the portion of holdings of T2 Capital instruments that is to be deducted by dividing the amount specified in (a) by the amount specified in (b):
 - (a) the amount of holdings required to be deducted pursuant to (1)(a);

- (b) the aggregate amount of direct and indirect holdings by the Authorised Firm of the capital instruments of Relevant Entities in which the Authorised Firm does not have a significant investment.

Exclusion in relation to Managing a PSIA

- 3.15.9** An Authorised Firm must exclude from T2 Capital any amount by which the total of the Profit Equalisation Reserve and the Investment Risk Reserve exceeds the Displaced Commercial Risk Capital Requirement calculated in accordance with IFR Rule 5.4.4.

3.16 Minority interests and instruments issued by Subsidiaries

Minority interests that qualify for inclusion in consolidated CET1 Capital

- 3.16.1** Minority interests must include the CET1 Capital instruments, plus the related retained earnings and share premium accounts, of a Subsidiary only where all of the following conditions are met:

- (a) the Subsidiary is one of the following:
 - (i) an Authorised Firm; or
 - (ii) a regulated entity,
- (b) the Subsidiary is a member of the Financial Group and included in the scope of consolidated supervision in accordance with chapter 8; and
- (c) those CET1 Capital instruments are owned by persons other than the Undertakings included in the Financial Group.

- 3.16.2** Minority interests that are funded directly or indirectly, through a special purpose entity or otherwise, by the Parent of the Authorised Firm or any member of its Financial Group must not qualify for inclusion in the consolidated CET1 Capital of the Financial Group.

- 3.16.3** An Authorised Firm must determine the amount of minority interests of a Subsidiary that is eligible for inclusion in its consolidated CET1 Capital by subtracting from the minority interests of that Subsidiary the result of multiplying the amount referred to in (a) by the percentage referred to in (b):

- (a) the CET1 Capital of the Subsidiary minus the lesser of the following:
 - (i) the amount of CET1 Capital of that Subsidiary required to meet the sum of the Subsidiary's CET1 Capital requirement (on a solo basis) of 60% of the Risk Capital Requirement and its Capital Conservation Buffer requirement of 25% of the Risk Capital Requirement; or
 - (ii) the amount of consolidated CET1 Capital that relates to that Subsidiary that is required on a consolidated basis to meet the sum of its Financial Group's CET1 Capital requirement of 60% of the Risk Capital Requirement and its Capital Conservation Buffer requirement of 25% of the Risk Capital Requirement;

- (b) the minority interests of the Subsidiary expressed as a percentage of all CET1 Capital instruments of that Undertaking plus the related retained earnings and share premium accounts.

Qualifying AT1, T1, T2 Capital and qualifying own funds

3.16.4 Qualifying AT1, T1, T2 Capital and qualifying Capital Resources must include the minority interest, AT1, T1 or T2 Capital instruments, as applicable, plus the related retained earnings and share premium accounts, of a Subsidiary, only where the following conditions are met:

- (a) the Subsidiary is one of the following:
 - (i) an Authorised Firm; or
 - (ii) a regulated entity,
- (b) the Subsidiary is a member of the Financial Group and included in the scope of consolidated supervision in accordance with chapter 8; and
- (c) those instruments are owned by persons other than the Undertakings included in the Financial Group.

Qualifying AT1 and T2 Capital issued by a special purpose entity

3.16.5 AT1 and T2 Capital instruments issued by an SPE, and the related retained earnings and share premium accounts, are included in qualifying AT1 or T2 Capital or qualifying Capital Resources, as applicable, only where the following conditions are met:

- (a) the SPE issuing those instruments is included fully in the Financial Group to which the Authorised Firm belongs;
- (b) the instruments, and the related retained earnings and share premium accounts, are included in qualifying AT1 Capital only where the conditions laid down in Rule 3.14.3(1) are satisfied;
- (c) the instruments, and the related retained earnings and share premium accounts, are included in qualifying T2 Capital only where the conditions laid down in Rule 3.15.3(1) are satisfied; and
- (d) the only asset of the SPE is its investment in the Capital Resources of any of its Parents or their Subsidiaries, which are included fully in the Financial Group to which the Authorised Firm belongs, the form of which satisfies the relevant conditions laid down in Rule 3.14.3(1) or Rule 3.15.3(1), as applicable.

Guidance

If the DFSA considers the assets of a special purpose entity to be minimal and insignificant for such an entity, the DFSA may consider waiving the condition specified in Rule 3.16.5(d).

Qualifying T1 Capital instruments included in consolidated T1 Capital

3.16.6 An Authorised Firm must determine the amount of qualifying T1 Capital of a Subsidiary that is included in consolidated T1 Capital of the Authorised Firm's Financial Group by subtracting from the qualifying T1 Capital of that Subsidiary the result of multiplying the amount referred to in (a) by the percentage referred to in (b):

- (a) the lesser of the following:
 - (i) the amount of T1 Capital of that Subsidiary required to meet the sum of the subsidiary's T1 Capital requirement (on a solo basis) of 80% of the Risk Capital Requirement and its Capital Conservation Buffer requirement of 25% of the Risk Capital Requirement; or
 - (ii) the amount of consolidated T1 Capital that relates to the Subsidiary that is required on a consolidated basis to meet the sum of its Financial Group's T1 Capital requirement of 80% of the Risk Capital Requirement and its Capital Conservation Buffer requirement of 25% of the Risk Capital Requirement;
- (b) the qualifying T1 Capital of the Subsidiary expressed as a percentage of all T1 Capital instruments of that Subsidiary plus the related retained earnings and share premium accounts.

Qualifying T1 Capital included in consolidated AT1 Capital

3.16.7 An Authorised Firm must determine the amount of qualifying T1 Capital of a Subsidiary that is included in consolidated AT1 Capital by subtracting from the qualifying T1 Capital of that Subsidiary included in consolidated T1 Capital, the minority interests of that Subsidiary that are included in consolidated CET1 Capital.

Qualifying Capital Resources included in consolidated Capital Resources

3.16.8 An Authorised Firm must determine the amount of qualifying Capital Resources of a Subsidiary that is included in consolidated Capital Resources of its Financial Group by subtracting from the qualifying Capital Resources of that Subsidiary, the result of multiplying the amount referred to in (a) by the percentage referred to in (b):

- (a) the lesser of the following:
 - (i) the amount of Capital Resources of the Subsidiary required to meet the sum of the Subsidiary's total Capital Requirement (on a solo basis) of 100% of the Risk Capital Requirement and its Capital Conservation Buffer requirement of 25% of the Risk Capital Requirement; or
 - (ii) the amount of Capital Resources that relates to the Subsidiary that is required on a consolidated basis to meet the sum of its Financial Group's total Capital Requirement of 100% of the Risk Capital Requirement and its Capital Conservation Buffer requirement of 25% of the Risk Capital Requirement;
- (b) the qualifying Capital Resources of the Subsidiary, expressed as a percentage of all Capital Resources instruments of the Subsidiary that are

included in its CET1, AT1 and T2 Capital items and the related retained earnings and share premium accounts.

Qualifying Capital Resources instruments included in consolidated T2 Capital

- 3.16.9** An Authorised Firm must determine the amount of qualifying Capital Resources of a Subsidiary that is included in consolidated T2 Capital by subtracting from the qualifying Capital Resources of that Subsidiary that are included in consolidated Capital Resources, the qualifying T1 Capital of that subsidiary that is included in consolidated T1 Capital of the Financial Group of the Authorised Firm.

3.17 Qualifying Holdings outside the financial sector

- 3.17.1** (1) Where an Authorised Firm has a Qualifying Holding in an Undertaking which is not one of the following:
- (a) an Undertaking that is a Relevant Entity; or
 - (b) an Undertaking that carries on activities that are:
 - (i) a direct extension of banking;
 - (ii) ancillary to banking, or
 - (iii) leasing, factoring, the management of unit trusts, the management of data processing services or any other similar activity;

and the amount of the holding exceeds 15% of the eligible total Capital Resources of the Authorised Firm, the Authorised Firm must comply with the requirements in (3).
- (2) The total amount of the Qualifying Holdings of an Authorised Firm in Undertakings other than those referred to in (1) that exceeds 60% of its Capital Resources are subject to the requirements in (3).
- (3) An Authorised Firm must apply the following requirements to Qualifying Holdings referred to in (1) and (2):
- (a) a risk weight of 1000% to the following:
 - (i) the amount of Qualifying Holdings referred to in (1) in excess of 15% of Capital Resources; and
 - (ii) the total amount of Qualifying Holdings referred to in (2) in excess of 60% of the Capital Resources of the Authorised Firm; and
 - (b) must not count Qualifying Holdings referred to in (1) and (2) where the amount of those holdings exceeds the percentages of Capital Resources laid down in (1) and (2).

- (4) As an alternative to applying a 1000% risk weight to the amounts in excess of the limits specified in (1) or (2), an Authorised Firm may deduct those amounts from CET1 Capital.
- (5) Shares of Undertakings to which (1) or (2) do not apply must not be included in calculating the eligible capital limits specified in (1) where any of the following conditions are met:
 - (a) those shares are held temporarily during a financial reconstruction or rescue operation,
 - (b) the holding of the shares is an underwriting position held for 5 working days or less; or
 - (c) those shares are held in the name of the Authorised Firm on behalf of others.

PART 5 – Calculating the Leverage Ratio

3.18 Leverage Ratios

3.18.1 This section applies to an Authorised Firm in Category 1, 2 (other than a Matched Principal) or 5.

Guidance

This section is relevant to an Authorised Firm that is required to report its Leverage Ratio to the DFSA under chapter 2, or to disclose its Leverage Ratio under chapter 11, of these Rules.

3.18.2 An Authorised Firm must calculate its Leverage Ratio in accordance with the following formula:

$$\text{Leverage Ratio} = \text{Capital Measure} \div \text{Exposure Measure}$$

where:

- (a) “Capital Measure” represents Tier 1 Capital of the Authorised Firm calculated in accordance with Rule 3.12.1; and
 - (b) “Exposure Measure” represents the value of exposures of the Authorised Firm calculated in accordance with Rule 3.18.3.
- 3.18.3** For the purpose of determining the Exposure Measure, the value of exposures of an Authorised Firm must be calculated in accordance with the International Financial Reporting Standards (IFRS) subject to the following adjustments:
- (a) on-balance sheet, non-derivative exposures must be net of specific allowances and valuation adjustments (e.g. credit valuation adjustments);
 - (b) physical or financial collateral, guarantees or credit risk mitigation purchased must not be used to reduce on-balance sheet exposures; and
 - (c) loans must not be netted with deposits;
 - (d) collateral must not be netted against a derivative position, whether or not netting is allowed;
 - (e) collateral posted against a derivative position must be added to the Exposure Measure if posting the collateral has resulted in a reduction in the Firm’s balance sheet;

- (f) if an Authorised Firm writes a credit derivative, it must include the notional value of the derivative in its Exposure Measure without allowing any offset;
- (g) if an Authorised Firm is a clearing member of a CCP and is contractually liable to reimburse its Clients or the CCP for losses, as a result of a change in the value of the trading transactions in the event of a default, it must:
 - (i) treat each transaction as if it were a derivative exposure entered into directly with the Client, including with regard to the provision or receipt of variation margin; and
 - (ii) include the transaction in its Exposure Measure; and
- (h) an item that has been deducted under section 3.13 from the Tier 1 Capital must also be deducted from the Exposure Measure, except where that item relates to a liability.

Guidance

1. The following Guidance is intended to illustrate how an Authorised Firm should calculate its Leverage Ratio under this section.
2. The Exposure Measure under Rule 3.18.3 should be calculated as the sum of:
 - a. on-balance sheet items; and
 - b. off-balance sheet items.
3. In relation to on-balance sheet items:
 - a. for SFTs, the exposure value should be calculated in accordance with IFRS and the netting requirements referred to in Rule 4.9.14;
 - b. for transactions giving rise to CCR Exposures (except for SFTs), including credit protection sold, the exposure value should be calculated in accordance with section 4.14A; and
 - c. for other on-balance sheet items, the exposure value should be calculated based on their balance sheet values in accordance with Rule 4.9.3.
4. In relation to off-balance sheet items:
 - a. for commitments that are unconditionally cancellable at any time by the Authorised Firm without prior notice, the exposure value should be the notional amount for the item multiplied by a CCF of 10%; and
 - b. for other off-balance sheet items, including:
 - i. direct credit substitutes;

- ii. certain transaction-related contingent items;
- iii. short-term self-liquidating trade-related contingent items and commitments to underwrite debt and equity securities;
- iv. note issuance facilities and revolving underwriting facilities;
- v. transactions, other than SFTs, involving the posting of securities held by the Authorised Firm as collateral;
- vi. asset sales with recourse, where the credit risk remains with the Authorised Firm;
- vii. other commitments with certain drawdown;
- viii. any other commitments; and
- ix. unsettled transactions,

the exposure value should be the notional amount for each of the items multiplied by a CCF of 100%.

- 5. For an Islamic Financial Institution, assets corresponding to Unrestricted PSIA's will fall within the Exposure Measure and, therefore, are considered for the purpose of the Leverage Ratio calculation.
- 6. Further Guidance about the method for completing forms relating to Leverage Ratios can be found in PRU.

Minimum Leverage Ratios

- 3.18.4** (1) An Authorised Firm must maintain a Leverage Ratio of at least 3%.
- (2) An Authorised Firm designated by the DFSA as a G-SIB must maintain a Leverage Ratio of at least the sum of 3% plus 50% of its HLA Ratio.
- (3) An Authorised Firm designated by the DFSA as a D-SIB must maintain a Leverage Ratio of at least 3.5%.
- (4) The DFSA may, by written notice, require an Authorised Firm to maintain a higher minimum Leverage Ratio than the percentage specified in (1), (2) or (3) if it considers it necessary to do so, and the Authorised Firm must comply with that requirement.

Guidance

- 1. In calculating its minimum Leverage Ratio under Rule 3.18.4(2) an Authorised Firm that, for example, is subject to a 2% HLA Ratio for being a G-SIB, should have a Leverage Ratio of at least 4%, consisting of 3% (the minimum Leverage Ratio) plus 1% (50% of its HLA Ratio).
- 2. Circumstances in which the DFSA may require a higher minimum Leverage Ratio under Rule 3.18.4(4) include where:
 - a. it is necessary due to the Authorised Firm's risk profile;
 - b. it has received a request from a regulator as part of measures to preserve financial stability; or
 - c. it considers it reasonably necessary due to any other circumstances.

4 CREDIT RISK

Introduction

Guidance

1. Chapter 4 deals with the prudential requirements relating to the management of Credit Risk by an Authorised Firm. Credit Risk refers to risk of incurring losses due to failure on the part of a borrower or a counterparty to fulfil their obligations in respect of a financial transaction.
2. This chapter aims to ensure that an Authorised Firm holds sufficient regulatory capital of acceptable quality so that it can absorb unexpected losses arising out of its Credit Risk exposures, should the need arise and that it continues to operate in a sustainable manner.
3. This chapter requires an Authorised Firm to:
 - a. appropriately apply a risk-weight to all on-balance sheet assets and off-balance sheet exposures for capital adequacy purposes. A risk-weight is based on a Credit Quality Grade aligned with the likelihood of counterparty default;
 - b. calculate the Credit Risk Capital Requirement for its on-balance sheet assets and off-balance sheet exposures; and
 - c. reduce the Credit Risk Capital Requirement for its on-balance sheet assets and off-balance sheet exposures where the exposure is covered fully or partly by some form of eligible Credit Risk mitigant.
4. Appendix 4 provides detailed requirements, parameters, calculation methodologies and formulae in respect of the primary requirements outlined in chapter 4.

PART 1 – Application

4.1 Application

4.1.1 This chapter applies to an Authorised Firm in Category 1, 2, 3A or 5.

Guidance

1. This chapter imposes systems and controls pertaining to Credit Risk, and prescribes the manner of calculation of the Credit Risk Capital Requirement (also referred to in this module as CRCOM).
2. Rules 3.8.2 and 3.8.3 provide that the CRCOM is a component in the calculation of the overall Risk Capital Requirement of an Authorised Firm, and that the CRCOM is to be calculated in accordance with this chapter 4.
3. The Rules in section 4.8 provide that the Authorised Firm's CRCOM is 8% of the Credit RWA of the firm, which in turn is calculated as the sum of:
 - a. the RWA for Credit Risk Exposures (CR Exposures);
 - b. the RWA for securitisation Exposures (SE Exposures);
 - c. the RWA for Counterparty Credit Risk Exposures (CCR Exposures); and

- d. the RWA for settlement risk Exposures (SR Exposures).
- 4. This chapter sets out the manner in which each of those components must be calculated, monitored and controlled by an Authorised Firm.
- 5. In addition to complying with the applicable Rules in this chapter, an Authorised Firm investing in or holding Islamic Contracts whether or not for the purpose of a PSIA will need to take account of the provisions under IFR Rules 5.4.6 and 5.4.7 to calculate the Credit Risk for those Islamic Contracts.

PART 2 – Credit Risk systems and controls

4.2 Application of this part

- 4.2.1** This part applies to an Authorised Firm in Category 1, 2, 3A or 5 with respect to both its Non-Trading Book and Trading Book transactions.

4.3 Credit Risk management systems

- 4.3.1** An Authorised Firm must implement and maintain comprehensive Credit Risk management systems which:

- (a) are appropriate to the firm's type, scope, complexity and scale of operations;
- (b) enable the firm to effectively identify, assess, monitor and control Credit Risk and to ensure that adequate Capital Resources are available to cover the risks assumed; and
- (c) ensure effective implementation of the Credit Risk strategy and policy.

Guidance

1. Credit Risk is the risk that a borrower or Counterparty fails to meet its obligations. It exists in both the Non-Trading Book and the Trading Book, and both on and off the balance sheet of an Authorised Firm.
2. Obviously, Credit Risk arises from loans but there are other sources of Credit Risk such as.
 - a. trade finance and acceptances;
 - b. interbank transactions;
 - c. commitments and guarantees;
 - d. interest rate, foreign exchange and Credit Derivatives (including swaps, options, forward rate agreements and financial futures);
 - e. bond and equity holdings; and
 - f. settlement of transactions.
3. The objective of the Credit Risk management system must be to ensure that every Authorised Firm holds adequate capital to cover Credit Risk and absorb any potential losses arising from that risk. Since Authorised Firms need to provide credit as part of their usual business, this needs to be achieved by effectively managing the Credit Risk assumed by the Authorised Firm as part of its credit business.
4. Failure to manage Credit Risk effectively could cause an Authorised Firm to face a situation of inadequate capital, which would threaten its safety and soundness. Such problems normally arise from:
 - a. lax credit standards for borrowers and Counterparties;
 - b. poor portfolio risk management; and

- c. failure to identify in good time changes in economic or other conditions that may impair the financial strength of borrowers and Counterparties.
- 5. Therefore, it is essential for Authorised Firms involved in the business of providing credit to design, implement and maintain comprehensive and effective systems to manage Credit Risk.

4.3.2 The Credit Risk management framework of an Authorised Firm must have at least the following principal elements effectively implemented to ensure that the Credit Risk Exposures of the Authorised Firm are of a sufficiently good quality:

- (a) an appropriate Credit Risk environment, defined by a documented Credit Risk strategy and a documented Credit Risk policy;
- (b) application of the Credit Risk strategy and policy, where appropriate, on a consolidated basis and at the level of individual subsidiaries;
- (c) sound processes for assuming and managing Credit Risk;
- (d) prudent lending controls and limits, including policies and processes for monitoring Exposures in relation to limits, and approvals of exceptions to limits;
- (e) adequate appropriately skilled human resources to manage the Credit Risk function;
- (f) independence of credit approval and review functions from credit initiation functions to avoid any real or potential conflicts of interest;
- (g) prudent procedures for approving credits, defined by a documented credit procedures manual;
- (h) effective systems for credit administration, measurement and monitoring; and
- (i) adequate controls over Credit Risk.

- 4.3.3**
- (1) An Authorised Firm must ensure that its Governing Body retains responsibility for the Credit Risk management framework and ensure it is appropriate for the nature, scale and complexity of operations, in the context of prevailing market and macro-economic conditions.
 - (2) An Authorised Firm must ensure that its senior management or an appropriate designated body, regularly reviews and understands the implications as well as the limitations of the risk management information that they receive from the Credit Risk management function, in order to evaluate the suitability and effectiveness of such information in enabling them to provide effective oversight over the Credit Risk management function.
 - (3) An Authorised Firm must ensure that its Governing Body regularly reviews and understands the implications as well as the limitations of Credit Risk management information and reports presented to it, to ensure that the contents and the format of such reports are suitable for effective Governing Body oversight.

- (4) An Authorised Firm must ensure that its Governing Body is responsible for carrying out regular stress testing on the credit portfolio which is appropriate for the nature, scale and complexity of the Credit Risks assumed by the Authorised Firm. An Authorised Firm must ensure that its Governing Body annually reviews the stress scenarios and takes action to address any perceived issues arising from those reviews.
- (5) An Authorised Firm must establish and enforce internal controls and practices so that deviations from policies, procedures, limits and prudential guidelines are promptly reported to the appropriate level of management.

Guidance

1. An Authorised Firm may structure its credit processes and Credit Risk management function in a manner which suits its or its Group's internal organisational structure, culture and internal practices, provided the key functions and components relevant to Credit Risk management, as mentioned above, are present, and there must be adequate segregation of functions responsible for critical Credit Risk management processes. In particular, the credit initiation function must be independent of the credit approval and review functions to avoid any potential conflicts of interest. In cases where an Authorised Firm finds it necessary to delegate small lending limits to staff in the front office for operational needs, there must be adequate safeguards, e.g. independent review of credits granted, to prevent abuse.
2. An Authorised Firm's senior management or an appropriately delegated body (such as a credit committee) should be responsible for effectively implementing the Credit Risk strategy and policy approved by the Governing Body of the Authorised Firm. Senior management or such a credit committee will need to establish adequate procedures to identify, quantify, monitor and control the Credit Risk inherent in the Authorised Firm's activities and at the level of both the overall portfolio and individual borrowers/Counterparties.
3. The appropriate level at which credit decisions are taken will vary according to the type of credit offered and the size and structure of the Authorised Firm. For some Authorised Firms, a credit committee may be appropriate, with formal terms of reference laid down. In other Authorised Firms, individuals may be given pre-assigned authority limits. It will usually be appropriate for the final credit approval authority to be given by staff reporting independently from those staff interacting with clients.
4. As part of its stress testing programme for Credit Risk measurement, an Authorised Firm should take into account the realistic recoveries available from security or Collateral under stressed market and macro-economic conditions.
5. Rule 4.3.3 (3) requires the Governing Body of an Authorised Firm to review the management information reports presented to it by the senior management of that firm and assess the reports in respect of their utility and effectiveness in enabling the Governing Body to effectively discharge their responsibilities towards effective oversight of the firm and its credit risk management.

4.3.4 An Authorised Firm must also consider whether it is prudent to set out specific provisioning requirements for country and transfer risks to which it is exposed.

Guidance

Guidance on country and transfer risk Exposure is set out in section A4.1 (Credit Risk systems and controls) in App4.

- 4.3.5** Where an Authorised Firm avails itself of Credit Risk mitigations, the Authorised Firm must have mechanisms in place to regularly assess the net realisable value of such mitigations taking into account prevailing market conditions.

Guidance

1. Section 4.13 sets out the principles and methodologies for the recognition of Credit Risk mitigation in the calculation of Credit RWA.
2. Further Guidance on Credit Risk systems and controls (including Credit Risk mitigation), and on the specific areas which the Credit Risk policy should cover, is set out in section A4.1.

4.4 Credit Risk strategy, policy, and procedures manual

Credit Risk strategy

- 4.4.1** (1) An Authorised Firm must implement and maintain a Credit Risk strategy, which prescribes its stated degree of risk tolerance, level of capital available for credit activities, business strategy for credit activities and Credit Risk management approach.
- (2) The strategy must be:
- (a) documented;
 - (b) approved by the Governing Body; and
 - (c) regularly reviewed and updated by the Authorised Firm at periodic intervals and at least annually, as appropriate to the nature, scale and complexity of its activities.

Guidance

1. An Authorised Firm's Credit Risk strategy should reflect the aim to achieve sound credit quality while ensuring profit and business growth. Therefore the Credit Risk strategy should address the Authorised Firm's approach towards the decision on an acceptable level of risk/reward relationship, after taking into account resource and capital costs.
2. An Authorised Firm's Credit Risk strategy should allow for economic cycles and their effects on the credit portfolio during different stages of an economic cycle. For example, it should cater for a higher incidence of defaults in the personal loan and credit card portfolios in times of economic recession.

Credit Risk policy

- 4.4.2** (1) An Authorised Firm must implement and maintain a Credit Risk policy which prescribes all the essential elements of the Credit Risk management system and associated processes.
- (2) The policy must be:
- (a) documented;
 - (b) approved by the Governing Body; and

- (c) regularly reviewed and updated by the Authorised Firm at periodic intervals and at least annually, as appropriate to the firm's current financial performance, credit market conditions in its main markets and its Capital Resources position as well the firm's nature, scale and complexity of its activities.
- (3) Any changes to the Credit Risk policy and how exceptions to the policy will be dealt with must be approved by the Governing Body or an appropriately delegated committee of senior management (such as a credit committee).
- (4) An Authorised Firm with one or more branches outside the DIFC must implement and maintain Credit Risk policies adapted to each local market and its regulatory conditions.

4.4.3 The Credit Risk policy must:

- (a) be consistent with the approved Credit Risk strategy, considering a range of factors, including but not limited to an approved degree of risk tolerance, capital allocated to Credit Risks, business strategy and market conditions in its main credit markets;
- (b) provide sound, well-defined Credit Risk norms and criteria for approval of credit applications;
- (c) clearly specify the Exposure limits, product types, business segments, nature of target borrowers and the nature of Credit Risk that the Authorised Firm wishes to incur;
- (d) set out, where appropriate, the amounts and terms and conditions under which Counterparties or clients may be eligible or ineligible for credit;
- (e) include minimum information that is required to be obtained for processing an application for credit;
- (f) include well defined criteria and policies for approving new Exposures as well as renewing and refinancing existing Exposures, identifying the appropriate approval authority for the size and complexity of the Exposures;
- (g) include effective credit administration policies, including continued analysis of a borrower's ability and willingness to repay under the terms of the debt, monitoring of documentation, legal covenants, contractual requirements and Collateral, and a classification system that is consistent with the nature, size and complexity of the Authorised Firm's activities or, at the least, with the asset grading system prescribed in Rule 4.5.4;
- (h) include comprehensive policies for reporting Exposures on an on-going basis;
- (i) include comprehensive policies for identifying and managing problem assets;
- (j) include a provisioning policy approved by the Governing Body which ensures that all loans are promptly and prudently provided for;

- (k) set out limits and approval processes involved for the approval of credit facilities that can be approved by the delegated authorities, and stipulate that the Governing Body retains responsibility for the governance of such limits;
- (l) require that major Credit Risk Exposures exceeding a specified amount or at a minimum all Large Exposures of the Authorised Firm are approved by the Authorised Firm's senior management or its designated body like credit committee; and
- (m) require that all Credit Risk Exposures that are especially risky or inconsistent with the approved credit strategy of the Authorised Firm are approved by the Authorised Firm's senior management or its designated body such as a credit committee.

4.4.4 In relation to conflicts of interest and Related Person transactions, the policy must:

- (a) set out adequate procedures for handling conflicts of interest relating to the provision and management of credit, including measures to prevent any Person directly or indirectly benefiting from the credit being part of the process of granting or managing the credit;
- (b) subject to Rule 4.4.5, prohibit Exposures to Related Persons on terms that are more favourable than those available to Persons who are not Related Persons; and
- (c) if Exposures to Related Persons are allowed on terms which are no more favourable than those available to Persons who are not Related Persons, set out procedures that:
 - (i) require such Exposures, and any write-off of such Exposures, exceeding specific amounts or otherwise posing special risks to the Authorised Firm, to be made subject to the prior written approval of the firm's Governing Body or the Governing Body's delegate; and
 - (ii) exclude Persons directly or indirectly benefiting from the grant or write off of such Exposures being part of the approval process.

4.4.5 The prohibition in Rule 4.4.4(b) does not apply to providing credit to a Related Person under a credit policy on terms (such as for credit assessment, tenor, interest rates, amortisation schedules and requirements for Collateral) that are more favourable than those on which it provides credit to Persons who are not Related Persons, provided the credit policy:

- (a) is an Employee credit policy that is widely available to Employees of the Authorised Firm;
- (b) is approved by the Authorised Firm's Governing Body or the Governing Body's delegate;
- (c) clearly sets out the terms, conditions and limits (both at individual and aggregate levels) on which credit is to be provided to such Employees; and
- (d) requires adequate mechanisms to ensure on-going compliance with the terms and conditions of that credit policy, including immediate reporting to

the Governing Body or the Governing Body's delegate where there is a deviation from or a breach of the terms and conditions or procedures applicable to the provision of such credit for timely and appropriate action.

Guidance

1. The requirements in these Rules do not prevent arrangements such as Employee loan schemes that allow more favourable and flexible loan terms to Employees of the Authorised Firm than those available under its normal commercial arrangements. However, such a loan scheme must comply with the requirements set out in these Rules, which are designed to address conflicts of interest that may arise in the grant, approval or management of such loans. Such conflicts are especially likely to arise where one or more of the Employees concerned are Directors, Partners or senior managers.
2. Generally, where an Authorised Firm has an Employee loan scheme under these Rules, the DFSA expects its Governing Body to have ensured, before it or its delegate approved that scheme, that the terms, conditions and particularly limits (both at individual and aggregate level) on which credit is to be provided to Employees under the scheme are adequate and effective in addressing the risks arising from such lending. The Authorised Firm should also be able to demonstrate to the DFSA that the procedures it has adopted relating to an Employee loan scheme are adequate to address any risks arising from such lending. The DFSA expects to have access to records relating to lending under an Employee loan scheme upon request or during its supervisory visits. Any significant breach of or deviation from the procedures adopted in relation to an Employee loan scheme may also trigger the reporting requirements to the DFSA under GEN Rule 11.10.7.

4.4.6 For the purposes of the Rules in this chapter, a Person is a "Related Person" of an Authorised Firm if the Person:

- (a) is, or was in the past 2 years:
 - (i) a member of a Group or Partnership in which the Authorised Firm is or was also a member; or
 - (ii) a Controller of the Authorised Firm or a Close Relative of such a Controller;
- (b) is, or was in the past 2 years, a Director, Partner or senior manager of the Authorised Firm or an entity referred to under (a)(i) or (ii), or a Close Relative of such a Director, Partner or senior manager; or
- (c) is an entity in which a Director, Partner or senior manager of the Authorised Firm or an entity referred to in (a)(i) or (a)(ii), or a Close Relative of such a Director, Partner or senior manager has a significant interest by:
 - (i) holding 20% or more of the shares of that entity, or a Parent of that entity, if that entity is a company; or
 - (ii) being entitled to exercise 20% or more of the voting rights in respect of that entity;

except that a Partner is not a Related Person where that Person is a limited partner of a Limited Partnership formed under the Limited Partnership Law of 2006 or any similar limited partnership constituted under the law of a country or territory outside the DIFC.

Credit procedures manual

4.4.7 An Authorised Firm must implement and maintain a documented credit procedures manual, which sets out the criteria and procedures for granting new credits, for approving extensions of existing credits and exceptions, for conducting periodic and independent reviews of credits granted and for maintaining the records for credits granted.

4.4.8 The credit procedures manual must establish:

- (a) sound, well-defined criteria for granting credit, including a thorough understanding of the borrower or Counterparty, the purpose and structure of the credit and its source of repayment;
- (b) well defined processes for approving new Exposures as well as renewing and refinancing existing Exposures;
- (c) effective credit administration processes, including continued analysis of a borrower's ability and willingness to repay under the terms of the debt, monitoring of documentation, legal covenants, contractual requirements and Collateral;
- (d) effective processes for classification and grading of credit assets consistent with the nature, size and complexity of the Authorised Firm's activities;
- (e) comprehensive processes for reporting Exposures on an ongoing basis; and
- (f) comprehensive processes for identifying problem assets, managing problem assets, monitoring their collections and for estimating required level of provisions.

Guidance

The same criteria should be applied to both advised and unadvised facilities and should deal with all Credit Risks associated with the Authorised Firm's business whether in the Non-Trading or Trading Book or on or off balance sheet.

4.5 Processes for credit assessment

- 4.5.1**
- (1) When utilising external credit rating agencies as part of its credit assessment processes, an Authorised Firm must:
 - (a) maintain an internal credit grading system; and
 - (b) stress test its capital position on at least an annual basis to consider the capital implications to the Authorised Firm of a significant reduction in the credit quality and associated reduction on credit ratings from credit rating agencies for its credit portfolio.
 - (2) An Authorised Firm must not solely use external credit rating agency credit ratings as a basis for its assessment of the risks associated with an Exposure, in particular in respect of a Large Exposure, and must at all times conduct its own credit assessment of such an Exposure.

Guidance

An Authorised Firm should closely monitor the adequacy of the internal credit assessment processes, in order to assess whether there is an upward bias in internal ratings.

4.5.2 An Authorised Firm must implement and maintain appropriate policies, processes, systems and controls to:

- (a) administer its credit portfolios, including keeping the credit files current, getting up-to-date financial information on borrowers and other Counterparties, funds transfer, and electronic storage of important documents;
- (b) ensure that the valuations of Credit Risk mitigants employed by the Authorised Firm are up-to-date, including periodic assessment of Credit Risk mitigants such as guarantees and Collateral;
- (c) review all material concentrations in its credit portfolio and report the findings of such reviews to the Governing Body; and
- (d) measure Credit Risk (including to measure Credit Risk of off-balance sheet products such as Derivatives in credit equivalent terms) and monitor the condition of individual credits to facilitate identification of problem credits and to determine the adequacy of provisions and reserves.

4.5.3 The Credit Risk management system and, in particular, the systems, policies and processes aimed at classification of credits, monitoring and identification of problem credits, management of problem credits and provisioning for them must include all the on-balance sheet and off-balance sheet credit Exposures of the Authorised Firm.

Guidance

An Authorised Firm should ensure that its loan portfolio is properly classified and has an effective early-warning system for problem loans.

- 4.5.4**
- (1) An Authorised Firm must establish clearly defined criteria for identifying its problem credits and/or impaired assets which ensure that credits are classified as impaired in all cases where there is some reason to believe that all amounts due (including principal and interest) will, or may, not be collected in accordance with the contractual terms of the loan agreement.
 - (2) For the purpose of (1), and subject to (3), an Authorised Firm must categorise its credits into five categories as detailed in the following table, where credits in the substandard, doubtful and loss categories must be considered as problem credits:

Standard	includes credits with no element of uncertainty about timely repayment of the outstanding amounts, including principal and interest. Credits are currently in regular payment status with prompt payments.
----------	--

Special mention	includes credits with deteriorating or potentially deteriorating credit quality which, may adversely affect the borrower's ability to make scheduled payments on time. The credits in this category warrant close attention by the Authorised Firm.
Substandard	includes credits which exhibit definitive deterioration in credit quality and impaired debt servicing capacity of the borrower.
Doubtful	includes credits which show strong credit quality deterioration, worse than those in substandard category, to the extent that the prospect of full recovery of all the outstanding amounts from the credit is questionable and consequently the probability of a credit loss is high, though the exact amount of loss cannot be determined yet.
Loss	includes credits which are assessed as uncollectable and credits with very low potential for recoverability of amounts due.

- (3) An Authorised Firm may also have in place a more detailed credit grading system provided it can address the categories detailed in (2).

Guidance

1. With respect to the ratings above, Authorised Firms should consider Exposures as classified special mention, substandard, doubtful and loss where the loans are contractually in arrears for a minimum number of days of 30, 60, 90- 120 and 120-180 days respectively. Authorised Firms should also consider the treatments as set out in Rule 4.5.7 (Evergreening).
2. Credits exhibiting the following categories should be included in the special mention category.
 - a. a declining trend in the operations of the borrower or in the borrower's ability to continue to generate cash required for repayment of the credit;
 - b. any signals which indicate a potential weakness in the financial position of the borrower, but not to the point at which repayment capacity is definitely impaired; or
 - c. business, economic or market conditions that may unfavourably affect the profitability and business of the borrower in the near to medium term.
3. Credits exhibiting the following categories should be included in the substandard category.
 - a. inability of the borrower to meet contractual repayment terms of the credit facility;
 - b. unfavourable economic and market conditions or operating problems that would affect the profitability and cash flow generation of the borrower;
 - c. weak financial condition or the inability of the borrower to generate sufficient cash flow to service the payments.;
 - d. difficulties experienced by the borrower in servicing its other debt obligations; or
 - e. breach of any financial covenants by the borrower.

- 4.5.5** An Authorised Firm must have detailed policies, processes and resources for managing problem credits which address the following:
- (a) monitoring of credits and early identification of credit quality deterioration;
 - (b) review of classification of problem credits; and
 - (c) ongoing oversight of problem credits, and for collecting on past due obligations.
- 4.5.6** An Authorised Firm must ensure that each and every credit which qualifies as a Large Exposure and is classified as an impaired credit is managed individually. This includes valuation, classification and provisioning for such credits on an individual item basis.
- 4.5.7** Any evergreening exercise involving refinancing of past due credits must not result in their being classified as a higher category. In particular, impaired credits cannot be refinanced with the aim of classifying them as standard or special mention credits.
- 4.5.8** An Authorised Firm's provisioning policy must specify the following minimum provisioning requirements:
- (a) for substandard assets – 20% of the unsecured portion of the credit;
 - (b) for doubtful assets – 50% of the unsecured portion of the credit; and
 - (c) for loss assets – 100% of the unsecured portion of the credit.
- 4.5.9** An Authorised Firm must, on a periodic basis, at a minimum monthly frequency, review its problem credits (at an individual level or at a portfolio level for credits with homogeneous characteristics) and review the asset classification, provisioning and write-offs for each of those problem credits.

PART 3 - CRCOM

4.6 Application

Guidance

1. As indicated in Rule 4.1.1, this chapter 4 (including this part 3) applies to Authorised Firms in Categories 1, 2, 3A and 5. However, the provisions in this part are applied in a differentiated manner in that Category 3A firms must, and Category 2 firms may, use the Simplified Approach under section 4.7.
2. The Credit Risk Capital Requirement (also referred to in this module as CRCOM) is a component of the calculation of the overall Capital Requirement of an Authorised Firm, as provided in Rules 3.8.2 and 3.8.3. The Rules in this part 3, supplemented by App4, govern the manner of calculation of the CRCOM.

4.7 Simplified Approach

Category 3A firms

- 4.7.1** (1) This Rule applies only to an Authorised Firm in Category 3A.
- (2) Subject to (3) and (4), an Authorised Firm must apply the Simplified Approach as prescribed in section A4.12 in App4.
- (3) An Authorised Firm is not required to apply the Simplified Approach if it obtains prior approval of the DFSA not to do so.
- (4) After obtaining approval under (3), a firm must not revert to the Simplified Approach without further prior approval from the DFSA.

Guidance

1. In effect, the Simplified Approach reduces undue regulatory burden on Category 3A firms to reflect more appropriately their risk profile.
2. In relation to (3) and (4), the DFSA may consider granting its approval for a change of approach if it is satisfied that there are no regulatory capital arbitrage opportunities. Firms should be able to demonstrate to the DFSA solid and reasonable grounds to be able to move from one approach to the other. For instance, in assessing whether or not to grant approval, the DFSA may consider whether or not there has been a material change in the business of the firm.

Category 2 firms

- 4.7.2** (1) This Rule applies only to an Authorised Firm in Category 2.
- (2) Subject to (3) and (4), an Authorised Firm may apply the Simplified Approach, as prescribed in section A4.12 in App4, upon obtaining prior approval to do so from the DFSA.
- (3) After obtaining approval under (2), a firm must not disapply the Simplified Approach without further prior approval from the DFSA.

- (4) The DFSA may revoke its approval under (2) and require a firm to disapply the Simplified Approach, where the DFSA considers that this is warranted by the firm's business model and risk profile.

Guidance

In relation to (3) and (4), the DFSA may consider granting its approval for a change of approach if it is satisfied that there are no regulatory capital arbitrage opportunities. Firms should be able to demonstrate to the DFSA solid and reasonable grounds to be able to move from one approach to the other. For instance, in assessing whether or not to grant approval, the DFSA may consider whether or not there has been a material change in the business of the firm.

4.8 Calculation of the CRCOM

- 4.8.1** (1) The Credit Risk Capital Requirement is calculated as follows:

$$\text{CRCOM} = 8\% \times \text{Credit RWA}$$

- (2) The Credit RWA of an Authorised Firm is the sum of:
- (a) its risk weighted assets (RWA) for all its Credit Risk Exposures (referred to in this module as "CR Exposures") calculated in accordance with Rules 4.8.2 and 4.8.3;
 - (b) its RWA for all its securitisation Exposures (referred to in this module as "SE Exposures") calculated in accordance with Rule 4.8.4 and section 4.14;
 - (c) its RWA for its Counterparty Credit Risk Exposures (referred to in this module as "CCR Exposures") as calculated in accordance with Rule 4.8.7 and section 4.14A; and
 - (d) its RWA for its settlement risk Exposures (referred to in this module as "SR Exposures") calculated in accordance with section A4.6.

Calculation of RWA for Credit Risk Exposures (CR Exposures)

- 4.8.2** An Authorised Firm must include in its calculation of RWA for CR Exposures:

- (a) any on-balance sheet asset; and
- (b) any off-balance sheet item;

but excluding:

- (c) any SE Exposure;
- (d) any securitised Exposure that meets the requirements for the recognition of risk transference in a Traditional Securitisation set out in section 4.14;
- (e) any CCR Exposure;
- (f) any SR Exposure; and

- (g) any Exposure classified as a position or instrument in the Trading Book in accordance with section A2.1.

4.8.3 To calculate its RWA for CR Exposures, an Authorised Firm must:

- (a) calculate the value of the Exposure (represented as “E”) for every on-balance sheet and every off-balance sheet asset in accordance with the Exposure measurement methodology specified in section 4.9 and recognising the effects of any applicable Credit Risk mitigation;
- (b) categorise that Exposure in accordance with the Rules in section 4.10;
- (c) allocate an applicable Credit Quality Grade and risk weight for that Exposure in accordance with the Rules in section 4.11 and 4.12;
- (d) calculate the RWA amount for that Exposure using the following formula:

$$RWA(CR) = E \times CRW$$

where:

- (i) “RWA(CR)” refers to the risk-weighted Exposure amount for that CR Exposure;
- (ii) “E” refers to the Exposure value or amount, for that CR Exposure; and
- (iii) “CRW” refers to the applicable risk weight for that CR Exposure determined in accordance with (b) and (c); and
- (e) add the RWA amounts calculated in accordance with (d) for all its CR Exposures.

Calculation of RWA for securitisation Exposures (SE Exposures)

4.8.4 To calculate its RWA for all its SE Exposures, an Authorised Firm must:

- (a) calculate the value of the Exposure for each of its SE Exposures in accordance with Exposure measurement methodology specified in section 4.9 and recognising the effects of any applicable Credit Risk mitigation;
- (b) allocate an applicable Credit Quality Grade for that SE Exposure in accordance with the Rules in section 4.11;
- (c) calculate the RWA amount for each SE Exposure, except for those SE Exposures which the Authorised Firm is required to include as deductions from any component of Capital Resources, using the following formula:

$$RWA(SE) = SE \times CRW$$

where:

- (i) “RWA(SE)” refers to the risk-weighted Exposure amount for that securitisation Exposure;

- (ii) “SE” refers to the Exposure value or amount for that SE Exposure calculated in accordance with (a); and
- (iii) “CRW” refers to the applicable risk weight for that SE Exposure determined in accordance with (b); and
- (d) add the RWA amounts calculated in accordance with (c) for all its SE Exposures to the RWA amounts calculated in accordance with Rule 4.8.5 in respect of its Early Amortisation Exposures.

4.8.5 To calculate its RWA for Early Amortisation Exposures, an Authorised Firm must:

- (a) calculate the value of the Exposure (EAE) for each of its Early Amortisation Exposures in accordance with Exposure measurement methodology specified in section 4.9 and recognising the effects of any applicable Credit Risk mitigation;
- (b) calculate the risk-weighted Exposure amount for each Early Amortisation Exposure using the following formula:

$$RWA(EAE) = EAE \times CRW$$

where:

- (i) “RWA(EAE)” refers to the risk-weighted Exposure amount for that Early Amortisation Exposure;
- (ii) “EAE” refers to the Exposure value or amount, for that Early Amortisation Exposure calculated in accordance with (a); and
- (iii) “CRW” refers to the applicable risk weight for the underlying Exposure type as if the Exposure had not been securitised; and
- (c) add the RWA amounts calculated in accordance with (b) for all its Early Amortisation Exposures.

4.8.6 The aggregate RWA amount for all of the SE Exposures of an Authorised Firm to a securitisation and Exposures arising from Credit Risk mitigation applied to those SE Exposures must not exceed the aggregate RWA amount corresponding to the underlying Exposures of the securitisation had they been on the balance sheet of the Authorised Firm and included in the calculation of the Credit RWA of the Authorised Firm. For avoidance of doubt, the aggregate RWA amount must not include any deduction for a gain-on-sale or a Credit-Enhancing Interest-Only Strip arising from the securitisation.

Calculation of RWA for Counterparty Credit Risk Exposures (CCR Exposures)

4.8.7 To calculate its RWA for its CCR Exposures, an Authorised Firm must:

- (a) calculate the value of each CCR Exposure in accordance with the methodology specified in section 4.14A;
- (b) categorise each CCR Exposure in accordance with the Rules in section 4.10 as if it were a CR Exposure;

- (c) allocate an applicable Credit Quality Grade and risk weight for each CCR Exposure based on the Counterparty in accordance with the Rules in section 4.11 and 4.12;
- (d) calculate the RWA amount for each CCR Exposure using the following formula:

$$\text{RWA(CCR)} = \text{CCRE} \times \text{CCRW}$$

where:

- (i) “RWA(CCR)” refers to the risk-weighted Exposure amount for that CCR Exposure;
- (ii) “CCRE” refers to the Exposure value or amount for that CCR Exposure; and
- (iii) “CCRW” refers to the applicable risk weight for that CCR Exposure determined in accordance with (b) and (c); and
- (e) add the RWA amounts calculated in accordance with (d) for all its CCR Exposures.

Guidance

The calculation of the CRW for Unsettled Transactions is set out in section A4.6.

4.9 Methodology for measurement of Exposures

- 4.9.1** An Authorised Firm must apply the Exposure measurement methodology set out in the Rules in this part to calculate the value or amount of an Exposure for any CR Exposure, SE Exposure, CCR Exposure or SR Exposure.

Guidance

1. The measurement methodology in this section prescribes the manner of calculation of Exposures for the purpose of determining the Credit RWA for Credit Risk (CR) Exposures as provided in Rule 4.8.3, for securitisation (SE) Exposures as provided in Rule 4.8.4 and for Counterparty Credit Risk (CCR) Exposures as provided in Rule 4.8.7.
2. Due regard should be given to the Guidance relating to prudent valuation in section 2.4 and related provisions in App2.5.
3. An Authorised Firm should consult with the DFSA on the appropriate treatment to apply in the measurement of E, for transactions that have not been addressed in this part.

- 4.9.2** An Authorised Firm must calculate E for any CR Exposure, SE Exposure or CCR Exposure, net of any individual impairment provision attributable to such Exposures, as determined in accordance with the International Financial Reporting Standards.

Measurement of E for on-balance sheet assets

- 4.9.3** For each on-balance sheet asset, E should be the carrying value of the asset as determined in accordance with the International Financial Reporting Standards.

Guidance

1. For any asset, E should be equal to the fair value of that asset presented in the balance sheet except that:
 - a. for any asset held at cost, E, should be equal to the cost of the asset presented in the balance sheet; and
 - b. for any debt security or loan classified and measured at fair value through other comprehensive income, E, should be equal to the fair value less provision for impairment of that debt security or loan, adjusted by deducting any unrealised fair value gains and adding back any unrealised fair value losses on revaluation (broadly equivalent to the amortised cost of the debt security or loan less any provision for impairment).
2. In the case of a lease where the Authorised Firm is exposed to residual value risk (i.e. potential loss due to the fair value of the leased asset declining below the estimate of its residual value reflected on the balance sheet of the Authorised Firm at lease inception), the Authorised Firm should calculate (i) an Exposure to the lessee equivalent to the discounted lease payment stream; and (ii) an Exposure to the residual value of the leased assets equivalent to the estimate of the residual value reflected in the balance sheet of the Authorised Firm.
3. Any foreign exchange transaction or translation gain or loss from a foreign currency-denominated on- balance sheet item as well as interest earned on a fixed income instrument should be allocated to the Exposure to which it accrues.

Measurement of E for off-balance sheet items other than CCR Exposures or SFTs

- 4.9.4** (1) For each off-balance sheet item other than a CCR Exposure, exchange-traded derivative, or an Exposure arising from an SFT, an Authorised Firm must calculate E by:
- (a) in the case of an Early Amortisation Exposure, multiplying the amount of investors' interest by the applicable CCF set out in Rules A4.2.1 and A4.2.2 in App4; and
 - (b) in all other cases, multiplying the notional amount of each item by:
 - (i) the applicable CCF set out in Rule A4.2.1 in App4 if that item is a CR Exposure; or
 - (ii) the applicable CCF set out in Rule A4.2.2 in App4 if that item is an SE Exposure.
- (2) For the purpose of (1)(a), an Authorised Firm must calculate the amount of investors' interest as the sum of:
- (a) investors' drawn balances related to the securitised Exposures; and
 - (b) E associated with investors' undrawn balances related to the SE Exposures, where E is determined by allocating the undrawn balances of securitised Exposures on a pro-rata basis based on the proportions of the Originator's and investors' shares of the securitised drawn balances.

Guidance

1. An Authorised Firm which is exposed to the risk of the underlying Securities in an OTC derivative, long settlement transaction or SFT which is in substance similar to a forward purchase or credit substitute should calculate E, for such an Exposure, in accordance with Rule 4.9.4(1).
2. For avoidance of doubt, where an Authorised Firm has provided unfunded credit protection via a total rate of return swap, E should be equal to the notional amount of the underlying reference credit for which the Authorised Firm is providing protection adjusted for any payments received from or made to the protection buyer and recognised in the profit and loss account of the Authorised Firm. Where an Authorised Firm has provided unfunded credit protection via a credit default swap, E should be equal to the notional amount of the underlying reference credit for which the Authorised Firm is providing protection.
3. The notional amount of an off-balance sheet item refers to the amount which has been committed but is as yet undrawn. The amount to which the CCF is applied is the lower of the value of the unused committed credit line, and the value which reflects any possible constraining availability of the facility, such as the existence of a ceiling on the potential lending amount which is related to an obligor's reported cash flow. If the facility is constrained in this way, the Authorised Firm must have sufficient line monitoring and management procedures to support this contention.
4. Any foreign exchange transaction or translation gain or loss from a foreign currency-denominated off- balance sheet item should be allocated to the Exposure to which it accrues.

Recognition of eligible financial Collateral for on-balance sheet assets and off-balance sheet items other than CCR Exposures

- 4.9.5** (1) An Authorised Firm which has taken eligible financial Collateral for any transaction other than an equity Exposure, an SE Exposure, CCR Exposure or SFT may recognise the effect of such Collateral in accordance with Rules 4.9.6 and 4.9.7.
- (2) An Authorised Firm must use either the:
- (a) Financial Collateral Simplified Approach (FCSA) which adopts the treatment under Rule 4.13.5 in relation to the composition of financial Collateral; or
 - (b) Financial Collateral Comprehensive Approach (FCCA) which adopts the treatment under Rule 4.13.6;
- to recognise the effect of eligible financial Collateral.
- (3) An Authorised Firm must apply the chosen approach consistently to its entire Non-Trading Book and must not use a combination of both approaches.
- 4.9.6** An Authorised Firm using the FCSA may recognise the effect of eligible financial Collateral in accordance with the Rules in section 4.13.
- 4.9.7** An Authorised Firm using the FCCA may calculate the CR Exposure adjusted for eligible financial Collateral (referred to in PIB as "E*"), in accordance with Rules in section A4.3 of App4 and substitute E* for E when calculating the Credit Risk-weighted Exposure amount for that CR Exposure under section 4.8.

Recognition of eligible financial Collateral for securitisation (SE) Exposures

- 4.9.8** An Authorised Firm that has taken eligible financial Collateral for an SE Exposure may recognise the effect of such Collateral in accordance with Rules 4.9.9 to 4.9.11.
- 4.9.9** An Authorised Firm calculating RWAs for SE Exposures must use either the FCSA or the FCCA approaches to recognise the effect of eligible financial Collateral. An Authorised Firm must apply the chosen approach consistently to the entire Non-Trading Book and must not use a combination of both approaches.
- 4.9.10** An Authorised Firm using the FCSA approach for an SE Exposure may recognise the effect of eligible financial Collateral in accordance with section 4.13 and Rule 4.14.70.
- 4.9.11** An Authorised Firm using the FCCA approach for an SE Exposure must calculate E*, the SE Exposure adjusted for eligible financial Collateral, in accordance with Rules in section A4.3 of App4 and substitute E* for E when calculating the RWA for SE Exposure under section 4.8.

Measurement of E for Counterparty Exposures arising from SFTs

- 4.9.12** An SFT must be treated as Collateralised lending, notwithstanding the wide range of structures which could be used for SFTs.
- 4.9.13** An Authorised Firm must calculate E, for a Counterparty Exposure arising from an SFT, other than an Exposure covered by a Qualifying Bilateral Netting Agreement, in accordance with Rules 4.9.14 to 4.9.18.
- 4.9.14** An Authorised Firm must determine E, for a Counterparty Exposure arising from an SFT which is not covered by a Qualifying Bilateral Netting Agreement as follows:
 - (a) in the case where the Authorised Firm has lent Securities to a Counterparty or sold Securities to a Counterparty with a commitment to repurchase those Securities at a specified price on a specified future date, the latest fair value of the Securities lent or sold; and
 - (b) in the case where the Authorised Firm has lent cash to a Counterparty through the borrowing of Securities from the Counterparty or paid cash for the purchase of Securities from a Counterparty with a commitment to resell those Securities at a specified price on a specified future date, the amount of cash lent or paid.
- 4.9.15** An Authorised Firm which has taken eligible financial Collateral for any SFT where the Counterparty Exposure is determined in accordance with Rule 4.9.14 may recognise the effect of such Collateral in accordance with Rules 4.9.16 to 4.9.19.
- 4.9.16** An Authorised Firm must use either the FCSA or the FCCA to recognise the effect of eligible financial Collateral for any SFT in the Non-Trading Book. The Authorised Firm must apply the chosen approach consistently to the entire Non-Trading Book and must not use a combination of both approaches. For a Counterparty Exposure arising from any SFT in the Trading Book, an Authorised Firm must only use the FCCA to recognise the effect of eligible financial Collateral.
- 4.9.17** An Authorised Firm using the FCSA may recognise the effect of eligible financial Collateral for any SFT in accordance with Rule A4.3.11 in App4.

4.9.18 An Authorised Firm which has taken eligible financial Collateral for any SFT that is not covered by a Qualifying Bilateral Netting Agreement and using the FCCA, must calculate E* in accordance with Rule A4.3.2 in App4, and substitute E* for E when calculating the Credit Risk-weighted Exposure amount for that CR Exposure under section 4.8.

4.9.19 An Authorised Firm which has taken eligible financial Collateral for an SFT that is covered by a Qualifying Bilateral Netting Agreement and using the FCCA, must calculate E* for all its CR Exposures to any single Counterparty covered by the Qualifying Bilateral Netting Agreement, in accordance with Rule A4.3.3 in App4, and substitute E* for E when calculating the Credit Risk-weighted Exposure amount for its CR Exposures to that Counterparty under section 4.8.

Exceptions to the measurement of E

4.9.20 An Authorised Firm may attribute a value of zero to E for:

- (a) any Counterparty Exposure arising from any Derivative transaction or SFT outstanding with a CCP and which has not been rejected by that CCP, provided that the Exposure is fully Collateralised on a daily basis; and
- (b) any Credit Risk Exposure arising from any Derivative transaction, SFT or spot transaction which an Authorised Firm has outstanding with a CCP for which the latter acts as a custodian on the Authorised Firm's behalf, provided that the Exposure is fully Collateralised on a daily basis.

Guidance

Credit Risk (CR) Exposures outstanding with a CCP would, for example, include credit Exposures arising from monies placed and from Collateral posted, with the Counterparty.

4.10 Categorisation of Credit Risk Exposures (CR Exposures)

Guidance

This section categorises Exposures for the purpose of determining the CRW for CR Exposures, as provided in Rule 4.8.3.

4.10.1 An Authorised Firm must categorise any CR Exposure into one of the following Exposure classes:

- (a) Exposures to central governments and Central Banks;
- (b) Exposures to Public Sector Entities;
- (c) Exposures to Multilateral Development Banks;
- (d) Exposures to international organisations;
- (e) Exposures to banks;
- (f) Exposures to corporates;
- (g) Retail Exposures;

- (h) Subordinated debt, equity and other capital instruments;
- (i) Real Estate Exposures;
- (j) Exposures in default; and
- (k) Other Exposures.

4.11 Credit Quality Grade and External Credit Assessments

Guidance

This section governs credit assessments of Exposures for the purpose of determining the CRW for Credit Risk (CR) Exposures as provided in Rule 4.8.3 and for securitisation (SE) Exposures as provided in Rule 4.8.4.

- 4.11.1** An Authorised Firm must assign a CR Exposure to a Credit Quality Grade based on the external credit assessment that is applicable to the CR Exposure in accordance with tables mapping the ratings from an ECAI to Credit Quality Grades, which will be published by the DFSA.
- 4.11.2** An Authorised Firm must only use an external credit assessment which is accessible to the public. An Authorised Firm may not use a credit assessment that is made available only to the parties to a transaction.
- 4.11.3** An Authorised Firm must only use external credit assessments by a recognised ECAI. The DFSA may impose conditions on the use of such external credit assessments.
- 4.11.4** An Authorised Firm must use its chosen recognised external credit rating agencies and their external credit assessments consistently for each type of Exposure, for both risk weighting and risk management purposes. Where an Authorised Firm has two external credit assessments which map into different Credit Quality Grades, it must assign the CR Exposure to the Credit Quality Grade associated with the higher risk weight. Where an Authorised Firm has three or more external credit assessments which map into two or more different Credit Quality Grades, it must assign the CR Exposure to the Credit Quality Grade associated with the higher of the two lowest risk weights.

Guidance

For illustration, if there are three external credit assessments mapping into Credit Quality Grades with risk weights of 0%, 20% and 50%, then the applicable risk weight is 20%. If the external credit assessments map into Credit Quality Grades with risk weights of 20%, 50% and 50%, then the applicable risk weight is 50%.

- 4.11.5** An Authorised Firm must not recognise the effects of Credit Risk mitigation if such mitigation is already reflected in the issue-specific external credit assessment of the CR Exposure.
- 4.11.6** Where a CR Exposure has an issue-specific external credit assessment from a recognised ECAI, an Authorised Firm must use such assessment. Where a CR Exposure does not have an issue-specific external credit assessment, an Authorised Firm must:

- (a) if there is an issue-specific external credit assessment for another Exposure to the same obligor, use the issue-specific assessment for the other Exposure only if the Exposure without an issue-specific assessment ranks pari passu with or is senior to the Exposure with the issue-specific assessment;
 - (b) if the obligor has an Issuer external credit assessment, use the Issuer assessment of the obligor only if the Exposure without an issue-specific assessment ranks pari passu with or is senior to any unsecured claim that is not subordinated to any other claim on the obligor; or
 - (c) in all other cases, apply a risk weight equal to the higher of the risk weight that is applicable to an unrated Exposure and the risk weight associated with the external credit assessment, if any, of the obligor or another Exposure to the same obligor.
- 4.11.7** Where a CR Exposure is risk-weighted in accordance with Rules 4.11.6(a) or (b), an Authorised Firm may use a domestic currency external credit assessment only if the CR Exposure is denominated in that domestic currency.
- 4.11.8** An Authorised Firm may use an external credit assessment to risk weight a CR Exposure only if the external credit assessment has taken into account and reflects the entire amount of Credit Risk Exposure the Authorised Firm has with regard to all payments owed to it.
- 4.11.9** An Authorised Firm must not use unsolicited external credit assessments to assign any CR Exposure to a Credit Quality Grade, unless:
- (a) it has assessed the quality of the unsolicited external credit assessments that it intends to use and is satisfied that these are comparable in performance with solicited external credit assessments and maintains relevant records and documents to be made available to the DFSA upon request; and
 - (b) it uses unsolicited external credit assessments consistently for each type of Exposures, for both risk weighting and risk management purposes.
- 4.11.10** An Authorised Firm must not use an external credit assessment which incorporates assumptions of implicit government support, unless the credit assessment refers to a bank owned by or set up and sponsored by that government.
- 4.11.11** By way of derogation, during the period from 1 January 2025 to and including 31 December 2029, an Authorised Firm is not required to comply with Rule 4.11.10.

4.12 Risk weights

Exposures to central governments and Central Banks

- 4.12.1** Subject to Rule 4.12.2, an Authorised Firm must risk-weight any Exposure to a central government or a Central Bank in accordance with the following table:

Credit Quality Grade	1	2	3	4	5	6	Unrated
Risk Weight	0%	20%	50%	100%	100%	150%	100%

- 4.12.2** (1) An Authorised Firm may apply a 0% risk weight to any Exposure to the central government or the Central Bank of the U.A.E. which is denominated and funded in the domestic currency.
- (2) For the purposes of (1), the government of an individual Emirate of the U.A.E. will be treated as a central government.
- (3) An Authorised Firm may apply a 0% risk weight to any Exposure to a central government or a Central Bank of a GCC member country if:
- (a) the Exposure is denominated and funded in the domestic currency of the central government or Central Bank of that country; and
 - (b) the national supervisor:
 - (i) permits the application of a 0% risk weight to such Exposures; and
 - (ii) applies the same treatment to Exposures to the central government and Central Bank of the U.A.E.
- (4) An Authorised Firm may apply a Risk Weight to any Exposure to a central government or a Central Bank of a non GCC member country that corresponds to one Credit Quality Grade higher than the grade that would otherwise apply in accordance with the table in Rule 4.12.1 if:
- (a) the central government or the Central Bank has a Credit Quality Grade of 3 or better;
 - (b) the Exposure is denominated and funded in the domestic currency of the central government or Central Bank of that country; and
 - (c) the national supervisor of the country in question permits the application of a 0% risk weight to such Exposures.

Exposures to Public Sector Entities (PSEs)

- 4.12.3** (1) An Authorised Firm must, subject to (2) and (3), risk-weight any Exposure to a PSE in accordance with the following table based on the external rating of the sovereign:

Credit Quality Grade	1	2	3	4	5	6	Unrated
Risk Weight	20%	50%	100%	100%	100%	150%	100%

- (2) Exposures to any PSE, which has been recognised by its national supervisor as exhibiting Credit Risks comparable to its central government, may be treated as claims on a central government in accordance with Rule 4.12.1.

- (3) Exposures to any PSE that exhibits risk characteristics of a commercial enterprise must be treated in accordance with Rules 4.12.11 to 4.12.15.

Guidance

1. As a consequence of Rule 4.12.2(1), an Authorised Firm may apply a 0% risk weight to an Exposure to a PSE referred to in Rule 4.12.3(2) if the PSE is incorporated in the U.A.E. and the Exposure is denominated and funded in the domestic currency of the U.A.E.
2. The DFSA considers that under Rule 4.12.3(2) a PSE could qualify for the same treatment as claims on its central government if it has specific revenue-raising powers and has specific institutional arrangements the effect of which is to reduce its risk of default. Conversely, if a PSE does not have such arrangements or if strict lending rules apply and a declaration of bankruptcy is not possible because of its special public status, the PSE should be treated in accordance with Rule 4.12.3(1).
3. A PSE will exhibit risk characteristics of a commercial enterprise under Rule 4.12.3(3), in particular, if the entity functions as a corporate in a competitive market even though the state, a regional authority or a local authority is a majority shareholder.

Exposures to Multilateral Development Banks (MDBs)

- 4.12.4** Subject to Rule 4.12.5, an Authorised Firm must risk-weight any Exposure to an MDB in accordance with the following table:

Credit Quality Grade	1	2	3	4	5	6	Unrated
Risk Weight	20%	30%	50%	100%	100%	150%	50%

- 4.12.5** An Authorised Firm must apply a 0% risk weight to any Exposure to one of the following MDBs:

- (a) the World Bank Group comprised of the International Bank for Reconstruction and Development, the Multilateral Investment Guarantee Agency, the International Finance Corporation and the International Development Association;
- (b) the Asian Development Bank;
- (c) the African Development Bank;
- (d) the European Bank for Reconstruction and Development;
- (e) the Inter-American Development Bank;
- (f) the European Investment Bank;
- (g) the European Investment Fund;
- (h) the Nordic Investment Bank;
- (i) the Caribbean Development Bank;
- (j) the Islamic Development Bank;

- (k) the Council of Europe Development Bank;
- (l) the International Finance Facility for Immunisation; and
- (m) the Asian Infrastructure Investment Bank.

Exposures to international organisations

- 4.12.6** An Authorised Firm must apply a 0% risk weight to any Exposure to the Bank for International Settlements, the International Monetary Fund, the European Central Bank, the European Union, the European Stability Mechanism or the European Financial Stability Facility.

Exposures to banks

- 4.12.7** (1) Subject to (2) and Rules 4.12.8, 4.12.9 and 4.12.18, an Authorised Firm must risk-weight any Exposure to a bank for which an external credit assessment by a recognised external credit rating agency is available in accordance with the following table:

Credit Quality Grade	1	2	3	4	5	6
Risk Weight	20%	30%	50%	100%	100%	150%

- (2) Exposures to a bank for which an external credit assessment by a recognised external credit rating agency is available must be assigned a risk weight in accordance with the following table if the Exposure:

- (a) has an original maturity of three months or less; or
- (b) has an original maturity of six months or less and arises from the movement of goods across national borders.

Credit Quality Grade	1	2	3	4	5	6
Risk Weight	20%	20%	20%	50%	50%	150%

- 4.12.8** (1) An Authorised Firm must risk-weight any Exposure to a bank for which a short-term external credit assessment by a recognised external credit rating agency is available in accordance with the following table:

Short-Term Credit Quality Grade	I	II	III	IV
Risk Weight	20%	50%	100%	150%

- (2) Any short-term credit assessment must only apply to the item the short-term credit assessment refers to, and it must not be used to derive risk weights for any other item, except in the following cases:

- (a) if a short-term rated facility is assigned a 50% risk weight, no unrated short-term Exposure shall be assigned a risk weight lower than 100%;
 - (b) if a short-term rated facility is assigned a 150% risk weight, then all unrated Exposures on that obligor, whether short-term or long-term, shall also be assigned a 150% risk weight.
- (3) An Authorised Firm may use short-term credit assessments for short-term assets and off-balance sheet items constituting Exposures to banks.
- 4.12.9** (1) For the purpose of Rules 4.12.7 and 4.12.8, an Authorised Firm must conduct due diligence to ensure that the external ratings appropriately and conservatively reflect the creditworthiness of the bank counterparty.
- (2) If the due diligence analysis reflects higher risk characteristics than implied by the Credit Quality Grade, the Authorised Firm must assign a risk weight at least one grade higher than the risk weight determined by the external rating.
- 4.12.10** (1) An Authorised Firm must classify Exposures to banks for which an external credit assessment by a recognised external credit rating agency is not available as Grade A, Grade B or Grade C in accordance with the following principles:
 - (a) where the bank has adequate capacity to meet its financial commitments in a timely manner for the projected life of the assets or exposures and irrespective of the economic cycles and business conditions, it may be classified Grade A. A counterparty bank classified into Grade A must meet or exceed the published minimum regulatory requirements and buffers (excluding liquidity standards) established by its national supervisor as implemented in the jurisdiction where it is incorporated or established, except for bank-specific minimum financial regulatory requirements or buffers that may be imposed through supervisory actions and not made public. If such minimum financial regulatory requirements (other than bank-specific minimum regulatory requirements or buffers) are not publicly disclosed or otherwise made available by the counterparty bank, then the counterparty bank must be classified as Grade B or lower;
 - (b) where the counterparty bank is subject to substantial credit risk, such as repayment capacities that are dependent on stable or favourable economic or business conditions, it may not be classified higher than Grade B. A counterparty bank classified as Grade B must meet or exceed the published minimum regulatory capital requirements (excluding buffers and liquidity standards) established by its national supervisor as implemented in the jurisdiction where it is incorporated or established, except for bank-specific minimum financial regulatory requirements that may be imposed through supervisory actions and are not made public. If such minimum regulatory requirements are not publicly disclosed or otherwise made available by the counterparty bank then the counterparty bank must be classified as Grade C;
 - (c) where the counterparty bank has material default risks and limited margins of safety, it must be classified Grade C. For this purpose a counterparty bank has material default risk and limited margins of safety where adverse business, financial or economic conditions are very likely

to lead, or have led, to an inability to meet its financial commitments. At a minimum, if any of the following triggers is breached, an Authorised Firm must classify the Exposure as Grade C:

- (i) the counterparty bank does not meet the criteria for being classified as Grade B with respect to its published minimum regulatory requirements; or
 - (ii) where the audited financial statements are required, the external auditor has issued an adverse audit opinion or has expressed substantial doubt about the counterparty bank's ability to continue as going concern in its financial statements or audited reports within the previous 12 months.
- (2) An Authorised Firm must, subject to (3), (4) and (5), risk-weight any Exposure to a bank for which an external credit assessment by a recognised external credit rating agency is not available in accordance with the following table.

Credit Risk Assessment	Grade A	Grade B	Grade C
Risk Weight	40%	75%	150%

- (3) An Authorised Firm may risk-weight any Exposure to a bank classified as Grade A at 30% if that bank has:
- (a) a Common Equity Tier 1 ratio which meets or exceeds 14%; and
 - (b) a Tier 1 leverage ratio which meets or exceeds 5%.
- (4) An Authorised Firm may risk-weight any Exposure to a bank for which an external credit assessment by a recognised external credit rating agency is not available in accordance with the following table if the Exposure:
- (a) has an original maturity of three months or less; or
 - (b) has an original maturity of six months or less and arises from the movement of goods across national borders.

Credit Risk Assessment	Grade A	Grade B	Grade C
Risk Weight	20%	50%	150%

- (5) Notwithstanding paragraphs (2) to (4), the risk weight assigned to any Exposure to a bank for which an external credit assessment by a recognised external credit rating agency is not available may not be less than the risk weight applicable to Exposures to the sovereign of the jurisdiction where the bank is incorporated or established if:
- (a) the Exposure:
 - (i) is not in the local currency of the jurisdiction of incorporation or establishment of the debtor bank; or

- (ii) for a borrowing booked in a branch of the debtor bank in a foreign jurisdiction, is not in the local currency of the jurisdiction in which the branch operates; and
- (b) the Exposure is not a self-liquidating, trade-related contingent items arising from the movement of goods with an original maturity of less than one year.

Guidance

1. An Exposure to a bank includes any claim (such as loans and senior debt, unless considered as subordinated debt, equity or other capital instrument under Rule 4.12.18) on a Regulated Financial Institution that is licensed to take deposits from the public and is subject to appropriate prudential standards and level of supervision. For internationally active banks, appropriate prudential standards and level of supervision should be in accordance with the global regulatory framework established by the Basel Committee on Banking Supervision.
2. An Authorised Firm may treat an Exposure to a securities firm as an Exposure to a bank if it can demonstrate to the DFSA that the securities firm is subject to prudential standards and a level of supervision equivalent to those applied to Financial Institutions referred to in Guidance item 1. Exposures to other securities firms, as well as Exposures to insurers should be treated as an Exposure to corporates in accordance with Rules 4.12.11 to 4.12.15.
3. The short-term Exposures referred to in 4.12.7(2) may include on-balance sheet exposures such as loans and off-balance sheet exposures such as self-liquidating trade-related contingent items. An Authorised Firm should not treat as short-term Exposures loans that are expected to be rolled over.
4. The treatment of Exposures under Rule 4.12.8(1) and the general preferential treatment for short-term Exposures set out in Rule 4.12.7(2) interact as follows:
 - a. if there is no short-term credit assessment, the general preferential treatment for short-term Exposures as specified in Rule 4.12.7(2) applies.
 - b. if there is a short-term credit assessment and the treatment under Rule 4.12.8(1) is more favourable as, or identical to, the preferential treatment of Exposures under Rule 4.12.7(2), an Authorised Firm should apply the risk weight under Rule 4.12.8(1) for that specific Exposure only.
 - c. if there is a short-term credit assessment and the treatment under Rule 4.12.8(1) is less favourable than the preferential treatment of Exposures under Rule 4.12.7(2), an Authorised Firm should:
 - i. not apply the risk weight under Rule 4.12.8(1); and
 - ii. assign the same risk weight as that applied by the specific short-term assessment to all unrated short-term claims.

Exposures to corporates

4.12.11 Subject to Rules 4.12.12, 4.12.13 and 4.12.18, an Authorised Firm must risk-weight any Exposure to a corporate for which an external credit assessment by a recognised external credit rating agency is available in accordance with the following table:

Credit Quality Grade	1	2	3	4	5	6
Risk Weight	20%	50%	75%	100%	150%	150%

4.12.12 (1) An Authorised Firm must risk-weight any Exposure to a corporate for which a short-term external credit assessment by a recognised external credit rating agency is available in accordance with the following table:

Short-Term Credit Quality Grade	I	II	III	IV
Risk Weight	20%	50%	100%	150%

(2) Any short-term credit assessment must only apply to the item the short-term credit assessment refers to, and it must not be used to derive risk weights for any other item, except in the following cases:

- (a) if a short-term rated facility is assigned a 50% risk weight, no unrated short-term Exposure shall be assigned a risk weight lower than 100%;
- (b) if a short-term rated facility is assigned a 150% risk weight, then all unrated unsecured Exposures on that obligor, whether short-term or long-term, shall also be assigned a 150% risk weight.

(3) An Authorised Firm may use short-term credit assessments for short-term assets and off-balance sheet items constituting Exposures to corporates.

4.12.13 (1) For the purpose of Rules 4.12.12 and 4.12.13, an Authorised Firm must conduct due diligence to ensure that the external ratings appropriately and conservatively reflect the creditworthiness of the corporate counterparty.

(2) If the due diligence analysis reflects higher risk characteristics than that implied by the Credit Quality Grade, the Authorised Firm must assign a risk weight at least one grade higher than the risk weight determined by the external rating.

4.12.14 An Authorised Firm must assign to any Exposure to a corporate for which an external credit assessment by a recognised external credit rating agency is not available, a risk weight of 100% or the risk weight that is applicable to an Exposure to the central government of the jurisdiction in which the corporate is incorporated or established, whichever is higher.

4.12.15 (1) An Authorised Firm must treat any Exposure to a corporate that is not a Real Estate Exposure as a specialised lending Exposure if it has any of the following characteristics, either in legal form or economic substance:

- (a) the Exposure is to an entity that was created specifically to finance and/or operate physical assets;

- (b) the borrowing entity has few or no other material assets or activities, and therefore little or no independent capacity to repay the obligation, apart from the income that it receives from the asset(s) being financed. The primary source of repayment of the obligation is the income generated by the asset(s), rather than the independent capacity of the borrowing entity; and
 - (c) the terms of the obligation give the lender a substantial degree of control over the asset(s) and the income that it generates.
- (2) An Authorised Firm must risk-weight any specialised lending Exposure for which a directly applicable external credit assessment by a recognised external credit rating agency is available in accordance with the following table.

Credit Quality Grade	1	2	3	4	5	6
Risk Weight	20%	50%	75%	100%	100%	150%

- (3) An Authorised Firm must risk-weight any specialised lending Exposure for which a directly applicable external credit assessment by a recognised external credit rating agency is not available as follows:
- (a) where the purpose is to finance the acquisition of physical assets (such as ships, aircraft, satellites, railcars or fleets) and where the repayment of the Exposure is dependent on the cash flows generated by the specific assets that have been financed and pledged or assigned to the lender (“Object Finance”), by assigning a risk weight of 100%;
 - (b) where the purpose is to provide short-term financing of reserves, inventories or receivables of exchange-traded commodities (such as crude oil, metals or crops) and where the Exposure will be repaid from the proceeds of the sale of the commodity and the obligor has no independent capacity to repay the Exposure (“Commodities Finance”), by assigning a risk weight of 100%;
 - (c) where the purpose is to finance a project for the development of large, complex and expensive installations such as power plants, chemical processing plants, mines, transportation infrastructure, environment, media or telecoms and where the revenues generated by a single project are primarily the source of repayment and security for the Exposure (referred to as “Project Finance”), by assigning a risk weight of 130% during the pre-operational phase and, subject to (5), 100% during the operational phase.
- (4) For the purpose of (3)(c) operational phase means the phase in which the entity that was created to finance the project has:
- (a) a positive cash flow that is sufficient to cover any remaining contractual obligation; and
 - (b) declining long-term debt.

- (5) Where a Project Finance Exposure is in the operational phase and is considered high quality in accordance with the criteria in (6), an Authorised Firm may assign a risk weight of 80%.
- (6) A Project Finance Exposure is considered high quality if:
 - (a) it is an Exposure to an entity that is able to meet its financial commitments in a timely manner and its ability to do so is assessed to be robust against adverse changes in the economic cycle and business conditions; and
 - (b) the following conditions are met:
 - (i) the entity is restricted from acting to the detriment of the creditors (including by not being able to issue additional debt without the consent of existing creditors);
 - (ii) the entity has sufficient reserve funds or other financial arrangements to cover the contingency funding and working capital requirements of the project;
 - (iii) the revenues are availability-based or subject to a rate-of-return regulation or take-or-pay contract;
 - (iv) the entity's revenue depends on one main counterparty and this main counterparty is a central government, PSE or a corporate entity with a risk weight of 80% or lower;
 - (v) the contractual provisions governing the Exposure to the entity provide for a high degree of protection for creditors in case of default of the entity;
 - (vi) the main counterparty or other counterparties which similarly comply with the eligibility criteria for the main counterparty will protect the creditors from the losses resulting from a termination of the project;
 - (vii) all assets and contracts necessary to operate the project have been pledged to the creditors to the extent permitted by applicable law; and
 - (viii) creditors may assume control of the entity in case of its default.
- (7) For the purpose of (6)(b)(iii), revenues are availability-based if:
 - (a) the entity is entitled to payments from its contractual counterparties once construction is completed, as long as contractual conditions are fulfilled;
 - (b) the payments are sized to cover operating and maintaining costs, debt service costs and equity returns as the entity operates the project; and
 - (c) the revenues are not subject to swings in demand, such as traffic levels, and are adjusted typically only for lack of performance or lack of availability of the assets to the public.

Retail Exposures

- 4.12.16** (1) An Authorised Firm must, subject to Rule 4.12.17, risk-weight Retail Exposures as follows:
- (a) Regulatory Retail Exposures that do not arise from Exposures to Transactors must be assigned a risk weight of 75%;
 - (b) Regulatory Retail Exposures that arise from Exposures to Transactors must be assigned a risk weight of 45%; and
 - (c) other Retail Exposures that do not qualify as Regulatory Retail Exposures must be assigned a risk weight of 100%.
- (2) For the purposes of (1), a Retail Exposure qualifies as a Regulatory Retail Exposure if it meets all of the following conditions:
- (a) the Retail Exposure takes the form of any of the following types of Exposure:
 - (i) revolving credit and lines of credit, including credit cards, charge cards and overdrafts;
 - (ii) personal term loans and leases, including instalment loans, vehicle loans and leases, student and educational loans and personal finance; or
 - (iii) small business credit facilities and commitments;

but excluding mortgage loans, derivatives and other securities (such as bonds or equities), whether listed or not;
 - (b) the gross value of the Retail Exposure (either individually or aggregated with all other Retail Exposures) to a single obligor or a group of Closely Related Counterparties does not exceed \$1 million; and
 - (c) the gross value of the Retail Exposure (either individually or aggregated with all other Retail Exposures) to a single obligor or a group of Closely Related Counterparties does not exceed 0.2% of the overall Retail Exposures.

Guidance

1. In deciding on whether an Exposure, including one arising from Islamic Financial Business arrangements, falls within the types of Exposures referred to in Rule 4.12.16(2)(a), an Authorised Firm should take a “substance over form” approach.
2. When calculating the ‘gross value of the Retail Exposure’ in Rule 4.12.16(2)(b) and (c) an Authorised Firm should not take into account any credit risk mitigation. In case of off balance sheet items, the amount should be calculated after applying credit conversion factors.
3. To apply the 0.2% threshold under Rule 4.12.16(2)(c), an Authorised Firm should, first, identify the full set of Retail Exposures and, second, identify the subset of Exposures that meet the criterion in Rule 4.12.16(2)(a) and do not exceed the threshold in Rule 4.12.16(2)(b) and, third, exclude any Exposures that have a value greater than 0.2% of the subset before exclusion.

- 4.12.17** (1) An Authorised Firm must apply a 1.5 times multiplier to the applicable risk weight calculated under Rule 4.12.16(1) to any unhedged Retail Exposure to an individual where the lending currency differs from the currency of the obligor's source of income, subject to a maximum risk weight of 150%.
- (2) For the purposes of (1), a Retail Exposure is hedged if:
- (a) the obligor:
 - (i) in its normal operating procedures, receives foreign currency income that matches the currency of a given Exposure, such as remittances, rental incomes or salaries; or
 - (ii) has a legal contract with a Financial Institution; and
 - (b) the hedge covers at least 90% of any instalment for the Exposure.
- (3) An Authorised Firm may apply a 1.2 times multiplier if the following conditions are met:
- (a) the official exchange rate between the lending currency and the currency of the obligor's source of income is fixed via an official currency peg; and
 - (b) each of the currencies in (a) are issued by a central government or Central Bank that has an external credit assessment by a recognised external credit rating agency of Credit Quality Grade 1.

Subordinated debt, equity and other capital instruments

- 4.12.18** (1) An Authorised Firm must categorise as an equity Exposure:
- (a) any instrument constituting an investment in subordinated debt, equity or other capital instrument if:
 - (i) the return of invested funds can be achieved only by the sale of the investment or sale of the rights to the investment or by the liquidation of the issuer;
 - (ii) it does not place an obligation on the issuer; and
 - (iii) it conveys a residual claim on the assets or income of the issuer;
 - (b) any instrument with the same structure as those permitted as T1 Capital if issued by an Authorised Firm;
 - (c) any instrument that places an obligation on the issuer and meets any of the following conditions:
 - (i) the issuer may defer indefinitely the settlement of the obligation;
 - (ii) the obligation requires (or permits at the issuer's discretion) settlement by issue of a fixed number of the issuer's equity shares;

- (iii) the obligation requires (or permits at the issuer's discretion) settlement by issue of a variable number of the issuer's equity shares and (all else being equal) any change in the value of the obligation is attributable to, comparable to, and in the same direction as, the change in the value of a fixed number of the issuer's equity shares; or
 - (iv) the holder has the option to require that the obligation be settled in equity shares, unless either:
 - (A) in the case of a traded instrument, the Authorised Firm has satisfied itself on reasonable grounds that the instrument trades more like the debt of the issuer than like its equity; or
 - (B) in the case of non-traded instruments, the Authorised Firm has satisfied itself on reasonable grounds that the instrument should be treated as a debt position;
 - (d) any debt obligation and other security, partnership, derivative or other vehicle structured in a way that the economic substance is similar to the exposures referred to in points (a), (b) and (c), including liabilities from which the return is linked to that of equities; and
 - (e) any equity Exposure that is recorded as a loan but arises from a debt/equity swap made as part of the orderly realisation or restructuring of the debt.
- (2) An Authorised Firm must not categorise as an equity Exposure:
 - (a) any equity investment that is structured with the intent of conveying the economic substance of debt holdings; or
 - (b) any equity investment constituting a securitisation Exposure.
- (3) An Authorised Firm must assign a risk weight of 250% to any equity Exposure, unless it is:
 - (a) an equity investment in accordance with paragraph (4); or
 - (b) required to be deducted or risk-weighted in accordance with PIB Part 3.
- (4) An Authorised Firm must assign a risk weight of 400% to any equity investment in unlisted companies that are invested for short-term resale purposes or are considered venture capital or similar investments which are subject to price volatility and are acquired in anticipation of significant future capital gains.
- (5) An Authorised Firm must risk-weight any subordinated debt or capital instrument that is not an equity Exposure at 150% unless it is required to be deducted or risk-weighted in accordance with PIB Part 3.

Real Estate Exposures

- 4.12.19** (1) An Authorised Firm must apply the risk weights in accordance with Rules 4.12.23 and 4.12.24 to Regulatory Real Estate Exposures.

- (2) An Authorised Firm must apply the risk weights in accordance with Rule 4.12.25 to Other Real Estate Exposures.
- (3) An Authorised Firm must apply the risk weights in Rule 4.12.26 to ADC Exposures.

4.12.20 A Real Estate Exposure is a Regulatory Real Estate Exposure if all of the following conditions are met:

- (a) the property securing the Exposure meets at least one of the following conditions:
 - (i) it has been fully completed;
 - (ii) it is forest or agricultural land;
 - (iii) it is Residential Real Estate under construction or land upon which Residential Real Estate is planned to be constructed where that plan has been approved by all authorities concerned and where any of the following conditions are met:
 - (A) the property does not have more than four residential housing units and will be the primary residence of the obligor and the lending to the individual is not indirectly financing ADC Exposures; or
 - (B) the central government or PSE involved have the legal powers and ability to ensure that the property under construction will be finished;
- (b) it is valued in accordance with the requirements in Rule 4.12.21;
- (c) the Exposure is secured by:
 - (i) a first lien held by the Authorised Firm on the property; or
 - (ii) a junior lien held by the Authorised Firm and the Authorised Firm holds the first lien and any sequentially lower ranking lien on that property; or
 - (iii) a junior lien held by the Authorised Firm in jurisdictions where junior liens provide the holder with a claim for Collateral that is legally enforceable and constitutes an effective credit risk mitigant and the following requirements are met:
 - (A) each party holding a lien on a property can initiate the sale of the property independently from other entities holding a lien on the property;
 - (B) where the sale of the property is not carried out by means of a public auction, parties holding a senior lien take reasonable steps to obtain a fair market value or the best price that may be obtained in the circumstances when exercising any power of sale on their own;

- (d) the security is enforceable in all relevant jurisdictions and the applicable legal framework must allow the Authorised Firm to realise the value of the property within a reasonable time frame following a default;
- (e) the Authorised Firm has assessed the borrower as able to repay in accordance with its Credit Risk policy required under Rule 4.4.2; and
- (f) all the information required at origination of the Exposure and for monitoring purposes is properly documented, including information on the ability of the borrower to repay and on the valuation of the property.

4.12.21 For the purpose of Rules 4.12.23 and 4.12.24, an Authorised Firm must calculate the loan-to-value (LTV) by dividing the loan amount by the value of the property in accordance with the following requirements:

- (a) the value of the property must be equal to the value measured at origination unless any of the below applies:
 - (i) if an extraordinary, idiosyncratic event occurs that results in a permanent reduction of value of the property, the Authorised Firm must, within a reasonable time, obtain an updated valuation and the value of the property is the value shown by the updated valuation;
 - (ii) if modifications are made to the property that unequivocally increase its value, and an updated valuation confirms the increase, the Authorised Firm may use the value shown by the updated valuation;
- (b) the LTV must be calculated prudently in accordance with the following requirements:
 - (i) the amount of the loan must include the outstanding loan amount and any undrawn committed amount of the mortgage loan, without taking into account credit risk adjustments and other capital deductions or any form of credit risk mitigation except for pledged deposit accounts with the Authorised firm that meet all requirements for on-balance sheet netting in accordance with Rules 4.13.17 to 4.13.19 and have been unconditionally and irrevocably pledged for the sole purposes of payment of the mortgage loan;
 - (ii) the value of the immovable property must be appraised independently from the Authorised Firm's mortgage acquisition, loan processing and loan decision process, using prudently conservative valuation criteria. The valuation must:
 - (A) exclude expectations on price increases;
 - (B) be adjusted to take into account the potential for the current market price to be significantly above the value that would be sustainable over the life of the loan;
 - (C) where a market value can be determined, not be higher than the market value; and

- (D) where the loan is financing the purchase of the property, not be higher than the effective purchase price;
- (c) the value of the property must not depend materially on the performance of the borrower.
- 4.12.22** (1) Subject to (2), for the purpose of Rules 4.12.23 to 4.12.25, a Real Estate Exposure is materially dependent on the cash flows generated by the property if the payment of the loan materially depends on the cash flows generated by the property securing the Exposure, rather than on the underlying capacity of the borrower to service the loan from other sources.
- (2) A Regulatory Real Estate Exposure is not materially dependent on the cash flows generated by the property if it is:
- (a) secured by a property which is the borrower's primary residence;
 - (b) an Exposure secured by an income-producing residential housing unit to an individual who has no more than two other mortgaged properties or housing units, to an individual;
 - (c) secured by a Residential Real Estate Exposure to associations or cooperatives of individuals that are regulated under national law and exist with the only purpose of granting its members the use of a primary residence in the property securing the loans; or
 - (d) secured by Residential Real Estate to public housing companies and not-for-profit associations regulated under national law that exist to serve social purposes and to offer tenants long-term housing.

Guidance

1. The primary source of the cash flows generated by a property are generally lease or rental payments or the sale of the property. The distinguishing characteristic of these Exposures compared to other Regulatory Real Estate Exposures is that both the servicing of the loan and the prospects for recovery in the event of default depend materially on the cash flows generated by the property securing the Exposure.
 2. The DFSA expects that the material dependence condition would predominantly apply to loans to corporates, SMEs or SPVs, but is not restricted to those borrower types. As an example, a loan may be considered materially dependent if more than 50% of the income from the borrower used in the Authorised Firm's assessment of its ability to service the loan is from cash flows generated by the residential property.
- 4.12.23** (1) An Authorised Firm must, subject to (3), risk-weight any Regulatory Residential Real Estate Exposure that is not materially dependent on cash flows generated by the property in accordance with the following table.

Loan-to-Value	LTV ≤ 50%	50% < LTV ≤ 60%	60% < LTV ≤ 80%	80% < LTV ≤ 90%	90% < LTV ≤ 100%	LTV > 100%
Risk Weight	20%	25%	30%	40%	50%	70%

- (2) An Authorised Firm must, subject to (3), risk-weight the whole of a Regulatory Residential Real Estate Exposure that is materially dependent on cash flows generated by the property in accordance with the following table.

Loan-to-Value	LTV ≤ 50%	50% < LTV ≤ 60%	60% < LTV ≤ 80%	80% < LTV ≤ 90%	90% < LTV ≤ 100%	LTV > 100%
Risk Weight	30%	35%	45%	60%	75%	105%

- (3) Where a party other than the Authorised Firm holds a senior lien and a junior lien by the Authorised Firm is recognised under Rule 4.12.20(c), the Authorised Firm must multiply the whole of the Regulatory Residential Real Estate Exposure risk weight determined under (1) and (2) respectively by 1.25, unless the LTV below or equal to 50%, in which case the Authorised Firm must not apply the multiplier.
- (4) For the purpose of defining the LTV bucket in (3), the loan amount of the junior liens must include all other loans secured with liens of equal or higher ranking than the Authorised Firm's lien. If there is insufficient information for ascertaining the ranking of the other liens, the Authorised Firm must rank the other liens pari passu with the junior lien held by it.

- 4.12.24** (1) An Authorised Firm must, subject to (3), risk-weight any Regulatory Commercial Real Estate Exposure that is not materially dependent on cash flows generated by the property in accordance with the following table.

Loan-to-Value	LTV ≤ 60%	LTV > 60%
Risk Weight	Min (60%, Risk Weight of Counterparty)	Risk Weight of Counterparty

- (2) An Authorised Firm must, subject to (3), risk-weight the whole of a Regulatory Commercial Real Estate Exposure that is materially dependent on cash flows generated by the property in accordance with the following table.

Loan-to-Value	LTV ≤ 60%	60% < LTV ≤ 80%	LTV > 80%
Risk Weight	70%	90%	110%

- (3) Where a party other than the Authorised Firm holds a senior lien and a junior lien by the Authorised Firm is recognised under Rule 4.12.20(c), the Authorised Firm must multiply the whole of the Regulatory Commercial Real Estate Exposure risk weight determined under (1) and (2) respectively by 1.25, unless the LTV ≤ 50% , in which case the Authorised Firm must not apply the multiplier.
- (4) For the purpose of (3), the loan amount of the junior liens must include all other loans secured with liens of equal or higher ranking than the Authorised Firm's lien. If there is insufficient information for ascertaining the ranking of the other

liens, the Authorised Firm must rank the other liens *pari passu* with the junior lien held by it.

- 4.12.25** (1) Subject to (2), an Authorised Firm must risk-weight any Other Real Estate Exposure that is not materially dependent on cash flows generated by the property in accordance with the table below:

Counterparty	Individual	Other Counterparties
Risk Weight	75%	Risk Weight of Counterparty

- (2) An Authorised Firm must risk-weight any Other Real Estate Exposure that is materially dependent on cash flows generated by the property at 150%.

- 4.12.26** (1) An Authorised Firm must, subject to (2), risk-weight any ADC Exposure at 150%.

- (2) An Authorised Firm may risk-weight an ADC Exposure to Residential Real Estate at 100% if:

(a) the Authorised Firm applies sound origination and monitoring standards which meet the requirements of section 4.4; and

(b) one of the following conditions is met:

(i) legally binding pre-sale or pre-lease contracts, for which the purchaser or tenant has made a substantial cash deposit which is subject to forfeiture if the contract is terminated, amount to a significant portion of total contracts; or

(ii) the borrower has substantial equity at risk.

- 4.12.27** (1) An Authorised Firm must, subject to (3), apply a 1.5 times multiplier to the applicable risk weight calculated under Rule 4.12.23 to any unhedged Residential Real Estate Exposure to an individual where the lending currency differs from the currency of the obligor's source of income, subject to a maximum risk weight of 150%.

- (2) For the purposes of (1), a Regulatory Residential Real Estate Exposure is hedged if:

(a) the obligor:

(i) in its normal operating procedures, receives foreign currency income that matches the currency of a given Exposure, such as remittances, rental incomes or salaries; or

(ii) has a legal contract with a Financial Institution; and

(b) the hedge covers at least 90% of any instalment for the Exposure.

- (3) An Authorised Firm may apply a 1.2 times multiplier if the following conditions are met:

- (a) the official exchange rate between the lending currency and the currency of the obligor's source of income is fixed via an official currency peg; and
- (b) each of the currencies in (a) are issued by a central government or Central Bank that has an external credit assessment by a recognised external credit rating agency of Credit Quality Grade 1.

Exposures in default

4.12.28 (1) Subject to (4), an Authorised Firm must risk-weight the unsecured part of any Exposure that is:

- (a) past due for more than 90 days; or
- (b) to a defaulted borrower,

net of specific provisions and partial write-offs in accordance with the following table:

Condition	Risk Weight
Where specific provisions are less than 20% of the outstanding amount of the Exposure	150%
Where specific provisions are not less than 20% of the outstanding amount of the Exposure	100%

- (2) For the purposes of (1)(b), and subject to (3) a defaulted borrower is a borrower in respect of whom any of the following events have occurred:
 - (a) the borrower is more than 90 days past due on any material credit obligation to the Authorised Firm. For overdrafts, days past due commence once an obligor has breached an advised limit or has been advised of a limit smaller than current outstandings;
 - (b) the Authorised Firm puts a material credit obligation of the borrower on a non-accrued status;
 - (c) the Authorised Firm recognises a specific provision resulting from a significant perceived decline in credit quality subsequent to the Authorised Firm taking on the Exposure;
 - (d) the Authorised Firm sells the credit obligation at a material credit-related economic loss;
 - (e) the Authorised Firm consents to the distressed restructuring of the credit obligation where this is likely to result in a diminished financial obligation caused by the material forgiveness, or postponement, of principal, interest or, where relevant, fees;

- (f) the borrower's bankruptcy or a similar order in respect of any of the borrower's credit obligations to the Authorised Firm or any member of the Authorised Firm's Group has been filed;
 - (g) the borrower has sought or has been placed in bankruptcy or similar protection where this would avoid or delay repayment of any of the credit obligations to the Authorised Firm or any member of the Authorised Firm's Group; or
 - (h) the Authorised Firm considers that the borrower is unlikely to pay its credit obligations in full without recourse by the Authorised Firm to actions such as realising security.
- (3) For Retail Exposures, an Authorised Firm may apply the definition of defaulted borrower in (2) at the level of a particular credit obligation, rather than in relation to the total obligations of the borrower and the default by a borrower on one obligation does not require the Authorised Firm to treat all other obligations to the Authorised Firm or any member of the Authorised Firm's Group as in default.
- (4) An Authorised Firm must risk-weight any unsecured Regulatory Residential Real Estate Exposure referred to in Rule 4.12.23(1) net of specific provisions and partial write-offs at 100% if a default has occurred in accordance with Rule 4.12.28(1)(a) or (b).

4.12.29 For the purposes of Rule 4.12.28(1) and (4), an Authorised Firm must calculate the unsecured portion of any Exposure as follows:

- (a) for an Authorised Firm using the FCSA:

$$\text{Unsecured Portion} = E - P - Cf$$

where:

- (i) E = E calculated in accordance with section 4.9;
- (ii) P = notional amount of eligible credit protection received; and
- (iii) Cf = fair value of eligible financial Collateral received; or

- (b) for an Authorised Firm using the FCCA:

$$\text{Unsecured Portion} = E^* - P$$

where -

- (i) E^* = E^* calculated in accordance with section 4.9; and
- (ii) P = notional amount of eligible credit protection received.

Other Exposures

4.12.30 (1) An Authorised Firm must, subject to (2) and (3), risk-weight any Exposure which does not fall within any of the categories in Rule 4.10.1(a) to (j) at 100%, with the exception of the following Exposures:

- (2) An Authorised Firm must assign:
 - (i) a 0% risk weight to:
 - (A) cash owned by the Authorised Firm or in transit;
 - (B) gold bullion held at the Authorised Firm or held in another banking institution to the extent that the gold bullion assets are backed by gold bullion liabilities; and
 - (ii) a 20% risk weight to cash items in the process of collection.
- (3) Where an Authorised Firm provides credit protection for a number of exposures subject to the condition that the nth-to-default among the exposures triggers payment and that this credit event terminates the contract, the risk weights of the Exposures included in the basket must be aggregated, excluding n-1 Exposures, up to a maximum of 1000% and multiplied by the nominal amount of the protection provided by the credit derivative to obtain the risk-weighted Exposure amount.
- (4) The n-1 Exposures to be excluded from the aggregation under (3) must be determined on the basis that they must include those Exposures each of which produces a lower risk-weighted Exposure amount than the risk-weighted Exposure amount of any of the Exposures included in the aggregation.

Guidance

Cash in transit referred to in Rule 4.12.30(2)(i)(A) includes receivable funds arising from transactions that are settled on a delivery versus payment basis which are outstanding up to and including the 4th business day after the settlement date. Cash items in the process of collection, referred to in Rule 4.12.30(2)(ii) include cheques, drafts and other items drawn on other banking institutions that are either payable immediately upon presentation or that are in the process of collection.

4.13 Credit Risk mitigation

Guidance

This section sets out the principles and methodologies for the recognition of Credit Risk mitigation in the calculation of Credit RWA.

General Requirements

- 4.13.1** (1) An Authorised Firm must not recognise the effects of Credit Risk mitigation unless:
- (a) all documentation relating to that mitigation is binding on all relevant parties and legally enforceable in all relevant jurisdictions;
 - (b) there is no material positive correlation between the credit quality of the Counterparty and the employed Credit Risk mitigation technique or with the resulting residual risk; and
 - (c) the Authorised Firm complies with the Rules set out in this section, as applicable.

- (2) Where the calculation of Credit RWA already takes into account the Credit Risk mitigant, the provisions of this section do not apply.
- (3) An Authorised Firm must, where it uses a specific Credit Risk mitigation technique for the purposes of calculating Credit RWA, use the same technique for Large Exposure reduction, where it is permitted and chooses to use mitigation under the requirements in chapter 4.

Guidance

An Authorised Firm should conduct sufficient legal review to verify this and have a well-founded legal basis to reach this conclusion, and undertake such further review as necessary to ensure continuing enforceability. The review should cover relevant jurisdictions such as the jurisdiction whose law governs the credit protection or Collateral agreement and the jurisdiction whose law governs the transaction subject to the credit protection or Collateral agreement. There should be sufficient written documentary evidence to adequately support the conclusion drawn and rebut any legal challenge. While an Authorised Firm may use either in-house or external legal counsel, it should consider whether or not in-house counsel opinion is appropriate. The senior management of the Authorised Firm should ensure that an officer of the Authorised Firm who is legally qualified and independent of the parties originating the transaction reviews the legal opinion and confirms that he is satisfied that an adequate review has been completed and that he agrees with the conclusions drawn. A record of these reviews should be kept and made available at the request of the DFSA.

4.13.2 Where an Authorised Firm uses multiple Credit Risk mitigation for a single Exposure, the Authorised Firm must divide the Exposure into portions covered by each mitigation and must calculate the Credit Risk-weighted Exposure amount of each portion separately. An Authorised Firm must apply the same approach when recognising eligible credit protection by a single protection provider where the eligible credit protection has differing maturities.

- 4.13.3**
- (1) An Authorised Firm must take all appropriate steps to ensure the effectiveness of the Credit Risk mitigation arrangements it employs and to address related risks.
 - (2) Where an Authorised Firm reduces or transfers Credit Risk by the use of Credit Risk mitigation, an Authorised Firm must employ appropriate and effective policies and procedures to identify and control other risks which arise as a consequence of the transfer.

Guidance

1. The use of techniques to reduce or transfer Credit Risk may simultaneously increase other risks (residual risks) which include legal, operational, liquidity and Market Risks. The DFSA expects an Authorised Firm to employ methods to identify and control these risks, including:
 - a. strategy;
 - b. consideration of the underlying credit;
 - c. valuation;
 - d. policies and procedures;
 - e. systems;
 - f. control of roll-off risks; and

- g. management of concentration risk arising from the use of Credit Risk mitigation and the interaction of such risk with the overall Credit Risk profile of the Authorised Firm.
2. In order to fulfil the above, an Authorised Firm should ensure a clearly articulated strategy for the use of Credit Risk mitigation as an intrinsic part of the general credit strategy of an Authorised Firm.
3. Where an Exposure is subject to Credit Risk mitigation, credit managers should continue to assess the Exposure on the basis of the obligor's creditworthiness. Credit managers should obtain and analyse sufficient financial information to determine the obligor's risk profile and its management and operational capabilities.
4. Collateral should be revalued frequently, and the unsecured Exposure should also be monitored frequently. Frequent revaluation is prudent, and the revaluation of marketable securities should occur on at least a daily basis. Furthermore, measures of the potential unsecured Exposure under collateralised transactions should be calculated under stressed and normal conditions. One such measure would take account of the time and cost involved if the obligor or Counterparty were to default and the Collateral had to be liquidated. Furthermore, the setting of limits for collateralised Counterparties should take account of the potential unsecured Exposure. Stress tests and scenario analysis should be conducted to enable the Authorised Firm to understand the behaviour of its portfolio of Credit Risk mitigation arrangements under unusual market conditions. Any unusual or disproportionate risk identified should be managed and controlled.
5. Clear policies and procedures should be established in respect of Collateral management, including:
 - a. the terms of Collateral agreements;
 - b. the types of Collateral and enforcement of Collateral terms (e.g. waivers of posting deadlines);
 - c. the management of legal risks;
 - d. the administration of agreement (e.g. detailed plans for determining default and liquidating Collateral); and
 - e. the prompt resolution of disputes, such as valuation of Collateral or positions, acceptability of Collateral, fulfilment of legal obligations and the interpretation of contract terms.
6. The policies and procedures referred to under Guidance note 1(d) should be supported by Collateral management systems capable of tracking the location and status of posted Collateral (including re-hypothecated Collateral), outstanding Collateral calls and settlement problems.
7. Where an Authorised Firm obtains credit protection that differs in maturity from the underlying credit Exposure, the Authorised Firm should monitor and control its roll-off risks, i.e. the fact that the Authorised Firm will be fully exposed when the protection expires, and the risk that it will be unable to purchase credit protection or ensure its capital adequacy when the credit protection expires.
8. Taking as Collateral large quantities of instruments issued by one obligor creates Concentration Risk. An Authorised Firm should have a clearly defined policy with respect to the amount of Concentration Risk it is prepared to run. Such a policy might, for example, include a cap on the amount of Collateral it would be prepared to take from a particular Issuer or market. The Authorised Firm should also take Collateral and purchased credit protection into account when assessing the potential concentrations in its overall credit profile.
9. Notwithstanding the presence of Credit Risk mitigation considered for the purposes of calculating Credit RWA amounts, an Authorised Firm should continue to undertake a full Credit Risk assessment of the underlying Exposure.

- 4.13.4** (1) An Authorised Firm must be able to satisfy the DFSA that it has systems in place to manage potential concentration of risk arising from its use of guarantees and Credit Derivatives.
- (2) An Authorised Firm must be able to demonstrate how its strategy in respect of its use of Credit Risk mitigation techniques, and in particular use of Credit Derivatives and guarantees interacts with its management of its overall risk profile.

Collateral

Guidance

In order to recognise the effects of Credit Risk mitigation of the types of Collateral set out in Rules 4.13.5 to 4.13.7, an Authorised Firm must ensure that the relevant requirements in Rule 4.13.8 are complied with.

- 4.13.5** (1) For an Authorised Firm using the FCSA, eligible financial Collateral comprises:
- (a) cash (as well as certificates of deposit or other similar instruments issued by the Authorised Firm) on deposit with the Authorised Firm;
 - (b) gold;
 - (c) any debt security:
 - (i) with an Original Maturity of one year or less that has a short-term Credit Quality Grade of 3 or better as set out in section 4.12; or
 - (ii) with an Original Maturity of more than one year that has:
 - (A) a Credit Quality Grade of 4 or better as set out in section 4.12 if it is issued by a central government, Central Bank or a PSE referred to in Rule 4.12.3(2); or
 - (B) a Credit Quality Grade of 3 or better as set out in section 4.12 if it is issued by any other entity;
 - (d) any debt security issued by a bank that does not have an external credit assessment by a recognised ECAI if it fulfils the following criteria:
 - (i) any debt security which is listed on a regulated exchange;
 - (ii) the debt security is classified as senior debt, not subordinated to any other debt obligations of its Issuer;
 - (iii) all other rated debt securities issued by the same Issuer which rank equally with the mentioned debt security have a long term or short term (as applicable) Credit Quality Grade by a recognised ECAI of “3” or better;

- (iv) the Authorised Firm is not aware of information to suggest that the issue would justify a Credit Quality Grade of below “3” as indicated in (iii) above; and
 - (v) the Authorised Firm can demonstrate to the DFSA that the market liquidity of the debt security is sufficient to enable the Authorised Firm to dispose the debt security at market price.
- (e) any equity security (including convertible bonds) that is included in a main index; or
- (f) any Unit in a Collective Investment Fund where:
 - (i) a price for the units is publicly quoted daily; and
 - (ii) all of the deposited property of the Fund is invested in instruments listed in this Rule,
 but excluding Collateral representing a re-securitisation under Rule 4.14.2(c).
- (2) Cash-funded credit-linked notes issued by an Authorised Firm against Exposures in the Non-Trading Book which fulfil the criteria for eligible Credit Derivatives must be treated as cash collateralised transactions.
- (3) Cash, mentioned in (1)(a), includes cash on deposit, certificates of deposit or other similar instruments issued by the Authorised Firm that are held as Collateral at a third-party bank in a non-custodial arrangement and that are pledged or assigned to the Authorised Firm. This is subject to the pledge or assignment being unconditional and irrevocable. Under the FCSA, the risk weight to be applied to the Exposure covered by such Collateral must be the risk weight of the third-party bank.

Guidance

1. For the purposes of Rule 4.13.5 and 4.13.6, eligible financial Collateral excludes any T1 Capital instrument or T2 Capital instrument issued by any entity in the Financial Group of the Authorised Firm, which is held by the Authorised Firm or any of its Financial Group entities as Collateral.
2. For an Authorised Firm using Units of a Fund under the FCSA approach, the use or potential use by that Fund of Derivative instruments solely to hedge investments listed in Rule 4.13.5 should not preclude the Units in that Fund from being recognised as eligible financial Collateral.

4.13.6 For an Authorised Firm using the FCCA, eligible financial Collateral comprises:

- (a) any instrument listed in Rule 4.13.5;
- (b) any equity Security (including a convertible bond) that is traded on a regulated exchange;
- (c) any Unit in a Collective Investment Fund which invests in equity securities referred to in (b), where:
 - (i) a price for the Units is publicly quoted daily; and

- (ii) all of the deposited property of the Fund is invested in instruments listed in this Rule and Rule 4.13.5.

4.13.7 In the case of any Counterparty Credit Risk Exposures in Rules 4.13.5. and 4.13.6 arising from an SFT which are included in the Trading Book, eligible financial Collateral includes all instruments which an Authorised Firm may include in its Trading Book.

Guidance

For an Authorised Firm using Units of a Fund under the FCSA approach, the use or potential use by that Fund of Derivative instruments solely to hedge investments listed in Rule 4.13.5 should not preclude the Units in that Fund from being recognised as eligible financial Collateral.

Requirements for Recognition of Collateral

4.13.8 An Authorised Firm must ensure that the following requirements are complied with before it recognises the effects of Credit Risk mitigation of any Collateral:

- (a) the legal mechanism by which Collateral is pledged, assigned or transferred must confer on the Authorised Firm the right to liquidate or take legal possession of the Collateral, in a timely manner, in the event of the default, insolvency or bankruptcy (or one or more otherwise-defined credit events set out in the transaction documentation) of the Counterparty (and, where applicable, of the custodian holding the Collateral);
- (b) the Authorised Firm has taken all steps necessary to fulfil those requirements under the law applicable to the Authorised Firm's interest in the Collateral for obtaining and maintaining an enforceable security interest by registering it with a registrar or for exercising a right to net or set off in relation to title transfer Collateral;
- (c) the credit quality of the Counterparty and the value of the Collateral do not have a material positive correlation;
- (d) securities issued by the Counterparty or any Closely Related Counterparty are not eligible;
- (e) the Authorised Firm has implemented procedures for the timely liquidation of Collateral to ensure that any legal conditions required for declaring default of Counterparty and liquidating the Collateral are observed, and that the Collateral can be liquidated promptly;
- (f) for an Authorised Firm using the FCSA, the Collateral has been marked to market and revalued at least every six months or more frequently if required by business or market circumstances;
- (g) the Authorised Firm has dedicated sufficient resource to the orderly operation of margin agreements with OTC derivatives and securities-financing Counterparties, as measured by the timeliness and accuracy of their outgoing margin calls and response time to incoming margin calls;
- (h) the Authorised Firm has implemented Collateral risk management policies to control, monitor and report:

- (i) the risk to which margin agreements expose them (such as the volatility and liquidity of the securities exchanged as Collateral);
- (ii) the concentration risk to particular types of Collateral;
- (iii) the reuse of Collateral (both cash and non-cash) including the potential liquidity shortfalls resulting from the reuse of Collateral received from Counterparties; and
- (iv) the surrender of rights on Collateral posed to Counterparties; and
- (i) where the Collateral is held by a custodian, the Authorised Firm has taken reasonable steps to ensure that the custodian segregates the Collateral from its own assets.

Guidance

In determining if there is a material positive correlation as referred to in Rule 4.13.8(c), an Authorised Firm should take into account several factors, including the consistency of any correlation over time, the incidental nature of any correlation (whether it will hold for over a one-year horizon) or whether any correlation may lead to larger or smaller swings in potential credit losses.

Guarantees

- 4.13.9** (1) An Authorised Firm may recognise the effects of Credit Risk mitigation of a guarantee only if it is provided by any of the following entities:
- (a) central government or Central Bank;
 - (b) MDB referred to in Rule 4.12.5;
 - (c) International Organisations referred to in Rule 4.12.6;
 - (d) PSE;
 - (e) prudentially regulated financial institutions;
 - (f) where credit protection is provided to a non-securitisation Exposure, any entity that has an external credit assessment by a recognised external credit rating agency, including credit protection provided by a Parent or a Subsidiary; or
 - (g) where credit protection is provided to a securitisation Exposure, any entity that has an external credit assessment by a recognised external credit rating agency of Credit Quality Grade 3 or above, including credit protection provided by a Parent or a Subsidiary.
- (2) An Authorised Firm must not recognise the effects of Credit Risk mitigation of a guarantee unless all of the following requirements are complied with:
- (a) the guarantee is an explicitly documented obligation assumed by the guarantor;
 - (b) the guarantee represents a direct claim on the guarantor;

- (c) the extent of the credit protection cover is clearly defined and incontrovertible;
 - (d) other than in the event of non-payment by the Authorised Firm of money due in respect of the guarantee if applicable, there is an irrevocable obligation on the part of the guarantor to pay out a pre-determined amount upon the occurrence of a credit event, as defined under the guarantee;
 - (e) the guarantee does not contain any clause, the fulfilment of which is outside the direct control of the Authorised Firm, that:
 - (i) would allow the guarantor to cancel the guarantee unilaterally;
 - (ii) would increase the effective cost of the guarantee as a result of deteriorating credit quality of the underlying Exposure;
 - (iii) could prevent the guarantor from being obliged to pay out in a timely manner in the event that the underlying obligor fails to make any payment due; or
 - (iv) could allow the maturity of the guarantee agreed ex-ante to be reduced ex-post by the guarantor;
 - (f) the Authorised Firm is able in a timely manner to pursue the guarantor for any monies outstanding under the documentation governing the transaction on the default of, or non-payment by, the underlying obligor without first having to take legal action to pursue the underlying obligor for payment; and
 - (g) the guarantee covers all types of payments that the underlying obligor is expected to make under the documentation governing the transaction, except in the case of accrued interest, accrued expenses or fees outstanding, where these are deemed immaterial.
- (3) In (1)(e) the term 'prudentially regulated financial institution' means a legal entity:
- (a) whose activities are primarily financial in nature and which is supervised by a regulator that imposes prudential requirements consistent with international norms; or
 - (b) which is included in a consolidated group where any substantial legal entity in the consolidated group whose activities are primarily financial in nature is supervised by a regulator that imposes prudential requirements consistent with international norms.

Guidance

1. Rule 4.13.9(2)(e) does not include any guarantee with a cancellation clause where it is provided that any obligation incurred or transaction entered into prior to any cancellation, unilateral or otherwise, continues to be guaranteed by the guarantor.

2. The guarantee payments may be in the form of the guarantor making a lump sum payment of all monies to the Authorised Firm or the guarantor assuming the future payment obligations of the Counterparty covered by the guarantee, as specified in the relevant documentation governing the guarantee.
3. Prudentially regulated financial institutions in Rule 4.13.9(3) include prudentially regulated insurance companies, broker/dealers, thrifts and futures commission merchants, and Qualifying CCPs.

4.13.10 In addition to the requirements in Rule 4.13.9, where an Authorised Firm has an Exposure that is protected by a guarantee or that is counter-guaranteed by a central government or Central Bank, a regional government or local authority, a PSE referred to in Rule 4.12.3(2), a MDB or an international organisation to which a 0% risk weight is assigned under section 4.12, an Authorised Firm may treat the Exposure as being protected by a direct guarantee from the central government or Central Bank in question, provided the following requirements are complied with:

- (a) the counter-guarantee covers all Credit Risk elements of the Exposure;
- (b) both the original guarantee and the counter-guarantee comply with all the requirements for guarantees set out in this section, except that the counter-guarantee need not be direct and explicit with respect to the original Exposure; and
- (c) the Authorised Firm is able to satisfy the DFSA that the cover is robust and that nothing in the historical evidence suggests that the coverage of the counter-guarantee is less than effectively equivalent to that of a direct guarantee by the entity in question.

Credit Derivatives

- 4.13.11** (1) An Authorised Firm may recognise the effects of Credit Risk mitigation of a Credit Derivative only if it is provided by any of the following entities:
- (a) central government or Central Bank;
 - (b) MDB referred to in Rules 4.12.4 to 4.12.5;
 - (c) International Organisations referred to in Rule 4.12.6;
 - (d) PSE;
 - (e) prudentially regulated financial institutions;
 - (f) where credit protection is provided to a non-securitisation Exposure, any entity that has an external credit assessment by a recognised external credit rating agency, including credit protection provided by a Parent or a Subsidiary; or
 - (g) where credit protection is provided to a securitisation Exposure, any entity that has an external credit assessment by a recognised external credit rating agency of Credit Quality Grade 3 or above, including credit protection provided by a Parent or a Subsidiary.

- (2) An Authorised Firm may recognise the effects of Credit Risk mitigation of only the following types of Credit Derivatives:
 - (a) credit default swaps;
 - (b) total return swaps;
 - (c) credit linked notes which are cash funded; and
 - (d) instruments that are composed of, or are similar in economic substance, to one or more of the Credit Derivatives in (a) to (c).
- (3) An Authorised Firm must not recognise the effects of Credit Risk mitigation of first-to-default and all other nth-to-default credit derivatives.
- (4) In (1)(e) the term ‘prudentially regulated financial institution’ means a legal entity:
 - (a) whose activities are primarily financial in nature and which is supervised by a regulator that imposes prudential requirements consistent with international norms; or
 - (b) which is included in a consolidated group where any substantial legal entity in the consolidated group whose activities are primarily financial in nature is supervised by a regulator that imposes prudential requirements consistent with international norms.

Guidance

- 1. Prudentially regulated financial institutions in Rule 4.13.11(4) include prudentially regulated insurance companies, broker/dealers, thrifts and futures commission merchants, or Qualifying CCPs.

- 4.13.12** (1) An Authorised Firm must not, subject to (2), recognise the effects of Credit Risk mitigation of any Credit Derivative unless all of the following requirements are complied with:
- (a) the terms and conditions of any credit protection obtained via a Credit Derivative must be set out in writing by both the Authorised Firm and the provider of credit protection;
 - (b) the Credit Derivative must represent a direct claim on the provider of credit protection;
 - (c) the extent of the credit protection cover is clearly defined and incontrovertible;
 - (d) other than in the event of non-payment by the Authorised Firm of money due in respect of the Credit Derivative, there is an irrevocable obligation on the part of the provider of the credit protection to pay out a pre-determined amount upon the occurrence of a credit event, as defined under the Credit Derivative contract;

- (e) the Credit Derivative contract must not contain any clause, the fulfilment of which is outside the direct control of the Authorised Firm, that:
 - (i) would allow the provider of credit protection to cancel the credit protection cover unilaterally;
 - (ii) would increase the effective cost of the credit protection cover as a result of deteriorating credit quality of the underlying Exposure;
 - (iii) could prevent the provider of credit protection from being obliged to pay out in a timely manner in the event that the underlying obligor fails to make any payment due; or
 - (iv) could allow the maturity of the credit protection agreed ex-ante to be reduced ex-post by the provider of credit protection;
- (f) the credit events specified by the contracting parties must at a minimum cover:
 - (i) failure to pay the amounts due under terms of the underlying Exposure that are in effect at the time of such failure (with a grace period, if any, that is closely in line with the grace period in the underlying Exposure);
 - (ii) bankruptcy, insolvency or inability of the underlying obligor to pay its debts, or its failure or admission in writing of its inability generally to pay its debts as they become due, and analogous events; and
 - (iii) restructuring of the underlying Exposure involving forgiveness or postponement of principal, interest or fees that results in a credit loss event (i.e. charge-off, specific provision or other similar debit to the profit and loss account) unless the underlying Exposure is to a body corporate where:
 - (A) unanimity is required to amend the maturity, principal, coupon, currency or seniority status of the underlying Exposure to the body corporate; and
 - (B) the body corporate is incorporated in a jurisdiction which has a well-established insolvency regime that allows for an entity to reorganise or restructure and provides for an orderly settlement of creditor claims;
- (g) the Credit Derivative must not terminate prior to the maturity of the underlying Exposure or expiration of any grace period required for a default on the underlying Exposure to occur as a result of a failure to pay;
- (h) a robust valuation process to estimate loss reliably must be in place in order to estimate loss reliably for any Credit Derivative that allows for cash settlement. There must be a clearly specified period for obtaining post-credit event valuations of the underlying obligation;

- (i) where the right or ability of the Authorised Firm to transfer the underlying Exposure to the credit protection provider is required for settlement, the terms of the underlying Exposure must provide that any required consent to such transfer may not be unreasonably withheld;
 - (j) the identity of the parties responsible for determining whether a credit event has occurred must be clearly defined. This determination must not be the sole responsibility of the credit protection provider. The Authorised Firm must have the right or ability to inform the credit protection provider of the occurrence of a credit event; and
 - (k) the underlying obligation and the reference obligation specified in the Credit Derivative contract for the purpose of determining the cash settlement value or the deliverable obligation or for the purpose of determining whether a credit event has occurred may be different only if:
 - (i) the reference obligation ranks pari passu with or is junior to the underlying obligation; and
 - (ii) the underlying obligation and reference obligation share the same obligor (i.e. the same legal entity) and legally enforceable cross- default or cross-acceleration clauses are in place.
- (2) Where the credit event under (1)(f) does not include the restructuring of the underlying obligation under (1)(f)(iii), an Authorised Firm may partly recognise the effects of Credit Risk mitigation of any Credit Derivative as follows:
- (a) if the amount that the protection provider has undertaken to pay is lower than or equal to the amount of the Exposure, the Authorised Firm may recognise 60% of the value of the Credit Risk mitigation; or
 - (b) if the amount that the protection provider has undertaken to pay is higher than the amount of the Exposure, the Authorised Firm may recognise the Credit Risk mitigation up to 60% of the Exposure value.

Guidance

1. An Authorised Firm should not recognise the effects of Credit Risk mitigation of a total return swap if it purchases credit protection through a total return swap and records the net payments received on the swap as net income, but does not record offsetting deterioration in the value of the underlying asset that is protected (either through reductions in its marked-to-market value or by an addition to reserves).
2. The DFSA would generally consider the requirements in (f) to have been complied with even if the requirements are not specifically set out so long as the obligations of the credit protection provider under the Credit Derivative contract would include those requirements.
3. The DFSA would generally consider the cash settlement methodology provided in the ISDA Credit Derivatives Definitions as satisfying the requirement for obtaining post-credit event valuations of the underlying obligation.

Currency mismatches

- 4.13.13** In a case where there is a currency mismatch between the credit protection and the underlying Exposure, an Authorised Firm using the FCCA or credit protection in the form of a guarantee or credit derivative must reduce the amount of the Exposure deemed to be protected by applying a haircut, as follows:

$$\text{Protected portion } G_A = G \cdot (1 - H_{FX})$$

where:

G = notional amount of the credit protection

H_{FX} = 8% (based on ten-business days holding period), unless the credit protection is not marked-to-market daily, in which case an Authorised Firm must scale H_{FX} in accordance with Rule A4.3.10

Guidance

Currency mismatches are allowed under all approaches. Under the FCSA there is no specific treatment for currency mismatches, given that a minimum risk weight of 20% (floor) is generally applied.

Maturity mismatches

- 4.13.14** (1) Subject to (2), an Authorised Firm may recognise the effects of Credit Risk mitigation for an Exposure where there is a maturity mismatch only if the Credit Risk mitigant has an Original Maturity of at least one year and a residual maturity of at least three months. For the purposes of calculating Credit RWA, a maturity mismatch occurs when the residual maturity of the Credit Risk mitigant is less than that of the underlying Exposure.
- (2) An Authorised Firm using FCSA must not recognise the effects of Credit Risk mitigation of any Collateral with a maturity mismatch.
- 4.13.15** (1) An Authorised Firm must determine the maturity of the underlying Exposure and the maturity of the Credit Risk mitigant conservatively. The residual maturity of the underlying Exposure must be gauged as the longest possible remaining time before the Counterparty is scheduled to fulfil its obligation, taking into account any applicable grace period.
- (2) In the case of Credit Risk mitigant, embedded options which may reduce the term of the credit protection must be taken into account so that the shortest possible residual maturity is used. Where a call is at the discretion of the protection seller, the residual maturity will be at the first call date. If the call is at the discretion of the Authorised Firm but the terms of the arrangement at origination of the Credit Derivative contain a positive incentive for the Authorised Firm to call the transaction before contractual maturity, the remaining time to the first call date will be deemed to be the residual maturity.
- 4.13.16** (1) An Authorised Firm must calculate the value of the Credit Risk mitigation adjusted for any maturity mismatch, P_a , using the following formula:

$$P_a = P \cdot \frac{t - 0.25}{T - 0.25}$$

where

P = value of the credit protection (e.g. Collateral amount, guarantee amount) adjusted for any haircuts

t = $\min \{T, \text{residual maturity of the Credit Risk mitigant expressed in years}\}$

T = $\min \{\text{five years, residual maturity of the Exposure expressed in years}\}$

- (2) For residual maturity of the Exposure in the case of a basket of Exposures with different maturities, an Authorised Firm must use the longest maturity of any of the Exposures as the maturity of all the Exposures being hedged.

Guidance

The positive incentive for an Authorised Firm to call the transaction before contractual maturity as referred in Rule 4.13.15 would be, for example, a situation wherein there is a step-up in cost in conjunction with a call feature or where the effective cost of cover remains the same even if credit quality remains the same or increases.

On-balance sheet Netting

- 4.13.17** (1) An Authorised Firm may recognise as eligible the Netting of an on-balance sheet Exposure against an offsetting on-balance sheet item if the related Netting agreement meets the condition in Rule 4.13.19.
- (2) Eligibility for Netting is limited to reciprocal cash balances between the Authorised Firm and its Counterparty. Only loans and deposits of the Authorised Firm may be subject to a modification of their Credit RWAs as a result of an on-balance sheet Netting agreement.
- 4.13.18** (1) Assets (loans) and liabilities (deposits) subject to recognised on-balance sheet Netting are to be treated as cash Collateral using the formula in A4.3.6, under which an Authorised Firm may use zero haircuts for Exposure and Collateral.
- (2) When a currency mismatch exists, an Authorised Firm must apply the standard supervisory haircut of 8% for currency mismatch.
- (3) When a maturity mismatch exists between the off-setting items, an Authorised Firm must apply the Rules 4.13.14 to 4.13.16 to address the maturity mismatch.
- (4) Net credit Exposure, after taking into account recognised Netting, will be subject to the applicable CRW for the Counterparty.
- 4.13.19** For an Authorised Firm to recognise an on-balance sheet Netting agreement for the purposes of Rule 4.13.17, all of the following conditions must be satisfied:
 - (1) (a) both the on-balance sheet Exposure (asset) and the offsetting on-balance sheet item (liability) are owing between the Authorised Firm and the same Counterparty;

- (b) the Authorised Firm nets the on-balance sheet Exposure (asset) and the offsetting on-balance sheet item (liability) in a way that is consistent with its legal rights against the Counterparty;
 - (c) a legal right of set-off exists;
 - (d) the agreement between the Authorised Firm and the Counterparty does not contain a Walkaway Clause;
 - (e) the Netting provided for in the agreement between the Authorised Firm and the Counterparty is effective and enforceable in the event of default, bankruptcy, liquidation or other similar circumstances affecting either the Counterparty or the Authorised Firm;
 - (f) the on-balance sheet Exposure (asset) and the offsetting on-balance sheet item (liability) are monitored, controlled and managed on a net basis; and
 - (g) the potential for roll-off Exposure is monitored and controlled where there is a maturity mismatch; and
- (2) it has, in respect of each relevant jurisdiction, a written and reasoned legal opinion which:
- (a) has been provided by an external source of legal advice of appropriate professional standing;
 - (b) confirms that the requirements of (1)(a)-(e) are met for all relevant jurisdictions; and
 - (c) is kept under review to ensure that it remains correct and up to date in the event of changes to the relevant laws.

Guidance

1. An Authorised Firm should assess whether any qualifications, assumptions or reservations contained in the legal opinion cast doubt upon the enforceability of the Netting agreement. If, as a result of the qualifications, assumptions or reservations, there is material doubt about the enforceability of the agreement, the Authorised Firm should assume that the requirements for Netting have not been met.
2. An Authorised Firm using a standard form Netting agreement and a supporting legal opinion should ensure that the relevant requirements in Rules 4.13.17 to 4.13.19 are met. A standard form Netting agreement is a form of agreement which is prepared by a reputable, internationally recognised industry association and is supported by its own legal opinion. Where additional clauses are added to a standard form Netting agreement, the Authorised Firm should satisfy itself that the amended Netting agreement continues to meet the legal and contractual requirements in Rules 4.13.17 to 4.13.19. For instance, in such cases, an Authorised Firm may wish to obtain a second legal opinion to confirm that the relevant requirements in Rules 4.13.17 to 4.13.19 are still satisfied.

4.14 Securitisation

Application

4.14.1 This section applies to an Authorised Firm which:

- (a) acts as an Originator in a securitisation;
- (b) transfers Credit Risk on a single item or on a pool of items by any of the legal transfer methods set out in Rule A4.10.1;
- (c) acts as a Sponsor in a securitisation; or
- (d) provides Credit Enhancement, liquidity support, or Underwriting or dealing facilities relating to the items being transferred.

Interpretation

4.14.2 For the purposes of this chapter and App4, “securitisation” includes Traditional Securitisation, Synthetic Securitisation and Re-securitisation, as defined below:

- (a) A Traditional Securitisation is a structure where the cash flow from an underlying pool of Exposures is used to service at least two different stratified risk positions or tranches reflecting different degrees of Credit Risk. Payments to the investors depend upon the performance of the specified underlying Exposures, as opposed to being derived from an obligation of the entity originating those Exposures. A Traditional Securitisation will generally assume the movement of assets off balance sheet.
- (b) A Synthetic Securitisation is a structure with at least two different stratified risk positions or tranches that reflect different degrees of Credit Risk where Credit Risk of an underlying pool of Exposures is transferred, in whole or in part, through the use of funded (e.g. credit-linked notes) or unfunded (e.g. credit default swaps) Credit Derivatives or guarantees that serve to hedge the Credit Risk of the portfolio. Accordingly, the investors’ potential risk is dependent upon the performance of the underlying pool. A Synthetic Securitisation may or may not involve the removal of assets off balance sheet.
- (c) A Re-securitisation Exposure is a securitisation Exposure in which the associated underlying pool of Exposures is tranching and at least one of the underlying Exposures is a securitisation Exposure. In addition, an Exposure to one or more Re-securitisation Exposures is a Re-securitisation Exposure.

Guidance

The DFSA would treat other techniques to achieve the financing or re-financing of assets which are legally transferred to a scheme, by packaging them into a tradable form through the issue of Securities which are secured on the assets and serviced from the cashflows which they yield as “securitisation”.

Systems and controls for the use of securitisations

4.14.3 An Authorised Firm must implement and maintain appropriate risk management systems to identify, manage, monitor and, where applicable, control all risks in

relation to a securitisation transaction whether the firm is an investor, Originator or Sponsor. In particular, such risk management systems should effectively address the following risks:

- (a) the liquidity and capital implications that may arise from the items returning to the balance sheet;
- (b) the Operational Risks that may arise under a securitisation; and
- (c) reputational risks that may arise as a result of its securitisation activities.

4.14.4 An Authorised Firm must have appropriate policies and procedures to ensure that the economic substance of the transaction is fully reflected in the process of managing the risks arising from such transactions. An Authorised Firm must have appropriate policies and procedures in place to document its systems and controls in relation to securitisation risks. These policies should include details on the capital effects of the securitisation as set out in this chapter.

4.14.5 An Authorised Firm must conduct periodic stress tests in relation to its securitisation activities and off balance sheet Exposures, including testing of future ability to transact securitisation as a means of Credit Risk mitigation or for liquidity purposes.

Guidance

1. The periodic stress testing in relation to securitisation activities referred to in Rule 4.14.5 should consider the firm-wide impact of those activities and Exposures in stressed market conditions and the implications for other sources of risk. Such stress tests should include both existing securitisation Exposures and transactions in the pipeline, as there is a risk of the pipeline transactions not being completed in a stressed market scenario.
2. The frequency and extent of stress testing to fulfil the requirements of Rule 4.14.5 should be determined on the basis of the materiality of the Authorised Firm's securitisation volumes and its off-balance sheet Exposures.
3. An Authorised Firm should have procedures in place to assess and respond to the results produced from the stress testing and these should be taken into account under the ICAAP.

4.14.6 In order to qualify for using the Rules specified in this section, and particularly the risk weighting approach outlined below, an Authorised Firm must demonstrate the following:

- (a) a comprehensive understanding of the risk characteristics of its individual securitisation Exposures, whether on balance sheet or off balance sheet, as well as the risk characteristics of the pools underlying securitisation Exposures;
- (b) ability to access the performance information on the underlying pools on an on-going basis in a timely manner; and
- (c) a thorough understanding of all structural features of a securitisation transaction that would materially impact the performance of the Authorised Firm's Exposure to the transaction, such as waterfall triggers, Credit Enhancements, liquidity enhancements, market value triggers and deal specific definitions of default.

Guidance

1. An Authorised Firm which is an investor, Originator or Sponsor of a Securitisation should fully understand the risks it has assumed in order to ensure that it can accurately determine the Capital Requirements for the Exposures arising from the securitisation in accordance with the Rules in this section.
2. For the purposes of Rule 4.14.6(b) information should include the percentage of loans 30,60, 90 days past due, default rates, prepayment rates, loans in foreclosure, property type, occupancy, average credit score etc. For Re-securitisations, Authorised Firms should have information relating to not only the underlying securitisation transactions but also the characteristics and performance of the underlying pools of such transactions.

4.14.7 Where an Authorised Firm is either an Originator or a Sponsor of a Traditional Securitisation or a Synthetic Securitisation:

- (a) the Authorised Firm intending to conduct the securitisation must notify the DFSA at least 30 days in advance of the proposed execution of the securitisation;
- (b) the Authorised Firm conducting the securitisation must calculate its Credit RWAs for all resultant Exposures from that securitisation, in accordance with section 4.8, provided the requirements of this section are met; and
- (c) the Authorised Firm conducting the securitisation must produce documentation reflecting the execution and economic substance of the transaction.

Guidance

The notification made to the DFSA under (a) should include, inter alia, amounts of assets subject to securitisation, amounts retained, details of securitisation including legal structure, rating, tranches, details of legal transfer and any Credit Risk mitigation applied and implications on the capital and liquidity position on the Authorised Firm.

Calculation of Credit RWA arising from securitisations

4.14.8 An Authorised Firm must calculate the Credit RWA amounts for Exposures arising from securitisations according to the requirements in this section.

Guidance

1. An Authorised Firm should apply the securitisation framework set out in this section for determining the regulatory Capital Requirements on Exposures arising from traditional and Synthetic Securitisations or similar structures that contain features common to both.
2. This section sets out the requirements for Originators, Authorised Firms which transfer Credit Risk from their balance sheets and Sponsors in a securitisation transaction involving Non-Trading Book Exposures. This section also sets out the methodologies for calculation of RWA amounts for securitisation Exposures. The Rules setting out the methodologies for calculation of Market Risk Capital Requirement amounts for securitisation Exposures held in the Trading Book are specified in chapter 5 and App5 of this module.
3. As securitisations may be structured in many different ways, an Authorised Firm engaging in the activities relating to securitisations (whether traditional or Synthetic) must ensure that the economic substance of the transaction is fully considered, and reflected, in determining the capital treatment of a securitisation, rather than relying on the legal form of the Securitisation.

4.14.9 An Authorised Firm is required, subject to Rule 4.14.12, to include all securitisation Exposures in its calculation of Credit RWAs relating to securitisations, including the following:

- (a) those arising from the provision of Credit Risk mitigants to a securitisation;
- (b) investments in asset backed Securities;
- (c) retention of a subordinated tranche;
- (d) extension of a liquidity facility; and
- (e) extension of Credit Enhancement.

4.14.10 An Authorised Firm must include in its calculation of Credit RWA all of its securitisation Exposures held in the Non-Trading Book, except for those securitisation Exposures which the Authorised Firm is required to include as deductions from T1 Capital and deductions from T2 Capital.

4.14.11 Repurchased securitisation transactions must be treated as retained securitisation Exposures.

Deductions

4.14.12 (1) An Authorised Firm may deduct SE Exposures which it has chosen not to treat in accordance with Rules 4.14.8 to 4.14.11 from Capital Resources - 100% from CET1.

(2) Credit-Enhancing Interest-Only Strips (net of the deductions from CET1 Capital required at Rule 4.14.13) are deducted 100% from CET1 Capital.

(3) Deductions from capital may be calculated net of specific provisions taken against relevant securitisation Exposures.

4.14.13 An Authorised Firm must include as deductions from CET1 Capital any increase in issued capital or reserves resulting from a securitisation, such as that associated with expected future margin income resulting in a gain-on-sale that is recognised as issued capital or reserves.

Guidance

Gain-on-sale arises when there has been an increase in equity of the Authorised Firm associated with recognising the discounted value of the expected future margin income as part of the regulatory capital.

4.14.14 An Authorised Firm must assign a securitisation Exposure to a Credit Quality Grade based on the external credit assessment (where available) that is applicable to the securitisation Exposure in accordance with relevant Rules in this chapter.

Implicit Support

4.14.15 An Originator or a Sponsor of a securitisation must not provide Implicit Support to a securitisation transaction with a view to reducing potential or actual losses to investors outside of its contractual obligations;

4.14.16 If an Originator fails to comply with Rule 4.14.15 in respect of a securitisation, it:

- (a) must include all the underlying Exposures of the securitisation in its calculation of Credit RWAs as if those Exposures had not been Securitised and were on the balance sheet of the Authorised Firm;
- (b) must not recognise any gain-on-sale of assets to the securitisation; and,
- (c) must disclose to investors that the Authorised Firm has provided non contractual support and the regulatory capital impact of doing so.

Requirements in order for a Traditional Securitisation to be excluded from the calculation of RWA

- 4.14.17** (1) An Authorised Firm which is an Originator or a Sponsor of a Traditional Securitisation may exclude Securitised Exposures from the calculation of Credit RWA amounts only if all of the conditions detailed in Rule A4.10.1 have been complied with.
- (2) An Authorised Firm meeting the requirements specified in Rule A4.10.1 must hold regulatory capital against any securitisation Exposures it retains.

Requirements in order for a Synthetic Securitisation to be excluded from the calculation of RWA

- 4.14.18** (1) An Authorised Firm which is an Originator or a Sponsor of a Synthetic Securitisation may recognise the effects of Credit Risk mitigation of the Synthetic Securitisation in calculating its SE Exposure RWAs, only if:
- (a) all of the conditions detailed in Rule A4.10.2 have been complied with;
 - (b) the effects of Credit Risk mitigation are obtained through eligible credit protection, eligible financial Collateral or both; and
 - (c) Credit Risk is transferred to third parties.
- (2) In relation to (b), the Credit Risk mitigation techniques used must meet the requirements of section 4.13.

Guidance

In relation to (1)(c) the transferor is deemed to have effective control over the transferred Credit Risk Exposures if it has the ability to repurchase the assets, or is obliged to retain the risk of the transferred assets. This does not include the retention of servicing rights.

- 4.14.19** (1) An Authorised Firm meeting the conditions in Rule 4.14.18 must still hold regulatory capital against any securitisation Exposures it retains.
- (2) The Authorised Firm may recognise the effects of Credit Risk mitigation of eligible financial Collateral pledged by any SPE, but it may not recognise any SPE which is an Issuer of securitisation Exposures as an eligible protection provider.

Operational requirements for use of external credit assessments

4.14.20 The external credit assessment used for determining the applicable risk weight for a CR Exposure must be determined by taking into account the entire amount of Credit Risk (principal and interest) an Authorised Firm is exposed to.

4.14.21 Credit assessments can only be considered from an ECAI, and must meet the following criteria:

- (a) any credit assessments used for the purposes of risk weighting must be publicly available;
- (b) the external credit rating agencies must have expertise and market acceptance in rating securitisations of the nature being used for risk weighting purposes;
- (c) Authorised Firms must apply external credit rating agency ratings consistently to all tranches of securitisations;
- (d) where an Exposure has two ratings from external credit rating agencies the less favourable rating must be used; and
- (e) where an Exposure has more than two assessments by external credit rating agencies the two most favourable ratings can be selected, the review of these assessments is then determined in line with (d).

4.14.22 Where any Credit Risk mitigation has been considered as part of any rating applied to a tranche of a securitisation, the risk weighting should be used and no additional capital recognition is permitted.

4.14.23 An Authorised Firm must treat any securitisation Exposure as an unrated Exposure where:

- (a) the external credit assessment incorporates the credit protection provided directly to the SPE by a protection provider which is not an eligible protection provider;
- (b) the external credit assessment is at least partly based on unfunded support provided by the Authorised Firm itself (e.g. if an Authorised Firm buys ABCP) where it provides an unfunded securitisation Exposure extended to the ABCP Programme, such as a liquidity facility or Credit Enhancement, and that Exposure plays a role in determining the credit assessment on the ABCP, the Authorised Firm must treat the ABCP as if it were not rated and continue to hold capital against the other securitisation Exposures it provides);
- (c) the Credit Risk mitigant is not obtained by the SPE but is separately obtained and applied to a specific securitisation Exposure (e.g. a particular tranche); or
- (d) the Credit Risk mitigation does not meet the eligibility criteria for mitigation specified in section 4.13.

4.14.24 Where Credit Risk mitigation is applied to a specific Exposure within a securitisation the Authorised Firm must treat the Exposure as unrated, and then use the mitigation as set out in section 4.13 should the Rules contained in that section apply.

4.14.25 An Authorised Firm must not use an external credit rating agency rating for risk weighting purposes where the assessment is at least partly based on unfunded support provided by the Authorised Firm itself.

4.14.26 The treatment outlined in Rule 4.14.24 also applies to Exposures in the Authorised Firm's Trading Book. An Authorised Firm's Capital Requirement for such Exposures held in the Trading Book can be no less than the amount required under the Non-Trading Book.

Calculation of RWA amounts for securitisation Exposures

4.14.27 (1) In order to calculate the RWA amount for a securitisation position, the relevant risk weight must be assigned to the Exposure value of the position in accordance with this section, based on the credit quality of the position.

(2) For the purposes of this Rule, the credit quality of a position must be determined by reference to the applicable credit quality assessment from a recognised external credit rating agency.

4.14.28 In cases where there are Exposures to different tranches in a securitisation, the Exposure to each tranche must be considered a separate securitisation position.

4.14.29 The Exposure value of an off-balance sheet securitisation position must, subject to A4.2.2, be its nominal value multiplied by a CCF of 100%, wherever applicable.

4.14.30 The Exposure value of a securitisation position arising from a financial derivative must be determined in accordance with Rules 4.6.14 to 4.6.21 dealing with treatment of financial derivatives.

Assigning risk weights

4.14.31 An Authorised Firm must assign a risk weight for any SE Exposure in accordance with the tables below, to calculate the Credit RWA amounts for that Exposure.

Risk Weights for Long-Term securitisation Exposures

Long Term rating category					
Credit Quality Grade	1	2	3	4	5 and above including unrated
Risk Weight to be applied to securitisation Exposures(excluding Re-securitisation Exposures)	20%	50%	100%	350%	1000% or Deduction from Capital Resources
Risk weight applied to Re-securitisation Exposures	40%	100%	225%	650%	1000% or Deduction from Capital Resources

Risk Weights for Short-Term securitisation Exposures

Short- term rating category				
Credit Quality Grade	I	II	III	IV and above including unrated
Risk Weight to be applied	20%	50%	100%	1000% or Deduction from Capital Resources
Risk Weight applied to Re-securitisation Exposures	40%	100%	225%	1000% or deduction from Capital Resources

4.14.32 (1) In respect of securitisation positions which are assigned a 1000% risk weighting pursuant to the tables in Rule 4.14.31, an Authorised Firm may as an alternative to including the position in its calculation of Credit RWA amounts, deduct from its CET1 Capital the Exposure value of such positions.

(2) For the purposes of this Rule, the calculation of the Exposure value may reflect eligible funded credit protection consistent with applicable Rules in this chapter.

4.14.33 For an Authorised Firm that is an Originator or Sponsor of a securitisation, the Credit RWA amounts calculated for its securitisation positions may be limited to the RWA amounts which would be calculated for the SE Exposures had they not been Securitised subject to the presumed application of a 150% risk weight to all past due items and items belonging to regulatory high risk categories.

4.14.34 [Not currently in use]

4.14.35 [Not currently in use]

Exceptions to deduction of unrated securitisation Exposures

4.14.36 In accordance with the tables under Rule 4.14.31, all unrated securitisation positions must be deducted or risk weighted at 1000% with the following exceptions:

- (a) most senior Exposure in a securitisation;
- (b) Exposures that are in a second loss position or better of an ABCP and meet the requirements of Rule 4.14.41; and
- (c) eligible liquidity positions.

Most senior Exposure in a securitisation

4.14.37 (1) Where an Authorised Firm holds or guarantees unrated securitisation Exposure from the most senior tranche in a securitisation, the Authorised Firm may apply the “look through” treatment, provided the composition of the underlying pool of Exposures securitised is known at all times and the Authorised Firm is able to determine the applicable risk weights for the underlying Exposures.

(2) An Authorised Firm applying the look-through treatment to an unrated securitisation Exposure, pursuant to (1), must apply to that securitisation Exposure the weighted average of the risk weights of the underlying Exposures determined in accordance with the Rules in this chapter, multiplied by a concentration factor.

(3) For the purposes of (2), the concentration factor is calculated as the sum of the nominal amounts of all the tranches in that securitisation divided by the sum of the nominal amounts of the tranches junior to, or pari-passu with, the tranche in which the position is held, including that tranche itself. The resulting weighted average risk weight must not be higher than 1000% or lower than the risk weight applicable to a more senior tranche which is rated.

(4) Where the Authorised Firm is unable to determine the risk weights for the underlying Exposures in accordance with this Rule, the unrated securitisation position will not be eligible for the relief and must be deducted from CET 1 Capital of the Authorised Firm.

4.14.38 An Authorised Firm wishing to apply the treatment referred to in Rule 4.14.37 must notify the DFSA, in writing, at least 30 days in advance, of the intention to adopt this treatment. The notification should include the treatments being adopted and the weightings applied under the provision.

4.14.39 The resulting weighted average risk weight must not be higher than 1000% or lower than the risk weight applicable to a more senior tranche which is rated.

4.14.40 An Authorised Firm must have systems and controls in place to monitor effectively the composition of Exposures where the look-through provision has been applied on an ongoing basis.

Exposures that are in a second loss position or better of an ABCP

4.14.41 An Authorised Firm may apply a risk weight of 100% or the highest risk weight assigned to any of the underlying Exposures in the ABCP Programme, whichever is higher, to an unrated securitisation Exposure arising from the ABCP Programme, provided the securitisation position complies with the following conditions:

- (a) the subject securitisation Exposure must be in a tranche which is economically in a second loss position or better and the First Loss Position must provide meaningful credit protection to the second loss tranche;
- (b) the associated Credit Risk of the securitisation Exposure is the equivalent of a Credit Quality Grade of III or better in the short-term rating category; and
- (c) the Authorised Firm must not hold a position in the First Loss Position.

Eligible liquidity positions

4.14.42 An Authorised Firm providing an unrated eligible liquidity facility may assign to the resulting securitisation Exposure the highest risk weight that would be applied to any of the underlying Exposures covered by the facility.

4.14.43 (1) An off balance sheet SE Exposure will receive a 100% CCF unless:

- (a) the Exposure qualifies as an eligible liquidity facility, or
 - (b) the Exposure is an eligible Servicer cash advance facility.
- (2) In relation to (1), an eligible Servicer cash advance facility is a facility provided to a securitisation in order to ensure uninterrupted flow of payments to investors. As long as the Servicer is entitled to full reimbursement and this right is senior to all other claims on cash flows from the underlying pool of Exposures, and where these facilities meet the requirements of 4.14.44 and are unconditionally cancellable at any time, any undrawn commitments can then have a 0% CCF applied.

4.14.44 (1) For the purposes of Rule 4.14.42, an Authorised Firm may treat an Exposure as an eligible liquidity facility provided the following requirements are met:

- (a) the liquidity facility documentation must clearly identify and limit the circumstances under which it may be drawn;
- (b) draws must be limited to the amount that is likely to be repaid from the liquidation of the underlying Exposures and any seller provided Credit Enhancements;
- (c) the facility must not provide credit support by covering for any losses incurred in the underlying pool of Exposures prior to drawdown;
- (d) the facility must not be structured to provide regular or permanent funding;
- (e) the facility must be subject to an asset quality test to preclude it being used to cover Credit Risk Exposures that are in default;

- (f) where the facility is used to fund externally rated Securities the facility can only be used to fund Securities that are externally rated Investment Grade at the time of funding;
 - (g) the facility cannot be drawn after all Credit Enhancements from which the liquidity facility would benefit have been exhausted; and
 - (h) repayment of draws of the facility cannot be subordinated to any interests of any note holder in the programme or be subject to deferral or waiver.
- (2) Where the Exposure meets the requirements as set out in (1), the following CCF will apply:
- (a) 50% to the eligible liquidity facility regardless of maturity; and
 - (b) 100% if an external rating of the liquidity facility is used for the risk weighting.

- 4.14.45** (1) An Authorised Firm which provides credit protection for a basket of reference Exposures through an unrated first-to-default Credit Derivative may apply to the securitisation Exposure the aggregate of the risk weights that would be assigned to the reference Exposures, provided that the resulting Capital Requirement does not exceed the notional amount of the credit protection.
- (2) An Authorised Firm which provides credit protection for a basket of reference Exposures through an unrated second-to-default Credit Derivative may apply the treatment referred to in (1), except that in aggregating the risk weights, the reference Exposure with the lowest risk-weighted amount may be excluded.

Overlapping Exposures

- 4.14.46** (1) Where an Authorised Firm has two or more overlapping Exposures to a securitisation, the firm must, to the extent that the positions overlap, include in its calculation of Credit RWA amounts only the Exposure, or portion of the Exposure, producing the higher Credit RWA amounts.
- (2) For the purposes of (1), overlapping Exposures result where an Authorised Firm provides two or more facilities (whether they are liquidity facilities or Credit Enhancements) in relation to a securitisation that can be drawn under various conditions with different triggers, with the result that the Authorised Firm provides duplicate coverage to the underlying Exposures. The facilities provided by the Authorised Firm may overlap since a draw on one facility may preclude (in part) a draw on the other facility.
- (3) Where the overlapping Exposures are subject to different conversion factors the Authorised Firm must apply the higher of the conversion factors to the Exposure.

Guidance

The firm may also recognise such an overlap between capital charges for Specific Risk in relation to positions in the trading book and capital charges for positions in the Non-Trading Book, provided that the firm is able to calculate and compare the capital charges for the relevant positions.

However, if overlapping facilities are provided by different Authorised Firms, each Authorised Firm must calculate Capital Requirement for the maximum amount of its Exposure.

Credit Risk mitigation

- 4.14.47** Where an Authorised Firm obtains credit protection on a securitisation Exposure, the calculation of Credit RWA amounts must be in accordance with the Rules in Credit Risk mitigation in section 4.13.
- 4.14.48** Where an Authorised Firm provides credit protection to a securitisation Exposure it must calculate a Capital Requirement as if it were an investor in the securitisation in line with section 4.13.
- 4.14.49** An Authorised Firm must not recognise any SPE which is an Issuer of securitisation Exposures, as an eligible credit protection provider. Guarantees provided must meet the requirements of section 4.13.
- 4.14.50** For the purpose of setting regulatory capital against a maturity mismatch, the Capital Requirement must be determined in accordance with section 4.13. When Exposures being hedged have different maturities, the longest maturity must be used. Maturity of credit protection must be calculated in accordance with section 4.13.

Capital Requirements for securitisations with Early Amortisation provisions

- 4.14.51** An Authorised Firm which is the Originator or Sponsor of a securitisation involving revolving Exposures as well as an Early Amortisation provision, must calculate an additional RWA amount in accordance with Rule 4.14.57 to address the possibility that its Credit Risk Exposure levels may increase following the operation of the Early Amortisation provision.

Guidance

1. This section sets out the methodology for calculation of the Credit RWA amount by an Originator, when it sells revolving Exposures into a securitisation that contains an Early Amortisation provision.
 2. Early Amortisation of the Securities describes the process whereby the repayment of the investors' interest is brought forward upon the occurrence of specified events. Events that are economic in nature by reference to the financial performance of the transferred assets are known as economic triggers.
- 4.14.52** (1) An Authorised Firm which is the Originator or Sponsor of a securitisation involving revolving Exposures, must calculate Credit RWA amounts in respect of the total Exposure related to a securitisation (both drawn and undrawn balances) when:
- (a) the Authorised Firm sells Exposures into a structure that contains an Early Amortisation feature; and
 - (b) the Exposures are of a revolving nature.
- (2) Where the underlying pool of a securitisation comprises revolving and term Exposures, an Authorised Firm must apply the amortisation treatment outlined below for determining applicable regulatory capital only to that portion of the underlying pool containing revolving Exposures.

4.14.53 An Authorised Firm which is the Originator of a Revolving Securitisation that includes economic triggers for Early Amortisation may regard the Exposures as transferred for the period up to the point of repayment, provided that:

- (a) during the amortisation period there is full sharing of interest, principal, expenses, losses and recoveries; and
- (b) the Authorised Firm's risk management system provides warning indicators when economic or non-economic triggers may be activated.

Guidance

Examples of such triggers include tax events, legal changes resulting in an Authorised Firm's non-performance in its role as a servicing agent, and triggers relating to the insolvency of the Originator.

4.14.54 An Authorised Firm is not required to calculate a Capital Requirement for Early Amortisation in the following situations:

- (a) replenishment structures where the underlying Exposures do not revolve and the Early Amortisation ends the ability of the Authorised Firm to add new Exposures;
- (b) where the risk associated with revolving assets containing amortisation features that mimic term structures, where the risk does not return to the Authorised Firm;
- (c) structures where the Authorised Firm securitises one or more credit lines and where investors remain fully exposed to future draws by borrowers so that the risk on the underlying facilities does not return to the Originator even after an Early Amortisation event has occurred; or
- (d) where the Early Amortisation clause is solely triggered by events not related to the performance of the Securitised assets or the Authorised Firm, such as material changes in tax laws or regulations.

4.14.55 For an Authorised Firm subject to the Capital Requirement referred to in Rule 4.14.51, the maximum Credit RWA calculated under that Rule must not exceed the greater of the following:

- (a) the RWA amounts calculated in respect of its positions in the investors' interest; or
- (b) the RWA amounts that would be calculated in respect of the Securitised Exposures, if those had not been securitised.

4.14.56 An Authorised Firm must deduct from its CET1 Capital any gain-on-sale and Credit-Enhancing Interest-Only Strips arising from any securitisation subject to the provisions of the Rules above.

Calculation of Credit RWA amounts for securitisation positions subject to Early Amortisation clause

4.14.57 In regard to securitisation positions subject to an Early Amortisation clause, the Credit RWA amounts for an Authorised Firm acting as the Originator are calculated as the product of the following:

- (a) the investors' interest
- (b) the appropriate CCF (in accordance with the table in Rule 4.14.61); and
- (c) the appropriate risk weight for the underlying Exposure type.

Guidance

In relation to Rule 4.14.57(c), the Authorised Firm should also consider whether a line, or facility, is committed or uncommitted. A line is considered to be uncommitted if it is unconditionally cancellable without prior notice by the Authorised Firm. They also differ according to whether the Securitised Exposures are committed retail credit lines or credit lines (such as revolving credit facilities).

- 4.14.58** (1) An Early Amortisation provision that does not satisfy the conditions for a Controlled Early Amortisation provision will be treated as a non-Controlled Early Amortisation provision.
- (2) For the purpose of (1), the conditions for a Controlled Early Amortisation provision are as follows:
- (a) the Authorised Firm must have an appropriate capital/liquidity plan in place to ensure that it has sufficient capital and liquidity available in the event of an Early Amortisation;
 - (b) throughout the duration of the transaction, including the amortisation period, there is the same pro rata sharing of interest, principal, expenses, losses and recoveries based on the firm's and investors' relative shares of the receivables outstanding at the beginning of each month;
 - (c) the firm must set a period for amortisation that would be sufficient for at least 90% of the total debt outstanding at the beginning of the Early Amortisation period to have been repaid or recognised as in default; and
 - (d) the pace of repayment should not be any more rapid than would be allowed by straight-line amortisation over the period set out in (c).
- 4.14.59** For uncommitted retail credit lines in securitisations containing Controlled Early Amortisation which is triggered by the Excess Spread level falling to a specified level, an Authorised Firm must compare the three month average Excess Spread level with the Excess Spread levels at which the Excess Spread is required to be trapped.
- 4.14.60** Where the securitisation does not require Excess Spread to be trapped, the trapping point is deemed to be 4.5 percentage points greater than the Excess Spread level at which Early Amortisation is triggered.
- 4.14.61** An Authorised Firm must divide the Excess Spread level by the transaction's Excess Spread trapping point to determine the appropriate segments and apply corresponding conversion factors as set out in the following table:

Controlled Early Amortisation features		
	Uncommitted	Committed
Retail Credit Lines	3 Month average Excess Spread CCF	90%
133.33% of trapping point or more	0%	
<133.33% to 100% of trapping point	1%	
<100% to 75% of trapping point	2%	
<75% to 50% trapping point	10%	
<50% to 25% of trapping point	20%	
<25%	40%	
Non-retail credit lines	90%	90%

Non-Controlled Early Amortisation

4.14.62 In regard to non-Controlled Early Amortisation, an Authorised Firm must apply the same steps as set out at Rules 4.14.59 to 4.14.61 and determine appropriate segments and apply the corresponding conversion factors as set out in the following table:

Non-Controlled Early Amortisation features		
	Uncommitted	Committed
Retail Credit Lines	3 Month average Excess Spread CCF	100%
133.33% of trapping point or more	0%	
<133.33% to 100% of trapping point	5%	
<100% to 75% of trapping point	15%	
<75% to 50% trapping point	50%	
<50% of trapping point	100%	
Non-retail credit lines	100%	100%

Transfers to Special Purpose Entities (SPEs)

4.14.63 An Authorised Firm need not include in its calculation of Capital Resources or Credit RWA amounts, assets transferred to:

- (a) an SPE; or
- (b) any Person, if the transfer is in connection with a securitisation under which the Issuer of the Securities is an SPE;

provided that:

- (c) the Authorised Firm does not own any share or proprietary interest in the SPE;
- (d) no more than one member of the Governing Body of the SPE is an officer, partner, or Employee of the Authorised Firm;
- (e) the SPE does not have a name that implies any connection with the Authorised Firm or any other member of the Authorised Firm's Group;
- (f) the Authorised Firm does not fund the SPE except where permitted under the requirements for Credit Enhancement below;
- (g) the Authorised Firm does not provide temporary finance to the SPE to cover cash shortfalls arising from delayed payments or non-performance of loans transferred except where it meets the requirements for liquidity support below;
- (h) the Authorised Firm does not bear any of the recurring expenses of the SPE; and
- (i) any agreements between the Authorised Firm and the SPE are at market rates and at arm's length.

4.14.64 Where an Originator acts as Underwriter for the Securities issued, the underlying items will not be regarded as being transferred until 90% of the total issuance has been sold to third parties.

Dealing

4.14.65 An Originator dealing in Securities which would attract a Credit Quality Grade of 4 or better and issued by an SPE must deduct any holdings in such Securities from its CET1 Capital unless the holding is subject to:

- (a) an ongoing limit of 3% of the Securities issued; and
- (b) a limit of 10% of the Securities issued for a period of five business days:
 - (i) immediately following close of the transaction; or
 - (ii) in the case of Revolving Securitisations only, at the beginning of the scheduled amortisation period.

4.14.66 An Authorised Firm acting as the Originator and holding in excess of the dealing limits in Rule 4.14.65 must either:

- (a) where the holding is less than 10%, deduct from its CET1 Capital the excess over the dealing limit; or
- (b) where the holding is greater than 10%, regard the transferred risks associated with the items as being back on its balance sheet.

4.14.67 An Authorised Firm acting as the Originator must not deal in the Securities during the amortisation period.

4.14.68 An Authorised Firm acting as the Sponsor dealing in the Securities issued by the SPE must include these Securities in the calculation of its Credit RWAs.

4.14.69 An Authorised Firm involved in Synthetic Securitisations must seek individual guidance on a case-by-case basis from the DFSA regarding the regulatory treatment of such transactions.

Recognition of eligible financial Collateral under FCSA Approach

4.14.70 An Authorised Firm which has taken eligible financial Collateral for an SE Exposure and is using the FCSA may recognise the effect of the eligible financial Collateral as follows:

- (a) break down the SE Exposure into:
 - (i) a collateralised portion with E equal to the latest fair market value of the eligible financial Collateral; and
 - (ii) an uncollateralised portion whose Exposure value equals the E of the SE Exposure less the latest fair market value of the eligible financial Collateral; and
- (b) apply the CRW that is applicable to the eligible financial Collateral to the collateralised portion calculated in accordance with (a)(i) to calculate the Credit RWA amount of the collateralised portion as though the Authorised Firm had a direct Exposure to the eligible financial Collateral; and
- (c) either:
 - (i) apply the CRW that is applicable to the SE Exposure to the uncollateralised portion calculated in accordance with (a)(ii) to calculate the Credit RWA amount of the uncollateralised portion; or
 - (ii) include the uncollateralised portion as a deduction from CET1 Capital.

Guidance

Collateral in the context of a SE Exposure refers to assets used to hedge the Credit Risk of a securitisation Exposure rather than the underlying Exposures of the securitisation, including Collateral pledged by an SPE.

4.14A Counterparty Credit Risk

4.14A.1 This section applies to Exposures in the Trading Book and the Non-Trading Book that give rise to Counterparty Credit Risk and that are not cleared through a CCP, including the following transactions:

- (a) OTC derivatives;
- (b) exchange-traded derivatives; and
- (c) long settlement transactions;

but excluding:

- (d) SFTs.

Guidance

1. Transactions that give rise to Counterparty Credit Risk exhibit the following abstract characteristics. The transactions:
 - a. generate a current exposure or market value;
 - b. have an associated random future market value based on market variables;
 - c. generate an exchange of payments or an exchange of a financial instrument (including commodities) against payment; and
 - d. are undertaken with an identified counterparty against which a unique probability of default can be determined.
2. Other common characteristics of the transactions in Rule 4.14A.1 include the following:
 - a. Collateral may be used to mitigate risk exposure and is inherent in the nature of some transactions;
 - b. short-term financing may be a primary objective in that the transactions mostly consist of an exchange of one asset for another (cash or securities) for a relatively short period of time, usually for the business purpose of financing; the two sides of the transactions are not the result of separate decisions but form an indivisible whole to accomplish a defined objective;
 - c. netting may be used to mitigate the risk;
 - d. positions are frequently valued (most commonly on a daily basis), according to market variables; and
 - e. remargining may be employed.

Calculating the CCR Exposure

4.14A.2 (1) An Authorised Firm must calculate the Exposure value for the transaction types referred to in Rule 4.14A.1 by applying the methodology set out in section A4.8 in accordance with this Rule.

- (2) If an Authorised Firm purchases protection through a Credit Derivative against a Non-Trading Book Exposure or against a CCR Exposure it may:
 - (a) calculate its Credit Risk Capital Requirement for the hedged Exposure in accordance with sections 4.12 and 4.13; and
 - (b) set the Exposure value for the Credit Derivative to zero.
- (3) An Authorised Firm may set the Exposure value for a sold credit default swap in the Non-Trading Book to zero provided that the credit default swap is:
 - (a) treated as credit protection provided by the Authorised Firm; and
 - (b) subject to Credit Risk Requirements for the underlying exposure for the full notional amount.
- (4) Subject to (5), the Exposure value for a given Counterparty is equal to the sum of the Exposure amounts for each Netting Set with that Counterparty.
- (5) The Exposure value for a given OTC Counterparty is the greater of:
 - (a) zero; and
 - (b) the difference between the sum of Exposures:
 - (i) across all Netting Sets with the Counterparty; and
 - (ii) the credit valuation adjustment for that Counterparty that has already been recognised by the Authorised Firm as an incurred write-down without taking into account any offsetting debit value adjustments which has already been excluded from CET1 Capital under Rule 3.13.5.

4.15 Concentration Risk

Applicability and limits

- 4.15.1** This section applies with respect to Trading Book transactions as calculated in App2 and Non-Trading Book transactions as calculated in section 4.8.
- 4.15.2** For the purposes of this section an Exposure that arises in the Trading Book is calculated by summing the following:
 - (a) the net positive position (long positions net of short positions) for each financial instrument as set out in Rules A4.11.10 to A4.11.28;
 - (b) the Authorised Firm's net Underwriting Exposures for any Counterparty; and
 - (c) any other Exposures arising from transactions, agreements and contracts that would give rise to Counterparty Credit Risk.
- 4.15.3** For the purposes of this section an Authorised Firm must:

- (a) identify its Exposures;
- (b) identify its Counterparties, including whether any are Closely Related to each other or Connected to the Authorised Firm;
- (c) measure the size of its Exposures;
- (d) establish the value of its Exposures;
- (e) determine the size of its Exposures as a proportion of its Tier 1 Capital;
- (f) identify whether it has Exposures which are subject to the requirements of section 4.13 (Credit Risk mitigation);
- (g) identify which, if any, of its Exposures are exempt in accordance with section A4.11 from the limits set out in Rules 4.15.4 to 4.15.7;
- (h) aggregate its Exposures to the same Counterparty or group of Closely Related Counterparties or group of Connected Counterparties;
- (i) monitor and control its Exposures on a daily basis within the Concentration Risk limits; and
- (j) notify the DFSA immediately of any breach of the limits set out in this section and confirm it in writing.

Large Exposure limits

4.15.4 A Large Exposure of an Authorised Firm means a total Exposure which is equal to or exceeds 10% of the firm's Tier 1 Capital, to any Counterparty, Connected Counterparty, group of Connected Counterparties, or group of Closely Related Counterparties, whether in the Authorised Firm's Trading Book or Non-Trading Book, or both.

- 4.15.5**
- (1) Subject to IFR Rule 5.4.15, an Authorised Firm must ensure that Exposures in its Non-Trading Book and, subject to Rule 4.15.6, Trading Book to a Counterparty or to a group of Closely Related Counterparties or to a group of Connected Counterparties, after taking into account the effect of any eligible Credit Risk mitigations, do not exceed 25% of its Tier 1 Capital, except as otherwise provided in (2) or required by the DFSA under (3).
 - (2) An Authorised Firm's Exposure must not exceed 15% of its Tier 1 Capital if the Authorised Firm is a G-SIB and the Exposure is to another G-SIB, or a subsidiary of a G-SIB, in or outside the DIFC.
 - (3) An Authorised Firm which is a D-SIB must, if required in writing by the DFSA, apply an Exposure limit of between 15% to 25% of its Tier 1 Capital as specified by the DFSA in the requirement, where the Exposure is to another D-SIB, or to a subsidiary of a D-SIB, in or outside the DIFC.

4.15.6 Where an Authorised Firm's Trading Book Exposure to a Counterparty or to a group of Closely Related Counterparties or to a group of Connected Counterparties, on its own or when added to any Non-Trading Book Exposure, is likely to exceed 25% of its Tier 1 Capital, the Authorised Firm must immediately give the DFSA written notice,

explaining the nature of its Trading Book Exposure and seeking specific guidance from the DFSA regarding the prudential treatment of any such Exposure.

4.15.7 Subject to IFR Rule 5.4.16 an Authorised Firm must ensure that the sum of its Large Exposures does not exceed 800% of its Tier 1 Capital.

Guidance

1. Exposures can arise in the Non-Trading Book and in the Trading Book from Credit Risk (for example on loans and advances) Counterparty Risk (for example, on unsettled trades and on Derivative contracts) and from Issuer risk (for example, on holdings of equities and bonds).
2. Some Derivatives contracts may result in an Authorised Firm being exposed to an Issuer as well as the Derivatives Counterparty. For example, a Derivative referenced on a Security may result in an Exposure to the Counterparty, to the transaction and to the Issuer of the underlying Security.
3. Examples of an Exposure are actual or potential claims on a Counterparty including contingent liabilities arising in the normal course of an Authorised Firm's business.
4. App4 includes further Rules and Guidance on:
 - a. fully and partially exempt Exposures, Exposures to undisclosed Counterparties, parental guarantees and capital maintenance agreements;
 - b. identification of Exposures;
 - c. identification of Closely Related and Connected Counterparties, and exemptions for Connected Counterparties;
 - d. measuring Exposures to Counterparties and Issuers in relation to Derivatives, equity indices, and other items; and
 - e. country risk Exposure.

Exclusions from the Large Exposure limits

- 4.15.8** (1) For the purposes of this section, Exposure excludes:
- (a) claims and other assets required to be deducted for the purposes of calculating an Authorised Firm's Tier 1 Capital;
 - (b) a transaction entered into by an Authorised Firm as depository or as agent that does not create any legal liability on the part of the Authorised Firm;
 - (c) claims resulting from foreign exchange transactions where an Authorised Firm has paid its side of the transaction and the countervalue remains unsettled during the 2 business days following the due payment or due delivery date. After 2 business days the claim becomes an Exposure;
 - (d) claims arising as a result of money transmission, payment services, clearing and settlement, correspondent banking or financial instruments clearing, settlement and custody services to clients, delayed receipts in funding and other Exposures arising from client activity which do not last longer than the following business day;

- (e) in the case of the services outlined in (d) intra-day Exposures to Financial Institutions who provide these services are excluded;
 - (f) claims resulting from the purchase and sale of Securities during settlement where both the Authorised Firm and the Counterparty are up to five business days overdue in settling. The five business days include the due payment or due delivery date. After five business days, the claim becomes an Exposure; and
 - (g) Exposures that are guaranteed by the Authorised Firms Parent in accordance with Rule 4.15.18.
- (2) For the purposes of this section, Exposure to a CCP which carry a 0% CCR in accordance with section 4.8 are excluded.

4.15.9 An Authorised Firm need not include fully exempt Exposures, as referred to in Rule A4.11.1 when monitoring compliance with the limits in Rules 4.15.5, 4.15.6 and 4.15.7.

Institutional exemption

- 4.15.10** (1) This Rule applies to an Authorised Firm in Category 2 and 3A.
- (2) For Exposures to a Financial Institution, or a group of Connected Counterparties one of which is a Financial Institution, the total amount of an Authorised Firm's Exposures may exceed 25% of its Tier 1 Capital, provided those institutions are Investment Grade (Credit Quality Grades 1 to 3) and subject to the following:
- (a) Exposures to any entities within the group of Connected Counterparties that are not Financial Institutions are limited to 25% of Tier 1 Capital after taking account of Credit Risk mitigation;
 - (b) the Exposures must not form part of the Tier 1 Capital of the Counterparty;
 - (c) the Counterparty Risk profile must be subject to review on at least an annual basis; and
 - (d) Exposures of this nature must not in any case exceed a maximum of US\$ 100 million or 100% of Tier 1 Capital, whichever is the lower.

Guidance

The DFSA will, in exceptional circumstances, consider an application to waive or modify the limits set out above. In such circumstances the Authorised Firm will have to make a submission to the DFSA as to why its specific circumstances would warrant a relaxation of the limits specified in (d) above.

Systems and Controls

- 4.15.11** (1) An Authorised Firm must implement and maintain systems and controls to identify its Exposures and effectively manage Concentration Risks as a result of its activities.

- (2) Such systems and controls in place must be proportionate to the nature, scale and complexity of the Authorised Firm and must include written policies and procedures to address Concentration Risks, both on and off balance sheet, which:
 - (a) are approved by the Governing Body on at least an annual basis; and
 - (b) include internal approval limits for Exposures as well as limits for the risks associated with specific sectors, geographic location and single economic risk factors.

Guidance

The DFSA expects the systems and controls to include:

- a. processes for the tiered approval of Exposures based on size, risk profile and complexity;
- b. mechanisms for identifying, recording and monitoring all Exposures with particular focus on Large Exposures;
- c. mechanisms in place for the monitoring and control of Exposures to Counterparties and Groups of Connected Counterparties;
- d. mechanisms for monitoring and recording Exposures within its Group;
- e. mechanisms to monitor Counterparties in the same economic sector and exposed to single economic risks;
- f. mechanisms to identify and control risks arising from single geographic jurisdictions; and
- g. mechanisms to identify risks arising from related activities or commodities.

Recognition of Credit Risk mitigations

- 4.15.12** (1) For the purposes of this section, an Authorised Firm may reduce the value of its Exposures, at its discretion, by any one or more of the following:
- (a) the amount of any specific provision made, where the provision relates to the risk of a credit loss occurring on that Exposure and is not held as part of a general provision or reserve against its Credit Risks;
 - (b) Netting its claims on and liabilities to a Counterparty, provided that the conditions in section 4.13 of Credit Risk mitigation are met;
 - (c) the amount of Collateral held against its Exposures, where that Collateral is of a type listed based on the FCSA and FCCA approaches and meeting the requirements under section 4.13;
 - (d) the amount of any eligible guarantees as permitted under section 4.13.9;
 - (e) the value of a Credit Derivative, where the Credit Derivative is an instrument included in Rule 4.13.11 and the transaction meets the conditions set out in that section; and

- (f) the effects of transactions transferring Credit Risks from the Authorised Firm to another party through securitisation, provided that the conditions in section 4.14 are met.
- (2) Where Credit Risk mitigation is used against an Exposure, an Authorised Firm must reduce the value of the original Exposure and recognise an equal Exposure to the Credit Risk mitigation provider, except where:
 - (a) a credit default swap is used; and
 - (b) neither the reference entity, nor the credit default swap provider, is a Financial Institution,
- (3) For the purposes of Exposure shifting under (2), the amount subject to shifting is:
 - (a) the value of the protected portion for an unfunded credit protection;
 - (b) where the FCSA is used, the market value of the collateral; and
 - (c) where the FCCA is used, the market value of the collateral adjusted by applying the standard supervisory haircuts to the FCCA.

4.15.13 An Authorised Firm intending to utilise any of the provisions contained in section 4.13 (Credit Risk mitigation) for the purposes of reducing Exposure values should have in place policies and procedures addressing the following:

- (a) risks arising from maturity mismatches between Exposures and any credit protection on those Exposures;
- (b) the Concentration Risk arising from the application of Credit Risk mitigation techniques, including indirect Large Exposures – for example to a single Issuer of Securities taken as Collateral; and
- (c) the conduct of stress testing on Credit Risk mitigation taken as Collateral.

4.15.14 Where an Authorised Firm has availed itself of the reductions to Exposure values as set out in A4.11 the Authorised Firm must calculate the Exposure as a percentage of its Tier 1 Capital on both a gross and net basis.

4.15.15 An Authorised Firm that avails itself of the reduction in its Exposure value through the application of Rule A4.11 must conduct periodic stress tests on its Exposures against the realisable value of any Collateral considered under with the FCSA or FCCA.

4.15.16 Where the value of the Collateral under the stress scenario is lower than the value applied under Rule 4.15.12 the lower value should be used when determining the Exposure value for the purposes of this section.

Guidance:

Such stress tests should include market value changes of underlying Collateral, risks relating to liquidity and realisation of such Collateral in stress scenarios. An assessment of the impact of any such changes on the Exposure value and the capital position of the Authorised Firm should be conducted. Stress testing of these positions should be conducted at least once a year.

4.15.17 An Authorised Firm must document its policy for the use of any of the exclusions in Rule 4.15.12.

Guidance

Such policy should include risks such as maturity mismatches, stress testing of Collateral values, indirect Exposures arising from Credit Risk mitigation, such as mitigation provided on Exposures by the same Counterparty.

Treatment of Parental Guarantees

4.15.18 An Authorised Firm may exclude an Exposure from the Concentration Risk limits set out in Rules 4.15.5 to 4.15.7 where:

- (a) the Authorised Firm's Parent guarantees that Exposure to a Counterparty or to a group of Closely Related Counterparties; and
- (b) the following conditions are met:
 - (i) the Counterparty or group of Closely Related Counterparties are not Connected to the Authorised Firm;
 - (ii) the guarantee is to be provided by the Authorised Firm's Parent, or regulated member of its Group;
 - (iii) the criteria for guarantees must be in line with the Credit Risk mitigation requirements as set out in section 4.13;
 - (iv) the entity providing the guarantee must be a bank regulated to standards acceptable to the DFSA;
 - (v) the total amount of guarantees provided to the Authorised Firm must be less than 10% of the Parent (or other) Authorised Firm's Tier 1 Capital;
 - (vi) the Parent must be rated as a Credit Quality Grade of 1 or 2 by a recognised credit rating agency;
 - (vii) the Authorised Firm must provide confirmation from the home state Financial Services Regulator that it is satisfied that the Parent Authorised Firm has sufficient resources to provide such guarantees and has no objection to the provision of such guarantees;
 - (viii) the Authorised Firm should provide an annual confirmation that there are no changes to the enforceability of such guarantees; and
 - (ix) the Authorised Firm must notify the DFSA when such guarantees represent 200%, 400% and 600% of Tier 1 Capital and the overall Large Exposure limit must not exceed 800%.

5 MARKET RISK

Introduction

Guidance

1. This chapter addresses the regulatory requirements in respect of managing the Market Risk exposures of an Authorised Firm. Market Risk refers to the risk of incurring losses on positions due to adverse changes in market prices or in underlying value drivers. This chapter aims to ensure that an Authorised Firm engaging in activities exposing the firm to risks associated with potential adverse movements in market prices adopts appropriate and effective risk management practices and holds regulatory capital of the right quality that is also commensurate with the risks involved.
2. This chapter includes requirements that an Authorised Firm:
 - a. implement a comprehensive Market Risk management framework to manage, measure and monitor Market Risk commensurate with the nature, scale and complexity of the firm's operations; and
 - b. calculate the Market Risk Capital Requirement and hold the same.
3. The chapter allows the use of standard pre-defined methodologies for estimating the capital requirement and also allows the use of DFSA-approved internal models to calculate elements of a firm's Market Risk Capital Requirement. The chapter covers Rules for determining Market Risk Capital Requirement on exposures involving interest rate risk, equity risk, foreign exchange risk, commodities risk, options risk, collective investment fund risk, securities underwriting risk and credit valuation adjustment risk.
4. Appendix 5 provides the detailed requirements, parameters, calculation methodologies and formulae in respect of the primary requirements outlined in chapter 5. Appendix 5 also provides detailed guidance on criteria for approval of internal models for calculation of Market Risk Capital Requirement, incorporation of incremental risk charges in internal models, if allowed and guidance on the required level of stress testing.

5.1 Application

5.1.1 This chapter applies to an Authorised Firm in Category 1, 2, 3A or 5 as follows:

- (a) sections 5.2 to 5.11 apply to an Authorised Firm in Category 1 or 2 (other than a Matched Principal);
- (b) sections 5.2 and 5.6 apply to an Authorised Firm in Category 2 (if it meets the definition of Matched Principal) or 3A; and
- (c) sections 5.2, 5.3 and 5.5 to 5.11 apply to an Authorised Firm in Category 5.

Guidance

1. Rule 5.3.1 provides that the Market Risk Capital Requirement of an Authorised Firm is calculated as the sum of a number of subsidiary Capital Requirements. Sections 5.4 to 5.10 set

out the manner in which each of those subsidiary Capital Requirements must be calculated, monitored and controlled by an Authorised Firm.

2. In addition to complying with the applicable Rules in chapter 5, an Authorised Firm investing in or holding Islamic Contracts whether or not for the purpose of a PSIA will need to take account of the provisions under IFR Rules 5.4.8 to 5.4.14 to calculate the Market Risk for those Islamic Contracts.

5.2 Market Risk systems and controls

- 5.2.1**
- (1) An Authorised Firm in Category 1, 2, 3A or 5 must implement and maintain a Market Risk policy which enables it to identify, assess, control and monitor Market Risk.
 - (2) The policy must be documented and include the Authorised Firm's risk appetite and how it identifies, assesses, mitigates, controls and monitors that risk.
 - (3) An Authorised Firm must:
 - (a) ensure that its risk management systems enable it to implement the Market Risk policy;
 - (b) identify, assess, mitigate, control and monitor its Market Risk; and
 - (c) review and update the policy at intervals that are appropriate to the nature, scale and complexity of its activities.

Guidance

Guidance in respect of what an Authorised Firm's Market Risk policy should include is provided in section A5.1.

5.3 Calculation of the Market Risk Capital Requirement

- 5.3.1** An Authorised Firm must calculate its Market Risk Capital Requirement as the sum of the following components:
- (a) Interest Rate Risk Capital Requirement;
 - (b) Equity Risk Capital Requirement;
 - (c) Foreign Exchange Risk Capital Requirement;
 - (d) Commodities Risk Capital Requirement;
 - (e) Option Risk Capital Requirement;
 - (f) Collective Investment Fund Risk Capital Requirement;

- (g) Securities Underwriting Capital Requirement; and
- (h) Credit Valuation Adjustment Risk Capital Requirement.

Guidance

1. Detailed Rules and Guidance in respect of the Market Risk Capital Requirement and each of its components in (a) to (g) are contained in this chapter.
2. Rules and Guidance in respect of calculating Market Risk for Islamic Contracts are contained in IFR chapter 5.

5.4 Interest Rate Risk Capital Requirement

5.4.1 An Authorised Firm in Category 1 or 2 (other than a Matched Principal) must calculate its Interest Rate Risk Capital Requirement in respect of Trading Book transactions:

- (a) by applying its internal Market Risk model which has been approved by the DFSA for this purpose; or
- (b) by applying the Rules set out in section A5.2.

5.5 Equity Risk Capital Requirement

5.5.1 An Authorised Firm in Category 1, 2 (other than a Matched Principal) or 5, must calculate its Equity Risk Capital Requirement in respect of Trading Book transactions:

- (a) by applying its internal Market Risk Model which has been approved by the DFSA for this purpose; or
- (b) by applying the Rules set out in section A5.3.

5.6 Foreign Exchange Risk Capital Requirement

5.6.1 An Authorised Firm in Category 1, 2, 3A or 5 must, subject to Rule 5.6.2, calculate its Foreign Exchange Risk Capital Requirement in respect of Trading Book and Non-Trading Book foreign exchange positions by:

- (a) applying its internal Market Risk model which has been approved by the DFSA for this purpose; or
- (b) applying the Rules in section A5.4.

5.6.2 An Authorised Firm need not calculate a Foreign Exchange Risk Capital Requirement if:

- (a) its Foreign Currency business, defined as the greater of the sum of its gross long positions and the sum of its gross short positions in all Foreign

Currencies, does not exceed 100% of Capital Resources as defined in chapter 3; and

- (b) its overall net open position as defined in Rule A5.4.4 does not exceed 2% of its Capital Resources as defined in chapter 3.

5.7 Commodities Risk Capital Requirement

5.7.1 An Authorised Firm in Category 1, 2 (other than a Matched Principal) or 5 must calculate its Commodities Risk Capital Requirement in respect of Trading Book and Non-Trading Book commodity positions by:

- (a) applying its internal Market Risk model which has been approved by the DFSA for this purpose; or
- (b) applying the Rules set out in section A5.5.

5.8 Option Risk Capital Requirement

5.8.1 An Authorised Firm in Category 1, 2 (other than a Matched Principal) or 5 must calculate an Option Risk Capital Requirement if it has positions in Options in its Trading Book by:

- (a) applying its internally developed Market Risk model which has been approved by the DFSA for this purpose; or
- (b) by applying the Rules set out in section A5.6.

5.9 Collective Investment Fund Risk Capital Requirement

5.9.1 An Authorised Firm in Category 1, 2 (other than a Matched Principal) or 5 must calculate its Collective Investment Fund Risk Capital Requirement in respect of Trading Book positions in Units in a Collective Investment Fund by:

- (a) applying its internally developed Market Risk model which has been approved by the DFSA for this purpose; or
- (b) applying the Rules set out in section A5.7.

5.10 Securities Underwriting Capital Requirement

5.10.1 This section applies to an Authorised Firm in Category 1, 2 (other than a Matched Principal) or 5 in respect of Trading Book Securities Underwriting positions.

Guidance

1. This section sets out a framework for calculating the amount of Capital Requirement when an Authorised Firm has commitments to underwrite an issue of Securities, and the associated risk management standards which an Authorised Firm Underwriting Securities must meet.
2. Underwriting is defined in the Glossary as an arrangement under which a party agrees to buy, before issue, a specified quantity of Securities in an issue of Securities on a given date and at a given price, if no other party has purchased or acquired them.

5.10.2 An Authorised Firm must establish and maintain such systems and controls to monitor and manage its Underwriting and sub-underwriting business as are appropriate to the nature, scale and complexity of its Underwriting and sub-underwriting business.

Guidance

1. An Authorised Firm should take reasonable steps to:
 - a. allocate responsibility for the management of its Underwriting and sub-underwriting business;
 - b. allocate adequate resources of the Authorised Firm to monitor and control its Underwriting and sub- underwriting business;
 - c. satisfy itself that its systems to monitor its Exposure to a Counterparty will calculate, revise and update its Underwriting Exposure to each Counterparty and its Capital Requirements;
 - d. satisfy itself of the suitability of each person who performs functions for it in connection with the Authorised Firm's Underwriting business, having regard to the person's skill and experience; and
 - e. satisfy itself that its procedures and controls to monitor and manage its Underwriting business address the capacity of sub-underwriters to meet sub-underwriting commitments.

- 5.10.3** (1) An Authorised Firm must calculate a Securities Underwriting Capital Requirement if it has a commitment to underwrite or sub-underwrite an issue of Securities.
- (2) An Authorised Firm has a commitment to underwrite or sub-underwrite an issue of Securities where:
- (a) it gives a commitment to an Issuer of Securities to underwrite an issue of Securities;
 - (b) it gives a commitment to sub-underwrite an issue of Securities; or
 - (c) it is a member of a syndicate or Group that gives a commitment to an Issuer to underwrite an issue of Securities or a commitment to sub-underwrite an issue of Securities.

5.10.4 An Authorised Firm must regard a commitment to underwrite an issue of Securities, subject to any right set out in Rule 5.10.6, as the initial commitment to underwrite from the earlier of:

- (a) the time the Authorised Firm signs an agreement with the Issuer of Securities to underwrite those Securities; or
- (b) the time the price and allocation of the issue are set.

5.10.5 Where the issue price has not been fixed, an Authorised Firm must use the highest estimate of the price and its allocation for the purpose of calculating its initial gross commitment.

5.10.6 If an Authorised Firm has at its discretion an irrevocable right to withdraw from an Underwriting commitment, exercisable within a certain period, the commitment commences when that right expires.

5.10.7 An Authorised Firm must calculate its Securities Underwriting Risk Capital Requirement by:

- (a) applying its internally developed Market Risk model which has been approved by the DFSA for this purpose; or
- (b) applying the Rules in section A5.8.

5.10A Credit Valuation Adjustment Risk Capital Requirement

5.10A.1 An Authorised Firm in Category 1, 2 (other than a Matched Principal) or 5 must calculate its Credit Valuation Adjustment Risk Capital Requirement in respect of Trading Book and Non-Trading Book transactions by applying the Rules set out in section A5.10.

5.11 Use of internal Market Risk models

5.11.1 An Authorised Firm in Category 1, 2 (other than a Matched Principal) or 5 may use an internal model to calculate its Market Risk Capital Requirement or any components of its Market Risk Capital Requirement, if its internal model and its use have been approved in writing by the DFSA.

Guidance

1. Guidance in respect of criteria for use of internally developed Market Risk models is provided in section A5.9.
2. Internal models cannot be used to calculate an Authorised Firm's Credit Valuation Adjustment Risk Capital Requirement.

5.11.2 If the DFSA approves the use of an internal model, it may:

- (a) impose, withdraw or amend at any time conditions in respect of the use of the internal model; and
- (b) withdraw approval if it forms the view that the internal model or its use is no longer suitable for the calculation of the Authorised Firm's Market Risk Capital Requirement or any component of it.

5.11.3 An Authorised Firm which uses an internal model in accordance with Rule 5.11.1 must have in place a rigorous and comprehensive stress-testing programme which meets the criteria set out in Rule A5.9.4.

5.11.4 An Authorised Firm that has received approval for the use of an internal model may only revert to calculating its Market Risk Capital Requirement or any component of it in accordance with App5 with the prior written consent of the DFSA.

Guidance

1. This section sets out the conditions under which an Authorised Firm is permitted to use an internal model to calculate its Market Risk Capital Requirement or any component of its Market Risk Capital Requirement other than its Credit Valuation Adjustment Risk Capital Requirement. An Authorised Firm that wishes to use an internal model to calculate any part of this requirement is required to apply to the DFSA. Internal models will commonly permit more extensive Netting of long and short positions and have greater risk sensitivity.
2. In assessing whether to give approval, the DFSA will consider an Authorised Firm's risk management standards; the quantitative model standards; the stress-testing and back-testing standards and the process surrounding the calculation of the appropriate regulatory Capital Requirement.
3. The DFSA will usually only give its approval for the use of an internal risk model if:
 - a. the use of the model to calculate the Market Risk Capital Requirement has been approved by another appropriate regulator or the DFSA is satisfied having been provided by the Authorised Firm with such opinions from independent experts as it may require, that the model adequately addresses Market Risk requirements;
 - b. use of the methodology is integrated into the governance and control framework of the Authorised Firm. Specifically, the Governing Body and senior management of the Authorised Firm receives and reviews appropriate reports in respect of the entity;
 - c. it is satisfied that the Authorised Firm's risk management system is conceptually sound and is implemented with integrity;
 - d. the Authorised Firm has sufficient numbers of staff skilled in the use of sophisticated models not only in the trading area but also in the risk control, audit, and if necessary, back office areas;
 - e. the Authorised Firm's models have a proven track record of reasonable accuracy in measuring risk; and
 - f. the Authorised Firm regularly conducts stress tests.
4. In determining whether an internal value at risk (VaR) model meets the standard for approval, the DFSA will apply the criteria set out in section A5.9, which are based on the Basel Market Risk Capital Amendment 1996 and Basel Revisions to the Basel II Market Risk framework 2009 and which can be grouped under the following headings:

- a. qualitative standards;
 - b. specification of Market Risk factors;
 - c. quantitative standards;
 - d. adjustments to Market Risk Capital Requirements;
 - e. stress testing; and
 - f. combination of internally developed models and the Standardised Methodology.
5. In addition to value-at-risk models, the DFSA recognises option risk aggregation models and interest rate ‘pre-processing’ or sensitivity models, as set out under the EU’s Capital Adequacy Directive (these are the so-called ‘CAD1 models’).
 6. Option risk aggregation models analyse and aggregate options risks for interest rate, equity, foreign exchange and commodity options.
 7. Interest rate pre-processing models are used to calculate weighted positions for inclusion in an Authorised Firm’s interest rate Market Risk Capital Requirement calculation under the Duration Method.

6 OPERATIONAL RISK

Introduction

Guidance

1. This chapter includes the detailed Rules and associated guidance in respect of a firm's obligation to manage effectively its exposures to Operational Risk. Operational Risk refers to the risk of incurring losses due to the failure of systems, processes, and personnel to perform expected tasks. Operational Risk losses also include losses arising out of legal risk. This chapter aims to ensure that an Authorised Firm has a robust Operational Risk management framework commensurate with the nature, scale and complexity of its operations and that it holds sufficient regulatory capital against Operational Risk exposures.
2. This chapter requires an Authorised Firm to:
 - a. design and implement an effective Operational Risk management system complete with appropriate systems and controls;
 - b. calculate the Operational Risk Capital Requirement and hold the same; and
 - c. hold adequate professional indemnity insurance cover.
3. This chapter includes, among others, specific Operational Risk management requirements relating to IT systems, information security, outsourcing, business continuity and disaster recovery and the management of Operational Risks in trading rooms.
4. Appendix 6 provides the detailed requirements, parameters, calculation methodologies and formulae for calculating the Operational Risk Capital Requirement specified in chapter 6.

6.1 Application

6.1.1 This chapter applies to an Authorised Firm as follows:

- (a) sections 6.1 to 6.9 apply to an Authorised Firm in any Category;
- (b) sections 6.10 and 6.11 apply only to an Authorised Firm in Category 1, 2, 3A or 5;
- (c) section 6.12 applies only to an Authorised Firm in Category 3B, 3C, 3D or 4 which undertakes one or more of the following Financial Services:
 - (i) Arranging Deals in Investments;
 - (ii) Managing Assets;
 - (iii) Advising on Financial Products;
 - (iv) Managing a Collective Investment Fund;

- (v) Providing Custody;
 - (vi) Insurance Intermediation;
 - (vii) Insurance Management;
 - (viii) Managing a Profit Sharing Investment Account (unrestricted);
 - (ix) Providing Trust Services;
 - (x) Providing Fund Administration;
 - (xi) Acting as the Trustee of a Fund;
 - (xii) Arranging Credit and Advising on Credit;
 - (xiii) Operating a Crowdfunding Platform;
 - (xiv) Operating an Employee Money Purchase Scheme;
 - (xv) Acting as the Administrator of an Employee Money Purchase Scheme;
or
 - (xvi) Arranging or Advising on Money Services; or
 - (xvii) Operating an Alternative Trading System, if it holds Client Assets; and
- (d) section 6.13 applies only to an Authorised Firm that provides Money Services, Account Information Services or Payment Initiation Services.

Guidance

Section 5.3 of the GEN module contains Rules and Guidance in relation to Systems and Controls, some of which may relate to the management of Operational Risk. The Corporate Governance rules in the GEN module set out overarching requirements in relation to Board responsibilities, including risk management. The Rules and Guidance in this section seek to complement the aforementioned requirements, while providing for a framework to address matters which directly relate to Operational Risk management.

6.2 Risk management framework and governance

- 6.2.1**
- (1) An Authorised Firm must implement and maintain an Operational Risk policy which enables it to identify, assess, control and monitor Operational Risk.
 - (2) The policy must be documented and provide for a sound and well-defined risk management framework to address the Authorised Firm's Operational Risk.
 - (3) An Authorised Firm must:
 - (a) ensure that its risk management systems enable it to implement the Operational Risk policy;

- (b) identify, assess, mitigate, control and monitor the risk; and
- (c) review and update the policy at intervals that are appropriate to the nature, scale and complexity of its activities.

6.2.2 An Authorised Firm must ensure that its Governing Body approves the Operational Risk policy in Rule 6.2.1.

Guidance

1. Some of the key aspects that an Authorised Firm should consider in its Operational Risk policy include:
 - a. the governance structures used to manage Operational Risk, including reporting lines and accountabilities;
 - b. risk assessment tools and how they are used;
 - c. the Authorised Firm's accepted Operational Risk appetite, permissible thresholds or tolerances for inherent and residual risk, and approved risk mitigation strategies and instruments;
 - d. the Authorised Firm's approach to establishing and monitoring thresholds or tolerances for inherent and residual risk Exposure;
 - e. risk reporting and MIS; and
 - f. appropriate independent review and assessment of the Authorised Firm's Operational Risk framework.
2. An Authorised Firm's Operational Risk policy should, amongst other things, include consideration of Principles for the Sound Management of Operational Risk, issued by the Basel Committee on Banking Supervision (BCBS) and the Guidelines on the management of Operational Risk in market-related activities issued by the European Banking Authority which are useful in relation to activities other than banking.

Governing Body responsibilities

1. The GEN Module contains Rules and Guidance regarding corporate governance requirements for Authorised Firms, including the responsibilities of an Authorised Firm regarding risk management.
2. In developing, implementing and maintaining an effective Operational Risk framework, an Authorised Firm's Governing Body should:
 - a. approve and review a risk appetite and tolerance for Operational Risk that articulates the nature, types and levels of Operational Risk that the Authorised Firm is willing to assume;
 - b. consider all relevant risks, the Authorised Firm's level of risk appetite, its current financial condition and its strategic direction. The Governing Body should monitor management adherence to the risk appetite and tolerance and provide for timely detection and remediation of breaches;
 - c. encourage a management culture, and develop supporting processes, which help to engender within the Authorised Firm an understanding by relevant Employees of the

- nature and scope of the Operational Risk inherent in the Authorised Firm's strategies and activities;
- d. provide senior management with clear guidance and direction regarding the principles underlying the Authorised Firm's Operational Risk management framework and approve the corresponding policies developed by senior management;
 - e. regularly review the Authorised Firm's Operational Risk policy to ensure that the Authorised Firm has identified and is managing the Operational Risk arising from external market changes and other environmental factors, as well as those Operational Risks associated with new strategies, products, activities, or systems, including changes in risk profiles and priorities (e.g. changing business volumes). Such review should also take into account the Operational Risk loss experience, the frequency, volume or nature of limit breaches, the quality of the control environment and the effectiveness of risk management or mitigation strategies;
 - f. ensure that the Authorised Firm's Operational Risk policy and framework is subject to effective independent review by audit or other appropriately-trained Persons;
 - g. ensure that management is incorporating industry best practice in managing Operational Risk; and
 - h. establish clear lines of management responsibility and accountability for implementing a strong control environment. The control environment should provide appropriate independence/separation of duties between Operational Risk control functions, business lines and support functions.

Senior Management Responsibilities

- 1. GEN 5.2 contains Rules and Guidance regarding senior management arrangements for Authorised Firms.
- 2. In relation to establishing and maintaining a robust Operational Risk framework, an Authorised Firm's senior management should:
 - a. translate the Operational Risk management framework established by the Governing Body into specific policies and procedures that can be implemented and verified within the different business units;
 - b. clearly assign authority, responsibility and reporting relationships to encourage and maintain accountability, and to ensure that the necessary resources are available to manage Operational Risk in line within the Authorised Firm's risk appetite and tolerance; and
 - c. ensure that the management oversight process is appropriate for the risks inherent in a business unit's activity.

6.3 Risk identification and assessment

6.3.1 An Authorised Firm must:

- (a) ensure that it identifies and assesses the Operational Risks inherent in all the Authorised Firm's products, activities, processes and systems;

- (b) ensure the inherent risks in (a) are understood by relevant Employees of the Authorised Firm;
- (c) systematically track Operational Risk events and any financial impact associated with such events; and
- (d) ensure that the tracking in (c) is consistent with the Operational Risk event types described in the Basel III framework.

Guidance

1. An Authorised Firm should record all Operational Risk events, including near misses and events which result in a positive financial outcome.
2. These Rules complement related Rules in section 5.3 of the GEN module relating to risk management systems and controls. For example, GEN Rule 5.3.6 requires an Authorised Firm to appoint an individual to advise its Governing Body and senior management as to risks.

6.3.2 An Authorised Firm must ensure that its Operational Risk policy in Rule 6.2.1:

- (a) includes an approval process for all new products, activities, processes and systems; and
- (b) incorporates the requirement in Rule 6.3.1(a).

Guidance

1. An Authorised Firm should have policies and procedures that address the process for review and approval of new products, activities, processes and systems. The review and approval process should include consideration of:
 - a. inherent risks in any new product, service, or activity;
 - b. resulting changes to the Authorised Firm's Operational Risk profile, appetite and tolerance, including changes to the risk of existing products or activities;
 - c. necessary controls, risk management processes, and risk mitigation strategies;
 - d. residual risk;
 - e. changes to relevant risk limits;
 - f. procedures and metrics to measure, monitor, and manage the risk of the new product or activity; and
 - g. appropriate investment in human resources and technology infrastructure.
2. Tools that an Authorised Firm may employ for identifying and assessing Operational Risk include:
 - a. internal loss data collection and analysis;
 - b. external data collection and analysis;
 - c. risk assessments;
 - d. business process mapping;

- e. risk and performance indicators; and
- f. scenario analysis.

6.4 Risk monitoring and reporting

6.4.1 An Authorised Firm must:

- (a) regularly monitor material Exposures to Operational Risk losses;
- (b) ensure that appropriate reporting mechanisms are in place at its Governing Body, senior management, and business line levels to support effective management of the Authorised Firm's Operational Risk; and
- (c) immediately notify the DFSA of any material Operational Risk event including notification of any resulting financial impact, positive or negative, associated with such event.

Guidance

1. GEN Rule 5.3.17 requires an Authorised Person to establish and maintain arrangements to provide its Governing Body and senior management with the information necessary to organise and control its activities, to comply with legislation applicable in the DIFC and to manage risks.
2. Rule 6.4.1 is intended to complement GEN Rule 5.3.17 and requires Authorised Firms to establish and maintain reporting mechanisms specifically addressing the Operational Risk matters.
3. The frequency of internal reporting of Operational Risks required by Rule 6.4.1(b) should reflect the risks involved and the pace and nature of changes in the Authorised Firm's operating environment.
4. The following lists some of the items that an Authorised Firm should consider including in its internal reporting of Operational Risks:
 - a. the results of monitoring activities;
 - b. assessments of the Operational Risk framework performed by control functions such as internal audit, compliance, risk management and/or external audit;
 - c. reports generated by (and/or for) supervisory authorities;
 - d. material breaches of the Authorised Firm's risk appetite and tolerance with respect to Operational Risk;
 - e. details of recent significant internal Operational Risk events and losses, including near misses or events that resulted in a positive return; and
 - f. relevant external events and any potential impact on the Authorised Firm and its Operational Risk framework, including Operational Risk capital.

6.5 Control and mitigation

Guidance

1. GEN Rule 5.3.1 requires an Authorised Person to establish and maintain systems and controls, including but not limited to financial and risk systems and controls that ensure that its affairs are managed effectively and responsibly by its senior management.
2. In complying with GEN Rule 5.3.1, an Authorised Firm should establish and maintain a strong control environment that uses policies, processes and systems, appropriate internal controls and appropriate risk mitigation and/or transfer strategies.
3. In establishing systems and controls to address Operational Risk an Authorised Firm should consider the following:
 - a. clear segregation of duties and dual control;
 - b. clearly established authorities and/or processes for approval;
 - c. close monitoring of adherence to assigned risk limits or thresholds;
 - d. safeguards for access to, and use of, the Authorised Firm's assets and records;
 - e. appropriate staffing level and training to maintain expertise;
 - f. ongoing processes to identify business lines or products where returns appear to be out of line with reasonable expectations; and
 - g. regular verification and reconciliation of transactions and accounts.

6.6 Information Technology (IT) systems

- 6.6.1** An Authorised Firm must establish and maintain appropriate and sound information technology infrastructure to meet its current and projected business requirements, under normal circumstances and in periods of stress, which ensures data and system integrity, security and availability and supports integrated and comprehensive risk management.

Guidance

1. IT systems include the computer systems and information technology infrastructure required for the automation of processes and systems, such as application software, operating system software, network infrastructure, and desktop, server and mainframe hardware.
2. An Authorised Firm should consider the following in establishing its systems and controls for the management of IT system risks:
 - a. governance and oversight controls that ensure technology, including outsourcing arrangements, is aligned with and supportive of the Authorised Firm's business objectives;
 - b. an Authorised Firm's organisation and reporting structure for technology operations, including adequacy of senior management oversight; and

- c. the appropriateness of the systems acquisition, development and maintenance activities, including the allocation of responsibilities between IT development and operational areas.
3. Additional requirements to manage cyber risk apply to an Authorised Firm under GEN section 5.5.

6.7 [Deleted]

6.8 Outsourcing

6.8.1 An Authorised Firm must establish and maintain appropriate systems and controls to manage its outsourcing risk.

Guidance

1. GEN Rules 5.3.21 and 5.3.22 set out the DFSA requirements on outsourcing by Authorised Firms. This section complements the requirements in the GEN module and contains guidance on managing the Operational Risk associated with outsourcing arrangements.
2. The assessment of outsourcing risk at an Authorised Firm may depend on several factors, including the scope and materiality of the outsourced activity, how well the Authorised Firm manages, monitors and controls outsourcing risk (including its general management of Operational Risk), and how well the service provider manages and controls the potential risks of the operation.
3. Factors that an Authorised Firm should consider in establishing outsourcing arrangements include the following:
 - a. the financial, reputational and operational impact on the Authorised Firm of the failure of a service provider to perform adequately the activity;
 - b. potential losses to an Authorised Firm's customers and counterparts in the event of a service provider failure;
 - c. the consequences of outsourcing the activity on the ability and capacity of the Authorised Firm to conform with regulatory requirements and changes in such requirements;
 - d. the interrelationship of the outsourced activity with other activities within the Authorised Firm;
 - e. the cost associated with the outsourcing;
 - f. any affiliation or other relationship between the Authorised Firm and the service provider;
 - g. the regulatory status of the service provider;
 - h. the degree of difficulty and time required to select an alternative service provider or to bring the business activity in-house, if necessary;
 - i. the complexity of the outsourcing arrangement. For example, the ability to control the risks where more than one service provider collaborates to deliver an end-to-end outsourcing solution; and

- j. any data protection, security and other risks which may be adversely affected by the geographical location of an outsourcing service provider. To this end, Specific Risk management expertise in assessing country risk related, for example, to political or legal conditions, could be required when entering into and managing outsourcing arrangements that are taken outside of the home country.

6.9 Business continuity and disaster recovery

Guidance

See GEN chapter 5 regarding requirements relating to an Authorised Firm's business continuity and disaster recovery arrangements.

6.10 Management of Operational Risks in trading activities

Guidance

This section complements the Rules and Guidance set out in other sections of this chapter with more specific guidance for the identification, assessment, control and monitoring of Operational Risks in trading activities. In this Guidance, reference to "trading activities" should be construed in its natural sense in the context of Financial Services and should include an Authorised Firm's activities in Dealing in Investments as Principal and Dealing in Investments as Agent. In addressing the Operational Risks arising from trading activities, an Authorised Firm should consider the following:

- a. staff members in support and control functions, comprising functions such as operations, settlement, finance, risk management, legal, compliance, internal and external audit, should have adequate representation and authority within the Authorised Firm's overall governance framework so as to be able to effectively challenge the activities undertaken by the front office;
- b. Operational Risk management systems should set criteria, indicators and thresholds enabling the identification of material incidents detected by internal control procedures. This should include tracking of Operational Risk losses in trading activities and analysis of those losses for possible interconnections (i.e. losses based on one event or root cause);
- c. high professional standards and a sound risk culture should be promoted within the Authorised Firm, particularly in the front office, in a way that supports professional and responsible behaviour. This should include, but is not limited to, developing and implementing appropriate policies and procedures, setting standards (often in the form of a "code of conduct") for relations between traders and their counterparts, and training procedures;
- d. there should be adequate segregation of duties between front office and the support and controls functions in charge of supporting, verifying and monitoring trade transactions;
- e. appropriate policies and procedures relating to leave requirements and staff movements should be developed, implemented and regularly monitored; in particular:
 - i. procedures establishing a minimum absence requirement of at least two consecutive weeks' leave for traders (via a vacation, "desk holiday" or other absence from the office or trading) so that traders are physically unable to mark or value their own books, this responsibility being carried out by a different person during those periods; and
 - ii. employees changing job positions between front, middle and back offices or IT should be properly tracked.

- f. terms of reference describing the activity of each trader or group of traders should be established. Adherence to these terms should be subject to monitoring by support and control functions;
- g. documentation requirements for trading activities should be properly defined so as to minimise legal uncertainties in enforceability of contracts with clients and Counterparties. This should include consideration of using contracts that are standardised as far as possible, particularly in OTC transactions;
- h. all trading positions, profits and losses, cash flows and calculations associated with a transaction should be clearly recorded in the Authorised Firm's management information systems with a documented audit trail. The audit trail should allow for the tracing of cash flows at a sufficiently granular level (e.g. traders, books, products and portfolios);
- i. appropriate procedures for confirmation of the terms and conditions of transactions with external Counterparties/clients should be established;
- j. appropriate processes and procedures should be implemented for the settlement of transactions. This should include consideration of the following elements.
 - i. the authorisation of inputs by the back office;
 - ii. payment/settlements carried out against independent documents;
 - iii. reconciliation between front office and back office systems; and
 - iv. reconciliation procedures independent of the processing functions.
- k. controls should include daily reconciliation of positions and cash flows across various internal systems and external parties. The reconciliations should include all events attached to the transactions including amendments, cancellation, exercises, resets and expiries;
- l. procedures and processes should be established to ensure accurate and timely monitoring and follow up of margin or Collateral calls;
- m. profit attribution is a key control for understanding the risk in a trading operation and therefore the control and support functions should have a good understanding of the various aspects that lead to P&L generation, particularly in relation to more complex products. Major implausibilities discovered within the P&L in the context of the trading mandate and market developments should be further analysed to see if they are caused by Operational Risk events; and
- n. control procedures should be established to monitor and escalate unusual transactions, anomalies in confirmation and reconciliation processes, errors in recording, processing and settling transactions, along with cancellations, amendments, late trades and off-market rates.

6.11 Operational Risk Capital Requirement

6.11.1 This section applies to an Authorised Firm in Category 1, 2, 3A or 5.

- 6.11.2** (1) An Authorised Firm must, subject to (2), use the Basic Indicator Approach as prescribed in App6 to calculate its Operational Risk Capital Requirement.
- (2) An Authorised Firm may, with the written approval of the DFSA, use the Standardised Approach or the Alternative Standardised Approach, both as prescribed in App6, to calculate its Operational Risk Capital Requirement if the DFSA is satisfied that:

- (a) its Governing Body and senior management, as appropriate, are actively involved in the oversight of its Operational Risk framework;
- (b) it has, in accordance with the requirements set out in this chapter, implemented and maintains an Operational Risk policy which provides for a sound and well-defined risk management framework to address the Authorised Firm's Operational Risk; and
- (c) it has dedicated sufficient resources in the use of the relevant approach in its major business lines and its control and audit functions.

6.11.3 An Authorised Firm seeking to apply the Standardised Approach or the Alternative Standardised Approach must develop specific policies and have documented criteria for mapping gross income for current business lines and activities into the Standardised Approach or the Alternative Standardised Approach, as prescribed in App6. The criteria must be reviewed and adjusted for new or changing business activities as appropriate.

6.11.4 Once an Authorised Firm has written approval to apply the Standardised Approach or Alternative Standardised Approach, it must not revert to the Basic Indicator Approach without DFSA approval.

6.11.5 (1) The DFSA may at any time by written notice require an Authorised Firm to adopt a specified approach to calculating its Operational Risk Capital Requirement where the DFSA considers that this is:

- (a) appropriate given the nature, size, complexity and risk profile of the Authorised Firm's business; or
- (b) necessary in the prevailing economic circumstances and it is in the interests of the DIFC.

(2) An Authorised Firm must comply with a requirement made under (1).

6.12 Professional indemnity insurance

6.12.1 This section applies to an Authorised Firm in Category 3B, 3C, 3D or 4 which undertake one or more of the Financial Services prescribed in Rule 6.1.1(c).

6.12.2 An Authorised Firm must:

- (a) take out and maintain professional indemnity insurance cover appropriate to the nature, size, complexity and risk profile of the Authorised Firm's business;
- (b) at least annually, provide the DFSA with a copy of the professional indemnity insurance cover in (a) covering the following twelve month period; and
- (c) notify the DFSA of any material changes to the cover in (a), including the level of cover, its renewal or termination.

Guidance

1. In complying with Rule 6.12.2, an Authorised Firm should take out and maintain a contract for professional indemnity insurance (PII) from a reputable and well-capitalised insurer and such contract should include cover in respect of claims for which the Authorised Firm may be liable as a result of the conduct of itself and its Employees and appropriate cover in respect of legal costs arising from a claim.
2. Pursuant to Rule 6.4.1(c), an Authorised Firm should notify the DFSA of any significant PII claim made. What amounts to a significant claim will depend on the nature size and complexity of the Authorised Firm and the DFSA would expect the Authorised Firm to treat a series of small single claims which are significant in aggregate as significant for the purposes of Rule 6.4.1(c).
3. An Authorised Firm can fulfil the requirements under this section by ensuring coverage of its activities under a group-wide PII policy, provided that policy covers the Authorised Firm and its activities and that policy meets the conditions specified in this section. Where the Authorised Firm's group PII cover does not meet the requirements specified under this section, the Authorised Firm will be required to obtain PII cover that meets those requirements.

6.13 Management of Operational Risk in Money Services

6.13.1 This section applies to an Authorised Firm that provides:

- (a) Money Services;
- (b) Account Information Services; or
- (c) Payment Initiation Services.

Definition of strong customer authentication

- 6.13.2** (1) In this section, “strong customer authentication” or “SCA” means authentication that is based on the use of two or more elements that are:
- (a) independent, in that breach of one element does not compromise the reliability of any other element; and
 - (b) designed in such a way as to protect the confidentiality of the authentication data.
- (2) The elements in (1)(a) must consist of two or more of the following:
- (a) something known only by the User (“knowledge”);
 - (b) something held only by the User (“possession”); or
 - (c) something inherent to the User (“inherence”).

Strong customer authentication and User security measures

- 6.13.3** (1) An Authorised Firm must, except as provided in Rule 6.13.4, apply strong customer authentication where a User:

- (a) accesses a Payment Account online, either directly or through an Account Information Service;
 - (b) initiates an electronic Payment Transaction; or
 - (c) carries out any action through a remote channel which may imply a risk of payment fraud or other abuses.
- (2) If a payer initiates a Payment Transaction directly or through a Payment Initiation Service, the Authorised Firm must apply SCA that includes elements which dynamically link the transaction to a specific amount and a specific payee.
- (3) If a multipurpose device is used in the SCA process, the Authorised Firm must adopt adequate security measures that mitigate the risk of the device being compromised.
- (4) The Authorised Firm must maintain adequate security measures to protect the confidentiality and integrity of Users' personal security credentials.

Guidance

1. An Authorised Firm should, as best practice in maintaining the integrity of strong customer authentication, try to ensure that:
 - (a) no element of 'knowledge', 'possession' or 'inherence' (as defined in Rule 6.13.2) can be derived from the disclosure of the authentication code cover;
 - (b) it is not possible to generate a new authentication code based on an old one;
 - (c) the authentication code cannot be forged;
 - (d) where the authentication through a remote channel has failed to generate an authentication code, it is not possible to identify which of the SCA elements was incorrect;
 - (e) a maximum of 5 failed consecutive authentication attempts within a given period result in the account being temporarily or permanently blocked;
 - (f) the duration and number of retries for a temporary block should be linked to the service offered and trigger a fraud risk alert; and
 - (g) the User is alerted before the block becomes permanent and a secure procedure is established to regain the use of the blocked payment instrument.
2. An Authorised Firm should, as best practice in maintaining the integrity of User security credentials (USC), endeavour:
 - (a) not to allow the USC to be fully readable when inputted by the User or by its own staff;
 - (b) to ensure that the USC always remain encrypted and no information relating to the USC is stored in plain text;
 - (c) to protect secret cryptographic material from unauthorised disclosure;
 - (d) to document the process used to encrypt or otherwise render the USCs unreadable;
 - (e) to adopt measures to mitigate the risk of unauthorised use of compromised USCs;
 - (f) to ensure secure delivery of the USC to the User, secure association of the USC with the User and the secure disposal of the USC once it is obsolete; and
 - (g) to immediately inform the User and the issuer of the USC (if another firm) in the event a USC is compromised under the firm's sphere of control.
3. Dynamic linking referred to in Rule 6.13.3(2) may include:

- (a) the User being made aware of the payment amount and the beneficiary;
 - (b) the authentication code generated using SCA being specific to the transaction amount and the beneficiary; and
 - (c) any change to the amount or the beneficiary resulting in the authentication code becoming invalid.
- 4. The security measures referred to in Rule 6.13.3(4) may include:
 - (a) the use of separated secure execution environments through the software installed on the device;
 - (b) mechanisms to ensure that the software or device has not been, and cannot be, altered by the User or a third party; and
 - (c) where alterations have taken place, mechanisms to mitigate the consequences.

Exceptions

6.13.4 An Authorised Firm is not required to apply strong customer authentication under Rule 6.13.3 when:

- (a) the User accesses its own payment account information unless:
 - (i) it is the first time the account is accessed; or
 - (ii) the account has not been accessed for 90 days or more;
- (b) the User makes a payment of a small amount;
- (c) the User makes a payment to a specified beneficiary on a list created by the User, or under a standing order, where strong customer authentication was applied when the list or standing order was created; or
- (d) a transfer is made between accounts held by the same User.

Guidance

In Rule 6.13.4(b) an example of a small amount may be a transaction not exceeding a certain value e.g. \$50 or five payments not exceeding an aggregated amount of \$150 over a specified period.

Technical standards

6.13.5 An Authorised Firm must develop and implement technical standards relating to:

- (a) the implementation of the requirements for strong customer authentication referred to in Rule 6.13.3;
- (b) procedures for applying the exclusions in Rule 6.13.4;
- (c) common and secure standards of communication for the purpose of identification, authentication, notification, and sharing information with Users and other service providers; and
- (d) if applicable, procedures, systems and controls that ensure the reliability and continuity of the interface made available by a Payment Account Provider.

Guidance

1. In developing technical standards referred to in 6.13.5(c) (common and secure standards of communication) an Authorised Firm should, as best practice and where applicable:
 - (a) apply secure identification when communicating between devices used for electronic payments;
 - (b) address the risk of a communication being misdirected to unauthorised parties;
 - (c) trace all payment transactions and interactions with all relevant parties to the Payment Service;
 - (d) use strong and widely recognised encryption techniques when exchanging data;
 - (e) keep the access sessions with the Payment Account Provider (PAP) as short as possible and terminate any session after the requested action is completed;
 - (f) ensure, when maintaining parallel network sessions, that the sessions are securely linked to relevant sessions established with the User to prevent the possibility that information communicated between them could be misrouted;
 - (g) include unambiguous unique references to the User, communication session, payment transaction and requested amount; and
 - (h) have systems and controls to prevent access to User information being available by the Payment Account Provider, beyond what is needed to provide the relevant service. The frequency of the access should also be agreed with the User.
2. In developing technical standards referred to in Rule 6.13.5(d) (procedures, systems and controls to ensure the reliability and continuity of the interface with the PAP) an Authorised Firm should:
 - (a) contractually ensure that the interface applies best practice standards of communication issued by the Security Standards Council;
 - (b) understand and document the interface technical specification such as the routines, protocols, and tools needed to interoperate with the systems of the PAP;
 - (c) contractually require the PAP to provide the firm with at least 3 months' notice ahead of any change to the interface, except in an emergency;
 - (d) adequately test the interface to ensure reliability and performance;
 - (e) establish key performance indicators and service level standards with the PAP; and
 - (f) request the PAP to provide adequate contingency measures in the event the interface is not available and test the measures for reliability.

Systems and controls to detect fraud

- 6.13.6** (1) An Authorised Firm must have in place transaction monitoring systems and controls to detect and prevent unauthorised or fraudulent Payment Transactions.
- (2) The systems referred to in (1) must be designed to take into account the following fraud risk factors:
- (a) compromised or stolen authentication elements;
 - (b) the amount of each payment transaction;
 - (c) known fraud scenarios in the provision of the particular Payment Service;
 - (d) analysis of Payment Transactions typical of the type of Users;

- (e) signs of malware infection in any sessions of the authentication procedure; and
- (f) if the firm provides the access device or software (the Payment Instrument), a log of the use of the access device or software and abnormal use.

Reporting of information about transactions and rates of fraud

6.13.7 An Authorised Firm must provide the following information to the DFSA at least quarterly:

- (a) rates of fraud or suspected fraud;
- (b) the total value of fraudulent transactions and the total value of all payment transactions;
- (c) the number of payment transactions and the average transaction value; and
- (d) a breakdown by percentage of transactions initiated through SCA and transactions initiated through each exception to the SCA requirements.

Reporting of information about Money Transmission

6.13.8 An Authorised Firm that provides Money Transmission must send the following information to the DFSA at least quarterly relating to transactions:

- (a) details of senders and recipients of transfers;
- (b) the amounts transferred;
- (c) dates of transfers; and
- (d) any other firms involved in transfers.

7 INTEREST RATE RISK IN THE NON-TRADING BOOK

Introduction

Guidance

1. This chapter relates to interest rate risk in the Non-Trading Book, also known as interest rate risk in the banking book (IRRBB). In relation to the Trading Book, an Interest Rate Risk Capital Requirement is imposed as a component of the Market Risk Capital Requirement under chapter 5.
2. IRRBB refers to the current or prospective risk to an Authorised Firm's capital resources and earnings arising from adverse movements in interest rates that affect its Non-Trading Book positions. When interest rates change, the present value and timing of future cash flows change. This in turn changes the underlying value of an Authorised Firm's assets, liabilities and off-balance sheet items and hence its economic value. Changes in interest rates also affect an Authorised Firm's earnings by altering interest rate-sensitive income and expenses, affecting its net interest income. Excessive IRRBB can pose a significant threat to an Authorised Firm's current Capital Resources and future earnings if not managed appropriately.
3. IRRBB may take different forms, for example:
 - a. Gap risk arises from the term structure of Non-Trading Book instruments and describes the risk arising from the timing of instruments' rate changes. The extent of gap risk depends on whether changes to the term structure of interest rates occur consistently across the yield curve (parallel risk) or differentially by period (non-parallel risk).
 - b. Basis risk describes the impact of relative changes in interest rates for financial instruments that have similar tenors but are priced using different interest rate indices.
 - c. Option risk arises from option derivative positions or from optional elements embedded in an Authorised Firm's assets, liabilities or off-balance sheet items, where the Authorised Firm or its customer can alter the level and timing of their cash flows. Option risk can be further categorised into automatic option risk and behavioural option risk.

7.1 Application

- 7.1.1** (1) This chapter (except for Rule 7.2.2A) applies to an Authorised Firm in Category 1.
- (2) Rules 7.2.2A and 7.2.4, and section 7.5 apply to an Authorised Firm in Category 2..
- 7.1.2** Where an Authorised Firm to which this chapter applies is part of a Financial Group, this chapter also applies on a consolidated basis in relation to all the entities within the Financial Group.

7.2 Measuring Non-Trading Book interest rate risk

- 7.2.1** (1) An Authorised Firm must carry out an evaluation of its Exposure to the interest rate risk arising from its Non-Trading Book activities, including an evaluation of its Exposures to risk arising from potential changes in interest rates that affect either or both of the following:
- (a) the economic value of its Non-Trading Book activities; or
 - (b) the earnings in respect of its Non-Trading Book activities.
- (2) An Authorised Firm with balance sheet positions in different currencies must measure its risk Exposures in each currency that accounts for more than 5% of its total Non-Trading Book assets or liabilities.

- 7.2.2** (1) The evaluation under Rule 7.2.1 must cover the effects of the interest rate scenarios specified in the left hand column of the table in (2) and calculated in accordance with corresponding formulae in the right hand column of that table.
- (2) This is the table referred to in (1):

scenario 1: parallel shock up	$\Delta R_c(t_k) = \bar{R}_{\text{parallel},c}$
scenario 2: parallel shock down	$\Delta R_c(t_k) = -\bar{R}_{\text{parallel},c}$
scenario 3: steepener shock (short rates down and long rates up)	$\Delta R_c(t_k) = -0.65 \cdot \bar{R}_{\text{short},c} \cdot e^{\frac{t_k}{4}} + 0.9 \cdot \bar{R}_{\text{long},c} \cdot \left(1 - e^{\frac{-t_k}{4}}\right)$
scenario 4: flattener shock (short rates up and long rates down)	$\Delta R_c(t_k) = 0.8 \cdot \bar{R}_{\text{short},c} \cdot e^{\frac{-t_k}{4}} - 0.6 \cdot \bar{R}_{\text{long},c} \cdot \left(1 - e^{\frac{-t_k}{4}}\right)$
scenario 5: short rates shock up	$\Delta R_c(t_k) = \bar{R}_{\text{short},c} \cdot e^{\frac{-t_k}{4}}$
scenario 6: short rates shock down	$\Delta R_c(t_k) = -\bar{R}_{\text{short},c} \cdot e^{\frac{-t_k}{4}}$

where:

c = the index that denotes currency

k = the index that denotes the buckets in accordance with the table in Rule 7.3.4

t_k = the bucket midpoint of bucket 'k', measured in years

$\Delta R_c(t_k)$ = the change in interest rate at the point ' t_k ' for currency 'c'

$\bar{R}_{\text{parallel},c}$ = the prescribed parallel interest rate shock for currency 'c' determined in accordance with column two of the table referred to in (3)

$\bar{R}_{\text{short},c}$ = the change in short interest rate at the point ' t_k ' for currency 'c'

$\bar{R}_{\text{long},c}$ = the change in long interest rate at the point ' t_k ' for currency 'c'

(3) For the purposes of (2), the interest rate shock scenarios for individual currencies are those set out in the table below:

Specified size of interest rate shocks (in bps)			
Currency	Parallel	Short	Long
AED	200	300	225
ARS	400	500	300
AUD	350	425	300
BRL	400	500	300
CAD	200	275	175
CHF	175	250	200
CNY	225	300	150
EUR	225	350	200
GBP	275	425	250
HKD	225	375	200
IDR	400	500	300
INR	325	475	225
JPY	100	100	100
KRW	225	350	225

MXN	400	500	200
RUB	400	500	300
SAR	275	375	250
SEK	275	425	200
SGD	175	250	225
TRY	400	500	300
USD	200	300	225
ZAR	325	500	300

Guidance

1. In carrying out the evaluation under Rule 7.2.1(1), the DFSA expects an Authorised Firm to consider the need to select a range of interest rate scenarios (in addition to the six interest rate scenarios prescribed by Rule 7.2.2(2)) to capture comprehensively the firm's risk profile, according to its ICAAP.
2. If an Authorised Firm's material currency is not listed in Rule 7.2.2(3), an Authorised Firm should consult the DFSA to determine applicable interest rate shocks for that currency.

- 7.2.2A** (1) An Authorised Firm must carry out an evaluation of its Exposure to the interest rate risk arising from its Non-Trading Book activities. An Authorised Firm with balance sheet positions in different currencies must measure its risk Exposures in each currency that accounts for more than 5% of its total Non-Trading Book assets or liabilities.
- (2) The evaluation under (1) must cover the effect of a sudden and unexpected parallel change in interest rates of 200 basis points in both directions.
- (3) To carry out effectively the stress testing requirements specified in (2), an Authorised Firm must include appropriate scenarios in its stress testing programmes for measuring its vulnerability to loss arising from the impact of adverse interest rate movements on its Non-Trading Book structure.

- 7.2.3** An Authorised Firm must immediately notify the DFSA if any evaluation under this section suggests that, as a result of the change in interest rates described in Rule 7.2.2, the economic value of the firm would decline by more than 15% of its T1 Capital.

Guidance

If an evaluation under this section suggests that the economic value of an Authorised Firm would decline by more than the level set out in Rule 7.2.3, the DFSA may consider taking appropriate action, including requiring the Authorised Firm to reduce its IRRBB Exposures (for example, by hedging),

raise additional capital, set constraints on the internal risk parameters used by the Authorised Firm or improve its risk management framework.

Frequency of measuring Non-Trading Book interest rate risk

- 7.2.4** An Authorised Firm must carry out the evaluations required by Rule 7.2.1 or Rule 7.2.2A, as applicable, as frequently as necessary for it to be reasonably satisfied that it has at all times a sufficient understanding of the degree to which it is exposed to the risks referred to in that Rule and the nature of that Exposure. In any case it must carry out those evaluations no less frequently than quarterly.

7.3 Standardised approach for interest rate risk in the Non-Trading Book (SA-IRRBB)

7.3.1 In this section:

- (a) “ Δ EVE” means the change in the economic value of equity (EVE) under an interest rate stress or shock scenario, calculated by subtracting EVE under the interest stress or shock scenario from EVE under the current interest rates;
- (b) “Fixed Rate Positions” means positions which generate cash flows that are certain until the point of contractual maturity, including fixed rate loans without embedded prepayment options, term deposits without redemption risk and other amortising products such as mortgages;
- (c) “Floating Rate Positions” means positions which generate cash flows that are not predictable past the next Repricing Date other than that the present value would be reset at par;
- (d) “Interest Rate-Sensitive Positions in the Non-Trading Book” means:
 - (i) assets, excluding assets which are deducted from CET1 Capital under Rule 3.13.7, fixed assets, including real estate and intangible assets, and equity Exposures in the Non-Trading Book;
 - (ii) liabilities, including non-remunerated deposits, but excluding CET1 Capital instruments; or
 - (iii) off-balance sheet items;
- (e) “Notional Repricing Cash Flow” means:
 - (i) any repayment of principal;
 - (ii) any repricing of principal where repricing is said to occur at the earliest date at which either the Authorised Firm or its Counterparty is entitled to unilaterally change the interest rate, or at which the rate on a floating rate instrument changes automatically in response to a change in an external benchmark; or

- (iii) any interest payment on a tranche of principal that has not yet been repaid or repriced; and
- (f) “Repricing Date” means the date of each repayment, repricing or interest payment of a Notional Repricing Cash Flow.

Overview

7.3.2 An Authorised Firm must carry out the evaluation in Rule 7.2.1(1)(a) under the SA-IRRBB by carrying out the following steps:

- (1) Step 1: Allocate Interest Rate-Sensitive Positions in the Non-Trading Book to one of the following categories, in accordance with the table in Rule 7.3.3:
 - (a) amenable to standardisation;
 - (b) not amenable to standardisation; or
 - (c) less amenable to standardisation.
- (2) Step 2: Slot cash flows arising from Interest Rate-Sensitive Positions in the Non-Trading Book that are amenable to standardisation and that are not amenable to standardisation, in accordance with Rules 7.3.4 to 7.3.11.
- (3) Step 3: Determine the add-on for positions that are less amenable to standardisation under the six prescribed interest rate scenarios for each material currency, in accordance with Rule 7.3.12.
- (4) Step 4: Determine the Δ EVE under the six prescribed interest rate scenarios for each material currency, in accordance with Rule 7.3.13(a) to (d); and
- (5) Step 5: Calculate the Δ EVE as the maximum of the worst aggregated reductions to EVE across the six prescribed interest rate shocks, in accordance with Rule 7.3.13(e).

Categories of Interest Rate-Sensitive Positions in the Non-Trading Book

7.3.3 This is the table referred to in Rule 7.3.2:

	Category
Amenable to standardisation	Fixed Rate Positions
	Floating Rate Positions
Not amenable to standardisation	Non-maturing deposits
	Fixed rate loans with retail borrowers subject to prepayment risk

	Term deposits by retail depositors subject to early redemption risk
Less amenable to standardisation	Automatic interest rate options

Guidance

1. Where a wholesale customer has a behavioural option that may change the pattern of Notional Repricing Cash Flows, such options should be included within the category of automatic interest rate options.
2. Term deposits lock in a fixed rate for a fixed term and would usually be hedged on that basis. However, term deposits may be subject to the risk of early withdrawal. Consequently, term deposits should only be treated as a Fixed Rate Position if the depositor has no legal right to withdraw the deposit or early withdrawal results in a significant penalty that at least compensates for the loss of interest between the date of withdrawal and the contractual maturity date and the economic cost of breaking the contract.
3. In the case of fixed rate loans with retail borrowers subject to prepayment risk and term deposits by retail depositors subject to early redemption risk the customer has an option which, if exercised, will alter the timing of an Authorised Firm's cash flows. The customer's exercise of the option is, among other factors, influenced by changes in interest rates. In the case of a fixed rate loan, the customer has an option to repay the loan early (i.e. prepay); and for a fixed term deposit, the customer may have an option to withdraw the deposit before the scheduled date.
4. Fixed rate loans with retail borrowers subject to prepayment risk include loan products where the economic cost of prepayment is never charged or charged only for prepayments above a certain threshold.
5. The most important automatic interest rate options likely to occur in the Non-Trading Book are caps and floors, which are often embedded in banking products. Swaptions, such as prepayment options on non-retail products, may also be treated as automatic interest rate options, as, in cases where such options are held by sophisticated financial market Counterparties, the option holder will almost certainly exercise the option if it is in their financial interest to do so. Any behavioural option positions with wholesale customers that may change the pattern of Notional Repricing Cash Flows should be considered as embedded automatic interest rate options.

Slotting of cash flows

- 7.3.4** (1) An Authorised Firm must, in accordance with Rules 7.3.5 to 7.3.11, project for each material currency 'c' identified in accordance with Rule 7.2.1(2) and for each interest rate scenario 'i' specified in the table in Rule 7.2.2(2) all future Notional Repricing Cash Flows arising from the Interest Rate-Sensitive Positions in the Non-Trading Book on to:
- (a) the time bucket as set out in the table in (2), according to their Repricing Dates; or
 - (b) the time bucket midpoints, as set out in the table in (2), by splitting up Notional Repricing Cash Flows between two adjacent maturity bucket midpoints.

(2) This is the table referred to in (1):

Time bucket intervals and midpoints (M = months; Y = years)			
	Bucket number (k)	Bucket interval	Bucket midpoint
Short-term rates	1	Overnight	0.0028Y
	2	>Overnight and <= 1M	0.0147Y
	3	>1M and<= 3M	0.1667Y
	4	>3M and<= 6M	0.375Y
	5	>6M and<= 9M	0.625Y
	6	>9M and<= 1Y	0.875Y
	7	>1Y and<= 1.5Y	1.25Y
	8	>1.5Y and<= 2Y	1.75Y
Medium-term rates	9	>2Y and<= 3Y	2.5Y
	10	>3Y and<= 4Y	3.5Y
	11	>4Y and<= 5Y	4.5Y
	12	>5Y and<= 6Y	5.5Y
	13	>6Y and<= 7Y	6.5Y
Long-term rates	14	>7Y and<= 8Y	7.5Y
	15	>8Y and<= 9Y	8.5Y
	16	>9Y and<= 10Y	9.5Y
	17	>10Y and<= 15Y	12.5Y
	18	>15Y and<= 20Y	17.5Y
	19	20Y	25Y

(3) For the purpose of projecting all future Notional Repricing Cash Flows under (1), an Authorised Firm may deduct commercial margins and other spread components from the Notional Repricing Cash Flows, using a prudent and transparent methodology.

Process for slotting positions that are amenable to standardisation

- 7.3.5** (1) An Authorised Firm must:
- (a) in respect of a Fixed Rate Position, allocate Notional Repricing Cash Flows either to the relevant time buckets or the time bucket midpoints closest to the contractual maturity of such cash flows; and
 - (b) in respect of a Floating Rate Position, treat such position as a series of coupon payments until the next Repricing Date and a par notional cash flow at the time bucket midpoint closest to the next reset date bucket, with no additional slotting of Notional Repricing Cash Flows to later time buckets or time bucket midpoints (other than the spread component which is not repriced).
- (2) For the purpose of (1), an Authorised Firm must ignore the optionality (whether sold or bought) arising from a position with an embedded automatic interest rate option and must treat the embedded automatic interest rate option as if it were an explicit automatic interest rate option, in accordance with Rule 7.3.12.
- (3) If an Authorised Firm is able to demonstrate that the optionality in (2) is not material, the Authorised Firm may choose not to perform the bifurcation in (2) and ignore the optionality.

Guidance

An example of a product with embedded automatic interest rate options under Rule 7.3.5(2) is a floating rate mortgage loan with embedded caps or floors. In this case Notional Repricing Cash Flows should be treated as a Fixed Rate Position until the next Repricing Date, thereby ignoring the option, which instead should be treated like a separate automatic interest rate option.

Process for slotting positions that are not amenable to standardisation

- 7.3.6** (1) An Authorised Firm must allocate non-maturing deposits (NMDs) to one of the following categories:
- (a) retail (transactional) NMDs;
 - (b) retail (non-transactional) NMDs; or
 - (c) wholesale NMDs.
- (2) For the purpose of (1), an Authorised Firm must:
- (a) treat a deposit:
 - (i) as a retail (transactional or non-transactional) NMD if it is made by:
 - (A) an individual; or

- (B) a small business customer, provided it is managed as a retail Exposure and the total aggregated liabilities raised from one small business customer are less than \$1 million; or
- (ii) as a wholesale NMD in any other case; and
- (b) treat a retail NMD as a:
 - (i) transactional NMD if regular transactions are carried out through that account or if the NMD is not interest bearing; or
 - (ii) non-transactional NMD in any other case.

Guidance

Retail deposits should be considered as held in a transactional account under Rule 7.3.6(2)(b) if regular transactions are carried out in that account (e.g. when salaries are regularly credited) or if the deposit is non-interest bearing. Other retail deposits should be considered as held in a non-transactional account. Deposits from legal entities, sole proprietorships or partnerships are captured in wholesale deposit categories.

- 7.3.7** (1) For each category in Rule 7.3.6(1), an Authorised Firm must allocate each position to one of the following sub-categories:
- (a) core portion, consisting of deposits that are found to remain undrawn with a high degree of likelihood, based on observed volume changes over the past ten years, and unlikely to reprice even under significant changes in the interest rate environment (Core Portion); or
 - (b) non-core portion, consisting of the portion not allocated to the Core Portion under (a) (Non-core Portion).
- (2) For the allocation in (1), an Authorised Firm must ensure that the core portion is no greater than the maximum set out in the table below:

NMD category	Maximum proportion of core deposits (%)
Retail (transactional)	90
Retail (non-transactional)	70
Wholesale	50

- 7.3.8** (1) For the purpose of Rule 7.3.4(1), the Notional Repricing Cash Flows of NMDs are:
- (a) for the Core Portion the initial Notional Repricing Cash Flows, based on an appropriate slotting procedure determined by the Authorised Firm, up

to the maximum average maturity per category, as specified in the table in (2); and

- (b) for the Non-core Portion the initial Notional Repricing Cash Flows with an overnight tenor.

(2) This is the table referred to in (1)(a):

NMD category	Maximum average maturity of core deposits (years)
Retail (transactional)	5
Retail (non-transactional)	4.5
Wholesale	4

7.3.9 For fixed rate loans with retail borrowers that are subject to prepayment risk, as determined in accordance with the table in Rule 7.3.3, an Authorised Firm must:

- (a) allocate each loan to a single portfolio of homogeneous loans 'p', denominated in currency 'c';
- (b) for each portfolio of homogeneous loans, determine and notify to the DFSA a monthly baseline conditional prepayment rate 'CPR', in currency 'c', under the prevailing term structure of interest rates; and
- (c) for each portfolio of homogeneous loans, determine the CPR in currency 'c' for interest rate scenario 'i', in accordance with the following formula:

$$CPR_{i,c}^p = \min(1, \gamma_i \cdot CPR_{0,c}^p)$$

where:

$CPR_{i,c}^p$ = the CPR in currency 'c' for interest rate scenario 'i' of portfolio 'p'

$CPR_{0,c}^p$ = the constant baseline CPR for portfolio 'p' in currency 'c'

γ_i = the prescribed scalar multiplier applied for interest rate scenario 'i', as given in the table below:

Scenario number 'i'	Interest rate scenarios	γ_i (scenario multiplier)
1	Parallel up	0.8
2	Parallel down	1.2
3	Steeper	0.8

4	Flattener	1.2
5	Short rate up	0.8
6	Short rate down	1.2

- (d) for each portfolio of homogeneous loans, determine the Notional Repricing Cash Flows, as follows:

- (i) where time buckets are used, in accordance with the following formula:

$$CF_{i,c}^p(k) = CF_{i,c}^s(k) + CPR_{i,c}^p \cdot N_{i,c}^p(k-1)$$

- (ii) where time bucket midpoints are used, in accordance with the following formula:

$$CF_{i,c}^p(t_k) = CF_{i,c}^s(t_k) + CPR_{i,c}^p \cdot N_{i,c}^p(t_k - 1)$$

where:

$CF_{i,c}^p(k)$ and $CF_{i,c}^p(t_k)$ = the Notional Repricing Cash Flows arising from a portfolio 'p' in currency 'c' under interest rate scenario 'i' with tenor that corresponds to bucket 'k'

$CF_{i,c}^s(k)$ and $CF_{i,c}^s(t_k)$ = the scheduled interest and principal repayments for interest rate scenario 'i' and currency 'c', adjusted to take into account prepayments

$N_{i,c}^p(k-1)$ and $N_{i,c}^p(t_k - 1)$ = the notional outstanding at time bucket $k-1$ and time bucket midpoint $t_k - 1$, respectively, for interest rate scenario 'i' and currency 'c'

- (e) where it has in place annual limits on uncompensated prepayments, apply these annual limits in the relevant cash flows referred to in (d).

7.3.10 An Authorised Firm may adjust the formulae in Rule 7.3.9(c) and (d) to reflect a base monthly CPR that varies over the life of each loan in the portfolio. In that case, it is denoted as for each time bucket 'k' or time bucket midpoint ' t_k ', in accordance with the table in Rule 7.3.4(2).

7.3.11 For term deposits by retail depositors that are subject to early redemption risk, as determined in accordance with the table in Rule 7.3.3, an Authorised Firm must:

- (a) allocate each deposit to a single portfolio of homogeneous deposits 'p', denominated in currency 'c';

- (b) for each portfolio of homogeneous deposits, determine and notify to the DFSA a baseline term deposit redemption ratio 'TDRR', in currency 'c', under the prevailing term structure of interest rates; and
- (c) for each portfolio of homogeneous deposits, determine the TDRR in currency 'c' for interest rate scenario 'i', in accordance with the following formula:

$$TDRR_{i,c}^p = \min(1, v_i \cdot TDRR_{0,c}^p)$$

where:

$TDRR_{i,c}^p$ = the conditional TDRR in currency 'c' for interest rate scenario 'i' of portfolio 'p'

$TDRR_{0,c}^p$ = the baseline TDRR for portfolio 'p' in currency 'c' under the current interest rates

v_i = the prescribed multiplier applied for interest rate scenario 'i', as given in the table below:

Scenario number 'i'	Interest rate scenarios	v_i (scenario multiplier)
1	Parallel up	1.2
2	Parallel down	0.8
3	Steepener	0.8
4	Flattener	1.2
5	Short rate up	1.2
6	Short rate down	0.8

- (d) for each portfolio of homogeneous deposits, determine the Notional Repricing Cash Flows, as follows:

- (i) where time buckets are used, in accordance with the following formula:

$$CF_{i,c}^p(1) = TD_{0,c}^p \cdot TDRR_{i,c}^p$$

- (ii) where time bucket midpoints are used, in accordance with the following formula:

$$CF_{i,c}^p(t_1) = TD_{0,c}^p \cdot TDRR_{i,c}^p$$

where:

$CF_{i,c}^p(1) =$	the Notional Repricing Cash Flows arising from portfolio 'p' in currency 'c' under interest rate scenario 'i' with a tenor that corresponds to bucket 'k', which are expected to be redeemed early and slotted into the overnight time bucket
$CF_{i,c}^p(t_1) =$	the Notional Repricing Cash Flows arising from portfolio 'p' for interest rate scenario 'i' and currency 'c', which are expected to be redeemed early and slotted into the overnight time bucket midpoint
$TD_{0,c}^p =$	the outstanding amount of term deposits for portfolio 'p' and currency 'c'

Add-on for positions that are less amenable to standardisation

- 7.3.12** (1) An Authorised Firm must calculate an add-on in currency 'c' for interest rate scenario 'i', $KAO_{i,c}$, for all automatic interest rate options (sold or bought), as determined in accordance with the table in Rule 7.3.3, except as provided under (4), in accordance with the following formula:

$$KAO_{i,c} = \sum_{o=1}^{n_c} \Delta FVAO_{i,c}^o - \sum_{q=1}^{m_c} \Delta FVAO_{i,c}^q$$

where:

- $n_c =$ the number of sold options, in currency 'c'
- $\Delta FVAO_{i,c}^o =$ the change in value of sold automatic option 'o' for interest rate scenario 'i', calculated in accordance with (2)
- $m_c =$ the number of bought options, in currency 'c'
- $\Delta FVAO_{i,c}^q =$ the change in value of bought automatic option 'q' for interest rate scenario 'i', calculated in accordance with (2)
- (2) An Authorised Firm must calculate the value change for each sold option 'o' (bought option 'q', respectively) for interest scenario 'i', $\Delta FVAO_{i,c}^o$ ($\Delta FVAO_{i,c}^q$, respectively), as the sum of:
- (i) an estimate of the value of the option to the option holder, given:
 - (A) a yield curve in currency 'c' under the interest rate scenario 'i'; and
 - (B) a relative increase in the implicit volatility of 25%; minus
 - (ii) the value of the sold option to the option holder, given the yield curve in currency 'c' at the valuation date.

- (3) An Authorised Firm must notify the DFSA of the methodology used to estimate the value of the option under (2)(i).
- (4) An Authorised Firm may choose to calculate an add-on only for bought automatic interest rate options that are used for hedging sold automatic interest rate options, provided that it adds to $KAO_{i,c}$ under (1) any change in market value that is included in its Capital Resources.

Calculation of the standardised EVE measure

7.3.13 An Authorised Firm must calculate the loss in the economic value of equity, EVE_{loss} , under scenario 'i' and currency 'c', as follows:

- (a) within each time bucket 'k' or time bucket midpoint ' t_k ', net all positive and negative Notional Repricing Cash Flows that have been slotted in accordance with Rules 7.3.4 to 7.3.11 to form a single long or short position, with the netted parts removed from the calculation, leading to a set of Notional Repricing Cash Flows, $CF_{i,c}(k)$ or $CF_{i,c}(t_k)$, $k \in \{1, 2, \dots, K\}$, respectively;
- (b) weigh the net Notional Repricing Cash Flows in each time bucket 'k' or time bucket midpoint ' t_k ', determined under (a) by the continuously compounded discount factor, $DF_{i,c}(k)$, which is calculated in accordance with the following formula:

$$DF_{i,c}(t_k) = \exp^{-R_{i,c}(t_k) \cdot t_k}$$

where:

$R_{i,c}(t_k)$ = the risk-free rate at bucket midpoint ' t_k ', including commercial margins and other spreads if the Authorised Firm has included commercial margins and other spread components in its cash flows, in currency 'c' under interest scenario 'i'

- (c) sum the risk-weighted net positions, calculated under (b) to determine the economic loss in currency 'c' under scenario 'i' (excluding automatic interest rate positions), $EVE_{i,c}^{nao}$, as follows:

- (i) where time buckets are used:

$$EVE_{i,c}^{nao} = \sum_{k=1}^{19} CF_{i,c}(k) \cdot DF_{i,c}(t_k)$$

- (ii) where time bucket midpoints are used:

$$EVE_{i,c}^{nao} = \sum_{k=1}^{19} CF_{i,c}(t_k) \cdot DF_{i,c}(t_k)$$

where:

$CF_{i,c}(k)$ and $CF_{i,c}(t_k)$ = the Notional Repricing Cash Flow for bucket 'k' or time bucket midpoint ' t_k ', respectively, in currency 'c' for interest rate scenario 'i', calculated in accordance with (a)

$DF_{i,c}(t_k)$ = the discount factor for time bucket midpoint ' t_k ', in currency 'c' for interest rate scenario 'i', calculated in accordance with (b)

- (d) obtain the ΔEVE in currency 'c' associated with interest rate scenario 'i' by subtracting $EVE_{i,c}^{nao}$, calculated under (c), from the economic value of equity under the current interest rate term structure, $EVE_{0,c}^{nao}$ and by adding the total measure for automatic interest rate risk, $KAO_{i,c}$, calculated under Rule 7.3.12, as follows:

- (i) where time buckets are used:

$$\sum_{k=1}^{19} CF_{0,c}(k) \cdot DF_{0,c}(t_k) - \sum_{k=1}^{19} CF_{i,c}(k) \cdot DF_{i,c}(t_k) + KAO_{i,c}$$

- (ii) where time bucket midpoints are used:

$$\sum_{k=1}^{19} CF_{0,c}(t_k) \cdot DF_{0,c}(t_k) - \sum_{k=1}^{19} CF_{i,c}(t_k) \cdot DF_{i,c}(t_k) + KAO_{i,c}$$

- (e) aggregate the economic value of equity losses $\Delta EVE_{i,c} > 0$ under a given interest rate scenario 'i', and taking the maximum loss across all interest rate scenarios, in accordance with the following formula:

$$\text{Standardised EVE risk measure} = \max_{i \in (1,2,\dots,6)} \left\{ \max \left(0; \sum_{c: \Delta EVE_{i,c} > 0} \frac{\Delta EVE_{i,c}}{\text{loss in currency } c} \right) \right\}$$

7.3.14 As an alternative to the use of the SA-IRRBB under Rule 7.3.1 to Rule 7.3.13, an Authorised Firm may use an internal measurement system to carry out the evaluation in Rule 7.2.1(1)(a) if the internal measurement system and its use have been approved in writing by the DFSA.

Guidance

1. The DFSA will usually only give its approval for the use of an internal measurement system if:
 - a. the use of the system to evaluate the Authorised Firm's Exposures to risks arising from potential changes in interest rates that affect the economic value of its Non-Trading Book activities has been approved by another appropriate regulator, or the DFSA is satisfied, having been provided by the Authorised Firm with such opinions from independent experts as it may require, that the system adequately measures all material sources of IRRBB and assesses the effect of market changes on the scope of its activities;
 - c. the use of the system is integrated into the governance and control framework of the Authorised Firm so that the Governing Body and senior management of the Authorised Firm receive and review appropriate reports in respect of the entity; and

- d. the DFSA is satisfied that the Authorised Firm's system is conceptually sound and is implemented with integrity.
2. In determining whether an internal measurement system meets the standard for approval, the DFSA will assess whether the system appropriately takes into account principle 4 (IRRBB measurement), principle 5 (behavioural and modelling assumptions) and principle 6 (data integrity and model governance) of chapter 31 of the Basel Committee on Banking Supervision's supervisory review process framework of 2019.

7.4 Non-Trading Book interest rate risk under chapter 10

- 7.4.1** In order to meet effectively the obligations specified under chapter 10 of PIB, which includes the need to address Non-Trading Book interest rate risk, an Authorised Firm is required to make a written record of its assessments made under Rules specified in chapter 10.

Guidance

1. This chapter 7 does not impose an explicit Capital Requirement relating to interest rate risk. Rather, the DFSA may impose an Individual Capital Requirement (ICR) under chapter 10 where it is of the view that the Authorised Firm's Capital Requirement is insufficient to address adequately all its risks, and in particular its Exposure to interest rate risk in the Non-Trading Book. The DFSA may also take other supervisory action, including requiring the Authorised Firm to reduce its IRRBB Exposures (for example, by hedging), raise additional capital, set constraints on the internal risk parameters used by the Authorised Firm or improve its risk management framework.
2. Sections 10.3 and 10.4 of chapter 10 require an Authorised Firm to submit IRAP and ICAAP assessments to the DFSA within 4 months of the end of the firm's financial year. The provisions also require the firm to ensure the assessments are documented in writing and to retain the records for at least 6 years.
3. An Authorised Firm's record of its approach to evaluating and managing interest rate risk in its Non-Trading Book, as part of its ICAAP should cover the following issues:
 - a. the internal definition of, and boundary between, "Non-Trading Book " and "Trading Book" in accordance with chapter 2 and App2;
 - b. the definition of economic value and its consistency with the method used to value assets and liabilities (e.g. discounted cashflows);
 - c. the size and the form of any internally selected interest rate scenario used to comply with Rule 7.2.1(1), together with its key modelling assumptions and their sensitivities;
 - d. the methodology and assumptions used to forecast the balance sheet when evaluating the impact on earnings under Rule 7.2.1(b);
 - e. the extent to which sensitivities to shocks can be scaled up on a linear basis without material loss of accuracy (i.e. covering both, convexity generally and the non-linearity of pay-off associated with basis and option risk);
 - f. the Authorised Firm's expectations for the exercise of options (explicit and implicit) by both, itself and its customers;

- g. the treatment of non-maturing deposits;
- h. the treatment of fixed rate assets (liabilities) where customers have a right to repay (redeem) early;
- i. the treatment of potential future business, including any related hedging;
- j. the analysis and impact of credit spread risk in the Non-Trading Book;
- k. the aggregation of multi-currency interest rate Exposures;
- l. the inclusion (or not) of non-interest bearing assets and liabilities (including capital and reserves);
- m. the degree of granularity employed (for example, offsets within a time bucket); and
- n. whether all future cash flows or only principal balances are included.

7.5 Systems and controls for Non-Trading Book interest rate risk

Non-Trading Book interest rate risk and credit spread risk policy

- 7.5.1** (1) An Authorised Firm must implement and maintain a policy which enables it to identify, assess, control and monitor its Non-Trading Book interest rate risk, including in the case of an Authorised Firm in Category 1, the risk arising from potential changes in interest rates that affect either or both of the following:
- (a) the economic value of its Non-Trading Book activities; or
 - (b) the earnings in respect of its Non-Trading Book activities.
- (2) An Authorised Firm must implement and maintain systems to monitor and assess the risks arising from potential changes in credit spreads in respect of its Non-Trading Book activities.
- (3) The policy in (1) must be documented and include an appropriate interest rate risk strategy as well as an enterprise-wide interest rate risk management framework appropriate to the nature, scale and complexity of its Non-Trading Book activities. The strategy and management framework must include:
- (a) an effective information system for accurate and timely identification, aggregation, monitoring and reporting of interest rate risk Exposures to the Governing Body and senior management of the Authorised Firm;
 - (b) effective tracking and reporting processes which ensure prompt action at the appropriate level of the Authorised Firm's senior management or, where necessary, Governing Body;
 - (c) a process for the measurement and monitoring of Non-Trading Book interest rate risk, using robust and consistent methods which enable

the Authorised Firm to implement the requirements set out in section 7.2 and section 7.3, as applicable;

- (d) a clearly defined risk appetite statement which sets out the aggregated level and types of interest rate risks Exposure in the Non-Trading Book that it is willing to accept, or intends to avoid, to achieve its business objectives;
 - (e) if the Authorised Firm uses any model for the purposes of this chapter, an effective model validation framework to ensure that the model has been adequately validated by a suitably qualified party which is independent of the development process so that it is conceptually sound and adequately captures all material risks. The validation framework must provide for:
 - (i) the evaluation of the conceptual and methodological soundness of the model;
 - (ii) the ongoing monitoring of the effectiveness and appropriateness of the use of the model;
 - (ii) an outcome analysis, including backtesting of key internal parameters; and
 - (iv) regular reviews at appropriate intervals of the model validation framework by the internal audit function; and
 - (f) a system for controlling and managing Non-Trading Book interest rate risk which enables it to comply with the overall risk management standards expected of an Authorised Firm and ensure continued compliance with the Rules in PIB.
- (4) An Authorised Firm must identify the Non-Trading Book interest rate risk impact of any new product, activity or service that it proposes to start and ensure that such impacts are duly addressed with adequate controls before the new product or activity is undertaken or introduced.
- (5) An Authorised Firm must review and update the policy in (1) at intervals that are appropriate to the nature, scale and complexity of its activities and market developments.

Guidance

1. The DFSA expects that an Authorised Firm's strategy towards IRRBB will set out the approach that the Authorised Firm will take towards management of the risk, including various quantitative and qualitative targets. It should be communicated to all relevant functions and staff within the organisation and be set out in the Authorised Firm's IRRBB policy.
2. The DFSA expects that an Authorised Firm's framework for managing IRRBB will address the following:

- a. the IRRBB management framework should be integrated into the Authorised Firm's enterprise-wide risk management framework, including but not limited to. integration with its daily risk management practices;
- b. the types of reports that the senior management and the Governing Body should receive under the framework. While these reports will vary based on an Authorised Firm's portfolio composition, the DFSA expects reports to include at least the following:
 - i. summaries of the Authorised Firm's aggregate interest rate risk Exposures and explanatory text that highlights the assets, liabilities, cash flows and strategies that are driving the level and direction of IRRBB;
 - ii. information demonstrating the Authorised Firm's compliance with its policies and limits;
 - iii. key modelling assumptions such as non-maturing deposit characteristics, prepayment on fixed loans and currency aggregation;
 - iv. results of stress tests, including sensitivity assessments to key assumptions and parameters; and
 - v. summaries of the reviews of IRRBB systems and controls and the adequacy of the measurement systems, including any findings of internal and external auditors or other equivalent external parties (such as consultants);
- c. an Authorised Firm's IRRBB measurement system should be clearly defined and consistent with the nature and complexity of its balance sheet structure;
- d. the risk appetite statement should clearly define authorised instruments, hedging strategies and risk-taking opportunities. The statement should include limits that are appropriate to the nature, size, complexity and capital adequacy of the Authorised Firm, as well as its ability to measure and manage its risks. Limits may be associated with specific scenarios of changes in interest rates or term structures, such as an increase or decrease of a particular size or a change in shape. The interest rate movements used in developing these limits should represent meaningful shock and stress situations taking into account historical interest rate volatility and the time required to mitigate those risk Exposures. Limits could be absolute in the sense that they should never be exceeded or whether, under specific circumstances, breaches of limits can be tolerated for a predetermined short period of time;
- e. the processes, procedures and limits should be clearly documented and should reflect a consideration of the interest rate risk associated with the balance sheet structure of the Authorised Firm, considering various asset and liability positions. These processes, procedures and limits should be reviewed and approved by appropriate levels of senior management; and
- f. the IRRBB framework should include measures to consider balancing cash flows and management of the risk's impact from new products or services through hedging using swaps or other derivatives. Any such major hedging or risk management initiatives should be approved in advance by the Asset Liability Committee (ALCO) or the Governing Body of the Authorised Firm.

Responsibilities of Governing Body

- 7.5.2**
- (1) An Authorised Firm must ensure that its Governing Body is responsible for monitoring the nature and level of Non-Trading Book interest rate risk assumed by the Authorised Firm and the process used to manage that risk.
 - (2) Without limiting the operation of (1), the responsibilities of an Authorised Firm's Governing Body in respect of the risk include:
 - (a) approving the Authorised Firm's Non-Trading Book interest rate risk policy, including its strategy and management framework;
 - (b) establishing and maintaining a senior management structure for the management of the risk and for ensuring compliance with the Authorised Firm's risk strategy;
 - (c) monitoring the Authorised Firm's overall Non-Trading Book interest rate risk profile on a regular basis and being aware of any material changes in the Authorised Firm's current or prospective profile; and
 - (d) ensuring that Non-Trading Book interest rate risk is adequately identified, assessed, mitigated, controlled and monitored.

Guidance

- 1. The Governing Body of the Authorised Firm may delegate responsibility for establishing IRRBB policies and strategies to the Asset and Liability Committee (ALCO) or an equivalent committee, which is the designated senior management committee for managing balance sheet structure and interest rate risk associated with it.
- 2. An Authorised Firm involved in banking activities or complex principal dealing activities should have a designated committee for design and implementation of IRRBB management.

8 GROUP RISK

Introduction

Guidance

1. This chapter deals with management of Group Risk exposure of an Authorised Firm. Group Risk refers to the risk of potential losses incurred by an Authorised Firm on account of its relationship with other members of its Financial Group, if it were to be part of one.
2. This chapter includes requirements that an Authorised Firm implement:
 - a. an effective management framework for Group Risk exposure;
 - b. a specified methodology for the calculation of Financial Group Capital Resources and Financial Group Capital Requirements.
3. This chapter also includes requirements limiting Financial Group exposures and restrictions on the ownership or control of deposit-taking firms.

8.1 Application

- 8.1.1** (1) This section and section 8.5 apply to an Authorised Firm in any Category.
- (2) Sections 8.2 to 8.4 apply only to an Authorised Firm in Category 1, 2 (other than a Matched Principal) or 5.

Guidance

1. Group membership may be a source of both strength and weakness to an Authorised Firm. The purpose of Group Risk requirements is to ensure that an Authorised Firm takes proper account of the risks related to the Authorised Firm's membership of a Group. The Group Risk requirements form a key part of the DFSA's overall approach to prudential supervision.
2. Section 8.5 imposes important restrictions on the ability of Authorised Firms in Category 3A, 3B, 3C, 3D or 4, and non-regulated Financial Institutions, to be a Parent of an Authorised Firm in Category 1 or 5, or of a firm carrying on similar activities outside of the DIFC.

Requirements by the DFSA

- 8.1.2** (1) The DFSA may require an Authorised Firm to:
- (a) form a Financial Group with any other entity within its Group; or
 - (b) include within its Financial Group any other entity within its Group;
- where the DFSA considers it necessary or desirable to do so in the interests of effective supervision of the Authorised Firm.
- (2) An Authorised Firm may, for the purposes of this section, exclude from its Financial Group any entity the inclusion of which would be misleading or inappropriate for the purposes of Financial Group supervision, provided the Authorised Firm has obtained the DFSA's prior written approval.

- (3) An Authorised Firm must provide to the DFSA, if requested, any of the following information in relation to its Group or Financial Group:
 - (a) details as to the entities within the Group or Financial Group;
 - (b) the structure of the Group or Financial Group; and
 - (c) the systems and controls in place to manage Group Risk.

Guidance

1. If more than one member of the same Group is subject to an obligation to provide information in respect of a position of the Group or Financial Group, one or more of those Authorised Firms may make application to the DFSA for an appropriate waiver or modification.
2. For the purposes of Rule 8.1.2, the DFSA would consider a range of factors when requiring an Authorised Firm to form a Financial Group. These factors would include regulatory risk factors, including but not limited to, (direct and indirect) participation, influence or contractual obligations, interconnectedness, intra group exposures, intra group services, regulatory status and legal framework.

8.2 Systems and controls requirements

8.2.1 An Authorised Firm in Category 1, 2 (other than a Matched Principal) or 5 that is a member of a Group must establish and maintain systems and controls for the purpose of:

- (a) monitoring the effect on the Authorised Firm of:
 - (i) its relationship with other members of its Group;
 - (ii) its membership in its Group; and
 - (iii) the activities of other members of its Group;
- (b) monitoring compliance with Financial Group supervision requirements below, including systems for the production of relevant data;
- (c) monitoring funding within the Group; and
- (d) monitoring compliance with Financial Group reporting requirements.

Guidance

For the purposes of the above requirement, an Authorised Firm may take into account its position within its Group. For instance, it would be reasonable for a small Authorised Firm within a larger Group to place some reliance on its parent to ensure that appropriate systems and controls are in place.

- 8.2.2** (1) An Authorised Firm must have systems and controls to enable it to determine and monitor:
- (a) its Financial Group Capital Requirement; and

- (b) whether the amount of its Financial Group Capital Resources is, and is likely to remain, greater than the amount of its Financial Group Capital Requirement.
- (2) Such systems and controls must include an analysis of:
 - (a) realistic scenarios which are relevant to the circumstances of the Financial Group; and
 - (b) the effects on the Financial Group Capital Requirement and on the Financial Group Capital Resources if those scenarios occurred.

8.3 Financial Group Capital Requirements and Financial Group Capital Resources

Guidance

By Rule 8.1.1(2), this section applies to an Authorised Firm in Category 1, 2 (other than a Matched Principal) or 5.

- 8.3.1** (1) The other Rules in section 8.3 do not apply to an Authorised Firm if:
- (a) the Authorised Firm's Financial Group is already the subject of Financial Group prudential supervision by the DFSA as a result of the authorisation of another Financial Group member;
 - (b) the DFSA has confirmed in writing, in response to an application from the Authorised Firm, that it is satisfied that the Authorised Firm's Group is the subject of consolidated prudential supervision by an appropriate regulator; or
 - (c) except where the DFSA has directed the inclusion of an entity pursuant to Rule 8.1.2, the percentage of total assets of Authorised Firms and Financial Institutions in the Financial Group is less than 40% of the total Financial Group assets.
- (2) If an Authorised Firm receives confirmation in writing from the DFSA in accordance with (1)(b), it must immediately advise the DFSA in writing if the circumstances upon which the confirmation was based change.
- 8.3.2** An Authorised Firm must ensure at all times that its Financial Group Capital Resources, as calculated in Rule 8.3.4, are equal to or in excess of its Financial Group Capital Requirement as calculated in Rule 8.3.3.

Guidance

If an Authorised Firm breaches Rule 8.3.2, the DFSA will take into account the full circumstances of the case, including any remedial steps taken by another regulator or the Authorised Firm, in determining what enforcement action, if any, it will take.

Financial Group Capital Requirement

- 8.3.3** (1) An Authorised Firm in Category 1, 2 (other than a Matched Principal) or 5 must calculate its Financial Group Capital Requirement by applying the accounting consolidation method, which calculates the Capital Requirement of the Financial Group based on the Financial Group's consolidated financial statements, and using applicable prudential Rules in PIB.
- (2) For the purposes of this Rule, the consolidated financial statements of the Financial Group must be prepared in accordance with International Financial Reporting Standards.

Financial Group Capital Resources

- 8.3.4** (1) An Authorised Firm in Category 1, 2 (other than a Matched Principal) or 5 must calculate its Financial Group Capital Resources by applying either of the following methods, excluding those amounts referred to in Rule 8.3.5:
- (a) the accounting consolidation method, which calculates the Capital Resources of the Financial Group based on the Financial Group's consolidated financial statements; or
 - (b) the aggregation method, which is the sum of:
 - (i) the Capital Resources of the Parent of the Financial Group;
 - (ii) subject to (2), the Capital Resources of any Authorised Firms and Financial Institutions included in the Financial Group; and
 - (iii) the Financial Group's proportionate share of Capital Resources in Financial Institutions not included in the Financial Group in which any member of the Financial Group has a participation.
- (2) For the purposes of (1)(b)(ii), an investment by one Financial Group member in another must not be included.

Guidance

The calculation of Financial Group Capital Resources is subject to the provisions in part 3 of chapter 3.

- 8.3.5** When calculating the Financial Group Capital Resources of a Financial Group, an Authorised Firm must not include Capital Resources or Adjusted Capital Resources (as the case may be) of subsidiaries or participations of that Financial Group to the extent that those Capital Resources or Adjusted Capital Resources:
- (a) exceed the entity requirement in respect of that subsidiary or participation, calculated in accordance with Rule 8.3.3; and
 - (b) are not freely transferable within the Financial Group.

Guidance

1. Because the Financial Group Capital Requirement set out in Rule 8.3.3 includes Capital Requirements in respect of Group entities, Capital Resources may be included in the calculation of Financial Group Capital Resources to the extent of those requirements. Capital that is surplus

to those requirements is, however, subject to an additional condition before it may be taken into account for the purposes of Financial Group capital adequacy.

2. In general, Capital Resources or Adjusted Capital Resources are considered not to be freely transferable if they are subject to a legal or constructive limitation on their transferability, whether that transfer would be made by dividend, return of capital or other form of distribution. Examples of relevant limitations might include obligations to maintain minimum Capital Requirements to meet domestic solvency requirements, or to comply with debt covenants.

8.3.6 Deductions for Qualifying Holdings under section 3.17 may be calculated based on the Group's total T1 and T2 Capital.

8.4 Financial Group Concentration Risk limits

8.4.1 An Authorised Firm in Category 1, 2 (other than a Matched Principal) or 5 must ensure that its Financial Group Exposure, including the Financial Group's PSIAu's, to a Counterparty or group of Closely Related Counterparties does not exceed 25% of its Group's Tier 1 Capital.

8.4.2 An Authorised Firm in Category 1, 2 (other than a Matched Principal) or 5 must ensure that the sum of its Financial Group Large Exposures, including the Financial Group's PSIAu's, to a Counterparty or group of Closely Related Counterparties does not exceed 800% of its Financial Group's Tier 1 Capital.

8.5 Restrictions on ownership or control

Guidance

By Rule 8.1.1(1), this section applies to an Authorised Firm in any Category.

Parents of Category 1 and 5 Authorised Firms

- 8.5.1**
- (1) No entity other than one of the following may be the Parent of, or any of the Parents of, an Authorised Firm in Category 1 or 5:
 - (a) another Authorised Firm in Category 1 or 5; or
 - (b) a Regulated Financial Institution licensed to carry on the activities of accepting deposits.
 - (2) An entity other than one referred to in (1)(a) or (b) may be the Parent of an Authorised Firm in Category 1 or 5 where the ultimate, or any intermediate, Parent is an entity of the type specified under 1(a) or (b).

Restrictions on Category 2, 3A, 3B, 3C, 3D and 4 Authorised Firms

- 8.5.2**
- (1) An Authorised Firm in Category 2 (if it meets the definition of Matched Principal), 3A, 3B, 3C, 3D or 4 must not, subject to (2), be a Parent of an entity that:
 - (a) is an Authorised Firm in Category 1 or 5 or an Insurer; or

- (b) carries on activities that would, if conducted in the DIFC, constitute Accepting Deposits, Managing a PSIA which is a PSIAu, or Effecting or Carrying out Contracts of Insurance.
- (2) An Authorised Firm in Category 2 (if it meets the definition of Matched Principal), 3A, 3B, 3C, 3D or 4 may own or control an entity referred to in (1) where it is itself a Subsidiary of:
 - (a) a Regulated Financial Institution licensed to carry on any one or more of the activities specified in (1)(b); or
 - (b) an Authorised Firm in Category 1 or 5 or an Insurer.

9 LIQUIDITY RISK

Introduction

Guidance

1. This chapter deals with management of Liquidity Risk by an Authorised Firm. Liquidity Risk refers to the risk of potential losses incurred by an Authorised Firm's failure to have liquid assets to ensure payment of all its liabilities as they fall due and be in a position to meet all payments required to sustain its business on a planned growth path.
2. This chapter requires an Authorised Firm to:
 - a. maintain and implement a Liquidity Risk policy;
 - b. identify, measure and monitor Liquidity Risk;
 - c. maintain a minimum level of High Quality Liquid Assets (HQLA);
 - d. determine quantitative limits on cumulative negative maturity mismatch in accordance with a specified methodology; and
 - e. maintain a minimum Net Stable Funding Ratio.

9.1 Application

- 9.1.1** (1) This chapter applies to an Authorised Firm in Category 1, 2 (other than a Matched Principal) or 5.
- (2) Only Rule 9.2.2(3) applies to an Authorised Firm in Category 2 (other than a Matched Principal).

Guidance

In accordance with Rules 3.2.2 or 3.2.4, an Authorised Firm is required to ensure that there is no significant risk that liabilities cannot be met as they fall due. With specific reference to liquidity, an Authorised Firm may meet its obligations in a number of ways, including:

- a. by holding sufficient immediately available cash or unencumbered readily marketable assets; and
- b. by securing an appropriate matching future profile of cashflows.

9.2 Liquidity Risk policy, systems and controls

- 9.2.1** (1) An Authorised Firm must establish and maintain a Liquidity Risk policy.

- (1A) An Authorised Firm must ensure the policy is in writing and is approved at least annually by its Governing Body.
- (1B) The policy must set out the level of Liquidity Risk the Authorised Firm is willing to tolerate, which must be in line with its business objectives, strategy and overall risk tolerance.
- (2) The policy must include systems and controls for intra-day, daily, short-term, medium-term and long-term management of Liquidity Risk appropriate to the nature, scale and complexity of the activities conducted by the firm.
- (3) The systems and controls referred to in (2) must include:
 - (a) a system for identifying and assessing Liquidity Risk in accordance with Rule 9.2.4;
 - (b) a system for the measurement and monitoring of Liquidity Risk using a robust and consistent method which enables the Authorised Firm to implement the requirements set out in Rule 9.2.5;
 - (c) a system for controlling Liquidity Risk which enables the Authorised Firm to implement the requirements set out in Rule 9.2.6;
 - (d) a system for collateral management and asset encumbrance which is able to adequately identify, monitor and manage the risks associated with these activities in accordance with Rule 9.2.8;
 - (e) a system for adequate allocation of liquidity costs, benefits and risks that meets the requirements set out in Rule 9.2.9; and
 - (f) a system to manage intra-day liquidity positions effectively and meet the requirements in Rule 9.2.10.
- (4) An Authorised Firm must ensure that it has risk management systems to implement the policy.

Guidance

- 1. The DFSA expects that an Authorised Firm's Liquidity Risk policy will set out the approach that the Authorised Firm will take to Liquidity Risk management, including various quantitative and qualitative targets. It should be communicated to all relevant functions and staff within the organisation.
- 2. The level of Liquidity Risk tolerance should ensure that the Authorised Firm manages its liquidity and funding risk prudently in normal times in a way that allows it to withstand periods of stress. The level of Liquidity Risk tolerance should be expressed in qualitative and quantitative terms that are clear enough for all levels of management to be able to understand the trade-off between risks and profits.
- 3. The DFSA expects that an Authorised Firm will integrate its Liquidity Risk policy within its overall risk management framework and that its policy will take into account the need to:
 - a. develop liquidity management processes and procedures to implement the Authorised Firm's stated Liquidity Risk tolerance;

- b. ensure that the Authorised Firm maintains sufficient liquidity resources at all times to meet its ongoing liquidity obligations and withstand a period of individual or market-wide stress;
- c. determine the structure, responsibilities and controls for managing Liquidity Risk and for overseeing the liquidity positions of all branches and subsidiaries in the jurisdictions in which the Authorised Firm is active, and outline these elements clearly in the Authorised Firm's liquidity policies;
- d. have in place adequate internal controls to ensure the integrity of its Liquidity Risk management processes;
- e. ensure that stress tests, contingency funding plans and holdings of liquid assets are effective and appropriate for the Authorised Firm's business model, funding strategy, complexity of its on- and off-balance sheet activities and funding mismatches. The Authorised Firm should also make appropriate assumptions in relation to the marketability of liquid assets under various stress scenarios;
- f. establish a set of reporting criteria, specifying the scope, manner and frequency of reporting to various recipients (such as the Governing Body, senior management and the asset/liability committee) and who is responsible for preparing the reports. The reporting should include a comprehensive system for projecting cash flows arising from assets, liabilities and off-balance sheet items, both consolidated and at the entity level, over an appropriate set of time horizons;
- g. establish the specific procedures and approvals necessary for exceptions to policies and limits, including the escalation procedures and follow-up actions to be taken for breaches of limits;
- h. monitor closely current trends and potential market developments that may present significant, unprecedented and complex challenges for managing Liquidity Risk so that appropriate and prompt changes to the liquidity management strategy can be made as needed;
- i. continuously review information on the Authorised Firm's liquidity developments and report regularly to the Governing Body; and
- j. maintain an independent and competent internal control function and conduct regular internal audit reviews to ensure the integrity and effectiveness of the Liquidity Risk policy.

- 9.2.2** (1) An Authorised Firm must ensure that its Governing Body is ultimately responsible for the Liquidity Risk assumed by the firm as well as the adequacy of systems, controls and processes used to manage that risk.
- (2) Without limiting the operation of (1), the responsibilities of an Authorised Firm's Governing Body in respect of Liquidity Risk include:
- (a) approving the Authorised Firm's Liquidity Risk policy;
 - (b) establishing and maintaining a senior management structure with clearly defined responsibilities and roles for the management of Liquidity Risk and for ensuring compliance with the Authorised Firm's Liquidity Risk policy;

- (c) ensuring the senior management in (b) and other relevant personnel have the necessary experience to manage Liquidity Risk;
- (d) monitoring the Authorised Firm's overall Liquidity Risk profile on a regular basis by receiving adequate reporting and being aware of any material changes in the Authorised Firm's current or prospective Liquidity Risk profile;
- (e) ensuring that Liquidity Risk is adequately identified, assessed, mitigated, controlled and monitored in accordance with the Authorised Firm's Liquidity Risk policy;
- (f) ensuring that the Liquidity Risk policy is documented; and
- (g) ensuring that the Liquidity Risk policy is reviewed at least annually.

Guidance

1. Senior management and the Governing Body of an Authorised Firm are expected to demonstrate a thorough understanding of the links between funding liquidity risk and market liquidity risk, as well as how other risks, including credit, market, operational and reputation risks, affect the Authorised Firm's overall Liquidity Risk policy.
2. Senior management should ensure that all business units with activities that have an impact on Liquidity Risk are aware of the Liquidity Risk policy and limits.
3. Senior management should ensure that the Liquidity Risk policy outlines clearly the structure, responsibilities and controls for managing Liquidity Risk in and across different jurisdictions, legal entities and branches. They should also ensure that the structure, responsibilities and controls take into account legal, operational, regulatory, reputational and other constraints on liquidity transfer.

Requirements imposed on a Category 2 firm

- (3) An Authorised Firm in Category 2 (other than a Matched Principal) must:
- (a) establish and maintain a senior management structure to manage Liquidity Risk;
 - (b) identify, assess, mitigate, control and monitor Liquidity Risk; and
 - (c) monitor the Authorised Firm's overall Liquidity Risk profile on a regular basis.

Guidance

In respect of Rule 9.2.2(2)(b), senior management are expected to:

- a. oversee the development, establishment and maintenance of procedures and practices that translate the goals, objectives and risk tolerances approved by the Governing Body into operating standards that are consistent with the Governing Body's intent and which are understood by the relevant members of an Authorised Firm's staff;
- b. adhere to the lines of authority and responsibility that the Governing Body has established for managing Liquidity Risk;

- c. oversee the establishment and maintenance of management information and other systems that identify, assess, control and monitor the Authorised Firm's Liquidity Risk; and
- d. oversee the establishment of effective internal controls over the Liquidity Risk management process.

- 9.2.3** (1) An Authorised Firm may delegate the day-to-day management of its Liquidity Risk to another entity in the same Group for management on a Group basis only if:
- (a) the Governing Body of the Authorised Firm:
 - (i) has formally approved the delegation;
 - (ii) keeps the delegation under review; and
 - (b) the Authorised Firm notifies the DFSA in writing of the delegation immediately upon its being made.
- (2) If an Authorised Firm delegates the management of its Liquidity Risk in accordance with (1), the requirements in this chapter continue to apply to the Authorised Firm.
- (3) An Authorised Firm must revoke a delegation referred to in (1) and bring day-to-day Liquidity Risk management back within the Authorised Firm if the DFSA requests it in writing to do so.

Guidance

If Liquidity Risk management is delegated as set out in Rule 9.2.3, responsibility for its effectiveness remains with the Authorised Firm's Governing Body.

Identifying Liquidity Risk

- 9.2.4** (1) An Authorised Firm must comply with the requirements in this Rule in implementing its systems and controls referred to in Rule 9.2.1(2) and (3).
- (2) An Authorised Firm must assess the cash flows for its assets, liabilities and off-balance sheet items under both normal market conditions and stressed conditions resulting from either general market turbulence or firm-specific difficulties.
- (3) An Authorised Firm must assess the extent to which committed facilities can be relied upon under stressed conditions identified in accordance with Rule 9.2A.3.
- (4) An Authorised Firm must consider potential liability concentrations when determining the appropriate mix of liabilities.
- (5) An Authorised Firm must identify the Liquidity Risk across all legal entities, branches and subsidiaries and in all jurisdictions in which it operates.
- (6) If an Authorised Firm has significant, unhedged liquidity mismatches in particular currencies, it must assess:

- (a) the volatilities of the exchange rates of the mismatched currencies;
- (b) likely access to the foreign exchange markets in normal and stressed conditions; and
- (c) the stability of deposits in those currencies with the Authorised Firm in stressed conditions.

Guidance

1. As part of the assessment for the purposes of Rule 9.2.4(2), an Authorised Firm should:
 - a. identify significant concentrations within its asset portfolio; and
 - b. value the assets conservatively, taking into account the likely deterioration in the value of assets under market-wide stress conditions.
2. For the purposes of Rule 9.2.4(4), an Authorised Firm should consider factors including:
 - a. the term structure of its liabilities;
 - b. the credit-sensitivity of its liabilities;
 - c. the mix of secured and unsecured funding;
 - d. concentrations among its liability providers or related Groups of liability providers;
 - e. reliance on particular instruments or products;
 - f. the geographical location of liability providers; and
 - g. reliance on intra-Group funding.
3. As appropriate, an Authorised Firm would be expected to consider the amount of funding required by:
 - a. commitments given;
 - b. standby facilities given;
 - c. wholesale overdraft facilities given;
 - d. proprietary derivatives positions; and
 - e. liquidity facilities given for securitisation transactions.

Measuring and monitoring Liquidity Risk

- 9.2.5** (1) An Authorised Firm must ensure that the method referred to in Rule 9.2.1(3)(b) for measuring Liquidity Risk is capable of:
- (a) measuring the extent of the Liquidity Risk it is incurring;
 - (b) tracking early warning indicators to aid the Liquidity Risk management processes;

- (c) dealing with the dynamic aspects of the Authorised Firm's liquidity profile;
 - (d) where appropriate, measuring the Authorised Firm's Exposure to Foreign Currency Liquidity Risk; and
 - (e) where appropriate, measuring the Authorised Firm's Exposure to PSIA and Islamic Contract Liquidity Risk.
- (2) An Authorised Firm must establish and maintain a system of management reporting which provides relevant, accurate, comprehensive, timely, forward looking and reliable Liquidity Risk reports to relevant functions within the Authorised Firm.
- (3) The method for measuring Liquidity Risk under (1)(a) must enable the Authorised Firm to forecast prospective cash flows for assets, liabilities, off-balance sheet commitments and contingent liabilities over a variety of time horizons, under both normal conditions and a range of stress scenarios, including scenarios of severe stress.

Guidance

1. An Authorised Firm, in measuring its Liquidity Risk under Rule 9.2.5(1)(a), should ensure that:
 - (a) the variety of time horizons cover changes in liquidity needs and funding capacity on an intra-day, daily, short-term, medium-term and long-term basis;
 - (b) it considers the vulnerabilities of cash flows to events, activities and business strategies;
 - (c) its dynamic cash flow forecasts are carried out at a sufficiently detailed level and include assumptions on the actions of key counterparties in response to changes in operating conditions;
 - (d) cash flows in all significant foreign currencies are measured on an aggregate basis, as well as at the individual currency level, taking into account stressed conditions affecting foreign exchange markets;
 - (e) it captures the impact of providing correspondent, custody and settlement activities on cash flows; and
 - (f) assumptions used to determine future liquidity and funding needs are realistic and reflect the complexities of the underlying businesses, products and markets.
2. Early warning indicators should be designed to assist the Authorised Firm to identify any negative trends in its liquidity position and to assist its management to assess and respond to mitigate its exposure to those trends.
3. Management information should include the following:
 - a. a cash-flow or funding gap report on an aggregate basis and by currency, legal entity and country;
 - b. a funding maturity schedule;

- c. a list of large providers of funding;
 - d. reports on Collateral and encumbered assets to enable compliance with Rule 9.2.8;
 - e. a liquidity costs, benefits and risks allocation report to assist compliance with Rule 9.2.9;
 - f. intra-day liquidity reports to assist compliance with Rule 9.2.10;
 - g. where appropriate, a schedule of Islamic funding sources;
 - h. a limit monitoring and exception report;
 - i. asset quality and trends;
 - j. earnings projections; and
 - k. the Authorised Firm's reputation in the market and the condition of the market itself.
4. An Authorised Firm should be able to generate critical liquidity reports on a daily basis, including in times of stress.
 5. Where an Authorised Firm is a member of a Group, it should be able to assess the potential impact on it of Liquidity Risk arising in other parts of the Group.
 6. Where an Authorised Firm has subsidiaries or branches, it should be able to monitor and control Liquidity Risk at the individual branch or subsidiary level and on a consolidated level taking into account legal, operational, regulatory, reputational and other relevant constraints.

Controlling Liquidity Risk

9.2.6 An Authorised Firm must ensure that the system referred to in Rule 9.2.1(3)(c):

- (a) enables the Authorised Firm's Governing Body and senior management to review compliance with limits set in accordance with Rule 9.2.7 and operating procedures; and
- (b) has appropriate approval processes, limits and other mechanisms designed to provide reasonable assurance that the Authorised Firm's Liquidity Risk management processes are adhered to.

- 9.2.7**
- (1) An Authorised Firm must hold sufficient liquidity resources and ensure that its Governing Body sets appropriate liquidity limits to manage its Liquidity Risk effectively under both day-to-day and stressed conditions.
 - (2) An Authorised Firm must periodically review and, where appropriate, adjust the limits referred to in (1) when its Liquidity Risk policy changes.
 - (3) An Authorised Firm must promptly escalate and resolve any policy or limit exceptions according to the processes described in its Liquidity Risk policy.

Guidance

An Authorised Firm should set limits to control its liquidity risk exposure and vulnerabilities. Limits and corresponding escalation procedures should be reviewed regularly. Limits should be relevant to the business in terms of its location, complexity of activity, nature of products, currencies and markets

served. If an Authorised Firm breaches a liquidity risk limit, it should implement a plan to review its exposure and reduce it to a level that is within the limit.

Management of collateralised and encumbered assets

- 9.2.8**
- (1) An Authorised Firm must prudently manage its collateral positions using a collateral management system.
 - (2) The Authorised Firm's collateral management system must be able to:
 - (a) distinguish between pledged and unencumbered assets, including during periods of liquidity stress;
 - (b) take into account the legal entity in which liquid assets reside; and
 - (c) identify, in a timely manner, the countries where assets are legally recorded and any restrictions imposed on their transfer or liquidation.
 - (3) An Authorised Firm must manage its encumbered balance sheet assets within prudent limits to minimise the impact on its liquidity position and funding cost.
 - (4) For the purposes of (3), the Authorised Firm's system supporting the management of encumbered assets must be able to provide information on:
 - (a) the current and expected level and types of asset encumbrance and related transactions;
 - (b) the nature of unencumbered assets including amount, location and credit quality;
 - (c) the capacity for further asset encumbrance, including available unencumbered assets and the potential liquidity that can be generated; and
 - (d) the expected amount, level and type of additional encumbrance that may result from stress scenarios.

Allocation of liquidity costs, benefits and risks

- 9.2.9**
- An Authorised Firm must ensure that the system referred to in Rule 9.2.1(3)(e):
- (a) incorporates liquidity costs, benefits and risks in internal pricing, performance measurement, and new product approval processes for all significant business activities both on- and off-balance sheet;
 - (b) assigns appropriate liquidity charges to positions, portfolios and transactions. The liquidity charge must incorporate factors relating to the holding period of assets and liabilities, market liquidity characteristics, stability of the funding source and any other relevant factor;

- (c) provides quantification and attribution of Liquidity Risk that is explicit, transparent and takes into account liquidity under stressed conditions; and
- (d) is reviewed periodically to reflect changing business and market conditions.

Intra-day liquidity

- 9.2.10** (1) An Authorised Firm must manage its intra-day liquidity positions prudently to ensure that it is able to meet its settlement and payment obligations in a timely manner under business as usual and stressed conditions, in all material currencies and active markets.
- (2) For the purposes of (1), an Authorised Firm must be reasonably able to:
- (a) identify and prioritise the most time critical payment and settlement obligations;
 - (b) measure daily gross liquidity inflows and outflows and any potential funding gaps;
 - (c) identify cash flow timings and shortfalls at different points in time during the day;
 - (d) manage the timing of cash outflows to give priority to time critical payments; and
 - (e) obtain sufficient intra-day funding, including intra-day liquidity facilities from correspondent banks or Central Banks.

9.2A Funding strategy, stress testing and contingency funding plan

Funding strategy

- 9.2A.1** (1) An Authorised Firm must develop a funding strategy that provides effective diversification in the sources and nature of its funding.
- (2) An Authorised Firm must ensure that the funding strategy is in writing and is approved by its Governing Body.
- (3) The funding strategy must be in line with the Authorised Firm's stated Liquidity Risk tolerance and supported by robust assumptions that are consistent with the Authorised Firm's budgeting and business planning process.
- (4) The funding strategy must be supported by systems that allow the Authorised Firm to identify, measure, manage and monitor funding positions.
- (5) An Authorised Firm must ensure that its funding strategy is reviewed regularly and at least annually, and is updated as necessary in light of changed funding conditions and any change in the Authorised Firm's strategy.

- (6) An Authorised Firm must notify the DFSA in writing immediately of any material changes to the Authorised Firm's funding strategy.

Guidance

1. The diversification under Rule 9.2A.1(1) should include, for example, different counterparties, instruments, currencies, geographies and markets.
2. The assumptions in Rule 9.2A.1(3) should be forward looking and take into account the macroeconomic and market conditions in which the Authorised Firm operates and any other factors that are likely to impact its funding position.
3. The DFSA expects that funding positions referred to in Rule 9.2A.1(4) would cover both present and projected positions across multiple time horizons generated from both on- and off-balance sheet items.
4. In order to formulate the funding strategy properly, an Authorised Firm should pay attention to other risks, including, for example, credit, market, operational and reputational risk and their impact on funding requirements.
5. An Authorised Firm should maintain an ongoing presence in its chosen funding markets and strong relationships with funds providers.

- 9.2A.2** (1) An Authorised Firm must assess market access under a variety of normal and stressed conditions.
- (2) An Authorised Firm must assess regularly its capacity to raise funds quickly including on a secured and unsecured basis.
- (3) An Authorised Firm must:
- (a) identify the main factors that affect its ability to raise funds; and
 - (b) monitor those factors closely to ensure that estimates of fund raising capacity remain valid.

Stress testing

- 9.2A.3** (1) An Authorised Firm must conduct stress tests regularly to identify sources of potential liquidity strain and to ensure that its exposures remain within its Liquidity Risk tolerance.
- (2) When using stress testing in accordance with (1), an Authorised Firm must:
- (a) use scenarios based on varying degrees of short-term and protracted institution-specific and market-wide stress (individually and in combination); and
 - (b) include a cash-flow projection for each scenario tested, based on reasonable estimates of the impact (both on and off-balance sheet) of that scenario on the Authorised Firm's funding needs and sources.

- (3) An Authorised Firm must fully document its stress test scenarios and related assumptions, and review the scenarios and assumptions, at least annually, to ensure they remain appropriate.

Guidance

1. An Authorised Firm should consider carefully the design of stress scenarios and the variety of shocks used. Regardless of how strong its current liquidity situation appears to be, it should take a conservative approach when setting stress testing assumptions. It should consider the potential impact of severe stress scenarios and how they would affect the following Liquidity Risk drivers as applicable to the firm's operations:
 - a. retail funding risk;
 - b. wholesale secured and unsecured funding risk;
 - c. risks arising from funding markets;
 - d. lack of diversification between funding types;
 - e. off-balance sheet funding risk;
 - f. risks arising from the firm's funding tenors;
 - g. risks associated with a deterioration of the firm's credit rating;
 - h. cross-currency funding risk;
 - i. risk that liquidity resources cannot be transferred across entities, sectors and countries;
 - j. funding risks resulting from estimates of future balance sheet growth;
 - k. reputational risk;
 - l. marketable and non-marketable assets risk; and
 - m. intra-day payment and settlement risk.
2. Market-wide stress scenarios under 9.2A.3(2)(a) should include:
 - a. a simultaneous drying up of market liquidity in several previously highly liquid markets;
 - b. severe constraints in accessing secured and unsecured funding;
 - c. restrictions on currency convertibility; and
 - d. severe operational or settlement disruptions affecting one or more payment or settlement systems.
3. The identification of the possible balance sheet and off-balance sheet impact referred to in Rule 9.2A.3(2)(b) should take into account:
 - a. possible changes in the market's perception of the Authorised Firm and the effects that this might have on the Authorised Firm's access to the markets, including:

- i. where the Authorised Firm funds its holdings of assets in one currency with liabilities in another, access to foreign exchange markets, particularly in less frequently traded currencies;
 - ii. access to secured funding, including by way of repurchase agreement transactions; and
 - iii. the extent to which the Authorised Firm may rely on committed facilities made available to it;
 - b. whenever applicable the possible effect of each scenario tested on currencies whose exchange rates are currently pegged or fixed; and
 - c. that:
 - i. general market turbulence may trigger a substantial increase in the extent to which persons exercise rights against the Authorised Firm under off-balance sheet instruments to which the Authorised Firm is party;
 - ii. access to OTC derivative and foreign exchange markets is sensitive to credit-ratings;
 - iii. Early Amortisation in asset securitisation transactions with which the Authorised Firm has a connection may be triggered;
 - iv. its ability to securitise assets may be reduced; and
 - v. there may be a potential need to buy back debt or honour non-contractual obligations to mitigate reputational risk.
- 4. An Authorised Firm is required to conduct stress tests regularly. The frequency with which an Authorised Firm should conduct stress tests will depend on the risks to the particular Authorised Firm. For some Authorised Firms, it may be adequate to conduct tests annually, but, for others, it may be necessary to conduct tests more frequently e.g. quarterly.

9.2A.4 An Authorised Firm must ensure that stress tests conducted under Rule 9.2A.3 enable it to analyse the impact of stress scenarios on its liquidity positions, as well as on the liquidity positions of its individual business lines.

- 9.2A.5** (1) An Authorised Firm must ensure that results of the stress tests are integrated into its strategic planning process and its day-to-day risk management practices.
- (2) An Authorised Firm must apply the results of the stress tests:
- (a) to adjust its liquidity management strategy, policies and positions, including to determine an appropriate buffer of HQLA;
 - (b) for the setting of internal limits; and
 - (c) for the purpose of the IRAP and ICAAP assessments under chapter 10, where applicable.

- (3) An Authorised Firm must incorporate the stress test results in assessing and planning for related potential funding shortfalls in its Contingency Funding Plan.
- (4) An Authorised Firm must ensure that the stress test results and vulnerabilities and any resulting actions are reported to, and discussed with, its Governing Body and the DFSA.

Guidance

If the DFSA considers that an Authorised Firm has not carried out effective stress tests under Rules 9.2A.3 to 9.2A.5, it may use its power under Article 75A of the Regulatory Law to require the Authorised Firm to maintain a buffer of liquid assets in addition to that required under section 9.3.

Contingency Funding Plan

Guidance

A Contingency Funding Plan, or CFP, is a compilation of policies, procedures and action plans for responding to severe disruptions to an Authorised Firm's ability to meet its liabilities as they fall due or its ability to fund some or all of its activities quickly and at a reasonable cost.

- 9.2A.6**
- (1) An Authorised Firm must have a documented Contingency Funding Plan (CFP) that sets out clearly its strategies for addressing liquidity shortfalls in emergency situations.
 - (2) An Authorised Firm must ensure that its CFP is in writing and is approved by its Governing Body.
 - (3) The CFP must be commensurate with an Authorised Firm's complexity, risk profile and scope of operations and its role in the financial systems in which it operates.
 - (4) The CFP must:
 - (a) list the events or circumstances that will lead the Authorised Firm to put any part of the plan into action;
 - (b) set out available potential contingency funding sources and the amount of funds an Authorised Firm estimates can be derived from these sources;
 - (c) estimate the lead time needed to tap additional funds from each of the contingency sources;
 - (d) set out the extent to which the plan relies upon:
 - (i) asset sales, using assets as Collateral on secured funding (including repurchase agreements), securitising its assets or otherwise reducing its assets;
 - (ii) modifying the structure of, or increasing, its liabilities; and
 - (iii) the use of committed facilities; and

- (e) contain clear administrative policies and procedures that will enable the Authorised Firm to manage the implementation of the plan, including:
 - (i) the roles and responsibilities of senior management, including who has the authority to invoke the CFP;
 - (ii) the names, location and contact details of members of the team responsible for implementing the plan;
 - (iii) the details of who is responsible for contact with the Authorised Firm's head office (if appropriate), analysts, investors, external auditors, media, significant customers, regulators and others; and
 - (iv) the mechanisms that enable senior management and the Governing Body to receive relevant, accurate, comprehensive, timely and reliable management information.

Guidance

1. The CFP should provide a framework with a high degree of flexibility so that an Authorised Firm can respond quickly in a variety of situations.
2. The CFP's design, plans and procedures should be closely integrated with the Authorised Firm's ongoing analysis of Liquidity Risk and with the results of the scenarios and assumptions used in stress tests.
3. The CFP should assist the Authorised Firm to manage a range of scenarios of severe liquidity stress that include both firm-specific and more generalised market-wide stress, as well as the potential interaction between them.
4. The CFP should, for each of the tested scenarios, demonstrate that the Authorised Firm has sufficient liquid financial resources to meet its liabilities over a range of different time periods, including intraday.

9.2A.7 An Authorised Firm must ensure that its CFP accounts for:

- (a) the impact of stressed market conditions on its ability to sell or securitise assets;
- (b) the link between asset liquidity and funding liquidity;
- (c) second round and reputational effects related to execution of contingency funding measures; and
- (d) the potential to transfer liquidity across Group entities, borders and lines of business, taking into account legal, regulatory, operational and time zone constraints.

9.2A.8 (1) An Authorised Firm must review and test its CFP regularly to ensure it is effective and operationally feasible.

- (2) For the purposes of (1), an Authorised Firm must review and update its CFP at least annually for approval by its Governing Body, or more frequently if required by business or market circumstances.

Guidance

1. Key aspects of CFP testing include ensuring that roles and responsibilities are appropriate and understood, confirming that contact information is up to date, proving the transferability of cash and collateral (especially across borders and entities) and ensuring that the necessary legal and operational documentation is in place to execute the plan at short notice.
2. An Authorised Firm should test key assumptions regularly, such as its ability to sell or repo certain assets or periodically draw down credit lines.

- 9.2A.9** An Authorised Firm must ensure that its CFP is consistent with its business continuity and disaster recovery arrangements and can operate in situations where business continuity arrangements have been invoked.

Guidance

1. See GEN chapter 5 regarding requirements relating to an Authorised Firm's business continuity and disaster recovery arrangements.
2. An Authorised Firm should ensure effective coordination between teams managing issues surrounding liquidity crises and business continuity. Liquidity crisis team members and alternates should have ready access to CFPs on-site and off-site.

9.3 Liquidity requirements

- 9.3.1**
- (1) This section applies to an Authorised Firm in Category 1 or 5.
 - (2) The Rules in this section apply, except as provided in (3), to an Authorised Firm on a solo basis.
 - (3) The DFSA may require an Authorised Firm to apply the requirements in this section to its Financial Group, if the Authorised Firm and its Financial Group are subject to consolidated supervision.

Global liquidity concession

- 9.3.2**
- (1) An Authorised Firm which carries on business in or from the DIFC through a Branch may apply to the DFSA for a global liquidity concession.
 - (2) An application for a global liquidity concession must be made in accordance with the requirements in section A9.1 of App9.
 - (3) If the DFSA grants a global liquidity concession to an Authorised Firm, that Authorised Firm need not comply with all or any of the requirements of this section as specified by the DFSA in the concession.
 - (4) The DFSA may specify the period for which a global liquidity concession is valid.

HQLA requirement

- 9.3.3** An Authorised Firm must maintain an adequate level of HQLA to meet its liquidity needs for, at a minimum, a 30 calendar day period under a severe stress scenario.

Guidance

Rules A9.2.2 to A9.2.9 in App9 set out the conditions that must be met for assets to be treated as HQLA.

Liquidity Coverage Ratio

- 9.3.4** An Authorised Firm must, except as provided in Rule 9.3.8, maintain a LCR of at least the level specified in the table below from the date specified in the table.

Table - Minimum LCR levels

Date	1st January 2015	1st January 2016	1st January 2017	1st January 2018	1st January 2019
Minimum LCR	60%	70%	80%	90%	100%

Guidance

Under Rule 9.3.4, an Authorised Firm must maintain a minimum level of LCR of 60% starting on 1 January 2015. The minimum requirement will be increased subsequently in each following year in equal annual steps of 10% to reach 100% on and from 1 January 2019. Rule 9.3.4 sets minimum levels and is not intended to limit the generality of the requirement in Rule 9.3.3.

- 9.3.5** An Authorised Firm must calculate its LCR using the following formula and in accordance with the Rules in section A9.2 of App9.

$$\text{LCR} = \frac{\text{Value of stock of HQLA}}{\text{Total Net Cash Outflows over the next 30 calendar days}}$$

Guidance

1. Section A9.2 of App9 sets out how the value of stock of HQLA and Total Net Cash Outflows are to be calculated.
2. An Authorised Firm active in multiple currencies should:
 - a. maintain HQLA consistent with the distribution of its liquidity needs by currency;
 - b. assess its aggregate foreign currency liquidity needs and determine an acceptable level of currency mismatches; and
 - c. undertake a separate analysis of its strategy for each currency in which it has material activities, considering potential constraints in times of stress.

Individual Liquidity Requirement

- 9.3.6** (1) The DFSA may by written notice to an Authorised Firm:
- (a) adjust the LCR Requirement or NSFR Requirement;
 - (b) adjust requirements under section A9.2 of App9 for calculating the Authorised Firm's stock of HQLA or the Total Net Cash Outflows, or under section A9.4 of App9 for calculating its ASF or RSF;
 - (c) alter the calculation methodologies or parameters for the purposes of the LCR Requirement or NSFR Requirement;
 - (d) disapply the LCR Requirement or NSFR Requirement; or
 - (e) impose additional requirements based on the DFSA assessment of the Liquidity Risk exposure of that Authorised Firm.
- (2) If the DFSA amends a requirement under (1)(a), (b), (c) or (e), the Authorised Firm must comply with the requirement as amended. If the DFSA disapplies a requirement under (1)(d), the Authorised Firm need not comply with that requirement.
- (3) The procedures in Schedule 3 to the Regulatory Law apply to a decision of the DFSA under (1)(a),(b),(c) or (e).
- (4) If the DFSA decides to exercise its power under (1)(a),(b),(c) or (e), the Authorised Firm may refer the matter to the FMT for review.

Liquid assets buffer

- 9.3.7** (1) An Authorised Firm must, except as provided under Rule 9.3.8, maintain a buffer of HQLA over the minimum level of LCR required under its LCR Requirement, appropriate to the nature, scale and complexity of its operations and in line with its Liquidity Risk tolerance.

- (2) In determining the size of its buffer of HQLA under (1), an Authorised Firm must also take into account the results of stress tests conducted under section 9.2A.

Guidance

1. For the purposes of Rule 9.3.7(2), an Authorised Firm should conduct its own stress tests to assess the level of liquidity it should hold beyond the minimum required under this section, and construct its own scenarios that could cause difficulties for its specific business activities. Such internal stress tests should incorporate longer periods than the one required under this section. Authorised Firms are expected to share the results of these additional stress tests with the DFSA.
2. As set out in the Guidance after Rule 9.2A.5, the DFSA may require an Authorised Firm to maintain an additional buffer of liquid assets in cases where the DFSA assesses that the Authorised Firm has failed to carry out stress tests effectively.

Monetisation of assets during periods of stress

- 9.3.8** During a period of financial or liquidity stress, an Authorised Firm may monetise part of its stock of HQLA and use the cash generated to cover cash outflows. Its level of HQLA may fall below the levels required under its LCR Requirement and Rule 9.3.7 to the extent necessary to deal with cash outflows during that period.

Notification if LCR Requirement not met

- 9.3.9** An Authorised Firm must notify the DFSA in writing immediately if it does not meet, or becomes aware of circumstances that may result in it not meeting, its LCR Requirement (including during a period of stress referred to in Rule 9.3.8).

Guidance

1. An Authorised Firm should in its notification clearly explain:
 - a. the reasons for not meeting the limits;
 - b. measures that have been taken and will be taken to ensure it meets its LCR Requirement; and
 - c. its expectations regarding the potential duration of the situation.
2. An Authorised Firm that makes a notification should discuss with the DFSA what, if any, further steps it should take to deal with the situation.

The Maturity Mismatch approach

Guidance

The Maturity Mismatch approach measures an Authorised Firm's liquidity by assessing the mismatch between its inflows (assets) and outflows (liabilities) within different timebands on a Maturity Ladder.

- 9.3.10** (1) An Authorised Firm in Category 1 or 5 must use the Maturity Mismatch approach, as set out in this section, to measure liquidity.

- (2) When using the Maturity Mismatch approach, an Authorised Firm must determine the net cumulative Maturity Mismatch position for each time band by:
 - (a) determining, in accordance with the Rules in sections A9.3 of App9, the inflows (assets), outflows (liabilities), liquid assets and funding capacity which are to be included in the relevant time bands in the Maturity Ladder; and
 - (b) subtracting outflows (liabilities) from inflows (assets) in each time band, and adding the eligible assets, in accordance with section A9.3 of App 9.

Measuring liquidity for Category 1 and Category 5

- 9.3.11**
- (1) An Authorised Firm in Category 1 or 5 must determine a net cumulative Maturity Mismatch position for each time band in respect of each of the following means of funding used by the Authorised Firm:
 - (a) PSIAus; and
 - (b) deposits.
 - (2) An Authorised Firm in Category 1 or 5 must calculate its liquidity by using the net cumulative Maturity Mismatch position separately for each means of funding used by the Authorised Firm as a percentage of the means of funding in each time band as follows:
 - (a) PSIAus net cumulative Maturity Mismatch % =

$$\frac{\text{Net cumulative Maturity Mismatch} \times 100}{\text{Total PSIAus}}$$
 - (b) Total deposit liabilities net cumulative Maturity Mismatch % =

$$\frac{\text{Net cumulative Maturity Mismatch} \times 100}{\text{Total deposits}}$$
 - (3) An Authorised Firm must ensure that its net cumulative Maturity Mismatch position for each means of funding used by the Authorised Firm in the sight – 8 days time band does not exceed negative 15%.
 - (4) An Authorised Firm must notify the DFSA in writing immediately if it exceeds or is likely to exceed the net cumulative Maturity Mismatch limit referred to in (3).

Net Stable Funding Ratio (NSFR) Requirement

- 9.3.12**
- (1) An Authorised Firm must maintain a Net Stable Funding Ratio (NSFR) of at least 100%.
 - (2) The NSFR under (1) must be calculated using the formula:

$$\text{NSFR} = \frac{\text{ASF} \times 100}{\text{RSF}}$$

where:

- (a) ASF (Available Stable Funding) is the amount, calculated in accordance with Rule A9.4.1, representing the relative stability of an Authorised Firm's available funding sources; and
- (b) RSF (Required Stable Funding) is the amount, calculated in accordance with Rule A9.4.2, representing the Liquidity Risk profile of an Authorised Firm's assets and OBS Exposures (or potential liquidity Exposures).

Guidance

1. The objective of the NSFR Requirement is to require an Authorised Firm to maintain a stable funding profile relative to the composition of its assets and off-balance sheet activities. A stable funding profile reduces the likelihood that disruptions to an Authorised Firm's regular sources of funding will erode its liquidity position in a way that would increase the risk of its failure and potentially lead to broader systemic stress. The NSFR Requirement limits over-reliance on short-term wholesale funding, encourages better assessment of funding risk across all on- and off-balance sheet items and promotes funding stability.
2. Section A9.4 of App9 sets out how an Authorised Firm's Available Stable Funding (ASF) and Required Stable Funding (RSF) are to be calculated.
3. If the DFSA considers that the Financial Services Regulator of the home state of an Authorised Firm that is a Branch has not fully implemented the Basel III NSFR requirements, it may use its power under Article 75A of the Regulatory Law to require the Authorised Firm to comply with appropriate NSFR requirements.

Notification if the NSFR Requirement not met

- 9.3.13** An Authorised Firm must notify the DFSA in writing immediately if it does not meet, or becomes aware of circumstances that may result in it not meeting, its NSFR Requirement.

Guidance

1. An Authorised Firm should explain clearly in its notification:
 - a. the reasons for it not meeting its NSFR Requirement;
 - b. measures that have been taken and will be taken to ensure it meets its NSFR Requirement; and
 - c. its expectations regarding the potential duration of the situation.
2. An Authorised Firm that makes a notification should discuss with the DFSA what, if any, further steps it should take to deal with the situation.

10 SUPERVISORY REVIEW AND EVALUATION PROCESSES

Introduction

Guidance

1. This chapter deals with the regulatory requirements arising out of the need for Authorised Firms to carry out a self-assessment of their risk which can be reviewed and assessed by the regulator. This chapter details the Rules stipulating the need to complete internal risk assessments by Authorised Firms in defined frequencies and the DFSA's role in reviewing the results of such assessments. In the case of Authorised Firms facing financial risks, the requirements in this chapter mandate completion of an internal capital adequacy assessment process. The DFSA will review the results of such internal risk assessments. This chapter also sets out how the DFSA may impose an additional capital requirement on a firm-specific basis in addition to the minimum requirement specified in chapter 3 of this module.
2. Appendix 10 provides detailed Guidance on the various components of the supervisory review and evaluation process and explains the role of the different parties in completing the required processes.

10.1 Application

10.1.1 This chapter applies to an Authorised Firm in Category 1, 2 or 5.

10.1.2 Where an Authorised Firm to which this chapter applies is part of a Financial Group, this chapter applies on a consolidated basis in relation to all the entities within the Financial Group.

10.1.3 In implementing the requirements prescribed in this chapter, an Authorised Firm must give due and appropriate regard to the provisions in App10.

Guidance

1. If an Authorised Firm is part of a Financial Group which is already subject to requirements similar to those prescribed in this chapter, the DFSA may consider a request for a waiver or modification in relation to the requirements of this chapter.
2. In appropriate cases, the DFSA may use its supervisory powers to require an Authorised Firm in a Category not specified in Rule 10.1.1, to comply with requirements equivalent to one or more of those specified in this chapter.

10.2 Overview

Guidance

1. These Rules are designed to implement key aspects of Pillar 2 of the revised framework of capital adequacy, commonly known as Basel III, published by the Basel Committee.

Internal Risk Assessment Process (IRAP)

2. An Authorised Firm in Category 1, 2 or 5 is required to carry out an IRAP. An IRAP is a comprehensive internal risk evaluation as detailed in section 10.3. More detail on the

establishment of an IRAP and the manner of carrying out an IRAP assessment is provided in App10.

Internal Capital Adequacy Assessment Process (ICAAP)

3. An Authorised Firm in Category 1, 2 or 5 is also required to carry out an ICAAP as detailed in section 10.4. This process enables such an Authorised Firm to determine and maintain an adequate amount and quality of capital, relative to its risk profile. More detail on the establishment of an ICAAP and the manner of carrying out an ICAAP assessment is provided in App10.

Supervisory Review and Evaluation Process (SREP)

4. The documented results of IRAP and ICAAP assessments are required to be submitted to the DFSA. The DFSA then applies a process known as the SREP as detailed in section 10.5. As part of the SREP, the DFSA will evaluate the quality, completeness and consistency of the IRAP and, where applicable, the ICAAP of the Authorised Firm, to form a view on the overall risk profile of the firm and whether the capital held by the firm is sufficient to deal with the risks. More detail concerning the SREP is provided in App10.
5. Following review of the IRAP and ICAAP of an Authorised Firm, the DFSA may engage in a dialogue with the firm to evaluate the assessment of risks and where relevant, additional capital which the DFSA considers that the firm should hold resulting from the IRAP, ICAAP or SREP.

Individual Capital Requirement (ICR)

6. Upon completing the SREP, the DFSA may impose an Individual Capital Requirement on an Authorised Firm in Category 1, 2 or 5 as detailed in section 10.6. The ICR may be imposed where the DFSA concludes that the firm should hold more capital to provide for its overall risks.

10.3 IRAP

10.3.1 [Deleted]

- 10.3.2** (1) An Authorised Firm must establish and maintain an IRAP which details the processes and procedures by which the firm will identify, assess, aggregate and monitor the risks faced by it.
- (2) The firm must conduct the IRAP assessment at least annually giving due regard to the Guidance in section A10.1 of App10.
- (3) The IRAP assessment conducted by the firm pursuant to (2) must be approved by its Governing Body and then submitted to the DFSA within four months from the end of the firm's financial year.
- (4) In addition to (2), the firm must conduct an IRAP assessment:
 - (a) whenever there is material change to the business, strategy, nature or scale of the activities of the firm which may have a significant impact on its risk profile or adequacy of its Capital Resources or Adjusted Capital Resources, as applicable; or
 - (b) as and when required by the DFSA.

- (5) An IRAP assessment conducted by the firm pursuant to (4) must be approved by its Governing Body and then submitted to the DFSA within two months, or such other period as may be specified by the DFSA, from the date of such material change or requirement.

10.3.3 The results of an IRAP assessment must be documented by the Authorised Firm in writing and include details of:

- (a) the risks identified;
- (b) the firm's strategies and plans to deal with those risks;
- (c) the firm's assessment of the adequacy of its Capital Requirement as calculated under PIB to address all the risks identified by its IRAP;
- (d) the details of any stress testing and scenario analysis carried out and the resultant impact on the Capital Requirement; and
- (e) any other relevant information, giving due regard to the Guidance in App10.

10.3.4 An Authorised Firm must retain the records of an IRAP assessment for at least six years.

10.4 ICAAP

10.4.1 [Deleted]

- 10.4.2**
- (1) An Authorised Firm must implement and maintain an ICAAP which details the processes and procedures by which the firm will assess and maintain adequate Capital Resources in relation to the risks faced by it.
 - (2) The firm must conduct an ICAAP assessment at least annually giving due regard to the Guidance in section A10.2 of App10.
 - (3) The ICAAP assessment conducted by the firm pursuant to (2) must be approved by its Governing Body and then submitted to the DFSA within four months from the end of the firm's financial year.
 - (4) In addition to (2), the firm must conduct an ICAAP assessment:
 - (a) whenever there is material change to the business, strategy, nature or scale of the activities of the firm which may have a significant impact on its risk profile or adequacy of its Capital Resources or Adjusted Capital Resources, as applicable; or
 - (b) as and when required by the DFSA.
 - (5) The ICAAP assessment conducted by the firm pursuant to (4) must be approved by its Governing Body and then submitted to the DFSA within two months from the date of such material change or requirement.

10.4.3 An Authorised Firm must ensure that an ICAAP assessment is documented in writing and includes details of:

- (a) the calculations and models used in the determination of the level of Capital Requirements which it considers will be adequate to cover all the risks identified by its ICAAP assessment;
- (b) the firm's strategies and plans to ensure availability of the level of capital determined by the ICAAP;
- (c) specifications of any models used in the ICAAP, including the underlying assumptions, parameters, and results of back-testing; and
- (d) any other relevant information, giving due and appropriate regard to the Guidance in App10.

10.4.4 An Authorised Firm must retain the records of an ICAAP assessment for at least six years.

10.5 SREP

Guidance

1. The DFSA may conduct a SREP to review and evaluate the assessments carried out by an Authorised Firm under its IRAP and ICAAP. Section A10.3 of App10 contains guidance in relation to a SREP.
2. The DFSA may engage with a firm in a dialogue where, following an SREP, the DFSA considers that it is or may be appropriate to impose an Individual Capital Requirement on the firm.
3. It is important that a firm cooperates in an open and co-operative manner with the DFSA in the course of its conduct of the dialogue.

10.6 Imposition of an Individual Capital Requirement

- 10.6.1**
- (1) This section applies to an Authorised Firm in Category 1, 2 or 5.
 - (2) The DFSA may, subject to (3) and (4), at any time by written notice to an Authorised Firm:
 - (a) impose an Individual Capital Requirement; or
 - (b) vary or withdraw an Individual Capital Requirement.
 - (3) The DFSA may act under (2) on its own initiative where the DFSA forms the view that the firm's Capital Requirement is insufficient to address adequately all its risks.
 - (4) The DFSA will, in addition to prescribing an Individual Capital Requirement, also specify in the notice the types and amounts of Capital Resources required to meet the Individual Capital Requirement.
 - (5) The procedures in Schedule 3 to the Regulatory Law apply to a decision of the DFSA under (2) made after a Licence has been granted.

- (6) If the DFSA decides to exercise its power under (2) after a Licence has been granted, the Authorised Firm may refer the matter to the FMT for review.

10.6.2 An Authorised Firm must have and maintain, at all times, Capital Resources of the types and amounts specified in the notice issued to it under Rule 10.6.1 to meet its Individual Capital Requirement.

11 DISCLOSURE REQUIREMENTS

Introduction

Guidance

1. This chapter specifies the disclosures required to be made by an Authorised Firm to enable market participants and potential counterparties to exercise market discipline in relation to the firm. Chapter 11 applies to banks, principal dealers and Islamic banks and specifies the disclosure requirements for such firms.
2. Appendix 11 provides a detailed list of disclosure requirements for Authorised Firms to which this chapter applies.

11.1 Application and general obligation of disclosure

Guidance

The purpose of the requirements in this chapter is to ensure that minimum public disclosures are made available to market participants to assist them in forming an opinion on the risk profile and capital adequacy of an Authorised Firm. The DFSA expects an Authorised Firm to convey its actual risk profile to market participants.

11.1.1 An Authorised Firm in Category 1, 2 (other than a Matched Principal) or 5 must make the disclosures as prescribed in this chapter.

11.1.2 (1) An Authorised Firm which is a member of a Financial Group must, subject to (2), ensure that the disclosures specified in App11 are made at Financial Group level.

(2) An Authorised Firm which is a Subsidiary of

(a) a Regulated Financial Institution; or

(b) another Authorised Firm which is in Category 1, 2 (other than a Matched Principal) or 5;

which is already subject to equivalent public disclosure requirements, does not need to comply with the requirements in this chapter to the extent that it meets those equivalent public disclosure requirements.

11.1.3 An Authorised Firm must disclose the quantitative indicators in Table 15 of App 11 if it is:

(a) a G-SIB; or

(b) in Category 1 or 5 and has a Leverage Ratio Exposure Measure of US\$ 200 Billion or more.

Guidance

An Authorised Firm does not have to disclose the information in Table 15 of App 11 if it is a subsidiary of another Regulated Financial Institution or Authorised Firm that has to disclose the information – see Rule 11.1.2.

11.2 Disclosure policy

11.2.1 An Authorised Firm must implement and maintain a written disclosure policy that:

- (a) sets out the firm's approach for determining which of the disclosures set out in App11 it needs to make;
- (b) details the processes and procedures and its internal controls in relation to such disclosure;
- (c) details the medium for disclosure that most appropriately meets the purposes of this chapter; and
- (d) is approved by the Governing Body of the firm.

11.2.2 An Authorised Firm must ensure that appropriate verification, whether internal or external, is performed in relation to any disclosure, and take all reasonable steps to ensure its accuracy and timeliness.

11.2.3 To the extent that any required disclosure is substantially similar to a disclosure required of the Authorised Firm under the International Financial Reporting Standards, a disclosure under such standards must be taken to meet the requirement for disclosure under this chapter.

11.3 Disclosure frequency, locations and process

Frequency

- 11.3.1**
- (1) The disclosures set out in this chapter must be made by the Authorised Firm at least once a year, other than disclosures of CET1 Capital, T1 Capital and T2 Capital, deductions from Capital Resources, Liquidity Coverage Ratio, Net Stable Funding Ratio and Leverage Ratios which must be made on a quarterly basis.
 - (2) Reporting deadlines must be in accordance with quarterly and annual reporting obligations under the DFSA Rulebook.

Locations

- 11.3.2**
- (1) An Authorised Firm must, subject to (2), make these disclosures either in its annual report or periodic financial statements.
 - (2) An Authorised Firm may disclose the items marked as quantitative in App11 in a medium or location other than its annual report or periodic financial statements, provided that:
 - (a) it has prior approval of the DFSA to do so;

- (b) the annual report or periodic financial statements contain clear references to the location of such disclosures; and
- (c) such disclosures are readily accessible by the market.

Guidance

An Authorised Firm has discretion to determine the form of the disclosures required, and may choose to use graphical and other representations where appropriate.

Omissions

- 11.3.3** (1) An Authorised Firm may omit certain disclosures if the omitted item is:
- (a) not material, in accordance with the concept of materiality under the International Financial Reporting Standards,
 - (b) proprietary in nature, and the disclosure of the relevant information to the public would undermine the Firm's competitive position or render the Firm's investments in products and systems less valuable, or
 - (c) confidential in nature, and the disclosure of the relevant information would violate or jeopardise confidentiality agreements with Clients or Counterparties.
- (2) Where in reliance upon (1)(b) or (c) an Authorised Firm omits an item that is marked as a quantitative disclosure in App11, it must disclose general qualitative information about the subject matter of that particular requirement, together with the reasons for the omission.

APP1 CATEGORIES OF AUTHORISED FIRMS

A1.1 Categorisation of Authorised Firms

Guidance

1. This section contains the table referred to in the Guidance notes at the commencement of section 1.3 of PIB. This table is for guidance purposes only.
2. The Financial Services described in the emboldened boxes in the table are the determinants for the prudential Category. The activities set out in the boxes in the table are Financial Services (see GEN chapter 2). The Financial Services that an Authorised Firm is authorised to carry on are specified on its Licence.
3. If a Person carries on any one or more of the Financial Services specified in an emboldened box under a particular Category, then the highest such Category is that Person's Category for the purposes of this module.
4. An exception to the above is an Islamic Financial Institution which Manages a PSIA which is an PSIAu. Such an institution falls in Category 5.

PRUDENTIAL – INVESTMENT, INSURANCE INTERMEDIATION AND BANKING (PIB)

Category 1	Category 2	Category 3A	Category 3B	Category 3C	Category 3D	Category 4	Category 5
Accepting Deposits	Dealing in Investments as Principal	Dealing in Investments as Agent	Providing Custody (only if for a Fund or of Crypto Assets)	Managing a Collective Investment Fund	Providing Money Services (other than issuing Stored Value or Money Transmission)	Arranging Deals in Investments	An Islamic Financial Institution which Manages a PSIAu
Managing a PSIAu	Providing Credit		Acting as the Trustee of a Fund	Managing Assets		Advising on Financial Products	
			Operating an Employee Money Purchase Scheme	Providing Trust Services as a trustee of an express trust		Arranging Custody	
			Acting as the Administrator of an Employee Money Purchase Scheme	Managing a PSIAr		Insurance Intermediation	
				Providing Custody (other than for a Fund and other than in relation to Crypto Assets)		Insurance Management	
				Providing Money Services (issuing Stored Value)		Operating an Alternative Trading System	
						Providing Fund Administration	

Providing Trust
Services other
than as a trustee of
an express trust

Arranging Credit
and Advising on
Credit

Operating a
Crowdfunding
Platform

Providing Money
Services (Money
Transmission)

Arranging or
Advising on
Money Services

APP2 GENERAL REQUIREMENTS

A2.1 Detail in the Trading Book

- A2.1.1** (1) This Appendix applies to an Authorised Firm which meets the criteria in Rule 2.2.1 and is thereby required to have a Trading Book.
- (2) An Authorised Firm which is required to have a Trading Book must:
- (a) have a Trading Book policy in accordance with section A2.2; and
 - (b) include positions in its Trading Book on a consistent basis in accordance with the policy and procedures set out in the Trading Book policy.
- (3) An Authorised Firm must include every position that is not included in its Trading Book in its Non-Trading Book.
- (4) An Authorised Firm must value every position included in its Trading Book and the Non-Trading Book in accordance with the relevant accounting standards and practices.

Value of Trading Book positions

- A2.1.2** (1) In calculating the value of positions for the purposes of Rule 2.2.1(c) an Authorised Firm must value:
- (a) equities and debt instruments at their market prices;
 - (b) derivatives according to the values of the underlying; and
 - (c) underwriting positions according to the market value of the underlying Securities.
- (2) An Authorised Firm must sum all long and short positions (ignoring the sign) to calculate its total Trading Book size.

Positions included in the Trading Book

- A2.1.3** An Authorised Firm must include in its Trading Book, subject to the Rules on trading intent and hedging Non-Trading Book Exposures:
- (a) each proprietary position in a Financial Instrument, commodity or commodity derivative which is held with trading intent as detailed in Rule A2.1.5(3);
 - (b) each position arising from Matched Principal broking and market making;
 - (c) each position taken in order to hedge another element of the Trading Book;
 - (d) each Exposure due to a repurchase agreement (repo), or Securities and commodities lending, which is based on a Security or commodity included in the Trading Book;

- (e) each Exposure due to a reverse repurchase agreement (reverse repo), or Securities and commodities borrowing transaction included in the Trading Book;
- (f) each Exposure arising from an Unsettled Transaction, free delivery or OTC derivative; and
- (g) each Exposure in the form of a fee, commission, interest, dividend or margin on an exchange-traded derivative directly related to the items included in the Trading Book.

Guidance

Whenever an Authorised Firm acts as principal (even in the context of activity normally described as ‘broking’ or ‘customer business’), positions should be assigned to the Trading Book. This applies even if the nature of the business means that the only risks being incurred by the Authorised Firm are Counterparty Risks (that is, no Market Risk Capital Requirements apply).

- A2.1.4**
- (a) An Authorised Firm must not include loans or traded loans in its Trading Book unless they have been used to hedge a Trading Book transaction.
 - (b) An Authorised Firm must not include in its Trading Book an Exposure relating to a direct holding of immovable property.

Trading intent

- A2.1.5**
- (1) An Authorised Firm must, subject to Rule A2.1.3, only include in its Trading Book:
 - (a) a position in a Financial Instrument, commodity or commodity derivative held with trading intent; or
 - (b) a position hedging other positions in the Trading Book.
 - (2) For the purpose of (1), such positions included in the Trading Book must be free of any restrictive covenants which limit their tradability or ability to be hedged.
 - (3) For the purpose of (1), a position in a Financial Instrument, commodity or commodity derivative is held with trading intent if:
 - (a) it is held with the intention of:
 - (i) benefiting in the short term from actual or expected differences between buying and selling prices or from other price or interest-rate variations;
 - (ii) selling it over the short term;
 - (iii) locking in arbitrage profits; or
 - (iv) market making;
 - (b) it is marked to market or marked to model regularly on a prudent and consistent basis, as part of the Authorised Firm’s internal risk management processes;

- (c) position-takers at the Authorised Firm have autonomy in entering into or changing transactions within pre-determined limits, or the position satisfies other criteria which the Authorised Firm applies to the composition of its Trading Book;
- (d) there is an appropriate documented trading strategy for the position, approved by senior management which includes the expected holding horizon; and
- (e) active monitoring of the position is undertaken using market information sources.

A2.1.6 Positions held with trading intent must comply with the following requirements:

- (a) trading intent must be evidenced in the strategies, policies and procedures established by the Authorised Firm to manage the position or its portfolio;
- (b) there must be clearly defined policies and procedures for active management of the position to ensure the following:
 - (i) the position is entered and/or managed on a trading desk;
 - (ii) position limits are set and monitored for appropriateness;
 - (iii) position-takers at the Authorised Firm have autonomy in entering into or changing transactions within pre-determined limits, or the position satisfies other criteria which the Authorised Firm applies to the composition of its Trading Book;
 - (iv) the position is marked-to-market or marked-to-model at least daily on a prudent and consistent basis as part of the Authorised Firm's internal risk management processes;
 - (v) where the position is marked-to-model, the parameters for the model are assessed on a daily basis;
 - (vi) the position is monitored against the documented trading strategy including the monitoring of turnover and stale positions in the Authorised Firm's Trading Book;
 - (vii) active monitoring of the position is undertaken using market information sources and an assessment made of the marketability or hedge-ability of the position or its component risks, including the assessment of the quality and availability of market inputs to the valuation process, level of markets turnover and sizes of positions traded in the market; and
 - (viii) positions and exceptions are reported to senior management as an integral part of the risk management process of the Authorised Firm.

Treatment of structural foreign exchange positions

A2.1.7 An Authorised Firm in Category 1 or 5 which has assumed a position in order to hedge partially or totally against the adverse effect of the exchange rate on its Capital Resources, in respect of an asset or any other item, may exclude such a position

from the calculation of its net open foreign exchange positions subject to the following:

- (a) the position is of a non-dealing nature;
- (b) the position does no more than protect the Capital Resources of the Authorised Firm; and
- (c) any exclusion of the position is applied consistently, with the treatment of the hedge remaining the same for the life of the asset or other item.

A2.1.8 In calculating its net open foreign exchange positions, an Authorised Firm may exclude any foreign exchange position related to:

- (a) items which are included as deductions from T1 Capital or deductions from T2 Capital, such as investments in unconsolidated subsidiaries; and
- (b) associated companies and joint ventures, denominated in foreign currencies, which are reported in the published accounts of an Authorised Firm at historic cost.

Repurchase and reverse repurchase agreements

A2.1.9 An Authorised Firm must include in its Trading Book an Exposure due to a repurchase agreement, reverse repurchase agreement, Securities and commodities borrowing, or Securities and commodities lending transactions if:

- (a) the Exposure is marked to market daily (cash borrowed or lent under a repurchase agreement or a reverse repurchase agreement may be included in the Trading Book even if not marked to market provided that the residual maturity of the borrowing or lending is one month or less);
- (b) the Collateral is adjusted to take account of changes in the value of the Securities or commodities involved;
- (c) the agreement or transaction provides for the Authorised Firm's claims to be automatically and immediately offset against its Counterparty's claims if the latter defaults; and
- (d) such agreements and transactions are confined to their accepted and appropriate use and artificial transactions, especially those not of a short-term nature, are excluded.

Guidance

Cash items include loans and deposits and the cash legs of repurchase, stock borrowing, reverse repurchase and stock lending transactions.

A2.1.10 Where the conditions under Rule A2.1.9 are not met, an Authorised Firm must, subject to Rule A2.1.3, include an Exposure arising under a repurchase agreement, reverse repurchase agreement, securities and commodities borrowing or securities and commodities lending in its Non-Trading Book.

Guidance

The Non-Trading Book treatment for such Exposures is set out in Rules in A4.8 and A4.9.

Hedging of a Trading Book Exposure by a non-financial instrument

- A2.1.11** (1) An Authorised Firm may hedge a Trading Book Exposure, completely or partially, by a non-financial instrument that is not listed in A2.1.3. The General Market Risk Exposure associated with the non-financial instrument may be incorporated into the calculation of General Market Risk in the Trading Book if:
- (a) the specific instrument is used to hedge an Exposure in an Authorised Firm's Trading Book;
 - (b) the hedge position satisfies the Netting rules contained in the relevant sections of the Market Risk chapter; and
 - (c) the hedge position is marked to market or marked to model and is valued regularly on a prudent and consistent basis.
- (2) For the purposes of (1), the non-financial instrument must be treated as attracting capital charges as if it were a Financial Instrument.

Guidance

1. If the conditions for incorporating non-financial instruments in the calculation of General Market Risk in the Trading Book under Rule A2.1.11 are not met, they will be treated as Non-Trading Book items.
2. For the purposes of section A2.1, a loan will attract General Market Risk (see chapter 5) and Counterparty Risk (see chapter 4) on the marked-to-market valuation.

- A2.1.12** (1) If an internal hedge meets the criteria specified in (2), an Authorised Firm may include it in the Trading Book without prejudice to the Capital Requirement application to the Non-Trading Book "leg" of the internal hedge.
- (2) Positions arising from internal hedges are eligible for Trading Book capital treatment, provided that they meet the criteria for trading intent specified in Rule A2.1.5 and the following criteria on prudent valuation:
- (a) the internal hedge is not primarily intended to avoid or reduce Capital Requirements which the Authorised Firm would be otherwise required to maintain;
 - (b) the internal hedge is properly documented and subject to specific internal approval and audit procedures;
 - (c) the internal hedge is dealt with at market conditions;
 - (d) the bulk of the Market Risk which is generated by the internal hedge is dynamically managed in the Trading Book within the limits approved by senior management; and
 - (e) the internal hedge is carefully monitored with adequate procedures.
- (3) Where an Authorised Firm hedges a Non-Trading Book Exposure using a Credit Derivative booked in the Trading Book, the Non-Trading Book

Exposure is not deemed to be hedged for the purpose of calculating its regulatory Capital Requirement, unless the Authorised Firm purchases from an eligible protection provider a Credit Derivative which complies with the requirements and meets the guidelines set out in the relevant section of chapter 4. Where eligible credit protection is purchased and is recognised as a hedge of the Non-Trading Book Exposure for the purpose of calculating its regulatory Capital Requirement, the Authorised Firm may exclude both the internal and external Credit Derivative hedge from the Trading Book for the purpose of calculating its regulatory Capital Requirement for the period of the hedge.

Guidance

An internal hedge is a position that materially or completely offsets the component risk element of a Non-Trading Book position or a set of positions.

Transfer of General Market Risk between the Trading Book and the Non-Trading Book

Guidance

1. General Market Risk arising from the Trading Book may hedge Non-Trading Book positions without reference to specific Financial Instruments.
2. An Authorised Firm may achieve the transfer of General Market Risk between the Trading Book and Non-Trading Book by entering into a notional legal agreement between the Trading Book and Non-Trading Book as if they were third parties.

A2.1.13 An Authorised Firm must ensure that:

- (a) a transfer of General Market Risk between its Trading Book and Non-Trading Book is subject to appropriate documentation and evidenced by a clear audit trail;
- (b) positions held in its Non-Trading Book that are being hedged by General Market Risk arising from positions in the Trading Book remain in the Non-Trading Book; and
- (c) the General Market Risk Exposure associated with the positions in the Non-Trading Book is incorporated into the calculation of General Market Risk in the Trading Book.

Guidance

An example of the application of Rule A2.1.13(c) is as follows:

- a. An Authorised Firm may have a fixed-rate loan portfolio in the Non-Trading Book. Although the Non-Trading Book does not attract a regulatory capital charge for interest rate risk, the portfolio is subject to interest rate risk. Firms may prefer to transfer this risk to the Trading Book where it may be actively managed.
- b. The Authorised Firm may transfer this interest rate risk by entering into, for example, a fixed versus floating rate swap between the Trading Book and the Non-Trading Book. The notional long and short positions created as result of the swap are recorded in the Trading Book, and the swap positions may be treated as financial instruments provided that appropriate documentation is in place (see Rule A2.1.14). The General Market Risk requirements associated with the swap legs are allocated to the appropriate Trading Book General Market Risk bucket and thus may reduce the overall General Market Risk requirement in the Trading Book.

- c. For an Authorised Firm to undertake such a transaction there should be existing positions in the Trading Book, which result in a sufficient General Market Risk requirement to offset the General Market Risk created as a result of the swap.

A2.1.14 Appropriate documentation under A2.1.13 must cover:

- (a) details of the instruments or Exposures being transferred and the method used to transfer; and
- (b) the pricing of the transfer.

Guidance

1. Separate documentation need not be produced for every transfer. If the same method is used for a number of transfers, a single document detailing the procedures will suffice. However, an Authorised Firm must still be able to distinguish transactions that have been undertaken for risk transfer purposes from other transactions.
2. Arm's-length prices must be used in any transfer. 'Arm's-length' means the prevailing market price for the particular transaction.

A2.2 Trading Book policy

Guidance

The requirement for a Trading Book policy is prescribed in Rule A2.1.1.

A2.2.1 A Trading Book policy must include clearly defined policies, procedures and methodologies by which the Authorised Firm:

- (a) defines its Trading Book and identifies positions to be included in its Trading Book;
- (b) allocates positions between the Non-Trading Book and the Trading Book;
- (c) actively manages and values its positions in the Trading Book;
- (d) measures its Trading Book risks; and
- (e) controls transfers of positions between the Non-Trading Book and the Trading Book.

A2.2.2 An Authorised Firm must obtain the approval of its Governing Body for its Trading Book policy. The Authorised Firm must review and where necessary update the policy at least annually. The Authorised Firm must obtain the approval of the Governing Body for all significant changes.

A2.2.3 An Authorised Firm must, without undue delay, notify the DFSA when its Governing Body approves the adoption of a Trading Book policy or approves any changes to the policy.

Guidance

1. At a minimum, the DFSA expects that a Trading Book policy will address the following:
 - a. the definition of Trading Book and trading strategy, including:

- i. the activities that the Authorised Firm considers to be trading and the types of positions that are to be allocated to the Trading Book for the purposes of calculating its regulatory Capital Requirements;
 - ii. the types of positions that are excluded from the Trading Book; and
 - iii. the procedures to ensure that the criteria by which positions are allocated to the Trading Book are adhered to on a consistent basis, including details on:
 - A. the organisational unit or department within the Authorised Firm responsible for monitoring adherence to the Trading Book policy;
 - B. the frequency of monitoring;
 - C. process and methodology of this monitoring; and
 - D. how the continuing appropriateness of allocations is confirmed;
- b. the extent of active management and valuation, including:
 - i. the extent to which a position can be marked-to-market daily by reference to an active, liquid two-way market;
 - ii. for positions which are marked-to-model, the extent to which the Authorised Firm can:
 - A. identify all the material risks of the position;
 - B. hedge the material risks of the position and, where the material risks of the position are hedged, the extent to which hedging instruments used have an active and liquid two-way market; and
 - C. derive reliable external estimates for the key assumptions and parameters used in the model;
 - iii. the extent to which the Authorised Firm can, and is required to, generate valuations for the position which can be validated externally by its auditors or by the DFSA in a consistent manner;
 - iv. the extent to which the Authorised Firm can, and is required to, maintain documents to support valuations of its Trading Book positions;
 - v. the basis for determining and maintaining valuation adjustments for the purposes of calculating regulatory Capital Requirements;
 - vi. the extent to which legal restrictions or other operational requirements would impede the ability of the Authorised Firm to effect an immediate liquidation of the position; and
 - vii. the extent to which the Authorised Firm can, and is required to, actively risk manage a position within its trading operations;
- c. transfers between Non-Trading and Trading Books, including:
 - i. the extent to which an Authorised Firm may transfer positions between the Non-Trading Book and the Trading Book and the criteria for such transfers;
 - ii. the procedures to effect such transfers; and
 - iii. the controls in place to prevent inappropriate transfers of positions between the Non-Trading Book and the Trading Book; and

- d. the following additional considerations:
 - i. whether there are any subsidiaries or offshore branches of the Authorised Firm undertaking transactions to be included in the Trading Book. If so, a list of such subsidiaries or branches shall be included, along with a description of the trading activities carried out by such entities;
 - ii. the treatment of inter-desk deals; and
 - iii. the identification and management of structural foreign exchange positions.
- 2. An Authorised Firm should prepare its Trading Book policy on a consolidated basis where the Financial Group either manages its trading risk centrally or employs the same risk management techniques across all the entities in the Financial Group. Where a Trading Book policy is prepared on a consolidated basis, an Authorised Firm should ensure that its application to the Authorised Firm and each of the other entities in the Financial Group is made clear and approved by the Governing Body of the Authorised Firm and the Governing Body of each of those entities.

A2.3 Risk management systems and controls for Trading Book

A2.3.1 An Authorised Firm must establish and maintain effective systems and controls to manage its Trading Book.

A2.3.2 An Authorised Firm must establish and maintain systems and controls sufficient to provide prudent and reliable valuation estimates. These systems and controls must include:

- (a) documented policies and procedures for the process of valuation (including clearly defined responsibilities of the various areas involved in the determination of the valuation, sources of market information and review of their appropriateness, frequency of independent valuation, timing of closing prices, procedures for adjusting valuations, month end and ad-hoc verification procedures); and
- (b) reporting lines for the department accountable for the valuation process that are clear and independent of the front office.

A2.4 Reporting to the DFSA

A2.4.1 In the following Table 1:

- (a) the column headed 'Domestic Firm or Branch' indicates whether the relevant form must be prepared and submitted by a Domestic Firm or Branch;
- (b) the column headed 'Authorised Firm Category' indicates which Category of Authorised Firm must prepare and submit the relevant form;
- (c) the column headed 'Frequency' indicates whether the relevant form must be prepared and submitted on a quarterly or annual basis; and
- (d) the column headed 'Semi-annual consolidated reporting for Financial Groups' indicates which Category of Authorised Firm of a Financial Group must prepare and submit the relevant form on a semi-annual basis.

Guidance

Table 1 is referred to in section 2.3, and forms part of the Rules.

Table 1 Authorised Firm Reporting Matrix

Form Number	Form Name	Domestic Firm or Branch	Authorised Firm Category	Frequency	Semi-annual consolidated reporting for Financial Groups
B10A	Assets	Domestic Firm and Branch	Category 1, 2, 3A, 3B, 3C, 3D and 4	Quarterly and annually	Category 1 and 2 (other than a Matched Principal)
B10B	Liabilities (Domestic)	Domestic Firm	Category 1, 2, 3A, 3B, 3C, 3D and 4	Quarterly and annually	Category 1 and 2 (other than a Matched Principal)
B10B	Liabilities (Branch)	Branch	Category 1, 2, 3A, 3B, 3C, 3D and 4	Quarterly and annually	Not applicable
B10C	Equity	Domestic Firm	Category 1, 2, 3A, 3B, 3C, 3D and 4	Quarterly and annually	Category 1 and 2 (other than a Matched Principal)
B10D	OBS Exposures	Domestic Firm and Branch	Category 1, 2, 3A, 3B, 3C, 3D and 4	Quarterly and annually	Category 1 and 2 (other than a Matched Principal)
B20A	Assets – IFI	Domestic Firm and Branch	Category 5	Quarterly and annually	Category 5
B20B	Liabilities (Domestic) – IFI	Domestic Firm	Category 5	Quarterly and annually	Category 5
B20B	Liabilities (Branch) – IFI	Branch	Category 5	Quarterly and annually	Not applicable
B20C	Equity - IFI	Domestic Firm	Category 5	Quarterly and annually	Category 5

Form Number	Form Name	Domestic Firm or Branch	Authorised Firm Category	Frequency	Semi-annual consolidated reporting for Financial Groups
B20D	OBS Exposures – IFI	Domestic Firm and Branch	Category 5	Quarterly and annually	Category 5
B20E	Analysis of Reserve Movement – IFI	Domestic Firm	Category 5	Quarterly and annually	Category 5
B30	Related Party Schedule	Domestic Firm and Branch	All Categories	Quarterly and annually	Not applicable
B40A	Profit and Loss	Domestic Firm and Branch	Category 1, 2, 3A, 3B, 3C, 3D and 4	Quarterly and annually	Category 1 and 2 (other than a Matched Principal)
B40B	Statement of Comprehensive Income	Domestic Firm and Branch	Category 1, 2, 3A, 3B, 3C, 3D and 4	Quarterly and annually	Category 1 and 2 (other than a Matched Principal)
B50A	Profit and Loss - IFI	Domestic Firm and Branch	Category 5	Quarterly and annually	Category 5
B50B	Statement of Comprehensive Income - IFI	Domestic Firm and Branch	Category 5	Quarterly and annually	Category 5
B100	Declaration by Authorised Firms	Domestic Firm and Branch	All Categories	Quarterly and annually	Not applicable
B110	Capital Ratios	Domestic Firm	Category 1, 2, 3A and 5	Quarterly and annually	Category 1, 2 (other than a Matched Principal) and 5

Form Number	Form Name	Domestic Firm or Branch	Authorised Firm Category	Frequency	Semi-annual consolidated reporting for Financial Groups
B120	Capital Resources	Domestic Firm	All Categories	Quarterly and annually	Category 1, 2 (other than a Matched Principal) and 5
B130	Credit Risk – Overview	Domestic Firm	Category 1, 2, 3A and 5	Quarterly and annually	Category 1, 2 (other than a Matched Principal) and 5
B130A	Credit Risk – Balance sheet Exposures	Domestic Firm	Category 1, 2, 3A and 5	Quarterly and annually	Category 1, 2 (other than a Matched Principal) and 5
B130B	Credit Risk – Counterparty Credit Risk Exposures	Domestic Firm	Category 1, 2, 3A and 5	Quarterly and annually	Category 1, 2 (other than a Matched Principal) and 5
B130C	Credit Risk - Securitisation	Domestic Firm	Category 1, 2, 3A and 5	Quarterly and annually	Category 1, 2 (other than a Matched Principal) and 5
B130D	Settlement Risk	Domestic Firm	Category 1, 2, 3A and 5	Quarterly and annually	Category 1, 2 (other than a Matched Principal) and 5
B140	Market Risk – Overview	Domestic Firm	Category 1, 2, 3A and 5	Quarterly and annually	Category 1, 2 (other than a Matched Principal) and 5
B140A	Market Risk – Interest Rate Risk	Domestic Firm	Category 1, 2 and 3A	Quarterly and annually	Category 1 and 2 (other than a Matched Principal)

Form Number	Form Name	Domestic Firm or Branch	Authorised Firm Category	Frequency	Semi-annual consolidated reporting for Financial Groups
B140B	Market Risk – IRR Maturity Approach	Domestic Firm	Category 1, 2 and 3A	Quarterly and annually	Category 1 and 2 (other than a Matched Principal)
B140C	Market Risk – IRR Duration Approach	Domestic Firm	Category 1, 2 and 3A	Quarterly and annually	Category 1 and 2 (other than a Matched Principal)
B140D	Market Risk - Equity	Domestic Firm	Category 1, 2, 3A and 5	Quarterly and annually	Category 1, 2 (other than a Matched Principal) and 5
B140E	Market risk – FX Risk	Domestic Firm	Category 1, 2, 3A and 5	Quarterly and annually	Category 1, 2 (other than a Matched Principal) and 5
B140F	Market Risk Options and Commodities	Domestic Firm	Category 1, 2, 3A and 5	Quarterly and annually	Category 1, 2 (other than a Matched Principal) and 5
B140G	Market risk – VAR	Domestic Firm	Category 1, 2, 3A and 5	Quarterly and annually	Category 1, 2 (other than a Matched Principal) and 5
B140H	Credit Valuation Adjustment	Domestic Firm	Category 1, 2, 3A and 5	Quarterly and annually	Category 1, 2 (other than a Matched Principal) and 5
B150	Operational Risk	Domestic Firm	Category 1, 2, 3A and 5	Quarterly and annually	Category 1, 2 (other than a Matched Principal) and 5

Form Number	Form Name	Domestic Firm or Branch	Authorised Firm Category	Frequency	Semi-annual consolidated reporting for Financial Groups
B180	Expenditure Based Capital Minimum	Domestic Firm	Category 2, 3A, 3B, 3C, 3D and 4 (if it is subject to section 3.7)	Quarterly and annually	Not applicable
B190	Leverage Ratio	Domestic Firm	Category 1, 2 (other than a Matched Principal) and 5	Quarterly and annually	Category 1, 2 (other than a Matched Principal) and 5
B210	Liquidity	Domestic Firm and Branch	Category 1 and 5	Quarterly only	Category 1 and 5
B220	Liquidity Coverage Ratio	Domestic Firm and Branch	Category 1 and 5	Quarterly and annually	Category 1 and 5
B230	Net Stable Funding Ratio	Domestic Firm	Category 1 and 5	Quarterly and annually	Category 1 and 5
B240	Funding Schedule	Domestic Firm and Branch	Category 1, 2 (other than a Matched Principal) and 5	Quarterly only	Category 1, 2 (other than a Matched Principal) and 5
B250	Funding Concentration	Domestic Firm and Branch	Category 1, 2 (other than a Matched Principal) and 5	Quarterly only	Category 1, 2 (other than a Matched Principal) and 5
B260	Interest Rate Risk in the Non-Trading Book	Domestic Firm and Branch	Category 1 and 2 (other than a Matched Principal)	Quarterly only	Category 1 and 2 (other than a Matched Principal)

Form Number	Form Name	Domestic Firm or Branch	Authorised Firm Category	Frequency	Semi-annual consolidated reporting for Financial Groups
B270	Currency Exposure	Domestic Firm and Branch	Category 1, 2, 3A and 5	Quarterly only	Category 1, 2 (other than a Matched Principal) and 5
B280	Outward and Inward Remittances	Domestic Firm and Branch	Category 1 and 5	Quarterly and annually	Not applicable
B310	Large Exposure	Domestic Firm and Branch	Category 1, 2, 3A and 5 and a Category 3C Authorised Firm that manages a Credit Fund	Quarterly only	Category 1, 2 (other than a Matched Principal) and 5
B320	Arrears and Provisions	Domestic Firm and Branch	Category 1, 2, 3A and 5 and a Category 3C Authorised Firm that manages a Credit Fund	Quarterly only	Category 1, 2 (other than a Matched Principal) and 5
B330	Forborne Exposures	Domestic Firm and Branch	Category 1, 2 (other than a Matched Principal) and 5	Quarterly only	Category 1, 2 (other than a Matched Principal) and 5
B340	Credit Activity	Domestic Firm and Branch	Category 1, 2, 3A and 5 and a Category 3C Authorised Firm	Quarterly only	Not applicable

Form Number	Form Name	Domestic Firm or Branch	Authorised Firm Category	Frequency	Semi-annual consolidated reporting for Financial Groups
			that manages a Credit Fund		
B350	Trade Finance Activity	Domestic Firm and Branch	Category 1, 2, 3A and 5 and a Category 3C Authorised Firm that manages a Credit Fund	Quarterly only	Not applicable
B360	Islamic Product Activity	Domestic Firm and Branch	Category 1, 2, 3A and 5	Quarterly only	Not applicable
B370	Investment Activity	Domestic Firm and Branch	Category 1, 2, 3A and 5	Quarterly only	Not applicable
B380	Investment Fair Value	Domestic Firm and Branch	Category 1, 2, 3A and 5	Quarterly only	Not applicable
B410	Advisory	Domestic Firm and Branch	All Categories	Quarterly and annually	Not applicable
B420	Asset Management	Domestic Firm and Branch	All Categories	Quarterly and annually	Not applicable
B430	Dealing overview and personnel	Domestic Firm and Branch	All Categories	Quarterly and annually	Not applicable
B440	Dealing and Arranging	Domestic Firm and Branch	All Categories	Quarterly and annually	Not applicable
B450	Staffing and Conduct	Domestic Firm and Branch	All Categories	Quarterly and annually	Not applicable

PRUDENTIAL – INVESTMENT, INSURANCE INTERMEDIATION AND BANKING (PIB)

Form Number	Form Name	Domestic Firm or Branch	Authorised Firm Category	Frequency	Semi-annual consolidated reporting for Financial Groups
B460	Money Services	Domestic Firm and Branch	Any Category of Authorised Firm licensed to Provide Money Services	Quarterly and annually	Not applicable
B470	Crowdfunding Intermediation	Domestic Firm	Any Category of Authorised Firm licensed to Operate a Crowdfunding Platform	Quarterly and annually	Not applicable
B510	Insurance Intermediation or Management	Domestic Firm and Branch	Any Category of Authorised Firm licensed as an Insurance Intermediary or Insurance Manager	Quarterly and annually	Not applicable

A2.5 Prudent Valuation Practices

Guidance

1. This following guidance sets out the DFSA’s expectations for prudent valuation practices, including adequate systems and controls and valuation methodologies, for the purpose of section 2.4 of PIB.

Systems and controls

2. Authorised Firms should establish and maintain adequate systems and controls sufficient to give management and the DFSA the confidence that their valuation estimates are prudent and reliable. These systems should be integrated with other risk management systems within the organisation (such as credit analysis). Such systems are expected to include:
 - a. documented policies and procedures for the process of valuation. This includes clearly defined responsibilities of the various areas involved in the determination of the valuation, sources of market information and review of their appropriateness, guidelines for the use of unobservable inputs reflecting the Authorised Firm’s assumptions of what market participants would use in pricing the position, frequency of independent valuation, timing of closing prices, procedures for adjusting valuations, end of the month and ad-hoc verification procedures; and
 - b. clear and independent (i.e. independent of front office) reporting lines for the department accountable for the valuation process. The reporting line should ultimately be to a main board executive director or equivalent.

Valuation methodologies

Marking-to-market

3. Marking-to-market is at least the daily valuation of positions at readily available close out prices in orderly transactions that are sourced independently. Examples of readily available close out prices include exchange prices, screen prices, or quotes from several independent reputable brokers.
4. Authorised Firms should mark-to-market as much as possible. The more prudent side of bid/offer should be used unless the institution is a significant market maker in a particular position type and it can close out at mid-market. Authorised Firms should maximise the use of relevant observable inputs and minimise the use of unobservable inputs when estimating fair value using a valuation technique. However, observable inputs or transactions may not be relevant, such as in a forced liquidation or distressed sale, or transactions may not be observable, such as when markets are inactive. In such cases, the observable data should be considered, but may not be determinative.

Marking-to-Model

5. Only where marking-to-market is not possible should Authorised Firms mark-to-model, but in such cases, the Authorised Firm should be able to demonstrate to the DFSA that the approach is prudent. Marking-to-model is defined as any valuation which has to be benchmarked, extrapolated or otherwise calculated from a market input. When marking-to-model, an extra degree of conservatism is appropriate. The DFSA will consider the following in assessing whether a mark-to-model valuation is prudent:

- a. senior management should be aware of the elements of the trading book or of other fair-valued positions which are subject to mark-to-model and should understand the materiality of the uncertainty this creates in the reporting of the risk/performance of the business;
- b. market inputs should be sourced, to the extent possible, in line with market prices (as discussed above). The appropriateness of the market inputs for the particular position being valued should be reviewed regularly;
- c. where available, generally accepted valuation methodologies for particular products should be used as far as possible;
- d. where the model is developed by the Authorised Firm itself, it should be based on appropriate assumptions, which have been assessed and challenged by suitably qualified parties independent of the development process. The model should be developed or approved independently of the front office. It should be independently tested. This includes validating the mathematics, the assumptions and the software implementation;
- e. there should be formal change control procedures in place and a secure copy of the model should be held and periodically used to check valuations;
- f. risk management should be aware of the weaknesses of the models used and how best to reflect those in the valuation output;
- g. the model should be subject to periodic review to determine the accuracy of its performance (e.g. assessing continued appropriateness of the assumptions, analysis of profit and loss versus risk factors, comparison of actual close out values to model outputs); and
- h. valuation adjustments should be made as appropriate, for example, to cover the uncertainty of the model valuation (see also valuation adjustments in Guidance notes 6 to 14.)

Independent price verification

6. Independent price verification is distinct from daily mark-to-market. It is the process by which market prices or model inputs are regularly verified for accuracy. While daily marking-to-market may be performed by dealers, verification of market prices or model inputs should be performed by a unit independent of the dealing room, at least monthly (or, depending on the nature of the market/trading activity, more frequently). It need not be performed as frequently as daily mark-to-market, since the objective, i.e. independent, marking of positions should reveal any error or bias in pricing, which should result in the elimination of inaccurate daily marks.
7. Independent price verification entails a higher standard of accuracy in that the market prices or model inputs are used to determine profit and loss figures, whereas daily marks are used primarily for management reporting in between reporting dates. For independent price verification, where pricing sources are more subjective, e.g. only one available broker quote, prudent measures such as valuation adjustments may be appropriate.

Valuation adjustments

8. As part of their procedures for marking-to-market, Authorised Firms should establish and maintain procedures for considering valuation adjustments. The DFSA expects Authorised Firms using third-party valuations to consider whether valuation adjustments are necessary. Such considerations are also necessary when marking-to-model.
9. The DFSA expects the following valuation adjustments to be formally considered at a minimum: unearned credit spreads, close-out costs, Operational Risks, early termination, investing and funding costs, and future administrative costs and, where appropriate, model risk.

Adjustment to the current valuation of less liquid positions for regulatory capital purposes

10. Authorised Firms should establish and maintain procedures for judging the necessity of and calculating an adjustment to the current valuation of less liquid positions for regulatory capital purposes. This adjustment may be in addition to any changes to the value of the position required for financial reporting purposes and should be designed to reflect the illiquidity of the position. The DFSA expects Authorised Firms to consider the need for an adjustment to a position's valuation to reflect current illiquidity whether the position is marked-to-market using market prices or observable inputs, third-party valuations or marked-to-model.
11. Bearing in mind that the assumptions made in relation to calculating VaR may not be consistent with the Authorised Firm's ability to sell or hedge out less liquid positions, where appropriate, Authorised Firms should take an adjustment to the current valuation of these positions, and review their continued appropriateness on an on-going basis. Reduced liquidity may have arisen from market events. Additionally, close-out prices for concentrated positions and/or stale positions should be considered in establishing the adjustment.
12. Authorised Firms should consider all relevant factors when determining the appropriateness of the adjustment for less liquid positions. These factors may include, but are not limited to, the amount of time it would take to hedge out the position/risks within the position, the average volatility of bid/offer spreads, the availability of independent market quotes (number and identity of market makers), the average and volatility of trading volumes (including trading volumes during periods of market stress), market concentrations, the aging of positions, the extent to which valuation relies on marking-to-model, and the impact of other model risks not included in Guidance note 10.
13. For complex products including, but not limited to, securitisation Exposures and n-th-to-default Credit Derivatives, Authorised Firms who are approved to use models to calculate Market Risk should explicitly assess the need for valuation adjustments to reflect two forms of model risk: the model risk associated with using a possibly incorrect valuation methodology; and the risk associated with using unobservable (and possibly incorrect) calibration parameters in the valuation model.
14. The adjustment to the current valuation of less liquid positions made under Guidance note 11 is likely to impact minimum Capital Requirements and may exceed those valuation adjustments made under the International Financial Reporting Standards and Guidance notes 8 and 9.

APP3 CAPITAL

A3.1 Stress and scenario testing

Guidance

1. Stress and scenario testing seek to anticipate possible losses or risks that might occur or become manifest. In applying them an Authorised Firm needs to decide how far forward to forecast and may want to consider the following factors:
 - a. how quickly it would be able to identify events or changes in circumstances that might lead to a loss occurring or risk crystallising; and
 - b. after the event or circumstance has been identified, how quickly and effectively the Authorised Firm could act to prevent or mitigate any resulting loss occurring or risk crystallising and to reduce its Exposure to any further adverse event or change in circumstance.
2. For example, the time horizon over which stress and scenario testing would need to be carried out for the Market Risk arising from the holding of investments would depend upon:
 - a. the extent to which there is a regular, open and transparent market in those assets, which would allow fluctuations in the value of the investment to be more readily and quickly identified; and
 - b. the extent to which the market in those assets is liquid (and would remain liquid in the changed circumstances contemplated in the stress or scenario test) which would allow the Authorised Firm, if needed, to sell its holding so as to prevent or reduce its Exposure to future price fluctuations.
3. Authorised Firms should focus on those scenarios and combinations of scenarios that are considered reasonably likely to occur. For this purpose other risks and losses include business risk, i.e. the potential impact of changes in business plans, future activities, and the business or economic environment.
4. In identifying what realistic combinations of losses or risks might occur or crystallise, an Authorised Firm should take into account scenarios in which expected correlations occur and where they might break down.
5. In identifying scenarios and assessing their impact, an Authorised Firm should take into account how changes in circumstances might impact upon:
 - a. the nature, scale and mix of future activities; and
 - b. the behaviour of Counterparties, and of the Authorised Firm itself, including the exercise of choices (including options embedded in financial instruments).
6. In determining whether it would have adequate financial resources in the event of each identified adverse scenario, an Authorised Firm should:
 - a. only include financial resources that could reasonably be relied upon as being available in the circumstances of the identified scenario; and

- b. consider any legal or other restriction on the purposes for which financial resources may be used, including any restriction on the transfer to the DIFC of assets held in other jurisdictions.

APP4 CREDIT RISK

A4.1 Credit Risk systems and controls

Guidance

1. Depending on an Authorised Firm's nature, scale, frequency, and complexity of Credit Risk granted or incurred, the Credit Risk policy of an Authorised Firm should address the following elements:
 - a. how, with particular reference to its activities, the Authorised Firm defines and measures Credit Risk;
 - b. the Authorised Firm's business aims in incurring Credit Risk including:
 - i. identifying the types and sources of Credit Risk to which the Authorised Firm wishes to be exposed (and the limits on that Exposure) and those to which the Authorised Firm wishes not to be exposed;
 - ii. specifying the level of diversification required by the Authorised Firm and the Authorised Firm's tolerance for risk concentrations and the limits on those Exposures and concentrations; and
 - iii. stating the risk-return that the Authorised Firm is seeking to achieve on Credit Risk Exposures;
 - c. types of facilities to be offered, along with ceilings, pricing, profitability, maximum maturities and maximum debt-servicing ratios for each type of lending;
 - d. a ceiling for the total loan portfolio, in terms, for example, of the loan-to-deposit ratio, undrawn commitment ratio, a maximum dollar amount or a percentage of capital base;
 - e. portfolio limits for maximum aggregate Exposures by country, industry, category of borrower/Counterparty (e.g. banks, non-bank financial institutions, corporates and retail), product (e.g. property lending), Groups of related parties and single borrowers;
 - f. limits, terms and conditions, approval and review procedures and records kept for Connected lending - all Authorised Firms should have a formal policy statement, endorsed by the Governing Body, on such lending covering these matters;
 - g. types of acceptable Collateral, loan-to-value ratios and the criteria for accepting guarantees; and
 - h. how Credit Risk is assessed both when credit is granted or incurred and subsequently, including how the adequacy of any security and other risk mitigation techniques are assessed;
 - i. the detailed limit structure for Credit Risk, which should:
 - i. address all key risk factors, including intra-Group Exposures;
 - ii. be commensurate with the volume and complexity of activity; and
 - iii. be consistent with the Authorised Firm's business aims, historical performance, and the level of capital the Authorised Firm is willing to risk;

- j. procedures for:
 - i. approving new products and activities which give rise to Credit Risk;
 - ii. regular risk position and performance reporting;
 - iii. limit exception reporting and approval; and
 - iv. identifying and dealing with problem Exposures;
 - k. the allocation of responsibilities for implementing the Credit Risk policy and for monitoring adherence to, and the effectiveness of, the policy; and
 - l. the required information systems, staff and other resources.
2. The Credit Risk policy should emphasize the principles of prudence and should be enforced consistently. The policy and its implementation should ensure that credit facilities are only granted to credit-worthy customers and that risk concentrations are avoided.
 3. The Credit Risk strategy and policy need to be clearly disseminated to, and understood by all relevant staff.
 4. The Credit Risk policy of an Authorised Firm should clearly specify the delegation of its credit approval authorities. Credit authority thus delegated should be appropriate for the products or portfolios assigned to the credit committee or individual credit officers and should be commensurate with their credit experience and expertise. An officer's credit authority may, however, be increased on the basis of his or her track record. An Authorised Firm should ensure that credit authority is required for acquiring any types of credit Exposures, including the use of Credit Derivatives for hedging or income generation.
 5. Credit authority delegated to the credit committee and each credit officer should be subject to regular review to ensure that it remains appropriate to current market conditions and the level of their performance.
 6. An Authorised Firm's remuneration policies should be consistent with its Credit Risk strategy. The policies should not encourage officers to generate short-term profits by taking an unacceptably high level of risk.

Counterparty Risk assessment

7. The Authorised Firm should make a suitable assessment of the risk profile of its Counterparties. The factors to be considered will vary according to both the type of credit and Counterparty such as whether the Counterparty is a company or a sovereign Counterparty and may include:
 - a. the purpose of the credit and the source of repayment;
 - b. an assessment of the skill, integrity and quality of management and overall reputation of the Counterparty;
 - c. the legal capacity of the Counterparty to assume the liability to the Authorised Firm;
 - d. an assessment of the nature and amount of risk attached to the Counterparty in the context of the industrial sector or geographical region or country in which it operates and the potential impact on the Counterparty of political, economic and market changes; this assessment may include consideration of the extent and nature of the Counterparty's other financial obligations;

- e. the Counterparty's repayment history as well as an assessment of the Counterparty's current and future capacity to repay obligations based on financial statements, financial trends, cash flow projections and the potential impact of adverse economic scenarios;
 - f. an analysis of the risk-return trade-off, with regard to the proposed price of the credit facility;
 - g. the proposed terms and conditions attached to the granting of credit, including on-going provision of information by the Counterparty, covenants attached to the facility, the adequacy and enforceability of Collateral and guarantees; and
 - h. the Authorised Firm's existing Exposure to the individual Counterparty, sector, country or product and the availability of credit given Exposure limits.
- 8. An Authorised Firm should document any variation from the usual credit policy.
 - 9. An Authorised Firm involved in loan syndications or consortia should not rely on other parties' assessments of the Credit Risks involved but should conduct a full assessment against its own Credit Risk policy.
 - 10. An Authorised Firm granting credit to obligors in other countries should be cognisant of the additional risks - country risk and transfer risk – involved in such credits. An Authorised Firm should therefore consider the environment - economic and political - in the relevant countries, the potential effect of changes thereto on the obligors' ability to service the credit and the contagion effects in regions where economies are closely related.
 - 11. The exception reporting should be adequately supported by a management reporting system whereby relevant reports on the credit portfolio are generated to various levels of management on a timely basis.
 - 12. Connected Counterparties should be identified and the procedures for the management of the combined Credit Risk considered. It may be appropriate for Authorised Firms to monitor and report the aggregate Exposure against combined limits in addition to monitoring the constituent Exposures to the individual Counterparties.
 - 13. An Authorised Firm should consider whether it needs to assess the credit-worthiness of suppliers of goods and services to whom it makes material prepayments or advances.

Risk assessment: Derivative Counterparties

- 14. An Authorised Firm should include in its Credit Risk policy an adequate description of:
 - a. how it determines with which derivative Counterparties to do business;
 - b. how it assesses and continues to monitor the credit-worthiness of those Counterparties;
 - c. how it identifies its actual and contingent Exposure to the Counterparty; and
 - d. whether and how it uses credit loss mitigation techniques, e.g. margining, taking security or Collateral or purchasing credit insurance.
- 15. In assessing its contingent Exposure to a Counterparty, the Authorised Firm should identify the amount which would be due from the Counterparty if the value, index or other factor upon which that amount depends were to change.
- 16. An Authorised Firm should clearly specify the delegation of its credit approval authorities. Credit authority thus delegated should be appropriate for the products or portfolios assigned to the credit committee or individual credit officers and should be commensurate with their credit experience and

expertise. An officer's credit authority may, however, be increased on the basis of his/her track record. An Authorised Firm should ensure that credit authority is required for acquiring any types of credit Exposures, including the use of Credit Derivatives for hedging or income generation.

17. Credit authority delegated to the credit committee and each credit officer should be subject to regular review to ensure that it remains appropriate to current market conditions and the level of their performance.

Credit approval procedures

18. An Authorised Firm should adhere closely to the "Know Your Customer" principle and should not lend purely on name and relationship without a comprehensive assessment of the credit quality of the borrower.
19. Credit decisions should be supported by adequate evaluation of the borrower's repayment ability based on reliable information. Sufficient and up-to-date information should continue to be available to enable effective monitoring of the account.
20. All credits should be granted on an arm's length basis. Credits to related borrowers should be monitored carefully and steps taken to control or reduce the risks of Connected lending.
21. An Authorised Firm should not rely excessively on Collateral or guarantees as Credit Risk mitigants. Such Credit Risk mitigants may provide secondary protection to the lender if the borrower defaults, the primary consideration for credit approval should be the borrower's debt-servicing capacity.
22. An Authorised Firm should be sensitive to rapid expansion of specific types of lending. Such trends may often be accompanied by deterioration of credit standards and thus merit increased focus on more marginal borrowers.
23. An Authorised Firm should ensure through periodic independent audits that the credit approval function is being properly managed and that credit Exposures comply with prudential standards and internal limits. The results of such audits should be reported directly to the Governing Body, the credit committee or senior management as appropriate.

Risk control

24. An Authorised Firm should consider setting credit limits for maximum Exposures to single and Connected Counterparties, as well as limits for aggregate Exposures to economic sectors, geographic areas, and on total Credit Risk arising from specific types of products. By limiting Exposures in these categories, an Authorised Firm can manage credit Exposure more carefully and avoid excessive concentrations of risk.
25. The Credit Risk policy of an Authorised Firm should include a policy to control and monitor Large Exposures and other risk concentrations. An Authorised Firm should carefully manage and avoid excessive risk concentrations of various kinds. These include Exposure to:
 - a. individual borrowers (in particular Exposure exceeding 10% of the firm's capital base);
 - b. Groups of borrowers with similar characteristics, economic and geographical sectors; and
 - c. types of lending with similar characteristics (e.g. those based on assets with similar price behaviour).
26. Notwithstanding the Concentration Risk limit specified as part of the prudential Rules on Large Exposures, Authorised Firms should exercise particular care in relation to facilities exceeding 10% of capital base.

27. Authorised Firms should recognise and control the Credit Risk arising from their new products and services. Well in advance of entering into business transactions involving new types of products and activities, they should ensure that they understand the risks fully and have established appropriate Credit Risk policies, procedures and controls, which should be approved by the Governing Body or its appropriate delegated committee. A formal risk assessment of new products and activities should also be performed and documented.
28. An Authorised Firm in Category 1, 2, 3A or 5 is also subject to concentration limits and notification requirements as spelt out in chapter 4.

Risk measurement

29. An Authorised Firm should measure its Credit Risk using a robust and consistent methodology, which should be described in its Credit Risk policy. The Authorised Firm should consider whether the measurement methodology should be back-tested and the frequency of any such back-testing.
30. An Authorised Firm should also be able to measure its total Exposure across the entire credit portfolio or within particular categories such as Exposures to particular industries, economic sectors or geographical areas.
31. Where an Authorised Firm is a member of a Group, the Group should be able to monitor credit Exposures on a consolidated basis.
32. An Authorised Firm should have the capability to measure its credit Exposure to individual Counterparties on at least a daily basis.
33. Authorised Firms should analyse their credit portfolios to identify material inter-dependencies which can exaggerate risk concentrations. The importance can be illustrated by the contagion effects that a substantial decline in property or stock prices may have on the default rate of those commercial and industrial loans which rely heavily on such types of Collateral.
34. Authorised Firms should establish a system of regular independent credit and compliance audits. These audits should be performed by independent parties, e.g. internal audit and compliance, which report to the Governing Body or the audit committee.
35. Credit audits should be conducted to assess individual credits on a sampling basis and the overall quality of the credit portfolio. Such audits are useful for evaluating the performance of account officers and the effectiveness of the credit process. They can also enable Authorised Firms to take early measures to protect their loans.

Risk monitoring

36. An Authorised Firm should implement an effective system for monitoring its Credit Risk, which should be described in its Credit Risk policy. The system may monitor the use of facilities, adherence to servicing requirements and covenants, and monitor the value of Collateral and identify problem accounts.
37. An Authorised Firm should consider the implementation of a system of management reporting which provides relevant, accurate, comprehensive, timely and reliable Credit Risk reports to relevant functions within the Authorised Firm.
38. Adequacy and sophistication of Credit Risk measurement tools required depends on the complexity and degree of the inherent risks of the products involved. An Authorised Firm should have information systems and analytical techniques that provide sufficient information on the risk profile and structure of the credit portfolio. These should be flexible to help Authorised Firm to identify

risk concentrations. To achieve this, an Authorised Firm system should be capable of analysing its credit portfolio by the following characteristics:

- a. size of Exposure;
 - b. Exposure to Groups of related borrowers;
 - c. products;
 - d. sectors, e.g. geographic, industrial;
 - e. borrowers' demographic profile for consumer credits, e.g. age or income group, if appropriate;
 - f. account performance;
 - g. internal credit ratings;
 - h. outstandings versus commitments; and
 - i. types and coverage of Collateral.
39. An Authorised Firm should have procedures for taking appropriate action according to the information within the management reports, such as a review of Counterparty limits.
 40. Particular attention should be given to the monitoring of credit that does not conform to usual Credit Risk policy, or which exceeds predetermined credit limits and criteria, but is sanctioned because of particular circumstances. Unauthorised exceptions to policies, procedures and limits should be reported in a timely manner to the appropriate level of management.
 41. Individual credit facilities and overall limits and sub-limits should be periodically reviewed in order to check their performance and appropriateness for both the current circumstances of the Counterparty and in the Authorised Firm's current internal and external economic environment. The frequency of review will usually be more intense for higher-risk counterparties or larger Exposures or in fluctuating economic conditions.
 42. An Authorised Firm should have in place a system for monitoring the overall quality of its Credit Risk Exposures under normal and stressful conditions. There should also be a reporting system which alerts management to aggregate Exposures approaching various pre-set portfolio limits.
 43. An Authorised Firm should be mindful of business and economic cycles and regularly stress-test their portfolios against adverse market scenarios. Adequate contingency planning should be developed in conjunction with stress-testing to address the possibility of crises developing in a very rapid fashion.
 44. Appropriate stress testing of credit Exposures can be an essential part of the credit management process. Examination of the potential effects of economic or industry downturns, market events, changes in interest rates, changes in foreign exchange rates and changes in liquidity conditions can provide valuable information about an Authorised Firm's Credit Risk. This information can be utilised to inform the Authorised Firm's on-going credit strategy.
 45. As new techniques for Credit Risk management, monitoring and reporting are developed, the Authorised Firm should ensure they are tested and evaluated before undue reliance is placed upon them.
 46. Where the account officer for a credit (or customer relationship manager, branch manager or similar) moves on, the incoming officer should carry out a take-over review. The review should cover inter

alia the credit- worthiness of the borrowers, the adequacy of the documentation, compliance with covenants, performance of each loan and the existence and value of any Collateral.

Problem Exposures

47. An Authorised Firm should have processes for the timely identification and management of problem Exposures. These processes should be described in the Credit Risk policy.
48. An Authorised Firm involved in providing credit should establish a dedicated unit to handle the recovery and work-out of problem loans and should establish policies for the referral of loans to this unit.
49. An Authorised Firm should ensure that its loan portfolio is properly classified and has an effective early-warning system for problem loans.
50. An Authorised Firm should develop and implement internal risk rating systems for managing Credit Risk. The rating system should be consistent with the nature, size and complexity of the Authorised Firm's activities. In using internal risk ratings, an Authorised Firm should seek to achieve a high granularity in the rating system and adopt multiple grades for loans that are not yet irregular and to develop the ability to track the migration of individual loans through the various internal credit ratings.
51. Depending on the size and nature of the Authorised Firm, it may be appropriate for problem Exposures to be managed by a specialised function, independent of the functions that originate the business or maintain the on-going business relationship with the Counterparty.
52. Exposures identified as problems or potential problems should be closely monitored by management, and an Authorised Firm should set out, for example, whether a loan grading system or a watch or problem list is used and, in the latter case, the criteria for adding an asset to or taking an asset off that list.
53. It is recommended that Authorised Firms establish a dedicated unit to handle the recovery and work-out of problem loans and put in place policies for the referral of loans to this unit.
54. An Authorised Firm should have adequate procedures for recovering Exposures in arrears or those which had provisions made against them. These should allocate responsibility both internally and externally for its arrears management and recovery and define the involvement of the Authorised Firm's solicitors.
55. Requirements relating to provisioning against loss on problem Exposures are covered in chapter 4.

Risk mitigation

56. Various methods can be used to mitigate Credit Risk, such as taking security or Collateral, obtaining a guarantee from a third party, purchasing insurance or Credit Derivatives. Authorised Firms should view these as complementary to, rather than a replacement for, thorough credit analysis and procedures.
57. In controlling Credit Risk, an Authorised Firm may utilise certain mitigation techniques. Normally, they include:
 - a. accepting Collateral, standby letters of credit and guarantees;
 - b. entering into Netting arrangements,
 - c. setting strict loan covenants; and

- d. using Credit Derivatives and other hedging instruments.
58. In determining which types of credit mitigation techniques should be used, firms should also consider:
- a. their own knowledge and experience in using such techniques;
 - b. cost-effectiveness;
 - c. type and financial strength of the Counterparties or Issuers;
 - d. correlation with the underlying credits;
 - e. availability, liquidity and realisability of the credit mitigation instruments;
 - f. the extent to which legally recognised documentation, e.g. ISDA Master Agreement, can be adopted; and
 - g. the degree of supervisory recognition of the mitigation technique.
59. While mitigation through Collateral and guarantees is usually dealt with at the time of granting of credits, Credit Derivatives and Netting are often employed after the credit is in place, or used to manage the overall portfolio risk. When the mitigation arrangements are in place they should then be controlled. Authorised Firms should have written policies, procedures and controls for the use of credit mitigation techniques. They should also ensure adequate systems are in place to manage these activities.
60. Authorised Firms should revalue their Collateral and mitigation instruments on a regular basis. The method and frequency of revaluation depends on the nature of the hedge and the products involved.
61. If an Authorised Firm takes security or Collateral, on credit facilities, appropriate policies and procedures should be documented covering:
- a. the types of security or Collateral considered;
 - b. procedures governing the valuation and revaluation of security or Collateral including the basis of valuation;
 - c. policies governing the taking of security or Collateral, including obtaining appropriate legal title; and
 - d. policies governing possession of security or Collateral.
62. The value of security and Collateral should be monitored at an appropriate frequency. For example, commercial property might be revalued annually, whereas Securities provided as Collateral should be marked to market usually on a daily basis. Residential property may not need to be revalued annually, but information should be sought as to general market conditions.
63. When taking Collateral in support of an Exposure, an Authorised Firm should ensure that legal procedures have been followed, to ensure the Collateral can be enforced if required.
64. An Authorised Firm should consider the legal and financial ability of a guarantor to fulfil the guarantee if called upon to do so.
65. An Authorised Firm should analyse carefully the protection afforded by risk mitigants such as Netting agreements or Credit Derivatives, to ensure that any residual Credit Risk is identified, measured, monitored and controlled.

66. An Authorised Firm providing mortgages at high loan-to-value ratios should consider the need for alternative forms of protection against the risks of such lending, including mortgage indemnity insurance, to protect against the risk of a fall in the value of the property.

Record keeping

67. The Authorised Firm should maintain appropriate records of:
- a. credit Exposures, including aggregations of credit Exposures, by:
 - i. Groups of Connected Counterparties; and
 - ii. types of Counterparty as defined, for example, by the nature or geographical location of the Counterparty;
 - b. credit decisions, including details of the facts or circumstances upon which a decision was made; and
 - c. information relevant to assessing current credit quality.
68. Credit records should be retained for at least six years, subject to any requirement in the Rules requiring such records to be kept for a longer period.
69. It is important that sound and legally enforceable documentation is in place for each credit agreement as this may be called upon in the event of a default or dispute. An Authorised Firm should therefore consider whether it is appropriate for an independent legal opinion to be sought on documentation used by the Authorised Firm. Documentation should be in place before the Authorised Firm enters into a contractual obligation or releases funds.

Country and transfer risk Exposure

70. Chapter 4 does not provide limits on the size of an Authorised Firm's Exposure to a particular country or region. However, an Authorised Firm which has Large Exposures in a country or region should include in its Credit Risk policy:
- a. the geographical areas in which the Authorised Firm does or intends to do business;
 - b. its definition of Credit Risk Exposure and transfer risks (such as exchange restrictions) associated with doing business in each country or region;
 - c. how to measure its total Exposure in each country or region and across several countries or regions;
 - d. the types of business the Authorised Firm intends to undertake in each country or region;
 - e. limits on Exposures to an individual country or region which the Authorised Firm deals with, and sub-limits for different types of business if appropriate;
 - f. the procedure for setting and reviewing country or regional limits; and
 - g. the process by which the Authorised Firm's actual country or regional Exposures will be monitored against limits and the procedure to be followed if the limits are breached.
71. When setting country or regional limits, an Authorised Firm should consider:
- a. the economic and political circumstances prevailing in the country or region;

- b. the transfer risks associated with any particular country or region;
- c. the type and maturity of business undertaken by the Authorised Firm in a particular country or region;
- d. the Authorised Firm's existing concentration of country or regional risk;
- e. the source of funding for the country or regional Exposure; and
- f. sovereign or other guarantees offered.

Provisioning

72. Depending upon the nature of the Authorised Firm and its business, the Authorised Firm's provisioning policy should set out:
- a. who has responsibility for reviewing the provisioning policy and approving any changes;
 - b. how frequently the policy should be reviewed;
 - c. when the review will take place, including the circumstances in which a review might be more frequent;
 - d. who has primary responsibility for ensuring the provisioning policy remains appropriate, including any division of responsibilities;
 - e. the areas of its business to which the provisioning policy relates - it should include both on balance sheet and off balance sheet Exposures and assets;
 - f. where it takes different approaches to different lines of its business and the key features of those differences;
 - g. who has responsibility for monitoring its asset portfolio on a regular basis in order to identify problem or potential problem assets and the factors it takes into account in identifying them;
 - h. whether a loan grading system or a watch or problem list is used and, in the latter case, the criteria for adding an asset to or taking an asset off that list;
 - i. the extent to which the value of any Collateral, guarantees or insurance which the Authorised Firm holds affects the need for or size of provision;
 - j. on what basis the Authorised Firm makes its provisions, including the extent to which the level of provisioning is left to managerial judgement or to a committee or involves specified formulae and the methodologies or debt management systems and other formulae used to determine provisioning levels for different business lines and the factors applied within these methodologies;
 - k. who is responsible for ensuring that the Authorised Firm's provisioning policy is being implemented properly, and the measures the Authorised Firm has in place if its provisioning policies are not adhered to;
 - l. who is responsible for the regular reviews of the Authorised Firm's specific and general provisions and who decides whether provision levels are satisfactory. The reviews should take account of changes in the status of the Exposures and potential losses and changes in the conditions associated with them;

- m. the reports used to enable management to ensure that the Authorised Firm's provisioning levels remain satisfactory, the frequency and purpose of those reports and their circulation;
 - n. the procedures for recovering Exposures in arrears or Exposures which have had provisions made against them, including who has responsibility both internally and externally for its arrears management and recovery and the involvement of the Authorised Firm's solicitors;
 - o. the procedures and methodologies for writing off and writing back provisions, including treatment of interest and who has the relevant responsibility for determining these;
 - p. the frequency of any review of its write off experience against provisions raised; such a review can help identify whether an Authorised Firm's policies result in over or under provisioning across the business cycle, and contribute to a general review of an Authorised Firm's provisioning policy and the design of any loan grading systems, Credit Risk models, and risk pricing; and
 - q. the Authorised Firm's procedures and methodologies for calculating and raising provisions for contingent and other liabilities, how frequently they should be reviewed and who has the relevant responsibilities. Other liabilities include the crystallisation of contingent liabilities such as acceptances, endorsements, guarantees, performance bonds, indemnities, irrevocable letters of credit and the confirmation of documentary credits.
73. Provisions may be general (against the whole of a given portfolio) or specific (against particular Exposures identified as bad or doubtful), or both. The DFSA expects contingent liabilities and anticipated losses to be recognised in accordance with the applicable accounting standards.
74. Appropriate systems and controls for provisions vary with the nature, scale and complexity of the credit granted. An Authorised Firm for which the extension of credit is a substantial part of its business is expected to have greater regard to developing, implementing and documenting a provisioning policy than an Authorised Firm for which Credit Risk is incidental to the operation of its business.
75. The DFSA recognises that the frequency with which an Authorised Firm reviews its provisioning policy once it has been established will vary from firm to firm. However, the DFSA expects an Authorised Firm to review its policy to ensure it remains appropriate for the business it undertakes and the economic environment in which it operates. The provisioning policy should be reviewed at least annually by the Governing Body.

A4.2 Credit conversion factors (CCFs) for calculating Exposures

CCFs for off-balance sheet CR Exposures

- A4.2.1** The applicable CCFs for off-balance sheet CR Exposures are provided in the table below.

	Description of Off-balance Sheet Item	CCF
(a)	Direct credit substitutes, in particular general guarantees of indebtedness (including standby letters of credit servicing as financial guarantees for loans and securities) and acceptances (including endorsements with the character of acceptances)	100%
(b)	Asset sales with recourse, where the Credit Risk remains with the Authorised Firm	100%
(c)	Securities lent by the Authorised Firm or securities posted by the Authorised Firm as Collateral, including instances where these arise out of repo-style transactions	100%
(d)	Forward asset purchases, forward deposits and partly paid shares and securities which represent commitments with certain drawdown	100%
(e)	Other items that are credit substitutes not included in any other category	100%
(f)	Note issuance facilities and revolving Underwriting facilities regardless of the maturity of the underlying facility	50%
(g)	Performance bonds, bid bonds, warranties and standby letters of credit related to particular transactions and other similar transactions related contingent items	50%
(h)	Commitments, regardless of the maturity of the underlying facility, unless they fall under another category	40%
(i)	Self-liquidating trade letters of credit arising from the movement of goods, in particular documentary credits collateralised by the underlying shipment, with an Original Maturity of less than one year	20%
(j)	Commitments that are unconditionally cancellable at any time without prior notice or that effectively provide for automatic cancellation due to deterioration in a borrower's creditworthiness	10%

Guidance

In cases where there is an undertaking to provide a commitment on another off-balance sheet Exposure, an Authorised Firm should apply the lower of the applicable CCFs. For example, if an Authorised Firm has a commitment to open self-liquidating trade letters of credit arising from the movement of goods, in particular documentary credits collateralised by the underlying shipment with an Original Maturity of less than one year, a 20% CCF should be applied instead of a 40% CCF.

CCFs for off-balance sheet SE Exposures

- A4.2.2** (1) The applicable CCFs for off-balance sheet SE Exposures are provided in the table below.

	Description of off-balance sheet item	CCF
(a)	Unrated eligible liquidity facilities	50%
(b)	Eligible Servicer cash advance facilities	0%
(c)	Others	100%

- (2) An Authorised Firm must notify the DFSA if it intends to provide eligible Servicer cash advance facilities and when there is a drawdown.

Guidance

Eligible Servicer cash advance facilities refers to undrawn Servicer cash advances or facilities that are contractually provided for and unconditionally cancellable without prior notice, so long as the Servicer is entitled to full reimbursement and this right is senior to other claims on cash flows from the underlying Exposures.

A4.3 Collateral calculations and haircuts

Core Market Participants

A4.3.1 For the purposes of this section, “core market participant” means:

- (a) any central government or Central Bank;
- (b) any PSE;
- (c) any MDB referred to in Rule 4.12.5;
- (d) any banking institution or securities firm;
- (e) any financial institution eligible for a 20% risk weight under section 4.12;
- (f) any Qualifying CCP;
- (g) any regulated mutual fund that is subject to capital or leverage requirements; or
- (h) any regulated pension fund.

Calculation of E* for collateralised transactions under FCCA

A4.3.2 An Authorised Firm using the FCCA must, subject to A4.3.3, calculate the Exposure amount after risk mitigation, E*, of a collateralised transaction other than transactions referred to in Rule 4.14A1(a) to (c), using the following formula:

$$E^* = \max\{0, [E \cdot (1 + H_e) - C \cdot (1 - H_c - H_{fx})]\}$$

where:

E = fair value of the Exposure, determined in accordance with section 4.9

H_e = haircut appropriate to the Exposure, determined in accordance with Rule A4.3.5

C = fair value of the eligible financial Collateral received

H_c = haircut appropriate to the Collateral, determined in accordance with Rule A4.3.5

H_{fx} = haircut appropriate for currency mismatch between the Collateral and Exposure, calculated in accordance with Rule 4.13.13

Guidance

Where the residual maturity of the Collateral is shorter than the residual maturity of the Exposure, the value of the Collateral received is adjusted under Rules 4.13.14 to 4.13.16.

- A4.3.3** (1) An Authorised Firm using the FCCA may, subject to (2), calculate the Exposure amount after risk mitigation, E^* , for any SFT covered by a Qualifying Bilateral Netting Agreement, using the following formula:

$$E^* = \max \left\{ 0; \sum_i E_i - \sum_i C_i + 0.4 \cdot \text{net exposure} + 0.6 \cdot \frac{\text{gross exposure}}{\sqrt{N}} + \sum_{FX} (E_{fx} \cdot H_{fx}) \right\}$$

where:

E_i = fair value of all cash lent and non-cash posted Collateral

C_i = fair value of all cash borrowed and non-cash received Collateral

$$\text{net exposure} = \left| \sum_S E_S H_S \right|$$

$$\text{gross exposure} = \sum_S E_S |H_S|$$

E_S = net fair value of each security issuance under the Netting Set (always a positive value)

H_S = haircut appropriate to E_S , determined in accordance with Rule A4.3.5 (H_S has a positive (negative) sign if the security is posted (received))

N = the number of security issues contained in the Netting Set (except that issuance where the value of E_S is less than one tenth of the value of the largest E_S in the Netting Set are not included in the count)

E_{fx} = the absolute value of the net position in each currency fx different from the settlement currency

H_{fx} = haircut appropriate for currency mismatch of currency fx , calculated in accordance with Rule 4.13.13

(2) An Authorised Firm may only recognise Netting across positions in the Trading Book and the Non-Trading Book if the netted transactions fulfil the following conditions:

- (i) all transactions are marked to market daily; and
- (ii) the Collateral used in the transactions are recognised as eligible financial Collateral in accordance with section 4.13.

A4.3.4 For the purpose of Rule A4.3.3, a qualifying bilateral netting agreement is a netting agreement with a Counterparty which:

- (a) provides the non-defaulting party the right to terminate and close out in a timely manner all transactions under the agreement upon an event of default, including in the event of insolvency or bankruptcy of the counterparty;
- (b) provide for the netting of gains and losses on transactions (including the value of any Collateral) terminated and closed out under it so that a single net amount is owed by one party to the other;
- (c) allow for the prompt liquidation or set-off of Collateral upon the event of default; and
- (d) is legally enforceable in each relevant jurisdiction upon the occurrence of an event of default and regardless of the counterparty's insolvency or bankruptcy.

A4.3.5 For the purpose of Rules A4.3.2 and A4.3.3, an Authorised Firm must use the supervisory haircuts set out in the table below (assuming daily remargining or revaluation and a ten-business day holding period) to determine the haircuts to the Collateral, H_e , and to the Exposure, H_c .

Eligible Financial Collateral		Supervisory Haircut		
Issue Rating for Debt Securities	Residual Maturity	Sovereigns	Other Issuers	Securitisation Exposure
Any debt security with a Credit Quality Grade of 1 or short- term Credit Quality Grade of "I"	≤ 1 year	0.5	1	2
	> 1 year, ≤ 3 years	2	3	8
	> 3 years, ≤ 5 years		4	
	> 5 years, ≤ 10 years	4	6	16
	> 10 years		12	
2-3 (long- and short-term) and unrated bank	≤ 1 year	1	2	4
	> 1 year, ≤ 3 years	3	4	12

securities (as defined in Rule 4.13.5(1)(d))	> 3 years, ≤ 5 years		6	
	> 5 years, ≤ 10 years	6	12	24
	> 10 years		20	
Any debt security with a Credit Quality Grade of 4	All	15	Not eligible	Not eligible
Any equity (including a convertible bond) included in a main index and gold		20		
Any other equity (including a convertible bond) listed on a regulated exchange		30		
Any security that the Authorised Firm lends, or posts as Collateral but not eligible financial Collateral		30		
Any Unit in a Collective Investment Fund		Highest haircut applicable to any security in which the fund can invest		
Cash in the same currency as the underlying Exposure		0		

Guidance

PSEs referred to in Rule 4.12.3(2) and MDBs referred to in Rule 4.12.5 should be treated as equivalent to sovereigns under this table. Other PSEs should be treated as equivalent to other issuers.

A4.3.6 Where the Collateral consists of a basket of eligible items, an Authorised Firm must calculate the supervisory haircut on the basket, H , as follows:

$$H = \sum_i a_i H_i$$

where:

a_i = the weight of the Collateral (as measured by units of currency) in the basket

H_i = the haircut applicable to the Collateral

Minimum holding periods, remargining or revaluation conditions

A4.3.7 The following table sets out the minimum holding periods and remargining or revaluation conditions for different types of transactions:

Transaction type	Minimum holding period	Remargining/ Revaluation Condition
SFTs	5 business days	daily remargining
Other capital market transactions	10 business days	daily remargining
Secured lending	20 business days	daily revaluation

A4.3.8 Under the following circumstances, an Authorised Firm must apply a higher minimum holding period:

- (a) for a Netting Set consisting of more than 5,000 transactions, the minimum holding period of risk is 20 business days;
- (b) for a Netting Set containing one or more transactions involving either illiquid Collateral or an OTC derivative that cannot be easily replaced, the minimum holding period of risk is 20 business days; and
- (c) if an Authorised Firm has experienced more than two margin call disputes on a particular Netting Set over the previous two quarters that have lasted longer than the applicable minimum holding period (before consideration of this provision), the Authorised Firm must use a minimum holding period that is at least double the supervisory floor for that netting set for the subsequent two quarters.

A4.3.9 Where the assumed minimum holding period is not met or remargining or revaluation conditions are not fulfilled, an Authorised Firm must calculate the applicable haircut, H, using the following formula:

$$H = H_{10} \sqrt{\frac{N_R + (T_{\min} - 1)}{10}}$$

where:

H_{10} = ten business-day haircut for Collateral

N_R = actual number of business days between remargining or revaluation, as applicable

$T_{\min}$ = the minimum holding period for the type of transaction, determined in accordance with Rules A4.3.7 and A4.3.8

Recognition of eligible financial Collateral under FCSA

A4.3.10 Subject to A4.3.11, an Authorised Firm which has taken eligible financial Collateral for a CR Exposure and is using the FCSA may recognise the effects of Credit Risk mitigation of the eligible financial Collateral as follows:

- (a) break down the Exposure into -

- (i) a collateralised portion with E equal to the latest fair value of the eligible financial Collateral; and
 - (ii) an uncollateralised portion with E equal to the E of the CR Exposure less the latest fair value of the eligible financial Collateral;
- and
- (b) for the purposes of calculating the Credit RWA amount pursuant to Rule 4.8.3, use:
 - (i) for the collateralised portion, the CRW that is applicable to the eligible financial Collateral as though the Authorised Firm had a direct Exposure to that Collateral; and
 - (ii) for the uncollateralised portion, the CRW that is applicable to the obligor.

A4.3.11 If the CRW determined in accordance with A4.3.10(b)(i) is less than 20%, an Authorised Firm must apply a CRW of 20% to the collateralised portion of the CR Exposure, except in the following cases:

- (a) a qualifying SFT where the Counterparty in the transaction is a core market participant, in which case the Authorised Firm may apply a risk weight of 0%;
- (b) a qualifying SFT where the Counterparty in the transaction is not a core market participant, in which case the Authorised Firm may apply a risk weight of 10%;
- (c) a transaction where there is no currency mismatch and the Collateral comprises:
 - (i) cash on deposit as set out in Rule 4.13.5(1)(a); or
 - (ii) Exposures to a central government, Central Bank or PSE, or a combination thereof, qualifying for a 0% risk weight under the Rules in section 4.12, and the latest fair value of such Collateral has been discounted by 20% for the purposes of determining the value of the collateralised portion of the CR Exposure in accordance with Rule A4.3.10(a)(i),

in which case the Authorised Firm may apply a CRW of 0%.

Guidance

An SFT is a qualifying SFT if the requirements in section A4.4 are met.

Recognition of eligible guarantees and eligible Credit Derivatives

A4.3.12 Where an Authorised Firm has an Exposure that is covered by an eligible guarantee or by an eligible Credit Derivative, the Authorised Firm may recognise the effect of Credit Risk mitigation as follows:

- (1) (a) break down the Exposure into -
 - (i) a protected portion; and

- (ii) an unprotected portion;
 - and
- (b) for the purposes of calculating the Credit RWA amount pursuant to Rule 4.8.3, use:
 - (i) for the protected portion, the CRW that is applicable to the protection provider as though the Authorised Firm had a direct Exposure to that provider; and
 - (ii) for the unprotected portion, the CRW that is applicable to the obligor.
- (2) Where a guarantee or Credit Derivative provides for a materiality threshold on payments below which no payment will be made in the event of loss, it is equivalent to a retained first loss position and must be deducted from the CET1 Capital.
- (3) Where losses are shared *pari passu* on a pro rata basis between the Authorised Firm and the protection provider, the protected portion of the Exposure receives the treatment applicable to eligible guarantees or Credit Derivatives with the remainder treated as unprotected.
- (4) Where an Authorised Firm transfers a part of the risk of an Exposure in one or more tranches, the Rules in section 4.14 apply.

A4.4 Qualifying securities financing transactions (SFTs)

A4.4.1 A qualifying SFT must comply with the following requirements:

- (a) both the Exposure and the Collateral are cash, or a Security issued by a central government, Central Bank or PSE qualifying for a 0% risk weight under the Rules in section 4.12;
- (b) both the Exposure and the Collateral are denominated in the same currency;
- (c) either the transaction is overnight or both the Exposure and the Collateral are marked-to-market daily and are subject to daily remargining;
- (d) following a Counterparty's failure to remargin, the time that is required between the last mark-to-market before the failure to remargin and the liquidation of the Collateral is considered to be no more than four business days;
- (e) the transaction is settled across a recognised settlement system for that type of transaction;
- (f) the documentation covering the agreement is standard market documentation for repos, reverse repos, Securities, lending transactions or Securities borrowing transactions in the Securities concerned;

- (g) the transaction is governed by documentation specifying that if the Counterparty fails to satisfy an obligation to deliver cash or Securities or to deliver margin, or otherwise defaults, then the transaction may be terminated immediately; and
- (h) upon any event of default, regardless of whether the Counterparty is insolvent or bankrupt, the Authorised Firm has the unfettered, legally enforceable right to immediately seize and liquidate the Collateral for the benefit of the Authorised Firm.

A4.5 [Deleted]

A4.6 Credit RWA - Unsettled Transactions

A4.6.1 This section applies to all securities, foreign exchange instruments and commodities transactions that give rise to a risk of delayed settlement or delivery including:

- (a) any transaction through a recognised clearing house that is subject to daily mark-to-market and payment of daily variation margins and that involves a mismatched trade;

but excluding;

- (b) any transaction that is subject to the treatment in section 4.14A; and
- (c) any SFT.

Guidance

1. An Authorised Firm is exposed to the risk associated with unsettled securities, commodities and foreign exchange transactions from trade date. Irrespective of the booking or the accounting of the transaction, unsettled transactions are taken into account for regulatory capital requirements purposes.
2. Transactions settled through a delivery-versus-payment system (DvP), providing simultaneous exchanges of securities for cash, expose an Authorised Firm to a risk of loss on the difference between the transaction valued at the agreed settlement price and the transaction valued at current market price (ie positive current exposure). PIB Rule A4.6.2 requires an Authorised Firm to calculate RWAs for such Exposures if the payments have not yet taken place five business days after the settlement date.
3. Transactions where cash is paid without receipt of the corresponding receivable (securities, foreign currencies, gold, or commodities) or, conversely, deliverables were delivered without receipt of the corresponding cash payment (non-DvP, or free deliveries) expose firms to a risk of loss on the full amount of cash paid or deliverables delivered. PIB Rule A4.6.3 requires an Authorised Firm that has made the first contractual payment/delivery leg to calculate RWAs for the Exposure if the second leg has not been received by the end of the business day. The RWAs increase if the second leg has not been received within five business days.
4. This section applies to transactions whether on balance sheet or off balance sheet. Where a transaction does not appear on balance sheet (i.e. settlement date accounting), the Authorised Firm

should apply a 100% conversion factor to determine the credit equivalent amount of the unsettled exposure amount.

5. An Authorised Firm should develop, implement and improve systems for tracking and monitoring the credit risk exposure arising from unsettled transactions and failed trades as appropriate so that they can produce management information that facilitates timely action. An Authorised Firm should closely monitor securities, commodities, and foreign exchange transactions that have failed, starting the first day they fail.

Unsettled Transactions

A4.6.2 An Authorised Firm must calculate the RWA for a transaction that remains unsettled after its due delivery date, using the following formula:

$$CRW = PCEA \cdot RM \cdot 12.5$$

where:

PCEA = the positive current exposure amount of the transaction representing the difference between the agreed settlement price for the instrument in question and its current market value, where the difference could involve a loss for the Authorised Firm

RM = the risk multiplier, determined in accordance with the table below

Number of business days after due settlement date	Risk Multiplier
0-4	0%
5-15	8%
16-30	50%
31-45	75%
46 or more	100%

Free deliveries

A4.6.3 For a transaction for which an Authorised Firm has:

- (a) delivered securities, foreign exchange instruments or commodities before receiving payment; or
- (b) paid for securities, foreign exchange instruments or commodities before receiving the item purchased; or
- (c) in the case of cross-border transactions, one day or more has elapsed since it made that payment or delivery,

the Authorised Firm must treat its Exposure to the Counterparty of the transaction in accordance with the table below.

Up to first contractual payment or delivery leg	From first contractual payment or delivery leg up to four business days after second contractual payment or delivery leg	From five business days post second contractual payment or delivery leg until extinction of the transaction
No capital charge	Treat as an Exposure risk weighted in accordance with PIB Rule 4.8.2	Treat as an Exposure risk weighted at 1250%

A4.6.4 If the amount of an Exposure resulting from a transaction in Rule A4.6.3 is not material, an Authorised Firm may apply a risk weight of 100% to that Exposure, except where a risk weight of 1250% in accordance with the last column of the table above is required.

A4.7 [Deleted]

A4.8 Standardised approach to Counterparty Credit Risk (SA-CCR)

A4.8.1 In this section:

- (a) “Basis Transaction” has the meaning given in Rule A4.8.5(2)(a);
- (b) “End Date” means the period of time between the calculation date (today) and the end date of the time period referenced by transaction ‘i’, measured in years; if a transaction references the value of another interest rate or credit transaction, such as a swaption or a bond option, the period must be determined on the basis of the underlying transaction;
- (c) “Margin Agreement” means a contractual agreement or provision to an agreement under which one Counterparty must supply Variation Margin to a second Counterparty when an Exposure of that second Counterparty to the first Counterparty exceeds a specified level;
- (d) “Margin Period of Risk” or “MPoR” means the time period from the last exchange of Collateral covering a Netting Set of transactions with a defaulting Counterparty until that Counterparty is closed out and the resulting market risk is re-hedged;
- (e) “Margined Netting Set” means a Netting Set covered by a Margin Agreement where a transaction is subject to exchange of Variation Margin;
- (f) “Net Independent Collateral Amount” or “NICA” means the amount of Collateral that an Authorised Firm may use to offset its Exposure on the default of the Counterparty, but excluding any Collateral an Authorised Firm has posted to a

segregated, bankruptcy-remote account that would be returned to the Authorised Firm upon the bankruptcy of the counterparty;

- (g) “Netting Set” means a group of transactions with a single Counterparty that are subject to a Qualifying Bilateral Netting Agreement; each transaction with a single Counterparty that is not subject to a Qualifying Bilateral Netting Agreement comprises its own Netting Set;
- (h) “Start Date” means the period of time between the calculation date (today) and the start date of the time period referenced by transaction ‘i’, measured in years; if a transaction references the value of another interest rate or credit transaction, such as a swaption or a bond option, the period must be determined on the basis of the underlying transaction; if the start date has already occurred, then the time period must be set to zero;
- (i) “Unmargined Netting Set” means a Netting Set that is not a Margined Netting Set, including Netting Sets covered by a one-way Margin Agreement, where the Authorised Firm posts, but does not collect Variation Margin;
- (j) “Volatility Transaction” has the meaning given in Rule A4.8.5(2)(b); and
- (k) “Variation Margin” means Collateral that is collected or posted to reflect the current Exposure resulting from changes in the mark-to-market value of a transaction.

Exposure value

- A4.8.2** (1) An Authorised Firm must calculate the CCR Exposure value of a Netting Set using the following formula:

$$\text{Exposure value} = \alpha \cdot (\text{RC} + \text{PFE})$$

where:

$$\alpha = 1.4$$

RC = the replacement cost calculated in accordance with Rule A4.8.3

PFE = the amount for potential future exposure calculated in accordance with Rule A4.8.6

- (2) The maximum Exposure value for a Margined Netting Set is the Exposure value of the same Netting Set as if it were an Unmargined Netting Set.
- (3) If multiple Margin Agreements apply to a single Netting Set, an Authorised Firm must divide each Margin Agreement into sub-Netting Sets, each aligning with its respective Margin Agreement and calculate the Exposure value of the original Netting Set by taking the sum of the Exposure value of each sub-Netting set.
- (4) An Authorised Firm may set to zero the Exposure value for sold options that are:
 - (a) not included in any netting or Margin Agreement; or

- (b) the only types of Exposures in a Netting Set, provided all premia have been paid upfront.
- (5) An Authorised Firm may cap the Exposure value of a credit derivative at the amount of unpaid premia if all the following conditions are met:
 - (a) the Authorised Firm is the protection seller;
 - (b) the credit derivative is not included in any netting or Margin Agreement.
- (6) Where eligible Collateral is taken outside a Netting Set but is available to an Authorised Firm to offset default losses from transactions subject to this section, an Authorised Firm must calculate the Exposure value as follows:
 - (a) if eligible Collateral is available to offset losses on one Netting Set only, an Authorised Firm must treat the eligible Collateral as an independent Collateral amount associated with that Netting Set; and
 - (b) if eligible Collateral is available to offset losses on more than one Netting Set, the eligible Collateral must be treated as Collateral taken under a single Margin Agreement applying to multiple Netting Sets.
- (7) If eligible Collateral is available to offset losses on transactions subject to this section and on other transactions, an Authorised Firm must include only the portion of eligible Collateral assigned to the transactions subject to this section.

Replacement Cost (RC)

- A4.8.3** (1) An Authorised Firm must calculate the replacement cost, RC, for an Unmargined Netting Set using the following formula:

$$RC = \max \{V - C_H(\text{one year}); 0\}$$

where:

V = the current market value of all the transactions in the Netting Set

$C_H(\text{one year})$ = the haircut value of net Collateral held, calculated in accordance with (3), with a holding period of one year

- (2) An Authorised Firm must calculate the replacement cost, RC, for a Margined Netting Set using the following formula:

$$RC = \max \{V - C_H(\text{MPoR}); TH + MTA - NICA; 0\}$$

where:

V = the total current market value of all the transactions in the Netting Set

$C_H(\text{MPoR})$ = the haircut value of net Collateral held, calculated in accordance with (3), with a holding period of MPoR

TH = the positive threshold before the Counterparty must send the Authorised Firm collateral

MTA = the minimum transfer amount applicable to the Counterparty

NICA = the net independent Collateral amount

- (3) For the purpose of (1) and (2), an Authorised Firm must calculate the haircut value of net Collateral, C_H , using the following formula:

$$C_H(t) = \begin{cases} C \cdot (1 - H(t)), & \text{if } C \geq 0 \\ C \cdot (1 + H(t)), & \text{if } C < 0 \end{cases}$$

where:

C = the current mark-to-market value of the net Collateral held, excluding Collateral posted by the Authorised Firm in a bankruptcy remote manner

t = the holding period applicable to the Collateral

H(t) = the haircut appropriate to the Collateral using a holding period of 't', determined in accordance with Rule A4.3.5

- (4) If a single Margin Agreement applies to multiple Netting Sets, an Authorised Firm must calculate RC using the following formula:

$$RC = \max \left\{ \sum_{NS \in MA} \max\{V_{NS}; 0\} - \max\{C_{MA}; 0\}; 0 \right\} + \max \left\{ \sum_{NS \in MA} \min\{V_{NS}; 0\} - \min\{C_{MA}; 0\}; 0 \right\}$$

where:

NS = the index that denotes the Netting Sets that are subject to the single Margin Agreement

V_{NS} = the current net mark-to-market value of all the transactions within the Netting Set

C_{MA} = the net haircut value of all currently available Collateral (including both NICA and VM) under the Margin Agreement

Guidance

1. There are two formulations of RC depending on whether the transactions with a Counterparty are unmargined or margined.
2. For unmargined transactions the RC intends to capture the loss that would occur if a Counterparty were to default and were closed out of its transactions immediately. For margined transactions the RC intends to capture the loss that would occur if a counterparty were to default at the present or at a future time, assuming that the close-out and replacement of transactions occur instantaneously.

3. In both cases, the haircut applicable to non-cash Collateral in the RC formulation represents the potential change in value of the Collateral during the appropriate time period (one year for unmargined transactions and the MPoR for margined transactions).
4. To calculate the net Collateral amount under Rule A4.8.3(3), an Authorised Firm should account Collateral received with a positive sign and the amount posted with a negative sign.
5. In the formula in Rule A4.8.3(2), $TH + MTA - NICA$ represents the largest exposure that would not trigger a Variation Margin call and it contains levels of Collateral that need always to be maintained. For example, without initial margin, the greatest exposure that would not trigger a Variation Margin call is the threshold plus any minimum transfer amount. In the adapted formulation, NICA is subtracted from $TH + MTA$. This makes the calculation more accurate by fully reflecting both the actual level of exposure that would not trigger a margin call and the effect of Collateral held and /or posted by a bank. The calculation is floored at zero, recognising that the Authorised Firm may hold NICA in excess of $TH + MTA$, which could otherwise result in a negative replacement cost.

Allocation of transactions to asset classes

- A4.8.4** (1) An Authorised Firm must allocate each transaction of a Netting Set to one of the following asset classes to determine the potential future exposure of the Netting Set under Rule A4.8.6:
- (a) interest rate asset class;
 - (b) foreign exchange asset class;
 - (c) credit asset class;
 - (d) equity asset class; or
 - (e) commodity asset class.
- (2) An Authorised Firm must conduct the allocation in (1) on the basis of the primary risk factor of the underlying instrument.
- (3) If a transaction is exposed to multiple risk factor referencing multiple asset classes, an Authorised Firm must:
- (a) take sensitivities and volatility of the underlying instrument into account for determining the primary risk factor; and
 - (b) assign the transaction to a single asset class except where the DFSA requires the Authorised Firm to assign the transaction to more than one asset class.

Guidance

1. Most transactions have one primary risk factor (for example, an interest curve for an interest rate swap, a reference entity for a credit default swap, a foreign exchange rate for a foreign exchange call option, etc.). When this primary risk factor is clearly identifiable, the transaction will fall into one the asset classes.
2. Under Rule A4.8.4(3)(b) the DFSA may require more complex trades (i.e. trades that may have more than one risk driver such as multi-asset or hybrid derivatives) to be allocated to more than one

asset class, resulting in the same position being included in multiple classes. In this case, for each asset class to which the position is allocated, the Authorised Firm should determine appropriately the sign and delta adjustment of the relevant risk driver.

Hedging sets

- A4.8.5** (1) An Authorised Firm must form hedging sets and allocate each transaction of a Netting Set to those hedging sets, as follows:
- (a) for transactions allocated to the interest rate asset class a separate hedging set must be formed for all transactions that reference interest rates of the same currency;
 - (b) for transactions allocated to the foreign exchange asset class a separate hedging set must be formed for all transactions that are based on the same currency pair;
 - (c) all transactions allocated to the credit asset class must form a single hedging set;
 - (d) all transactions allocated to the equity asset class must form a single hedging set;
 - (e) for transactions allocated to the commodities asset class a separate hedging set must be formed for each of the following four broad categories:
 - (i) energy;
 - (ii) metals;
 - (iii) agricultural; and
 - (iv) other commodities,

unless the DFSA requires the Authorised Firm to use more refined definitions of commodities when it is significantly exposed to the basis risk of different products within those commodity types.
- (2) Within each asset class an Authorised Firm must form separate hedging sets for transactions that reference:
- (a) the basis between two risk factors from the same asset class and that are denominated in a single currency ('Basis Transactions'); or
 - (b) the volatility of a risk factor ('Volatility Transactions').
- (3) For the purpose of (2)(a), an Authorised Firm must form a separate hedging set for each specific pair of risk factors.

Guidance

1. Examples of specific bases of Basis Transactions include three-month SOFR versus six-month SOFR, three-month SOFR versus three-month T-Bill, one-month SOFR versus overnight indexed swap rate, Brent Crude oil versus Henry Hub gas.
2. Transactions with two floating legs that are denominated in different currencies (such as cross-currency swaps) are not subject to the treatment in Rule A4.8.5(2)(a) and should be treated as non-basis foreign exchange transactions.
3. For hedging sets consisting of Basis Transactions, long and short positions should be determined with respect to the basis.
4. Examples of Volatility Transactions include variance and volatility swaps and options on realised or implied volatility. Volatility hedging sets follow the hedging set construction under Rule A4.8.5(1). For example, all equity volatility transactions form a single hedging set.

Potential future exposure

- A4.8.6** (1) An Authorised Firm must calculate the potential future exposure, PFE, of a Netting Set using the following formula:

$$\text{PFE} = \text{multiplier} \cdot \text{AddOn}^{\text{agg}}$$

where:

multiplier = the multiplier factor that allows for the recognition of over-collateralisation or negative mark-to-market value of the transactions, calculated using the formula in (3)

AddOn^{agg} = the aggregate add-ons for each asset class, calculated using the formula in (4)

- (2) An Authorised Firm must calculate the PFE of multiple Netting Sets that are subject to a single Margin Agreement, as referred to in Rule A4.8.3(4), as the sum of the PFE of all the individual Netting Sets as if they were not subject to any form of Margin Agreement.
- (3) For the purpose of (1) an Authorised Firm must calculate the multiplier as follows:

$$\text{multiplier} = \min \left\{ 1; \text{Floor} + (1 - \text{Floor}) \cdot \exp \left(\frac{V - C_H}{2 \cdot (1 - \text{Floor}) \cdot \text{AddOn}^{\text{agg}}} \right) \right\}$$

where:

Floor = 5%

V = the total current market value of all the transactions in the Netting Set

C_H = C_H(one year) for Unmargined Netting Sets and C_H(MPoR) for Margined Netting Sets (refer to Rule A4.8.3(1) and (2), respectively)

$AddOn^{agg}$ = the aggregated add-on factors for each asset class, calculated in accordance with (4)

- (4) For the purpose of (1) an Authorised Firm must calculate the $AddOn^{agg}$, as follows:

$$AddOn^{agg} = \sum_a AddOn^a$$

where:

a = the index that denotes the asset class (interest rate, foreign exchange, credit, equity and commodity) included in the calculation of the potential future exposure of the Netting Set

$AddOn^a$ = the add-on for asset class 'a'

- (5) For each asset class 'a', an Authorised Firm must calculate the add-on factor, $AddOn^a$, as the sum of all hedging set level add-ons within asset class 'a' as follows:

$$AddOn^a = \sum_j AddOn_j^a$$

where:

j = the index that denotes all hedging sets of an asset class 'a', formed in accordance with Rule A4.8.5

$AddOn_j^a$ = add-on for hedging set 'j' within asset class 'a'

Guidance

Diversification benefits across asset classes are not recognised. Instead, the respective add-ons for each asset class are aggregated using the formula in A4.8.6(5) (where the sum is across the asset classes).

Add-on for transactions in the interest rate asset class

- A4.8.7** (1) An Authorised Firm must map transactions within each hedging set of the interest rate asset class to the appropriate maturity bucket in the table below on the basis of the End Date of each transaction.

Maturity bucket (k)	End Date (in years)
1	> 0 and ≤ 1
2	> 1 and ≤ 5
3	> 5

- (2) An Authorised Firm must calculate the add-on for each hedging set 'j' of the interest rate asset class, $AddOn_j^{IR}$, using the following formula:

$$\text{AddOn}_j^{\text{IR}} = \text{SF}^{\text{IR}} \cdot \text{EffNot}_j^{\text{IR}}$$

where:

$\text{SF}^{\text{IR}} =$ the supervisory factor for the interest rate asset class, determined in accordance with Rule A4.8.18

$\text{EffNot}_j^{\text{IR}} =$ the effective notional amount of hedging set 'j', calculated in accordance with (3)

$j =$ the index that denotes all hedging sets of the interest rate asset class, formed in accordance with Rule A4.8.5

- (3) For the purpose of (2), an Authorised Firm must calculate the effective notional amount for hedging set 'j', $\text{EffNot}_j^{\text{IR}}$, by aggregating the effective notional amounts of each maturity bucket using the following formula:

$$\text{EffNot}_j^{\text{IR}} = \sqrt{(D_{j,1}^{\text{IR}})^2 + (D_{j,2}^{\text{IR}})^2 + (D_{j,3}^{\text{IR}})^2 + 1.4 \cdot D_{j,1}^{\text{IR}} \cdot D_{j,2}^{\text{IR}} + 1.4 \cdot D_{j,2}^{\text{IR}} \cdot D_{j,3}^{\text{IR}} + 0.6 \cdot D_{j,1}^{\text{IR}} \cdot D_{j,3}^{\text{IR}}}$$

where:

$D_{j,k}^{\text{IR}} =$ the effective notional amount for maturity bucket 'k' of hedging set 'j', calculated in accordance with (5)

- (4) By way of derogation from (3), an Authorised Firm may calculate the effective notional amount for hedging set 'j', $\text{EffNot}_j^{\text{IR}}$, by aggregating the effective notional amounts of each maturity bucket using the following formula:

$$\text{EffNot}_j^{\text{IR}} = |D_{j,1}^{\text{IR}}| + |D_{j,2}^{\text{IR}}| + |D_{j,3}^{\text{IR}}|$$

where:

$D_{j,k}^{\text{IR}} =$ the effective notional amount for maturity bucket 'k' of hedging set 'j', calculated in accordance with (5)

- (5) For the purpose of (3) and (4), an Authorised Firm must calculate the effective notional amount of bucket 'k' of hedging set 'j', $D_{j,k}^{\text{IR}}$, using the following formula:

$$D_{j,k}^{\text{IR}} = \sum_{i \in I(j,k)} \delta_i \cdot \text{AdjNot}_i^{\text{IR}} \cdot \text{MF}_i$$

where:

$I(j,k) =$ the set of all interest rate transactions belonging to maturity bucket 'k' and hedging set 'j'

$j =$ the index that denotes all the hedging sets of the interest rate asset class, formed in accordance with Rule A4.8.5

δ_i = the supervisory delta for the transaction 'i', calculated in accordance with Rule A4.8.12

MF_i = the maturity factor of the transaction 'i', calculated in accordance with Rule A4.8.15

$AdjNot_i^{IR}$ = the adjusted notional amount for transaction 'i', calculated in accordance with Rule A4.8.13(a)

Guidance

1. The calculation of the add-on for the interest rate asset class captures the risk of interest rate transactions of different maturities being imperfectly correlated. It does this by allocating trades to maturity buckets, in which full offsetting of long and short positions is permitted, and by using an aggregation formula that only permits limited offsetting between maturity buckets. This allocation of transactions to maturity buckets and the process of aggregation are only used in the interest rate asset class.
2. The formula for the aggregation of the effective notional amounts across the three maturity buckets in Rule A4.8.7(3) allows for the partial offsetting. An Authorised Firm may elect not to recognise any offsetting across maturity buckets. In this case it must aggregate the effective notional amounts using the formula in Rule A4.8.7(3).
3. Under this Rule, an Authorised Firm may treat inflation derivatives (which SA-CCR does not specifically assign to a particular asset class) in the same manner as they treat interest rate derivatives and subject them to the same 0.5% supervisory factor. Derivatives referencing inflation rates for the same currency should form a separate hedging set.

Add-on for transactions in the foreign exchange asset class

- A4.8.8** (1) An Authorised Firm must calculate the add-on for each hedging set of the foreign exchange asset class, $AddOn_j^{FX}$, using the following formula:

$$AddOn_j^{FX} = SF^{FX} \cdot EffNot_j^{FX}$$

where:

SF^{IR} = the supervisory factor for the interest rate asset class, determined in accordance with Rule A4.8.18

$EffNot_j^{IR}$ = the effective notional amount of hedging set 'j', calculated in accordance with (2)

j = the index that denotes all hedging sets of the foreign exchange asset class, formed in accordance with Rule A4.8.5

- (2) For the purpose of (1), an Authorised Firm must calculate the effective notional amount of hedging set 'j', $EffNot_j^{FX}$, using the following formula:

$$\text{EffNot}_j^{\text{FX}} = \sum_{i \in I(j)} \delta_i \cdot \text{AdjNot}_i^{\text{FX}} \cdot \text{MF}_i$$

where:

$I(j)$ = the set of all interest rate transactions belonging to hedging set 'j'

δ_i = the supervisory delta for the transaction 'i', calculated in accordance with Rule A4.8.12

MF_i = the maturity factor of the transaction 'i', calculated in accordance with Rule A4.8.15

$\text{AdjNot}_i^{\text{FX}}$ = the adjusted notional amount for transaction 'i', calculated in accordance with Rule A4.8.13(b)

j = the index that denotes all the hedging sets of the foreign exchange asset class, formed in accordance with Rule A4.8.5

Guidance

The steps to calculate the add-on for the foreign exchange asset class are similar to the steps for the interest rate asset class, except that there is no allocation of trades to maturity buckets (which means that there is full offsetting of long and short positions within the hedging sets of the foreign exchange derivative asset class).

Add-on for transactions in the credit asset class

- A4.8.9** (1) An Authorised Firm must divide the transactions of each hedging set of the credit asset class into different categories with each category containing all the transactions referencing the same entity or index.
- (2) An Authorised Firm must calculate the add-on for hedging set 'j' of the credit asset class, $\text{AddOn}_j^{\text{CR}}$, using the following formula:

$$\text{AddOn}_j^{\text{CR}} = \sqrt{\left(\sum_k \rho_k^{\text{CR}} \cdot \text{AddOn}_k^{\text{CR}} \right)^2 + \sum_k \left(1 - (\rho_k^{\text{CR}})^2 \right) \cdot (\text{AddOn}_k^{\text{CR}})^2}$$

where:

k = the index that denotes the categories of the hedging set formed under (1)

ρ_k^{CR} = the supervisory correlation parameter for category 'k', determined in accordance with Rule A4.8.18

$\text{AddOn}_k^{\text{CR}}$ = the add-on factor for category 'k', calculated in accordance with (3)

$j =$ the index that denotes all the hedging sets of the credit asset class, formed in accordance with Rule A4.8.5

- (3) For the purpose of (2), an Authorised Firm must calculate the add-on factor for category 'k', $AddOn_k^{CR}$, using the following formula:

$$AddOn_k^{CR} = SF_k^{CR} \cdot EffNot_k^{CR}$$

where:

$SF_k^{CR} =$ the supervisory factor for the rating class of category 'k', determined in accordance with Rule A4.8.18

$EffNot_k^{CR} =$ the supervisory factor for the rating class of category 'k', calculated in accordance with (4)

- (4) For the purpose of (3), an Authorised Firm must calculate the effective notional amount for category 'k', $EffNot_k^{CR}$, using the following formula:

$$EffNot_k^{CR} = \sum_{i \in I(j)} \delta_i \cdot AdjNot_i^{CR} \cdot MF_i$$

where:

$I(k) =$ the set of all transactions belonging to category 'k' within hedging set 'j'

$\delta_i =$ the supervisory delta for the transaction 'i', calculated in accordance with Rule A4.8.12

$MF_i =$ the maturity factor of the transaction 'i', calculated in accordance with Rule A4.8.15

$AdjNot_i^{CR} =$ the adjusted notional amount for transaction 'i', calculated in accordance with Rule A4.8.13

Guidance

The calculation of the add-on for the credit asset class only gives full recognition of the offsetting of long and short positions for derivatives that reference the same entity (e.g. the same corporate issuer of bonds). Partial offsetting is recognised between derivatives that reference different entities.

Add-on for transactions in the equity asset class

- A4.8.10** (1) An Authorised Firm must divide the transactions of each hedging set of the equity asset class into different categories with each category containing all the transactions referencing the same entity or index.
- (2) An Authorised Firm must calculate the add-on for hedging set 'j' of the equity asset class, $AddOn_j^{EQ}$, using the following formula:

$$\text{AddOn}_j^{\text{EQ}} = \sqrt{\left(\sum_k \rho_k^{\text{EQ}} \cdot \text{AddOn}_k^{\text{EQ}}\right)^2 + \sum_k \left(1 - (\rho_k^{\text{EQ}})^2\right) \cdot (\text{AddOn}_k^{\text{EQ}})^2}$$

where:

$k =$ the index that denotes the categories of the hedging set established under Rule (1)

$\rho_k^{\text{EQ}} =$ the supervisory correlation parameter for category 'k', determined in accordance with Rule A4.8.18

$\text{AddOn}_k^{\text{EQ}} =$ the add-on factor for category 'k', calculated in accordance with (3)

$j =$ the index that denotes all the hedging sets of the equity asset class, formed in accordance with Rule A4.8.5

- (3) For the purpose of (2), an Authorised Firm must calculate the add-on factor for category 'k', $\text{AddOn}_k^{\text{EQ}}$, using the following formula:

$$\text{AddOn}_k^{\text{EQ}} = \text{SF}_k^{\text{EQ}} \cdot \text{EffNot}_k^{\text{EQ}}$$

where:

$\text{SF}_k^{\text{EQ}} =$ the supervisory factor for the rating class of category 'k', determined in accordance with the table in Rule A4.8.18

$\text{EffNot}_k^{\text{EQ}} =$ the effective notional amount for the rating class of category 'k', calculated in accordance with (4)

- (4) For the purpose of (3), an Authorised Firm must calculate the effective notional amount for category 'k', $\text{EffNot}_k^{\text{EQ}}$, using the following formula:

$$\text{EffNot}_k^{\text{EQ}} = \sum_{i \in I(j)} \delta_i \cdot \text{AdjNot}_i^{\text{EQ}} \cdot \text{MF}_i$$

where:

$I(k) =$ the set of all transactions belonging to category 'k' within hedging set 'j'

$\delta_i =$ the supervisory delta for the transaction 'i', calculated in accordance with Rule A4.8.12

$\text{MF}_i =$ the maturity factor of the transaction 'i', calculated in accordance with Rule A4.8.15

$\text{AdjNot}_i^{\text{EQ}}$ = the adjusted notional amount for transaction 'i', calculated in accordance with Rule A4.8.13(c)

j = the index that denotes all the hedging sets of the equity asset class, formed in accordance with Rule A4.8.5

Guidance

The calculation of the add-on for the equity asset class is very similar to the calculation of the add-on for the credit asset class. It only gives full recognition of the offsetting of long and short positions for derivatives that reference the same entity (e.g. the same corporate issuer of shares). Partial offsetting is recognised between derivatives that reference different entities.

Add-on for transactions in the commodity asset class

- A4.8.11** (1) An Authorised Firm must calculate the add-on for each hedging set 'j' of the commodity asset class, $\text{AddOn}_j^{\text{COM}}$, using the following formula:

$$\text{AddOn}_j^{\text{COM}} = \sqrt{\left(\sum_k \rho^{\text{COM}} \cdot \text{AddOn}_j^{\text{COM}}\right)^2 + \sum_k (1 - (\rho^{\text{COM}})^2) \cdot (\text{AddOn}_j^{\text{COM}})^2}$$

where:

ρ^{COM} = the supervisory correlation parameter, determined in accordance with the table in Rule A4.8.18

$\text{AddOn}_j^{\text{COM}}$ = the add-on factor for hedging set 'j', calculated in accordance with (2)

j = the index that denotes all the hedging sets of the commodity asset class, formed in accordance with Rule A4.8.5

- (2) For the purpose of (1), an Authorised Firm must calculate the add-on for each hedging set 'j' of the commodities asset class, $\text{AddOn}_j^{\text{COM}}$, using the following formula:

$$\text{AddOn}_j^{\text{COM}} = \text{SF}^{\text{COM}} \cdot \text{EffNot}_j^{\text{COM}}$$

where:

SF^{COM} = the supervisory factor for the commodities asset class, determined in accordance with the table in Rule A4.8.18

$\text{EffNot}_j^{\text{COM}}$ = the effective notional amount of hedging set 'j', calculated in accordance with Rule (3)

j = the index that denotes all hedging sets of the commodities asset class, formed in accordance with Rule A4.8.5

- (3) For the purpose of (2), an Authorised Firm must calculate the effective notional amount' of hedging set 'j', denoted by $\text{EffNot}_j^{\text{COM}}$, using the following formula:

$$\text{EffNot}_j^{\text{COM}} = \sum_{i \in I(j)} \delta_i \cdot \text{AdjNot}_i^{\text{COM}} \cdot \text{MF}_i$$

where:

- $I(j)$ = the set of all transactions belonging to hedging set 'j'
- δ_i = the supervisory delta for the transaction 'i', determined in accordance with the table in Rule A4.8.18
- MF_i = the maturity factor of the transaction 'i', calculated in accordance with Rule A4.8.15
- $\text{AdjNot}_i^{\text{COM}}$ = the adjusted notional amount for transaction 'i', calculated in accordance with Rule A4.8.13

Guidance

The calculation of the add-on for the commodity asset class is similar to the calculation of the add-on for the credit and equity asset classes. It recognises the full offsetting of long and short positions for derivatives that reference the same type of underlying commodity. It also allows partial offsetting between derivatives that reference different types of commodities, however, this partial offsetting is only permitted within each of the four hedging sets of the commodity asset class, where the different commodity types are more likely to demonstrate some stable, meaningful joint dynamics. Offsetting between hedging sets is not recognised (eg a forward contract on crude oil cannot hedge a forward contract on corn).

Supervisory delta

A4.8.12 For the purpose of Rules A4.8.7 to A4.8.11, an Authorised Firm must determine the supervisory delta, δ , as follows:

- (a) for transactions that are not options or collateralised debt obligations the supervisory delta must be determined in accordance with the table below:

	Transaction is a long position in the primary risk factor	Transaction is a short position in the primary risk factor
δ_i	+1	-1

- (b) for the purpose of (a), a transaction is:
- (i) a long position in the primary risk factor if the market value of the transaction increases when the risk factor increases; and
 - (ii) a short position in the primary risk factor if the market value of the transaction decreases when the risk factor increases;

- (c) for options the supervisory delta must be calculated using the applicable formula in the table below:

	Bought	Sold
Call options	$\delta_i = \Phi \left(\frac{\ln \left(\frac{P_i}{K_i} \right) + 0.5 \cdot \sigma_i^2 \cdot T_i}{\sigma_i \cdot \sqrt{T_i}} \right)$	$\delta_i = -\Phi \left(\frac{\ln \left(\frac{P_i}{K_i} \right) + 0.5 \cdot \sigma_i^2 \cdot T_i}{\sigma_i \cdot \sqrt{T_i}} \right)$
Put options	$\delta_i = -\Phi \left(\frac{\ln \left(\frac{P_i}{K_i} \right) + 0.5 \cdot \sigma_i^2 \cdot T_i}{\sigma_i \cdot \sqrt{T_i}} \right)$	$\delta_i = \Phi \left(\frac{\ln \left(\frac{P_i}{K_i} \right) + 0.5 \cdot \sigma_i^2 \cdot T_i}{\sigma_i \cdot \sqrt{T_i}} \right)$

where:

Φ = the standard normal cumulative distribution function

P_i = the underlying price of option 'i'

K_i = the strike price of option 'i'

σ_i = the supervisory volatility of option 'i', determined in accordance with the table in Rule A4.8.18

T_i = the time interval between the calculation date and the latest contractual exercise date as referenced by transaction 'i', measured in years

- (d) for options in the interest rate asset class for which the supervisory delta cannot be calculated in accordance with (c) because the underlying price of the option or the strike price of the option is negative, the supervisory delta must be calculated using the applicable formula in the table below:

	Bought	Sold
Call options	$\delta_i = \Phi \left(\frac{\ln \left(\frac{P_i + \lambda_j}{K_i + \lambda_j} \right) + 0.5 \cdot \sigma_i^2 \cdot T_i}{\sigma_i \cdot \sqrt{T_i}} \right)$	$\delta_i = -\Phi \left(\frac{\ln \left(\frac{P_i + \lambda_j}{K_i + \lambda_j} \right) + 0.5 \cdot \sigma_i^2 \cdot T_i}{\sigma_i \cdot \sqrt{T_i}} \right)$

Put options	$\delta_i = -\Phi \left(\frac{\ln \left(\frac{P_i + \lambda_j}{K_i + \lambda_j} \right) + 0.5 \cdot \sigma_i^2 \cdot T_i}{\sigma_i \cdot \sqrt{T_i}} \right)$	$\delta_i = \Phi \left(\frac{\ln \left(\frac{P_i + \lambda_j}{K_i + \lambda_j} \right) + 0.5 \cdot \sigma_i^2 \cdot T_i}{\sigma_i \cdot \sqrt{T_i}} \right)$
-------------	---	--

where:

λ_j = the shift parameter that represents the lowest possible extent to which interest rates in the respective currency can become negative

all other terms are the ones set out in (c)

- (e) for tranches of collateralised debt obligations and a nth-to-default transaction in the credit asset class, the supervisory delta must be set according to the table below:

	Purchased (long protection)	Sold (short protection)
CDO tranches	$\delta_i = \frac{15}{(1 + 14 \cdot A_i) \cdot (1 + 14 \cdot D_i)}$	$\delta_i = -\frac{15}{(1 + 14 \cdot A_i) \cdot (1 + 14 \cdot D_i)}$

where:

A_j = the attachment point of the tranche or $(n - l)/m$ for a nth-to-default transaction on a pool of 'm' reference names for transaction 'i'

D_j = the detachment point of the tranche or n/m for a nth-to-default transaction on a pool of 'm' reference names for transaction 'i'

Guidance

1. The supervisory delta is used to ensure that the effective notional amounts take into account the direction of the trade, i.e. whether the trade is long or short, by having a positive or negative sign. It also takes into account whether the trade has a non-linear relationship with the underlying risk factor (which is the case for options and collateralised debt obligation tranches).
2. To calculate the supervisory delta for options an Authorised Firm should use, whenever appropriate, the forward (rather than the spot) value of the underlying in order to account for the risk-free rate as well as for possible cash flows prior to the option expiry (such as dividends).

Adjusted notional amount

A4.8.13 For the purpose of Rules A4.8.7 to A4.8.11, an Authorised Firm must calculate the adjusted notional amount, AdjNot_i^a , as follows:

- (a) for transactions allocated to the interest rate or credit asset class, the adjusted notional amount for transaction 'i' must be calculated using the following formula:

$$\text{AdjNot}_i^{\text{IR,CR}} = N_i \cdot \max \left\{ \left(\frac{e^{-0.05 \cdot S_i} - e^{-0.05 \cdot E_i}}{0.05} \right); \frac{10}{\text{OneBusinessYear}} \right\}$$

where:

N_i = the notional amount of transaction 'i', converted to USD, using the exchange rate on the calculation date

S_i = the Start Date of transaction 'i'

E_i = the End Date of transaction 'i'

- (b) for transactions allocated to the foreign exchange asset class, the adjusted notional amount for transaction 'i' must be set as the notional amount of the foreign currency leg of the transaction, converted to dollars; if both legs of the transaction are denominated in a foreign currency, the notional amount of each leg is converted to dollars and the adjusted notional amount must be set as the leg with the larger dollar value; and
- (c) for transactions allocated to the equity or commodity asset class, the adjusted notional amount for transaction 'i' must be calculated as the product of the current price of one unit of the stock or commodity and the number of units referenced by transaction 'i', except where transaction 'i' is an equity volatility transaction, in which case the adjusted notional amount must be calculated as the product of the underlying volatility or variance referenced by transaction 'i' and the contractual notional amount of transaction 'i'.

A4.8.14 For the purpose of calculating the adjusted notional amount of transaction 'i' referred to in Rule A4.8.13, an Authorised Firm must determine the notional amount of transaction 'i', as follows:

- (a) if the notional amount of a transaction is specified by the contract and fixed until maturity, the notional amount is the one specified;
- (b) if the notional amount is expressed as a formula of market values, the current market values must be used to determine the notional amount;
- (c) for a transaction allocated to the interest rate or credit asset class that has variable notional amounts specified by the contract, such as an amortising and accreting swap, the notional amount is the time-weighted average notional amount over the remaining life of the transaction;
- (d) for a leveraged swap, the transaction must be converted to the notional of the equivalent unleveraged swap; and
- (e) for a transaction that has multiple exchanges of the principal, the notional amount is multiplied by the number of exchanges of principal in the transaction.

Maturity factor

A4.8.15 For the purpose of Rules A4.8.7 to A4.8.11, an Authorised Firm must calculate the maturity factor, MF_i , as follows:

- (a) for transactions included in an Unmargined Netting Set, the maturity factor must be calculated as follows:

$$MF_i = \sqrt{\min \left(\max \left(M_i, \frac{10}{\text{OneBusinessYear}} \right); 1 \right)}$$

where:

M_i = the period between the calculation day (today) and the latest date when the transaction 'i' may still be active, measured in years; if a transaction 'i' has another derivative contract as its underlying and may be physically exercised into the underlying contract, then M_i the contract is the time period between the calculation date and the final settlement date of the underlying transaction. For a transaction that is structured such that on specified dates any outstanding exposure is settled and the terms are reset so that the fair value of the contract is zero, the remaining maturity, M_i , equals the time until the next reset date

OneBusinessYear = one year expressed in business days

- (b) for transactions included in a Margined Netting Set, the maturity factor must be calculated using the following formula:

$$MF_i = 1.5 \cdot \sqrt{\frac{MPOR_i}{\text{OneBusinessYear}}}$$

where:

$MPOR_i$ = the margin period of risk of the Netting Set, subject to the minimum holding periods specified in Rule A4.8.16

OneBusinessYear = one year expressed in business days

A4.8.16 (1) For the purpose of determining the margin period of risk of transaction 'i', $MPOR_i$ referred to in Rule A4.8.15, an Authorised Firm must apply the following floors:

- (a) ten business days for non-centrally cleared transactions subject to daily remargining; and
- (b) the sum of nine business days and the remargining period for non-centrally cleared transactions that are not subject to daily remargining.

- (2) Under the following circumstances, an Authorised Firm must apply a higher minimum holding period:
- (a) for a Netting Set consisting of more than 5,000 transactions at any point during a quarter, the minimum holding period of risk is 20 business days;
 - (b) for a Netting Set containing one or more transactions involving either illiquid Collateral or an OTC derivative that cannot be easily replaced, the minimum holding period of risk is 20 business days; and
 - (c) if an Authorised Firm has experienced more than two margin call disputes on a particular Netting Set over the previous two quarters that have lasted longer than the applicable minimum holding period (before consideration of this provision), the Authorised Firm must use a minimum holding period that is at least double the supervisory floor for that Netting Set for the subsequent two quarters.

Guidance

1. In Rule A4.8.16 (2)(b), the terms ‘illiquid Collateral’ and ‘OTC derivatives that cannot be easily replaced’ should be determined in the context of stressed market conditions and characterised by the absence of continuously active markets where a Counterparty would, with two or fewer days, obtain multiple price quotations that would not move the market or represent a price reflecting a market discount (in the case of Collateral) or premium (in the case of an OTC derivative). Examples of situations where transactions are deemed illiquid for this purpose include, transactions that are not marked daily and transactions that are subject to specific accounting treatment for valuation purposes (for example OTC derivatives referencing securities whose fair value is determined by models with inputs that are not observed in the market).
2. Under Rule A4.8.16 (2)(c), an Authorised Firm should take into account only Variation Margin call disputes.

Effective notional for options

- A4.8.17** (1) An Authorised Firm must calculate the effective notional amount under Rule A4.8.12 for single-payment options as follows:
- (a) for Asian options, the underlying price, P_i , must be set equal to the current value of the average used in the payoff;
 - (b) for American and Bermudan options, the latest allowed exercise date must be set as the exercise date, T_i ;
 - (c) for Bermudan swaptions, the Start Date must be equal to the earliest allowed exercise date, while the End Date must be equal to the End Date of the underlying swap;
 - (d) for digital options, the payoff of each digital option (bought or sold) with strike, K_i , must be approximated via a collar combination of bought and sold European options of the same type (call or put) with strikes set equal to $0.95 \cdot K_i$ and $1.05 \cdot K_i$; the size of the position in the collar components must be such that the digital payoff is reproduced exactly outside the region between

the two strikes; the effective notional amount for each component European option must be calculated separately using the formulae in Rule A4.8.12 with T_i set equal to the exercise date of the digital option and P_i set equal to the current value of the underlying for the digital option; the absolute value of the digital option effective notional amount must be capped by the ratio of the digital payoff to the relevant supervisory factor; and

- (e) if an option can be represented as a combination of European option payoffs, each European option component must be calculated separately.
- (2) For the purpose of (1) an Authorised Firm may represent multiple payment options as a combination of single payment options as follows:
 - (a) interest rate caps or floors may be represented as the portfolio of individual caplets or floorlets, where each caplet or floorlet is treated as a European option on the floating interest rate over a specific coupon period; and
 - (b) for each caplet or floorlet, the Start Date and exercise date, T_i , must be set equal to the start of the coupon period and the End Date must be set equal to the end of the coupon period.

Supervisory parameters

- A4.8.18** (1) When calculating the PFE add-on for each asset class, an Authorised Firm must, subject to (2), determine, the supervisory factor, the supervisory correlation parameter and the supervisory volatility in accordance with the table below:

Asset Class		Supervisory factor (SF) (%)	Correlation (ρ) (%)	Supervisory option volatility (σ) (%)
Interest rate		0.5	n/a	50
Foreign exchange		4.0	n/a	15
Credit single name	Credit Quality Grade 1	0.38	50	100
	Credit Quality Grade 2	0.38		
	Credit Quality Grade 3	0.42		
	Credit Quality Grade 4	0.54		
	Credit Quality Grade 5	1.01		
	Credit Quality Grade 6	6.0		

Credit index	Investment Grade	0.38	80	80
	Sub Investment Grade	1.06		
Equity single name		32	50	120
Equity index		20	80	75
Comm odity	Electricity	40	40	150
	Oil/Gas	18		70
	Metals			
	Agricultural			
	Other			

- (2) An Authorised Firm must multiply the applicable supervisory factor by:
- (i) one-half for a hedging set consisting of Basis Transactions; and
 - (ii) five for a hedging set consisting of Volatility Transactions.

A4.9 Exposures to Central Counterparties (CCPs)

- A4.9.1** An Authorised Firm may determine the Exposure value of a Credit Risk Exposure outstanding with a CCP in accordance with A4.9.2, provided that the CCP's Counterparty Credit Risk Exposure with all participants in its arrangements are fully collateralised on a daily basis and the CCP is a qualifying CCP.
- A4.9.2** An Authorised Firm may attribute an Exposure value of zero, for purposes of calculating the CRCOM to Derivative contracts and deferred settlement transactions, or to other Exposures arising in respect of those contracts or transactions (excluding an Exposure arising from Collateral held with CCPs as part of its default fund) where such Exposures are outstanding with a CCP and have not been rejected by the CCP.
- A4.9.3** An Authorised Firm which purchases Credit Derivative protection against a Non-Trading Book Exposure or against a Counterparty Credit Risk Exposure, must compute Market Risk Capital Requirements for the hedged asset in accordance with the relevant Rules in chapter 5.
- A4.9.4** An Authorised Firm must assign an Exposure value of zero for purposes of determination of CRCOM, for credit default swaps sold by it, where such credit default swaps are treated as credit protection provided by the Authorised Firm and subject to a Capital Requirement for Credit Risk for the full amount.

Credit default products

- A4.9.5** (1) The Protection Buyer relies on the Protection Seller to pay the Credit Event Payment if a Credit Event occurs, and therefore must compute Credit RWAs for the Counterparty Risk involved.
- (2) The Protection Seller is exposed to the Protection Buyer only if there are premium or interest rate-related payments outstanding.
- A4.9.6** In the case of Total Return Swaps, since each party relies on the other for payment, each party must compute Credit RWAs.
- A4.9.7** No Credit RWA applies to credit-linked notes unless a coupon or interest payment is outstanding.
- A4.9.8** (1) An Authorised Firm shall determine the Exposure value for clearing Exposures to non-qualifying CCPs as the amount in the table below:

Type of Clearing Exposure	Exposure Amount
Segregated initial margin	Nil
Non-segregated initial margin	Nominal amount of initial margin posted
Pre-funded default fund contributions	Nominal amount of the funded contribution
Unfunded default fund contributions	Nil
Equity stakes	Nominal amount

- (2) For the purposes of (1), a “non-qualifying CCP” is a CCP that does not meet the criteria set out in the definition of a Qualifying CCP in Rule 1.2.1.

A4.10 Securitisation

- A4.10.1** An Authorised Firm which is an Originator or a Sponsor of a Traditional Securitisation may exclude securitised Exposures from the calculation of Credit RWA amounts only if all of the following conditions have been complied with:
- (a) except as provided in (g), (i) and (k), significant Credit Risk associated with the securitised Exposures has been transferred from the Originator to third parties;
- (b) the Authorised Firm does not maintain effective or indirect control over the underlying Exposures;

- (c) the assets are legally isolated from the Authorised Firm in order to ensure the assets are beyond the reach of the Authorised Firm in the event of bankruptcy or receivership;
- (d) the Securities issued are not the obligations of the Authorised Firm;
- (e) the Securities are issued pursuant to the securitisation by an SPE and the holders of the beneficial interests in that entity have the right to pledge or exchange them without restriction;
- (f) where a securitisation includes a Clean-Up Call, Clean-Up Calls must satisfy the conditions set out in Rule A4.10.3.
- (g) the documentation of the securitisation does not contain any clauses that:
 - (i) require the Authorised Firm systematically to alter the underlying Exposures such that the pool's weighted average credit quality is improved unless this is achieved by selling Exposures to independent and unaffiliated third parties which are not Connected to the Authorised Firm or Related Persons of the Authorised Firm in accordance with Rule 4.4.6 at market prices;
 - (ii) allow for increases in a retained First Loss Position or Credit Enhancement provided by the Authorised Firm after the securitisation's inception;
 - (iii) other than step-up features incorporated in relation to the underlying Exposures of the securitisation, increase the yield payable to parties other than the Authorised Firm, such as investors and third-party providers of Credit Enhancements, in response to deterioration in the credit quality of the underlying Exposures in the pool; or
 - (iv) other than Clean-Up Calls, oblige the Authorised Firm to repurchase any of the underlying Exposures, at any time, except where that obligation arises from the exercise of a representation or warranty given by the Authorised Firm. The Authorised Firm may give a representation or warranty solely in respect of the nature or existing state of facts of any underlying Exposure, that is capable of being verified, at the time of its transfer.
- (h) the transfer of the underlying Exposures or the transfer of risk through sub-participation does not contravene the terms and conditions of any underlying agreement in respect of the underlying Exposures and where applicable, all the necessary consents for the transfer or sub-participation have been obtained;
- (i) the documentation of the securitisation specifies that, if cash flows relating to the underlying Exposures are rescheduled or renegotiated, the SPE to which the Exposures have been transferred and not the Authorised Firm, would be subject to the rescheduled or renegotiated terms;

- (j) the Authorised Firm receives a fixed amount of consideration for the underlying Exposures;
- (k) the Authorised Firm holds not more than 20% of the aggregate original amount of all Securities issued by the SPE, except where such holdings comprise entirely of Securities that have a Credit Quality Grade of 1 as set out in Rules in sections 4.11 and 4.12, and all transactions with the SPE are conducted at arm's length and on market terms and conditions;
- (l) where the assets relate to the Islamic Financial Business of an Authorised Firm, a written confirmation from the appointed Shari'a Supervisory Board that the securitisation complies with Shari'a; and
- (m) each of the points (a) to (l) must be evidenced and confirmed by a legal opinion from a qualified legal counsel.

Guidance

1. An Authorised Firm is deemed to have effective control over the transferred Exposures if :
 - a. it is able to repurchase from the transferee the previously transferred Exposures in order to realise their benefits; or
 - b. it is obligated to retain the risk of the transferred Exposures.
2. In this regard, an Authorised Firm acting as a Servicer in respect of the transferred Exposures will not necessarily constitute effective control of the Exposures.
3. In respect of Rule A4.10.1(j), the amount of consideration received in the form of a fixed amount of Securities in the SPE would generally be regarded as meeting this requirement if the transaction is conducted at arm's length and on market terms and conditions. Also, this requirement does not preclude excess cash from being channelled to the Authorised Firm after all claims connected with the Securities issued by the SPE have been paid out.

A4.10.2 An Authorised Firm which is an Originator or a Sponsor of a Synthetic Securitisation may recognise the credit protection obtained through the Synthetic Securitisation in its calculation of Credit RWA amounts only if all of the following conditions have been complied with:

- (a) significant Credit Risk associated with the underlying Exposures has been transferred from the Originator to third parties;
- (b) the instrument used to transfer the underlying Credit Risks must not contain terms or conditions that limit in any way the amount of Credit Risk transferred, including, but not limited to, clauses that:
 - (i) materially limit the credit protection or Credit Risk transference (e.g. significant materiality thresholds below which credit protection is deemed not to be triggered even if a credit event occurs or those that allow for the termination of the protection due to deterioration in the credit quality of the underlying Exposures);

- (ii) require the Authorised Firm to alter the underlying Exposures to improve the weighted average credit quality of the pool;
 - (iii) increase the cost of credit protection to the Authorised Firm in response to deterioration in the credit quality of the underlying Exposures;
 - (iv) increase the yield payable to parties other than the Authorised Firm, such as investors and third-party providers of Credit Enhancements, in response to a deterioration in the credit quality of the underlying Exposures; or
 - (v) provide for increases in a retained First Loss Position or Credit Enhancement provided by the originating bank after the transaction's inception.
- (c) an Authorised Firm must provide an external legal opinion from a qualified legal counsel that confirms each of the points (i-v) and the enforceability of the contracts in all relevant jurisdictions;
- (d) where the assets relate to the Islamic Financial Business of an Authorised Firm, a written confirmation from the appointed Shari'a Supervisory Board that the securitisation complies with Shari'a;
- (e) where the securitisation includes a Clean-Up Call it must meet the requirements of Rule A4.10.3;
- (f) in the case where the risks associated with the underlying Exposures are transferred to an SPE:
 - (i) the Securities issued by the SPE are not obligations of the Authorised Firm;
 - (ii) the holders of the beneficial interests in that SPE have the right to pledge or exchange their interests without restriction; and
 - (iii) the Authorised Firm holds not more than 20% of the aggregate original amount of all Securities issued by the SPE, except where such holdings consist entirely of Securities that have a Credit Quality Grade of 1 in accordance with Rules in sections 4.11 and 4.12, and all transactions with the SPE are conducted at arm's length and on market terms and conditions;
- (g) the Authorised Firm has, on an on-going basis, a comprehensive understanding of the risk characteristics of its individual securitisation Exposures, whether on- or off-balance sheet, as well as the risk characteristics of the pools underlying its securitisation Exposures;
- (h) the Authorised Firm is able to access performance information on the underlying Exposures on an on-going basis in a timely manner. Such information may include, as appropriate, Exposure type, percentage of loans 30, 60 and 90 days past due, default rates, prepayment rates, loans in foreclosure, property type,

occupancy, average credit score or other measures of creditworthiness, average loan-to-value ratio, and industry and geographic diversification. For Re-securitisations, the Authorised Firm should have information not only on the underlying securitisation tranches, such as the Issuers' names and credit quality, but also on the characteristics and performance of the pools underlying the securitisation tranches; and

- (i) the Authorised Firm has a thorough understanding of all structural features of a securitisation transaction that would materially impact the performance of the transaction, such as the contractual waterfall and waterfall-related triggers, Credit Enhancements, liquidity enhancements, market value triggers, and deal-specific definitions of default.

Guidance

In relation to Rules A4.10.1 and A4.10.2, significant Credit Risk will be considered to have been transferred by the Originator of a securitisation if:

- a. the RWA amounts of the mezzanine securitisation positions held by the Originator in the securitisation do not exceed 50% of the RWA amounts of all mezzanine securitisation positions existing in this securitisation; and
- b. where there are no mezzanine securitisation positions in a given securitisation and the Originator can demonstrate that the Exposure value of the securitisation positions that would be subject to deduction from Capital Resources or a 1250% risk weight exceeds a reasonable estimate of the expected loss on the Securitised Exposures by a substantial margin, the Originator does not hold more than 20% of the Exposures values of the securitisation positions that would be subject to deduction from Capital Resources or a 1250% risk weight.

Operational requirements for the treatment of Clean-Up Calls

A4.10.3 Where a Clean-Up Call is included within a securitisation, the Authorised Firm which has the ability to exercise the Clean-Up Call must ensure that:

- (a) the exercise of the Clean-Up Call must not be mandatory, in form or substance;
- (b) the Clean-Up Call must not be structured to avoid allocating losses to Credit Enhancements, or positions held by investors or in any way structured to provide Credit Enhancement; and
- (c) the Clean-Up Call must only be exercisable when 10% or less of the original underlying Exposures or Securities issued in that securitisation remains, or in the case of a Synthetic Securitisation, when 10% or less of the original reference portfolio value remains.

A4.10.4 Where the conditions listed in Rule A4.10.3 are not met the Authorised Firm must hold capital against the Exposures as follows:

- (a) for a Traditional Securitisation the underlying Exposures must be treated as if they had not been securitised;
- (b) Authorised Firms must not include any gain-on-sale in any element or component of their Capital Resources;

- (c) for Synthetic Securitisations, the Authorised Firm must hold capital against the entire amount of securitised Exposures; and
- (d) where a Synthetic Securitisation incorporates a call that is not a Clean-Up Call, the Authorised Firm must treat the transaction in accordance with the relevant Credit Risk mitigation techniques in section 4.13.

A4.10.5 An Authorised Firm must treat a currency mismatch or a maturity mismatch between the underlying Exposure being hedged and the Credit Risk mitigation obtained through the Synthetic Securitisation in accordance with Rules in sections 4.13 and A4.3.

A4.11 Concentration Risk

Exempt Exposures

A4.11.1 (1) An Authorised Firm may treat the following Exposures as exempt from the Concentration Risk limits in chapter 4 if they are to Counterparties not Connected to the Authorised Firm:

- (a) Exposures to central governments, Central Banks and PSEs referred to in Rule 4.12.3(2), which receive a Credit Quality Grade rating 1 or 2 in accordance with Rule 4.12.1;
- (b) Exposures to international organisations referred to in Rule 4.12.6 and MDBs referred to in Rule 4.12.5;
- (c) asset items or Exposures carrying the explicit guarantees of either (a) or (b) where the claims on the entity providing the guarantee would receive a 0% weighting (Credit Quality Grade rating of 1);
- (d) Exposures for which the Authorised Firm has Collateral in the form of cash deposits or certificates of deposit, including certificates of deposit issued by the Authorised Firm, held by the Authorised Firm, or held by the Authorised Firm's Parent Regulated Financial Institution or a Subsidiary of the Authorised Firm, but only if:
 - (i) the Authorised Firm and its Parent Regulated Financial Institution or the Subsidiary of the Authorised Firm concerned are subject to consolidated supervision;
 - (ii) the enforceability requirements in section 4.13 (Credit Risk mitigation) are met; and
 - (iii) material holdings in Regulated Financial Institutions and other Exposures which have been deducted from an Authorised Firm's Tier 1 Capital as required in chapter 3.

- (2) If an Authorised Firm obtains credit protection relating to an exempt Exposure under (1)(a), the Authorised Firm must nevertheless apply the Large Exposure

limits to the Exposure to the credit protection provider, notwithstanding that the original Exposure is exempt.

- (3) An Authorised Firm must report Exposures which are exempt under (1)(a), (b) and (c) to the DFSA.

Guidance:

1. In order to be applicable under (c) the guarantees must meet the requirements of section 4.13.9 in relation to Credit Risk mitigation.
2. An Authorised Firm can only treat Exposures as Collateralised provided the conditions of Rules 4.13.5 to 4.13.8 (relating to Credit Risk mitigation) are met. Item (d) also includes cash received under a credit linked note issued by the Authorised Firm and loans and deposits of a Counterparty to or with the Authorised Firm which are subject to an on balance sheet Netting agreement recognised under section 4.13 (Credit Risk mitigation).
3. The DFSA may consider a waiver for other sovereign Exposures where there is a local regulatory requirement to hold assets with a national regulatory authority. Authorised Firms will be required to apply for a waiver of the Large Exposure requirements in this regard and will be considered by the DFSA on a case by case basis.
4. The DFSA may, where it considers it appropriate, allow two or more Exposures to a sovereign not to be treated as connected or related if the sole reason for linking them is being controlled or economically dependent on the sovereign (including eligible PSEs). An eligible PSE is a PSE referred to in Rule A4.11.1(a).

A4.11.2 Where Exposures to a client are guaranteed by a third party, or secured by Collateral issued by a third party, an Authorised Firm may:

- (a) provided the Collateral meets the requirements of section 4.13 (Credit Risk mitigation), and would be assigned a lower risk weight under section 4.12, treat that portion of the Exposure which is secured by Collateral as an Exposure to the third party. An Authorised Firm must treat the portion secured by Collateral as having being incurred to the third party providing the Collateral rather than to the Client for the purposes of considering the limits as set out at Rule 4.15.5; or
- (b) provided the guarantee meets the requirements of section 4.13 (Credit Risk mitigation), and would be assigned a lower risk weight under section 4.12, treat that portion of the Exposure which is guaranteed as an Exposure to the third party. An Authorised Firm must treat the portion guaranteed as having being incurred to the third party rather than to the Client for the purposes of considering the limits as set out at Rule 4.15.5. When considering the guarantee there must not be any maturity mismatch between the guarantee and the underlying Exposure.

A4.11.3 If an Exposure is partially guaranteed by an Authorised Firm's Parent Regulated Financial Institution, and would be assigned a lower risk weight under section 4.12, only that part of the Exposure subject to the guarantee is exempt from the Concentration Risk limits in Rule 4.15.5. When considering the treatment of this Rule an Authorised Firm may also consider the exemptions permitted under Rule 4.15.18 relating to parental guarantees.

Identification of Counterparties

A4.11.4 When calculating the Exposures of an Authorised Firm, the Authorised Firm must include Trading Book Exposures and Non-Trading Book Exposures to:

- (a) an individual Counterparty;
- (b) group of Closely Related Counterparties;
- (c) Connected Counterparties; and,
- (d) Transactions, schemes or Funds.

Guidance

1. An individual Counterparty is a natural or legal person, which include governments, local authorities, public sector enterprises (PSEs), trusts, corporations, unincorporated businesses and non-profit-making bodies.
2. Examples of a Counterparty include:
 - a. the customer or borrower;
 - b. where the Authorised Firm is providing a guarantee, the person guaranteed;
 - c. for a Derivatives contract, the person with whom the contract was made;
 - d. for most exchange-traded contracts involving a central clearing mechanism, that central clearing mechanism; and
 - e. where a bill held by an Authorised Firm has been accepted by another Financial Institution, the acceptor.

Group of Closely Related Counterparties

A4.11.5 (1) For Concentration Risk purposes, Persons are Closely Related if:

- (a) the insolvency or default of one of them is likely to be associated with the insolvency or default of the others;
 - (b) it would be prudent when assessing the financial condition or creditworthiness of one to consider that of the others;
 - (c) there is, or is likely to be, a close relationship between the financial performance of those Persons; or
 - (d) they are assessed to be economically interdependent.
- (2) Persons are to be assessed as being economically interdependent for the purposes of (1)(d) where one or more of the following factors are present:
- (a) 50% or more of the annual receipts or expenditures of one is derived from transactions with the other;

- (b) a guarantee issued by one in favour of the other party is likely to result in the provider default if called;
 - (c) significant output is sold to the other party and an alternative buyer is not easily found;
 - (d) single source of funds to repay loans for both parties with no alternative; or
 - (e) common reliance on the same funding provider that is hard to replace.
 - (3) An Authorised Firm need not assess economic interdependence under (2) for a transaction where the sum of all Exposures to one individual counterparty is less than 5% of the Authorised Firm's T1 Capital.
 - (4) Persons who are Closely Related to each other are also Connected with each other.
 - (5) The requirement to aggregate Exposures where Persons are Closely Related does not apply where an Authorised Firm can demonstrate to the DFSA that it is unlikely that Persons who are Closely Related will default at the same time and the DFSA has, by notice in writing, disapplied the requirement.
- A4.11.6** (1) A single group of Closely Related Counterparties means, in relation to an Authorised Firm, all the Persons to which the Authorised Firm has an Exposure and which are Closely Related to each other.
- (2) An Authorised Firm must treat two or more Persons as falling within a group of Closely Related Counterparties if the Authorised Firm has Exposures to them all and any loss to the Authorised Firm on any of the Exposures to one is likely to be associated with a loss to the Authorised Firm with respect to at least one Exposure to each of the others.

Guidance:

Two or more Counterparties between whom there is no relationship of control as described in Rules A4.11.5 and A4.11.6 will be regarded as constituting a single risk if they are so interconnected that, if one of them were to experience financial problems, in particular funding or repayment difficulties, the other or all of the others would also be likely to encounter funding or repayment difficulties.

Connected Counterparties

- A4.11.7** (1) For Concentration Risk purposes, and in relation to a Person, a Connected Counterparty means another Person to whom the first Person has an Exposure and who fulfils one of the following conditions:
- (a) he is Connected to the first Person;
 - (b) he is an Associate of the first Person;
 - (c) the same Persons significantly influence the Governing Body or senior management of each of them;

- (d) one of those Persons has an Exposure to the other that was not incurred for the clear commercial advantage of both of them and which is not on arm's length terms; or
- (e) one of those Persons:
 - (i) has influence over the appointment or dismissal of the Board or senior management of the other party;
 - (ii) is entitled to exercise, or controls the exercise of 50% or more of the voting rights in the other party including through voting agreements; or
 - (iii) has effective control over the other party under the criteria adopted by the International Financial Reporting Standards.
- (2) A Person is not to be treated as a Connected Counterparty under (1), where an Authorised Firm can satisfy the DFSA that effective governance and controls are in place to mitigate risks resulting from effective control of one Person by the other and the DFSA has, by notice in writing, specified that they are not to be so treated.

Guidance:

A group of Connected Counterparties would be considered to be such where the entities share the same ultimate owner even though they may not be formally structured as a Group.

Exposures to transactions, schemes or Funds

- A4.11.8** (1) Where an Authorised Firm has an Exposure to a transaction, scheme, Fund, or other Exposure to a pool of underlying Exposures, the Authorised Firm must assess the Exposure to determine whether the Exposure is to a group of Closely Related Counterparties in its economic substance.
- (2) Where the Exposure is to a group and the Authorised Firm can demonstrate that the Exposure amount to each underlying asset of the structure is less than 0.25% of its Tier 1 Capital, the Authorised Firm may assign the Exposure amount to the structure itself.
 - (3) An Authorised Firm need not look through a structure for the purposes of (2) if the Authorised Firm can demonstrate that its Exposure to each asset in the structure is less than 0.25% of its Tier 1 Capital.
 - (4) Where the Exposure is to a group and the Authorised Firm can demonstrate that the Exposure amount to each underlying asset of the structure is equal to or exceeds 0.25% of its Tier 1 Capital, the Authorised Firm must look through the structure and identify each underlying asset and add that Exposure to the other Exposures of the same counterparty.
 - (5) If an Authorised Firm looks through the structure to identify underlying assets and one or more underlying assets cannot be identified, the Authorised Firm must

aggregate and assign that Exposure to a single "unknown customer" to which the Large Exposure limits apply.

- (6) If the Exposure to the underlying assets in a structure depends on the hierarchy of loss distribution to investors in the event of the winding up of the structure, an Authorised Firm must treat its Exposure to the structure:
 - (a) as the pro rata share of the firm's Exposure multiplied by the value of the underlying asset, where all investors rank equally; or
 - (b) as the pro rata share of the firm's investment in the structure multiplied by the lower of the tranche value or the nominal value of the underlying assets in the structure, where the seniority differs.
- (7) An Authorised Firm must aggregate its Exposures to any additional risks inherent in the structure itself, if that would result in Exposures to the same third party.

Guidance:

1. When considering this Rule the Authorised Firm should consider the following factors:
 - a. the structure, independence and control of the transaction, including governance arrangements;
 - b. the inter relatedness of the underlying Exposures;
 - c. beneficial owners of the underlying Exposures and whether they could be deemed Connected or Closely Related; and
 - d. whether the transactions are conducted on an arm's length basis.
2. An Authorised Firm should look through the structure to determine whether there are any Counterparties or Exposures that should be considered a Concentration Risk.

Connected Counterparty exemptions

- A4.11.9** (1) This Rule applies to an Authorised Firm in Category 2 and 3A.
- (2) An Authorised Firm may treat as exempt from the Concentration Risk limits in chapter 4 an Exposure to a Counterparty or Counterparties Connected to the Authorised Firm if all of the following conditions are met:
 - (a) the Authorised Firm has given the DFSA written notice one month in advance of its intention to use the exemption and explained how it will ensure that it will still meet the Concentration Risk limits on a continuing basis when using the exemption;
 - (b) the total amount of the Exposures that an Authorised Firm is treating as exempt under this Rule does not exceed 50% of the Authorised Firm's Tier 1 Capital;

- (c) the Authorised Firm makes and retains a record that identifies each Exposure it has treated in this way;
- (d) the Authorised Firm is subject to consolidated supervision;
- (e) the Counterparty is:
 - (i) an Authorised Firm which is the subject of consolidated supervision; or
 - (ii) a member of the Authorised Firm's Group which is the subject of consolidated supervision to the satisfaction of the DFSA; and
- (f) the Exposure satisfies one or more of conditions (i) to (iii):
 - (i) it is a loan made by the Authorised Firm with a maturity of one year or less in the course of the Authorised Firm carrying on a treasury role for other members of its Group;
 - (ii) it is a loan to the Parent of the Authorised Firm made in the course of a business carried on by the Authorised Firm of lending to its parent cash that is surplus to the needs of the Authorised Firm, provided that the amount of that surplus fluctuates regularly; or
 - (iii) it arises from the Authorised Firm or a Counterparty Connected to the Authorised Firm operating a central risk management function for Exposures arising from Derivatives contracts.

Measuring Exposure to Counterparties and Issuers

- A4.11.10** Rules A4.11.12 to A4.11.28 apply to both Non-Trading Book and Trading Book Exposures.
- A4.11.11** When calculating an Exposure, an Authorised Firm must include accrued interest and dividends due.
- A4.11.12** An Authorised Firm must not offset Non-Trading Book and Trading Book Exposures.
- A4.11.13** A net short position is not an Exposure for the purposes of Concentration Risk.
- A4.11.14** (1) Subject to (2), the value of an Authorised Firm's Exposure to a Counterparty, whether in its Non-Trading Book or its Trading Book, is the amount at risk calculated in accordance with chapter 4.
- (2) For the purposes of calculating the value of an Authorised Firm's Exposure to a Counterparty under (1), the CCF factors taken into account under A4.2 are subject to a minimum floor of 10%.

Exposures to Issuers

- A4.11.15** An Authorised Firm must calculate the value of an Exposure to the Issuer of a Security which is held in the Authorised Firm's Non-Trading Book as the sum of the excess, where

positive, of the book value of all long positions over all short positions (the net long position), for each identical instrument issued by that Issuer.

A4.11.16 For the purposes of Rule A4.11.15, short positions in one Security may be used to offset long positions in a non-identical Security issued by the same Issuer if:

- (a) both Securities are denominated in the same currency; and
- (b) where both Securities are:
 - (i) fixed rate or index-linked, and are within the same residual maturity time band; or
 - (ii) floating rate.

A4.11.17 An Authorised Firm must calculate the value of an Exposure to the Issuer of a Security or a credit derivative used as a hedge that is held in the Authorised Firm's Trading Book by calculating the excess of the current market value of all long positions over all short positions in all the Securities or credit derivatives issued by that Issuer, based on the seniority of those Securities or credit derivatives in the liquidation of the Issuer.

A4.11.18 An Authorised Firm must not offset an Exposure to one Issuer against an Exposure to another even where the Issuers are in a group of Closely Related Counterparties.

A4.11.19 An Authorised Firm must include as a long position a commitment by it to buy:

- (a) a debt Security or an equity at a future date; and
- (b) under a note issuance facility, at the request of the Issuer, a Security that is unsold on the issue date.

A4.11.20 An Authorised Firm must include as a short position a commitment by it to sell a debt Security or equity at a future date.

A4.11.21 Where the equity leg of an equity swap is based on the change in value of an individual equity, it is treated as an Exposure to the Issuer of the equity.

Guidance

An interest rate leg of an equity swap, or interest rate or currency swap does not generate an Exposure to an Issuer.

A4.11.22 An Authorised Firm must, when determining its Exposure to an Issuer arising from an Option, use an Exposure value based on the level of change in the price that will result from the default of the underlying instrument.

A4.11.23 (1) An Authorised Firm must treat the value of:

- (a) a call Option as equal to market value; and
- (b) a put Option as equal to the strike price minus the market value.

- (2) An Authorised Firm must aggregate the resulting option Exposures to each underlying counterparty and if there is a negative net Exposure after aggregation of all option Exposures, the option Exposure must be set to nil.

A4.11.24 An Authorised Firm must, for the purposes of Concentration Risk, treat an Exposure to an Issuer arising from an index or basket of debt Securities or a non-broad-based equity index or basket, as a series of Exposures to the Issuers of the underlying instruments or equities in accordance with the procedures in chapter 4.

Guidance

Broadly based equity indices should not be broken down into their constituent stocks. A position related to a broadly based equity index does not generate an Exposure to any Issuer.

A4.11.25 An Authorised Firm which receives cash on a repurchase agreement must treat the cash as if it is on its balance sheet and in accordance with sections 4.9 and 4.13. Any Collateral received against repurchase agreements or Securities and commodities borrowing must also be treated as a balance sheet item under sections 4.9 and 4.13.

A4.11.26 An Authorised Firm must treat a reverse repurchase agreement or Securities and commodities lending in its Non-Trading Book as a collateralised loan and the Collateral it holds as an asset, provided that the Collateral is eligible financial Collateral as defined in Rule 4.13.5. If the Collateral is not such an eligible financial Collateral, the Authorised Firm must treat the transaction as an unsecured loan to the Counterparty.

A4.11.27 An Authorised Firm with repurchase agreements and reverse repurchase agreements in its Trading Book has an Exposure to:

- (a) the Issuer of the Security it has sold in a repurchase agreement; and
- (b) the Counterparty where the Securities or cash given by the Authorised Firm exceed the Securities or cash it receives (i.e. there is a net margin given by the Authorised Firm) in a repurchase agreement or reverse repurchase agreement.

A4.11.28 An Authorised Firm must calculate in accordance with section 5.10 an Exposure to the Issuer arising from the Underwriting or sub-underwriting of a new Issue of Securities.

A4.12 The Simplified Approach for Category 2 and 3A firms

A4.12.1 This section applies only to an Authorised Firm in Category 2 or 3A for the purposes of section 4.7.

A4.12.2 An Authorised Firm that applies the Simplified Approach must comply with the requirements of chapter 4 with the variations as prescribed below:

Risk Weights

Exposures to central governments and Central Banks

A4.12.3 Under the Simplified Approach, Rules 4.12.1 is replaced by Rule A4.12.4.

A4.12.4 Subject to Rule 4.12.1, an Authorised Firm must risk-weight any Exposure to a central government or a Central Bank on the basis of the consensus country risk scores of export credit agencies (referred to in this section as “ECA”) participating in the OECD’s “Arrangement on Officially Supported Export Credits” and in accordance with the table below.

ECA Risk Scores	0-1	2	3	4 to 6	7
Risk Weights	0%	20%	50%	100%	150%

Guidance

The consensus country risk classification for the purpose of the “Arrangement on Officially Supported Export Credits” is published by the OECD. At the time of making of these Rules, the classification was available on the OECD’s website (<http://www.oecd.org>).

Exposures to banks

A4.12.5 Under the Simplified Approach, Rules 4.12.7 to 4.12.10 are replaced by the following Rule A4.12.6.

A4.12.6 An Authorised Firm must risk-weight any Exposure to a bank on the basis of the consensus ECA country risk scores as referred to in A4.12.4 for the jurisdictions in which they are incorporated, in accordance with the following table:

ECA Risk Scores	0-1	2	3	4 to 6	7
Risk Weights	20%	50%	100%	100%	150%

Exposures to corporates

A4.12.7 Under the Simplified Approach, Rules 4.12.11 to 4.12.15 are replaced by the following Rule A4.12.8.

A4.12.8 An Authorised Firm must apply a 100% risk weight to any Exposure to a corporate.

Credit Risk mitigation - Collateral

A4.12.9 Under the Simplified Approach, Rules 4.13.5 to 4.13.7 are replaced by the following Rules A4.12.10 to A4.12.12.

A4.12.10 An Authorised Firm may only use the financial Collateral Simplified Approach (FCSA) in its treatment of recognised Collateral for the purposes of calculating the Credit RWA for its Exposures booked in its Non-Trading Book.

A4.12.11 (1) For an Authorised Firm using the FCSA, eligible financial Collateral comprises:

- (a) cash (as well as certificates of deposit or other similar instruments issued by the Authorised Firm) on deposit with the Authorised Firm;
 - (b) gold;
 - (c) any debt securities issued by sovereigns (including a central government or Central Bank) of a jurisdiction that has an ECA country risk score of 4 or better; and
 - (d) any debt securities issued by a PSE referred to in Rule 4.12.3(2) and that is from a jurisdiction that has an ECA country risk score of 4 or better.
- (2) Cash-funded credit-linked notes issued by an Authorised Firm against Exposures in the Non-Trading Book which fulfil the criteria for eligible Credit Derivatives must be treated as cash collateralised transactions.
- (3) Cash, mentioned in (1)(a) includes cash on deposit, certificates of deposit or other similar instruments issued by the Authorised Firm that are held as Collateral at a third-party bank in a non-custodial arrangement and that are pledged or assigned to the Authorised Firm. This is subject to the pledge or assignment being unconditional and irrevocable. Under the FCSA, the risk weight to be applied to the Exposure covered by such Collateral must be the risk weight of the third-party bank.

A4.12.12 In the case of any Counterparty Credit Risk Exposures in Rules A4.12.10 and A4.12.11, arising from an SFT which is included in the Trading Book, eligible financial Collateral includes all instruments which an Authorised Firm may include in its Trading Book.

Credit Risk mitigation - Guarantees

A4.12.13 Under the Simplified Approach, Rules 4.13.9 and 4.13.10 are replaced by the following Rules A4.12.14 to A4.12.17.

A4.12.14 An Authorised Firm may recognise guarantees provided by the following eligible guarantors:

- (a) the Bank for International Settlements, the International Monetary Fund, the European Central Bank, and the European Commission;
- (b) the MDBs referred to in Rule 4.12.5;
- (c) PSEs; and
- (d) other entities eligible for a CRW of 20% or better and with a lower risk weight than the Counterparty.

A4.12.15 For the purpose of calculating the risk weight of a guaranteed Exposure, an Authorised Firm must assign the guaranteed portion the risk weight of the eligible guarantors. The uncovered portion of the Exposure must be assigned the risk weight of the underlying Counterparty.

- A4.12.16** An Authorised Firm can apply a 0% risk weight to any portions of Exposures guaranteed by central governments or Central Banks of a GCC member country where the guarantee is denominated in the domestic currency of that country, and the Exposure is funded in that same domestic currency.
- A4.12.17** An Authorised Firm must treat any materiality thresholds on payments below which no payment will be made in the event of loss as retained First Loss Positions and must deduct the full amount from its Capital Resources.

APP5 MARKET RISK

A5.1 Market Risk systems and controls

Guidance

1. In accordance with section 5.2, an Authorised Firm is required to have a Market Risk policy. The Market Risk policy should address all aspects of Market Risk whether arising from assets, liabilities or the mismatch between assets and liabilities and whether off or on-balance sheet. Such a policy would be expected to include the following information:
 - a. how, with particular reference to its activities, the Authorised Firm defines and measures Market Risk;
 - b. the Authorised Firm's investment or trading strategy distinguishing, as applicable, between its Trading and Non-Trading Books;
 - c. the detailed limit structure for Market Risk which should:
 - i. address all key risk factors;
 - ii. be commensurate with the volume and complexity of activity; and
 - iii. be consistent with the Authorised Firm's strategy, historical performance, and the overall level of earnings or capital the Authorised Firm is willing to risk;
 - d. procedures for:
 - i. approving new products and activities that give rise to Market Risk;
 - ii. regular risk position and performance reporting;
 - iii. limit exception reporting and approval; and
 - iv. reporting and controlling of off-market trades, if these are permitted;
 - e. where internal models are used to set Capital Requirements (as provided for in section 5.3), the methods and assumptions used in these models and how the models are tested; and
 - f. the allocation of responsibilities for implementing the Market Risk policy and for monitoring adherence to, and the effectiveness of, the policy.
2. An Authorised Firm should measure its Market Risk using a robust and consistent methodology. The appropriate method of measurement will depend upon the nature of the products traded. The Authorised Firm should consider whether the measurement methodologies should be tested, for example, through back-testing, and the frequency of such testing.
3. An Authorised Firm should be able to measure its Market Risk Exposure both across risk types, such as interest rate, foreign exchange and commodities, and across the entire portfolio.
4. Where an Authorised Firm is a member of a Group, which is subject to consolidated supervision, the Group should be able to monitor Market Risk Exposures on a consolidated basis (chapter 8).

5. An Authorised Firm should have the capability to assess the impact of any new transaction on its Market Risk position on an on-going basis, and should be capable of carrying out a full measurement of its positions at least daily.
6. An Authorised Firm should implement an effective system for monitoring its Market Risk. This system should be independent of those within the Authorised Firm who are responsible for taking Market Risk.
7. An Authorised Firm should implement a system of management reporting which provides relevant, accurate, comprehensive, timely and reliable Market Risk reports to relevant functions within the Authorised Firm. These reports should:
 - a. alert senior management's attention to the size of Exposures and the relationship between these Exposures and limits;
 - b. cover exceptions to the Authorised Firm's Market Risk policy;
 - c. present the results from stress tests undertaken; and
 - d. analyse and explain any changes to the level and nature of Market Risk and any remedial action proposed or taken.
8. An Authorised Firm should have procedures, including stop-loss procedures, for taking appropriate action according to the information within the management reports.
9. An Authorised Firm should ensure that there are controls and procedures for reporting any trades booked at off-market rates.
10. An Authorised Firm should ensure that risk monitoring is subject to a periodic independent check. Models used to determine or interpolate specific Market Risk factors should be independently reviewed or otherwise validated.
11. Particular attention should be given to the monitoring of Market Risk that does not conform to the usual Market Risk policy, or which exceeds predetermined Market Risk limits and criteria, but is sanctioned because of particular circumstances in accordance with the Authorised Firm's procedures. Unauthorised exceptions to policies, procedures and limits should be reported in a timely manner to the appropriate level of management along with any remedial action proposed or taken.
12. Market Risk limits should be periodically reviewed in order to check their suitability for current market conditions and the Authorised Firm's overall risk appetite.
13. An Authorised Firm should use a model or some form of analytical tool to assess risk in complex instruments or across portfolios. An Authorised Firm which wishes to use such a model to determine part of its financial resources requirement, should refer to section 5.3.
14. An Authorised Firm should also use stress testing to determine the potential effects of economic downturns, market events, changes in interest rates, foreign exchange or liquidity conditions.
15. An Authorised Firm should set an appropriate limit structure to control its Market Risk Exposure. The degree of granularity within the limit structure, or how hierarchical it is, will depend on the nature of the products traded (for example, whether the underlying risks are linear or non-linear) and the scale of the Authorised Firm's overall business (for example, whether the Authorised Firm is an active market maker). An Authorised Firm should set limits on risks such as simple price or rate risk as well as on the factors, Delta, Gamma, Vega, Rho, and Theta, arising from options positions.
16. Limits should also be imposed against net or gross positions, and in relation to maximum allowable loss ('stop-loss'), value at risk, maturity gap, and illiquid or volatile markets.

17. An Authorised Firm should provide a process for the identification, timely reporting and subsequent action in respect of exceptions to limits. An Authorised Firm should also ensure that limit breaches and action arising from exceptions are monitored. An Authorised Firm may also consider whether it is appropriate to set intermediate thresholds that alert management when limits are approached, triggering review or other appropriate action, or both.
18. Various methods can be used to hedge Market Risk. An Authorised Firm should document the appropriate products to be used to hedge Exposure and identify individuals within the Authorised Firm or Group responsible for monitoring hedge performance.
19. An Authorised Firm should ensure that it makes and maintains appropriate prudential records which show and explain the Authorised Firm's transactions, disclose its financial position and Exposure to Market Risk and enable it to demonstrate compliance with the DFSA rules. In particular, an Authorised Firm should have data history to enable it to perform back-testing of methods and assumptions used for stress and scenario testing and for value-at-risk models. Market Risk records should be retained for at least six years.

A5.2 Interest Rate Risk Capital Requirement

Guidance

Section A5.2 presents the method for the calculation of Specific Risk and General Market Risk in respect of the Interest Rate Risk Capital Requirement as referred to in Rule 5.4.1(b).

- A5.2.1** An Authorised Firm which calculates its Interest Rate Risk Capital Requirement in accordance with Rule 5.4.1(b) must apply the Rules in this section.
- A5.2.2** An Authorised Firm must calculate its Interest Rate Risk Capital Requirement as the sum of the two following separate charges:
- (a) Specific Risk of each net position as calculated in accordance with Rule A5.2.13; and
 - (b) General Market Risk calculated in accordance with Rule A5.2.15.
- A5.2.3** An Authorised Firm must calculate its Interest Rate Risk Capital Requirement in Trading Book positions in all fixed-rate and floating-rate debt Securities and instruments which behave like them, including:
- (a) non-convertible preference shares;
 - (b) futures or forwards on a debt security or on interest rates;
 - (c) swaps (or contracts for differences) whose value is based on interest rates;
 - (d) the cash leg of a repurchase or a reverse repurchase agreement;
 - (e) forward foreign exchange contracts or currency futures;
 - (f) interest rate legs of equity swaps;
 - (g) interest rate legs of equity futures or forwards; and

- (h) interest rate legs of equity based options treated under internal models in section 5.3.

Guidance

Where these positions will require the derivation of notional positions before they can be included in the calculation of Specific Risk and General Market Risk requirements, an Authorised Firm must derive the notional positions in accordance with Rules A5.2.5 to A5.2.12.

- A5.2.4** (1) An Authorised Firm may net, by value, long and short positions in the same debt instrument in its Trading Book to generate the individual net position in that instrument.
- (2) Instruments are considered to be the same for the purposes of (1) where:
- (a) the issuer is the same;
 - (b) the instruments have equivalent standing in liquidation; and
 - (c) the currency, coupon and maturity are the same.

Derivation of notional positions for certain instruments (including interest rate derivatives)

- A5.2.5** (1) The interest rate risk measurement must include all interest rate derivatives and off-balance sheet instruments in the Trading Book that react to changes in interest rates, including forward rate agreements other forward contracts, futures, interest rate and cross-currency swaps and forward foreign exchange positions.
- (2) Derivatives must be converted into positions in the relevant underlying instruments and are subject to Specific and General Market Risk requirements set out in Rules A5.2.13 and A5.2.15. The amounts used in the calculation must be the market values of the principal amount of the underlying instrument or of the notional underlying instrument.
- (3) The manner in which an Authorised Firm must derive a notional position (in the currency concerned) for certain instruments (including interest rate derivatives) is set out in Rules A5.2.6 to A5.2.12.

Futures on interest rates and forward rate agreements

- A5.2.6** A future on an interest rate and a forward rate agreement must be treated as two notional zero coupon government Securities as follows:
- (a) where an Authorised Firm sells an interest rate future or buys a forward rate agreement:
 - (i) the notional short position has a maturity equal to the time to expiry of the future (or the settlement date of the forward rate agreements) plus the maturity of the borrowing period; and

- (ii) the notional long position has a maturity equal to the time to expiry of the future (or the settlement date of the forward rate agreement); and
- (b) where an Authorised Firm buys an interest rate future or sells a forward rate agreement:
 - (i) the notional short position has a maturity equal to the time to expiry of the future (or the settlement date of the forward rate agreement); and
 - (ii) the notional long position has a maturity equal to the time to expiry of the future (or the settlement date of the forward rate agreement) plus the maturity of the deposit period.

Futures and forwards on a single debt security

A5.2.7 A future and a forward on a single debt Security must be treated as a notional debt Security and a notional zero coupon government Security as follows:

- (a) where an Authorised Firm has bought the future or forward:
 - (i) a notional long position in the underlying Security with a maturity:
 - (A) in the case of a fixed rate bond, equal to the underlying Security; and
 - (B) in the case of a floating rate bond, at the time to the next reset; and
 - (ii) a notional short position in a zero coupon government Security with a maturity equal to the time to expiry of the futures contract; and
- (b) where an Authorised Firm has sold the future or forward:
 - (i) a notional short position in the underlying Security with a maturity:
 - (A) in the case of a fixed rate bond, equal to the underlying Security; and
 - (B) in the case of a floating rate bond, at the time to the next reset; and
 - (ii) a notional long position in a zero coupon government Security with a maturity equal to the time to expiry of the futures contract.

Future or forward on a basket of debt securities

A5.2.8 A future and a forward on a basket of debt Securities must be treated as a set of notional positions in the constituent debt Securities.

Interest rate and currency swaps

A5.2.9 An interest rate and a currency swap must be treated as two notional government Securities as follows:

- (a) where the Authorised Firm is receiving fixed rate interest and paying floating rate interest:
 - (i) a notional long position with a maturity equal to the length of the swap; and
 - (ii) a notional short position with a maturity equal to the period remaining to the next interest rate reset date;
- (b) where the Authorised Firm is paying fixed rate interest and receiving floating rate interest:
 - (i) a notional short position with a maturity equal to the length of the swap; and
 - (ii) a notional long position with a maturity equal to the period remaining to the next interest rate reset date;
- (c) where the Authorised Firm is receiving fixed rate interest and paying fixed rate interest:
 - (i) a notional long position with a maturity equal to the length of the swap; and
 - (ii) a notional short position with a maturity equal to the length of the swap.
- (d) where the Authorised Firm is receiving floating rate interest and paying floating rate interest:
 - (i) a notional long position with a maturity equal to the period remaining to the next interest date reset date; and
 - (ii) a notional short position with a maturity equal to the period remaining to the next interest rate reset date; and
- (e) the two notional government Securities must have a coupon equal to the rate of interest payable or receivable on the leg.

Guidance

A currency swap is also subject to a Foreign Exchange Risk Capital Requirement (see section 5.6).

Dual currency bonds

A5.2.10 A dual currency bond must be treated as two positions as follows:

- (a) a debt Security denominated in the currency in which the dual currency bond is issued; and
- (b) a foreign exchange forward for the purchase of the redemption currency (see section 5.6).

Cash legs of repos

A5.2.11 The forward cash leg of a repo must be treated as a notional short position in a government Security with a maturity equal to that of the repo and coupon equal to the repo rate.

Guidance

If a Security is repo'd, the Authorised Firm continues to calculate an Interest Rate Risk Capital Requirement on the Security because, although legal ownership transfers to the Counterparty, the economic benefit or loss remains with the Authorised Firm.

Cash legs of reverse repos

A5.2.12 (1) The forward cash leg of a reverse repo must be treated as a notional long position in a government security with a maturity equal to that of the reverse repo and coupon equal to the repo rate.

(2) An Authorised Firm may exclude from the interest rate maturity framework (for both Specific and General Market Risk) long and short positions (both actual and notional) in identical derivative instruments with exactly the same issuer, coupon, currency and maturity. A fully-matched position in a future or forward and its corresponding underlying instrument may also be fully offset, and thus excluded from the calculation.

Guidance

If a Security is reverse repo'd, the Authorised Firm does not calculate an Interest Rate Risk Capital Requirement on the Security because, although the firm obtains the legal title, the economic benefit or loss remains with the original holder.

Specific Risk

Guidance

In respect of interest rate risk, a capital charge for Specific Risk is designed to protect against an adverse movement in the price of an individual Security owing to factors related to the individual issuer.

A5.2.13 (1) An Authorised Firm must calculate its Specific Risk as the sum of the market values of the individual net positions (whether they are long or short) multiplied by the appropriate risk percentage in (3).

(2) An Authorised Firm must not offset between different issues.

(3) An Authorised Firm must determine the appropriate risk percentage by reference to the following table:

Issuer	Credit Quality Grades	Residual Term to Maturity	Risk Percentage
Sovereign Debt	1	Any	0.00%

This category includes – (a) all forms of government debt, including bonds, treasury bills and other short-term instruments; and (b) Securities issued by PSEs which qualify for a 0% risk weight for Credit Risk. An Exposure to any debt Security issued by– (i) the central government or monetary authority; or (ii) other central governments with a Credit Quality Grade of 3 or better as set out in chapter 4, which is denominated in the domestic currency and funded in the same currency must be assigned a 0% Specific Risk charge. The DFSA may, at its discretion, assign a higher risk charge other than the above to Securities issued by certain governments, especially in cases where the Securities are denominated in a currency other than that of the issuing government.	2 or 3	6 months or less	0.25%
		More than 6 and up to 24 months	1.00%
		More than 24 months	1.60%
	4 or 5	Any	8.00%
	6	Any	12.00%
	Unrated	Any	8.00%
Qualifying Debt This category includes – (a) any Security that is issued by an MDB; (b) any Security (including one issued by a PSE) which has a Credit Quality Grade of 3 or better as set out in chapter 4; and (c) any unrated Security issued by a PSE which belongs to a country with a Credit Quality Grade of 1 as set out in chapter 4.		6 months or less	0.25%
		More than 6 and up to 24 months	1.00%
		More than 24 months	1.60%
Other For securities which have a high yield to redemption relative to government debt securities issued in the same country, the DFSA will require the Authorised Firm: (a) to apply a higher Specific Risk charge to such instruments; or (b) to disallow offsetting for the purpose of defining the extent of General Market Risk between such instruments and any other debt instruments.	4	Any	8% or such other percentage as the DFSA may direct.
	5 or 6	Any	12% or such other percentage as the DFSA may direct.
	Unrated	Any	8% or such other percentage as the DFSA may direct.

Guidance

- Offsetting is not permitted since differences in coupon rates, liquidity, and call features, for example, signify that prices may diverge in the short run.

2. The “Other” category will receive the same Specific Risk requirement as a private-sector borrower under the CRCOM, 8%. However, since this may, in certain cases, considerably underestimate the Specific Risk for debt Securities which have a high yield to redemption relative to government debt Securities, the DFSA has the right to apply to such Securities a Specific Risk percentage higher than 8%.

A5.2.14 [Not currently in use]

General Market Risk

- A5.2.15** (1) An Authorised Firm must calculate its General Market Risk on a currency by currency basis, irrespective of where the individual instruments are physically traded or listed. The calculations for each currency must then be added together to determine the amount of the Authorised Firm’s General Market Risk requirement.
- (2) An Authorised Firm must calculate its General Market Risk requirement for each currency by applying either:
- (a) the simplified framework set out in Rule A5.2.16;
 - (b) the Maturity Method set out in Rule A5.2.17; or
 - (c) with the consent of the DFSA, the Duration Method set out in Rule A5.2.19.

Simplified Framework

A5.2.16 In applying the simplified framework, an Authorised Firm must calculate its General Market Risk requirement for each currency by taking the following steps:

- (a) allocating the individual net positions to one of the time bands in the table below, as follows:
 - (i) fixed-rate instruments are allotted their time bands based upon the residual time to maturity; and
 - (ii) floating-rate instruments are allocated to time bands based upon the time remaining to the re-determination of the coupon;
- (b) adding the market values of the individual net positions within each band irrespective of whether they are long or short positions to produce a gross position figure;
- (c) multiplying the amount in (b) by the risk percentage for the relevant maturity band in the table below; and
- (d) adding the calculations in (c) to arrive at the General Market Risk requirement.

Zone	Time band	Risk percentage
------	-----------	-----------------

	Coupon of 3% or more	Coupon of less than 3%	
A	0 ≤ 1 month	0 ≤ 1 month	0.00%
	> 1 ≤ 3 months	> 1 ≤ 3 months	0.20%
	> 3 ≤ 6 months	> 3 ≤ 6 months	0.40%
	> 6 ≤ 12 months	> 6 ≤ 12 months	0.70%
B	> 1 ≤ 2 years	> 1.0 ≤ 1.9 years	1.25%
	> 2 ≤ 3 years	> 1.9 ≤ 2.8 years	1.75%
	> 3 ≤ 4 years	> 2.8 ≤ 3.6 years	2.25%
C	> 4 ≤ 5 years	> 3.6 ≤ 4.3 years	2.75%
	> 5 ≤ 7 years	> 4.3 ≤ 5.7 years	3.25%
	> 7 ≤ 10 years	> 5.7 ≤ 7.3 years	3.75%
	> 10 ≤ 15 years	> 7.3 ≤ 9.3 years	4.50%
	> 15 ≤ 20 years	> 9.3 ≤ 10.6 years	5.25%
	> 20 years	> 10.6 ≤ 12.0 years	6.00%
		> 12.0 ≤ 20.0 years	8.00%
		> 20 years	12.50%

Guidance

The risk percentages in the table above are designed to reflect the price sensitivity of the positions to changes in the interest rate.

Maturity Method

A5.2.17 Under the Maturity Method, the following steps must be carried out:

- the maturity weighted position for each instrument must be calculated by multiplying the market value of each individual long or short net position by the appropriate risk percentage per the table in Rule A5.2.16;
- the sum of the weighted long and the sum of the weighted short positions in each maturity band must be calculated;
- these weighted long and short positions must be matched within a maturity band to give the total matched weighted position in the maturity band and the total unmatched weighted position which will be long or short in the maturity band;

- (d) the matched weighted positions in all maturity bands must be summed;
- (e) the unmatched weighted positions in all the maturity bands must then be matched within a zone leaving an unmatched position for the zone (which will either be short or long); and
- (f) the unmatched positions in each zone must be matched with the unmatched positions in other zones leaving the residual unmatched weighted position.

A5.2.18 The General Market Risk requirement for each currency must be calculated as the sum of the following:

- (a) 10% of the matched weighted positions in each maturity band;
- (b) 40% of the matched weighted position in zone A;
- (c) 30% of the matched weighted position in zones B and C;
- (d) 40% of the matched weighted position between zones A and B, and between zones B and C;
- (e) 100% of the matched weighted position between zones A and C; and
- (f) 100% of the residual unmatched weighted positions.

Guidance

A worked example under the Maturity Method of the General Market Risk requirement calculation is as follows:

PRUDENTIAL – INVESTMENT, INSURANCE INTERMEDIATION AND BANKING (PIB)

Zone	Maturity Band		Individual Net Positions		Risk percentages	Weighted Individual Net Positions		By Maturity Band		By Zone		Between Zones	
	Coupon ≥3%	Coupon <3%	Long	Short		Long	Short	Matched	Unmatched	Matched	Unmatched	Matched	Unmatched
A	≤ 1 month	≤ 1 month	\$100	-\$50	0.00%	\$0.00	\$0.00	\$0.00	\$0.00				
	1 - 3 months	1 - 3 months	\$200	-\$100	0.20%	\$0.40	-\$0.20	\$0.20	\$0.20	\$0.00	\$1.30		
	3 - 6 months	3 - 6 months	\$300	-\$200	0.40%	\$1.20	-\$0.80	\$0.80	\$0.40				
	6 - 12 months	6 - 12 months	\$400	-\$300	0.70%	\$2.80	-\$2.10	\$2.10	\$0.70				
B	1 - 2 years	1 - 1.9 years	\$100	-\$200	1.25%	\$1.25	-\$2.50	\$1.25	-\$1.25			Zone 1&2	
	2 - 3 years	1.9 - 2.8 years	\$200	-\$300	1.75%	\$3.50	-\$5.25	\$3.50	-\$1.75	\$0.00	-\$5.25	\$1.30	
	3 - 4 years	2.8 - 3.6 years	\$300	-\$400	2.25%	\$6.75	-\$9.00	\$6.75	-\$2.25				Zones 1&3
													\$0.00
C	4 - 5 years	3.6 - 4.3 years	\$100	-\$100	2.75%	\$2.75	-\$2.75	\$2.75	\$0.00			Zone 2&3	
	5 - 7 years	4.3 - 5.7 years	\$200	-\$200	3.25%	\$6.50	-\$6.50	\$6.50	\$0.00			\$3.95	
	7 - 10 years	5.7 - 7.3 years	\$300	-\$100	3.75%	\$11.25	-\$3.75	\$3.75	\$7.50				
	10 - 15 years	7.3 - 9.3 years	\$100	-\$200	4.50%	\$4.50	-\$9.00	\$4.50	-\$4.50	\$4.50	\$8.25		
	15 - 20 years	9.3 - 10.6 years	\$200	-\$100	5.25%	\$10.50	\$5.25	\$5.25	\$5.25				
	> 20 years	10.6 - 12 years	\$300	-\$300	6.00%	\$18.00	-\$18.00	\$18.00	\$0.00				
		12 - 20 years			8.00%								
		> 20 years			12.50%								
								\$55.35			\$4.30		

Total General Market Risk requirement = 10% (\$55.35) + 40% (\$0.00) + 30% (\$0.00 + \$4.50) + 40% (\$1.30 + \$3.95) + 100% (\$4.30) + 100% (\$0.00) = \$13.29

Duration Method

A5.2.19 An Authorised Firm with the necessary capability may, with the written consent of the DFSA, use the Duration Method, which produces a more accurate measure for General Market Risk than the Maturity Method. An Authorised Firm must elect and use the Duration Method on a continuous basis and will be subject to supervisory monitoring of the systems used.

A5.2.20 Under the Duration Method, the following steps must be carried out:

- (a) the duration weighted position for each instrument must be calculated by multiplying the market value of each individual long or short net position by the Modified Duration in years and the assumed interest rate change in the table below;
- (b) the sum of the weighted long and the sum of the weighted short positions in each time band must be calculated;
- (c) these weighted long and short positions must be matched within a maturity band to give the total matched weighted position in the maturity band and the total unmatched weighted position which will be long or short in the maturity band;
- (d) the matched weighted positions in all maturity bands must be summed;
- (e) the unmatched weighted positions in all the maturity bands must then be matched within a zone leaving an unmatched position for the zone (which will either be short or long); and
- (f) the unmatched positions in each zone must be matched with the unmatched positions in other zones leaving the residual unmatched weighted position.

Zone	Modified Duration	Assumed move in interest rates (percentage points)
A	$0 \leq 1$ month	1.00
	$> 1 \leq 3$ months	1.00
	$> 3 \leq 6$ months	1.00
	$> 6 \leq 12$ months	1.00
B	$> 1.0 \leq 1.9$ years	0.90
	$> 1.9 \leq 2.8$ years	0.80
	$> 2.8 \leq 3.6$ years	0.75
C	$> 3.6 \leq 4.3$ years	0.75
	$> 4.3 \leq 5.7$ years	0.70
	$> 5.7 \leq 7.3$ years	0.65
	$> 7.3 \leq 9.3$ years	0.60
	$> 9.3 \leq 10.6$ years	0.60
	$> 10.6 \leq 12.0$ years	0.60
	$> 12.0 \leq 20.0$ years	0.60
	> 20 years	0.60

A5.2.21 For the purposes of this section Modified Duration is calculated as follows:

$$\text{Modified Duration} = \frac{\text{duration (D)}}{(1 + r)}$$

$$D = \frac{\sum_{t=1}^m \frac{tC_t}{(1+r)^t}}{\sum_{t=1}^m \frac{C_t}{(1+r)^t}}$$

where:

r = yield to maturity

C_t = cash payment in time t

m = total maturity

A5.2.22 The General Market Risk requirement for each currency must be calculated as the sum of the following:

- (a) 5% of the matched weighted positions in each time band;
- (b) 40% of the matched weighted position in zone A;
- (c) 30% of the matched weighted position in zones B and C;
- (d) 40% of the matched weighted position between zones A and B, and between zones B and C;
- (e) 100% of the matched weighted position between zones A and C; and
- (f) 100% of the residual unmatched weighted positions.

Guidance

A worked example of the General Market Risk requirement calculation under the Duration Method is as follows:

PRUDENTIAL – INVESTMENT, INSURANCE INTERMEDIATION AND BANKING (PIB)

Zone	Modified Duration	Individual Net Positions		Assumed move in	Modified Duration	Weighted Individual Net Positions		By Timeband		By Zone		Between Zones	
	(years)	Long	Short	(%p.a)	(years)	Long	Short	Matched	Unmatched	Matched	Unmatched	Matched	Matched
A	< 1 month	\$100.00	-\$50.00	1.00%	0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$1.30		
	1 to 3 months	\$200.00	-\$100.00	1.00%	0.20	\$0.40	-\$0.20	\$0.20	\$0.20				
	3 to 6 months	\$300.00	-\$200.00	1.00%	0.40	\$1.20	-\$0.80	\$0.80	\$0.40				
	6 to 12 months	\$400.00	-\$300.00	1.00%	0.70	\$2.80	-\$2.10	\$2.10	\$0.70				
B	1 to 1.9 years	\$100.00	-\$200.00	0.90%	1.40	\$1.26	-\$2.52	\$1.26	-\$1.26	\$0.00	-\$5.27	Zone 1&2	
	1.9 to 2.8 years	\$200.00	-\$300.00	0.80%	2.20	\$3.52	-\$5.28	\$3.52	-\$1.76			\$1.30	
	2.8 to 3.6 years	\$300.00	-\$400.00	0.75%	3.00	\$6.75	-\$9.00	\$6.75	-\$2.25				Zones 1&3
													\$0.00
C	3.6 to 4.3 years	\$100.00	-\$100.00	0.75%	3.65	\$2.74	-\$2.74	\$2.74	\$0.00	\$4.50	\$8.89	Zone 2&3	
	4.3 to 5.7 years	\$200.00	-\$200.00	0.70%	4.65	\$6.51	-\$6.51	\$6.51	\$0.00			\$3.97	
	5.7 to 7.3 years	\$300.00	-\$100.00	0.65%	5.80	\$11.31	-\$3.77	\$3.77	\$7.54				
	7.3 to 9.3 years	\$100.00	-\$200.00	0.60%	7.50	\$4.50	-\$9.00	\$4.50	-\$4.50				
	9.3 to 10.6 years	\$200.00	-\$100	0.60%	9.75	\$11.70	-\$5.85	\$5.85	\$5.85				
	10.6 to 12 years	\$0.00	\$0.00	0.60%	11.00	\$0.00	\$0.00	\$0.00	\$0.00				
	12 to 20 years	\$300.00	-\$300.00	0.60%	14.50	\$26.10	-\$26.10	\$26.10	\$0.00				
	Over 20 years	\$0.00	\$0.00	0.60%	22.00	\$0.00	\$0.00	\$0.00	\$0.00				
								\$64.10			\$4.92		

Total General Market Risk requirement =

$$5\% (\$64.10) + 40\% (\$0) + 30\% (\$4.50) + 40\% (\$5.27) + 100\% (\$0) + 100\% (\$4.92) = \$11.58$$

A5.3 Equity Risk Capital Requirement

Guidance

Section A5.3 presents the method for the calculation of Equity Risk Capital Requirement for the purpose of Rule 5.5.1(b).

A5.3.1 An Authorised Firm which calculates its Equity Risk Capital Requirement in accordance with Rule 5.5.1(b) must apply the Rules in this section.

A5.3.2 An Authorised Firm must calculate its Equity Risk Capital Requirement by:

- (a) identifying all applicable positions within the scope of the requirement, including notional positions derived from certain instruments;
- (b) Netting positions where they meet the conditions for Netting set in Rule A5.3.19;
- (c) calculating an Equity Risk Capital Requirement for each individual position using the standard method in accordance with Rule A5.3.23 or the simplified method in accordance with Rule A5.3.31;
- (d) in the case of a forward, future, option or company issued warrant on an equity, basket of equities or equity index, adding an Interest Rate Risk Capital Requirement; and
- (e) summing the Capital Requirements calculated in accordance with (c) and (d).

Guidance

For the purposes of Rule A5.3.2(d), an Authorised Firm is required to calculate the applicable Interest Rate Risk Capital Requirement in accordance with Rule A5.2.13 and the other applicable Rules in section A5.2.

- A5.3.3** (1) For the purposes of Rule A5.3.2(a) an Authorised Firm must calculate an Equity Risk Capital Requirement for long and short Trading Book positions in equities and instruments which exhibit behaviour similar to equities including but not limited to:
- (a) depository receipts;
 - (b) futures or forwards on an equity, baskets of equities or equity indices;
 - (c) net underwriting commitments; and
 - (d) investments in unleveraged Collective Investment Funds.
- (2) An Authorised Firm should calculate either an Equity Risk Capital Requirement or an Option Risk Capital Requirement for a Trading Book position in:
- (a) an equity hedging an option;
 - (b) an equity hedging a company-issued warrant;

- (c) an option on an equity, basket of equities, equity index or equity future provided it is in the money by at least the risk percentage stipulated in A5.3.31; and
- (d) a company issued warrant which relates to an equity, basket of equities or equity index provided it is in the money by at least the risk percentage stipulated in A5.3.31.

Guidance

1. If an Authorised Firm has an investment in a leveraged Collective Investment Fund, it should seek guidance from the DFSA in respect of the appropriate prudential treatment.
2. In respect of options that are out of the money, an Authorised Firm must apply the requirements of section 5.8.

A5.3.4 An Authorised Firm must calculate either an Equity Risk Capital Requirement or an Option Risk Capital Requirement for a Trading Book position in the equity leg of an equity swap in accordance with Rule A5.3.12.

A5.3.5 An Authorised Firm must calculate either an Equity Risk Capital Requirement or Interest Rate Risk Capital Requirement for a Trading Book position in a Convertible in accordance with Rules A5.3.6 and A5.3.7.

A5.3.6 An Authorised Firm must treat a Convertible as the underlying equity into which it converts, where:

- (a) the first date at which conversion can take place is less than three months ahead, or the next such date (where the first has passed) is less than a year ahead; and
- (b) the Convertible is trading at a premium of less than 10% to the underlying equity.

A5.3.7 An Authorised Firm which treats a Convertible as an equity must make an adjustment to the capital component as follows:

- (a) an addition equal to any loss on conversion; or
- (b) a deduction equal to any profit on conversion (subject to a maximum reduction to zero).

A5.3.8 An Authorised Firm must not calculate an Equity Risk Capital Requirement for a Trading Book position in:

- (a) material holdings deducted under chapter 3 for the purposes of calculating an Authorised Firm's Capital Resources;
- (b) the interest rate leg of an equity swap, equity future or forward, or equity based option; or
- (c) a non-Convertible preference security.

Derivation of notional positions

A5.3.9 An Authorised Firm must, before Netting, derive a notional position for a depository receipt, a swap, a future, a forward, an option and a company issued warrant in the calculation of its Equity Risk Capital Requirement.

Depository receipts

A5.3.10 An Authorised Firm must treat a depository receipt as a notional position in the underlying equity.

A5.3.11 A position in a depository receipt must only be netted against a position in the underlying equity if the equity is deliverable against the depository receipt.

Equity swaps

A5.3.12 An Authorised Firm must treat an equity swap as two notional positions: an interest rate leg and an equity leg, as follows:

- (a) the interest rate leg must be included in the Interest Rate Risk Calculation and treated as a notional government Security in accordance with the provisions for interest rate swaps in section 5.4; and
- (b) the equity leg must be treated as a long or short position in:
 - (i) where the payout or receipt of funds is based on, respectively, the appreciation or depreciation in price of the underlying equities, a future; or
 - (ii) where the payout is the appreciation in price of the underlying equities, an option, in which case the Authorised Firm must calculate an Option Risk Capital Requirement in accordance with section 5.8.

Equity futures and forwards

A5.3.13 An Authorised Firm must treat a future or forward on a single equity as a notional position in the underlying equity. In addition, an interest rate leg must be included in the interest rate risk calculation in section 5.4 as a notional government security.

A5.3.14 An Authorised Firm must treat a future or forward on a single country equity index as either:

- (a) notional positions in the constituent equities; or
- (b) a single notional position.

A5.3.15 Where Rule A5.3.14(b) applies, an Authorised Firm must apply the highest risk percentage to the single notional position that would apply to any one of its constituents.

A5.3.16 An Authorised Firm must treat a future or forward on a multiple country equity index as either:

- (a) notional positions in the constituent equities; or

- (b) a number of notional positions being one for each of the countries which is represented in the index, in the proportion of that country's representation in the index.

A5.3.17 Where Rule A5.3.16(b) applies, an Authorised Firm must apply the highest risk percentage to each notional position that would apply to any one of its constituents.

Equity options and company issued warrants

A5.3.18 An Authorised Firm must treat an option or company issued warrant on an equity, basket of equities or equity index that is eligible to be included in the equity method as a notional position in the underlying equity or equities as follows:

- (a) a purchased call option and a written put option must be treated as a long position; and
- (b) a purchased put option and a written call option must be treated as a short position.

Netting

Guidance

1. Before calculating the Equity Risk Capital Requirement, positions may be netted in order to produce the individual net position.
2. Since the Netting of positions for Equity Risk Capital Requirement purposes does not involve legal or contractual issues, this material appears here rather than in the Netting section of the Credit Risk chapter.

A5.3.19 (1) An Authorised Firm may only net equity positions when:

- (a) long and short (including notional) positions are in the same tranche of the same equity; and
 - (b) long and short (including notional) positions are in different tranches of the same equity where the tranches enjoy the same rights in all respects and become fungible within one hundred and eighty days, and thereafter the equity of one tranche can be delivered in settlement of the equity of the other tranche.
- (2) For the purposes of (1)(a), an equity is the same as another, only if they enjoy the same rights in all respects and are fungible with each other.

Calculation of the Equity Risk Capital Requirement

Guidance

There are two methods for calculating the Equity Risk Capital Requirement: the standard method and the simplified method. The standard method requires two separate calculations. The first is Specific Risk and the second is General Market Risk. The simplified method is easier to calculate but usually results in a higher Capital Requirement than the standard method. In addition, Authorised Firms must calculate an Interest Rate Risk Capital Requirement for a forward, a future, an option or a company issued warrant.

A5.3.20 (1) An Authorised Firm must allocate an equity position or notional position to the country in which the equity is listed.

- (2) An equity listed in more than one country must be allocated to one of the countries in which it is listed.

A5.3.21 An Authorised Firm must allocate an unlisted equity to the country in which it is issued.

The concentration test

A5.3.22 An Authorised Firm must apply either the standard method or simplified method to an equity position, except that where an individual net position exceeds 20% of the sum of the long and short positions (ignoring the sign) of its country portfolio, the simplified method must be applied to the excess.

Guidance

The part of the individual net position that does not exceed 20% may be treated under the simplified or standard method.

Standard method

A5.3.23 Under the standard method, the total Equity Risk Capital Requirement is the sum of the Specific Risk requirements for all individual net equity positions and the General Market Risk requirements calculated separately for each country.

Specific risk

A5.3.24 Specific Risk must be calculated for each net position in an individual equity.

A5.3.25 The Specific Risk of each individual net equity position is its market value (ignoring the sign) multiplied by 8%.

A5.3.26 [Not currently in use]

A5.3.27 [Not currently in use]

A5.3.28 [Not currently in use]

A5.3.29 An Authorised Firm must calculate General Market Risk on a country-by-country basis.

A5.3.30 An Authorised Firm must calculate the General Market Risk for each country in the following way:

- (a) all individual net positions are multiplied by 8%;
- (b) long and short positions in each country portfolio are netted; and
- (c) if the net equity position is negative, the sign must be reversed.

Simplified method

A5.3.31 The Equity Risk Capital Requirement for each country is the sum of the market value of all individual net positions (ignoring the sign) multiplied by the appropriate risk percentage in the table below:

	Percentage risk
Single equities	16%
Broad-based indices (not broken down into constituent equities)	8%
All other indices (not broken down into constituent equities)	16%

A5.3.32 For the purposes of Rule A5.3.31, a broad-based index means an index specified in the table under (c) or an index that satisfies the following criteria:

- (a) the index contains at least 20 shares;
- (b) the weighting of the largest company is not greater than 20% of the total index; and
- (c) the weighting of the largest five companies is not greater than 60% of the total index.

Australia	All Ordinaries
Austria	Austrian Traded Index
Belgium	BEL 20
Canada	TSE 35, TSE 100, TSE 300
France	CAC 40, SBF 250
Germany	DAX
European	Dow Jones Stoxx 50 Index, FTSE Eurotop 300, MSCI Euro Index
Hong Kong	Hang Seng
Italy	MIB 30
Japan	Nikkei 225, Nikkei 300, TOPIX
Korea	Kospi
Netherlands	AEX
Singapore	Straits Times Index
Spain	IBEX 35
Sweden	OMX
Switzerland	SMI
UK	FTSE 100, FTSE Mid 250, FTSE All Share
US	S&P 500, Dow Jones Industrial Average, NASDAQ Composite, Russell 2000

A5.4 Foreign Exchange Risk Capital Requirement

Guidance

Section A5.4 presents the method for the calculation of Foreign Exchange Risk Capital Requirement for the purpose of Rule 5.6.1(b).

A5.4.1 An Authorised Firm which calculates its Foreign Exchange Risk Capital Requirement in accordance with Rule 5.6.1(b) must apply the Rules in this section.

A5.4.2 An Authorised Firm must calculate its Foreign Exchange Risk Capital Requirement by using the standard method as follows:

- (a) calculating its net open position in each currency and in gold;
- (b) calculating its overall net open position in accordance with Rule A5.4.4; and
- (c) multiplying the overall net open position by the percentage provided in Rule A5.4.5.

Measuring the net open position in a single currency

A5.4.3 An Authorised Firm must calculate its net open position in each currency, and in gold, by summing:

- (a) the net spot position, being all asset items less all liability items, including accrued interest, denominated in the currency in question;
- (b) the net forward position, being all amounts to be received less all amounts to be paid under forward foreign exchange transactions, including currency futures and the principal on currency swaps not included in the spot position;
- (c) guarantees and similar instruments that are certain to be called and are likely to be irrecoverable;
- (d) net future income/expenses not yet accrued but already fully hedged, at the discretion of the Authorised Firm; and
- (e) any other item representing a profit or loss in Foreign Currencies.

Measuring the overall net open position

- A5.4.4** (1) An Authorised firm must convert the net position in each Foreign Currency and in gold at spot rates into the reporting currency.
- (2) The overall net open position is measured by aggregating:
- (a) the sum of the net short positions or the sum of the net long positions, whichever is the greater; plus
 - (b) the net position (short or long) in gold, regardless of sign.

A5.4.5 The Foreign Exchange Risk Capital Charge is 8% of the overall net open position.

Guidance

1. An example of how to calculate the overall net open position is as follows:

YEN	EURO	GB	Saudi Riyal	\$	Gold
+50	+100	+150	-20	-180	-35
TOTAL +300			TOTAL -200		TOTAL 35

2. The Foreign Exchange Risk Capital Charge would be 8% of the higher of either the net long currency positions or the net short currency positions (i.e. 300) plus the net position in gold (35) = $335 \times 8\% = 26.8$.
3. Forward currency and gold positions will normally be valued at current spot market exchange rates. Using forward exchange rates would be inappropriate since it would result in the measured positions reflecting to some extent current interest rate differentials. However, an Authorised Firm which bases its normal management accounting on net present values is expected to use the net present values of each position, discounted using current interest rates and valued at current spot rates, for measuring its forward currency and gold positions.

Treatment of accrued interest and expenses, forwards, and structural positions

- A5.4.6** (1) An Authorised Firm must include interest accrued and accrued expenses as a position.
- (2) If an Authorised Firm includes future income/expenses it must do so on a consistent basis and not include only those expected future flows that reduce its position.
- (3) An Authorised Firm may exclude any positions which it has deliberately taken in order to hedge partially or totally against the adverse effect of the exchange rate on its Capital Resources, from the calculation of net open currency positions, if each of the following conditions is met:
- (a) the positions are of a structure that is of a non-dealing nature;
 - (b) the Authorised Firm has notified the DFSA in writing of its intention to rely upon this Rule; and
 - (c) any exclusion of the position must be applied consistently, with the treatment of the hedge remaining the same for the life of the assets or other items.
- (4) An Authorised Firm need not include positions related to:
- (a) items which are deducted from its capital when calculating its Capital Resources, including investments in non-consolidated subsidiaries; or
 - (b) other long-term participations denominated in Foreign Currencies which are reported in the published accounts at historic cost.

A5.5 Commodities Risk Capital Requirement

Guidance

Section A5.5 presents the method for the calculation of Commodities Risk Capital Requirement for the purpose of Rule 5.7.1(b).

- A5.5.1** An Authorised Firm which calculates its Commodities Risk Capital Requirements in accordance with Rule 5.7.1(b) must apply the Rules in this section.

Calculation of Commodities Risk Capital Requirement

- A5.5.2** (1) An Authorised Firm must calculate its Commodities Risk Capital Requirement by applying the Maturity Ladder approach in Rule A5.5.5 or the Simplified Approach in Rule A5.5.6 to all Non-Trading and Trading Book:
- (a) commodity positions;
 - (b) commodity derivatives and off-balance sheet positions that are affected by changes in commodity prices, having derived notional commodity positions; and
 - (c) other positions against which no other Market or Credit Risk Capital Requirement has been applied.
- (2) An Authorised Firm must determine notional commodity positions by converting the commodity derivatives into notional underlying commodity positions and assigning appropriate maturities in accordance with Rule A5.5.3.

Treatment of Commodity derivatives

A5.5.3 An Authorised Firm must:

- (a) incorporate all futures and forward contracts relating to individual commodities in the measurement system as notional amounts and assigned a maturity with reference to the expiry date;
- (b) incorporate commodity swaps where one leg is a fixed price and the other the current market price as a series of positions equal to the notional amount of the contract, with one position corresponding to each payment on the swap and slotted into the Maturity Ladder accordingly. The positions will be long positions if the Authorised Firm is paying fixed and receiving floating, and short positions if the Authorised Firm is receiving fixed and paying floating; and
- (c) incorporate commodity swaps where the legs are in different commodities in the relevant Maturity Ladder. No offsetting will be allowed in this regard except where the commodities belong to the same sub-category.

- A5.5.4** (1) Subject to (2), an Authorised Firm must not net positions in different commodities for the purpose of calculating open positions.
- (2) An Authorised Firm may net positions in different commodities where those commodities:
- (a) are deliverable against each other; and
 - (b) are in one or more sub-categories of the same category;

Guidance

1. For the purposes of Rule A5.5.4, an example of a category is oil. An example of a sub-category is Brent.

2. For the Simplified Approach and the Maturity Ladder approach, long and short positions in each commodity may be reported on a net basis for the purposes of calculating open positions.

Maturity ladder approach

- A5.5.5** (1) An Authorised Firm which uses the Maturity Ladder approach to calculate the Commodities Risk Capital Requirement must:

- (a) express each commodity position (spot and forward) in terms of the standard unit of measurement and net long and short positions maturing on the same day or maturing within ten business days of each other in the case of contracts traded in markets with daily delivery dates;
- (b) allocate the positions remaining after taking the steps in (a) to the appropriate maturity band in the following table:

Band	Maturity of Position
1.	0-1 month
2.	1 – 3 months
3.	3 – 6 months
4.	6 – 12 months
5.	1 – 2 years
6.	2 – 3 years
7.	Over 3 years

- (c) calculate the spread charge each time long and short positions are matched within each band. In each instance, the spread charge equals the matched amount multiplied first by the spot price for the commodity and then by a spread rate of 1.5%;
 - (d) calculate a carry charge for each position that is carried across to another maturity band. In each instance, the carry charge equals the carried position multiplied first by the spot price for the commodity, then by the carry rate of 0.6% and finally by the number of bands by which the position is carried;
 - (e) repeat (c) if necessary;
 - (f) calculate the outright charge by multiplying all remaining unmatched positions (long plus short, ignoring the sign) by the spot price for the commodity, then by 15%; and
 - (g) sum the totals for Bands 2 to 7 referred to in (b), to reach the total requirement.
- (2) For the purposes of (1)(b), an Authorised Firm must:
 - (a) allocate physical stocks to the first maturity band; and
 - (b) set a separate Maturity Ladder for each commodity.

Guidance

The table below illustrates the calculation of the Commodity Risk Capital Requirement on an individual commodity using the Maturity Ladder approach.

After a firm has carried out the pre-processing required by A5.5.5(1)(a), it follows the steps required by Rule A5.5.5(1)(a) to (g). The spread rate is 1.5%, the carry rate is 0.6% and the outright rate is 15%. The example assumes that the spot price for the commodity is \$20					
Band	A5.5.5(1)(b) Allocate remaining positions to appropriate maturity bands	A5.5.5(1)(c) Match within bands. Each matched amount incurs a spread charge	A5.5.5(1)(d) Carry across bands. Each carried amount incurs a carry charge	A5.5.5(1)(e) Match within band. Each matched amount incurs a spread charge	A5.5.5(1)(f) Remaining unmatched position(s) incur an outright charge
0 ≤ 1 month					
>1 month ≤ 3 months	1100 long 800 short	800 matched	Remaining 300 long carried ↓		
>3 months ≤ 6 months					
>6 months ≤ 1 year					
>1 year ≤ 2 years	400 short	Nothing matched		400 matched	100 long remains unmatched
>2 years ≤ 3 years			200 long carried ↑		
> 3 years	200 long	Nothing matched			
Spread charges	800x\$20x1.5%	+	400x \$20x1.5%	=	\$3
Carry charges	300x\$20x0.6%x3	+	200x\$20x0.6%x2	=	\$156
Outright charge	100x\$20x15%			=	<u>\$300</u>
Total	\$816				

Simplified Approach

A5.5.6 An Authorised Firm using the Simplified Approach to calculate the Commodities Risk Capital Requirement must sum:

- (a) 15% of the net position multiplied by the spot price for the commodity; and
- (b) 3% of the gross position (long plus short, ignoring the sign) multiplied by the spot price of the commodity.

A5.6 Option Risk Capital Requirement

Guidance

Section A5.6 presents the method for the calculation of Option Risk Capital Requirement for the purpose of Rule 5.8.1(b).

A5.6.1 An Authorised Firm which calculates its Option Risk Capital Requirement in accordance with Rule 5.8.1(b) must apply the Rules in this section.

Calculation of Option Risk Capital Requirement

- A5.6.2** (1) An Authorised Firm may use the Simplified Approach set out in Rule A5.6.3 to calculate its Option Risk Capital Requirement only if:
- (a) it does not write options; or
 - (b) where it writes options, all written options are hedged by perfectly matched long positions in exactly the same options.
- (2) An Authorised Firm which writes options must, unless (1) applies, use the advanced approach known as the Delta-plus method set out in Rule A5.6.5 to calculate its Option Risk Capital Requirement.

Simplified Approach

A5.6.3 An Authorised Firm using the Simplified Approach must treat the positions for the options and the associated underlying instrument, cash or forward, and calculate the capital charge for each position, by reference to the following table:

Position	Treatment
Long cash and long put or short cash and long call.	The capital charge is the market value of the underlying instrument multiplied by the sum of Specific and General Market Risk percentages for the underlying Instrument less the amount the option is in the money, if any, bounded at zero.
Long call or long put.	The capital charge will be the lesser of: • the market value of the underlying instrument multiplied by the sum of Specific and General Market Risk percentages for the underlying instrument; or • the market value of the options.

Guidance

As an example of how the calculation would work, if a holder of 100 shares currently valued at \$10 each holds an equivalent put option with a strike price of \$11, the capital charge would be: $\$1,000 \times 16\%$ (i.e., 8% specific plus 8% General Market Risk) = \$160, less the amount the option is in the money $(\$11 - \$10) \times 100 = \$100$, i.e., the capital charge would be \$60. A similar methodology applies for options whose underlying instrument is a Foreign Currency, an interest rate related instrument or a commodity.

- A5.6.4** (1) For the purposes of Rule A5.6.3, the Specific Risk percentage for:
- (a) a currency option is 8%; and
 - (b) an option on commodities is 15%.
- (2) For the purposes of Rule A5.6.3, in the case of an option with a residual maturity of more than six months, the strike price must be compared with the forward, not current price, or if the Authorised Firm is unable to do this, then the money amount must be taken to be zero.

Delta-plus method

Guidance

The Delta-plus method uses the sensitivity parameters or “Greek letters” associated with options to measure their Option Risk Capital Requirement. Under this method, the Delta-equivalent position of each option becomes part of the standardised methodology set out in sections 5.4 to 5.7 with the Delta-equivalent amount subject to the applicable General Market Risk requirements. Separate capital charges are then applied to the Gamma and Vega risks of the option positions.

- A5.6.5** (1) An Authorised Firm that writes or purchases options may include Delta-weighted options positions within the standardised methodology set out in sections A5.2 to A5.5. Such options must be reported as a position equal to the market value of the underlying instrument multiplied by the Delta.
- (2) An Authorised Firm is also required to measure and Vega risks in order to calculate the total capital charge. These sensitivities will be calculated according to an approved proprietary options pricing model.
- (3) Delta-weighted positions with debt Securities or interest rates as the underlying instrument must be inserted into the interest rate timebands, as set out in section A5.2. A two-legged approach must be used as for other derivatives, requiring one entry at the time the underlying instrument takes effect and a second at the time the underlying instrument matures. Floating rate instruments with caps or floors must be treated as a combination of floating rate Securities and a series of European-style options.
- (4) The capital charge for options with equities as the underlying instrument must also be based on the Delta-weighted positions which must be incorporated in the measure of Equity Risk Capital Requirement described in section A5.3. For purposes of this calculation, each national market must be treated as a separate underlying instrument.
- (5) The capital charge for options on commodities, foreign currency (including gold) positions must be based on the method set out in section 5.8. For Delta risk, the net Delta-based equivalent of the commodities, foreign currency

including gold) options must be incorporated into the measurement of the Exposure for the respective currency (or gold) position.

- (6) Individual net Delta positions as described above must be treated as the underlying instrument in accordance with sections A5.4 to A5.5.

A5.6.6 An Authorised Firm using the Delta-plus method must calculate its Market Risk Capital Requirement for options by:

- (a) calculating the Delta-weighted position of each option in accordance with Rule A5.6.7 and adding these Delta-weighted positions to the net positions in the relevant risk category referred in sections A5.2 to A5.6 for the purpose of calculating the Specific Risk and General Market Risk Capital Requirements;
- (b) calculating the Capital Requirement for Gamma risk of its option positions (including hedge positions) based on the options pricing model of the an Authorised Firm, in accordance with Rules A5.6.8 to A5.6.9;
- (c) calculating the Capital Requirement for Vega risk of its option positions (including hedge positions) based on the options pricing model of an Authorised Firm, in accordance with Rule A5.6.10; and
- (d) summing the Capital Requirements determined in (b) and (c).

A5.6.7 An Authorised Firm must calculate its Delta-weighted position for each option as follows:

Delta-weighted position = Market value of the underlying instrument or commodities x Delta

A5.6.8 In addition to the capital charges referred to in Rule A5.6.5, arising from Delta risk, an Authorised Firm must calculate the Gamma for each option position, including hedge positions in the following way:

- (a) for each individual option a “Gamma impact” must be calculated as:

$$\text{Gamma impact} = \frac{1}{2} \times \text{Gamma} \times \text{VU}^2$$

where VU = Variation of the underlying instrument of the option;

- (b) VU must be calculated as follows:
 - (i) for interest rate options if the underlying instrument is a bond, the market value of the underlying instrument should be multiplied by the risk weights set out in section 5.4 for the underlying instrument. An equivalent calculation should be carried out where the underlying instrument is an interest rate, again based on the assumed changes in the corresponding yield in Rule A5.2.16;
 - (ii) for options on equities and equity indices, the market value of the underlying instrument should be multiplied by 8%;
 - (iii) for foreign exchange and gold options, the market value of the underlying instrument should be multiplied by 8%; and

- (iv) for options on commodities, the market value of the underlying instrument should be multiplied by 15%; and
- (c) for the purpose of this calculation the following positions must be treated as the same underlying instrument:
 - (i) for interest rates, each timeband as set out in Rule A5.2.16;
 - (ii) for equities and stock indices, each national market;
 - (iii) for foreign currencies and gold, each currency pair and gold; and
 - (iv) for commodities, positions in the same individual commodity as defined in section A5.5 for Commodities Risk Capital Requirement.

A5.6.9 An Authorised Firm must calculate its Capital Requirement for Gamma risk by:

- (a) calculating the net Gamma impact in respect of each underlying financial instrument or commodity by aggregating the individual Gamma impacts for each option position in respect of that underlying financial instrument or commodity (which may be either positive or negative); and
- (b) aggregating the absolute value of the net Gamma impacts that are negative.

Guidance

1. The underlying financial instrument or commodity should be taken to be the asset which would be received if the option were exercised. In addition, the notional value should be used for items where the market value of the underlying financial instrument or commodity could be zero (e.g. caps and floors, swaptions). Certain notional positions in zero-specific-risk securities do not attract Specific Risk, e.g. interest rate and currency swaps, Forward Rate Agreement (FRA), forward foreign exchange contracts, interest rate futures and futures on an interest rate index. Similarly, options on such zero-specific-risk securities also bear no Specific Risk. For the purposes of this sub-paragraph:
 - a. the specific and general risk weights in respect of options on interest rate-related instruments are determined in accordance with section A5.2;
 - b. the specific and general risk weights in respect of options on equities and equity indices are determined in accordance with section A5.3;
 - c. the risk weight in respect of foreign currency and gold options is 8%; and
 - d. the risk weight in respect of options on commodities is 15%.

For options with a residual maturity of more than 6 months, the strike price should be compared with the forward, and not current, price. Where an Authorised Firm is unable to do this, the in-the-money amount would be zero.
2. An Authorised Firm which trades in exotic options (e.g. barriers, digitals) would use either the scenario approach or the Internal Models Approach (IMA) to calculate its Market Risk Capital Requirement for such options, unless it is able to demonstrate to the DFSA that the Delta-plus method is appropriate. In the case of options on futures or forwards, the relevant underlying is that on which the future or forward is based (e.g. for a bought call option on a June 3-month bill future, the relevant underlying is the 3-month bill).

A5.6.10 An Authorised Firms must calculate its Capital Requirement for Vega risk by:

- (a) multiplying the sum of the Vegas for all option positions in respect of the same underlying financial instrument or commodity, as defined in the Rule 5.6.8(c), by a proportional shift in volatility of $\pm 25\%$; and
- (b) aggregating the absolute value of the individual Capital Requirements which have been calculated for Vega risk.

A5.7 Collective Investment Fund Risk Capital Requirement

Guidance

This section presents the method for the calculation of Collective Investment Fund Risk Capital Requirement for the purpose of Rule 5.9.1(b).

A5.7.1 An Authorised Firm which calculates its Collective Investment Fund Risk Capital Requirement in accordance with Rule 5.9.1(b) must apply the Rules in this section.

A5.7.2 An Authorised Firm must calculate its Collective Investment Fund Risk Capital Requirement by:

- (a) identifying all applicable positions in Funds within the scope of the requirement, including notional positions derived from certain instruments;
- (b) identifying the positions in Funds which will be subject to the risk Capital Requirements specified under this section;
- (c) converting on a daily basis net positions in every Fund to the Authorised Firm's base currency at the prevailing spot foreign exchange rate;
- (d) calculating a Collective Investment Fund Risk Capital Requirement for each individual position in a Fund; and
- (e) summing the resultant Capital Requirements calculated in (d).

A5.7.3 (1) For the purposes of Rule A5.7.2, an Authorised Firm must calculate its Collective Investment Fund Risk Capital Requirement for all Trading Book positions in Funds, unless they are covered under one of the Fund look through methods and included in the risk Capital Requirement calculations for the relevant underlying Investments or subject to an Option Risk Capital Requirement.

- (2) An Authorised Firm must also calculate its Collective Investment Fund Risk Capital Requirement for notional positions arising from Trading Book positions in options or warrants on Funds.

Calculation of the Collective Investment Fund Risk Capital Requirement

Guidance

There are two main approaches for calculating the Collective Investment Fund Risk Capital Requirement. The first approach involves directly calculating a risk Capital Requirement for any position in any Fund. The second approach involves using a look-through method which involves calculating the risk Capital Requirements for the positions or Exposures in underlying assets or investments of the Fund, using the relevant or applicable risk Capital Requirement calculation methods. As the name suggests, a look-through method involves looking through the Fund to identify the

underlying positions and trying to calculate the capital required to address the risk of loss arising from volatility in market prices of such underlying positions.

- A5.7.4** Without prejudice to other provisions in this section, a position in a Fund is subject to a Collective Investment Fund risk capital charge (General Market Risk and Specific Risk) of 32%, subject to Rules A5.7.5 and A5.7.6.

Look through methods

- A5.7.5** An Authorised Firm may determine the Collective Investment Fund Risk Capital Requirement for positions in Funds, using the standard Collective Investment Fund look-through method, provided the relevant positions meet the criteria set out in Rule A5.7.6.

- A5.7.6** An Authorised Firm may use the standard Fund look-through method, only if the positions are in Funds which meet the following eligibility criteria:

- (a) the Fund's prospectus or equivalent document must include:
 - (i) the categories of assets the Fund is authorised to invest in;
 - (ii) if investment limits apply, the relative limits and the methodologies to calculate them;
 - (iii) if leverage is allowed, the maximum level of leverage; and
 - (iv) if investment in OTC financial derivatives or repo-style transactions are allowed, a policy to limit Counterparty Risk arising from these transactions;
- (b) the Fund must publish half-yearly accounts and annual reports to enable an assessment to be made of the assets and liabilities, income and operations over the reporting period;
- (c) the Units of the Fund are redeemable in cash, out of the Fund's assets, on a daily basis at the request of the Unitholder;
- (d) investments in the Fund must be segregated from the assets of the Fund Manager; and
- (e) there must be adequate risk assessment, by the investing firm, of the Fund.

- A5.7.7** [Not currently in use]

Standard collective investment fund look through method: General

- A5.7.8** In the case of an Authorised Firm being aware of the underlying assets or investments of the Fund on a daily basis, the Authorised Firm may look through to those underlying investments in order to calculate the Market Risk Capital Requirement (General Market Risk and Specific Risk) for those positions in accordance with the methods set out in the relevant section of chapter 5 for calculating the relevant Market Risk Capital Requirement.

- A5.7.9** In this method, positions in Funds must be treated as positions in the underlying investments of the Fund. Netting is permitted between positions in the underlying investments of the Fund and other positions held by the Authorised Firm, as long as

the Authorised Firm holds a sufficient quantity of Units to allow for redemption/creation in exchange for the underlying investments.

Standard collective investment fund look through method: Index or Basket Funds

- A5.7.10** (1) An Authorised Firm may calculate the risk Capital Requirements for positions in Funds in accordance with the methods set out in various sections of chapter 5 applicable to various underlying assets or investments, on assumed positions representing those necessary to replicate the composition and performance of the externally generated index or fixed basket of equities or debt securities, subject to the following conditions:
- (a) the Fund's mandate is to replicate the composition and performance of an externally generated index or fixed basket of equities or debt securities; and
 - (b) a minimum correlation of 0.9 between daily price movements of the Fund and the index or basket of equities or debt securities it tracks, exists over the previous six month period.
- (2) Correlation as referred to in (1)(b) means the correlation coefficient between daily returns on the Fund and that on the index or basket of equities or debt securities it tracks.

A5.8 Securities Underwriting Risk Capital Requirement

Guidance

Section A5.8 presents the method for calculating a net Underwriting position or reduced net Underwriting position, which is then included in the calculation of Market Risk Capital Requirements as specified in this chapter. Section A5.8 also deals with Concentration Risk.

- A5.8.1** (1) An Authorised Firm which calculates its Securities Underwriting Risk Capital Requirement in accordance with Rule 5.10.7(b) must apply the Rules in this section.
- (2) An Authorised Firm which underwrites or sub-underwrites an issue of Securities must, for the purposes of calculating its Market Risk Capital Requirement:
- (a) identify commitments to underwrite or sub-underwrite which give rise to an underwriting position;
 - (b) identify the time of initial commitment; and
 - (c) calculate the net Underwriting position, reduced net Underwriting position or the net Underwriting Exposure.
- A5.8.2** An Authorised Firm must include the net Underwriting position or reduced net Underwriting position where:
- (a) debt Securities are being underwritten;

- (b) equities are being underwritten; or
- (c) Warrants are being underwritten.

Guidance

1. Sub-underwriting is a commitment given by one firm to someone other than the Issuer or seller of the Securities to sub-underwrite all or part of an issue of Securities. The net Underwriting position calculated in Rule A5.8.6 will also be used in calculating the net Underwriting Exposure under Rules A5.8.11 to A5.8.14.
2. The net Underwriting position or reduced net Underwriting position arising from Underwriting or sub-underwriting a rights or Warrants issue should be calculated using the current market price of the underlying Security for the purposes of the Equity Risk Capital Requirement or Option Risk Capital Requirement. However, the risk Capital Requirements will be limited to the value of the net Underwriting position calculated using the initial issue price of the rights or Warrants. Where there is no market price because the rights or Warrants are in relation to a new class of Securities and the initial price has not been set the net Underwriting position or reduced net Underwriting is the amount of the commitment.

Commitment to underwrite Securities

- A5.8.3** (1) For the purpose of Rule A5.8.1, an Authorised Firm underwrites or sub-underwrites an issue of Securities where:
- (a) it gives a commitment to an Issuer of Securities to underwrite an issue of Securities;
 - (b) in the case of new Securities (as defined in Rule A5.8.4) it gives a commitment to a seller of Securities to underwrite a sale of those Securities;
 - (c) it gives a commitment to a person, other than the Issuer of Securities or, if (b) applies, the seller of the Securities, to sub-underwrite an issue of Securities; or
 - (d) it is a member of a syndicate or Group that gives a commitment of the type described in (a) to (c).
- (2) Unless a Rule deals with an issue of Securities separately or the context otherwise requires, a provision of section A5.8 that deals with Underwriting also applies to sub-underwriting.

Guidance

Block trades, including bought deals, and private placements are not within the scope of this chapter because they involve an outright purchase by an Authorised Firm of the relevant Securities.

- A5.8.4** For the purposes of Rule A5.8.3(b), an Authorised Firm must treat Securities as being new if they are:
- (a) Securities that, prior to the allotment following the Underwriting, were not in issue; or
 - (b) Securities that have not previously been offered for sale or subscription to the public and have not been admitted to trading on a market operated by an Authorised Market Institution or an overseas investment exchange.

Grey market transactions

- A5.8.5** (1) An Authorised Firm that buys and sells Securities before issue is dealing in the grey market for the purposes of this section.
- (2) The other Rules in section A5.8 do not apply to an Authorised Firm with respect to its dealings in the grey market unless the Authorised Firm:
- (a) has an Underwriting commitment to the Issuer in respect of those Securities; or
 - (b) has a sub-underwriting commitment in respect of those Securities and is using the grey market solely for the purpose of reducing that sub-underwriting commitment.
- (3) The other Rules in section A5.8 do not apply to an Authorised Firm with respect to its dealings in the grey market if the transaction is undertaken by the proprietary trading part of the Authorised Firm or is undertaken for proprietary trading purposes.

Guidance

In Rule A5.8.5, the grey market is the market in which dealers "buy" and "sell" securities ahead of issue. In reality the dealers are buying and selling promises to deliver the securities when issued.

Calculating the net underwriting position

- A5.8.6** An Authorised Firm must calculate a net Underwriting position from the date of initial commitment until the Underwriting process ends, as the initial gross commitment adjusted for:
- (a) Underwriting or sub-underwriting commitments obtained from others since the time of initial commitment;
 - (b) purchases or sales of the Securities since the time of initial commitment;
 - (c) any allocation of Securities granted or received, arising from the commitment to underwrite the Securities, since the time of initial commitment; and
 - (d) in the case of sales in the grey market, as defined in Rule A5.8.5, any sales of the Securities as at the time of initial commitment or since the time of initial commitment is subject, in both cases, to the following conditions:
 - (i) any sales of the Securities as at the time of initial commitment must be confirmed in writing at the time of initial commitment;
 - (ii) sales must be net of any purchases in the grey market; and
 - (iii) any allocation of Securities granted or received, arising from the commitment to underwrite the Securities, since the time of initial commitment.
- A5.8.7** If the allocation of Securities has not been fixed, an Authorised Firm must calculate the gross amount of its commitment, for the purpose of Rule A5.8.6, by reference to the maximum amount it has committed to underwrite until the time the allocation is

set. An Underwriting commitment may only be reduced under this Rule on the basis of a formal agreement.

Guidance

Allocations may arise, after date of initial commitment, from the agreement to underwrite. For example, obligations or rights may be allocated to or from the Issuer, the underwriting Group or syndicate.

Time of initial commitment

- A5.8.8** (1) Subject to (2), the time of initial commitment is the earlier of:
- (a) in the case of Underwriting, the time the Authorised Firm agrees with the Issuer of Securities to underwrite those Securities;
 - (b) in the case of Underwriting falling under Rule A5.8.3(b), the time the Authorised Firm agrees with the seller of Securities to underwrite those Securities;
 - (c) in the case of sub-underwriting, the time the Authorised Firm agrees with the Person referred to Rule A5.8.3(c) to sub-underwrite those Securities;
 - (d) in the case of Rule A5.8.3(d), the time the Group or syndicate in question (or a member of that Group or syndicate on behalf of the others) agrees with the Issuer or other Person to whom the commitment is given as referred to in Rule A5.8.3(d) to underwrite or sub-underwrite the Securities in question; or
 - (e) if the firm at that time has a commitment (whether legally or binding or not), the time the price and allocation of the issue or offer are set.
- (2) If an Authorised Firm has an irrevocable and unfettered right to withdraw from an underwriting commitment, exercisable within a certain period, the commitment commences (and thus the time of initial commitment occurs) when that right expires.

Working day 0

- A5.8.9** For the purposes of section A5.8, working day 0 is the business day on which a firm that is Underwriting or sub-underwriting becomes unconditionally committed to accepting a known quantity of Securities at a specified price.

Guidance

For debt issues and securities which are issued in a similar manner, working day 0 is the later of the date on which the securities are allotted and the date on which payment for them is due. For equity issues and securities which are issued in a similar manner, working day 0 is the later of the date on which the offer becomes closed for subscriptions and the date on which the allocations are made public. For rights issues, working day 0 is the first day after the date on which the offer becomes closed to acceptances for subscription.

Calculating the reduced net underwriting position

- A5.8.10** To calculate the reduced net Underwriting position an Authorised Firm must apply the reduction factors in the table below to the net Underwriting position calculated under Rule A5.8.6 as follows:

- (a) in respect of debt Securities, an Authorised Firm must calculate two reduced net Underwriting positions; one for inclusion in the Authorised Firm's interest rate Specific Risk calculation, the other for inclusion in its interest rate General Market Risk calculation; and
- (b) in respect of equities, an Authorised Firm must calculate only one reduced net Underwriting position, and then include it in the simplified equity method.

Reduction Factors

	Debt Issue		Equity Issue	
Underwriting Timeline	General Market Risk	Specific Risk	General Market Risk	Specific Risk
Date of initial commitment until working day 0	0%	100%	90%	90%
Working day 1	0%	90%	90%	90%
Working day 2	0%	75%	75%	75%
Working day 3	0%	75%	75%	75%
Working day 4	0%	50%	50%	50%
Working day 5	0%	25%	25%	25%
Working day 6 and onwards	0%	0%	0%	0%

Large Exposure risk from underwriting securities: Calculating the net underwriting Exposure

- A5.8.11** For Concentration Risk purposes, the total amount of an Authorised Firm's Trading Book Exposures to any Person must include net Underwriting Exposure to that Person.
- A5.8.12** An Authorised Firm must include any other Exposures arising out of underwriting (including any Counterparty Exposures to any sub-underwriters) for the purposes of calculating the total amount of its Trading Book Exposures to a Person for Concentration Risk purposes.
- A5.8.13** An Authorised Firm, before entering into a new Underwriting commitment, must be able to recalculate the Concentration Risk capital component to the level of detail necessary to ensure that the firm's Capital Resources Requirement does not exceed the firm's Capital Resources.
- A5.8.14** An Authorised Firm must calculate the net Underwriting Exposure to an Issuer by applying the relevant reduction factors in the table below to its net Underwriting position calculated under Rule A5.8.6.

Time	Reduction factor to be applied to net underwriting position
Initial commitment to working day 0	100%
Working day 0	100%
Working day 1	90%
Working day 2	75%
Working day 3	75%
Working day 4	50%
Working day 5	25%
Working day 6 onwards	0%

Guidance

The effect of the Rule and the table above is that there is no concentration limit for net Underwriting Exposures between initial commitment and the end of working day 0.

Risk management

- A5.8.15** (1) An Authorised Firm must take reasonable steps to establish and maintain such systems and controls to monitor and manage its Underwriting and sub-underwriting business as are appropriate to the nature, scale and complexity of its Underwriting and sub-underwriting business.
- (2) In particular, an Authorised Firm must have systems to monitor and control its Underwriting Exposures between the time of the initial commitment and working day one in the light of the nature of the risks incurred in the markets in question.

Guidance

An Authorised Firm should take reasonable steps to:

- allocate responsibility for the management of its Underwriting and sub-underwriting business;
- allocate adequate resources to monitor and control its Underwriting and sub-underwriting business;
- satisfy itself that its systems to monitor Exposure to Counterparties will calculate, revise and update its Exposure to each Counterparty arising from its Underwriting or sub-underwriting business;
- satisfy itself of the suitability of each person who performs functions for it in connection with the firm's Underwriting and sub-underwriting business having regard to the person's skill and experience; and
- satisfy itself that its procedures and controls to monitor and manage its Underwriting business address, on an on-going basis, the capacity of sub-underwriters to meet sub-underwriting commitments.

A5.9 Use of internal models for Market Risk

A5.9.1 Criteria for use of internally developed Market Risk models

Guidance

Qualitative standards

1. Any internal model used for purposes of chapter 5 of PIB should be conceptually sound and implemented with integrity and, in particular, all of the following qualitative requirements should be met:
 - a. any internal model used to calculate Capital Requirements for equity risk, interest rate risk, foreign exchange risk or commodities risk should be closely integrated into the daily risk management process set out in (b) and serve as the basis for reporting risk Exposures to senior management;
 - b. the Authorised Firm should have a risk control unit that is independent from business trading units and reports directly to senior management. The unit should be responsible for designing and implementing any internal model used for purposes of chapter 5. The unit should conduct the initial and on-going validation of any internal model used for purposes of chapter 5. The unit should produce and analyse daily reports on the output of any internal model used for calculating Capital Requirements for position risk, foreign exchange risk and commodities risk, and on the appropriate measures to be taken in terms of trading limits;
 - c. the Authorised Firm's management body and senior management should be actively involved in the risk control process and the daily reports produced by the risk control unit are reviewed by a level of management with sufficient authority to enforce both reductions of positions taken by individual traders as well as in the Authorised Firm's overall risk Exposure;
 - d. the Authorised Firm should have sufficient numbers of staff skilled in the use of sophisticated internal models, and including the ones used for purposes of chapter 5, in the trading, risk-control, audit and back-office areas;
 - e. the Authorised Firm should have established procedures for monitoring and ensuring compliance with a documented set of internal policies and controls concerning the overall operation of its internal models, and including the ones used for purposes of chapter 5;
 - f. any internal model used for purposes of chapter 5 should have a proven track record of reasonable accuracy in measuring risks;
 - g. the Authorised Firm should frequently conduct a rigorous programme of stress testing, including reverse stress tests, which encompasses any internal model used for purposes of chapter 5 and the results of these stress tests should be reviewed by senior management and reflected in the policies and limits it sets. This process should particularly address illiquidity of markets in stressed market conditions, Concentration Risk, one way markets, event and jump-to-default risks, non-linearity of products, deep out-of-the-money positions, positions subject to the gapping of prices and other risks that may not be captured appropriately in the internal models. The shocks applied should reflect the nature of the portfolios and the time it could take to hedge out or manage risks under severe market conditions; and
 - h. the Authorised Firm should conduct, as part of its regular internal auditing process, an independent review of its internal models, and including the ones used for purposes of chapter 5.

2. The review referred to in (h) of Guidance note 1 above, should include the activities both of the business trading units and of the independent risk-control unit. At least once a year, the Authorised Firm should conduct a review of its overall risk management process. The review should consider the following:
 - a. the adequacy of the documentation of the risk-management system and process and the organisation of the risk-control unit;
 - b. the integration of risk measures into daily risk management and the integrity of the management information system;
 - c. the process the Authorised Firm employs for approving risk-pricing models and valuation systems that are used by front and back-office personnel;
 - d. the scope of risks captured by the risk-measurement model and the validation of any significant changes in the risk-measurement process;
 - e. the accuracy and completeness of position data, the accuracy and appropriateness of volatility and correlation assumptions, and the accuracy of valuation and risk sensitivity calculations;
 - f. the verification process the Authorised Firm employs to evaluate the consistency, timeliness and reliability of data sources used to run internal models, including the independence of such data sources; and
 - g. the verification process the Authorised Firm uses to evaluate back-testing that is conducted to assess the models' accuracy.
3. As techniques and best practices evolve, Authorised Firms should apply those new techniques and practices in any internal model used for purposes of chapter 5.

Specification of Market Risk factors

4. Any internal model used to calculate Capital Requirements for equity position risk, interest rate risk, foreign exchange risk, commodities risk and any internal model for correlation trading should meet all of the following requirements:
 - a. the model must accurately reflect, on a continuous basis, all material price risks, including General Market Risks and, where approval has been granted in relation to Specific Risk, Specific Risks arising on the underlying portfolio, and should ensure that sufficient risk factors are properly specified; and
 - b. the model should capture a sufficient number of risk factors, depending on the level of activity of the Authorised Firm in the respective markets. The risk factors in the model should be sufficient to capture the risks inherent in the Authorised Firm's portfolio of on and off-balance sheet trading positions. The Authorised Firm should at least incorporate those risk factors in its model that are incorporated into its pricing model. The risk-measurement model should capture nonlinearities for options and other products as well as correlation risk and basis risk. Where proxies for risk factors are used they should show a good track record for the actual position held. Although an Authorised Firm will have some discretion in specifying the risk factors for its internal models, the DFSA expects that such models will meet the criteria specified in the following paragraphs.
5. Any internal model used to calculate Capital Requirements for position risk, foreign exchange risk or commodities risk should meet all of the following requirements:
 - a. the model should incorporate a set of risk factors corresponding to the interest rates in each currency in which the Authorised Firm has interest rate sensitive on or off balance sheet positions. The Authorised Firm should model the yield curves using one of a number of generally accepted approaches, for example, by estimating forward rates of zero-coupon yields. For material Exposures to interest-rate risk in the major currencies

and markets, the yield curve should be divided into a minimum of six maturity segments, to capture the variations of volatility of rates along the yield curve. The model should also capture the risk of less than perfectly correlated movements between different yield curves. The risk measurement system should incorporate separate risk factors to capture spread risk, for example, between bonds and swaps;

- b. the model should incorporate risk factors corresponding to gold and to the individual foreign currencies in which the Authorised Firm's positions are denominated. For Collective Investment Funds the actual foreign exchange positions of the Fund should be taken into account. Authorised Firms may rely on third party reporting of the foreign exchange position of the Fund, where the correctness of this report is adequately ensured. If an Authorised Firm is not aware of the foreign exchange positions of a Fund, this position should be carved out of the model and treated separately;
 - c. the model should use a separate risk factor at least for each of the equity markets in which the Authorised Firm holds significant positions. At a minimum, this will include a risk factor that is designed to capture market-wide movements in equity prices, for example, a market index. Positions in individual securities or in sector indices could be expressed in "beta-equivalents" relative to this market-wide index. A relatively more-detailed approach would be to have risk factors corresponding to various sectors of the overall equity market, for instance, industry sectors or cyclical and non-cyclical sectors. The most extensive approach would be to have risk factors corresponding to the volatility of individual equity issues;
 - d. the model should use a separate risk factor at least for each commodity in which the Authorised Firm holds significant positions. The model must also capture the risk of less than perfectly correlated movements between similar, but not identical, commodities and the Exposure to changes in forward prices arising from maturity mismatches. It should also take account of market characteristics, notably delivery dates and the scope provided to traders to close out positions; For more actively traded portfolios, the model should also take account of variation in the "convenience yield" between derivative positions such as forwards and swaps and cash positions in the commodity; and
 - e. the Authorised Firm's internal model should conservatively assess the risk arising from less liquid positions and positions with limited price transparency under realistic market scenarios. In addition, the internal model should meet minimum data standards. Proxies should be appropriately conservative and should be used only where available data is insufficient or is not reflective of the true volatility of a position or portfolio.
6. Authorised Firms may, in any internal model used for purposes of chapter 5, use empirical correlations within risk categories and across risk categories only if the Authorised Firm's approach for measuring correlations is sound and implemented with integrity.

Quantitative standards

7. The DFSA will usually only approve an internal Value-at-Risk (VaR) model or its use when the VaR model meets the following quantitative criteria:
 - a. VaR should be computed at least on a daily basis;
 - b. in calculating the value-at-risk, a 99th percentile, one-tailed confidence interval is to be used;
 - c. in calculating VaR, an instantaneous price shock equivalent to a 10 day movement in prices is to be used, i.e., the minimum "holding period" will be 10 trading days;
 - d. an effective historical observation period of at least one year except where a shorter observation period is justified by a significant upsurge in price volatility; and

- e. its data set is updated by the Authorised Firm no less frequently than once every month and is reassessed whenever market prices are subject to material changes.
- 8. An Authorised Firm may use VaR numbers calculated according to shorter holding periods than 10 days scaled up to 10 days by an appropriate methodology that is reviewed periodically.

Qualitative standards

- 9. In addition to 7 and 8:
 - a. no particular type of model is prescribed. So long as each model used captures all the material risks run by the Authorised Firm, the Authorised Firm will be free to use models based, for example, on variance-covariance matrices, historical simulations, or Monte Carlo simulations;
 - b. an Authorised Firm will have discretion to recognise empirical correlations within broad risk categories, for example, interest rates, exchange rates, equity prices and commodity prices, including related options volatilities in each risk factor category;
 - c. an Authorised Firm's models should accurately capture the unique risks associated with options within each of the broad risk categories; and
 - d. an Authorised Firm should calculate, on a daily basis, its Market Risk Capital Requirement or any component for which an internal model is used, expressed as the higher of (a) its previous day's VaR number measured according to the parameters specified in this section and (b) an average of the daily VaR measures on each of the preceding sixty business days, multiplied by a multiplication factor.
- 10. The DFSA will usually set a multiplication factor of 3 that must be used by the Authorised Firm where all the qualitative and quantitative criteria are satisfied. This will be imposed as a condition on the approval and may be varied by the DFSA should circumstances require.
- 11. In addition to the calculation of VaR, an Authorised Firm using internal VaR models, an Authorised Firm should at least on a weekly basis, calculate a 'Stressed VaR' of the current portfolio, in accordance with the requirements set out in Guidance note 7, with VaR model inputs calibrated to historical data from a continuous 12-month period of significant financial stress relevant to the Authorised Firm's portfolio. The choice of such historical data should be subject to at least annual review by the Authorised Firm, which should notify the outcome to the DFSA.
- 12. An Authorised Firm using an internal model should calculate Capital Requirement for the relevant risk categories, as the sum of points (a) and (b):
 - a. the higher of:
 - i. its previous day's VaR number calculated in accordance with Guidance note 7 of this guidance under A5.9.1; or
 - ii. an average of the daily value-at-risk numbers calculated in accordance with Guidance note 7 of A5.9.1, on each of the preceding sixty business days (VaRavg), multiplied by the multiplication factor referred in Guidance note 8 of A5.9.1; plus
 - b. the higher of:
 - i. its latest available stressed-value-at-risk number calculated in accordance with Guidance note 9 of A5.9.1 (sVaRt-1); or
 - ii. an average of the stressed VaR numbers calculated in accordance with Guidance note 9 of A5.9.1 during the preceding sixty business days (sVaRavg), multiplied by the multiplication factor (ms) according to Guidance note 10.

Regulatory back-testing and multiplication factors

13. The results of the Stressed VaR calculations referred to in Guidance note 11 of A5.9.1 should be scaled up by the multiplication factors given below.
14. The multiplication factor referred above is defined as the sum of 3 and an addend between 0 and 1 in accordance with 1. That addend should depend on the number of violations for the most recent 250 business days as evidenced by the Authorised Firm's back-testing of the VaR as set out in Guidance note 7 above.

Number of violations	addend
Fewer than 5	0.00
5	0.40
6	0.50
7	0.65
8	0.75
9	0.85
10 or more	1.00

15. An Authorised Firm should count daily violations on the basis of back-testing on hypothetical and actual changes in the portfolio's value. A violation for this purpose is defined as a one-day change in the portfolio's value that exceeds the related one-day VaR number generated by the Authorised Firm's model. For the purpose of determining the addend the number of violations should be assessed at least on a quarterly basis and should be equal to the higher of the number of violations under hypothetical and actual changes in the value of the portfolio.
16. Back-testing on hypothetical changes in the portfolio's value should be based on a comparison between the portfolio's end-of-day value and, assuming unchanged positions, its value at the end of the subsequent day. Back-testing on actual changes in the portfolio's value should be based on a comparison between the portfolio's end-of-day value and its actual value at the end of the subsequent day excluding fees, commissions, and net interest income.
17. The DFSA may in individual cases limit the addend to that resulting from violations under hypothetical changes, where the number of violations under actual changes does not result from deficiencies in the internal model.
18. In order to enable the DFSA to monitor the appropriateness of the multiplication factors on an ongoing basis, the Authorised Firm should notify promptly, and in any case no later than within five working days, any violations that result from its back-testing programme.

Internal Validation

19. Authorised Firms should have processes in place to ensure that all their internal models used for purposes of chapter 5 have been adequately validated by suitably qualified parties independent of the development process to ensure that they are conceptually sound and adequately capture all material risks. The validation should be conducted when the internal model is initially developed and when any significant changes are made to the internal model. The validation should also be conducted on a periodic basis but especially where there have been any significant structural changes in the market or changes to the composition of the portfolio which might lead to the internal model no longer being adequate. As techniques and best practices for internal validation evolve, Authorised Firms should apply these advances. Internal model validation should not be limited to back-testing, but should, at a minimum, also include the following:
 - a. tests to demonstrate that any assumptions made within the internal model are appropriate and do not underestimate or overestimate the risk;

- b. in addition to the regulatory back-testing programmes, Authorised Firms should carry out their own internal model validation tests, including back-testing, in relation to the risks and structures of their portfolios; and
 - c. the use of hypothetical portfolios to ensure that the internal model is able to account for particular structural features that may arise, for example material basis risks and Concentration Risk.
- 20. The Authorised Firm should perform back-testing on both actual and hypothetical changes in the portfolio's value.

Requirements for modelling Specific Risk

- 21. An internal model used for calculating Capital Requirements for Specific Risk and an internal model for correlation trading should meet the following additional requirements:
 - a. it explains the historical price variation in the portfolio;
 - b. it captures concentration in terms of magnitude and changes of composition of the portfolio;
 - c. it is robust to an adverse environment;
 - d. it is validated through back-testing aimed at assessing whether Specific Risk is being accurately captured. If the Authorised Firm performs such back-testing on the basis of relevant sub-portfolios, these must be chosen in a consistent manner;
 - e. it captures name-related basis risk and should in particular be sensitive to material idiosyncratic differences between similar, but not identical, positions; and
 - f. it captures event risk.

Exclusions from Specific Risk models

- 22. An Authorised Firm may choose to exclude from the calculation of its Specific Risk Capital Requirement using an internal model those positions for which it fulfils a Capital Requirement for Specific Risk in accordance with relevant sections of chapter 5.
- 23. An Authorised Firm may choose not to capture default and migration risks for debt instruments in its internal model where it is capturing those risks through internal models for incremental default and migration risk.

A5.9.2 Incremental risk charge (IRC) model

Guidance

- 1. An Authorised Firm that uses an internal model for calculating Capital Requirements for Specific Risk of interest rate risk Exposures should also have an internal incremental default and migration risk (incremental risk charge, or IRC) model in place to capture the default and migration risks of its Trading Book positions that are incremental to the risks captured by the VaR measure as specified in Guidance note 7 of A5.9.1. An Authorised Firm should demonstrate that its internal model meets soundness standards comparable to the Internal Ratings Based (IRB) approach for Credit Risk under the assumption of a constant level of risk, and adjusted where appropriate to reflect the impact of liquidity, concentrations, hedging and optionality.

Scope of the internal IRC model

- 2. The internal IRC model should cover all positions subject to a Capital Requirement for specific interest rate risk, including those subject to a 0% Specific Risk capital charge under Rule A5.2.13, but should not cover securitisation positions and n-th-to-default Credit Derivatives.

3. The Authorised Firm may, subject to approval by the DFSA, choose to include consistently all listed equity positions and derivatives positions based on listed equities. The permission will be granted only if such inclusion is consistent with how the Authorised Firm internally measures and manages risk.

Parameters of the internal IRC model

4. Authorised Firms should use the internal model to calculate a number which measures losses due to default and internal or external ratings migration at the 99.9 % confidence interval over a time horizon of one year. Authorised Firms should calculate this number at least weekly.
5. Correlation assumptions should be supported by analysis of objective data in a conceptually sound framework. The internal model should appropriately reflect issuer concentrations. Concentrations that can arise within and across product classes under stressed conditions should also be reflected.
6. The internal IRC model should reflect the impact of correlations between default and migration events. The impact of diversification between, on the one hand, default and migration events and, on the other hand, other risk factors should not be reflected.
7. The internal model should be based on the assumption of a constant level of risk over the one-year time horizon, implying that given individual Trading Book positions or sets of positions that have experienced default or migration over their liquidity horizon are re-balanced at the end of their liquidity horizon to attain the initial level of risk. Alternatively, an Authorised Firm may choose to consistently use a one-year constant position assumption.
8. The liquidity horizons should be set according to the time required to sell the position or to hedge all material relevant price risks in a stressed market, having particular regard to the size of the position. Liquidity horizons should reflect actual practice and experience during periods of both systematic and idiosyncratic stresses. The liquidity horizon should be measured under conservative assumptions and should be sufficiently long that the act of selling or hedging, in itself, would not materially affect the price at which the selling or hedging would be executed.
9. The determination of the appropriate liquidity horizon for a position or set of positions is subject to a floor of three months.
10. The determination of the appropriate liquidity horizon for a position or set of positions should take into account an Authorised Firm's internal policies relating to valuation adjustments and the management of stale positions. When an Authorised Firm determines liquidity horizons for sets of positions rather than for individual positions, the criteria for defining sets of positions should be defined in a way that meaningfully reflects differences in liquidity. The liquidity horizons should be greater for positions that are concentrated, reflecting the longer period needed to liquidate such positions. The liquidity horizon for a securitisation warehouse should reflect the time to build, sell and securitise the assets, or to hedge the material risk factors, under stressed market conditions.

Recognition of hedges in the internal IRC model

11. Hedges may be incorporated into an Authorised Firm's internal model to capture the incremental default and migration risks. Positions may be netted when long and short positions refer to the same financial instrument. Hedging or diversification effects associated with long and short positions involving different instruments or different securities of the same obligor, as well as long and short positions in different issuers, may only be recognised by explicitly modelling gross long and short positions in the different instruments. Authorised Firms should reflect the impact of material risks that could occur during the interval between the hedge's maturity and the liquidity horizon as well as the potential for significant basis risks in hedging strategies by product, seniority in the capital structure, internal or external rating, maturity, vintage and other differences in the instruments. An Authorised Firm should reflect a hedge only to the extent that it can be maintained even as the obligor approaches a credit or other event.

12. For positions that are hedged via dynamic hedging strategies, a rebalancing of the hedge within the liquidity horizon of the hedged position may be recognised, provided that the Authorised Firm:
 - a. chooses to model rebalancing of the hedge consistently over the relevant set of Trading Book positions;
 - b. demonstrates that the inclusion of rebalancing results in a better risk measurement; and
 - c. demonstrates that the markets for the instruments serving as hedges are liquid enough to allow for such rebalancing even during periods of stress. Any residual risks resulting from dynamic hedging strategies must be reflected in the Capital Requirement.

Additional content of the internal IRC model

13. The internal model to capture the incremental default and migration risks should reflect the nonlinear impact of options, structured Credit Derivatives and other positions with material nonlinear behaviour with respect to price changes. The Authorised Firm should also have due regard to the amount of model risk inherent in the valuation and estimation of price risks associated with such products.
14. The internal model should be based on data that are objective and up-to-date.
15. As part of the independent review and validation of their internal models used for purposes of this chapter, an Authorised Firm should in particular do all of the following:
 - a. validate that its modelling approach for correlations and price changes is appropriate for its portfolio, including the choice and weights of its systematic risk factors;
 - b. perform a variety of stress tests, including sensitivity analysis and scenario analysis, to assess the qualitative and quantitative reasonableness of the internal model, particularly with regard to the treatment of concentrations. Such tests should not be limited to the range of events experienced historically; and
 - c. apply appropriate quantitative validation, including relevant internal modelling benchmarks.
16. The internal model should be consistent with the Authorised Firm's internal risk management methodologies for identifying, measuring, and managing trading risks.
17. Authorised Firms should document their internal models so that their correlation and other modelling assumptions are transparent to the DFSA.
18. The internal model should conservatively assess the risk arising from less liquid positions and positions with limited price transparency under realistic market scenarios. In addition, the internal model should meet minimum data standards. Proxies should be appropriately conservative and may be used only where available data is insufficient or is not reflective of the true volatility of a position or portfolio.

Not fully compliant IRC approaches

19. If an Authorised Firm uses an internal model to capture incremental default and migration risks that does not comply with all requirements specified in Guidance notes 4 to 18 of A5.9.2, but that is consistent with the Authorised Firm's internal methodologies for identifying, measuring and managing incremental default and migration risks, it should be able to demonstrate that its internal model results in a Capital Requirement that is at least as high as if it were based on a model in full compliance with the requirements of the Guidance notes referred above. The DFSA will review compliance with the previous sentence at least annually.

A5.9.3 Internal model for correlation trading

Guidance

1. The DFSA will grant permission to use an internal model for calculating the Capital Requirement for a correlation trading portfolio only to Authorised Firms that have obtained the DFSA's approval to use an internal model for Specific Risk of interest rate risk Exposures and that meet the requirements for internal models specified earlier in this section.
2. Authorised Firms should use this internal model to calculate a number which adequately measures all price risks at the 99.9 % confidence interval over a time horizon of one year under the assumption of a constant level of risk, and adjusted where appropriate to reflect the impact of liquidity, concentrations, hedging and optionality. Authorised Firms should calculate this number at least weekly.
3. The following risks should be adequately captured by the model referred to in Guidance note 1 of A5.9.3:
 - a. the cumulative risk arising from multiple defaults, including different ordering of defaults, in tranching products;
 - b. credit spread risk, including the gamma and cross-gamma effects;
 - c. volatility of implied correlations, including the cross effect between spreads and correlations;
 - d. basis risk, including both of the following:
 - i. the basis between the spread of an index and those of its constituent single names; and
 - ii. the basis between the implied correlation of an index and that of bespoke portfolios;
 - e. recovery rate volatility, as it relates to the propensity for recovery rates to affect tranche prices;
 - f. to the extent the comprehensive risk measure incorporated benefits from dynamic hedging, the risk of hedge slippage and the potential costs of rebalancing such hedges; and
 - g. any other material price risks of positions in the correlation trading portfolio.
4. An Authorised Firm should use sufficient market data within the model referred to in Guidance note 1 in order to ensure that it fully captures the salient risks of those Exposures in its internal approach in accordance with the requirements set out in this guidance in A5.9.3. It should be able to demonstrate to the DFSA through back testing or other appropriate means that its model can appropriately explain the historical price variation of those products.
5. The Authorised Firm should have appropriate policies and procedures in place in order to separate the positions for which it holds permission to incorporate them in the Capital Requirement in accordance with this guidance in A5.9.3 from other positions for which it does not hold such permission.
6. With regard to the portfolio of all the positions incorporated in the model referred to in Guidance note 1, the Authorised Firm should regularly apply a set of specific, predetermined stress scenarios. Such stress scenarios should examine the effects of stress to default rates, recovery rates, credit spreads, basis risk, correlations and other relevant risk factors on the correlation trading portfolio. The Authorised Firm should apply stress scenarios at least weekly and report at least quarterly to the DFSA the results, including comparisons with the Authorised Firm's Capital Requirement in accordance with this point. Any instances where the stress test results materially exceed the Capital Requirement for the correlation trading portfolio should be reported to the DFSA in a timely manner.

7. The internal model should conservatively assess the risk arising from less liquid positions and positions with limited price transparency under realistic market scenarios. In addition, the internal model should meet minimum data standards. Proxies should be appropriately conservative and may be used only where available data is insufficient or is not reflective of the true volatility of a position or portfolio.

Stress testing

- A5.9.4** (1) For the purposes of Rule 5.11.3, an Authorised Firm's internal model must meet the following criteria:
- (a) the Authorised Firm's stress scenarios must cover a range of factors that can create extraordinary losses or gains in trading portfolios, or make the control of risk in those portfolios very difficult. These factors include low-probability events in all major types of risks, including the various components of market, credit, and Operational Risks;
 - (b) the Authorised Firm's stress tests must be both of a quantitative and qualitative nature, incorporating both Market Risk and liquidity aspects of market disturbances. Quantitative criteria must identify plausible stress scenarios to which the Authorised Firm could be exposed. Qualitative criteria must emphasise that two major goals of stress testing are to evaluate the capacity of the Authorised Firm's capital to absorb potential large losses and to identify steps the Authorised Firm can take to reduce its risk and conserve capital; and
 - (c) the Authorised Firm must combine the use of supervisory stress scenarios with stress tests developed by the Authorised Firm itself to reflect their Specific Risk characteristics. Information is required in three broad areas:
 - (i) supervisory scenarios requiring no simulations by the Authorised Firm - the Authorised Firm must have information on the largest losses experienced during the reporting period available for supervisory review. This loss information must be compared to the level of capital that results from an Authorised Firm's internal measurement system;
 - (ii) supervisory scenarios requiring a simulation by the Authorised Firm – the Authorised Firm must subject its portfolio to a series of simulated stress scenarios and provide the DFSA with the results (e.g., the sensitivity of the Authorised Firm's Market Risk Exposure to changes in the assumptions about volatilities and correlations); and
 - (iii) scenarios developed by the Authorised Firm itself to capture the specific characteristics of its portfolio.
- (2) In addition to the scenarios prescribed under (1)(c), an Authorised Firm must also develop its own stress tests which it identifies as most adverse, based on the characteristics of its portfolio, for example, problems arising in a key region of the world combined with a sharp move in oil prices. The Authorised Firm must also provide the DFSA with a description of the methodology used to identify and carry out the scenarios as well as with a description of the results derived from these scenarios.

A5.10 Credit Valuation Adjustment Risk Capital Requirement

Guidance

This section presents the method for the calculation of the Credit Valuation Adjustment Risk Capital Requirement under Rule 5.10A.1.

A5.10.1 In this section:

- (a) “Covered Transaction” means:
 - (i) a derivative transaction but excluding derivatives transacted directly with a Qualifying CCP; and
 - (ii) a SFT if:
 - (A) it is fair-valued under the Authorised Firm’s applicable accounting framework; and
 - (B) the Authorised Firm’s CVA Risk arising from the transaction is material;
- (b) “CVA Risk” means the risk of losses arising from changing credit valuation adjustment (CVA) values in response to changes in Counterparty credit spreads and market risk factors that drive prices of Covered Transactions; and
- (c) “Eligible CVA Hedge” means a transaction used for the purpose of mitigating the Counterparty credit spread component of CVA Risk, and managed as such and that is either:
 - (i) a single-name credit default swap or a contingent credit default swap which references:
 - (A) the Counterparty directly;
 - (B) an entity legally related to the Counterparty, where legally related refers to cases where the reference name and the Counterparty are either a parent and its subsidiary or two subsidiaries of a common parent; or
 - (C) an entity that belongs to the same sector and region as the Counterparty; or
 - (ii) an index credit default swap.

General provisions

A5.10.2 An Authorised Firm must calculate its Credit Valuation Adjustment Risk Capital Requirement for its entire portfolio of Covered Transactions and Eligible CVA Hedges by using the following approaches:

- (a) the basic approach for credit valuation adjustment risk (BA-CVA) under Rules A5.10.4 to A5.10.12; or

- (b) the alternative approach under Rule A5.10.14, provided that the Authorised Firm meets the conditions set out in Rule A5.10.13.

Guidance

1. The BA-CVA calculations may be performed either via the reduced version or the full version. An Authorised Firm can choose whether to implement the full version or the reduced version at its discretion. However, all Authorised Firms must calculate the reduced version as the reduced version is also part of the full version as a conservative means to limit hedging recognition.
2. The Credit Valuation Adjustment Risk Capital Requirement is calculated for an Authorised Firm's CVA portfolio on a standalone basis. The CVA portfolio includes CVA for an Authorised Firm's entire portfolio of covered transactions and Eligible CVA Hedges.

A5.10.3 An Authorised Firm must:

- (a) exclude positions in hedging instruments entered into with third parties that are Eligible CVA Hedges from the calculation of its Market Risk Capital Requirement under Rule 5.3.1; and
- (b) include positions in hedging instruments entered into with third parties that are not Eligible CVA Hedges in the calculation of its Market Risk Capital Requirement under Rule 5.3.1.

Guidance

1. Eligible CVA Hedges can be external (i.e. with an external Counterparty) or internal (i.e. with one of the Authorised firm's trading desks). All external CVA hedges (eligible and ineligible) that are Covered Transactions must be included in the CVA calculation of the Counterparty providing the hedge.
2. An internal CVA Hedge involves two perfectly offsetting positions: one of the Authorised Firm's CVA desk and the opposite position of the trading desk:
 - a. If an internal CVA hedge is not an Eligible CVA Hedge, then both positions belong to the Trading Book where they cancel each other, so there is no impact on either the CVA portfolio or the trading book.
 - b. If an internal CVA hedge is an Eligible CVA Hedge, then the CVA desk's position is part of the CVA portfolio where it is capitalised as set out in this section, while the trading desk's position is part of the trading book where it is capitalised as set out in Rule 5.3.1.

Reduced version of BA-CVA (hedges are not recognised)

- A5.10.4** If an Authorised Firm does not use any Eligible CVA Hedges to hedge CVA Risk it must calculate its Credit Valuation Adjustment Risk Capital Requirement, using the following formula:

$$DS_{BA-CVA} \cdot K_{reduced}$$

where:

$DS_{BA-CVA} = 0.65$, the discount scalar

$K_{reduced}$ is calculated in accordance with the following formula:

$$K_{\text{reduced}} = \sqrt{\left(\rho \cdot \sum_C \text{SCVA}_C\right)^2 + (1 - \rho^2) \cdot \sum_C \text{SCVA}_C^2}$$

where:

SCVA_C = the Credit Valuation Adjustment Risk Capital Requirement for Counterparty 'C' on a standalone basis, calculated in accordance with Rule A.5.10.5

ρ = 50%, the supervisory correlation parameter

C = the index that denotes all counterparties for which the firm uses BA-CVA to calculate its Credit Valuation Adjustment Risk Capital Requirement

A5.10.5 For the purposes of Rule A5.10.4, an Authorised Firm must calculate SCVA_C , using the following formula:

$$\text{SCVA}_C = \frac{1}{\alpha} \cdot \text{RW}_C \cdot \sum_{\text{NS}} M_{\text{NS}} \cdot \text{EAD}_{\text{NS}} \cdot \text{DF}_{\text{NS}}$$

where:

RW_C = the risk weight for Counterparty 'C'. Counterparty 'C' must be mapped to one of the risk weights based on a combination of sector and credit quality and determined in accordance with the table in Rule A5.10.6

NS = the index that denotes all the netting sets with a given Counterparty

C = the index that denotes all the Counterparties for which the firm uses BA-CVA to calculate its Credit Valuation Adjustment Risk Capital Requirement

M_{NS} = the effective maturity for the netting set NS calculated as the weighted average remaining maturity of the exposures, where M is at least one year and the notional amount of each exposure is used for weighting the maturity

EAD_{NS} = the exposure at default (EAD) of the netting set NS with Counterparty 'C', calculated in the same manner as the Authorised Firm calculates it for determining its Counterparty Credit Risk Capital Requirement under Rule 4.8.7

DF_{NS} = the supervisory discount factor, calculated as follows:

$$\frac{1 - e^{-0.05 \cdot M_{\text{NS}}}}{0.05 \cdot M_{\text{NS}}}$$

α = 1.4

A5.10.6 For the purposes of Rule A5.10.5, an Authorised Firm must set the value of RW_C in accordance with the table below:

Sector of Counterparty	Credit quality of Counterparty	
	Investment Grade	High yield and not rated
Sovereigns including central banks and multilateral development banks	0.5%	2.0%
Local government, government-backed non-financials, education and public administration	1.0%	4.0%
Financials including government-backed financials	5.0%	12.0%
Basic materials, energy, industrials, agriculture, manufacturing, mining and quarrying	3.0%	7.0%
Consumer goods and services, transportation and storage, administrative and support service activities	3.0%	8.5%
Technology, telecommunications	2.0%	5.5%
Health care, utilities, professional and technical activities	1.5%	5.0%
Other sector	5.0%	12.0%

Full version of BA-CVA (hedges are recognised)

A5.10.7 If an Authorised Firm uses one or more Eligible CVA Hedges to hedge CVA Risk it must calculate its Credit Valuation Adjustment Risk Capital Requirement, using the following formula:

$$DS_{BA-CVA} \cdot K_{full}$$

where:

DS_{BA-CVA} = 0.65, the discount scalar

$$K_{full} = \beta \cdot K_{reduced} + (1 - \beta) \cdot K_{hedged}$$

where:

β = 0.25, the supervisory floor

$K_{reduced}$ = calculated in accordance with Rule A5.10.4

K_{hedged} = calculated in accordance with Rule A5.10.8

A5.10.8 For the purposes of Rule A5.10.7, an Authorised Firm must calculate K_{hedged} using the following formula:

$$K_{\text{hedged}} = \sqrt{\left(\rho \cdot \sum_C (\text{SCVA}_C - \text{SNH}_C) - \text{IH} \right)^2 + (1 - \rho^2) \cdot \sum_C (\text{SCVA}_C - \text{SNH}_C)^2 + \sum_C \text{HMA}_C}$$

where:

ρ = 50%, the supervisory correlation parameter

C = the index that denotes all counterparties for which the firm uses BA-CVA to calculate its Credit Valuation Adjustment Risk Capital Requirement

SCVA_C = the Credit Valuation Adjustment Risk Capital Requirement for Counterparty 'C' on a standalone basis, calculated in accordance with Rule A5.10.5

SNH_C = the quantity that gives recognition to the reduction in CVA Risk of Counterparty 'C' arising from the Authorised Firm's use of single-name hedges of credit spread risk, calculated in accordance with Rule A5.10.9

IH = the quantity that gives recognition to the reduction in CVA Risk across all Counterparties arising from the Authorised Firm's use of index hedges, calculated in accordance with Rule A5.10.10

HMH_C = the quantity representing hedging misalignment, calculated in accordance with Rule A5.10.11

A5.10.9 For the purposes of Rule A5.10.8, an Authorised Firm must calculate SNH_C , using the following formula:

$$\text{SNH}_C = \sum_{h \in C} r_{hc} \cdot \text{RW}_h \cdot M_h^{\text{SN}} \cdot B_h^{\text{SN}} \cdot D_h^{\text{SN}}$$

where:

r_{hc} = the supervisory correlation between the credit spread risk of Counterparty 'C' and the credit spread of a single-name hedge 'h' of Counterparty 'C' determined in accordance with the table Rule A5.10.12

M_h^{SN} = the remaining maturity of a single-name hedge 'h' recognised as an Eligible CVA Hedge

B_h^{SN} = the notional of a single-name hedge 'h' recognised as an Eligible CVA Hedge (for single-name contingent credit default swaps, the notional must be determined by the current market value of the reference portfolio or instrument)

D_h^{SN} = the supervisory discount factor for a single-name hedge, calculated as:

$$\frac{1 - e^{-0.05 \cdot M_h^{\text{SN}}}}{0.05 \cdot M_h^{\text{SN}}}$$

RW_h = the supervisory risk weight of a single-name hedge 'h' recognised as an Eligible CVA Hedge. Hedge 'h' must be mapped to one of the risk weights based on a combination of sector and credit quality of the reference credit spread of the hedging instrument in accordance with the table in Rule A5.10.6

h = the index that denotes all single-name Eligible CVA hedges that an Authorised Firm has taken out to hedge the CVA Risk of a Counterparty

A5.10.10 For the purposes of Rule A5.10.8, an Authorised Firm must calculate IH, using the following formula:

$$IH = \sum_i RW_i \cdot M_i^{ind} \cdot B_i^{ind} \cdot DF_i^{ind}$$

where:

M_i^{ind} = the remaining maturity of index hedge 'i' recognised as an Eligible CVA Hedge

B_i^{ind} = the notional of the index hedge 'i' recognised as an Eligible CVA Hedge

DF_i^{ind} = the supervisory discount factor for a single-name hedge, calculated as:

$$\frac{1 - e^{-0.05 \cdot M_i^{ind}}}{0.05 \cdot M_i^{ind}}$$

RW_h = the supervisory risk weight of the index hedge 'i' recognised as an Eligible CVA Hedge. RW_h must be based on a combination of sector and credit quality of all the index constituents, as determined in Rule A5.10.6 and adjusted as follows:

- (a) for an index where all index constituents belong to the same sector and are of the same credit quality, the Authorised Firm must multiply the relevant value in the table in Rule A5.10.6 by 0.7;
- (b) for an index spanning multiple sectors or with a mixture of investment grade constituents and other grade constituents, the Authorised Firm must calculate the name-weighted average of the risk weights from the table in Rule A5.10.6 and multiply the result by 0.7

i = the index that denotes all index Eligible CVA Hedges that an Authorised Firm has taken out to hedge CVA Risk

A5.10.11 For the purposes of Rule A5.10.8, an Authorised Firm must calculate HMA_C , using the following formula:

$$HMA_C = \sum_{h \in h} (1 - r_{hc}^2) \cdot (RW_h \cdot M_h^{SN} \cdot B_h^{SN} \cdot D_h^{SN})^2$$

where all the terms are the ones set out in Rule A5.10.9

A5.10.12 For the purposes of Rule A5.10.9, an Authorised Firm must set the value of r_{hc} in accordance with the table below:

Single-name hedge 'h' of Counterparty 'C'	Value of r_{hc}
references to Counterparty 'C' directly	100%
has legal relation with Counterparty 'C'	80%
shares sector and region with Counterparty 'C'	50%

Alternative approach

A5.10.13 An Authorised Firm whose aggregate notional amount of non-centrally cleared derivatives is less than or equal to 110 billion USD may choose to calculate its Credit Valuation Adjustment Risk Capital Requirement using the alternative approach in Rule A5.10.14, instead of using the BA-CVA, unless the DFSA mandates otherwise.

A5.10.14 An Authorised Firm using the alternative approach must calculate its Credit Valuation Adjustment Risk Capital Requirement by dividing the amount of RWAs for Counterparty Credit Risk Exposures calculated in accordance with Rule 4.8.7 by 12.5.

A5.10.15 For the purpose of Rule A5.10.14 an Authorised Firm must:

- (a) not recognise the effect of CVA hedges; and
- (b) apply the alternative approach to its entire portfolio of Covered Transactions and Eligible CVA Hedges.

Guidance

The DFSA will require an Authorised Firm to use BA-CVA if it determines that CVA Risk resulting from the Authorised Firm's derivative positions materially contributes to its overall risk.

APP6 CALCULATING THE OPERATIONAL RISK CAPITAL REQUIREMENT

Guidance

1. Section 6.11 of PIB provides that an Authorised Firm in Categories 1, 2, 3A and 5 must use the Basic Indicator Approach to calculate its Operational Risk Capital Requirement, unless the firm has approval from the DFSA to use the Standardised Approach or Alternative Standardised Approach. In this App6:
 - a. the Basic Indicator Approach is prescribed in section A6.1;
 - b. the Standardised Approach is prescribed in section A6.2; and
 - c. the Alternative Standardised Approach is prescribed in section A6.3.
2. The application of various components of the rules on Operational Risk to Authorised Firms licensed to carry out various financial services is detailed in the table below:

Financial Services	Prudential Category	Capital requirement	Systems and controls requirement	PII cover
Accepting Deposits	1	YES	YES	NO
Providing Credit	2	YES	YES	NO
Dealing in Investments as Principal	2	YES	YES	NO
Dealing in Investments as Agent	3A	YES	YES	NO
Managing Assets	3C	NO	YES	YES
Managing a Collective Investment Fund	3C	NO	YES	YES
Managing a Restricted PSIA	3C	NO	YES	YES
Providing Custody	3B or 3C	NO	YES	YES
Providing Trust Services as a trustee of an express trust	3C	NO	YES	YES
Acting as the Trustee of a Fund	3B	NO	YES	YES
Operating an Employee Money Purchase Scheme	3B	NO	YES	YES
Acting as the Administrator of an Employee Money Purchase Scheme	3B	NO	YES	YES
Providing Money Services (issuing Stored Value)	3C	NO	YES	NO

Financial Services	Prudential Category	Capital requirement	Systems and controls requirement	PII cover
Providing Money Services (other than issuing Stored Value)	3D or 4	NO	YES	NO
Arranging Deals in Investments	4	NO	YES	YES
Advising on Financial Products	4	NO	YES	YES
Arranging Custody	4	NO	YES	NO
Insurance Intermediation	4	NO	YES	YES
Insurance Management	4	NO	YES	YES
Providing Fund Administration	4	NO	YES	YES
Operating an Alternative Trading System	4	NO	YES	NO* (*unless it holds Client Assets)
Providing Trust Services other than as a trustee of an express trust	4	NO	YES	YES
Arranging Credit and Advising on Credit	4	NO	YES	YES
Operating a Crowdfunding Platform	4	NO	YES	YES
Arranging or Advising on Money Service	4	NO	YES	YES
Managing an Unrestricted PSIA	5 or 1	YES	YES	NO

A6.1 Basic Indicator Approach

- A6.1.1** (1) An Authorised Firm which uses the Basic Indicator Approach must calculate its Operational Risk Capital Requirement equal to the average over the previous three years of a fixed percentage (denoted alpha) of positive annual gross income.
- (2) In (1), if the figure for annual gross income in any of the previous three years is zero or negative an Authorised Firm must exclude such amounts from the calculation of the average.

- (3) The Operational Risk Capital Requirement in (1) must be calculated according to the following formula:

$$K_{BIA} = [\sum (GI_{1...n} \times \alpha)] / n$$

where:

K_{BIA} = the Operational Risk Capital Requirement under the Basic Indicator Approach

GI = the gross annual income, where positive, over the previous three years

n = number of the previous three years for which gross income is positive

α = 15%,

- (4) For the purpose of (1), “gross income” is net interest income plus net non-interest income and must:
- (a) be gross of any provisions (e.g. for unpaid interest);
 - (b) be gross of operating expenses, including fees paid to outsourcing service providers;
 - (c) exclude realised profits/losses from the sale of Securities in the Non-Trading Book; and
 - (d) exclude extraordinary or irregular items as well as income derived from insurance recoveries.

Guidance

1. In A6.1.1(1), the three year average should be calculated on the basis of the last three yearly observations at the end of the Authorised Firm’s financial year. When audited figures are not available, business estimates may be used.
2. If an Authorised Firm does not have sufficient income data to meet the three year requirement (e.g. a start-up), it may use its forecasted gross income projections for all or part of the three year time period. For example, if an Authorised Firm has two positive yearly gross incomes of USD 20 each and the final yearly observation shows a negative figure of USD 5, then the average should be calculated as USD 20 being USD 40 (sum of positive figures) divided by 2 (number of years for which positive figures are available).
3. Net interest income in A6.1.1(4) is the interest income minus interest expense. Guidance on what constitutes interest income and interest expense can be found in the PRU module.
4. Net non-interest income in A6.1.1(4) includes the income from fees and commissions, net income from trading Securities, net income from investment Securities, income from Islamic contracts and other operating income minus fee and commission expense. Guidance on non-interest income can be found in the PRU module.
5. In A6.1.1(4)(ii), outsourcing fees paid by the Authorised Firm should be excluded whereas any outsourcing fee received by the Authorised Firm should be included as part of the gross income.
6. When income from revaluation of trading items is included in the income statement, such revaluation income should be included in the calculation of the gross income.

A6.2 Standardised Approach

A6.2.1 An Authorised Firm which uses the Standardised Approach:

- (a) must calculate its Operational Risk Capital Requirement equal to the average annual gross income over the previous three years for each of the business lines in Rule A6.2.3 multiplied by the fixed percentage (denoted beta) in Rule A6.2.3;
- (b) subject to (c), may in any given year offset negative a Capital Requirement (resulting from negative gross income) in any business line against positive capital charges in other business lines without limit; and
- (c) must input to the numerator a value of zero if the aggregate Operational Risk Capital Requirement across all business lines within a given year is negative.

A6.2.2 An Authorised Firm which uses the Standardised Approach must calculate its Operational Risk Capital Requirement according to the following formula:

$$K_{SA} = \{ \sum_{\text{years } 1-3} \max[\sum(GI_{1-8} \times B_{1-8}), 0] \} / 3$$

where:

K_{SA} = the capital charge under the Standardised Approach

GI_{1-8} = annual gross income in a given year, as defined above in the Basic Indicator Approach, for each of the eight business lines

B_{1-8} = a fixed percentage set out in Rule A6.2.3.

A6.2.3 An Authorised Firm which uses the Standardised Approach must calculate its Operational Risk Capital Requirement using the following beta values:

Business Line	Beta Factor
Corporate finance	18%
Trading and sales	18%
Retail banking	12%
Commercial banking	15%
Payment and settlement	18%
Agency services	15%
Asset management	12%
Retail brokerage	12%

Guidance

The gross income is measured for each business line, not the whole institution, i.e. in corporate finance, the indicator is the gross income generated in the corporate finance business line.

A6.3 Alternative Standardised Approach

A6.3.1 An Authorised Firm which uses the Alternative Standardised Approach must calculate its Operational Risk Capital Requirement in accordance with the Standardised Approach in section A6.2 as modified by replacing annual gross income for its retail banking and commercial banking business lines with loans and advances multiplied by a fixed factor 'm'.

A6.3.2 Subject to Rule A6.3.4, an Authorised Firm which uses the Alternative Standardised Approach must calculate its Operational Risk Capital Requirement for its retail banking business line according to the following formula:

$$K_{RB} = B_{RB} \times m \times LA_{RB}$$

Where:

K_{RB} = the Capital Requirement for the retail banking business line

B_{RB} = the beta for the retail banking business line

LA_{RB} = total outstanding retail loans and advances (non-risk weighted and gross of provisions), averaged over the past three years

m = 0.035

A6.3.3 Subject to Rule A6.3.4, an Authorised Firm which uses the Alternative Standardised Approach must calculate its Operational Risk Capital Requirement for its commercial banking business line according to the following formula:

$$K_{CB} = B_{CB} \times m \times LA_{CB}$$

Where:

K_{CB} = the Capital Requirement for the commercial banking business line

B_{CB} = the beta for the commercial banking business line

LA_{CB} = total outstanding commercial banking loans and advances (non-risk weighted and gross of provisions), averaged over the past three years

m = 0.035

A6.3.4 An Authorised Firm may, in calculating its Operational Risk Capital Requirement under Rule A6.3.1:

- (a) aggregate retail and commercial banking using a single beta of 15%; or
- (b) if it is unable to disaggregate gross income in its six other business lines, aggregate the total gross income for these six business lines using a beta of 18%, with negative gross income treated as described in the Standardised Approach.

Guidance

1. For the purposes of the Alternative Standardised Approach, total loans and advances in the retail banking business line consist of the total drawn amounts in the following credit portfolios: retail, SMEs treated as retail, and purchased retail receivables.
2. For commercial banking, total loans and advances consist of the drawn amounts in the following credit portfolios: corporate, sovereign, bank, specialised lending, SMEs treated as corporate and purchased corporate receivables. The book value of securities held in the Non-Trading Book should also be included.
3. The three year average should be calculated on the basis of the last three yearly observations at the end of the Authorised Firm's financial year. When audited figures are not available, business estimates may be used.
4. If an Authorised Firm does not have sufficient income data to meet the three year requirement (e.g. a start-up) it may use its forecasted gross income projections for all or part of the three year time period.
5. In accordance with Rule 6.11.3 of PIB, an Authorised Firm seeking to apply the Standardised Approach or the Alternative Standardised Approach must develop specific policies and have documented criteria for mapping gross income for current business lines and activities into the Standardised Approach or the Alternative Standardised Approach. The criteria must be reviewed and adjusted for new or changing business activities as appropriate. The principles for business line mapping are set out below.

Principles for business line mapping

6. All activities should be mapped into the eight level 1 business lines in a mutually exclusive and jointly exhaustive manner.
7. Any activity which cannot be readily mapped into the business line framework, but which represents an ancillary function to an activity included in the framework, should be allocated to the business line it supports. If more than one business line is supported through the ancillary activity, an objective mapping criteria should be used.
8. When mapping gross income, if an activity cannot be mapped into a particular business line then the business line yielding the highest charge should be used. The same business line equally applies to any associated ancillary activity.
9. An Authorised Firm may use internal pricing methods to allocate gross income between business lines, provided that total gross income for the firm (as would be recorded under the Basic Indicator Approach) still equals the sum of gross income for the eight business lines.
10. The mapping of activities into business lines for Operational Risk capital purposes should be consistent with the definitions of business lines used for regulatory capital calculations in other risk categories, i.e. credit and Market Risk. Any deviations from this principle should be clearly motivated and documented.
11. The mapping process used should be clearly documented. In particular, written business line definitions should be clear and detailed enough to allow third parties to replicate the business line mapping. Documentation should, among other things, clearly motivate any exceptions or overrides and be kept on record.
12. Processes should be in place to define the mapping of any new activities or products and the mapping process to business lines should be subject to independent review.

Table – Mapping of Business Lines

Level 1	Level 2	Activity Groups
Corporate Finance	Corporate Finance	Mergers and acquisitions, underwriting, privatisations, securitisation, research, debt (government, high yield), equity, syndications, IPO, secondary private placements.
	Municipal/Government Finance	
	Merchant Banking	
	Advisory Services	
Trading and Sales	Sales	Fixed income, equity, foreign exchanges, commodities, credit, funding, own position securities, lending and repos, brokerage, debt, prime brokerage.
	Market Making	
	Proprietary Positions	
	Treasury	
Retail Banking	Retail Banking	Retail lending and deposits, banking services, trust and estates.
	Private Banking	Private lending and deposits, banking services, trust and estates, investment advice.
	Card Services	Merchant/commercial/corporate cards, private labels and retail.
Commercial Banking	Commercial Banking	Project finance, real estate, export finance, trade finance, factoring, leasing, lending, guarantees, bills of exchange.
Payment and Settlement	External Clients	Payments and collections, funds transfer, clearing and Settlement. Note: Payment and settlement losses related to an Authorised Firm's own activities would be incorporated in the loss experience of the affected business line.
Agency Services	Custody	Escrow, depository receipts, securities lending (customers) corporate actions.
	Corporate Agency	Issuer and paying agents.
	Corporate Trust	-
Asset Management	Discretionary Fund Management	Pooled, segregated, retail, institutional, closed, open, private Equity.
	Non-Discretionary Fund Management	Pooled, segregated, retail, institutional, closed, open.
Retail Brokerage	Retail Brokerage	Execution and full service.

APP7 NOT USED

APP8 NOT USED

APP9 LIQUIDITY

A9.1 Application for a global liquidity concession

- A9.1.1** (1) This Rule applies to an Authorised Firm carrying on business in or from the DIFC through a Branch that applies for a global liquidity concession under Rule 9.3.2.
- (2) The Authorised Firm must demonstrate to the DFSA that:
- (a) its Branch complies with all applicable liquidity systems and controls requirements in sections 9.2 and 9.2A;
 - (b) its head office is established in a jurisdiction where there are no legal constraints imposed by the home supervisor or any other authority on the provision of liquidity to its branch; and
 - (c) its head office is subject to equivalent or more restrictive liquidity requirements, than those imposed by the DFSA.
- (3) The DFSA may, when considering an application from the Authorised Firm for a global liquidity concession, impose additional or alternative conditions to those specified in (2) or disapply a condition in (2).

Guidance

1. An application for a global liquidity concession pursuant to Rule 9.3.2 should include at least the following:
 - a. a clear description of the home supervisor's requirements for managing, monitoring and controlling liquidity risk;
 - b. a clear description of the systems and controls used by the head office to ensure the adequacy of the Branch's liquidity;
 - c. a written assurance from the Authorised Firm's head office that it will:
 - i. ensure that adequate liquidity is available at all times to support the branch;
 - ii. notify the DFSA, at the same time as it notifies its home supervisor, of any material issues concerning its exposure to Liquidity Risk and issues in relation to its compliance with applicable liquidity limits, including its liquidity coverage ratio; and
 - iii. in the event of a liquidity crisis, provide the DFSA with all relevant information on the whole Authorised Firm's liquidity, and a list of any known constraints on the head office, legal or otherwise, providing the branch with liquidity.
 - d. a notification from the Authorised Firm's home supervisor:
 - i. expressing no objection to the branch obtaining the DFSA's liquidity concession or acknowledging the branch application to the DFSA for a global liquidity concession; and

- ii. providing information about, and confirming, the quality of Liquidity Risk systems and controls and the liquidity exposures at the Authorised Firm's head office.
 2. Under Rule A9.1.1(2)(b), the DFSA will consider liquidity transfer restrictions (e.g. ring-fencing measures, non-convertibility of local currency, foreign exchange controls) imposed under applicable laws, regulations or supervisory requirements in jurisdictions in which a banking group operates which affect the availability of liquidity by inhibiting the transfer of HQLA and fund flows within the Group.
- A9.1.2** (1) An Authorised Firm that has been granted a global liquidity concession must provide the DFSA with ongoing assurance about its Liquidity Risk by:
- (a) submitting to the DFSA at least quarterly a copy of the LCR calculation for the Authorised Firm, as submitted by its head office to its home supervisor;
 - (b) notifying the DFSA immediately of the results of every assessment which its home supervisor conducts which relates to the quality of liquidity systems and controls at its head office;
 - (c) notifying the DFSA in writing immediately of any adverse finding or action taken by its home supervisor;
 - (d) notifying the DFSA in writing immediately of any potential change in the branch funding strategy, business model or material potential change in its balance sheet structure; and
 - (e) notifying the DFSA in writing immediately of any changes relating to its compliance with the conditions referred to in Rule A9.1.1.
- (2) The DFSA may at any time, based on its assessment of the Liquidity Risk exposures of an Authorised Firm, by written notice adjust or exclude any of the requirements in (1), impose additional requirements or cancel the global liquidity concession granted to the Authorised Firm.

A9.2 The Liquidity Coverage Ratio

Guidance

1. The objective of the LCR is to promote short-term resilience of an Authorised Firm's liquidity risk profile. The LCR aims to ensure that an Authorised Firm maintains an adequate level of unencumbered HQLA that can be converted into cash to meet its liquidity needs for a 30 calendar day period under a severe liquidity stress scenario.
2. The LCR is calculated under Rule 9.3.5 using the following formula:

$$\text{LCR} = \frac{\text{Value of stock of HQLA}}{\text{Total Net Cash Outflows over the next 30 calendar days}}$$
3. The LCR has two components:
 - a. Value of the stock of HQLA in stressed conditions; and
 - b. Total Net Cash Outflows, calculated according to the scenario parameters outlined in this section.

4. The stress scenario entails both institution-specific and systemic shocks including:
 - a. the run-off of a proportion of retail deposits;
 - b. a partial loss of unsecured wholesale funding capacity;
 - c. a partial loss of secured, short-term financing with certain collateral and counterparties;
 - d. additional contractual outflows that would arise from a downgrade in the Authorised Firm's public credit rating, where applicable, by up to and including three notches, including collateral posting requirements;
 - e. increases in market volatility that affect the quality of collateral or potential future exposure of derivative positions and so require larger collateral haircuts or additional collateral, or lead to other liquidity needs;
 - f. unscheduled draws on committed but unused credit and liquidity facilities that the Authorised Firm has provided to its clients; and
 - g. the potential need for the Authorised Firm to buy back debt or honour non-contractual obligations to mitigate reputational risk.

A9.2.1 An Authorised Firm must calculate its LCR on an ongoing basis and separately for each significant currency. An Authorised Firm must report to the DFSA its aggregate LCR calculation in USD.

Guidance

A currency is considered significant if the aggregate liabilities denominated in that currency amount to 5% or more of the Authorised Firm's total liabilities.

High Quality Liquid Assets (HQLA)

Guidance

Assets that meet the conditions in Rules A9.2.2 to A9.2.9 are considered to be HQLA. Those assets are considered to be HQLA as they can be converted easily and immediately into cash at little or no loss of value. To qualify as HQLA, assets should be liquid in markets during a time of stress. In determining whether or not the market for an asset can be relied upon to raise liquidity during a time of stress, the following factors should be taken into account:

1. fundamental characteristics, for example:
 - a. low risk: high credit standing of the issuer and a low degree of subordination, low duration, low legal risk, low inflation risk, denomination in a convertible currency with low foreign exchange risk;
 - b. ease and certainty of valuation;
 - c. low correlation with risky assets, not subject to wrong-way risk; and
 - d. listing on a developed and recognised exchange.
2. market-related characteristics, for example:

- a. active and sizable market, including active outright sale or repo markets at all times. This can be demonstrated through:
 - i. historical evidence of market breadth and market depth (low bid-ask spreads, high trading volumes, large and diverse number of market participants); or
 - ii. existence of robust market infrastructure (presence of multiple committed market makers);
- b. low price volatility, including historical evidence of relative stability of market terms (e.g. prices and haircuts) and volumes during stressed periods; or
- c. flight to quality, i.e. that historically the market has shown a tendency to move into these types of high quality assets in a systemic crisis.

HQLA – general operational requirements

A9.2.2 To be eligible as HQLA, assets in the portfolio of HQLA must be appropriately diversified in terms of type of assets, type of issuer and specific counterparty or issuer.

A9.2.3 To be eligible as HQLA, assets must meet the following requirements:

- (a) the assets must be under the control of the specific function or functions charged with managing the liquidity of the Authorised Firm who must have the continuous authority and legal and operational capability to monetise any asset in the stock; and
- (b) a representative portion of the assets in the stock of HQLA must be monetised via outright sale or via a repurchase agreement periodically and at least annually by the Authorised Firm to test its access to the market, the effectiveness of its processes for monetisation, the availability of the assets, and to minimise the risk of negative signalling during a period of actual stress.

A9.2.4 To be eligible as HQLA, an asset must also meet the following requirements:

- (a) the asset must be unencumbered and free of legal, regulatory, contractual or other restrictions that affect the ability of the Authorised Firm to liquidate, sell, transfer, or assign the asset;
- (b) the asset must not be pledged, either explicitly or implicitly, to secure, collateralise or credit-enhance any transaction, nor be designated to cover operational costs (such as rents and salaries); and
- (c) an asset received in a reverse repo or securities financing transactions that is held at the Authorised Firm, is eligible for inclusion in the stock of HQLA only if the asset has not been rehypothecated and is legally and contractually available for the Authorised Firm's use.

Guidance

1. The requirements in Rules A9.2.2 to A9.2.4 are intended to ensure that the stock of HQLA is managed in such a way that the Authorised Firm can, and is able to demonstrate that it can, immediately use the assets as a source of contingent funds that is available to convert into cash

to fill funding gaps between cash inflows and outflows at any time during the 30-day stress period, with no restriction on the use of the liquidity generated.

2. Under Rule A9.2.3(a), the control of the HQLA may be evidenced either by:
 - (a) maintaining assets in a separate pool managed by the identified liquidity management function (typically the treasurer) with the sole intent to use it as a source of contingent funds; or
 - (b) demonstrating that the relevant function can monetise the asset at any point in the 30-day stress period and that the proceeds are available to the function throughout the 30-day stress period without directly conflicting with a stated business or risk management strategy.
3. Operational capability to monetise assets referred to in Rule A9.2.3(b), requires procedures and appropriate systems to be in place. This includes providing the liquidity management function with access to all necessary information to monetise any asset at any time. Monetisation of the asset should be executable operationally within the standard settlement period for the asset class in the relevant jurisdiction.

Caps on different types of HQLA

- A9.2.5**
- (1) Assets eligible to be included in the stock of HQLA for the purpose of the LCR calculation are classified under the following two categories:
 - (a) Level 1 HQLA, consisting of the highest quality and most liquid assets; and
 - (b) Level 2 HQLA, including Level 2A HQLA and Level 2B HQLA, consisting of other high quality liquid assets.
 - (2) When calculating the total stock of HQLA, an Authorised Firm must apply the following caps in respect of each category of assets:
 - (a) Level 1 HQLA can be included in the total stock of HQLA without any limit (i.e. up to 100% of HQLA);
 - (b) Total Level 2 HQLA, including both Level 2A HQLA and Level 2B HQLA, can comprise only up to 40% of the total stock of HQLA; and
 - (c) Level 2B HQLA can comprise only up to 15% of the total stock of HQLA within the overall 40% limit on Level 2 HQLA in (b).
 - (3) The caps on Level 2 HQLA and Level 2B HQLA must be determined after applying the haircuts required under Rules A9.2.7 and A9.2.8, and after unwinding the amounts of HQLA involved in short-term secured funding, secured lending and collateral swap transactions maturing within 30 calendar days that involve the exchange of HQLA.
 - (4) The assets to be included in each category of HQLA must be restricted to assets being held or owned by the Authorised Firm on the first day of the stress period, irrespective of their residual maturity.

Guidance

1. The following Guidance is intended to illustrate how Rule A9.2.5 should be applied in practice.
2. Under Rule A9.2.5(3) the adjusted amounts of HQLA should be calculated as the amount of HQLA that would result after unwinding those short-term secured funding, secured lending and collateral swap transactions that involve the exchange of any HQLA for any other HQLA.
3. The calculation of the stock of HQLA under Rule A9.2.5 can be expressed as the following formula:

Stock of HQLA = Level 1 HQLA + Level 2A HQLA + Level 2B HQLA – Adjustment for 15% cap – Adjustment for 40% cap

Where:

- a. Adjustment for 15% cap = Max (Adjusted Level 2B HQLA – 15/85 x (Adjusted Level 1 HQLA + Level 2A HQLA), Adjusted Level 2B HQLA – 15/60 x (Adjusted Level 1 HQLA, 0)
- b. Adjustment for 40% cap = Max ((Adjusted Level 2A HQLA + Adjusted Level 2B HQLA – Adjustment for 15% cap) – 2/3 x Adjusted Level 1 HQLA, 0)

Level 1 HQLA

A9.2.6 (1) Level 1 HQLA must be valued at market value.

(2) Level 1 HQLA consists of:

- (a) banknotes and coin;
- (b) central bank reserves, to the extent that such reserves are capable of being drawn down immediately in times of stress;
- (c) marketable securities representing claims on or claims guaranteed by sovereigns, central banks, Public Sector Entities (PSEs), the Bank for International Settlements, the International Monetary Fund, the European Central Bank and European Commission or Multilateral Development Banks (MDBs), and that satisfy all of the following conditions:
 - (i) they are assigned a zero % risk-weight according to Chapter 4 and App4 of this Module;
 - (ii) they are traded in large, deep and active repo or cash markets characterised by a low level of concentration;
 - (iii) they have a proven record as a reliable source of liquidity in the markets (repo or sale) even during stressed market conditions; and
 - (iv) they are not an obligation of a financial institution or any of its associated entities;
- (d) in the case of sovereigns that are not eligible for zero % risk-weight, sovereign or central bank debt securities issued in domestic currencies by the sovereign or central bank in the country in which the liquidity risk is

being taken or in the Authorised Firm's home jurisdiction, where those securities satisfy all of the conditions in paragraph (c) (ii)(iii) and (iv);

- (e) in the case of sovereigns that are not eligible for zero % risk-weight, domestic sovereign or central bank debt securities issued in foreign currencies, up to the amount of the Authorised Firm's stressed net cash outflows in that specific foreign currency stemming from the Authorised Firm's operations in the jurisdiction where the Authorised Firm's liquidity risk is being taken, where those securities satisfy all of the conditions in paragraph (c) (ii)(iii) and (iv); and
- (f) any other types of assets approved by the DFSA under Rule A9.2.9 as being eligible to be Level 1 HQLA.

Level 2A HQLA

- A9.2.7** (1) Level 2A HQLA must be valued at market value and subject to a 15% haircut.
- (2) Level 2A HQLA consists of:
- (a) marketable securities representing claims on or guaranteed by sovereigns, central banks, PSEs or MDBs that satisfy all of the following conditions:
 - (i) they are assigned a 20% risk-weight according to Chapter 4 and App4 of this Module;
 - (ii) they are traded in large, deep and active repo or cash markets characterised by a low level of concentration;
 - (iii) they have a proven record as a reliable source of liquidity in the markets (repo or sale) even during stressed market conditions (i.e. maximum decline of price or increase in haircut over a 30-day period during a relevant period of significant liquidity stress not exceeding 10%); and
 - (iv) they are not an obligation of a financial institution or any of its associated entities;
 - (b) corporate debt securities (including commercial paper) and covered bonds that satisfy all of the following conditions:
 - (i) in the case of corporate debt securities: they must not be issued by a financial institution or any of its associated entities and must include only plain vanilla assets (i.e. not include complex structured products or subordinated debt) whose valuation is readily available based on standard methods and does not depend on private knowledge;
 - (ii) in the case of covered bonds: they must not be issued by the Authorised Firm itself or any of its associated entities;

- (iii) the assets must have a Credit Quality Grade of 1 from a recognised ECAI or, if the assets do not have a credit assessment by a recognised ECAI, they must be internally rated as having a probability of default (PD) corresponding to a Credit Quality Grade of 1;
- (iv) they must be traded in large, deep and active repo or cash markets characterised by a low level of concentration; and
- (v) they must have a proven record as a reliable source of liquidity in the markets (repo or sale) even during stressed market conditions (i.e. maximum decline of price or increase in haircut over a 30-day period during a relevant period of significant liquidity stress not exceeding 10%); and
- (c) any other types of assets approved by the DFSA under Rule A9.2.9 as being eligible to be Level 2A HQLA.

Level 2B HQLA

- A9.2.8** (1) Level 2B HQLA must be valued at market value and subject to an appropriate haircut, as specified in (2), for each type of asset.
- (2) Level 2B HQLA consists of:
- (a) residential mortgage backed securities that satisfy all of the following conditions, subject to a 25% haircut:
 - (i) they are not issued by, and the underlying assets have not been originated by, the Authorised Firm itself or any of its affiliated entities;
 - (ii) they have a Credit Quality Grade of 1 from a recognised ECAI;
 - (iii) they are traded in large, deep and active repo or cash markets characterised by a low level of concentration;
 - (iv) they have a proven record as a reliable source of liquidity in the markets (repo or sale) even during stressed market conditions, (i.e. maximum decline of price or increase in haircut over a 30-day period during a relevant period of significant liquidity stress not exceeding 20%);
 - (v) the underlying asset pool is restricted to residential mortgages and does not contain structured products;
 - (vi) the underlying mortgages are “full recourse” loans (i.e. in the case of foreclosure the mortgage owner remains liable for any shortfall in sales proceeds from the property) and have a maximum loan-to-value ratio (LTV) of 80% on average at issuance; and
 - (vii) the securitisations are subject to “risk retention” regulations which

require issuers to retain an interest in the assets they securitise;

- (b) corporate debt securities (including commercial paper) that satisfy all of the following conditions, subject to a 50% haircut:
 - (i) they are not issued by a financial institution or any of its affiliated entities;
 - (ii) they have a Credit Quality Grade of 2 or 3 from a recognised ECAI or, in the case the assets do not have a credit assessment by a recognised ECAI, are internally rated as having a probability of default (PD) corresponding to a Credit Quality Grade of 2 or 3;
 - (iii) they are traded in large, deep and active repo or cash markets characterised by a low level of concentration; and
 - (iv) they have a proven record as a reliable source of liquidity in the markets (repo or sale) even during stressed market conditions, (i.e. maximum decline of price or increase in haircut over a 30-day period during a relevant period of significant liquidity stress not exceeding 20%);
- (c) equity shares that satisfy all of the following conditions, subject to a 50% haircut:
 - (i) they are not issued by a financial institution or any of its affiliated entities;
 - (ii) they are exchange traded and centrally cleared;
 - (iii) they are a constituent of the major stock index in the home jurisdiction, or where the liquidity risk is taken, as decided by the supervisor in the jurisdiction where the index is located;
 - (iv) they are denominated in the domestic currency of an Authorised Firm's home jurisdiction or in the currency of the jurisdiction where an Authorised Firm's liquidity risk is taken;
 - (v) they are traded in large, deep and active repo or cash markets characterised by a low level of concentration; and
 - (vi) they have a proven record as a reliable source of liquidity in the markets (repo or sale) even during stressed market conditions, (i.e. maximum decline of price or increase in haircut over a 30-day period during a relevant period of significant liquidity stress not exceeding 40%); and
- (d) any other types of assets approved by the DFSA under Rule A9.2.9 as being eligible to be Level 2B HQLA.

Approval of other types of HQLA

- A9.2.9** (1) The DFSA may approve other types of assets (in addition to those specified in Rules A9.2.6 to A9.2.8) as being eligible to be included in the stock of HQLA for the purposes of the calculation of the LCR.
- (2) If the DFSA approves assets under (1), it must specify whether they are to be classified as Level 1 HQLA or Level 2 HQLA and the haircut, if any, to be applied to them.

Guidance

The DFSA may use its discretion under Rule A9.2.9 to approve other types of assets as HQLA including, but not limited to, Shar'iah compliant financial products. When the DFSA approves assets it may define the conditions that the assets must satisfy to be treated as HQLA. It must specify whether the assets are to be treated as Level 1 HQLA or Level 2A or 2B HQLA.

Other provisions relating to LCR calculation

- A9.2.10** For the purpose of calculating the LCR, if an eligible asset within HQLA becomes ineligible (e.g. due to a rating downgrade), an Authorised Firm is allowed to keep the asset in its stock of HQLA for an additional 30 calendar days to allow time to adjust its stock as needed or replace the asset.
- A9.2.11** For the purpose of calculating a consolidated LCR for a Financial Group, where applicable, qualifying HQLA held to meet statutory liquidity requirements at a legal entity or sub-consolidated level may be included in the stock at the consolidated level only to the extent that the related risks are also reflected in the consolidated LCR. Any surplus of HQLA held at the legal entity can be included in the consolidated stock of HQLA only if those assets would also be freely available to the consolidated parent entity in times of stress.
- A9.2.12** An Authorised Firm must be able to meet its liquidity needs in each currency in which it has a significant exposure. The currencies of the stock of HQLA of an Authorised Firm must be similar in composition to its liquidity needs by currency.

Total Net Cash Outflow

- A9.2.13** (1) An Authorised Firm must calculate its Total Net Cash Outflow over the following 30 calendar days in accordance with the following formula:

$$\begin{aligned} &\text{Total Net Cash Outflows over the next 30 calendar days} \\ &= \\ &\quad \text{total expected cash outflows} \\ &- \\ &\quad \text{whichever is the lesser amount of total expected cash inflows or 75\% of} \\ &\quad \text{total expected cash outflows} \end{aligned}$$

- (2) Total expected cash outflows are calculated by multiplying the outstanding balances of various categories or types of liabilities and off-balance sheet

commitments by the rates at which they are expected to run off or be drawn down.

- (3) Total expected cash inflows are calculated by multiplying the outstanding balances of various categories of contractual receivables by the rates at which they are expected to flow in.
- (4) To ensure a minimum level of HQLA holdings at all times, total cash inflows are subject to an aggregate cap of 75% of total expected cash outflows.

A9.2.14 An Authorised Firm must not double-count items. That is, for assets included as part of the eligible stock of HQLA, the associated cash inflows arising from such assets must not be counted as cash inflows for the purpose of calculating the net cash outflows over the next 30 calendar days.

Cash Outflows

A9.2.15 The following table specifies, for each of the various categories or types of liabilities and off-balance sheet commitments, the rates at which they are expected to run off or be drawn down for the purpose of calculating the LCR.

Cash Outflows	
Item	Factor
A. Retail Deposits:	
Demand deposit and term deposits (less than 30 days maturity):	
• Stable deposits	5%
• Less stable retail deposits	10%
Term deposits with residual maturity greater than 30 days	0%
B. Unsecured Wholesale Funding:	
Demand and term deposits (less than 30 days maturity) provided by small business customers:	
• Stable deposits	5%
• Less stable deposits	10%
Small business customers - Term deposits with residual maturity greater than 30 days with no legal right to withdraw or a withdrawal with a significant penalty	0%
Operational deposits or operational accounts generated by clearing, custody and cash management activities:	25%
• Portion covered by deposit insurance	5%
Cooperative banks or institutions offering Islamic financial services in an institutional network (qualifying deposits with the centralised institution)	25%

Cash Outflows	
Item	Factor
Non-financial corporates, sovereigns, central banks, multilateral development banks and PSEs:	40%
If the entire amount is fully covered by a deposit protection scheme	20%
Other legal entity customers	100%
C. Secured Funding:	
Secured funding transactions with a central bank counterparty or backed by Level 1 HQLA with any counterparty	0%
Secured funding transactions backed by Level 2A HQLA, with any counterparty	15%
Secured funding transactions backed by non-Level 1 HQLA or non-Level 2A HQLA, with domestic sovereigns, multilateral development banks, or domestic PSEs as a counterparty	25%
Backed by RMBS eligible for inclusion in Level 2B HQLA	25%
Backed by other Level 2B HQLA	50%
All other secured funding transactions	100%
D. Additional Requirements:	
Derivatives cash outflows or Shari'a compliant hedging	100%
Liquidity needs (e.g. collateral calls) related to financing transactions, derivatives and other contracts	100%
Market valuation changes on non-Level 1 HQLA posted collateral securing derivatives	20%
Excess collateral held by a bank related to derivative transactions that could contractually be called at any time by its counterparty	100%
Liquidity needs related to collateral contractually due from the reporting bank on derivatives transactions	100%
Increased liquidity needs related to derivative transactions that allow collateral substitution to non-HQLA assets	100%
Market valuation changes on derivatives transactions (largest absolute net 30-day collateral flows realised during the preceding 24 months)	100%
ABCP, SIVs, Conduits, etc:	
Loss of funding on Asset Backed Securities, covered bonds and other structured financing instruments	100%
Loss of funding on ABCP, SIVs, SPVs, etc	100%
Undrawn committed credit and liquidity facilities:	

Cash Outflows	
Item	Factor
• Credit and Liquidity Facilities: Retail and small and medium-sized enterprise clients	5%
• Credit Facilities: Non-financial corporates, sovereigns and central banks, PSEs, MDBs	10%
• Liquidity Facilities: Non-financial corporates, sovereigns and central banks, PSEs, MDBs	30%
• Credit and Liquidity Facilities: Banks subject to prudential supervision	40%
• Credit Facilities: Other financial institutions (include securities firms, insurance companies, fiduciaries and beneficiaries)	40%
• Liquidity Facilities: Other financial institutions (include securities firms, insurance companies, fiduciaries and beneficiaries)	100%
• Credit and Liquidity Facilities: Other legal entity customers	100%
• Other contractual obligations to financial institutions	100%
• Other contractual obligations to retail and non-financial corporate clients	100%
Other contingent funding obligations:	
• Non-contractual obligations related to potential liquidity draws from joint ventures or minority investments in entities	100%
• Trade finance-related obligations (including letters of credit and guarantees)	3%
• Unconditionally revocable "uncommitted" credit and liquidity facilities	5%
• Guarantees and letters of credit unrelated to trade finance obligations	10%
Non-contractual obligations:	
• Debt-buy back requests (incl. related conduits)	100%
• Structured products	10%
• Managed funds	10%
• Other non-contractual obligations	100%
Outstanding debt securities with remaining maturity > 30 days	100%
Non contractual obligations where customer short positions are covered by other customers' collateral	50%
Other contractual cash outflows	100%

Guidance

1. The following Guidance sets out the DFSA’s views about how the Table to Rule A9.2.15 should be applied to different items.

Retail Deposits:

2. Retail deposits should include deposits from individuals placed with an Authorised Firm. Deposits from legal entities, sole proprietorships or partnerships should be included in wholesale deposit categories. Deposits may include demand deposits and term deposits, unless otherwise excluded.
3. Under COB section 4.2, an Authorised Firm can only accept deposits from individuals who are Professional Clients.
4. Deposits from individuals are divided under the Table into ‘stable’ and ‘less stable’ deposits. Stable deposits should include the portion of deposits that are fully covered by an effective deposit insurance scheme or by a public guarantee that provides equivalent protection and where:
 - a. the depositor has other established relationships with the Authorised Firm that make deposit withdrawal highly unlikely; or
 - b. the deposits are in transactional accounts (e.g. accounts where salaries are automatically credited).
5. If an Authorised Firm is not able to readily identify which retail deposits would qualify as “stable” according to paragraph 4, it should place the full amount in the “less stable” buckets.
6. Less stable deposits should consist of the portion of deposits that do not meet the conditions in paragraph 4 and also include types of deposits more likely to be withdrawn in a time of stress. These should include high-value deposits (i.e. deposits above any deposit insurance limit), deposits from customers who do not have established relationships with an Authorised Firm that make the deposit withdrawal unlikely, deposits from sophisticated or high net worth individuals, deposits where the internet is integral to the design, marketing and use of the account (on-line accounts) and deposits with promotional interest rates (i.e. that are heavily rate-driven).
7. Cash outflows related to retail term deposits with a residual maturity or withdrawal notice period of greater than 30 days should be excluded from total expected cash outflows only if the depositor has no legal right to withdraw deposits within the 30-day period of the LCR, or if early withdrawal results in a significant penalty that is materially greater than the loss of interest. If an Authorised Firm allows a depositor to withdraw such deposits despite a clause that says the depositor has no legal right to withdraw, the entire category of these funds should be treated as demand deposits.

Unsecured wholesale funding:

8. Unsecured wholesale funding should consist of liabilities and general obligations raised from non-natural persons (i.e. legal entities, including sole proprietorships and partnerships) and not collateralised by legal rights to specifically designated assets owned by the Authorised Firm accepting the deposit in the case of bankruptcy, insolvency, liquidation or resolution. Obligations related to derivative contracts should be excluded from this category.
9. The wholesale funding included in the LCR should consist of all funding that is callable within the LCR’s period of 30 days or that has its earliest possible contractual maturity date within this period (such as maturing term deposits and unsecured debt securities), as well as funding with an undetermined maturity. This should include all funding with options that are exercisable at the investor’s discretion within the 30-day period.
10. Wholesale funding that is callable by the funds provider subject to a contractually defined and binding notice period longer than the 30-day period should not be included.

11. Unsecured wholesale funding provided by small and medium-sized enterprise customers should be treated as deposits from individuals where:
 - a. the deposits and other extensions of funds made by non-financial small and medium-sized enterprise customers are managed as retail accounts and are generally considered as having similar liquidity risk characteristics to retail accounts; and
 - b. the total aggregated funding raised from a small and medium-sized enterprise customer is less than USD 1 million (on a consolidated basis where applicable).

Operational deposits

12. Operational deposits should consist of those deposits where customers place, or leave, deposits with an Authorised Firm to facilitate their access and ability to use payment and settlement systems and otherwise make payments. Balances can be included only if the customer has a substantive dependency on the Authorised Firm and the deposit is required for such activities.
13. Qualifying activities in this context refer to clearing, custody or cash management activities where the customer is reliant on the Authorised Firm to perform these services as an independent third-party intermediary in order to fulfil its normal banking activities over the next 30 days. These services should be provided to institutional customers under a legally binding agreement and the termination of such agreements should be subject either to a notice period of at least 30 days or to significant switching costs to be borne by the customer if the operational deposits are moved before 30 days.
14. Qualifying operational deposits generated by such an activity should consist of deposits which are:
 - a. by-products of the underlying services provided by the Authorised Firm;
 - b. not offered by the Authorised Firm in the wholesale market in the sole interest of offering interest income; and
 - c. held in specifically designated accounts and priced without giving an economic incentive to the customer to leave excess funds on these accounts.
15. Any excess balances that could be withdrawn without jeopardising these clearing, custody or cash management activities should not qualify as operational deposits.

Liquidity facilities

16. A liquidity facility should consist of any committed, undrawn back-up facility that would be used to refinance the debt obligations of a customer in situations where such a customer is unable to roll over that debt in financial markets. The amount of any commitment to be treated as a liquidity facility should consist of the amount of the outstanding debt issued by the customer (or proportionate share of a syndicated facility) maturing within a 30-day period that is backstopped by the facility. Any additional capacity of the facility should be treated as a committed credit facility. General working capital facilities for corporate entities (e.g. revolving credit facilities in place for general corporate or working capital purposes) should not be classified as liquidity facilities, but as credit facilities.
17. Despite paragraph 16, any facilities provided to hedge funds, money market funds and special purpose funding vehicles, or other vehicles used to finance an Authorised Firm's own assets, should be captured in their entirety as a liquidity facility to a financial institution.

Unrestricted PSIA's and other Shari'a compliant products

18. For the purposes of calculating cash outflows, Unrestricted PSIA's should be treated similarly to the relevant category of deposits specified in the Table. The appropriate run-off factor for a PSIA will depend on the contractual withdrawal rights of the investment account holders and whether it is a retail or wholesale account.

19. For commodity Murabaha transactions, a run-off factor of 100% should be applied to the balance of the Murabaha payable, if the remaining term of the contract does not exceed 30 days. If early withdrawal of the original amount is allowed at the discretion of the Authorised Firm with no mark-up, then the applicable run-off factor will be the same as that for the relevant category of deposit or Unrestricted PSIA under the Table.

Cash Inflows

- A9.2.16** (1) When considering its available cash inflows, an Authorised Firm may include contractual inflows from outstanding exposures only if they are fully performing and there is no reasonable basis to expect a default within the 30-day period. Contingent inflows are not included in total net cash inflows.
- (2) Where an Authorised Firm is overly reliant on cash inflows from one or a limited number of wholesale counterparties, the DFSA may set an alternative limit on the level of cash inflows that can be included in the LCR.
- A9.2.17** (1) The DFSA may allow an Authorised Firm to recognise as cash inflow, access to a parent entity's funds via a committed funding facility if the Authorised Firm is a subsidiary of a foreign bank. In such instances, the committed funding facility from the parent entity must meet both of the following criteria:
- (a) the facility must be an irrevocable commitment and must be appropriately documented; and
 - (b) the facility must be quantified.
- (2) A committed funding facility from a parent entity referred to in (1) can be recognised as a cash inflow only from day 16 of the LCR scenario. The cash inflow from a parent entity can be sufficient in size to cover only net cash outflows against items with a maturity or next call date between days 16 and 30 of the LCR.
- A9.2.18** The following table specifies, for each of the various categories and types of contractual receivables, the rates at which they are expected to flow in for the purpose of the calculation of the LCR:

Cash Inflows	
Item	Factor
Maturing secured lending (incl. reverse repos and securities borrowing), backed by the following as collateral:	
• Level 1 HQLA	0%
• Level 2A HQLA	15%
• Level 2B HQLA - eligible RMBS	25%
• Level 2B HQLA - Other assets	50%
• Margin lending backed by all other collateral	50%
• All other assets	100%

Cash Inflows	
• Credit or liquidity facilities provided to the reporting Bank	0%
• Operational deposits or operational accounts held at other financial institutions (including deposits held at centralised institution of network of co-operative banks or operational accounts held at institutions offering Islamic financial services)	0%
Other inflows by counterparty	-
• Amounts receivable from retail counterparties	50%
• Amounts receivable from non-financial wholesale counterparties, from transactions other than those listed in the above inflow categories	50%
• Amounts receivable from financial institutions and central banks, from transactions other than those listed in the above inflow categories	100%
• Net derivative receivables	100%
• Net Shari'a compliant hedging cash inflows	100%
• Other contractual cash inflows	100%

Guidance

Maturing secured lending, including reverse repos and securities borrowing

1. An Authorised Firm should assume that maturing reverse repurchase or securities borrowing agreements secured by Level 1 HQLA will be rolled over and will not give rise to any cash inflows (zero %). Maturing reverse repurchase or securities borrowing agreements secured by Level 2 HQLA should be modelled as cash inflows, equivalent to the relevant haircut for the specific assets. An Authorised Firm is assumed not to roll-over maturing reserve repurchase or securities borrowing agreements secured by non-HQLA assets and can assume it will receive 100% of the cash related to those agreements. Collateralised loans extended to customers for the purpose of taking leveraged trading positions, i.e. margin loans, should be modelled with a 50% cash inflow from contractual inflows made against non-HQLA collateral.
2. An exception to paragraph 1 is the situation where, if the collateral obtained through reverse repo, securities borrowing or collateral swaps, which matures within the 30-day period, is re-used (i.e. rehypothecated) and is tied up for 30 days or longer to cover short positions. An Authorised Firm should then assume that such reverse repo or securities borrowing arrangements will be rolled over and will not give rise to any cash inflows (zero %), reflecting its need to continue to cover the short position or to repurchase the relevant securities.
3. An Authorised Firm should manage its collateral so that it is able to fulfil obligations to return collateral whenever the counterparty decides not to roll-over any reverse repo or securities lending transaction. This is especially the case for non-HQLA collateral, since such outflows are not captured in the LCR framework.

Lines of credit

4. Lines of credit, liquidity facilities and other contingent funding facilities that an Authorised Firm holds at other institutions for its own purposes should be assumed to be able to be drawn and so such facilities should receive a zero % inflow rate.

Inflows by counterparty

5. All inflows should be taken only at the latest possible date, based on the contractual rights available to counterparties. Inflows from loans that have no specific maturity should not be included, with the exception of minimum payments of principal, fee or interest associated with an open maturity loan.
6. Operational deposits: a zero % inflow rate should apply to deposits held at other financial institutions for operational purposes.

Other cash inflows

7. Other contractual cash inflows: other contractual cash inflows should be included under this category. Cash inflows related to non-financial revenues should not be taken into account in the calculation of the net cash outflows for the purposes of the LCR. These items should receive an inflow rate of 100%.

A9.3 The Maturity Mismatch approach

Including inflows (assets) and outflows (liabilities) in the timebands

- A9.3.1**
- (1) Outflows (liabilities) must be included in the Maturity Ladder according to their earliest contractual maturity.
 - (2) Contingent liabilities may be excluded from the maturity mismatch ratio calculation only if there is a likelihood that the conditions necessary to trigger them will not be fulfilled within the ratio period.
 - (3) Inflows (assets) must be included in the Maturity Ladder according to their latest contractual maturity.

Including liquid assets in the Maturity Ladder

- A9.3.2**
- (1) Liquid assets are included at market value in the relevant section of the Maturity Ladder.
 - (2) An asset is regarded as liquid if:
 - (a) prices are regularly quoted for the asset;
 - (b) the asset is regularly traded;
 - (c) the asset may readily be sold, including by repurchase agreement, either on an exchange, or in a deep and liquid market for payment in cash; and

- (d) settlement is according to a prescribed timetable rather than a negotiated timetable.
- (3) Deleted.
- (4) The discount factor to be applied to types of marketable assets when completing the maturity mismatch ratio calculation must be determined by reference to the following table and Rules A9.2.6 to A9.2.9 (which define which assets are eligible to be included in each HQLA level):

Liquid Assets	Discount
Level 1 HQLA	0%
Level 2A HQLA	15%
Level 2B HQLA - Eligible asset backed securities	25%
Level 2B HQLA - Other HQLA	50%
Non-HQLA eligible trading assets that are Investment Grade	60%

- (5) The DFSA may vary the discounts to reflect the conditions of a particular market or institution.

A9.4 The Net Stable Funding Ratio (NSFR)

Guidance

- An Authorised Firm's NSFR is the amount of Available Stable Funding (ASF) relative to the amount of Required Stable Funding (RSF). The NSFR should be at least 100% at all times (see Rule 9.3.12). The NSFR is calculated under Rule 9.3.12 using the following formula:

$$\text{NSFR} = \frac{\text{ASF} \times 100}{\text{RSF}}$$

- For the purposes of the NSFR Requirement, an Authorised Firm's ASF and RSF are calibrated to reflect the presumed degree of stability of liabilities and liquidity of assets.
- This Appendix sets out how the ASF and RSF are to be calculated.

Available Stable Funding (ASF)

- A9.4.1** (1) An Authorised Firm must calculate its ASF by:
- assigning the carrying value of each liability and capital instrument to the applicable ASF Category set out in the table;
 - adjusting the carrying value of each liability and capital instrument by multiplying it by the applicable ASF Factor as set out in the table; and
 - adding together each adjusted carrying value.

ASF Categories and associated ASF Factors	
ASF Factor	Components of ASF Category
100%	(a) the total amount of regulatory capital, before the application of capital deductions, excluding the proportion of Tier 2 instruments with residual maturity of less than one year; (b) the total amount of any capital instrument not included in (a) that has an effective residual maturity of one year or more, but excluding any instruments with explicit or embedded options that, if exercised, would reduce the expected maturity to less than one year; and (c) the total amount of secured and unsecured borrowings and liabilities (including term deposits) with effective residual maturities of one year or more. Cash flows falling below the one-year horizon but arising from liabilities with a final maturity greater than one year do not qualify for the 100% ASF Factor.
95%	“stable” non-maturity (demand) deposits, term deposits and/or PSIAus with residual maturities of less than one year provided by retail and small business customers.
90%	“less stable” non-maturity (demand) deposits, term deposits and/or PSIAus with residual maturities of less than one year provided by retail and small business customers.
50%	(a) funding (secured and unsecured) with a residual maturity of less than one year provided by non-financial corporate customers; (b) operational deposits or operational accounts; (c) funding with residual maturity of less than one year from sovereigns, public sector enterprises (PSEs), and multilateral and national development banks; (d) other funding (secured and unsecured) not included in the categories above with residual maturity between six months to less than one year, including funding from Central Banks and financial institutions; and (e) funding with residual maturity of less than one year provided by non-financial corporate customers.
0%	(a) all other liabilities and equity categories not included in the above categories, including other funding with residual maturity of less than six months from Central Banks and financial institutions; (b) other liabilities without a stated maturity. This category may include short positions and open maturity positions. Two exceptions can be recognised for liabilities without a stated maturity:

	(i) deferred tax liabilities, which should be treated according to the nearest possible date on which such liabilities could be realised; and (ii) minority interest, which should be treated according to the term of the instrument, usually in perpetuity. These liabilities would then be assigned either a 100% ASF Factor, if the effective maturity is one year or greater, or 50%, if the effective maturity is between six months and less than one year; (c) NSFR derivative liabilities (net of NSFR derivative assets) if NSFR derivative liabilities are greater than NSFR derivative assets; (d) Net NSFR Shari'a compliant hedging liabilities (if NSFR Shari'a compliant hedging liabilities are greater than NSFR Shari'a compliant hedging assets); and (e) "trade date" payables arising from purchases of financial instruments, foreign currencies and commodities that: (i) are expected to settle within the standard settlement cycle or period that is customary for the relevant exchange or type of transaction; or (ii) have failed to, but are still expected to, settle.
--	--

Guidance

1. An Authorised Firm's ASF represents the relative stability of its funding sources, including the contractual maturity of its liabilities and the differences in the propensity of different types of funding providers to withdraw their funding.
2. The calibration of the ASF reflects the stability of liabilities, taking into account a number of factors:
 - a. funding tenor – longer-term liabilities are assumed to be more stable than short-term liabilities; and
 - b. funding type and counterparty – it is assumed that short-term deposits or PSIAus (i.e. maturing in less than one year) provided by retail customers and funding provided by small business customers are behaviourally more stable than wholesale funding of the same maturity from other counterparties.
3. For ASF purposes, when determining the maturity of an equity or liability instrument, investors are assumed to redeem a call option at the earliest possible date. For funding with options exercisable at the firm's discretion, the DFSA will take into account reputational factors that may limit the firm's ability not to exercise the option. In particular, where the market expects certain liabilities to be redeemed before their legal final maturity date, the firm should assume such behaviour for the purpose of the NSFR Requirement and include these liabilities in the corresponding ASF Category. For long-dated liabilities, only the portion of cash flows falling at or beyond the six-month and one-year time horizons should be treated as having an effective residual maturity of six months or more and one year or more, respectively.

4. Derivative liabilities are to be calculated first based on the replacement cost for derivative contracts (obtained by marking to market) where the contract has a negative value. When an eligible bilateral netting contract is in place that meets the conditions specified in the instructional guidelines for Form B300 set out in PRU Section 1.48, the replacement cost for the set of derivative exposures covered by the contract will be the net replacement cost.
5. In calculating NSFR derivative liabilities, collateral posted in the form of variation margin in connection with derivative contracts, regardless of the asset type, must be deducted from the negative replacement cost amount.
6. Carrying value is the amount at which a liability or equity instrument is recorded before the application of any regulatory deductions, filters or other adjustments.
7. Regulatory capital consists of the sum of:
 - a. T1 Capital (going-concern capital):
 - i CET1; and
 - ii AT1; and
 - b. T2 Capital (gone-concern capital).
8. The Guidance in paragraphs 4 to 6 and 12 to 15 after Rule A9.2.15 should be used to determine whether a deposit or PSIAu is to be treated as “stable”, “less stable” or “operational”.

Required Stable Funding (RSF)

- A9.4.2** (1) An Authorised Firm must calculate its RSF by adding together:
- (a) the adjusted carrying values of its assets, calculated in accordance with (2); and
 - (b) the adjusted carrying values of its off-balance sheet (OBS) Exposures (or potential liquidity Exposures), calculated in accordance with (3).
- (2) An Authorised Firm must calculate the adjusted carrying values of its respective assets by:
- (a) assigning the carrying value of each asset to the applicable RSF Category set out in Table 1 to this Rule;
 - (b) adjusting the carrying value of each asset by multiplying it by the applicable RSF Factor set out in Table 1; and
 - (c) adding together each adjusted carrying value.
- (3) An Authorised Firm must calculate the adjusted carrying values of its respective OBS Exposures (or potential liquidity Exposures) by:
- (a) assigning the carrying value of each Exposure to one of the OBS-RSF Categories set out in Table 2 to this Rule;
 - (b) adjusting the carrying value of each asset by multiplying it by the applicable OBS-RSF Factor for that OBS-RSF Category, as set out in Table 2; and

(c) adding together each adjusted carrying value.

Table 1 - RSF Factors and Categories	
RSF Factor	Components of RSF Category
0%	(a) coins and banknotes immediately available to meet obligations; (b) all Central Banks reserves (including required reserves and excess reserves); (c) all claims on Central Banks with residual maturities of less than six months; and (d) trade date receivables arising from sales of financial instruments, foreign currencies and commodities that: (i) are expected to settle within the standard settlement cycle or period that is customary for the relevant exchange or type of transaction; or (ii) have failed to, but are still expected to, settle.
5%	Unencumbered assets included in Rule A9.2.6(2) (Level 1 HQLA), excluding assets receiving a 0% RSF Factor as specified above.
10%	Unencumbered loans to financial institutions with residual maturities of less than six months, if the loan is secured against Level 1 HQLA included in Rule A9.2.6, and where the bank has the ability freely to hypothecate the received collateral for the life of the loan.
15%	All loans to financial institutions with a residual maturity of less than six months, other than those receiving a 10% RSF Factor as specified above. Unencumbered assets as defined in Rule A9.2.7(2) (Level 2 HQLA).
50%	(a) unencumbered assets included in Rule A9.2.8(2) (Level 2B HQLA), excluding any haircuts required under that Rule; (b) any HQLA as defined in Rules A9.2.6 to A9.2.8 that are encumbered for a period of between six months and less than one year; (c) all loans to financial institutions and Central Banks with residual maturity of between six months and less than one year; (d) operational deposits, that is, deposits held at other financial institutions for operational purposes, that are subject to the 50% ASF Factor set out in the table in Rule A9.4.1; and (e) all other non-HQLA not included in the above categories that have a residual maturity of less than one year, including loans to non-financial

	corporate clients, loans to retail customers (i.e. natural persons) and small business customers, and loans to sovereigns and PSEs.
65%	(a) unencumbered residential mortgages with a residual maturity of one year or more that would qualify for a 50% or lower risk weight under section 4.12 (Risk Weights); and (b) other unencumbered loans not included in the above categories, excluding loans to financial institutions, with a residual maturity of one year or more that would qualify for a 50% or lower risk weight under section 4.12 (Risk Weights).
85%	(a) cash, securities or other assets posted as initial margin for derivative contracts or Shari'a compliant hedging contracts and cash or other assets provided to contribute to the default fund of a Central Counterparty (CCP). Where securities or other assets posted as initial margin for derivative contracts would otherwise receive a higher RSF Factor, they should retain that higher factor; (b) other unencumbered performing loans that do not qualify for the 50% or lower risk weight under section 4.12 and have residual maturities of one year or more, excluding loans to financial institutions; (c) unencumbered securities with a remaining maturity of one year or more and exchange-traded equities, that are not in default and do not qualify as HQLA; and (d) physical traded commodities, including gold.
100%	(a) all assets that are encumbered for a period of one year or more; (b) NSFR derivative assets (net of NSFR derivative liabilities) if NSFR derivative assets are greater than NSFR derivative liabilities; (c) NSFR Shari'a compliant hedging assets net of NSFR Shari'a compliant hedging liabilities if NSFR Shari'a compliant hedging assets are greater than NSFR Shari'a compliant hedging liabilities; (d) all other assets not included in the above categories, including non-performing loans, loans to financial institutions with a residual maturity of one year or more, non-exchange-traded equities, fixed assets, items deducted from regulatory capital, retained interest, insurance assets, subsidiary interests and defaulted securities; (e) 20% of derivative liabilities (i.e. negative replacement cost amounts), before deducting variation margin posted; and (f) 20% of Shari'a compliant hedging liabilities.

Table 2 - OBS-RSF Factors and Categories	
OBS-RSF Factor	Components of OBS-RSF Category
3%	Trade finance related obligations (including guarantees and letters of credit).
5%	(a) irrevocable and conditionally revocable credit and liquidity facilities to any client; and (b) unconditionally revocable credit and liquidity facilities.
10%	(a) the following non-contractual obligations: (i) structured products where customers anticipate ready marketability, such as adjustable rate notes and variable rate notes; and (ii) managed funds that are marketed with the objective of maintaining a stable value; and (b) guarantees and letters of credit unrelated to trade finance obligations.
100%	The following non-contractual obligations: (a) potential requests for debt repurchases of the bank's own debt or that of related conduits, securities investment vehicles and other such financing facilities; and (b) other non-contractual obligations not mentioned above.

Guidance

1. An Authorised Firm's RSF represents the liquidity risk profile of its assets and OBS Exposures (or potential liquidity Exposures).
2. When determining its RSF, an Authorised Firm should:
 - a. include financial instruments, foreign currencies and commodities for which a purchase order has been executed; and
 - b. exclude financial instruments, foreign currencies and commodities for which a sales order has been executed, even if such transactions have not been reflected in the balance sheet under a settlement-date accounting model, provided that:
 - i. the transactions are not reflected as derivatives or secured financing transactions in the firm's balance sheet; and
 - ii. the effects of such transactions will be reflected in the firm's balance sheet when settled.
3. For secured funding arrangements, use of balance sheet and accounting treatments should generally result in a firm excluding, from their assets, securities which they have borrowed in securities financing transactions (such as reverse repos and collateral swaps) where they do not

have beneficial ownership. In contrast, a firm should include securities they have lent in securities financing transactions where they retain beneficial ownership. A firm should also not include any securities they have received through collateral swaps if those securities do not appear on their balance sheet. Where a firm has encumbered securities in repos or other securities financing transactions, but has retained beneficial ownership and those assets remain on its balance sheet, the firm should allocate such securities to the appropriate RSF Category.

4. SFT with a single counterparty should be measured net when calculating the NSFR, provided that the netting conditions set out in Rules 4.9.13 to 4.9.20 are met.
5. Derivative assets are calculated first based on the replacement cost for derivative contracts (obtained by marking to market) where the contract has a positive value. When an eligible bilateral netting contract is in place that meets the conditions as specified in the instructional guidelines for Form B300 set out in PRU Section 1.48, the replacement cost for the set of derivative exposures covered by the contract will be the net replacement cost.
6. Shari'a compliant hedging liabilities (e.g. Islamic swaps) are calculated first based on the replacement cost for the Shari'a compliant hedging contracts (obtained by marking to market), such as ISDA/IIFM Tahawwut Master Agreement (TMA), where the contract has a negative value. When an eligible bilateral netting contract is in place, the replacement cost for the set of Shari'a compliant hedging exposures covered by the contract will be the net replacement cost.
7. In calculating NSFR derivative assets, collateral received in connection with derivative contracts should not offset the positive replacement cost amount, regardless of whether or not netting is permitted under the firm's operative accounting or risk-based framework, unless it is received in the form of cash variation margin and meets the conditions as specified in the instructional guidelines for Form B300 set out in PRU section 1.48. Any remaining balance sheet liability associated with: (a) variation margin received that does not meet the criteria above; or (b) initial margin received; may not offset derivative assets and should be assigned a 0% ASF Factor.
8. In calculating NSFR Shari'a compliant hedging liabilities, collateral posted in the form of variation margin that follows Shari'a principles in connection with Shari'a compliant hedging contracts as in the TMA contract, regardless of the asset type, must be deducted from the negative replacement cost amount.

APP10 SUPERVISORY REVIEW AND EVALUATION PROCESSES

A10.1 IRAP

Guidance

Application

1. This Guidance is relevant to an Authorised Firm to whom section 10.3 applies (that is, a firm in Category 1, 2 or 5) in regard to an Internal Risk Assessment Process (referred to in this Guidance as an IRAP).
2. The following Guidance generally assumes that the Rules relating to capital adequacy in PIB apply to an Authorised Firm on a solo basis. However, the Guidance is to be read as also applying where the capital adequacy requirements in these modules apply to the Financial Group of an Authorised Firm on a consolidated basis.

Purpose of the IRAP

3. The IRAP is an internal process of an Authorised Firm which enables it to identify, assess, aggregate and monitor its risks adequately. The objective of the IRAP is to develop a comprehensive and detailed risk profile for the firm. The IRAP should help the firm ensure that sound risk management systems are in place, address any weaknesses in its risk management framework, and maintain adequate internal capital relative to its risk profile.
4. An Authorised Firm should ensure that the IRAP forms an integral part of the firm's risk management framework and decision-making processes. The IRAP should cover all activities of the Authorised Firm and should be proportionate to the nature and complexity of the firm's activities.
5. The Authorised Firm should be able to demonstrate to the DFSA that its internal risk assessment is comprehensive and adequate relative to the nature of risks posed by its business activities and its operating environment.
6. The DFSA does not prescribe any specific approach for the IRAP and, consequently, an Authorised Firm can choose to implement an IRAP which is proportionate to the nature, size and complexity of the business activities.
7. The IRAP should be subject to adequate internal controls and reviews by internal audit to ensure the integrity and objectivity of the process. The IRAP should consider the quality and effectiveness of the Authorised Firm's risk management framework while determining its risk profile.
8. The IRAP should:
 - a. identify and outline all related parties of the Authorised Firm, and list the types of transactions that occur between those related parties and the firm;
 - b. identify the most significant risks to which the firm is exposed, which should, at a minimum, include the risks identified in Guidance note 9;
 - c. identify each of the firm's major business lines and prepare a comprehensive list of the major risks to which each of the businesses are exposed;
 - d. identify the controls and risk management measures used to address the risks referred to in b. and c. and assess the strength of such controls and systems; and

- e. consider the impact of an economic or industry downturn on its future earnings, taking into account its business plans.
- 9. The IRAP should, in addition to the aforementioned factors:
 - a. estimate, with the aid of historical data, where available, the range and distribution of possible losses which might arise from each of those risks and consider using stress tests to provide risk estimates;
 - b. consider the extent to which the firm's Capital Requirement adequately addresses the type of risks referred to under Guidance note 8 (b) and (c); and
 - c. estimate the expected change in the firm's risk profile on the basis of projections of the firm's business activities for the next 3 to 5 years.
- 10. If the firm's IRAP is based on this Guidance, it may enable the DFSA more easily to review the IRAP as part of its SREP. However, the DFSA may decide to rely on an IRAP that is not consistent with the elements of this Guidance, owing to specific reasons and/or circumstances which necessitate an alternative approach.

Guidance on risks to be covered as part of the IRAP

- 11. An Authorised Firm should consider the following risks, where relevant, in its IRAP:
 - a. Credit Risk, including Large Exposures and Concentration Risks;
 - b. Market Risk;
 - c. Liquidity Risk;
 - d. for Islamic Financial Business involving PSIAs, displaced commercial risk;
 - e. interest rate risk in the Non-Trading Book ;
 - f. Operational Risk;
 - g. internal controls and systems; and
 - h. reputational risk.
- 12. This Guidance is merely an indicative list of risk categories, which does not preclude an Authorised Firm from assessing other risks that it considers significant (for example, securitisation risks and residual risks). Likewise, certain categories of risks might not be relevant to all Authorised Firms completing the IRAP. In this case, the IRAP should clearly indicate why the risk is considered minimal or not relevant. The IRAP should also consider all risks arising from any non-regulated activities of the Authorised Firm, if they are seen as material to the risk profile of the firm.

A10.2 ICAAP

Guidance

Application

- 1. This Guidance is relevant to an Authorised Firm to whom section 10.4 applies (that is, a firm in Category 1, 2 or 5) in regard to an Internal Risk Capital Adequacy Assessment Process (referred to in this Guidance as an ICAAP).

Purpose and process of the ICAAP

2. The ICAAP is an internal process of an Authorised Firm which enables it to determine and maintain the amount and quality of capital that is adequate in relation to the Authorised Firm's risk profile as assessed in the IRAP. Authorised Firms are encouraged to maintain capital over and above the regulatory minimum capital. The ICAAP, in conjunction with the IRAP, should be embedded in the Authorised Firm's business and organisational processes.
3. When assessing its capital needs, an Authorised Firm should take into account the impact of economic cycles, and sensitivity to other external risks and factors. For larger or more complex institutions, this may mean developing an appropriate stress testing and scenario testing framework.
4. The DFSA does not prescribe any specific approach for the ICAAP and, consequently, an Authorised Firm can choose to implement an ICAAP which is proportionate to the nature, size and complexity of the business activities.
5. In completing an ICAAP, an Authorised Firm should:
 - a. estimate the amount of capital required to absorb potential losses, if any, for the significant risks identified in the IRAP (particularly risks which lead to financial losses);
 - b. perform reasonable and proportionate sensitivity tests to analyse the impact of variation in the risk parameters of significant risks identified in the IRAP on the profitability and the capital position of the Authorised Firm;
 - c. estimate, using the range and distribution of possible losses estimated from historical data, the level of capital required reasonably to cover likely losses;
 - d. estimate the Capital Resources required to address potential variation in the Authorised Firm's Capital Requirement arising from planned growth in business levels or any significant deviation in growth from plans; and
 - e. document the ranges of capital required for each of the factors identified above and enable the Governing Body and the senior management to form an overall view on the amount and quality of capital which that Authorised Firm should hold.
6. The DFSA does not require an Authorised Firm to implement ICAAP through sophisticated models and the DFSA has no prescribed approach for developing an internal capital model for the firm's ICAAP assessment. However, an Authorised Firm should be able to demonstrate:
 - a. the confidence levels set and whether these are linked to its corporate strategy;
 - b. the time horizons set for the different types of business that it undertakes;
 - c. the extent of historic data used and back-testing carried out;
 - d. that it has in place a process to verify the correctness of the model's outputs; and
 - e. that it has the skills and resources to operate, maintain and develop the model.
7. If an Authorised Firm's internal model makes explicit or implicit assumptions in relation to correlations within or between risk types, or in relation to diversification benefits between business lines, the firm should be able to explain to the DFSA, with the support of empirical evidence, the basis of those assumptions. An Authorised Firm's model should also reflect the past experience of both the firm and the sectors in which it operates.
8. The assumptions required to aggregate risks that are modelled and the confidence levels adopted should be considered by the Authorised Firm's senior management. An Authorised Firm should also consider whether any relevant risks, including systems and control risks, are not captured by the model.

9. An Authorised Firm using an internal capital model should validate the assumptions of the model through a comprehensive stress testing programme. In particular this validation should:
 - a. test correlation assumptions (where risks are aggregated in this way) using combined stresses and scenario analyses;
 - b. use stress tests to identify the extent to which the firm's risk models omit non-linear effects, for instance the behaviour of derivatives in Market Risk models; and
 - c. consider not just the effect of parallel shifts in interest rate curves, but also the effect of the curves becoming steeper or flatter.
10. Any internal assessment of capital adequacy should address diversification benefits and transferability of Capital Resources between members of the Financial Group. It should also describe the distribution of the capital required by the Financial Group across all entities, including the Authorised Firm.

A10.3 Supervisory Review and Evaluation Process (SREP)

Guidance

Application

1. This Guidance is relevant to an Authorised Firm to whom section 10.5 applies (that is, a firm in Category 1, 2 or 5) in regard to a Supervisory Review and Evaluation Process (referred to in this Guidance as a SREP).

Introduction

2. This Guidance covers the evaluation criteria and methodology (referred to as a SREP) that the DFSA may use when reviewing and evaluating the IRAP and ICAAP of an Authorised Firm.

The SREP in detail

3. A SREP of an IRAP and an ICAAP forms an integral part of the overall supervisory approach of the DFSA. A SREP is expected to enable assessment of the effectiveness, completeness and quality of an IRAP and ICAAP in relation to the overall risk profile of the Authorised Firm. It leverages from information collected and assessments carried out as part of the wider supervisory regime, including desk-based reviews, on-site risk assessments, discussions with the firm's management, and reviews completed by internal and external auditors.
4. The SREP is structured to provide consistency of treatment across Authorised Firms, taking into consideration the differences in risk profiles, business strategies and management. An essential element of the SREP is the qualitative assessment of each type of risk and its management within the overall context of the firm's internal governance.
5. The DFSA's assessment of the individual risk profile of an Authorised Firm will provide the context for evaluation of the firm's IRAP and/or ICAAP. The evaluation in turn will be used by the DFSA to augment its understanding of the overall risk profile of a firm. Also, the DFSA might involve such a firm in a formalised discussion of risks and capital adequacy, which might lead to a requirement for additional capital.
6. The SREP may be used as a regulatory tool for Authorised Firms which are required to perform an IRAP and/or ICAAP. The SREP for each Authorised Firm will be proportionate in terms of the size, scale and complexity of its business and its impact on financial sector stability. The DFSA will cooperate actively with other supervisory authorities whenever an Authorised Firm is part of a Group and is prudentially regulated on a consolidated basis.

7. The SREP evaluation cycle will be determined in the discretion of the DFSA and be based on the risk assessment, developments in the risk profile and changes in the Authorised Firm's strategy or products. The SREP is as far as possible aligned with the risk assessment process to ensure that a recent risk assessment is available for the SREP evaluation process.
8. It is envisaged that the DFSA will use a range of supervisory tools of qualitative or quantitative nature to perform the SREP. The SREP is not intended as, and should not constitute, a parallel or secondary IRAP or ICAAP. Its purpose is to evaluate the quality, completeness and consistency of the IRAP or ICAAP of the Authorised Firm.

Review of the IRAP and ICAAP Assessment

9. Upon receipt of an IRAP or ICAAP the DFSA would normally:
 - a. subject the data provided to an initial analysis for completeness and accuracy followed by a more detailed comparison with the relevant data held on file at the DFSA about the Authorised Firm;
 - b. determine if there are material changes compared with previous submissions;
 - c. determine if the submitted data contains indicators of a possible material change in the firm's risk profile;
 - d. address and discuss any information gaps or anomalies with the firm; and
 - e. form an assessment about content and quality of the submission which will be integrated into the overall supervisory approach.

Evaluation of the IRAP and ICAAP

10. The SREP evaluation of the IRAP and, where applicable, the ICAAP covers all activities of an Authorised Firm and takes all relevant data collected during the supervisory process into account. The SREP evaluation process will use desk based reviews, firm visits and meetings to arrive at a final view.
11. As part of the SREP, the DFSA will consider:
 - a. the completeness of the IRAP and, where applicable, the ICAAP by ensuring that it covers all business areas, internal governance and all risk categories of the Authorised Firm;
 - b. the soundness and quality of the IRAP and, where applicable, the ICAAP process in relation to the firm's size, business complexity and risk profile;
 - c. soundness of qualitative calibration and quantitative methodology whenever employed by the firm;
 - d. execution of the IRAP and, where applicable, the ICAAP in terms of consistency, quality and documentation;
 - e. adequacy of internal controls and quality assurance processes on the IRAP and, where applicable, the ICAAP; and
 - f. adequacy of management information and whether the management had responded adequately and in a timely manner to such information.
12. Based on the SREP, the DFSA will form an assessment which will be communicated to the Authorised Firm and flow into the overall supervisory approach. The action required resulting from the IRAP and ICAAP will be communicated to the firm as part of a risk mitigation programme.

13. Where the DFSA does not agree with the results of the firm's ICAAP results, the DFSA will involve the firm in a dialogue to reconcile any difference in view to arrive at a consensus estimate of the capital level required to address all risks identified either by the firm or by the DFSA in its SREP. Such an estimate will be specified by the DFSA as the Individual Capital Requirement for the firm. Where consensus is not possible the DFSA may impose an Individual Capital Requirement on a firm.

APP11 PUBLIC DISCLOSURE REQUIREMENTS

Table 1 – Scope of application

Qualitative Disclosures	(a)	The name of the Authorised Firm.
	(b)	In the case of a Financial Group, a list of all the entities forming part of the Financial Group and a brief description of each of those entities. In addition, a description of differences in the basis of consolidation for regulatory purposes compared to that required under the International Financial Reporting Standards. The description must include a brief description of the entities: (i) that are fully consolidated; (ii) that are consolidated on a pro-rata basis; (iii) that are equity-accounted; (iv) that are included as deductions from any of the components of Capital Resources; (v) from which surplus capital is recognised, if any; and (vi) that are not consolidated and not deducted.
	(c)	Any restrictions or impediments on transfer of funds or regulatory capital within the Financial Group.

Table 2 – Capital

Qualitative Disclosures	(a)	A description of the terms and conditions and main features of all capital instruments included within every component of Capital Resources – CET 1 Capital, AT1 Capital and T2 Capital.
Quantitative Disclosures	(b)	(i) Amounts of every element eligible for inclusion in CET1 Capital; (ii) Regulatory adjustments to CET1 Capital; (iii) Deductions from CET1 Capital; and (iv) Amount of total CET1 Capital.
	(c)	(i) Amounts of every element eligible for inclusion in AT1 Capital; (ii) Regulatory adjustments to AT1 Capital; (iii) Deductions from AT1 Capital; and (iv) Amount of total AT1 Capital.
	(d)	(i) Amounts of every element eligible for inclusion in T2 Capital; (ii) Regulatory adjustments to T2 Capital; (iii) Deductions from T2 Capital; and (iv) Amount of total T2 Capital.
	(e)	Amount of eligible Capital Resources.

Table 3 – Capital Adequacy

Qualitative Disclosures	(a)	A description of the overall capital management system and approach to assessing the adequacy of its capital to support current and future activities. This should include description of systems, controls and processes for capital management and capital mobilisation plans for the medium term.
Quantitative Disclosures	(b)	(i) Amount of CRCOM; (ii) Amount of Credit RWA for each asset class giving rise to CR Exposures, for SE Exposures, CCR Exposures and SR Exposures; and (iii) Amount of Credit RWAs for Early Amortisation Exposures, included in SE Exposures, if any.
	(c)	Market Risk Capital Requirement for each component of Market Risk as listed in Rule 5.1.3, calculated using: (i) Rules prescribed in chapter 5; (ii) Internal Models Approach; or (iii) both (i) and (ii).
	(d)	Operational Risk Capital Requirement calculated under the following approaches, where applicable: (i) Basic Indicator Approach; (ii) Standardised Approach; (iii) Alternative Standardised Approach; or (iv) a combination of any of the above.
	(e)	Capital Requirement at the solo and at the Financial Group level.
	(f)	(i) CET1 Capital ratio as a percentage of total RWAs; (ii) T1 Capital ratio as a percentage of total RWAs; (iii) Capital Resources as a percentage of total RWAs; and (iv) These ratios need to be disclosed at both the Authorised Firm level and at the Financial Group level.
	(g)	The ratios referred to in (f) must be disclosed for each significant entity in the case of a Financial Group.

Table 4 - Credit Risk – general disclosures

Qualitative Disclosures	(a)	A description of the policies of the Authorised Firm in relation to : (i) past due and impaired loans in accordance with the International Financial Reporting Standards; (ii) assessment of the level of individual and collective impairment provisions in accordance with the International Financial Reporting Standards; (iii) Credit Risk management; and (iv) the nature of the Exposures within each asset class. For each asset class: (i) the name of each recognised external credit rating agency which ratings are used by the Authorised Firm, and the reasons for any changes in the use of a recognised external credit rating agency; (ii) the types of Exposure for which ratings of each recognised external credit rating agency are used; (iii) a description of the process used to transfer public issue ratings onto comparable assets in the Non-Trading Book; and (iv) the alignment of the alphanumeric scale of each recognised external credit rating agency used by the Authorised Firm with relevant risk weights. For Exposures to Banks, a description of the methodology used to classify Exposures into one of the three risk weight Grades.
Quantitative Disclosures	(b)	Total gross credit Exposures, and average gross credit Exposures over the reporting period, broken down by major types of credit Exposure.
	(c)	Geographic distribution of credit Exposures, broken down in significant areas by major types of credit Exposure.
	(d)	Industry or Counterparty-type distribution of credit Exposures, broken down by major types of credit Exposure.
	(e)	Residual contractual maturity broken down by major types of credit Exposure.
	(f)	By major industry or Counterparty type: (i) amount of classified loans; (ii) amount of past due loans; (iii) individual and collective impairment provisions; and (iv) charges for individual impairment provisions and charge-offs during the period.

	(g)	By significant geographic area: (i) amount of classified loans; (ii) amount of past due loans; and (iii) individual and collective impairment provisions, where feasible.
	(h)	Reconciliation of changes in the provisions for loan impairment, and separate disclosures for charge-offs and recoveries that are recorded directly to the income statement.
	(i)	An analysis by risk-weights (including deducted Exposures) for the total rated and unrated credit Exposures after taking into account the effects of Credit Risk mitigation.

Table 5 – Credit Risk mitigation disclosures

Qualitative Disclosures	(a)	A description of the following items with respect to Credit Risk mitigation: (i) policies and procedures for, and an indication of the extent to which the Authorised Firm makes use of, on-balance sheet Netting; (ii) policies and procedures for Collateral valuation and management; (iii) the main types of Collateral taken by the Authorised Firm; (iv) the main types of guarantor or Credit Derivative Counterparty and their creditworthiness; and (v) information about Market Risk or Credit Risk concentrations within the mitigation taken
Quantitative Disclosures	(b)	For each separately disclosed asset class, the extent to which credit Exposures are covered by eligible financial Collateral, after the application of haircuts.
	(c)	For each separately disclosed asset class, the amount by which credit Exposures have been reduced by eligible credit protection.

Table 6 - General disclosures for Exposures related to Counterparty Credit Risk

Qualitative Disclosures	(a)	A description of the following items in relation to Counterparty Credit Risk: (i) methodologies used to assign economic capital and credit limits for Counterparty credit Exposures; (ii) policies for securing Collateral and establishing credit reserves; (iii) policies with respect to Exposures that give rise to general or specific wrong-way risk; and (iv) impact of the amount of Collateral the Authorised Firm would have to provide given a credit rating downgrade.
Quantitative Disclosures	(b)	(i) Gross positive fair value of transactions, Netting benefits, netted current credit Exposure, and amount and type of Collateral held;

		(ii) the total CCR Exposure value calculated under Rule 4.14A.2; and (iii) The notional value of Credit Derivative hedges, and the distribution of current credit Exposure by types of credit Exposure.
	(c)	Credit Derivative transactions that create Exposures to Counterparty Credit Risk (notional value), segregated between use for the credit portfolio of the Authorised Firm and the intermediation activities of the firm, including the distribution of Credit Derivatives used, analysed further in terms of protection bought and sold within each type of Credit Derivative.

Table 7 – Securitisation Exposures

Qualitative Disclosures	(a)	A description of the following items with respect to securitisation (including Synthetic Securitisation): (i) objectives of the Authorised Firm in relation to its securitisation, including the extent to which the securitisation transfers Credit Risk of the underlying securitised Exposures away from the Authorised Firm to other entities and including the types of risks assumed and retained with Re-securitisation activity; (ii) the nature of other risks (e.g. Liquidity Risk) inherent in securitised assets (iii) the various roles played by the Authorised Firm in the securitisation process and an indication of the extent of the involvement of the firm in each of them; (iv) the processes in place to monitor changes in the Credit Risk and Market Risk of securitisation Exposures (e.g., how the behaviour of the underlying assets impacts securitisation Exposures) including how those processes differ for Re-securitisation exposures. (v) the Authorised Firm's policy governing the use of Credit Risk mitigation to mitigate the risks retained through securitisation and Re-securitisation Exposures; (vi) the regulatory capital approaches applied to the securitisation activities of the Authorised Firm, including the type of securitisation Exposures to which each approach applies; and (vii) where an Authorised Firm provides Implicit Support to a securitisation, a statement that it has provided non- contractual support and a description of the capital impact of doing so.
	(b)	A list of: (i) the types of SPEs that the Authorised Firm, as a Sponsor, uses to securitise third party Exposures, indicating whether the firm has Exposure to these SPEs, either on or off-balance sheet; and (ii) entities that the firm manages or advises that invest either in the securitisation Exposures that the firm has securitised or in SPEs that the firm Sponsors.
	(c)	A summary of the accounting policies of the Authorised Firm for securitisation, including:

	(i) whether the securitisation is treated as sales or financings; (ii) recognition of gain–on-sale; (iii) methods and key assumptions (including inputs) for valuing positions retained or purchased; (iv) changes in methods and key assumptions from the previous period and the impact of such changes; (v) treatment of Synthetic Securitisation if this is not covered by other accounting policies (e.g. on Derivatives); (vi) how Exposures intended to be securitised (e.g. in the pipeline or warehouse) are valued and whether they are recorded in the Non-Trading Book or the Trading Book; and (vii) policies for recognising liabilities on the balance sheet for arrangements that could require the Authorised Firm to provide financial support for securitised assets.
	(d) In the Non-Trading Book, the names of recognised external credit rating agencies used for securitisations and the types of securitisation Exposure for which each agency is used.
	(e) An explanation of significant changes to any of the quantitative information (e.g. amounts of assets intended to be securitised, movement of assets between Non-Trading Book and Trading Book) since the last reporting period.
	(f) The total amount of outstanding Exposures securitised by the Authorised Firm and defined under the securitisation framework set out in chapter 4, broken down in terms of traditional and Synthetic, and by Exposure type, separately for securitisations of third-party Exposures for which the firm acts only as Sponsor.
	(g) For Exposures securitised by the Authorised Firm and defined under the securitisation framework set out in chapter 4: (i) the amount of securitised assets that are classified or past due under the PIB Rules, broken down by Exposure type; and (ii) losses recognised by the firm during the current period broken down by Exposure type.
	(h) The total amount of outstanding Exposures intended to be securitised broken down by Exposure type.
	(i) Summary of securitisation of the current period, including the total amount of Exposures securitised by Exposure type, and the recognised gain or loss on sale by Exposure type.
	(j) Aggregate amount of: (i) on-balance sheet securitisation Exposures retained or purchased broken down by Exposure type; and (ii) off-balance sheet securitisation Exposures broken down by Exposure type.

	(k)	Aggregate amount of securitisation Exposures retained or purchased and the associated capital charges, broken down between securitisation and Re-securitisation Exposures and further broken down into a meaningful number of risk weight bands for each regulatory capital approach. Exposures included as deductions from T1 Capital, credit-enhancing interest only strips and other Exposures included as deductions from T1 Capital and deductions from T2 Capital must be disclosed separately by Exposure type.
	(l)	For securitisation subject to the Early Amortisation treatment, the following items by Exposure type for securitised facilities: (i) the aggregate drawn Exposures attributed to the interests of the seller and the investor; (ii) the aggregate capital charges incurred by the Authorised Firm against its retained (i.e. the seller's) shares of the drawn balances and undrawn lines; and (iii) the aggregate capital charges incurred by the firm against the shares of drawn balances and undrawn lines of the investor.
	(m)	Aggregate amount of Re-securitisation Exposures retained or purchased broken down according to: (i) Exposures to which Credit Risk mitigation is applied and those not applied; and (ii) Exposures to guarantors broken down according to guarantor credit worthiness categories or guarantor name.
Quantitative disclosures: Trading Book	(n)	The total amount of outstanding Exposures securitised by the Authorised Firm and defined under the securitisation framework set out in chapter 4, broken down in terms of traditional and Synthetic, and by Exposure type, separately for securitisations of third-party Exposures for which the firm acts only as Sponsor.
	(o)	The total amount of outstanding Exposures intended to be securitised broken down by Exposure type.
	(p)	Summary of securitisation of the current period, including the total amount of Exposures securitised by Exposure type, and the recognised gain or loss on sale by Exposure type.
	(q)	Aggregate amount of Exposures securitised by the Authorised Firm for which the firm has retained some Exposures and which is subject to the Market Risk approach, broken down in terms of traditional and Synthetic, by Exposure type.
	(r)	Aggregate amount of: (i) on-balance sheet securitisation Exposures retained or purchased broken down by Exposure type; and (ii) off-balance sheet securitisation Exposures broken down by Exposure type.
	(s)	Aggregate amount of securitisation Exposures retained or purchased separately for:

		(i) securitisation Exposures retained or purchased subject to the comprehensive risk measure for Specific Risk; and (ii) securitisation Exposures subject to the securitisation framework for Specific Risk broken down into a meaningful number of risk weight bands for each regulatory capital approach.
	(t)	Aggregate amount of: (i) the Capital Requirements for the securitisation Exposures (Re-securitisation or securitisation), subject to the securitisation framework broken down into a meaningful number of risk weight bands for each regulatory capital approach; and (ii) securitisation Exposures that are included as deductions from CET1 Capital, credit enhancing interest-only strips and other Exposures included as deductions from AT1 Capital and deductions from T2 Capital disclosed separately by Exposure type.
	(u)	For securitisation subject to the Early Amortisation treatment, the following items by Exposure type for securitised facilities: (i) the aggregate drawn Exposures attributed to the interests of the seller and the investor; (ii) the aggregate capital charges incurred by the Authorised Firm against its retained (i.e. the seller's) shares of the drawn balances and undrawn lines; and (iii) the aggregate capital charges incurred by the firm against the shares of drawn balances and undrawn lines of the investor.
	(v)	Aggregate amount of Re-securitisation Exposures retained or purchased broken down according to: (i) Exposures to which Credit Risk mitigation is applied and those not applied; and (ii) Exposures to guarantors broken down according to guarantor creditworthiness categories or guarantor name.

Table 8 – Market Risk Disclosures

Qualitative Disclosures	(a)	A description of risk management objectives and policies covering all Market Risk Exposures.
Quantitative Disclosures	(b)	The Capital Requirements for the following risks as set out in chapter 5 of the PIB module: (i) Interest Rate Risk; (ii) Equity Position Risk; (iii) Foreign Exchange Risk; (iv) Commodity Risk; (v) Option Risk; (vi) Collective Investment Fund Risk; and (vii) Securities Underwriting Risk.

Table 9 – Market Risk – disclosures for the internal models approach

Qualitative Disclosures	(a)	A description of the valuation methodologies employed by the Authorised Firm.
	(b)	A description of the soundness standards on which the internal capital adequacy assessment of the Authorised Firm is based, as well as the methodologies used to achieve a capital adequacy assessment that is consistent with those soundness standards.
	(c)	For each portfolio covered by the internal models approach: (i) the characteristics of the models used; (ii) a description of stress testing applied to the portfolio; and (iii) a description of the approach used for back testing and validating the accuracy and consistency of the internal models and modelling processes.
	(d)	The scope of approval by the DFSA.
	(e)	A description of the methodologies used and the risks measured through the use of internal models for the incremental risk capital charge and the comprehensive risk capital charge. Included in the qualitative description should be: (i) the approach used by the Authorised Firm to determine liquidity horizons; (ii) the methodologies used to achieve a capital assessment that is consistent with the required soundness standard; and (iii) the approaches used in the validation of the models.
Quantitative Disclosures	(f)	For trading portfolios under the internal models approach: (i) the high, mean and low VaR values over the reporting period and period-end; (ii) the high, mean and low stressed VaR values over the reporting period and period-end; (iii) the high, mean and low incremental and comprehensive risk capital charges over the reporting period and period-end; and (iv) a comparison of VaR estimates with actual gains or losses experienced by the Reporting Firm, with analysis of outliers in back test results.

Table 10 – Operational Risk

Qualitative Disclosures	(a)	A description of the regulatory approach or approaches to the calculation of Operational Risk Capital Requirements.
-------------------------	-----	---

Table 11 – Interest Rate Risk in the Non-Trading Book

Qualitative Disclosures	(a)	A description of the key assumptions made by the Authorised Firm including assumptions regarding loan prepayments and behaviour of non-maturity deposits, and frequency with which interest rate risk in the Non-Trading Book is measured, in addition to the general disclosures set out in Chapter 9 in respect of interest rate risk in the Non-Trading Book.
Quantitative Disclosures	(b)	The changes in earnings or economic value (or relevant measure used by the authorised firm) for upward and downward rate shocks according to the internal method of the Authorised Firm for measuring interest rate risk in the Non-Trading Book, broken down by currency, where applicable.

Table 12 – Liquidity Coverage Ratio Information

Qualitative Disclosures	(a)	Governance and organisation of Liquidity Risk management.		
	(b)	Risk tolerance and strategy in relation to Liquidity Risk management.		
	(c)	Scope and nature of Liquidity Risk reporting and measurement systems.		
	(d)	The techniques used for mitigating Liquidity Risk and the process of monitoring the effectiveness of the mitigants in place.		
	(e)	Overview of how stress testing is used.		
	(f)	Outline of the Contingency Funding Plan.		
	(g)	A declaration approved by the governing body on the adequacy of Liquidity Risk management arrangements in place with regard to the firm's profile and strategy.		
	(h)	A Liquidity Risk statement approved by the governing body describing the firm's overall Liquidity Risk profile associated with the business strategy, including how the Liquidity Risk profile of the firm interacts with the risk tolerance set by the management body and:		
		(i) the main drivers of the LCR results and the evolution of the contribution of inputs to the LCR's calculation over time;		
		(ii) intra-period changes as well as changes over time;		
		(iii) the composition of HQLA;		
		(iv) concentration of funding sources;		
		(v) derivatives exposures and potential collateral calls;		
		(vi) currency mismatch in the LCR;		
		(vii) a description of the degree of centralisation of liquidity management and interaction between the group's units; and		
		(viii) other inflows and outflows in the LCR calculation that are not captured in the LCR common template but which the firm considers to be relevant for its liquidity profile.		
	(i)		TOTAL UNWEIGHTED	TOTAL WEIGHTED

Quantitative Disclosures			VALUE (average)	VALUE (average)
		HIGH-QUALITY LIQUID ASSETS		
	1	Total high-quality liquid assets (HQLA)		
		CASH OUTFLOWS		
	2	Retail deposits/PSIAs and deposits/PSIAs from small business customers, of which:		
	3	Stable deposits/PSIAs		
	4	Less stable deposits/PSIAs		
	5	Unsecured wholesale funding, of which:		
	6	Operational deposits or accounts (all counterparties) and deposits/PSIAs in networks of cooperative banks		
	7	Non-operational deposits/PSIAs (all counterparties)		
	8	Unsecured debt		
	9	Secured wholesale funding		
	10	Additional requirements, of which:		
	11	Outflows related to derivative exposures, Shari'a compliant hedging instrument exposures and other collateral requirements		
	12	Outflows related to loss of funding on debt products		
	13	Credit and liquidity facilities		
	14	Other contractual funding obligations		

	15	Other contingent funding obligations		
	16	TOTAL CASH OUTFLOWS		
	CASH INFLOWS			
	17	Secured lending (e.g. reverse repos) or Shari'a compliant secured financing		
	18	Inflows from fully performing exposures		
	19	Other cash inflows		
	20	TOTAL CASH INFLOWS		
	TOTAL ADJUSTED VALUE			
	21	TOTAL HQLA		
	22	TOTAL NET CASH OUTFLOWS		
	23	LIQUIDITY COVERAGE RATIO (%)		

Guidance

The following Guidance is intended to assist firms to complete the above table.

The figures to be presented are averages of the daily observations of individual line items over the financial reporting period (i.e. the average of components and the average LCR over the most recent three months of daily positions, irrespective of the financial reporting schedule). The averages are calculated after the application of any haircuts, inflow and outflow rates and caps, where applicable.

Unweighted figures of HQLA are calculated at market value. Weighted figures of HQLA are calculated after the application of the respective haircuts but before the application of any caps on Level 2B and Level 2 assets.

Unweighted inflows and outflows are calculated as outstanding balances. Weighted inflows and outflows are calculated after the application of the inflow and outflow rates.

In completing the LCR disclosure table, Authorised Firms should follow the instructions below. The numbers in the instructions correspond to the relevant row number in the quantitative disclosure section of the table.

1. Sum of all eligible high-quality liquid assets (HQLA), as defined in section A9.2 in App 9, before the application of any limits, excluding assets that do not meet the operational requirements, and including, where applicable, assets qualifying under alternative liquidity approaches.
2. Retail deposits/PSIAs and deposits/PSIAs from small business customers are the sum of stable deposits/PSIAs, less stable deposits/PSIAs and any other funding sourced from (i) natural

persons and/or (ii) small business customers (as defined by paragraph 231 of the Basel II framework).

3. Stable deposits/PSIAs include deposits/PSIAs placed with a bank by a natural person and unsecured wholesale funding provided by small business customers, defined as “stable” in Rule A9.2.15 in App 9.
4. Less stable deposits/PSIAs include deposits/PSIAs placed with a bank by a natural person and unsecured wholesale funding provided by small business customers, not defined as “stable” in Rule A9.2.15 in App 9.
5. Unsecured wholesale funding is defined as those liabilities and general obligations from customers other than natural persons and small business customers that are not collateralised.
6. Operational deposits or accounts include deposits/PSIAs from bank clients with a substantive dependency on the bank where deposits/PSIAs are required for certain activities (i.e. clearing, custody or cash management activities). Deposits in institutional networks of cooperative banks include deposits of member institutions with the central institution or specialised central service providers.
7. Non-operational deposits/PSIAs are all other unsecured wholesale deposits or PSIAs, both insured and uninsured.
8. Unsecured debt includes all notes, bonds and other debt securities issued by the bank, regardless of the holder, unless the bond is sold exclusively in the retail market and held in retail accounts.
9. Secured wholesale funding is defined as all collateralised liabilities and general obligations.
10. Additional requirements include other off-balance sheet liabilities or obligations.
11. Outflows related to derivative exposures and other collateral requirements include expected contractual derivatives cash flows on a net basis. These outflows also include increased liquidity needs related to: downgrade triggers embedded in financing transactions, derivative and other contracts; the potential for valuation changes on posted collateral securing derivatives and other transactions; excess non-segregated collateral held at the bank that could contractually be called at any time; contractually required collateral on transactions for which the counterparty has not yet demanded that the collateral be posted; contracts that allow collateral substitution to non-HQLA assets; and market valuation changes on derivatives or other transactions.
12. Outflows related to loss of funding on secured debt products include loss of funding on: asset-backed securities, covered bonds and other structured financing instruments; and asset-backed commercial paper, conduits, securities investment vehicles and other such financing facilities.
13. Credit and liquidity facilities include drawdowns on committed (contractually irrevocable) or conditionally revocable credit and liquidity facilities. The currently undrawn portion of these facilities is calculated net of any eligible HQLA if the HQLA: (a) have already been posted as collateral to secure the facilities; or (b) are contractually obliged to be posted when the counterparty draws down the facility.
14. Other contractual funding obligations include contractual obligations to extend funds within a 30-day period and other contractual cash outflows not previously captured under Rule A9.2.15 in App 9.
15. Other contingent funding obligations, as defined in Rule A9.2.15 in App 9.
16. Total cash outflows: sum of rows 2–15.
17. Secured lending includes all maturing reverse repurchase and securities borrowing agreements.

18. Inflows from fully performing exposures include both secured and unsecured loans or other payments that are fully performing and contractually due within 30 calendar days from retail and small business customers, other wholesale customers, operational deposits and deposits held at the centralised institution in a cooperative banking network.
19. Other cash inflows include derivatives cash inflows and other contractual cash inflows.
20. Total cash inflows: sum of rows 17–19.
21. Total HQLA (after the application of any cap on Level 2B and Level 2 assets).
22. Total net cash outflows (after the application of any cap on cash inflows).
23. Liquidity Coverage Ratio (after the application of any cap on Level 2B and Level 2 assets and caps on cash inflows).

Table 13 – Net Stable Funding Ratio Information

Quantitative Disclosures	(a)	In currency amount	Unweighted value by residual maturity				Weighted value	
			No maturity	< 6 months	6 months to < 1yr	≥ 1yr		
		ASF Item						
		1	Capital					
		2	Regulatory Capital					
		3	Other capital					
		4	Retail deposits/PSIAs and deposits/PSIAs from small business customers:					
		5	Stable Deposits/PSIAs					
		6	Less stable deposits/PSIAs					
		7	Wholesale funding:					
		8	Operational deposits / operational accounts					
		9	Other wholesale funding					
10	Liabilities with matching interdependent assets							
11	Other liabilities:							

	12	NSFR derivative liabilities and net liabilities for Shari'a compliant hedging contracts					
	13	All other liabilities and equity not included in the above categories					
	14	Total ASF					
	RSF Item						
	15	Total NSFR high-quality liquid assets (HQLA)					
	16	Deposits/PSIAs held at other financial institutions for operational purposes					
	17	Performing loans and securities (including Shari'a compliant securities):					
	18	Performing loans to financial institutions secured by Level 1 HQLA					
	19	Performing loans to financial institutions secured by non-Level 1 HQLA and unsecured performing loans to financial institutions					
	20	Performing loans to non- financial corporate clients, loans to retail and small business customers, and loans to sovereigns, Central Banks and PSEs, of which:					
	21	With a risk weight of less than or equal to 50%					

			under section 4.12.					
	22		Performing residential mortgages, of which:					
	23		With a risk weight of less than or equal to 50% under section 4.12.					
	24		Securities that are not in default and do not qualify as HQLA, including exchange-traded equities					
	25		Assets with matching interdependent liabilities					
	26		Other Assets					
	27		Physical traded commodities, including gold					
	28		Assets posted as initial margin for derivative contracts/Shari'a compliant hedging contracts and contributions to default funds of CCPs					
	29		NSFR derivative assets					
	30		NSFR derivative liabilities before deduction of variation margin posted					
	31		All other assets not included in the above categories					
	32		Off-balance sheet items					
	33		Total RSF					

		34	Net Stable Funding Ratio (%)	
--	--	----	------------------------------	--

Guidance

The following Guidance is intended to assist firms to complete the above table.

Data must be presented as quarter-end observations.

Both unweighted and weighted values of the NSFR components must be disclosed unless otherwise indicated. Weighted values are calculated as the values after ASF or RSF Factors are applied.

In completing the NSFR disclosure table, Authorised Firms should follow the instructions below. The numbers in the instructions correspond to the relevant row numbers in the quantitative disclosures section of the table.

1. Capital is the sum of rows 2 and 3.
2. Regulatory capital before the application of capital deductions.
3. Total amount of any capital instruments not included in row 2.
4. Retail deposits/PSIAs and deposits from small business customers, as defined in Rule A9.2.15 in App 9.
5. Stable deposits/PSIAs comprise “stable” (as defined in Rule A9.2.15 in App 9) non-maturity (demand) deposits/PSIAs and/or term deposits/PSIAs provided by retail and small business customers.
6. Less stable deposits/PSIAs comprise “less stable” (as defined in Rule A9.2.15 in App 9) non-maturity (demand) deposits/PSIAs and/or term deposits/PSIAs provided by retail and small business customers.
7. Wholesale funding is the sum of rows 8 and 9.
8. Operational deposits/PSIAs: As defined in section A9.2 in App 9, including deposits/PSIAs in institutional networks of cooperative banks.
9. Other wholesale funding including funding (secured and unsecured) provided by non-financial corporate customer, sovereigns, public sector enterprises (PSEs), multilateral and national development banks, central banks and financial institutions.
10. Liabilities with matching interdependent assets.
11. Other liabilities are the sum of rows 12 and 13.
12. In the unweighted cells, report NSFR derivatives liabilities as calculated according to section 9.4 of App 9. There is no need to differentiate by maturities. The weighted value under NSFR derivative liabilities is cross-hatched given that it will be zero after the 0% ASF Factor is applied.
13. All other liabilities and equity not included in above categories.
14. Total ASF is the sum of all weighted values in rows 1, 4, 7, 10 and 11.

15. Total HQLA as defined in section A9.2 in App 9 (encumbered and unencumbered), without regard to LCR operational requirements and LCR caps on Level 2 and Level 2B assets that might otherwise limit the ability of some HQLA to be included as eligible in calculation of the LCR, where:
 - a. encumbered assets include assets backing securities or covered bonds; and
 - b. unencumbered means free of legal, regulatory, contractual or other restrictions on the ability of the bank to liquidate, sell, transfer or assign the asset.
16. Deposits/PSIAs held at other financial institutions for operational purposes as defined in section A9.2 in App 9.
17. Performing loans and securities are the sum of rows 18, 19, 20, 22 and 24.
18. Performing loans to financial institutions secured by Level 1 HQLA, as defined in section A9.2 in App 9.
19. Performing loans to financial institutions secured by non-Level 1 HQLA and unsecured performing loans to financial institutions.
20. Performing loans to non-financial corporate clients, loans to retail and small business customers, and loans to sovereigns, central banks and PSEs.
21. Performing loans to non-financial corporate clients, loans to retail and small business customers, and loans to sovereigns, central banks and PSEs with risk weight of less than or equal to 50% under section 4.12.
22. Performing residential mortgages.
23. Performing residential mortgages with risk weight of less than or equal to 50% under section 4.12.
24. Securities that are not in default and do not qualify as HQLA including exchange-traded equities.
25. Assets with matching interdependent liabilities.
26. Other assets are the sum of rows 27 to 31.
27. Physical traded commodities, including gold.
28. Cash, securities or other assets posted as initial margin for derivative contracts and contributions to default funds of CCPs.
29. In the unweighted cell, report NSFR derivative assets, as calculated according to section A9.4 in App 9. There is no need to differentiate by maturities. In the weighted cell, if NSFR derivative assets are greater than NSFR derivative liabilities (as calculated according to section A9.4 in App 9), report the positive difference between NSFR derivative assets and NSFR derivative liabilities.
30. In the unweighted cell, report derivative liabilities as calculated according to report 20% of derivatives liabilities unweighted value (subject to 100% NSFR).
31. All other assets not included in the above categories.
32. Off-balance sheet items.
33. Total RSF is the sum of all weighted value in rows 15, 16, 17, 25, 26 and 32.

34. Net stable funding ratio (%).

Table 14 – Leverage Ratios

Qualitative Disclosures	(a)	The source of material differences between the bank's total balance sheet assets in their financial statements and on-balance sheet exposures in the common disclosure template in Form B300.
Quantitative Disclosures	(b)	A comparison of the Authorised Firm's total accounting asset amounts and Leverage Ratio exposures using the summary comparison template in Form B300.
	(c)	A breakdown of the main Leverage Ratio regulatory elements using the common disclosure template in Form B300.

Table 15 – Indicators for assessing systemic importance of firms

Quantitative Disclosures	(a)	Cross-jurisdictional activity: (i) cross-jurisdictional claims; and (ii) cross-jurisdictional liabilities.
	(b)	Size – the total of its Leverage Ratio Exposure Measure.
	(c)	Interconnectedness: (i) intra-financial system assets; (ii) intra-financial system liabilities; and (iii) securities outstanding.
	(d)	Substitutability: (i) assets under custody; (ii) payments activity; and (iii) underwritten transactions in debt and equity markets.
	(e)	Complexity: (i) notional amount of over-the-counter (OTC) derivatives;

		(ii) level 3 assets within the meaning of IFRS 7; and (iii) trading and available-for-sale securities.
--	--	---